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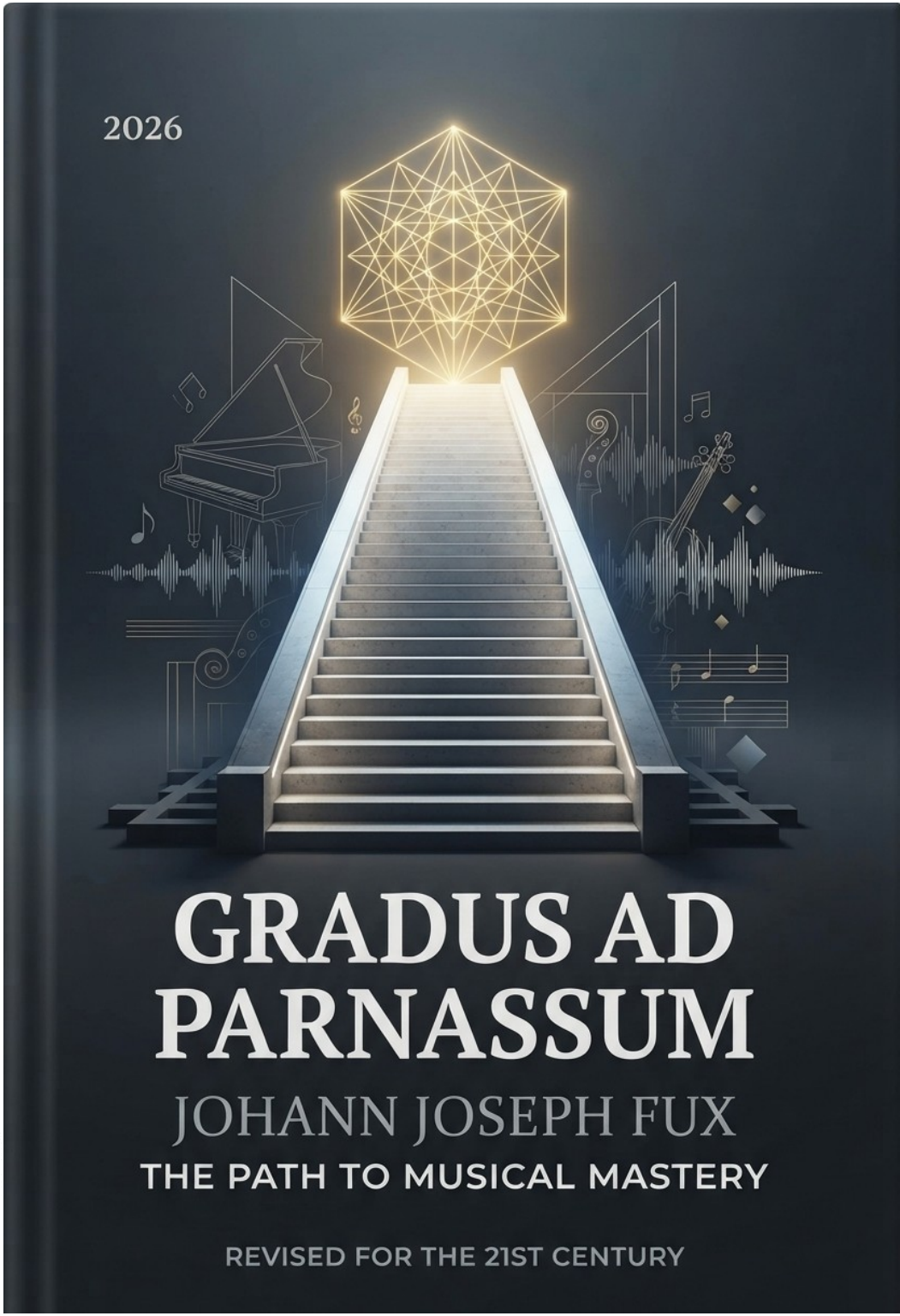

GRADUS
AD
PARNASSUM,
Sive
MANUDUCTIO
AD
COMPOSITIONEM MUSICÆ REGULAREM,
Methodo novâ, ac certâ, nondum antè tam exacto ordine in lucem edita :

Elaborata à

JOANNE JOSEPHO FUX,

Sacræ Cæſareæ, ac Regiæ Catholicæ Majeſtatis CAROLI VI. Romanorum Imperatoris

SUPREMO CHORI PRÆFECTO.

VIENNÆ AUSTRIÆ,
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GRADUS AD PARNASSUM,

or

A Guide
to the
Regular Composition
of Music,

in a new and reliable method, never
before published in such exact order:

Prepared by
JOHANN JOSEPH FUX,

Chief Kapellmeister of

His Sacred Imperial and Royal Catholic Majesty
CHARLES VI,
Emperor of the Romans.

Vienna, Austria,
Printed by Johann Peter Van Ghelen,
Printer to His Sacred Imperial and Royal Catholic Majesty, 1725.

AUGUSTISSIMO, INVICTISSIMO,
AC
POTENTISSIMO PRINCIPI

CAROLO VI.

ROMANORUM
IMPERATORI,
GERMANIÆ, HISPANIÆ,
HUNGARIÆ, BOHEMIÆ
REGI,
ARCHIDUCI AUSTRIÆ, &c. &c.
DOMINO DOMINO CLEMENTISSIMO.

AUGUSTE CÆSAR

SI flumina ad Oceanum, unde originem trahunt, cursum suum reflectunt: si terra rore matutino madefacta eundem humorem, tanquam in grati animi tesseram, ad æthera denuò

To the Most August, Invincible, and Mighty Prince

CHARLES VI,

Emperor of the Romans,
King of Germany, Spain,
Hungary, Bohemia,
Archduke of Austria, etc., etc.,
My Most Clement Lord:
AUGUST CAESAR,

If rivers turn back toward the ocean from which they draw their origin; if the earth, moistened by the morning dew, sends that same moisture back to the heavens

mittit; quò aliàs Opuſculum hoc ſe vertat, niſi ad TE AUGUSTE CÆSAR? Tuum enim eſt jure Dominii; quia partus clientis. Tuum eſt origine; quia Inclytorum Anteceſſorum Tuorum ſub Auſpiciis Muſica mea initium ſumpſit, & incrementum traxit. Tu, excelſe Monarcha, animum addidiſti: Tu ſumptus ſuppeditâſti: clementiſſimo Benignitatis Tuæ annutu in lucem prodit. Quidquid itaque emolumenti ſtudioſa juventus exinde cæperit, totum MAJESTATI TUÆ in acceptis referendum eſt. Quam ob rem non dedignaberis benignis oculis aſpicere, quod omni ex parte tuum eſt: & cujus antè actæ vitæ labores tam clementi aſſenſu comprobâſti (quantæ autem molis eſt, exquiſito, ſelectoque ſapori ſatisfacere Tuo) ejusdem poſtremum fermè, quaſi Cygni cantum, eâdem benignitate Te complexurum eſſe, minimè diffido. Hâc ſpe fretus, ad ſacratos provolutus pedes, emorior

SAC. CÆS. REGIÆQUE CATHOL.
MAJESTATIS TUÆ

Clientum infimus
Joannes Joſephus Fux, Styrus.

grateful acknowledgment—then where else should this little work turn itself, if not to you, August Caesar?

For it is yours by right of dominion, since it is the offspring of your servant. It is yours by origin as well, for under the auspices of your illustrious ancestors my music first began and grew.

You, exalted Monarch, encouraged my spirit; you supplied the means; by the most gracious nod of your benevolence it now comes forth into the light. Whatever benefit the studious youth may gain from it must therefore be credited entirely to Your Majesty.

For this reason, you will not disdain to look kindly upon what is in every respect yours; and since you have approved the labors of my earlier life with such gracious assent (and how great a task it is to satisfy your refined and distinguished taste!), I do not at all doubt that you will embrace with the same kindness this final work—almost like the swan's last song.

With this hope, prostrate at your sacred feet, I remain

The humblest of Your Sacred Imperial and Royal Catholic Majesty's servants,

Johann Joseph Fux of Styria.

PRÆFATIO
AD
LECTOREM.

MIrabuntur fortaſſis nonnulli, cùm tot præſtantiſſimorum Virorum exſtent monumenta, qui de Muſica perquàm doctè, & abundanter ſcripſerunt, cur ego ad hoc ſcribendi genus me contulerim, hoc maximè tempore, quo, Muſicâ ferè arbitrariâ factâ, Compoſitores nullis præceptis, nullisque inſtitutis obſtringi volentes, Legum, ac Scholæ nomen ad mortis inſtar exhorreſcunt; quibus vellem cognita eſſet mens mea: extitiſſe certè permultos, & doctrinâ, & auctoritate claros Scriptores, qui de Muſica ſpeculativa quàm opulentiſſima Scripta reliquerunt, de Muſica autem activa ſatìs parca, nec omnino clara; plerumque quibusdam Paradigmatibus contenti, minimè ſolliciti de invenienda methodo facili, quâ pedetentim tyrones tanquam per ſcalam ſcandere, atque ad artis hujus adeptionem

PREFACE TO THE READER

Perhaps some will wonder, since so many monuments of most excellent men exist—men who have written about music with great learning and in great abundance—why I have devoted myself to this kind of writing, especially at this time, when music has become almost arbitrary, and composers, unwilling to be bound by any rules or institutions, shudder at the very name of "law" and "school" as though it were death itself. To such people I would wish my intention to be known.

Certainly, many writers distinguished both for learning and authority have left behind most abundant works on speculative music, but on practical music they have written rather sparingly, and not altogether clearly. For the most part they were content with certain paradigms, and were not at all concerned with discovering an easy method by which beginners might gradually climb, as if by a ladder, and thus attain mastery of this art.

nem pervenire poſſent. Nec me deterrent aſperrimi Scholæ oſores, neque temporum corruptela.

Ægrotantibus etenim, ac non bonâ valetudine gaudentibus medicina paratur. Quanquam labor meus non eò tendit, nec tantum mihi roboris arrogo, ut quaſi torrenti extra limites præcipitanter erranti curſum inhibere, Compoſitoresque de licentioſa ſcribendi hæreſi ad reſipiſcentiam revocare me poſſe confidam. Per me liceat cuique ſequi ſuum conſilium. Inſtituti mei eſt, præſidium ferre adoleſcentibus ad hanc facultatem adſpirantibus, quorum permultos novi, etiamnum noſco, mirificâ indole præditos, diſciplinæque hujus ſtudio ſummè flagrantes, qui & argento deſtituti, & Magiſtro, deſiderium ſuum, tanquam diuturnam deſperatamque virtutis ſitim explere non valent. Quapropter jam plures ante annos animo ſcrutari cœpi, nulli neque labori, neque ſtudio parcens in reperienda methodo facili, ſimili prorsùs, quâ tenera ætas primò literas noſcere, pòſt ſyllabizare, plures deinde ſyllabas conjungere, poſtremò legere, & ſcribere docetur; nec temerè. Cùm enim eâ uſus in dandis lectionibus advertiſſem, Studioſos parvo temporis ſpatio mirabilem hac in arte progreſſum feciſſe, ratus opem haud contemnendam huic facultati me allaturum, ſi non ſolùm hanc, in gratiam ſtudioſæ juventutis publicam redderem, verùm etiam, quidquid per triginta annorum propè continuam praxim, tribus A. A. Romanorum Imperatoribus inſerviens (de quo duntaxat mihi liceat gloriari) adeptus ſum, cum Republica

Neither the harshest enemies of the schools nor the corruption of the times deters me. For medicine is prepared for those who are sick and not in good health. Yet my work does not aim at this, nor do I claim such strength for myself that I would hope to restrain, as it were, a torrent rushing beyond its bounds, or to recall composers from the heresy of licentious writing to a sound mind.

Let each one follow his own judgment, so far as I am concerned. My purpose is to bring help to young people who aspire to this discipline—many of whom I have known, and still know, endowed with marvelous talent and burning with the greatest zeal for this study, yet lacking both money and a teacher, and thus unable to satisfy their desire, as though it were a long-standing and desperate thirst for excellence.

Therefore, many years ago I began to search in my mind—sparing neither labor nor study—for an easy method, entirely similar to that by which tender youth is first taught to know letters, then to syllabize, then to join several syllables, and finally to read and write. Nor was this done rashly. For when I observed in my teaching that students made remarkable progress in this art in a short time, I thought I would bring no small assistance to this discipline if I not only made this method public for the sake of eager youth, but also faithfully shared with the musical community whatever I have gained through nearly thirty years of continuous practice while serving three Roman Emperors (a fact of which I may be permitted to boast).

blica Muſica fideliter communicarem. Accedit etiam, quod Plato apud Ciceronem monet, ajens: non ſolùm nobis nati ſumus: ortûsque noſtri partem patria vendicat, partem parentes, partem amici. Cæterùm facilè advertes dilecte Philomuſe, hoc Tractatu, ſpeculativæ parti perparùm: activæ verò, utpote magìs neceſſariæ (virtutis enim laus omnis in actione conſiſtit) plus operæ me dediſſe.

Poſtremò ad faciliorem captum, utque veritas magìs eluceſceret, ſecundam partem dialogicè tractandam aſſumpſi, ubi per Aloyſium, Magiſtrum, clariſſimum illud Muſicæ lumen Prænestinum, vel ùt alii volunt, Præeſtinum intelligo, cui totum, quidquid in hoc genere ſcientiæ in me eſt, in acceptis refero, cujusque memoriam, dum vivam, maximo pietatis officio proſequi nunquam deſinam. Joſephi autem nomine Diſcipulum Muſicæ artis addiſcendæ cupidum compello. Noviſſimè ne te offendat ſtyli humilitas; non enim alio, quàm jure poſtliminii mihi arrogo latinitatem; malo etiam intelligi, quàm Orator videri. Vale, fruere, & indulge.

APPENDIX.

SI quid, benevole Lector, à recto ſcribendi ordine alieni deprehenderis, æquo animo te laturum confido, poſtquàm intellexeris, & crebris infirmitatibus, nonnunquam per aliquot menſium, quin imò

There is also this consideration, which Plato, as quoted by Cicero, reminds us of: "We are not born for ourselves alone; our country claims a part of our birth, our parents another part, and our friends another."

Moreover, dear lover of music, you will easily notice that in this treatise I have given little attention to the speculative part, but more to the practical, as being more necessary—for all praise of virtue consists in action.

Finally, for easier understanding and that the truth might shine forth more clearly, I have chosen to treat the second part in dialogue form, where by "Aloysius, the Master," I mean that most illustrious light of music, the Palestrinian—or, as others prefer, the Praenestinian— to whom I attribute all that knowledge of this subject that I possess, and whose memory, as long as I live, I shall never cease to honor with the greatest devotion. By the name "Joseph," I designate the student eager to learn the art of music.

Lastly, let the humility of my style not offend you; for I claim Latin for myself only by the right of restoration. I prefer to be understood rather than to appear an orator. Farewell, enjoy, and be indulgent.

APPENDIX

If, kind reader, you should discover anything that departs from the proper order of writing, I trust you will bear it with a patient spirit, once you understand that frequent illnesses —sometimes lasting several months, indeed even an entire year—and the interruptions of my official duties have often broken the continuity of my work. I had neither the leisure to review what

imò integri anni ſpatium interruptum, & reconvaleſcentem; officii verò munere ſæpiſſimè interpellatum eſſe. De novo ſemper præcedentia revidendi nec otium erat, nec ægro animo imponendum. Inde etiam non obſtante ſedulâ amicorum reviſione nonnulli errores in Litteris, & Muſica irrepsêre; quos ex Erratis in fine Libri annexis corrigendos reperies.

had already been written anew each time, nor was it reasonable to impose such a burden on a weakened mind. Hence it has happened that, despite the diligent corrections of friends, certain errors have slipped into both the text and the music. You will find them listed and corrected in the Errata at the end of the book.

LIBRI

LIBRI PRIMI

CAPUT I.

Uſica nomen genericum eſt, latiſſimè patens, ſpeciesque continens quamplurimas: ùt Muſicam cæleſtem, terreſtrem, naturalem, artificialem, hiſtoricam, &c. Ego brevitati, utilitatique potiùs, quàm curioſitati inſerviens, de artificiali tractare duntaxat propoſitum habeo. Ea duplex eſt: *ſpeculativa* & *practica:* de prima, nempe *ſpeculativa* hoc Libro nobis ſermo erit, eaque ſolummodo, quæ ad pleniorem Praxis adeptionem neceſſaria videbuntur, breviter attingam; ſecundam partem, nempe *practicam* diffuſiùs tractandam ad Librum ſecundum remittam. Cùm autem Muſica noſtra circa ſonum, tanquam objectum ſuum verſetur, de eo priùs agendum eſt. Sit ergo

CAPUT II.

De Sono.

MErſennus *lib.* 1. *Harmon. propo.* 2. *ex mente Ariſtot.* ſonum ait eſſe motum aëris: quod de quovis aëre intelligi utique non poteſt, quia aër ſibi relictus ſinè controverſia movetur, & tamen juxta eundem Ariſtotelem, à Merſenno ibidem citatum: *aër eſt expers ſoni;* unde conſequitur, qualem-

BOOK ONE

CHAPTER I

"Music" is a general name, very broad in scope, and containing many different kinds: celestial music, earthly music, natural, artificial, historical, and so on. Favoring brevity and usefulness rather than curiosity, I intend to treat only of artificial music.

This is twofold: speculative and practical. Of the first, namely speculative music, we shall speak in this book, and I will touch only briefly on those matters that seem necessary for a fuller attainment of practice. The second part, that is, the practical, I will treat more fully in the second book. But since our music concerns itself with sound as its object, we must first speak of sound. Let therefore be

CHAPTER II

On Sound

Mersenne, Harmonics, Book I, Proposition 2, following Aristotle, says that sound is the motion of air. This cannot be understood of just any motion of air, for air left to itself undoubtedly moves, and yet, according to the same Aristotle as cited by Mersenne, air is devoid of sound. From this it follows that not every motion of air is sound, but that air acted upon by some external force—enclosed and compressed—is the efficient cause of sound.

lemcumque aëris motum non eſſe ſonum, ſed aërem vi quadam externâ affectum, incluſum, compreſſumque cauſam eſſe ſoni efficientem. Phyſicorum autem hæc ſit indagatio; Muſicus enim non conſiderat ſonum ùt ſic, ſive (ùt ajunt) in abſtracto, ſed relativè, & ſub certa comparatione ſumptum, quâ ſcilicet ſonus à ſono ratione acuminis & gravitatis differt. Cùm autem comparatio hæc potiſſimùm fiat numerorum adjumento; ideò ſit

CAPUT III.

De Numeris, eorumque Proportionibus, & Differentiis.

PEr Numeros intelligo Elementa Arithmeticæ :
1. 2. 3. 4. 5. 6. 7. 8. 9.

Quibus tanquam ancillis, ad intervalla ſua reperienda, Muſicus utitur, comparando numerum cum numero, eorum diſtantias, differentias, exceſſus, & defectus inquirendo.

Multifariæ dantur numerorum ſpecies. Nos autem relictis aliis, quæ ad noſtrum inſtitutum non faciunt, ſequentes ſolummodo commemorabimus, nempe : *rationales*, *irrationales*; *radicales*, & *irradicales*. *Rationales* ſunt, quæ quodam numero dividi poſſunt, ità ut nihil inde remaneat, ùt 6. cum 2. & 3. : 12. cum 2. 3. 4. & 6. *Irrationales* illæ, quæ non ità metiri, aut dividi poſſunt, ùt 3. 5. 7. 9. reſpectu 2. *Radicales* ſunt illæ, quæ in minores numeros redigi non poſſunt, ſalvâ proportione, quam conſtituunt; *Irradicales* autem è converſo.

Conſtat autem ex *proportionibus*, ſeu comparatione unius quantitatis ad aliam per ſe oriri varias diſtantias, & differentias. Quæritur modò, quid ſit hæc *Proportio ?* Definitur ab Euclide, & aliis, quòd ſit *duarum quantitatum unius ad alteram habitudo in eodem genere propinquo*. *Quantitas*, de qua hìc mentio fit, duplex eſt : *continua*, quæ eſt in quodam corpore, ùt lineâ, aſſere, ſuperficie, &c., & *diſcreta*, quæ reperitur in aliqua multitudine, ùt populo, numeris, &c. Per habitudinem intelligitur relatio illa, quæ habetur de una quantitate ad aliam; dicitur prætereà : *in eodem genere propinquo*; quo cavetur, ne proportio ſtatuatur de quantitate continua ad diſcretam : ùt de linea ad numerum; ſed de linea ad lineam, de numero ad numerum.

Quam-

But this inquiry belongs to the physicists. The musician does not consider sound as such, or "in the abstract," as they say, but relatively, and under a certain comparison—namely, how one sound differs from another in sharpness or gravity (highness or lowness). And since this comparison is chiefly made by means of numbers, therefore let there be

CHAPTER III

On Numbers, Their Proportions, and Their Differences

By "numbers" I mean the elements of arithmetic:
1, 2, 3, 4, 5, 6, 7, 8, 9.
By means of these, as though they were handmaids, the musician discovers his intervals, comparing number with number and examining their distances, differences, excesses, and defects.

There are many kinds of numbers.But leaving aside the others, which do not pertain to our present purpose, we shall mention only the following: rational, irrational, radical, and irradical. Rational numbers are those that can be divided by some number so that nothing remains—for example, 6 by 2 and 3; 12 by 2, 3, 4, and 6. Irrational numbers are those that cannot be measured or divided in this way, such as 3, 5, 7, and 9 with respect to 2.Radical numbers are those that cannot be reduced to smaller numbers without destroying the proportion they constitute; irradical numbers are the opposite.

It is clear that from proportions, or from the comparison of one quantity to another, various distances and differences arise of themselves. The question now is: what is this Proportion? Euclid and others define it as the relation of one quantity to another of the same proximate kind. The quantity spoken of here is twofold: continuous, which exists in some body—such as a line, a beam, a surface, etc.—and discrete, which is found in some multitude, such as a population or numbers. By "relation" is understood that connection which one quantity has to another. It is further said "of the same proximate kind," which prevents proportion from being established between a continuous and a discrete quantity—for example, between a line and a number—but only between line and line, number and number.

Quamvis mediante quantitate continuâ, dimetiendo lineam, chordas, &c. Muſicus propoſitum ſuum conſequi poſſit, quia tamen & iſta dimenſio ultimatim ad numeros eſt reducenda, nempe quot partibus una linea alteram excedat longitudine, quotque ea brevior ſit : ideò de ſola proportione, quæ ex quantitate diſcreta, ſive ex comparatione numeri ad numerum oriri ſolet, hoc loco diſſerendum duxi.

Definitionem proportionis ſuperiùs adductam, generalem, & qualicunque proportioni convenientem eſſe patet : unde ſingularis nulli diſciplinæ præter Muſicam, conveniens ſtatuenda eſt ; ea verò talis eſt : *Proportio Muſicalis* eſt habitudo, ſive relatio numeri ſonori ad numerum ſonorum. Definitio ex ſuprà dictis per ſe clara ulteriori explicatione non indiget.

Duplex eſſe poteſt hæc proportio *æqualitatis*, & *inæqualitatis*. Proportio *æqualitatis* oritur ex comparatione numeri æqualis ad numerum æqualem, ùt 1. ad 1., 2. ad 2., 3. ad 3. &c. Proportio *inæqualitatis* producitur comparatione numeri inæqualis ad inæqualem, exempli gratiâ, 1. ad 2., 3. ad 4., 5. ad 8. &c.

Intervalla Muſices, ſeu conſonantias, & diſſonantias ab ſonorum diverſitate produci conſtat ; inde colligitur, proportiones illas, quarum termini, ſive extrema æqualitate numerorum nituntur, neque aptas eſſe ad producenda intervalla, neque uniſonum eandem ob rationem intervallorum nomine comprehendi poſſe. Aliud judicium de uniſono in praxi formandum eſt, quod exequar, cùm de quovis conſono, & diſſono ſingillatim dicturus ſum. Porrò ex comparatione numeri inæqualis ad inæqualem quamplurima poſſunt exiſtere proportionum genera, quæ Muſicus ad quinque reſtringere conſuevit : *Genus multiplex*, *ſuperparticulare*, *ſuperpartiens*, *multiplex ſuperparticulare*, *multiplex ſuperpartiens*. Unum quodque ex his generibus infinitas propemodum comprehendere ſpecies, Capita ſequentia demonſtrabunt.

CAPUT IV.

De Genere multiplici.

GEnus multiplex eſt *proportio numeri ad numerum, quorum major includit minorem bis, ter, quater, &c.* Si

Although by means of continuous quantity—by measuring a line, strings, etc.—the musician can achieve his purpose, nevertheless this measurement must ultimately be reduced to numbers: namely, by how many parts one line exceeds another in length, and by how many parts the other is shorter. Therefore, I have judged it best to speak here only of proportion arising from discrete quantity, that is, from the comparison of number to number.

The definition of proportion given above is general and applies to any kind of proportion. From it, a definition suited to no discipline except Music must be established; and it is this: Musical proportion is the relation of a sounding number to the number of sounds. This definition, from what has been said, is clear in itself and needs no further explanation.

This proportion may be of two kinds: equality and inequality. The proportion of equality arises from comparing an equal number to an equal number, as 1 to 1, 2 to 2, 3 to 3, etc. The proportion of inequality is produced by comparing an unequal number to an unequal number—for example, 1 to 2, 3 to 4, 5 to 8, etc.

Musical intervals—whether consonances or dissonances—are produced by the diversity of sounds. From this it follows that those proportions whose terms or extremes rest upon numerical equalityare neither suited to produce intervals nor can unison, for the same reason, be included under the name of interval. A different judgment of unison must be formed in practice, which I shall explain when I speak individually of each consonance and dissonance. Moreover, from the comparison of unequal numbers with unequal numbers, many kinds of proportions may arise, which musicians customarily reduce to five: the multiple, the superparticular, the superpartient, the multiple-superparticular, and the multiple-superpartient. Each of these genera contains, as it were, an infinite number of species, as the following chapters will show.

CHAPTER IV

On the Multiple Genus

The multiple genus is the proportion of one number to another, in which the greater contains the lesser twice, three times, four times, and so on.

minor numerus comprehendatur à majori bis, vocabitur *dupla*; ſi ter, *tripla*; ſi quater, *quadrupla*; & ſic in infinitum. Vide ſis, quarumdam Specierum exempla.

Dupla :

2. 4. 6. 8. 10. 12. &c.
1. 2. 3. 4. 5. 6,

Tripla :

3. 6. 9. 12. 15. &c.
1. 2. 3. 4. 5.

Quadrupla :

4. 8. 12. 16. 20. &c.
1. 2. 3. 4. 5.

Hoc modo infinitæ poſſunt formari hujus generis ſpecies, ùt experienti patebit.

CAPUT V.

De ſecundo Proportionum Genere.

QUod *ſuperparticulare* vocatur. Antequam hoc genus explicandum aggrediar, exponendum eſt, quid ſit pars *aliquota*, & pars *aliquanta*. Pars *aliquota*, quæ etiam appellatur multiplicativa, eſt, quæ multiplicata ſemper redintegrat ſuum totum. E. G. 3. eſt pars aliquota numerorum 6. 9. 12. 15. 18., quia 3. ſumptus bis efficit adæquatè 6. ſuum totum; idem ternarius multiplicatus in ſe, exhibet integrè 9. ſuum totum; & ſic de cæteris. Pars *aliquanta*, ſive aggregativa eſt, quæ ſumpta quocunque modo, non adæquat ſuum totum, ſed ſuperet illud, vel non attingat, neceſſe eſt. E. G. 3. eſt pars aliquanta numeri 5.; quia 5. continet 3. ſemel, & inſuper duas ejus partes, id eſt binarium, qui ſemel ſumptus, non ſufficit ad redintegrationem ternarii, ſumptus bis, excedit illum; quod requiritur ad eſſentiam partis aliquantæ, ſive aggregativæ.

His ſuppoſitis dico : Genus ſuperparticulare eſt *Proportio, cujus numerus major continet minorem ſemel, & inſuper ejus par-*

If the lesser number is contained in the greater twice, the proportion is called double; if three times, triple; if four times, quadruple; and so on to infinity. See, if you please, examples of some of these species.

Double:
2. 4. 6. 8. 10. 12.
1. 2. 3. 4. 5. 6. etc.

Triple:
3. 6. 9. 12. 15.
1. 2. 3. 4. 5. etc.

Quadruple:
4. 8. 12. 16. 20.
1. 2. 3. 4. 5. etc.

In this way infinitely many species of this kind may be formed, as experience will show.

CHAPTER V

On the Second Genus of Proportions

This is called the superparticular genus. Before I proceed to explain this genus, it is necessary to set forth what is meant by an aliquot part and an aliquant part. An aliquot part, which is also called a multiplicative part, is one which, when multiplied, always restores its whole. For example, 3 is an aliquot part of the numbers 6, 9, 12, 15, 18, because 3 taken twice makes exactly 6, its whole; the same 3, multiplied by itself, yields exactly 9, its whole; and so with the others.

An aliquant part, or aggregative part, is one which, taken in any manner, does not equal its whole, but must either exceed it or fall short of it. For example, 3 is an aliquant part of the number 5, because 5 contains 3 once, and in addition two of its parts—that is, the number 2—which, taken once, does not suffice to restore the whole of 3, and taken twice exceeds it; which is required for the nature of an aliquant or aggregative part.

With these points established, I say: the superparticular genus is a proportion in which the greater number contains the lesser once and in addition an aliquot part of it.

partem aliquotam. E. G. in proportione hac $\frac{3}{2}$. Seſquialtera numerus major comprehendit minorem ſemel, & ſimul ejus partem aliquotam ; quod per diviſionem majoris numeri cum minore clarè patet $\frac{3}{2}.|1\frac{1}{2}.$ ubi ternarius continet binarium, & inſuper ejus partem dimidiam, quæ bis ſumpta efficit adæquatè binarium, ſuum totum.

Infinitæ hujus generis poſſunt eſſe ſpecies, quæ pro diverſitate numerorum etiam diverſas ſortiuntur denominationes : ùt *Proportio ſeſquialtera, ſeſquitertia, ſeſquiquarta, ſeſquiquinta, ſeſquiſexta, &c.*

Antequam ad formandi hujus generis modum me accingam, ſcire neceſſe eſt, quid ſibi velit particula *Seſqui :* quæ vox nempe medietatem alicujus totius ſignificat. Hinc patet, hanc particulam nulli alteri hujus generis ſpeciei apponi poſſe, nisì ſeſquialteri, in qua ſola numerus major includit minorem ſemel, & inſuper ejus dimidiam partem ; nam in ſeſquitertia proportione $\frac{4}{3}.$ major numerus continet minorem quidem ſemel, verùm non & dimidiam, ſed tertiam duntaxat ejus partem, $\frac{4}{3}.|1\frac{1}{3}.$ In hac proportione $\frac{5}{4}.$ pars ex diviſione reſtans, non dimidia, ſed quarta pars eſt, $\frac{5}{4}.|1\frac{1}{4}.$, & ſic in omnibus aliis ſpeciebus. Quia tamen particulam hanc omnibus hujus generis ſpeciebus præponi, uſu, vel potiùs abuſu jam receptum eſt, ad evitandam confuſionem nihil immutandum conſultiùs exiſtimavi.

Formatur autem genus iſtud facili hoc modo. Accipiantur duo numeri relictâ unitate, eo, quo in tabella Pythagorica ordine ſe ſequuntur, ponendo majorem ſuprà, & minorem infrà, ùt : $\frac{3}{2}.\ \frac{4}{3}.\ \frac{5}{4}.\ \frac{6}{5}.\ \frac{7}{6}.\ \frac{9}{8}.$ &c. En ſex ſpecie diverſas rationes, ſive proportiones ; quarum prima *ſeſquialtera*, ſecunda *ſeſquitertia*, tertia *ſeſquiquarta*, quarta *ſeſquiquinta*, quinta *ſeſquiſexta*, ſexta *ſeſquioctava* vocatur, & quarum quælibet denominationem ſuam accipit à ſuo denominatore, ſive minore termino infrà poſito.

Omnes rationes ſuis terminis radicalibus conſtantes etiam extra terminos radicales conſiſtere poſſunt, quod duplicando rationis terminos efficitur. E. G. duplicentur termini radicales rationis ſeſquialterius (nisì Grammaticorum legibus neglectis, quod in vocibus Technicis ſolemus, *ſeſquialteræ* dicere libeat) $\frac{3}{2}.$ hac ratione $\frac{6}{4}.$, quæ ratio eſt ejusdem ſeſquialteræ extra terminos ſuos radicales. Hæc ratio rurſus duplicata exhi-

For example, in the proportion 3/2, called sesquialter, the greater number contains the lesser once and also an aliquot part of it; which is clearly shown by dividing the greater by the lesser: 3/2 = 1 1/2. Here the number 3 contains 2 once, and in addition its half-part, which taken twice makes exactly 2, its whole.

There can be infinitely many species of this genus, which, according to the diversity of numbers, also receive diverse names: the sesquialter proportion, sesquitertian, sesquiquartan, sesquiquintan, sesquisextan, etc.

Before I set myself to the method of forming this genus, it is necessary to know what the particle "sesqui" signifies. This word denotes the half of some whole. Hence it is clear that this particle can properly be applied to no species of this genus except the sesquialter, in which alone the greater number contains the lesser once and in addition its half-part. For in the sesquitertian proportion 4/3, the greater number indeed contains the lesser once, but not its half, only its third part: 4/3 = 1 1/3. In the proportion 5/4, the part remaining from the division is not a half, but a fourth: 5/4 = 1 1/4; and so in all the species of this kind. However, because it has now become customary—indeed, by misuse—to prefix this particle to all the species of this genus, I have judged it more prudent to change nothing, in order to avoid confusion.

This class is formed in the following easy way. Let two numbers be taken, leaving aside the unit, in the order in which they follow one another in the Pythagorean table, placing the greater above and the smaller below, as: 3/2, 4/3, 5/4, 6/5, 7/6, 9/8, etc. Here are six ratios of different kinds, or proportions: the first is called sesquialtera, the second sesquitertia, the third sesquiquarta, the fourth sesquiquinta, the fifth sesquisexta, the sixth sesquioctava; and each of them receives its name from its denominator, that is, from the smaller term placed below.

All ratios that consist of their radical terms can also exist outside their radical terms, which is accomplished by doubling the terms of the ratio. For example, let the radical terms of the sesquialteral ratio (unless, neglecting the laws of grammarians—as we are accustomed to do in technical vocabulary—it pleases us to say sesquialtera) 3/2 be doubled. Thus we obtain 6/4, which is the same sesquialteral ratio outside its radical terms. This ratio, doubled again, yields

bet eandem ſpeciem ſeſquialteræ, $\frac{12}{8}$. quæ quodam reſpectu ſpecies compoſitæ dici poſſunt, quod idem de aliis hujus generis ſpeciebus intelligendum eſt. En Exempla.

Proportio *ſeſquialtera* ſuis terminis radicalibus conſtans.

3.
2.

Eadem proportio extra terminos radicales.

6. 12. 24. &c.
4. 8. 16.

Proportio *ſeſquitertia* terminis radicalibus conſtans.

4.
3.

Eadem proportio extra terminos radicales.

8. 16. 32. &c.
6. 12. 24.

Proportio *ſeſquiquarta* radicalis.

5.
4.

Eadem proportio extra terminos radicales.

10. 20. 40. &c.
8. 16. 32.

Proportio *ſeſquiquinta* radicalibus terminis.

6.
5.

Eadem proportio extra terminos radicales.

12. 24. 48. &c.
10. 20. 40.

Id quod de infinitis aliis ſpeciebus, obſervato eodem ſuprà dicto aſcendendi modo, intelligendum eſt.

CAPUT VI.

De tertio Proportionis Genere.

TErtium Proportionum genus, ſuperpartiens dicitur, *cujus major terminus comprehendit minorem ſemel, & inſuper aliquot partes ejus non aliquotas, ſed aliquantas, ſeu aggregativas.*

Pro diverſitate medii termini differentialis, qui intervenire poteſt inter hujus generis proportiones, variæ etiam oriuntur ſpecies, & denominationes. Si terminus medius differen-

the same species of sesquialtera, 12/8. These may in a certain sense be called composite species, and the same is to be understood of the other species of this class. Here are examples.

The sesquialteral proportion consisting of its radical terms:

3
2

The same proportion outside its radical terms:

6, 12, 24
4, 8, 16, etc.

The sesquitertial proportion consisting of its radical terms:

4
3

The same proportion outside its radical terms:

8, 16, 32
6, 12, 24, etc.

The sesquiquartal proportion in its radical terms:

5
4

The same proportion outside its radical terms:

10, 20, 40
8, 16, 32, etc.

The sesquiquintal proportion in its radical terms:

6
5

The same proportion outside its radical terms:

12, 24, 48
10, 20, 40, etc.

The same is to be understood of infinitely many other species, following the same method of ascent described above.

Chapter VI.

On the Third Kind of Proportion.

The third kind of proportion is called superpartient.

In this kind of proportion, the larger term contains the smaller term once, and in addition contains some number of its parts—not aliquot parts (i.e., not whole-number divisors), but non-aliquot, "aggregative" parts.

Different species and names of this proportion arise depending on the differential term, the number of parts by which the larger exceeds the smaller.

rentialis erit binarius, proportio ſuper*bi*partiens dicenda erit, ùt in hac proportione $\frac{5}{3}$. 2., ubi terminus differentialis binarius eſt.

Si terminus differentialis fuerit ternarius, ùt in hac proportione $\frac{8}{5}$.3. proportio ſuper*tri*partiens dicetur.

Sì terminus differentialis fuerit quaternarius, exempli gratiâ $\frac{9}{5}$.4. ſuper*quatri*partiens vocabitur proportio, & ſic de innumeris aliis.

In formando hoc genere ſequens norma tenenda eſt: accipiatur ex tabella Pythagorica numerus tertius, qui eſt 3. pro minore proportionis termino, & ommiſſo numero 4. naturali ordine hunc proximè ſequente, aſſumatur quinarius pro majori proportionis termino, hac ratione : $\frac{5}{3}$. & vocabitur ſuperbipartiens tertias, quæ prima ſpecies eſt hujus generis.

Particula *ſuper* indicat numerum minorem proportionis à majori comprehendi ſemel totum, & inſuper aliquas ejus partes. Quid autem per particulam *bi* intelligatur, ex paulò antè dictis conſtat. Porrò ad pleniorem intelligentiam hujus generis diverſarum ſpecierum, denominationumque juvat dividere majorem proportionis terminum cum ſuo minore, E. G. $\frac{5}{3}$.| $1\frac{2}{3}$., ubi fractionis numerator, nempe binarius indicat terminum differentialem proportionis, ſeu quòd numerus major proportionis contineat ſuum minorem ſemel, & inſuper ejus duas partes : denominator autem fractionis ternarius docet, quales ſint illæ partes ſupercontentæ, nempe duæ partes tertiæ ; ideo modò adducta proportio $\frac{5}{3}$. rectè vocatur ſuperbipartiens tertias. Ex quo clarè patet, quòd fractionis numerator indicet terminum differentialem proportionis, quotque partes minoris termini præter totum ſupercontineat major ; & ejus denominator qualitatem partium minoris termini proportionis ex diviſione reſtantium. Videamus modò varias hujus generis ſpecies, & denominationes, quibus veritas hucuſque dictorum magìs eluceſcat.

Proportionis ſuperbipartientis tertias ſpecies, ſuis terminis radicalibus conſtans.

5.
3.

Eædem proportiones extra terminos radicales.

10. 15. 20.
6. 9. 12.

• If the differential term is 2, the proportion is called superbipartient, as in the proportion 5/3, where the differential term is 2.
• If the differential term is 3, as in 8/5, the proportion is called supertripartient.
• If the differential term is 4, as in 9/5, the proportion is called superquadripartient, and so on for infinitely many others.

How to Form This Kind of Proportion: Take from the Pythagorean table the third number, which is 3, as the smaller term.Skip the next natural number (4), and take the following one, 5, as the larger term. Thus we obtain the proportion 5/3, which is called superbipartient of thirds, the first species of this genus.

The prefix super- indicates that the smaller number is contained once in the larger, plus some of its parts. The prefix bi- indicates that the larger contains two extra parts of the smaller.

To understand this more clearly, divide the larger term by the smaller:
5 ÷ 3 = 1 + 2/3

Here:
• The numerator 2 of the fraction 2/3 is the differential term: the larger exceeds the smaller by two parts.
• The denominator 3 tells us what kind of parts these are: thirds.

Thus 5/3 is rightly called superbipartient thirds.
The numerator always shows how many parts of the smaller term are added beyond the whole.
The denominator shows what kind of parts they are.

Examples of Superbipartient Thirds

Radical (fundamental) form:
5 : 3
Equivalent forms outside the radical terms:
10 : 6
15 : 9
20 : 12

Fortaſſis nonnemini videri poſſet, has proportiones ſuperiori nempe $\frac{5}{3}$.(2. minimè æquivalentes eſſe, cùm in illa terminus differentialis ſit binarius, & partes ex diviſione remanentes ſint tertiæ, hujus verò $\frac{10}{6}$.4. terminus differentialis ſit quaternarius & partes ex diviſione reſtantes ſint ſextæ; verùm hoc minimè obeſſe, quin dictæ extra terminos radicales conſiſtentes proportiones ſint ejuſdem ſpeciei cum prima, E. G. $\frac{5}{3}$. per auream reductionis regulam demonſtratur.

Quæratur numerus, qui utrumque proportionis terminum dividendo, totum exhauriat, nullâ parte remanente. E. G. hujus proportionis $\frac{10}{6}$.(2. communis diviſor eſt numerus binarius, quia dividit & numerum denarium, & ſenarium, ut nihil remaneat, ex decem producit 5., ex 6. 3., quæ proportio eſt ſuperbipartiens *tertias* numeris radicalibus conſtans, quod idem in reliquis uſuvenit.

(3	(4
15.	20.
9.	12.
5.	5.
3.	3.

Hac ratione omnes proportiones extra terminos radicales ad proportionem terminis radicalibus conſtantem reduci poſſunt, tanquam ad ſuam originem, ùt ſequentes ſpecies ulteriùs manifeſtabunt.

Proportio ſuperbipartiens *quintas* ſuis terminis radicalibus conſtans.

7.
5.

Eadem ſpecies extra terminos radicales.

(2	(3
14.	21.
10.	15.
7.	7.
5.	5.

Proportio ſuperbipartiens *ſeptimas* radicalis.

9.
7.

Eadem

Perhaps it might seem to someone that these proportions are by no means equivalent to the upper one, namely 5:3, since in that one the differential term is two, and the parts remaining from the division are thirds, whereas in this one, 10:6, the differential term is four and the parts remaining from the division are sixths; but this by no means prevents the said proportions existing outside their radical (lowest) terms from being of the same species as the first, which is demonstrated, for example, for 5:3 by the golden rule of reduction.

Let a number be sought which, by dividing both terms of the proportion, exhausts the whole, with no part remaining. For example, the common divisor of this proportion 10:6 is the number two, because it divides both the number ten and the number six so that nothing remains; from ten it produces 5, and from six it produces 3, which proportion is a superbipartient third consisting of radical numbers, which same thing happens in the rest.

Multiplier: (3)	Multiplier: (4)
15	20
9	12
—	—
5	5
3	3

By this method, all proportions outside their radical terms can be reduced to a proportion consisting of radical terms, as if to their origin, as the following species will further manifest.

The superbipartient fifth proportion consisting of its radical terms.

7
5

The same species outside its radical terms.

Multiplier: (2)	Multiplier: (3)
14	21
10	15
—	—
7	7
5	5

The radical superbipartient seventh proportion.

9
7

Eadem extra terminos radicales.

(2 (3
18. 27.
14. 21.
9. 9.
7. 7.

Proportio ſuperbipartiens *nonas* intrà terminos radicales.

11.
9.

Eadem ſpecies extra terminos radicales.

22. 33.
18. 27.

Proportio ſuperbipartiens *undecimas* terminis radicalibus conſtans.

13.
11.

Eadem Proportio extra terminos radicales.

26. 39.
22. 33.

Hæc ſufficiant de Proportione ſuperbipartiente: progrediamur ad Proportionem ſupertripartientem.

Proportio *ſupertripartiens* quartas terminis radicalibus conſtans.

7.
4.

Eadem ſpecies extra terminos radicales.

14. 21.
8. 12.

Proportio ſupertripartiens *quintas* intrà radicales.

8.
5.

Eadem Proportio extra terminos radicales.

16. 24.
10. 15.

Proportio ſupertripartiens *ſeptimas* in terminis radicalibus.

10.
7.

Eadem Proportio extra terminos radicales.

20. 30.
14. 21.

The same outside its radical terms.
Multiplier: (2) (3)
18 27
14 21
9 9
7 7
The superbipartient ninths proportion within its radical terms.
11
9
The same species outside its radical terms.
22 33
18 27
The superbipartient elevenths proportion consisting of its radical terms.
13
11
The same proportion outside its radical terms.
26 39
22 33
Let these suffice concerning the superbipartient proportion: let us proceed to the supertripartient proportion.
The supertripartient fourths proportion consisting of its radical terms.
7
4
The same species outside its radical terms.
14 21
8 12
The supertripartient fifths proportion within radical terms.
8
5
The same proportion outside its radical terms.
16 24
10 15
The supertripartient sevenths proportion in radical terms.
10
7
The same proportion outside its radical terms.
20 30
14 21

Proportio ſuperquatripartiens *quintas* terminis radicalibus.

9.
5.

Eadem Proportio extra terminos radicales.

18. 27.
10. 15.

Proportio ſuperquatripartiens *ſeptimas* terminis radicalibus conſtans.

11.
7.

Eadem Proportio extra terminos radicales.

22. 33.
14. 21.

Proportio ſuperquatripartiens *nonas* in terminis radicalibus.

13.
9.

Eadem Proportio extra radicales terminos.

26. 39.
18. 27.

Hìc in memoriam revocanda ſunt ea, quæ ſuprà ad proportionem ſuperbipartientem tertias de aurea reductionis regula diximus, quâ experimentum inſtituitur, an quædam proportio extra terminos radicales conſtans ſit ejuſdem ſpeciei cum ſua proportione originali, ſive terminis radicalibus conſtante; quæ regula quidem locum habet in proportionibus, quarum uterque terminus numeris rationalibus conſtat, ſive quæ per communem quendam diviſorem commenſurari poſſunt; quia autem contingit, ut quædam proportiones extra terminos radicales numeris irrationalibus conſtent, eæque ob rationem ad originem ſuam per communem diviſorem reduci non poſſint, ùt in his proportionibus : $\frac{22}{14}:\frac{33}{21}$ ſuperquatripartientibus ſeptimas, alia experimenti regula, quæ cruciata vocari ſolet, adhibenda erit.

Ponatur quædam proportio terminis ſuis radicalibus, & eadem proportio extra terminos radicales, numerisque irrationalibus conſtans, E. G. ſupradicta ſuperquatripartiens ſeptimas. 11. 33.
7. 21.

Multiplicetur antecedens rationis, ſuis terminis radicalibus conſtantis cum ſequente rationis extra terminos radicales poſitæ,

The superquadripartient fifths proportion consisting of its radical terms.

9
5

The same proportion outside its radical terms.

18 27
10 15

The superquadripartient sevenths proportion consisting of its radical terms.

11
7

The same proportion outside its radical terms.

22 33
14 21

The superquadripartient ninths proportion consisting of its radical terms.

13
9

The same proportion outside its extra-radical terms.

26 39
18 27

Here we must recall to memory those things which were said above regarding the superbipartient thirds proportion concerning the rule of golden reduction, by which an experiment is undertaken to see whether a certain proportion existing outside its radical terms is of the same species as its original proportion, or the one consisting of radical terms. This rule indeed holds true in proportions where both terms consist of rational numbers, or those which can be measured through some common divisor. Because, however, it happens that certain proportions outside their radical terms consist of irrational numbers, and for that reason cannot be reduced to their origin through a common divisor—as in these proportions: 22/14 : 33/21 superquadripartient sevenths—another rule of experiment, which is usually called the cross-rule, will have to be applied.

Let a certain proportion be set up in its radical terms, and the same proportion outside its radical terms, consisting of irrational numbers, for example, the above-mentioned superquadripartient sevenths.

11 \ / 33
7 / \ 21

Let the antecedent of the preceding ratio, consisting of its radical terms, be multiplied with the consequent of the following ratio placed outside its radical terms,

ſitæ ; item multiplicetur hujus antecedens cum ſequente rationis, ſuis terminis radicalibus conſtantis : ſi ambæ hâc multiplicatione productæ Summæ fuerint numero æquales, evidens ſignum erit, dictas duas proportiones ejuſdem eſſe ſpeciei. En experimentum.

11. 33.
7. 21.

231. 231.

Quod de omnibus aliis proportionibus, quæ numeris irrationalibus conſtant, intelligendum puta.

CAPUT VII.

De quarto Proportionis Genere, multiplex ſuperparticulare nuncupato.

GEnus hoc compoſitum tot comprehendit ſpecies, quot ejus ſimplex ſuperparticulare; eſtque *Proportio, cujus terminus major includit minorem bis, ter, quater, &c., & ſimul unam ejus partem aliquotam ;* & conſtruitur ex ſpeciebus generis ſuperparticularis ſui ſimplicis, hoc modo. Accipiatur E. G. prima generis ſuperparticularis ſpecies ſuis numeris radicalibus conſtans, nempe ſeſquialtera $\frac{3}{2}$., adde numerum minorem majori, & habebis 5. ; deinde numerum minorem ejuſdem ſeſquialteræ, nempe 2. ſuppone majori ; & exiſtet ſequens proportio $\frac{5}{2}$., quæ eſt prima ſpecies hujus generis, vocaturque dupla ſeſquialtera ; quia ejus numerus major continet minorem bis, & inſuper unam ejus partem aliquotam, ùt ex diviſione conſtat $\frac{5}{2}$. | $2\frac{1}{2}$. Videantur ſpecies hujus in exemplis.

Dupla ſeſquialtera terminis radicalibus conſtans.

5.
2.

Eadem Proportio extra terminos radicales.

10. 15. 20. &c.
4. 6. 8.

Tripla ſeſquialtera.

7. 14. 21.
2. 4. 6.

likewise, let the antecedent of this [second ratio] be multiplied with the consequent of the following ratio consisting of its radical terms: if both sums produced by this multiplication are equal in number, it will be an evident sign that the said two proportions are of the same species. Behold the experiment.

11 \ / 33
7 / \ 2
231 231

think the same is to be understood for all other proportions that consist of irrational numbers.

CHAPTER 7.

On the Fourth Kind of Proportions, called multiple superparticular.

This composite genus contains as many species as its simple superparticular; and it is a Proportion in which the greater term includes the lesser twice, three times, four times, etc., and also one aliquot part of it. It is constructed from the species of the simple superparticular genus in the following way. Let, for example, the first species of the superparticular genus be taken with its radical numbers, namely the sesquialteral ratio 3/2. Add the lesser number to the greater, and you will have 5; then place the lesser number of that same sesquialteral ratio, namely 2, under the greater; and the resulting proportion will be 5/2, which is the first species of this genus, and is called the double sesquialteral; because its greater number contains the lesser twice, and in addition one aliquot part of it, as is clear from the division 5/2 = 2 1/2. Let the species of this genus be seen in the examples.

Double sesquialteral, consisting of radical terms.

5
2

The same proportion outside the radical terms.

10 15 20
4 6 8 etc.

Triple sesquialteral.

7 14 21
2 4 6

Quadrupla ſeſquialtera.

9. 18. 27.
2. 4. 6.

Si numerum minorem à majore comprehendi quinquies, ſexies, contingat, cum una parte ejus aliquota, quintupla, vel ſexdupla, ſeſquialtera erit proportio.

Secunda hujus generis ſpecies, *dupla, ſeſquialtera* appellata, terminis radicalibus conſtans.

7.
3.

Species hæc eo prorsùs modo, quo prima hujus generis ſpecies conficitur, addendo terminos radicales proportionis ſuperparticularis ſeſquitertiæ, $\frac{4}{3}$. & dabit 7. ; infrà hunc ponatur terminus minor dictæ proportionis ſeſquitertiæ, nempe 3. & habebitur ſequens proportio $\frac{7}{3}$., quæ operatio in conficiendis omnibus aliis hujus generis ſpeciebus adhibenda erit.

Eadem Proportio extra terminos radicales.

14. 21.
6. 9.

Dupla ſeſquiquarta terminis radicalibus conſtans.

9.
4.

Eadem Proportio extra terminos radicales.

18. 27.
8. 12.

Dupla ſeſquiquinta terminis radicalibus conſtans.

11.
5.

Eadem Proportio extra terminos radicales.

22. 33.
10. 15.

Tripla ſeſquialtera.

7. 14. 21.
2. 4. 6.

Tripla ſeſquitertia.

10. 20. 30.
3. 6. 9.

Tripla ſeſquiquarta.

13. 26. 39.
4. 8. 12.

Tri-

Quadruple sesquialteral.

9 18 27
2 4 6

If the lesser number happens to be contained in the greater five times, six times, together with one aliquot part of it, the proportion will be quintuple or sextuple sesquialteral.

The second species of this genus, called double sesquitertial, consisting of radical terms.

7
3

This species is formed in exactly the same way as the first species of this genus, by adding the radical terms of the superparticular proportion sesquitertial, 4/3, which gives 7; beneath this is placed the lesser term of the said sesquitertial proportion, namely 3; and the resulting proportion is 7/3. This operation is to be used in constructing all the other species of this genus.

The same proportion outside the radical terms.

14 21
6 9

Double sesquiquartal, consisting of radical terms.

9
4

The same proportion outside the radical terms.

18 27
8 12

Double sesquiquintal, consisting of radical terms.

11
5

The same proportion outside the radical terms.

22 33
10 15

Triple sesquialteral.

7 14 21
2 4 6

Triple sesquitertial.

10 20 30
3 6 9

Triple sesquiquartal.

13 26 39
4 8 12

Tripla ſeſquiquinta.
16. 32. 48.
5. 10. 15.

Ubi animadvertendum eſt, primum exemplum cujuſvis triplæ hìc poſitæ conſtare terminis radicalibus, reliqua duo primum immediatè ſequentia eſſe extra terminos radicales. Non morabor hìc diutiùs, qui enim hucuſque dicta animo benè complexus fuerit, minimè laborabit in conficienda Quadrupla, Quindupla, Sexdupla, &c. earumque Speciebus, cùm intrà, tum extra terminos radicales reperiendis.

CAPUT VIII.

De quinto Proportionis Genere, multiplex ſuperpartiens dicto.

IN Genere hoc *major terminus comprehendit minorem bis, ter, quater, &c, cum aliqua ejus parte aliquanta*, componiturque ex terminis radicalibus proportionis, generis ſuperpartientis ſui ſimplicis; illius primæ ſpeciei, terminos radicales addendo. E. G. $\frac{5}{3}$. qui per additionem faciunt 8., minore verò termino dictæ proportionis infrà majorem poſito, conſtituunt hanc proportionem $\frac{8}{3}$., primam ſpeciem hujus generis, & vocatur dupla ſuperpartiens tertias; ſequuntur plura exempla ſpeciem hanc declarantia.

Dupla ſuperbipartiens *tertias.*
8. 16. 24.
3. 6. 9.
Dupla ſuperbipartiens *quintas.*
12. 24. 36.
5. 10. 15.
Dupla ſuperbipartiens *ſeptimas.*
16. 32. 48.
7. 14. 21.

Secunda Species.

Dupla ſupertripartiens vocatur, reliquam denominationem ex partibus reſtantibus ſumens.

Triple sesquiquintal.
16 32 48
5 10 15

It should be noted that the first example of each triple proportion placed here consists of the radical terms, while the two immediately following are outside the radical terms. I will not linger longer here; for whoever has well grasped what has been said thus far will have no difficulty in constructing the quadruple, quintuple, sextuple, etc., and their species, both within and outside the radical terms.

CHAPTER VIII.

On the fifth kind of Proportion, called the multiple Superpartient.

In this genus the greater term contains the lesser twice, three times, four times, etc., together with some aliquant part of it; and it is constructed from the radical terms of the proportion belonging to the simple superpartient genus, by adding the radical terms of its first species. For example, 5/3, which by addition make 8; and by placing the lesser term of that proportion beneath the greater, they form the proportion 8/3, the first species of this genus, and it is called the double superpartient of thirds. Several examples follow to illustratethis species.

Double superbipartient of thirds.
8 16 24
3 6 9

Double superbipartient of fifths.
12 24 36
5 10 15

Double superbipartient of sevenths.
16 32 48
7 14 21

Second Species.

It is called the double supertripartient, taking the remainder of its name from the parts that remain.

Cùm omnia Muſices intervalla ex diviſione octavæ, tanquam genitricis vel immediatè, vel mediatè ortum habeant, ab hac initium fieri, ordo poſtulat; ſit ergo

CAPUT IX.

De Diviſione.

OFficium Diviſionis eſt, dividere totum in ſuas partes: id, quod efficit vulgaris diviſio in numeris ſimplicibus. In dividendis rationibus, ſive numeris relatis, alia diviſio, ùt jam dixi, adhibenda eſt, cujus munus eſt ex una proportione efficere duas, vel plures, id, quod triplici modo fieri poteſt; ideo triplex etiam ſtatuenda eſt Diviſio: *Arithmetica*, *Harmonica*, & *Geometrica*.

De Diviſione Arithmetica.

Diviſio *Arithmetica* fortaſſis ità dicta, quia numeri per eam producti naturali ordine, nullo interjecto intervallo ſe conſequuntur; fit interpoſitione medii termini, æqualiter ab utroque extremo diſtantis, ità ut efficiat differentias æquales, & proportiones inæquales, majorem proportionem inter numeros minores, minorem inter majores conſtituendo. Paradigma rem clariorem reddet. Sit proportio dupla hac ratione: 4. 2., ternarius, qui inter hæc duo extrema intrare poteſt, diviſor arithmeticus eſt; dividit enim extrema ità, ut æqualiter ab utroque diſtent, facitque differentias æquales: 4. 3. 2. etenim quo intervallo quaternarius diſtat à ternario, nempe unitate, eodem diſtat ternarius à binario, exhibetque proportiones inæquales, ſeſquitertiam, & ſeſquialteram: ſeſquitertiam inter 4. & 3. quæ proportio minor eſt ſeſquialterâ, & ſeſquialteram inter 3. & 2., quæ major eſt ſeſquitertiâ.

Seſquitertia. Seſquialtera.

4. 3. 2.

1. 1.

Differentiæ æquales.

Quid faciendum, ſi extrema alicujus rationis ità conjuncta reperiantur, ut medium recipere nequeant? E. G. 2. 1. ℟. extrema duplicanda ſunt, hac ratione: 4. 2. Si quis plures medios terminos, & proportiones facere cupiat, triplican-

Since all musical intervals arise from the division of the octave, as from their parent, either immediately or mediately, order requires that we begin from this.

CHAPTER IX.
On Division.

The function of division is to divide a whole into its parts: this is what ordinary division accomplishes in simple numbers. In dividing ratios, or rational numbers, another kind of Division, as I have already said, must be used, whose task is to produce from one proportion two or more; this can be done in three ways. Therefore a threefold Division must be established: Arithmetic, Harmonic, and Geometric.

On Arithmetic Division.

Arithmetic Division is perhaps so called because the numbers produced by it follow one another in natural order, with no interval inserted between them, by the interposition of a middle term equally distant from each extreme, so that it produces equal differences and unequal proportions—establishing a greater proportion between the smaller numbers and a lesser proportion between the greater ones. An example will make the matter clearer.

Let there be a double proportion in this form: 4. 2. The number 3, which can enter between these two extremes, is the arithmetic divisor; for it divides the extremes so that they are equally distant from each other, and it makes the differences equal: 4. 3. 2. For by whatever interval the four is distant from the three—namely by one—the three is distant from the two by the same amount, and it yields unequal proportions: the sesquitertian and the sesquialteran. The sesquitertian between 4 and 3, which is the smaller proportion; and the sesquialteran between 3 and 2, which is the greater proportion.

Sesquitertian. Sesquialteran.

4. 3. 2.

1. 1.

Equal differences.

What must be done if the extremes of some ratio are found to be so close that they cannot receive a middle term? For example: 2. 1. The rule: the extremes must be doubled, in this manner: 4. 2. If someone wishes to make more middle terms and more proportions, the extremes must be tripled.

da erunt extrema, E. G. 2. 1., adhibeatur triplicatio iſtorum extremorum, & habebitur eadem proportio dupla, hac ratione conſtituta : 6. 3. ubi duo media intrare poſſunt, quinarius, & quaternarius : 6. 5. 4. 3., efficiunturque tres differentes proportiones: ſeſquiquinta inter 6. & 5., ſeſquiquarta inter 5. & 4., ſeſquitertia inter 4. & 3. Ex quo colligitur, quò major ſit diſtantia extremorum, eò plures medios terminos intrare poſſe, ùt in proportione ſexdupla: 6. 1., quæ apta eſt recipere quatuor terminos medios, 5. 4. 3. 2., producitque quinque differentes proportiones hac ratione : 6. 5. 4. 3. 2. 1. ùt conſtat ex jam dictis. Datur etiam alius arithmeticè dividendi modus ; verùm ego brevitati, ùt præfatus ſum, ſtudens, hanc ad meum inſtitutum ſufficere arbitror.

CAPUT X.

De Diviſione Harmonica.

DIviſio *Harmonica* fit per medium terminum, ſive diviſorem, qui efficit proportiones inæquales, majores inter majores numeros, minores inter minores conſtituendo ; ità ut differentiæ ſint pari modo inæquales. Exemplo clarior fiet definitio : Accipiatur octava jam arithmeticè diviſa, hac ratione : 4. 3. 2.; multiplicentur duo extrema per medium terminum : 4. per 3., & habebitur 12. : 2. per eundem medium terminum 3., & prodibit alterum extremum proportionalitatis harmonicæ 12. 6., porrò multiplicentur inter ſe extrema proportionalitatis arithmeticæ, nempe 4. per 2., & reſultabit 8., qui eſt verus diviſor harmonicus dividens proportionem duplam hoc modo : 12. 8. 6., producitque duas proportiones diſſimiles : ſeſquialteram inter 12. & 8., & ſeſquitertiam inter 8. & 6., majorem inter terminos majores, minorem inter terminos minores exhibendo, facitque differentias eodem modo diſſimiles, ùt in ſequenti exemplo.

Seſquialtera. Seſquitertia.

12. 8. 6.

4. 2.

Differentiæ diſſimiles.

Pari ratione res inſtituenda eſt in dividendis cæterorum generum, ſpecierumque proportionibus. Sit dividenda harmoni-

For example: 2. 1. Let the tripling of those extremes be applied, and the same double proportion will be obtained, arranged in this form: 6. 3. Here two middle terms can enter, namely 5 and 4: 6. 5. 4. 3. And three different proportions are produced: the sesquiquintan between 6 and 5, the sesquiquartan between 5 and 4, and the sesquitertian between 4 and 3. From this it is inferred that the greater the distance of the extremes, the more middle terms can enter. Thus in the sixfold proportion 6. 1., which is able to receive four middle terms—5. 4. 3. 2.—it produces five different proportions in this form: 6. 5. 4. 3. 2. 1., as is clear from what has already been said. There is also another method of dividing arithmetically; but since I am, as I said before, aiming at brevity, I consider this sufficient for my purpose.

CHAPTER X.

On Harmonic Division.

Harmonic Division is made through a middle term, or divisor, which produces unequal proportions, establishing the greater proportions between the greater numbers and the lesser between the lesser numbers, so that the differences are likewise unequal. An example will make the definition clearer. Let the octave already divided arithmetically be taken in this form: 4. 3. 2. Let the two extremes be multiplied by the middle term: 4 by 3, and we obtain 12; 2 by the same middle term 3, and the other extreme of the harmonic proportionality will be produced, 12. 6. Next, let the extremes of the arithmetic proportionality be multiplied together, namely 4 by 2, and the result will be 8, which is the true harmonic divisor dividing the double proportion in this way: 12. 8. 6. This produces two unlike proportions: the sesquialteran between 12 and 8, and the sesquitertian between 8 and 6, showing the greater proportion between the greater terms and the lesser between the lesser terms, and producing differences likewise unequal, as in the following example.

Sesquialtera. Sesquitertia.

12. 8. 6.

4. 2.

Unequal differences.

By the same method the matter must be arranged in dividing the proportions of other genera and species.

monicè generis ſuperparticularis proportio ſeſquialtera 3. 2., dividatur primò arithmeticè hoc modo: 6. 5. 4., multiplicentur per medium hoc arithmeticum duo extrema: ſex per 5. habebitur 30., rursùs 4. per 5. dabit 20., & habebimus duo extrema proportionalitatis harmonicæ hac ratione; 30. 20. ad reperiendum medium harmonicum multiplicentur inter ſe extrema proportionalitatis arithmeticæ, 6. per 4. dat 24., qui eſt diviſor harmonicus quæſitus, hac ratione: 30. 24. 20., & ſic de cæteris.

CAPUT XI.

De Diviſione Geometrica.

DIviſio *Geometrica* eſt illa, cujus medius terminus inter duo extrema poſitus, conſtruit duas proportiones æquales, & differentias inæquales. Hinc patet, hanc diviſionem locum habere non poſſe, nisì in genere multiplici. Inſtituitur autem hæc diviſio hoc modo: accipiatur proportio quædam ex genere multiplici ſuis numeris radicalibus conſtans, E. G. dupla 2. 1. multiplicetur quodvis extremorum in ſe, nempe, 2. per 2., & habebitur 4., 1. per 1. reſtat 1., ex qua multiplicatione oritur ſequens proportio quadrupla 4. 1., porrò ad reperiendum medium, ſive diviſorem Geometricum ſcire oportet, quid numerus quadratus, quid ejuſdem radix. Dico ergo, numerum ſimplicem eſſe radicem ſui ipſius in ſe multiplicati. E. G. binarius eſt radix quaternarii, quia hic naſcitur ex binario in ſe multiplicato, ex quo infertur, binarium, utpote radicem quaternarii, eſſe verum diviſorem dictæ proportionis quadruplæ, geometricè dividendæ: 4. 2. 1., ex qua diviſione oriuntur duæ æquales proportiones, nempe duplæ, cum inæqualibus differentiis, nam inter 4. & 2. differentia eſt binarius, & 2. ab 1. differt unitate.

Inde concludo, quòd cujuſvis Proportionis geometricè diviſæ major numerus ſit quadratus, & radix ejus ſit diviſor geometricus; pro majori claritate placet hìc apponere aliquot numeros quadratos cum eorum radicibus.

Radices.	2.	3.	4.	5.	6.
Quadrati.	4.	9.	16.	25.	36.

Ex quo ulteriùs colligitur, quamlibet generis multiplicis ſpeciem dicto modo poſſe geometricè dividi: ùt tripla, quadrupla, quindupla, &c.

Let the superparticular proportion sesquialtera 3. 2. be divided harmonically. First let it be divided arithmetically in this way: 6. 5. 4. Let the two extremes be multiplied by this arithmetic mean: six by 5 gives 30; again, 4 by 5 gives 20; and we shall have the two extremes of the harmonic proportionality in this form: 30. 20. To find the harmonic mean, let the extremes of the arithmetic proportionality be multiplied together: 6 by 4 gives 24, which is the harmonic divisor sought, in this form: 30. 24. 20. And so with the rest.

CHAPTER XI.
On Geometric Division.

Geometric Division is that in which the middle term, placed between two extremes, constructs two equal proportions and unequal differences. Hence it is clear that this division can have no place except in the multiplicative genus. This division is carried out in this way: let some proportion from the multiplicative genus be taken, consisting of its radical numbers, for example the double 2, 1. Let either extreme be multiplied by itself, namely 2 by 2, and we obtain 4; 1 by 1 remains 1. From this multiplication arises the following quadruple proportion: 4. 1. Next, to find the mean or geometric divisor, one must know what a square number is and what its root is. I say therefore that a simple number is the root of itself multiplied by itself. For example, 2 is the root of 4, because 4 is born from 2 multiplied by itself. From this it follows that 2, as the root of 4, is the true divisor of the said quadruple proportion to be divided geometrically: 4. 2. 1. From this division arise two equal proportions, namely doubles, with unequal differences, for between 4 and 2 the difference is 2, and 2 differs from 1 by 1.

From this I conclude that in any proportion divided geometrically, the greater number is a square, and its root is the geometric divisor. For greater clarity, it is useful to set down here some square numbers with their roots.

Roots:	2.	3.	4.	5.	6.
Squares:	4.	9.	16.	25.	36.

From this it is further gathered that any species of the multiplicative genus can be divided geometrically in the said manner: as the triple, quadruple, quintuple, etc.

CAPUT XII.

De Muliplicatione Rationum.

Multiplicatio Rationum aliud non eſt, quàm translatio duarum vel plurium proportionum terminis radicalibus conſtantium ad terminos remotos, manentibus iisdem ſpecie proportionibus; nihil intereſt, an rationes ſint diverſi, vel ejuſdem generis, modò una non ſit majoris, & altera minoris inæqualitatis, quo caſu operatio uſuvenire non poſſet. Videatur methodus, quæ in multiplicandis duabus proportionibus tenenda eſt. Sint E. G. multiplicandæ, ratio dupla, & ſeſquialtera hac ratione poſitæ : $\frac{2}{1}$. $\frac{3}{2}$., multiplicetur antecedens duplæ cum antecedente ſeſquialteræ, & emerget 6.; ulteriùs, multiplicetur antecedens rationis ſeſquialteræ cum ſequente rationis duplæ, 3. per 1. habetur 3., qui comparatus cum 6. ex prima multiplicatione orto, producit rationem duplam remotioribus terminis conſtantem, nempe 6. 3., poſtea multiplicetur antecedens duplæ cum ſequente ſeſquialteræ : 2. per 2. evenit 4., qui comparatus cum 6., efficit rationem ſeſquialteræ hac ratione : $\frac{6}{3}$. $\frac{6}{4}$., quô conficitur, quòd hæc multiplicatio non immutet proportiones, ſed à terminis radicalibus ad terminos remotiores transferat. Volenti autem unâ operatione multiplicare tres proportiones, alia methodus uſurpanda eſt : ſint multiplicandæ, dupla, ſeſquialtera, & ſuperbipartiens tertias.

2. 3. 5.
1. 2. 3.

Multiplicetur antecedens duplæ cum antecedente ſeſquialteræ, & habebitur 6., ulteriùs multiplicetur ſequens duplæ cum antecedente ſeſquialteræ, & emerget 3., poſtea multiplicetur ſequens duplæ cum ſequente ſeſquialteræ, 1. per 2. prodibit 2., qui numeri hoc ordine ponantur.

6. 3. 2.

Modò multiplicentur hi tres numeri per antecedens rationis ſuperbipartientis tertias, 6. per 5. habetur 30., 3. per 5. oritur 15., 2. per 5. reſultat 10., poſtremò multiplicetur iterum minimus trium numerorum ex prima multiplicatione productorum cum ſequente rationis ſuperbipartientis tertias, nempe, 2. per 3. & habebimus 6., qui numeri remoti omnes hoc ordine collocati

30. 15. 10. 6.

effi-

CHAPTER XII.

On the Multiplication of Ratios.

The Multiplication of Ratios is nothing other than the transfer of two or more proportions, consisting of their radical terms, to more remote terms, the proportions themselves remaining of the same species. It makes no difference whether the ratios are of different or of the same genus, provided that one is not of greater and the other of lesser inequality, in which case the operation could not be carried out. Let the method to be followed in multiplying two proportions be examined. Let, for example, the double ratio and the sesquialteran ratio be multiplied, placed in this form: 2/1. 3/2. Let the antecedent of the double be multiplied with the antecedent of the sesquialteran, and 6 will emerge. Next, let the antecedent of the sesquialteran be multiplied with the consequent of the double: 3 by 1 gives 3, which, compared with the 6 arising from the first multiplication, produces a double ratio consisting of more remote terms, namely 6. 3. Afterwards, let the antecedent of the double be multiplied with the consequent of the sesquialteran: 2 by 2 gives 4, which, compared with 6, produces the sesquialteran ratio in this form: 6/3. 6/4. From this it is understood that this multiplication does not change the proportions, but transfers them from their radical terms to more remote ones. But if someone wishes in one operation to multiply three proportions, another method must be used. Let the proportions to be multiplied be the double, the sesquialteran, and the superbipartiens tertias:

2. 3. 5.
1. 2. 3.

Let the antecedent of the double be multiplied with the antecedent of the sesquialteran, and 6 will be obtained. Next, let the consequent of the double be multiplied with the antecedent of the sesquialteran, and 3 will emerge. Afterwards, let the consequent of the double be multiplied with the consequent of the sesquialteran: 1 by 2 produces 2. Let these numbers be placed in this order:

6. 3. 2.

Now let these three numbers be multiplied by the antecedent of the superbipartiens tertias ratio: 6 by 5 gives 30; 3 by 5 gives 15; 2 by 5 results in 10. Finally, let the smallest of the three numbers produced in the first multiplication be multiplied again by the consequent of the superbipartiens tertias ratio, namely 2 by 3, and we shall have 6. All these remote numbers placed in this order:

30. 15. 10. 6.

efficiunt easdem tres proportiones, quas ante multiplicationem dabant numeri ſimplices, ſeu radicales, nempe duplam inter 30. & 15., ſeſquialteram inter 15. & 10., & ſuperbipartientem tertias inter 10. & 6. Hæc ſufficiant de Multiplicatione. Pergamus ad

CAPUT XIII.

De Additione Rationum.

EX hucuſque dictis conſtat per Diviſionem ex una plures effici proportiones: per Multiplicationem eaſdem quidem retineri, ad terminos tamen remotos translatas. Videndum eſt deinceps, quo modo per additionem duarum, vel plurium proportionum fiat una ſola. Inſtituatur primò additio duarum proportionum, E. G. ſeſquialteræ cum ſeſquitertia: ponantur utriuſque proportionis termini majores à ſiniſtris, & minores à dextris hac ratione:

3. 2.
4. 3.

Multiplicetur minor terminus ſeſquialteræ cum minore ſeſquitertiæ, & habebitur 6., rursùs multiplicetur major terminus ſeſquialteræ cum majore ſeſquitertiæ, & habebitur 12., quo facto fiet ex duabus dictis proportionibus una ſola, nempe dupla:

12.
6.

2.
1.

Par ratio tenenda eſt in addendis plurium proportionum rationibus. Sint E. G. duæ ſeſquialteræ, & duæ ſeſquitertiæ addendæ ſupradicto modo collocatæ.

3. 2.
3. 2.
4. 3.
4. 3.

Multiplicentur inter ſe minores termini harum proportionum à dextris poſitarum: 2. per 2. habetur 4., 4. per 3. reſultat 12., 12. per 3. producitur 36., quod ultimum productum erit minor terminus proportionis per additionem oriundæ.

produce the same three proportions that the simple or radical numbers gave before multiplication, namely the double between 30 and 15, the sesquialteran between 15 and 10, and the superbipartiens tertias between 10 and 6. Let this suffice concerning Multiplication. Let us proceed to

CHAPTER XIII.

On the Addition of Ratios.

From what has been said thus far, it is clear that by Division severalratios are produced from one; by Multiplication they are indeed retained, but transferred to more remote terms. It must next be seen in what way from the addition of two or more ratios a single one is made. Let the addition of two ratios be first established, for example the sesquialteral (3:2) with the sesquitertial (4:3). Let the greater terms of each ratio be placed on the left, and the lesser on the right, in this manner:

3. 2.
4. 3.

Let the lesser term of the sesquialteral be multiplied by the lesser term of the sesquitertial, and 6 is obtained; again let the greater term of the sesquialteral be multiplied by the greater term of the sesquitertial, and 12 is obtained. When this is done, from the two said ratios one alone is produced, namely the double ratio:

12.
6.

2.
1.

The same rule must be observed in adding the ratios of several proportions. Let, for example, two sesquialteral ratios and two sesquitertial ratios be added, arranged in the manner described above.

3. 2.
3. 2.
4. 3.
4. 3.

Let the lesser terms of these ratios placed on the right be multiplied together: 2 times 2 gives 4, 4 times 3 results in 12, 12 times 3 produces 36, and this last product will be the lesser term of the ratio arising from the addition.

dæ. Poſtea multiplicentur pari ratione majores termini dictarum proportionum, 3. per 3. oritur 9., 9. per 4. producitur 36., 36. per 4. exhibet 144., quod productum dabit majorem terminum proportionis quæſitæ, hac ratione:

144. 36.

Quæ ratio quadrupla eſt, ùt palàm fiet per reductionem.

(6.
144. 36.

(6.
24. 6.

4. 1.

CAPUT XIV.

De Subtractione Rationum.

Cùm continens majus eſſe debeat ſuo contento, conſtat hanc operationem neque in duabus proportionibus quantitate ſimilibus locum habere, neque majorem proportionem à minori ſubduci poſſe; ex quo infertur, ſcitu neceſſarium eſſe, quâ ratione una proportio ſit alterâ major, vel minor. Dico ergo, quòd in genere ſuperparticulari, & ſuperpartiente, quò majores termini alicujus proportionis, eò minor ſit proportio, atque minus intervallum ab ea productum: hanc ob rationem, ſeſquialtera major eſt ſeſquitertiâ, hæc major ſeſquiquartâ, & ſic de cæteris hujus generis proportionibus, quarum termini ſemper numeris aſcendendo creſcunt. Par ratio in genere ſuperpartiente tenenda eſt. Proportio ſuperbipartiens tertias $\frac{5}{3}$. major eſt ſupertripartiente quintas, nempe, $\frac{8}{5}$. quia illa dat ſextam majorem, hæc minorem. Contrà uſuvenit in genere multiplici, in quo eò major eſt proportio, quò majores ſunt termini: inde major eſt tripla, quàm dupla, quadrupla major quàm tripla, &c.

Ex dictis facilè patet, quæ proportio ſubtrahi, & qualis illa eſſe debeat, de qua quis ſubtrahere velit. Ex ipſa Etymologia intelligitur ſubtractionis officium, nempe diminuere intervallum, de quo ſubtrahitur. Methodus autem ſubtrahendi eſt iſta: ponatur major proportio aggregatim à ſiniſtris, & minor à dextris, eodem modo collocata; multiplicetur antecedens majoris proportionis cum ſequente proportionis minoris,

Afterwards let the greater terms of the said ratios be multiplied in the same manner: 3 times 3 gives 9, 9 times 4 produces 36, 36 times 4 yields 144, and this product will give the greater term of the desired ratio, in this form:

144. 36.

This ratio is quadruple, as will be clear by reduction.

(6.
144. 36.
(6.
24. 6.
4. 1.

CHAPTER XIV.

On the Subtraction of Ratios.

Since the containing term must be greater than the term contained, it is clear that this operation can have no place in two ratios that are equal in quantity, nor can a greater ratio be subtracted from a lesser; from which it follows that it is necessary to know in what manner one ratio is greater or lesser than another. I say therefore that in the superparticular and superpartient genera, the greater the terms of a ratio, the smaller the ratio itself, and the smaller the interval produced from it. For this reason the sesquialteral ratio is greater than the sesquitertial, the latter greater than the sesquiquartal, and so on with the other ratios of this kind, whose terms always increase in ascending number. The same rule must be observed in the superpartient genus. The superbipartient third (5/3) is greater than the supertripartient fifth (8/5), because the former gives a greater sixth, the latter a smaller one. The opposite occurs in the multiple genus, in which the greater the terms, the greater the ratio: hence the triple is greater than the double, the quadruple greater than the triple, etc.

From what has been said it is easy to see which ratio may be subtracted, and what kind it must be from which one wishes to subtract. From the very etymology the function of subtraction is understood, namely to diminish the interval from which something is subtracted. The method of subtracting is this: let the greater ratio be placed on the left, and the lesser on the right, arranged in the same manner; let the antecedent of the greater ratio be multiplied by the consequent of the lesser ratio, and the consequent of the greater by the antecedent of the lesser; the product will give the ratio diminished by subtraction, and will show the difference between the two ratios.

ris, & ſequens majoris cum antecedente minoris proportionis; productum dabit proportionem per ſubtractionem diminutam, monſtrabitque differentiam inter duas illas proportiones intercedentem. Paradigma rem clariùs explicabit. Sit E. G. de duplâ ſubtrahenda ſeſquialtera; multiplicetur atecedens duplæ cum ſequente ſeſquialteræ, & viciſſim antecedens ſeſquialteræ cum ſequente duplæ, & ſic operatio erit peracta. Sed pro majori facilitate ducantur duæ virgulæ in formam crucis de antecedente unius rationis ad ſequens alterius, & viciſſim hac ratione.

2. 3.
1. 2.

4. 3.
Differentia.

Pari ratione operatio inſtituenda eſt in omnibus aliis proportionibus ſubtrahendis, ùt videtur in ſequentibus exemplis.

4. 16. 5. 9. 3. 4.
3. 15. 4. 8. 2. 3.

60. 48. 40. 36. 9. 8. Differentiæ.

10. 8. 10. 9.

5. 4.

Quorſum autem tot proportionum genera, tot ſpecies? Cui uſui ſint triplices illæ Diviſiones? Quid Multiplicatio, Additio, & Subtractio ſibi volunt? Quamvis ex dictis huic quæſtioni ut plurimùm ſatisfieri poſſit, tamen ex dicendis pleniùs ſatisfactum iri arbitror, cùm de quovis Muſices intervallo in particulari dicturus ſum, & initio ab Octava ducto, ex qua tanquam Matre omnes conſonantes, & diſſonantes generari demonſtravero.

CAPUT XV.

De Octava.

Octava ſic dicta, quia octo conſtat ſonis; omnium intervallorum ſimplicium maximum eſt; græcè *Diapaſon* appellata, quod vocabulum univerſitatem ſignificat, eò quòd Octava, univerſa poſſibilia comprehendat intervalla; actu ſimplicia, potentiâ verò compoſita. Verùm cùm Octava, utpote continens

An example will make the matter clearer. Let the sesquialteral ratio be subtracted from the double; multiply the antecedent of the double by the consequent of the sesquialteral, and conversely the antecedent of the sesquialteral by the consequent of the double, and the operation will be completed. But for greater ease let two small lines be drawn in the form of a cross from the antecedent of one ratio to the consequent of the other, and conversely, in this manner:

2. 3.
X
1. 2.
4. 3.
Difference.

In the same manner the operation must be carried out in all other ratios to be subtracted, as appears in the following examples.

4. 16. 5. 9. 3. 4.
X X X
3. 15. 4. 8. 2. 3.
60. 48. 40. 36. 9. 8. Differences.
10. 8. 10. 9.
5. 4.

But to what purpose are so many genera and species of ratios? What use are those threefold divisions? What do Multiplication, Addition, and Subtraction intend? Although from what has been said this question may for the most part be answered, yet I think it will be more fully satisfied by what is to be said when I speak in particular of each musical interval, beginning with the Octave, from which, as from a Mother, all consonances and dissonances are generated, as I shall demonstrate.

CHAPTER XV.

On the Octave.

The Octave is so called because it consists of eight sounds; it is the greatest of all simple intervals; in Greek it is called Diapason, a word signifying universality, because the Octave contains all possible intervals —simple in act, composite in power. But since the Octave, as the containing interval,

terorum intervallorum ſub contento comprehendi non poſſit, & Octava quaſi aliud non ſit, quàm uniſoni repetitio, ideò ſeptem tantùm diſtinctos infrà Octavam ſonos contineri conſtat; id quod Marcus Tull. Cicero *in ſomnio Scipio.* ſimilitudine ductâ à ſonis orbium cœleſtium hiſce elegantiſſimè expreſſit: Quis inquam, quis eſt, qui complet aures meas tantus, & tam dulcis ſonus? *Hic eſt*, inquit ille, *qui intervallis conjunctus imparibus, ſed tamen pro rata portione diſtinctus, impulſu, & motu ipſorum orbium efficitur, & acuta cum gravibus temperans, varios æquabiliter concentus efficit. Nec enim ſilentio tanti motus incitari poſſunt, & natura fert, ut extrema ex altera parte graviter, ex altera autem acutè ſonent; quam ob cauſam ſummus ille cœli ſtelliferi curſus, cujus converſio eſt concitatior, acuto, & excitato movetur ſono: graviſſimo hic lunaris, atque infimus: nam terra nona immobilis manens imâ ſede ſemper hæret, complexa medium mundi locum. Illi autem octo curſus, in quibus vis eadem eſt duorum, Mercurii, & Veneris, ſeptem efficiunt diſtinctos intervallis ſonos, qui numerus rerum omnium ferè modus eſt. Quod docti homines nervis imitati, atque Cantibus aperuêre ſibi reditum ad hunc locum, &c.*

Hujus rei veritatem per Octavæ diviſionem experturi ſumus, cùm primò ipſius Octavæ naturam examinaverimus.

Octava exiſtentiam ſuam habet in genere multiplici, ejus termini conſtitutivi ſunt 1. 2., hanc ob cauſam intervallorum omnium perfectiſſimum judicandum eſt, cùm primus ejus terminus ſit ipſum principium, nempe unitas, ſecundus, numerus binarius, unitati proximus. Quem ordinem à Deo in omnibus operibus ſuis, ſive rebus creatis mirabili, obſervatum eſſe, non obſcurè animadverti poteſt. Hinc cœleſtes illi Spiritus intelligentiæ præſtantiâ, principio ſuo proximi, omnia cætera creata perfectione antecellunt. His proximè nobilitate accedit homo, rationis particeps, quem nempe idem rerum omnium molitor, & architectus Deus paulò minùs ab Angelis minuit. Hunc ſequuntur cætera animantia rationis expertia. Hæc excipiunt vegetabilia, & ultimô inanimata. Quas res, quò longiùs à factore, ſive principio ſuo recedunt nobilitate, eò imperfectiores eſſe, quis non videt? Pari ratione in Muſices intervallis, illorumque perfectione, vel imperfectione ſe res habet; etenim quò majores termini cujuſdam proportio-

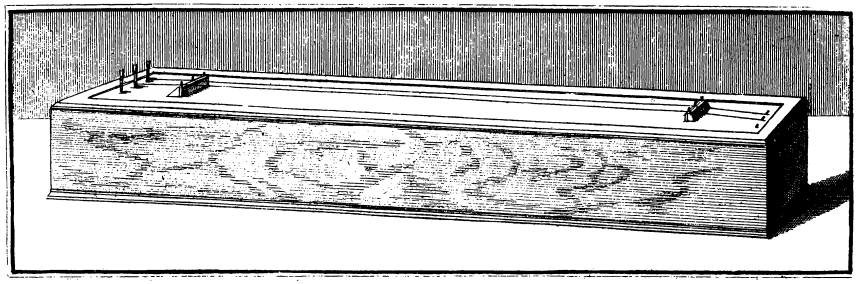
cannot be contained under another, and since the Octave is, as it were, nothing other than a repetition of the unison, it is clear that only seven distinct sounds are contained within the Octave. This Marcus Tullius Cicero, in the *Dream of Scipio*, expressed most elegantly by a comparison drawn from the sounds of the celestial spheres: *"Who, I ask, who is it that fills my ears with so great and so sweet a sound? This, he said, is that which, joined by unequal intervals yet distinguished in due proportion, is produced by the impulse and motion of the spheres themselves, and tempering the high with the low, produces various harmonies in equal measure. For such great motions cannot be stirred in silence, and nature requires that the extremes sound low on one side and high on the other; for this reason the highest course of the starry heaven, whose revolution is the swiftest, moves with a sharp and excited sound; the lunar sphere, being the lowest, with the gravest; for the earth, the ninth, remaining immovable in the lowest seat, always clings to the center of the world. But those eight courses, in which the force of two—Mercury and Venus—is the same, produce seven sounds distinguished by intervals, which number is the measure of almost all things. Which learned men, imitating with strings and songs, opened for themselves a return to this place,"* etc.

We shall test the truth of this by the division of the Octave, after we have first examined the nature of the Octave itself.

The Octave has its existence in the multiple genus; its constitutive terms are 1 and 2. For this reason it must be judged the most perfect of all intervals, since its first term is the very beginning, namely unity, and its second thenumber two, nearest to unity. This order, wonderfully observed by God in all His works and created things, may easily be noticed. Hence those heavenly spirits, excelling in intelligence and nearest to their principle, surpass all other creatures in perfection. Next to them in nobility comes man, partaker of reason, whom the same Maker and Architect of all things made a little lower than the angels. After these follow the other living creatures lacking reason. After these come plants, and lastly inanimate things. The farther things recede from their Maker, or principle, in nobility, the more imperfect they become. The same holds in musical intervals and their perfection or imperfection; for the greater the terms of a ratio, and the farther they are from unity, theirprinciple, the more imperfect will be the intervals arising from them.

portionis, quòque remotiores ſunt ab unitate, ſuo principio, eò imperfectiora erunt intervalla inde exorta.

Ut redeat oratio, unde nonnihil digreſſa eſt, fortaſſis nonnemo percunctabitur, cur termini proportionis hujus, 1. 2. Octavam ſonent? Ratio in promptu eſt; cùm enim proportio hæc dupla ſit, & chorda una ſit duplò longior alterâ, duplò etiam graviorem edat ſonum, neceſſe eſt. Ut autem veritas hæc non tantùm rationi, verùm etiam auribus patefiat, experimentum inſtitui poteſt in inſtrumento ſequentis figuræ, quod à Latinis Regula, à Græcis autem *Monochordon*, id eſt inſtrumentum unius chordæ appellatur. Verùm haud paulò faciliorem experimenti faciendi facultatem Polychordon exhibet, ſeu inſtrumentum duarum chordarum ſequentis figuræ:



Haud operæ pretium fuerit, in deſcribenda conficiendi hujus inſtrumenti methodo pluribus immorari, quam vel ſolo figuræ hujus contuitu à quovis Organopæo facilè conſtrui poſſe, confido. Itaque ad uſûs demonſtrationem progredior. Concordentur ambæ craſſitudine æquales inſtrumenti hujus chordæ accuratè in uniſono, & metiatur poſtmodum circino alterutra harum, dividaturque in tot partes, quot terminus major proportionis illius indicat, cujus ſonos quis probare intendit: minor terminus proportionis monſtrat, quâ parte equuleus ſubjiciendus ſit, fulcimenti genus, quo chordæ nituntur, Germanis Ephippium, vel Sattel vulgò dictum. Sint E. G. probandi portionis duplæ ſoni, 1. 2., dividatur ſecundùm majorem proportionis terminum in duas partes alterutra chordarum, alterâ in ſua longitudine relictâ; minor terminus proportionis 1. indicat, quâ parte, nempe mediâ, ſuppoſito equuleo chorda dividenda ſit; quo facto chorda unius partis ad chordam duarum partium Octavam ſonabit. Pari modò inſiſtendum eſt in probandis ſonis proportionis ſeſquialteræ 3. 2.: dividatur ſecundùm

To return to the point from which the discourse has somewhat digressed, someone may perhaps ask why the terms of this ratio, 1 and 2, produce the Octave. The reason is clear: since this ratio is double, and one string is twice as long as the other, it must produce a sound twice as low. But that this truth may be made evident not only to reason but also to the ears, an experiment may be made on the instrument of the following figure, which the Latins call the Rule, and the Greeks the Monochord, that is, the instrument of one string. But the Polychord, or instrument of two strings, of the following figure, offers a much easier means of making the experiment:

It would not be worth the effort to dwell at length on the method of making this instrument, since from the figure alone any organ-maker can easily construct it. Therefore I proceed to its use for demonstration. Let both strings of this instrument, equal in thickness, be tuned accurately in unison; then let one of them be measured with compasses and divided into as many parts as the greater term of the ratio whose sounds one wishes to test indicates; the lesser term of the ratio shows at what point the little bridge (a kind of support on which the strings rest, called by the Germans Ephippium or Tattel) must be placed. Let, for example, the sounds of the double ratio 1:2 be tested; let one of the strings be divided into two parts according to the greater term of the ratio, the other left at its full length; the lesser term 1 indicates at what point—namely the middle—the bridge must be placed; when this is done, the string of one part will sound the Octave to the string of two parts. In the same way one must proceed in testing the sounds of the sesquialteral ratio 3:2: let one of the strings be divided into three parts according to

numerum majorem hujus proportionis in tres partes una ex dictis duabus chordis, ſecunda pars, quam indicat terminus minor, producet ſonos quæſitos, nempe quintam ; quæ methodus adhibenda eſt in inquirendis omnium aliarum proportionum ſonis, cum hac obſervatione, ut in tot partes dividatur alterutra chordarum, quot indicat numerus major proportionis, & abbrevietur chorda altera, ſubmittendo equuleum ab ea parte, quam terminus minor ejuſdem proportionis indicat.

Ex iis, quæ ſuprà Cap. nono, & decimo de Diviſione Arithmetica, & Harmonica dicta ſunt, facilè intelligi poteſt, quo modo Octava, ſive *Diapaſon* dividatur, quòd nempe hac ratione: 6. 4. 3. ſit harmonicè diviſa, quia exhibetur quinta in parte gravi, & quarta in parte acuta ; arithmeticè autem diviſa hoc modo : 4. 3. 2. dat quartam in parte gravi, & quintam in parte acuta. Hæc ergo duo intervalla oriuntur ex prima Octavæ diviſione, quorum utrumque ſeorſum examinandum eſt.

CAPUT XVI.

De Quinta, ſeu Diapente.

Quinta, græcè *Diapente*, ſic dicta, quia naturali ordine in genere Diatonico quintum occupat locum ; locum ſuum habet in genere ſuperparticulari, nempe ſeſquialtera hac ratione : 3. 2., & eſt quidem perfecta conſonans, minùs tamen perfecta, quàm Octava, quia hujus termini, nempe, 1. 2. per additionem uniti dant 3., illius verò, nempe, 2. 3. eodem modo uniti dant 5., qui numerus remotior eſt ab unitate ſuo principio, quàm numerus 3. Ad quem uſque numerum iſta perfectionis prærogativa ſe extendat, haud abs re fuerit, inveſtigare. Meâ quidem ſententiâ haud incommodè pro limite ſtatuetur numerus ſenarius, non tam, quia pro primo numero perfecto habetur, quàm quòd natura ipſamet ad producendas conſonantias ejuſmodi inclinet, quarum termini per additionem uniti, numerum ſenarium non excedunt, ùt experiri licet in uniſono 1. ad 1. octavâ, 2. ad 1. quintâ, 3. ad 2. duodecimâ, 3. ad 1. decimaquintâ, 4. ad 1., è quibus omnibus tactâ alterutrâ chordarum, alteram etſi intactam ſuâ ſponte reſonare, aut certè contremiſcere compertum habemus, non aliâ certè ratione, quàm, quòd illarum termini infrà ſenarium ſe con-

the greater term of the ratio, the other left at its full length; the lesser term 1 indicates at what point—namely the middle—the bridge must be placed; when this is done, the string of one part will sound the Octave to the string of two parts. In the same way one must proceed in testing the sounds of the sesquialteral ratio 3:2: let one of the strings be divided into three parts according to the greater term of the ratio; the second part, indicated by the lesser term, will produce the desired sound, namely the fifth. This method must be used in investigating the sounds of all other ratios, with this observation: that one of the strings must be divided into as many parts as the greater term of the ratio indicates, and the other shortened by placing the bridge at the point indicated by the lesser term.

From what was said above in Chapters IX and X concerning arithmetic and harmonic division, it may easily be understood how the Octave, or Diapason, is divided: namely, in this manner 6, 4, 3 it is harmonically divided, because it exhibits the fifth in the lower part and the fourth in the upper; but arithmetically divided thus 4, 3, 2 it gives the fourth in the lower part and the fifth in the upper. These two intervals therefore arise from the first division of the Octave, each of which must be examined separately.

CHAPTER XVI.

On the Fifth, or Diapente.

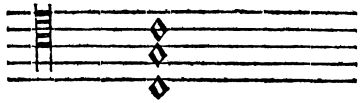
The Fifth, in Greek Diapente, is so called because in the natural order of the Diatonic genus it occupies the fifth place. It has its place in the superparticular genus, namely the sesquialteral ratio in this form: 3. 2. And it is indeed a perfect consonance, yet less perfect than the Octave, because the terms of the latter, namely 1 and 2, when united by addition give 3, while the terms of the Fifth, namely 2 and 3, united in the same way give 5 — a number farther removed from unity, its principle, than the number 3. Up to what number this prerogative of perfection extends is not unworthy of investigation. In my opinion the number six may conveniently be set as the limit, not so much because it is held to be the first perfect number, but because nature herself seems inclined to produce consonances of such kind whose terms, when united by addition, do not exceed the number six. This may be observed in the unison 1 to 1, the octave 2 to 1, the fifth 3 to 2, the twelfth 3 to 1, the fifteenth 4 to 1; in all of which, when one of the strings is struck, the other, though untouched, is found to resonate of its own accord, or at least to tremble —certainly for no other reason than that their terms, remaining within the number six, are not far removed from unity, their principle. So much for the Fifth.

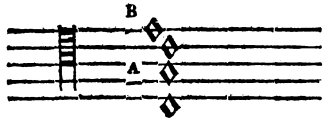
continentes non procul abſint ab unitate, ſuo principio. Hæc de Quinta.

CAPUT XVII.

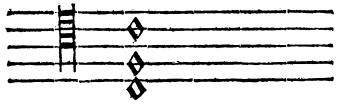
De Quarta, ſeu Diateſſeron.

SEcundum intervallum ex Diviſione Octavæ oriens eſt *Quarta*, græcè *Diateſſeron*, exiſtentiam ſuam habens in genere ſuperparticulari, his terminis 4. 3., quæ eſt ſeſquitertia proportio. Celebris, & plena difficultatis eſt quæſtio, an quarta ſit conſonans? Affirmant omnes Pythagorici, aliique Authores, & doctrinæ, & auctoritatis gloriâ clariſſimi; verùm antequam hac de re mentem aperiam meam, edicant velim, qualem intelligant Quartam, an ex diviſione Octavæ harmonica, vel arithmetica ortam? Si ex diviſione harmonica E. G.



Quis huic Paradigmati quartam nequaquam, verùm quintam, & octavam ineſſe non videt? Intervalla enim baſi, & fundamento, non item partibus mediis ad partes metienda ſunt. Aliàs enim ſequenti exemplo pari ratione ſextam majorem ineſſe dixerint,

ubi notula A.
cum notula B.

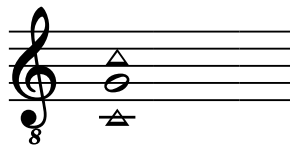
ſextam majorem efficere videtur. Quis autem mortalium, etiam vel in arte hoſpes huic ſiſtemati ſextam ineſſe affirmet ob rationem jam adductam? Si autem intelligant quartam è Diviſione arithmetica productam hac ratione:



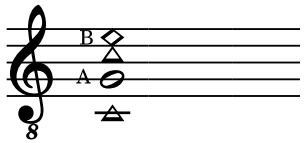
neſcio, quo pacto eam inter conſonantias referre poſſint; niſi quis credere velit, ab antiquitate non ſolum ſpeculatione, (eò quòd ab immediata Octavæ diviſione hoc intervallum proveniat) ſed etiam uſu pro conſonantia habitam fuiſſe. Id quidem, ſi cui pertendere libeat, nihil morabor. Verùm uſus à

CHAPTER XVII.
On the Fourth, or Diatesseron.

The second interval arising from the division of the Octave is the Fourth, in Greek Diatesseron, having its existence in the superparticular genus, with these terms 4. 3., which is the sesquitertial ratio. A celebrated and difficult question is whether the Fourth is a consonance. All the Pythagoreans affirm it, as do many other authors most illustrious in doctrine and authority. But before I reveal my own opinion on this matter, I would have them declare what kind of Fourth they understand — whether one arising from the harmonic division of the Octave, or from the arithmetic division. If from the harmonic division, e.g.

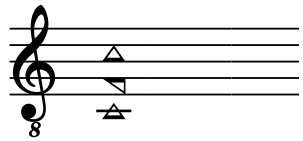


Who does not see that in this example there is not a Fourth at all, but rather a Fifth and an Octave? For intervals must be measured from the base and foundation, not from the middle parts to the other parts. Otherwise, by the same reasoning, in the following example they would say that a major Sixth is present:



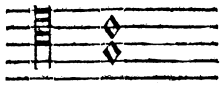
where note A
with note B

seems to produce a major Sixth. But who among mortals, even one a stranger to the art, would affirm that this system contains a Sixth for the reason just given? But if they understand the Fourth produced from the arithmetic division in this manner:

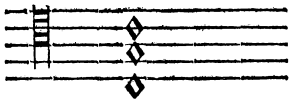


I do not know how they can reckon it among the consonances, unless one wishes to believe that antiquity considered it a consonance not only in speculation (because this interval arises from the immediate division of the Octave) but also in practice. If anyone wishes to maintain that, I shall not oppose him. But the usage received through so many centuries

tot ſæculis receptus, certè adverſari videtur, ad quem nos conformemus oportet; cùm ejus uſum ab uſu reliquarum diſſonantiarum hodie minimè diſcrepare compertum ſit, neque ejus uſurpandæ, nisì per Syncopen, vel ligaturam ratio ſuppetat. Certum eſt equidem Quartam reliquis diſſonantiis minùs diſſonare, auribus item tolerabilem magìs eſſe, ejus tamen perceptione auditum non omnino acquieſcere. E. G.



ſed adhuc quintam inferiùs deſiderare palàm eſt. E. G.



Ex hucuſque dictis perſpicuum fit, ex diviſione Octavæ produci quartam, & quintam. Nunc videndum eſt, qualia intervalla ex diviſione ipſius quintæ oriantur.

CAPUT XVIII.

De Diviſione Quintæ, ſive Diapentes.

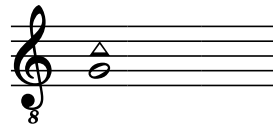
EX diviſione Quintæ, ſeu *Diapentes*, gignuntur duo intervalla ſeſquiquarta: 5. 4. vel ditonus, & ſeſquiquinta: 6. 5. vel ſemiditonus. Per diviſionem harmonicam ditonus collocatur in parte gravi, & ſemiditonus in parte acuta: vice versâ accidit per diviſionem arithmeticam.

Hinc patet, Quintam conſtare tertiâ majore, & minore, quæ exiſtentiam ſuam habent, ùt patet, in genere ſuperparticulari, ſuntque conſonantiæ, ſed imperfectæ, vel quia earum termini uniti per additionem numerum ſenarium excedunt, vel quia natura ad earum productionem minimè concurrit.

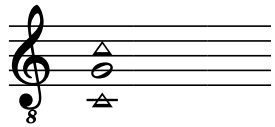
Ordo modò poſtularet, ut de Quarta, ſive *Diateſſeron* dividenda commemoraremus; cùm autem intervallum hoc dividi nequeat, quòd ejus diviſor, ſive medius terminus, ſeptenarius numerus eſſe deberet: 8. 7. 6., cui utpote numero myſtico in dividendis rationibus uſus eſſe nequit (ùt periculum ſumpturo manifeſtum fiet) ideo ad diviſionem tertiæ majoris, ſive *ditoni* nobis progrediendum eſt.

CA-

seems certainly to contradict this, and to that usage we must conform; for its use is found today not to differ at all from the use of the other dissonances, nor is there any reason to employ it except through syncopation or suspension. It is certain indeed that the Fourth is less dissonant than the other dissonances, and more tolerable to the ear, yet the hearing does not entirely rest satisfied with its perception. For example:



but it is clear that one still desires the Fifth below. For example:



From what has been said thus far it is clear that from the division of the Octave the Fourth and the Fifth are produced. Now it must be seen what intervals arise from the division of the Fifth itself.

CHAPTER XVIII
On the Division of the Fifth, or the Diapente.

From the division of the Fifth, or Diapente, two intervals arise: the sesquiquarta, 5:4, or the ditone; and the sesquiquinta, 6:5, or the semiditone. By harmonic division the ditone is placed in the lower part, and the semiditone in the upper part; the reverse happens through arithmetic division.

Hence it is clear that the Fifth consists of a major and a minor third, which, as is evident, have their existence in the superparticular genus, and are consonances, though imperfect— either because their terms, when added together, exceed the number six, or because nature contributes the least to their production.

Order would now require that we speak of the Fourth, or Diatesseron, to be divided; but since this interval cannot be divided—because its divisor, or middle term, would have to be the number seven, 8:7:6, which, being a mystical number, cannot be used in dividing ratios (as will become clear to anyone who tries it)—therefore we must proceed to the division of the major third, or ditone.

CAPUT XIX.

De diviſione Tertiæ majoris, ſeu Ditoni.

EX diviſione Tertiæ majoris 5. 4. producuntur duo intervalla: ſeſquioctava, ſive tonus major: 9. 8., & ſeſquinona, ſive tonus minor 10. 9., quæ intervalla diſſonantia ſunt, eò quòd auribus ingratum exhibeant ſonum, fortaſſis, quòd eorum termini uniti numerum ſenarium duplicatum excedant, & proinde talium ſonorum ictus difficillimè uniantur.

Atque hæc ſunt intervalla, quæ per diviſionem generantur. Cætera, ùt ſexta major, ſive *Hexachordon majus*, ſexta minor, ſeu *Haxachordon minus*, & omnia intervalla compoſita per additionem, ſemitonium majus, & minus, & comma per ſubtractionem reperiuntur, ùt infrà videbimus. Nunc ſit

CAPUT XX.

De formatione Sextæ majoris, & minoris.

SExta major ſic dicta, quia in genere Diatonico ſextum occupat locum; componitur ex conjunctione tertiæ majoris, & quartæ minoris, hoc modo:

5. 4.
4. 3.

20. 12.

5. 3.

Ex quo apparet, quòd ſuam exiſtentiam habeat in genere ſuperpartiente, nempe: ſuperbipartiente tertias, eſtque prima ſpecies hujus generis, & formari etiam poteſt eo, quo modo ſuprà dictum eſt Cap. 7., ſumendo nempe tertium numerum ex tabula Pythagorica pro minore, & relicto quaternario, quinarium pro majore termino. E. G. $\frac{5}{3}$, quæ proportio exhibet ſextam majorem. Poteſt etiam confici addendo proportionem ſeſquinonam ſeſquialteræ, hac ratione:

3. 2.
10. 9.
(6

30. 18.

5. 3.

CHAPTER XIX
On the Division of the Major Third, or Ditone.

From the division of the major third 5:4 two intervals are produced: the sesquioctave, or major tone, 9:8; and the sesquinona, or minor tone, 10:9. These intervals are dissonant, because they present to the ears an unpleasant sound—perhaps because their terms, when added together, exceed twice the number six, and therefore the beats of such sounds can hardly be united.

And these are the intervals that are generated by division. The rest—such as the major sixth, or the greater Hexachord; the minor sixth, or the lesser Hexachord; and all composite intervals formed by addition; the major and minor semitone; and the comma—are found by subtraction, as we shall see below. Now follows

CHAPTER XX
On the Formation of the Major and Minor Sixth.

The major sixth—so called because in the diatonic genus it occupies the sixth place—is composed from the conjunction of the major third and the minor fourth, in this manner:

5:4
4:3

20 :12

5:3

From this it is evident that it has its existence in the superpartient genus, namely the "superbipartient of the thirds," and is the first species of this genus. It may also be formed in the manner described above in Chapter 7, namely by taking the third number from the Pythagorean table as the lower term, and, leaving aside the four, taking the five as the upper term. For example, 5:3, which is the ratio that yields the major sixth. It may also be constructed by adding the sesquinonal ratio to the sesquialteral ratio, in this way:

3:2
10:9
———
30:18
———
5:3

Tametſi intervallum hoc ſuum locum habeat in genere ſuperpartiente, cujus proportiones omnes apud Pythagoricos intervalla diſſonantia producebant, ſextæ tamen tractu temporis inter conſonantias imperfectas numeratæ hodiedum ùt tales in compoſitione adhibentur. Hæc de Sexta majore.

Sexta minor, ùt ſuprà dictum eſt, conſtruitur ex terminis conſtitutivis Sextæ majoris $\frac{5}{3}$, addendo ternarium quinario, unde generatur 8. pro majori termino; accipiatur terminus major Sextæ majoris pro minore, & habebitur hæc proportio $\frac{8}{5}$, quæ forma eſt ipſius Sextæ minoris. Per additionem autem fit hoc modo: addatur ratio ſeſquitertiæ, ſive quartæ rationi ſeſquiquintæ, ſeu tertiæ minori, hac ratione:

4. 3.
6. 5.
(3.

24. 15.

8. 5.

quæ ratio eſt Sextæ minoris. Hìc quæri poteſt, cùm tertiæ, & ſextæ, majores, & minores conſonantiæ imperfectæ ſint, quænam ex illis imperfectiores exiſtimandæ ſint? Mihi quidem, ſi earum terminos conſtitutivos ſpectemus, certum eſt, tertiam minorem imperfectiorem eſſe ſuâ majore, quia illius termini 6. 5. per additionem uniti, 11. exhibent; hujus verò, nempe tertiæ 5. 4., pari modo uniti, 9. efficiunt. De Sextis idem ſit judicium. Verùm cùm hac in re non tam ratio terminorum haberi debeat, quàm generum, & tertiæ exiſtentiam ſuam habeant in genere ſuperparticulari, quod præſtantius eſt ſuperpartiente, cui Sextæ ſubjectæ ſunt, ideo hæ promiſcuè tertiis imperfectiores tenendæ ſunt. Confirmatur hæc opinio, quòd aures harmoniâ à tertiis editâ affatim contentæ, nihil ampliùs deſiderare videantur,

ùt : non idem

3.*tia major.* *minor.*

in ſextis, quarum harmoniam, utpote non finitam certiſſimum illud aurium judicium non adeò

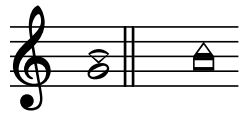
pro-

Although this interval has its place in the superpartient genus—whose proportions, among the Pythagoreans, always produced dissonant intervals—yet the sixths, through the passage of time, have been counted among the imperfect consonances, and are still used as such in composition today. So much for the major sixth.

The minor sixth, as said above, is constructed from the constitutive terms of the major sixth, 5:3, by adding three to five, from which the number 8 is produced as the upper term. Let the upper term of the major sixth be taken as the lower, and this ratio 8:5 will be obtained, which is the form of the minor sixth. By addition it may also be formed thus: add the sesquitertial ratio, or the fourth, to the sesquiquintal ratio, or the minor third, in this manner:

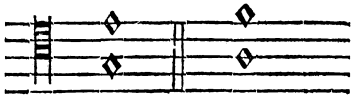
4:3
6:5
———
24:15
———
8:5

which is the ratio of the minor sixth. Here one may ask: since the major and minor thirds and sixths are imperfect consonances, which of them ought to be judged the more imperfect? For my part, if we consider their constitutive terms, it is certain that the minor third is more imperfect than the major, because its terms 6:5, when added, yield 11, whereas those of the major third, 5:4, similarly added, yield 9. The same judgment applies to the sixths. But since in this matter the ratio of the terms is not so much to be considered as the genera, and since the thirds have their existence in the superparticular genus—which is superior to the superpartient genus to which the sixths belong—therefore the sixths must be held indiscriminately more imperfect than the thirds. This opinion is confirmed by the fact that the ears, once satisfied by the harmony produced by the thirds, seem to desire nothing further,

DO:  not the same

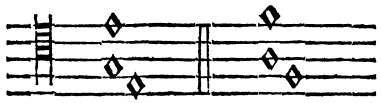
major third minor third

whereas in the case of the sixths, whose harmony is not complete, that most certain judgment of the ears

probare  ſed adhuc

major. *minor.*

octavam infra requirere experimur hac ratione:



Reſtat adhuc hoc Cap. inquirendum de Septima, ſive Heptachordo majore, & minore, quæ pari ratione per additionem reperiuntur, ſequenti methodo: addatur rationi ſeſquialteræ, ſeu quintæ ratio ſeſquiquarta, ſive tertia major, hoc modo:

3. 2.
5. 4.
15. 8.

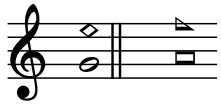
& habetur ratio ſuperſeptempartiens octavas, quæ forma eſt Septimæ majoris, intervalli diſſonantis: ad reperiendam autem rationem Septimæ minoris, addatur rationi ſeſquialteræ ratio ſeſquiquintæ:

3. 2.
6. 5.
18. 10.
9. 5.

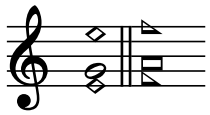
& oritur ratio ſuperquatripartiens quintas, conſtituens rationem Septimæ minoris, intervalli itidem diſſonantis.

Ultimum hoc loco reſtat, ut de ratione Tritoni, & Quintæ falſæ inquiramus, quæ duo intervalla diſſona ſunt. Ratio Tritoni reperitur addendo duos tonos ſeſquioctavos cum tono ſeſquinono, hac ratione:

9. 8.
9. 8.
10. 9.
(9.
180. 576.
(2.
90. 64.
45. 32.

does not so readily approve them, but we find that

major minor

they still require an octave below, in this manner:



It remains in this chapter to inquire concerning the Seventh, or the major and minor Heptachord, which are found by addition in the same way, by the following method: add to the sesquialteral ratio, or the fifth, the sesquiquartal ratio, or the major third, thus:

3:2
5:4
15:8

and the ratio "superseptempartient of the octaves" is obtained, which is the form of the major seventh, a dissonant interval. To find the ratio of the minor seventh, add to the sesquialteral ratio the sesquiquintal ratio:

3:2
6:5
18:10
9:5

and the ratio "superquatripartient of the fifths" arises, which is the ratio of the minor seventh, likewise a dissonant interval.

Finally, it remains here to inquire into the ratio of the tritone and the false fifth, which are two dissonant intervals. The ratio of the tritone is found by adding two sesquioctave tones to a sesquinonal tone, in this manner:

9:8
9:8
10:9
———
180:576
———
90:64
———
45:32

Quæ ratio eſt ſuperpartiens trigeſimas duas, forma Tritoni.

Quinta falſa autem componitur per additionem ſemitonii majoris cum quarta.

8. 3.
16. 15.

64. 45.

Quæ ratio eſt ſuperoctodecimpartiens quadrageſimas quintas.

Jam repertis omnibus intrà octavam contentis intervallis in genere Diatonico, exceptis ſemitonio majore, & minore, & commate, cùm hæc non tam per additionem, quàm ſubtrahendo quærenda ſint, novum Caput ſtatuendum videtur.

CAPUT XXI.

De formandis Semitonio majore, & minore, & Commate.

SEmitonium majus eſt differentia, vel illud intervallum, quo ſeſquitertia ſuperat ſeſquiquartam, quod reperitur per ſubtractionem unius rationis ab altera, hoc modo:

Seſqui 4. 3. *altera.*
Seſqui 5. 4. *quarta.*

Diffe 16. 15. *rentia.*
Semitonium majus.

Quæ ratio ſeſquiquinta decima conſtituit ſemitonium majus.

Semitonium minus eſt ipſa differentia per ſubtractionem reperta inter ſeſquiquartam, & ſeſquiquintam. E. G.

Seſqui 5. 4. *quarta.*
Seſqui 6. 5. *quinta.*

Seſquivi 25. 24. *geſima quarta.*
Differentia.

Comma eſt illa differentia, quâ tonus major ſuperat tonum minorem, ùt patet per ſubtractionem ſeſquinonæ à ſeſquioctava:

Tonus 9. 8. *major.*
Tonus 10. 9. *minor.*

81. 80.
Coñma.

Hæc

This is the ratio "superpartient of the thirty-seconds," the form of the tritone.

The false fifth, however, is composed by adding the major semitone to the fourth:

8:3
16:15
64:45

which is the ratio "superoctodecimpartient of the forty-fifths."

Now that all the intervals contained within the octave in the diatonic genus have been discovered—except the major and minor semitone and the comma, since these are to be sought not by addition but by subtraction—it seems proper to establish a new chapter.

CHAPTER XXI

On the Formation of the Major and Minor Semitone, and of the Comma.

The major semitone is the difference, or that interval, by which the sesquitertial ratio exceeds the sesquiquartal ratio, which is found by subtracting one ratio from the other, thus:

Sesquialteral: 4:3
Sesquiquartal: 5:4 *(note: not so sure about this)*
Difference 16:15
Major semitone.

This ratio, the sesquiquintadecimal, constitutes the major semitone.

The minor semitone is the very difference found by subtracting the sesquiquartal ratio from the sesquiquintal ratio. For example:

Sesquiquartal: 5:4 (the fourth)
Sesquiquintal: 6:5 (the fifth)
Sesquivigesimal fourth 25:24
Difference.

The comma is that difference by which the major tone exceeds the minor tone, as is clear from subtracting the sesquinonal ratio from the sesquioctave:

Major tone 9:8
Minor tone 10:9
81:80
Comma.

Hæc ergo ſunt omnia intervalla ſimplicia, quæ naſcuntur vel immediatè, vel mediatè ex Octavæ diviſione à Ptolomæo facta (quæ ſinè controverſia omnium optima eſt,) & quæ eo, quo producuntur ordine, cum ſuis terminis conſtitutivis, & nominibus in tabula ſequenti poſita videas :

Intervalla ſimplicia.

Octava :	2.	1.	*Diapaſon, ſeu dupla.*
Quinta :	3.	2.	*Diapente, ſeu ſeſquialtera.*
Quarta :	4.	3.	*Diateſſeron, vel ſeſquitertia.*
Tertia major :	5.	4.	*Ditonus, vel ſeſquiquarta.*
Tertia minor :	6.	5.	*Semiditonus, vel ſeſquiquinta.*
Tonus major :	9.	8.	*Seſquioctava.*
Tonus minor :	10.	9.	*Seſquinona.*
Sexta major :	5.	3.	*Hexachordon majus, vel ſuperbipartiens tertiam.*
Sexta minor :	8.	5.	*Hexachordon minus, vel ſuperbipartiens 5.*
Semitonium maj.	16.	15.	*Seſquidecima quinta.*
Semitonium min :	25.	24.	*Seſquivigeſima quarta.*
Comma :	81.	80.	*Seſquioctogeſima.*
Tritonus :	45.	32.	*Supertredecimpartiens trigeſimas duas.*
Quinta falſa :	64.	45.	*Superoctodecimpartiens quadrageſimas quintas.*

Quamquam nonnulla ex his intervallis quodammodo compoſita dici poſſint, quia conjunctis intervallis vel neceſſariò conſtrui debent : ùt ſexta major, ex tertia majore, & quarta; ſexta minor ex tertia minore, & quarta: vel poſſunt ſaltèm, ùt quinta ex tertia majore, & minore, &c. propriè tamen illa tantùm compoſitorum nomine comprehenduntur, quæ cum Octavâ conjuncta eam extenſione excedunt: ùt nona, decima, undecima, duodecima, &c., quæ neque naturâ, neque uſu, paucis exceptis, à ſuis ſimplicibus diſcrepant. Videamus porrò, qua ratione componenda ſint.

These, then, are all the simple intervals that arise either immediately or mediately from the division of the octave made by Ptolemy (which, without controversy, is the best of all), and which, in the order in which they are produced, together with their constitutive terms and names, you may see in the following table:

Simple Intervals.

Octave : 2:1	Diapason, or the double.
Fifth : 3:2	Diapente, or the sesquialteral.
Fourth : 4:3	Diatesseron, or the sesquitertial.
Major third : 5:4	Ditone, or the sesquiquartal.
Minor third : 6:5	Semiditone, or the sesquiquintal.
Major tone : 9:8	Sesquioctave.
Minor tone : 10:9	Sesquinona.
Major sixth : 5:3	Greater hexachord, or supertripartient of the third.
Minor sixth : 8:5	Lesser hexachord, or superbipartient of the fifth.
Major semitone : 16:15	Sesquidecima quinta.
Minor semitone : 25:24	Sesquivigesima quarta.
Comma : 81:80	Sesquioctogesimal.
Tritone : 45:32	Supertredecimpartient of the thirty-seconds.
False fifth : 64:45	Superoctodecimpartient of the forty-fifths.

Although some of these intervals may in a certain sense be called composite—because they must necessarily be constructed from combined intervals, as the major sixth from the major third and the fourth, the minor sixth from the minor third and the fourth, or at least may be so constructed, as the fifth from the major and minor thirds, etc. —yet properly only those are included under the name of composite intervals which, when joined to the octave, exceed it in extension: such as the ninth, tenth, eleventh, twelfth, etc., which, except in a few cases, differ neither by nature nor by use from their simple counterparts.Let us now see by what method they are to be composed.

CAPUT XXII.

De Intervallis compoſitis, & eorum componendorum methodo.

Dictum eſt alioquin, intervalla compoſita propriè appellari ſolita eſſe ea, quæ cum Octava componuntur, quod fit adjungendo quoddam intervallum Octavæ per additionem: volenti E. G. componere nonam minorem, addenda eſt ratio ſemitonii majoris: 16. 15. rationi Octavæ, hoc modo:

16. 15
2. 1.
32. 15.

Ad componendam nonam majorem addatur tonus major Octavæ, hac ratione:

2. 1.
9. 8.
18. 8.
9. 4.

In reperienda decima minore pari ratione adjungenda eſt Octavæ tertia minor:

2. 1.
6. 5.
12. 5.

& ſic de cæteris.

Si autem Octavæ bis poſitæ adjungatur aliquod intervallum, ùt in exemplo:

2. 1.
2. 1.
9. 8.
36. 8.
18. 4.

Intervallum bis compoſitum ratio 18. 4. nonæ majoris bis compoſitæ dicenda erit.

Pari modo, ſi adjungatur Octavæ ter poſitæ quodvis intervallum, illud ter compoſitum dicetur, ut ſi quis habere deſide-

CHAPTER XXII

On Composite Intervals and the Method of Constructing Them.

It has already been said that composite intervals are properly those which are formed together with the octave, which is done by adding some interval to the octave. For example, one who wishes to form the minor ninth must add the ratio of the major semitone, 16:15, to the ratio of the octave, in this manner:

16:15
2:1
32:15

To form the major ninth, let the major tone be added to the octave, in this manner:

2:1
9:8
18:8
9:4

To find the minor tenth, in like manner the minor third must be added to the octave:

2:1
6:5
12:5

—and so with the rest.

If, however, an interval is added to the octave taken twice, as in the example:

2:1
2:1
9:8
36:8
18:4

the interval thus twice compounded, with the ratio 18:4, will be called the major ninth twice compounded.

Likewise, if any interval is added to the octave taken three times, it will be called thrice compounded.

ſideret decimam ter compoſitam, adjungat Octavæ ter poſitæ tertiam majorem :

2. 1.
2. 1.
2. 1.
5. 4.

40. 4.

& habetur ratio decupla, quæ forma eſt decimæ majoris ter compoſitæ.

En in ſequenti tabula intervalla primò ſive ſemel compoſita, relictis cæteris (nempe ne in odioſam prolixitatem effundamur) bis, ter, quater compoſitis. Etenim in memoriam revocatis ſuprà dictis, facilis erit eorum inveniendorum ratio.

Intervalla primò compoſita.

Nona minor :	32.	15.	*Diapaſon, & Semitonium maj.*
Nona major :	9.	4.	*Diapaſon, & tonus major.*
Decima minor :	12.	5.	*Diapaſon, & Semiditonus.*
Decima major com.	5.	2.	*Diapaſon, & Ditonus.*
Tritonus compoſ.	45.	16.	*Diapaſon, & Tritonus.*
Undecima compoſ.	8.	3.	*Diapaſon, & Diateſſeron.*
Quinta falſa comp.	128.	45.	*Diapaſon, & Semitonium maj.*
Duodecima :	3.	1.	*Diapaſon, & Diapente.*
Sexta minor comp.	16.	5.	*Diapaſon, & Hexachordon min.*
Sexta major comp.	10.	3.	*Diapaſon, & Hexachordon maj.*
Septima minor com.	9.	5.	*Diapaſon, & Heptachordon min.*
Septima major com.	15.	4.	*Diapaſon, & Heptachordon maj.*
Octava compoſ.	4.	1.	*Bisdiapaſon.*

Hæc pro meo inſtituto, dilecte Philomuſe, ſufficere arbitror, quibus tanquam face prælatâ vaſtiſſimæ hujus Scientiæ tenebras perſcrutari, eorumque adjumento, ſciendi aviditatem tuam explere poſſis. Haud me fugit, pro rei amplitudine me rem vix attigiſſe. Neque plus temporis huic Scriptioni dandum exiſtimabam, ne in Cenſuram Marci Tull. Cicer. in-

Thus, if one wishes to have the major tenth thrice compounded, let him add the major third to the octave taken three times:

2:1
2:1
2:1
5:4
40:4

and the ratio 10:1 is obtained, which is the form of the major tenth thrice compounded.

Here, in the following table, are the intervals compounded once (or simply), leaving aside the others (lest we fall into tedious prolixity) that are compounded twice, thrice, or four times. For, recalling what has been said above, the method of finding them will be easy.

Intervals compounded once.

Minor ninth : 32:15	Octave & Major semitone.
Major ninth : 9:4	Octave & Major tone.
Minor tenth : 12:5	Octave & Semiditone.
Major tenth : 5:2	Octave & Ditone.
Composite tritone : 45:16	Octave & Tritone.
Composite eleventh : 8:3	Octave & Fourth.
Composite false fifth : 128:45	Octave & Major semitone.
Twelfth : 3:1	Octave & Fifth.
Composite minor sixth : 16:5	Octave & Minor sixth.
Composite major sixth : 10:3	Octave & Major sixth.
Composite minor seventh : 9:5	Octave & Minor seventh.
Composite major seventh : 15:4	Octave & Major seventh.
Composite octave : 4:1	Double octave.

These things, dear Philomuse, I judge sufficient for my purpose—by which, as by a torch carried before you, you may explore the darkness of this vast science, and with their help satisfy your eagerness for knowledge. I am well aware that, considering the magnitude of the subject, I have scarcely touched upon it. Nor did I think more time should be spent on this writing, lest I fall under the censure of Marcus Tullius Cicero,

currerem, qui *lib.* 1. *de Officiis* monet, *ne nimis magnum ſtudium, multamque operam in res obſcuras conferamus, eaſdemque non neceſſarias.* Introducto enim novo hodiernæ Muſicæ ſyſtemate mutatum eſſe illud antiquum, corruiſſéque cum ſuis Tetrachordis tria illa Genera Græca, *Diatonicum*, *Cromaticum*, & *Enarmonicum*, quis eſt, qui ignoret? quo factum eſt, ut pleraque Muſicæ intervalla proportionibus ſuis rationalibus conſiſtentia, aurium judicio potiùs, quàm ratione nitantur. Quid? objiciet fortaſſis nonnemo: itaque Muſica saltèm hodierna ſcilicet certis, neque unquam fallentibus principiis deſtituta, ac fallaci aurium judicio duntaxat nixa, Scientiæ nomine indigna erit. Ad quod reſpondeo, id dignitati Muſicæ noſtræ, infallibilitatique neutiquam obeſſe. Etenim ſi utriſque tonis majori, & minori, ſemitoniis majori, minorique, & aliquibus adhuc minoribus intervallis cuilibet ſingillatim ſua peculiaris etiam ſtatuatur clavis, id quod multi Muſices periti, & veteres, & recentiores ſuis Claviaturis rebus ipſis monſtravêre, utique omnia hodie uſitata intervalla, maximè ſecundùm Ptolomæi diviſionem, ſuis proportionibus rationalibus probè conſiſtentia haberi poſſent. Cùm autem Claviaturæ uſum difficultate plenum perſpexiſſent, & nihilominùs penuriæ intervallorum conſulturi ad amplificationem eorum, & acquiſitionem conſonantiarum imperfectarum animum adjeciſſent, tonum, & ſemitonium in duas partes æquales dividere tentaverunt. Poſtquam autem numeris id effici non poſſe, compererunt, aurium judicium in ſubſidium vocavêre, auferendo ab uno intervallo quandam quaſi inſenſibilem quantitatem, eamque alteri adjiciendo, per quam induſtriam, ſublatâ Claviaturæ difficultate effecêre, ut Muſica noſtra ex intervallorum egeſtate, tanquam carcere eluctata, in vaſtiſſimum modulationis campum hodie lætiſſimè excurrere valeat; modò Compoſitores, & Organœdi inter rationis limites ſe contineant, ne inſcitiâ, vel tranſponendi aviditate tranſverſum acti in Organo maximè per intervalla Cromatica licentiùs evagari audeant. Quo facto evitari vix poteſt, quin locò grati, amœnique concentûs aſperrimus ululatus, offenſionisque plenus auribus obtrudatur.

Quantum inde rei Muſicæ beneficii, ſplendorisque acceſſerit, quantiſque propterea primum Autorem (quæ gloria Ariſtoxeno antiquo Philoſopho, optimo jure attribuitur) laudibus

who in Book I of *De Officiis* warns us not to devote too much study and labor to obscure matters, and those not necessary.

For who does not know that, with the introduction of the new modern musical system, the ancient one has been altered, and that the three Greek genera—Diatonic, Chromatic, and Enharmonic—have collapsed along with their tetrachords? Hence it has come about that most musical intervals, though consistent in their rational proportions, rely more on the judgment of the ear than on reason.

"What then?" someone may object. "Will modern music, lacking fixed and unfailing principles, and resting only on the deceptive judgment of the ears, be unworthy of the name of science?" To this I reply that such a thing in no way harms the dignity or certainty of our music. For if to each of the major and minor tones, the major and minor semitones, and even to certain smaller intervals, a peculiar key were assigned—something which many skilled musicians, both ancient and modern, have shown in their keyboard instruments—then indeed all intervals used today, especially according to Ptolemy's division, could be held as perfectly consistent with their rational proportions.

But when musicians perceived that the use of the keyboard was full of difficulty, and yet, wishing to remedy the scarcity of intervals and to enlarge them and acquire imperfect consonances, they attempted to divide the tone and semitone into two equal parts. When they discovered that this could not be done by numbers, they called upon the judgment of the ears for aid, subtracting from one interval a certain almost imperceptible quantity and adding it to another. By this device, having removed the difficulty of the keyboard, they caused our music—freed, as it were, from the prison of interval-poverty—to run today most joyfully into the vast field of modulation, provided that composers and organists keep themselves within the bounds of reason, and do not, through ignorance or eagerness to transpose, wander too freely on the organ, especially through chromatic intervals. For if they do, it can scarcely be avoided that, instead of a pleasant and agreeable harmony, a harsh howling full of offense will be forced upon the ears.

How much benefit and splendor has thereby accrued to the art of music, and how greatly we ought therefore to praise its first author (a glory rightly attributed to the ancient philosopher Aristoxenus), I think no one even slightly versed in music can fail to perceive.

bus efferamus, neminem Muſicâ vel leviter tinctum latere exiſtimo.

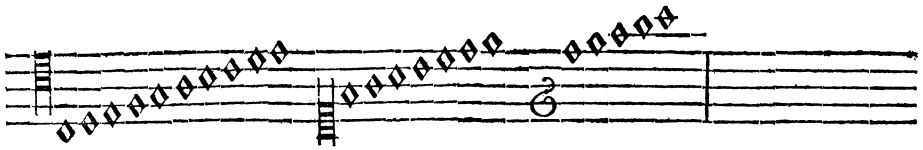
Quamvis ea, quæ ſubjiciam, de hodierno nempe Muſicæ Syſtemate, potiùs ad librum ſecundum, partem nempe practicam ſpectare videantur, ut tamen libri ſequentis exordium mox à præſcribendis lectionibus ducatur, ea hujus libri extremo fini inſerere conſultiùs duxi. Sit ergo

CAPUT POSTREMUM.

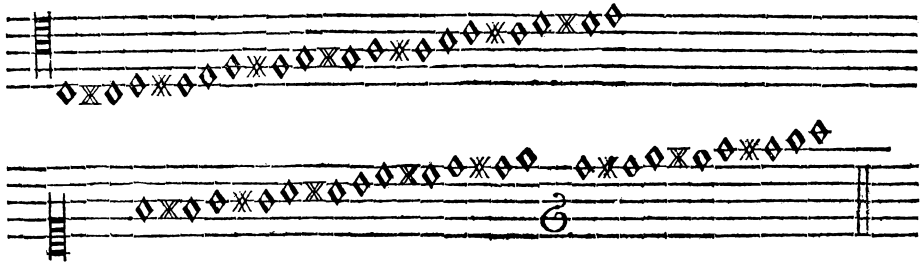
De hodierno Muſicæ Syſtemate.

SUblatâ Tonorum, & Semitoniorum diſparitate tria genera græca abolita fuiſſe, ſuprà commonuni, idque hîc in memoriam revocâſſe juverit: quam ob rem aliud Syſtema neceſſitas introduxit, illaque tria genera ad duo reſtrinxit, *Diatonicum* nempe, & *Cromaticum*, quod in Græcorum locum ſuceſſit, atque in præſenti oſtenſurus ſum ſequenti extenſione expreſſa:

Genus Diatonicum modernum.



Genus Cromaticum modernum.

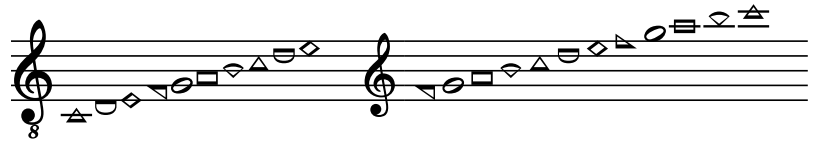
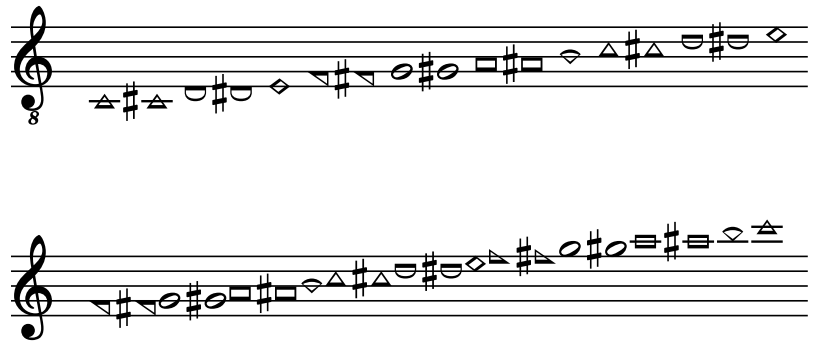


Cùm autem hominum inſatiabili cupiditati, in varietatem, novitatemque nunquam non imminenti, nondum utroque horum Generum ſatisfactum eſſet, Compoſitores duo illa promiſcuè in ſuis Compoſitionibus adhibere conſueverunt, ità ut hodie genus mixtum plerumque uſu receptum ſit, quod ego (quia tempori omnino ſerviendum eſt) minimè improbo;

Although the things I shall now add—concerning the modern musical system—seem rather to belong to the second book, that is, the practical part, yet so that the beginning of the next book may proceed at once to the prescribed lessons, I have judged it more convenient to place them at the end of this book. Let therefore follow

CHAPTER THE LAST
On the Modern Musical System.

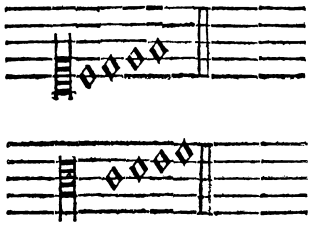
Since, with the removal of the inequality of tones and semitones, the three Greek genera have been abolished—as we remarked above—it will be useful to recall that fact here. For this reason necessity introduced another system, and restricted those three genera to two: namely the Diatonic and the Chromatic, which has taken the place of the Greek genera, and is shown here in its modern extension:

Modern Diatonic Genus

Modern Chromatic Genus

But since the insatiable desire of men—always pressing toward variety and novelty—was not yet satisfied by either of these genera, composers began to use the two promiscuously in their compositions, so that today a mixed genus is for the most part commonly employed. This I do not at all disapprove (because one must in some measure serve the times).

Compoſitores tamen admonitos velim, ne hoc genere mixto in Compoſitionibus à Cap.la abſque Organo decantari ſolitis utantur; aliàs certi ſint, ſe ſperatum effectum nunquam habituros; nec enim huic ſtylo præter genus Diatonicum purum, conſultum eſſe poteſt. Atque hanc magni momenti admonitionem, uſu multiplici, experientiâque doctus magnopere omnibus commendatam eſſe cuperem.

Jam ſuppoſitâ tonorum, & ſemitoniorum æqualitate ad intervalla hodie uſitata progrediamur, initium ſumendo

De Uniſono.

Uniſonum intervallorum nomine comprehendi non poſſe, ex eo patet, quòd ſuum eſſe in proportione æqualitatis habeat : ùt 1. ad 1., inter cujus extrema nulla intercedit ratione acuminis differentia. Quia tamen ejus extrema perfectiſſimè omnium conſonantiarum conſonare perſpicuum eſt, præſtantemque in praxi uſum habet, tantum abeſt, ut è conſonantiarum numero expungatur, ut potiùs conſonantiarum perfectiſſimam ſtatuendam, Practicus nemo dubitare poſſit. Etymologia ipſa explicat uniſonum, quòd ſit duarum chordarum, vel vocum idem ſonus, in notulis ita expreſſus :

Hujus, utì cæterarum conſonantiarum, & diſſonantiarum uſum libro ſequenti explanabimus, in præſenti earum eſſentiam contemplari libeat, incipiendo à Secunda, naturali ordine aſcendendo.

De Secunda.

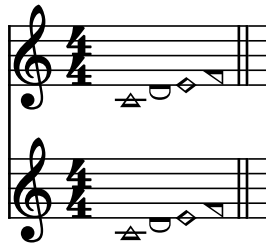
Secunda duplex eſt, major, & minor. Minor conſtat ſemitonio; major tono, ùt

Yet I would warn composers not to use this mixed genus in compositions intended to be sung a cappella without the organ; otherwise they may be certain that they will never achieve the effect they hope for. For to this style nothing is suitable except the pure diatonic genus. And this admonition—of great importance, and taught me by much use and experience—I would earnestly commend to all.

Now, with the equality of tones and semitones assumed, let us proceed to the intervals used today, beginning with

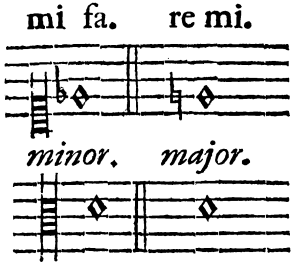
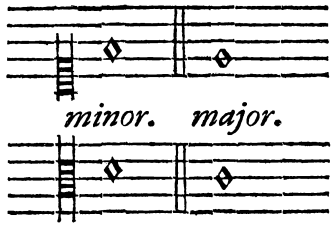
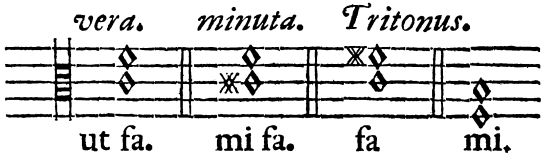
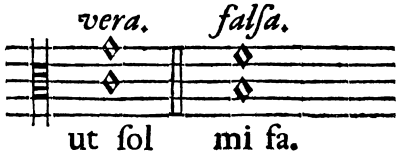
On the Unison.

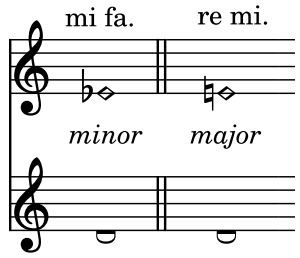
That the unison cannot be included under the name of intervals is clear, since it exists in the proportion of equality, 1 to 1, between whose extremes no difference of acuteness intervenes. Yet because its extremes sound together with the most perfect consonance of all, and because it has excellent use in practice, it is far from being removed from the number of consonances; rather, no practical musician can doubt that it should be considered the most perfect of consonances. The etymology itself explains *unisonus*: the same sound of two strings or voices, expressed in notation thus:



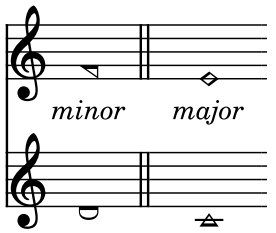
The use of this, as of the other consonances and dissonances, we shall explain in the following book; for the present let us consider their essence, beginning with the Second, ascending in natural order.

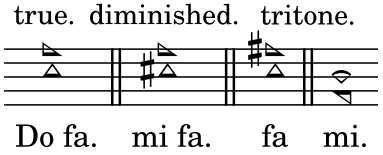
On the Second.

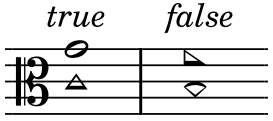
The second is twofold, major and minor. The minor consists of a semitone; the major of a tone, as:

Tertia pari modo duplex eſt : major & minor ; hæc conſtat tono , & ſemitonio. V. G. : re fa : major duobus tonis, V. G. ut mi:

Quarta triplex eſt : Quarta vera, minuta, & major, ſive tritonus. Quarta vera conſtat duobus tonis, & ſemitonio. V. G. ut fa. Minuta conſtat duobus ſemitoniis, & tono, V. G. mi fa, mi fa. Tritonus conſtat tribus tonis, V. G. fa mi. Paradigmata rem clariùs monſtrabunt :

Quinta duplex eſt, vera, & falſa. Vera conſtat tribus tonis, & ſemitonio. V. G. ut ſol, re la. Quinta falſa conſtat duobus ſemitoniis, & tono. V. G. mi fa, re mi fa.

Hîc item adnotandum eſt, à recentioribus quibusdam authoritate gravibus viris quintam *ſuperfluam* ſuis in Compoſitioni-



The third likewise is twofold, major and minor. The minor consists of a tone and a semitone, e.g. re–fa; the major of two tones, e.g. ut–mi:

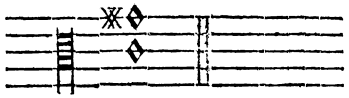
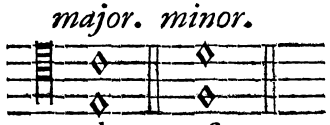


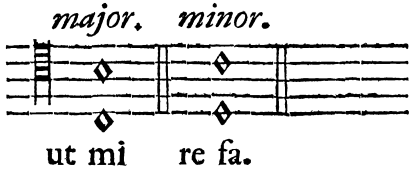
The fourth is threefold: the true fourth, the diminished fourth, and the augmented fourth, or tritone. The true fourth consists of two tones and a semitone, e.g. ut–fa. The diminished consists of two semitones and a tone, e.g. mi–fa, mi–fa. The tritone consists of three tones, e.g. fa–mi. The paradigms will show the matter more clearly:

The fifth is twofold, true and false. The true consists of three tones and a semitone, e.g. ut–sol, re–la. The false fifth consists of two semitones and a tone, e.g. mi–fa, re–mi–fa.

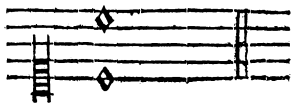


Here it should also be noted that certain more recent and authoritative musicians have introduced the *superfluous fifth* into their compositions, whose use—when rightly handled—is by no means to be despised, as I myself have often experienced in long practice (of which in the next book).

tionibus fuiſſe introductam, cujus uſum, dum ritè tractetur, haud quaquam contemnendum eſſe, cùm ipſe diutino meo uſu ſæpè numero expertus fuerim; (de quo libro ſequenti) ideo eam quoque adjicere, & explicare viſum eſt.

Quinta *ſuperflua* conſtat tribus tonis, & duobus ſemitoniis, hac ratione:

Sexta quoque duplex eſt; major, & minor. Major conſtat quatuor tonis, & ſemitonio. V. G. ut la. Minor conſtat tribus tonis, & duobus ſemitoniis. V. G. re fa.

Hîc quoque de Sexta ſuperflua, quam hodierni Compoſitores frequentiſſimè adhibent, dicendum foret: mihi autem uſus ejus nunquam probari potuit, ob rationem libro ſequenti dicendam, quam tamen, ut noſcatur, nec tam ejus uſurpandæ, quàm vitandæ gratiâ appono.

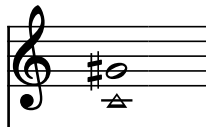
Septima eodem modo duplex eſt, major, & minor. Major conſtat quinque tonis, & ſemitonio. V. G. ut mi. Minor verò quatuor tonis, & duobus ſemitoniis: re fa.

Octava conſtat quinque tonis, & duobus ſemitoniis; nec alterari poteſt, vel ampliari, vel conſtringi. E. G.

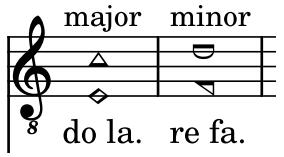
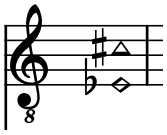


Omnia

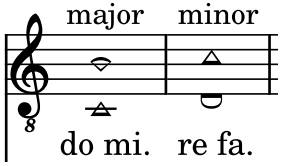
Therefore I have hought it proper to add and explain it as well.

The superfluous fifth consists of three tones and two semitones, thus:

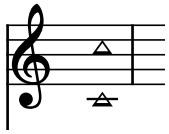


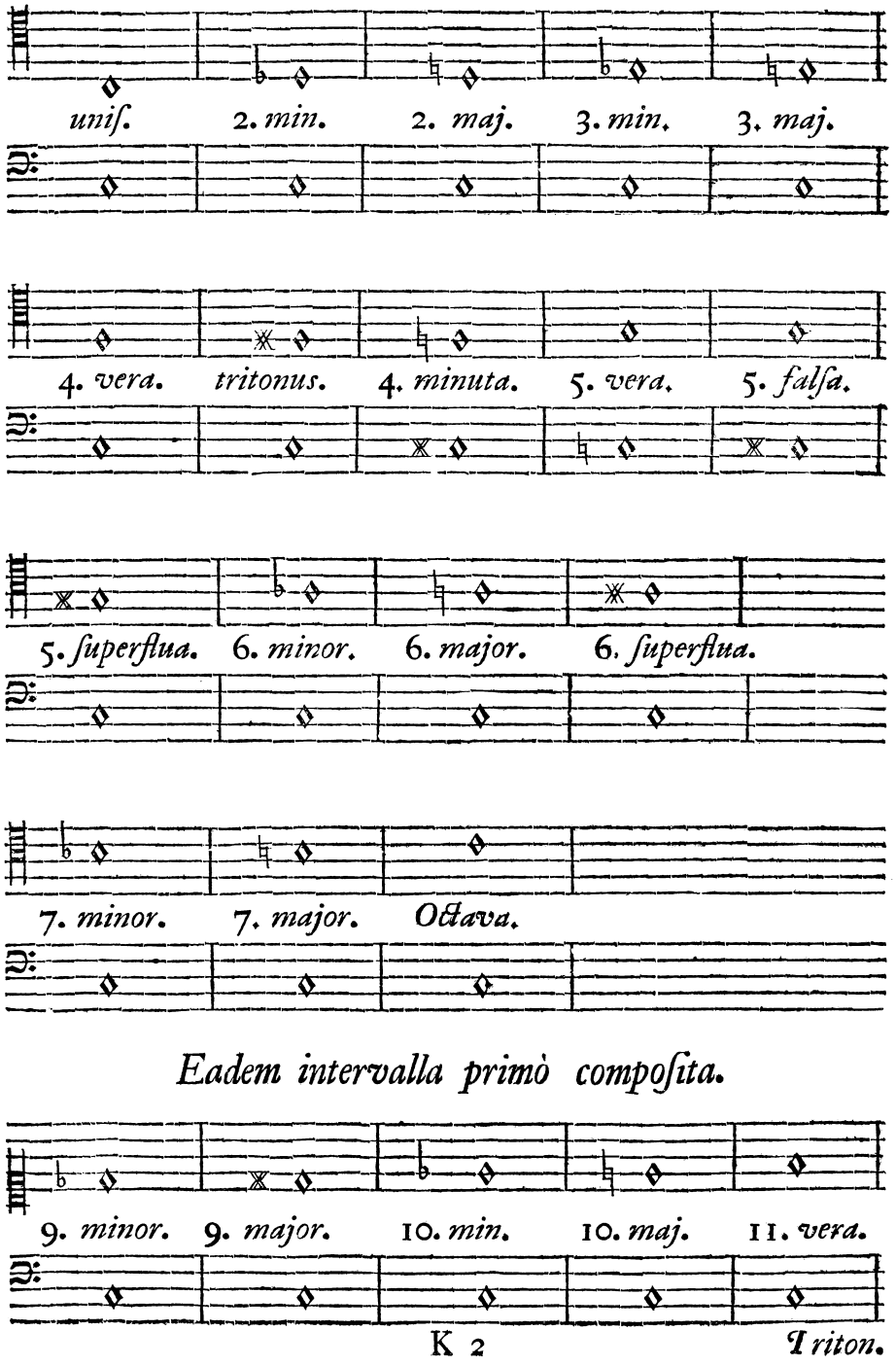
The sixth likewise is twofold, major and minor. The major consists of four tones and a semitone, e.g. ut–la. The minor consists of three tones and two semitones, e.g. re–fa.

Here also something should be said of the *superfluous sixth*, which modern composers use very frequently; but its use has never pleased me, for a reason to be given in the next book. Nevertheless, that it may be known—not so much for use as for avoidance—I append it:
{fig}

it consists of five tones.

The seventh likewise is twofold, major and minor. The major consists of five tones and a semitone, e.g. ut–mi. The minor of four tones and two semitones, e.g. re–fa.

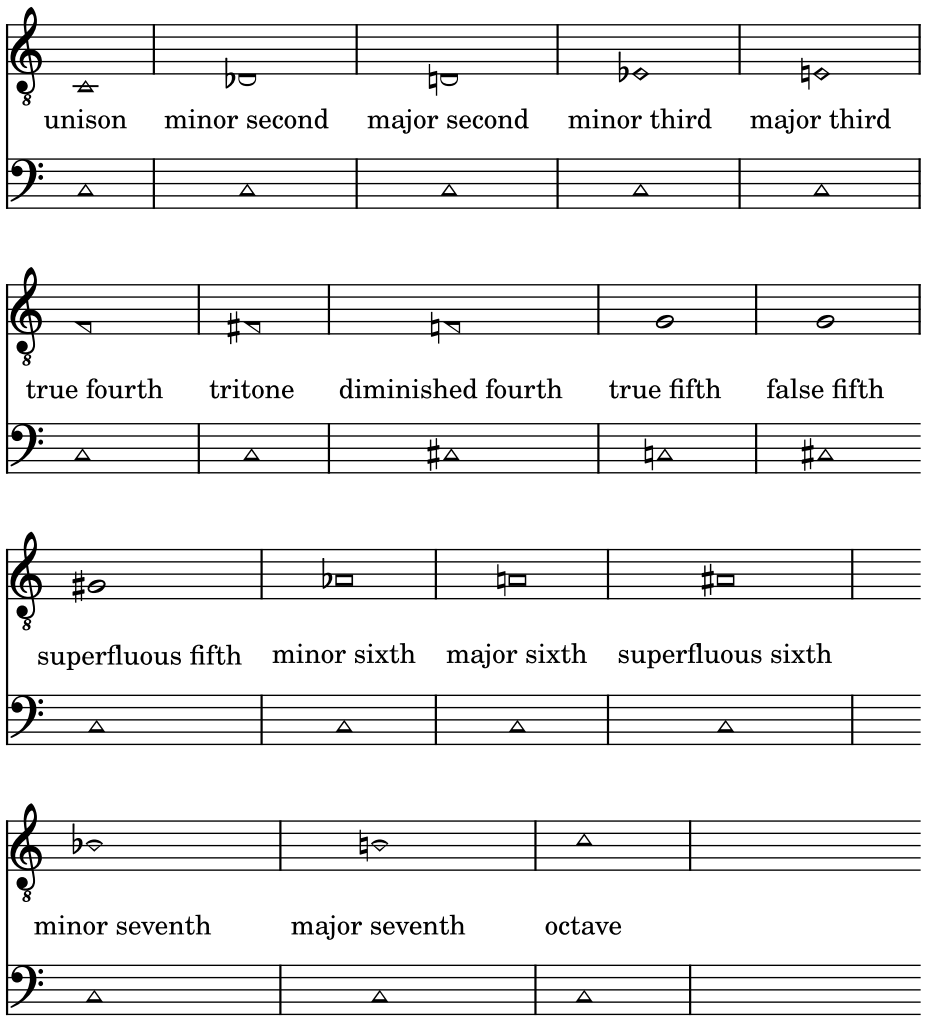


The octave consists of five tones and two semitones, and cannot be altered, enlarged, or contracted. For example:

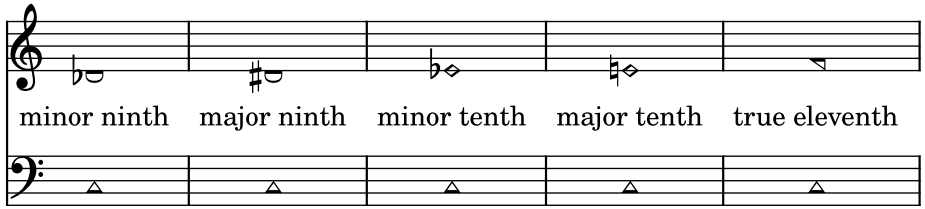
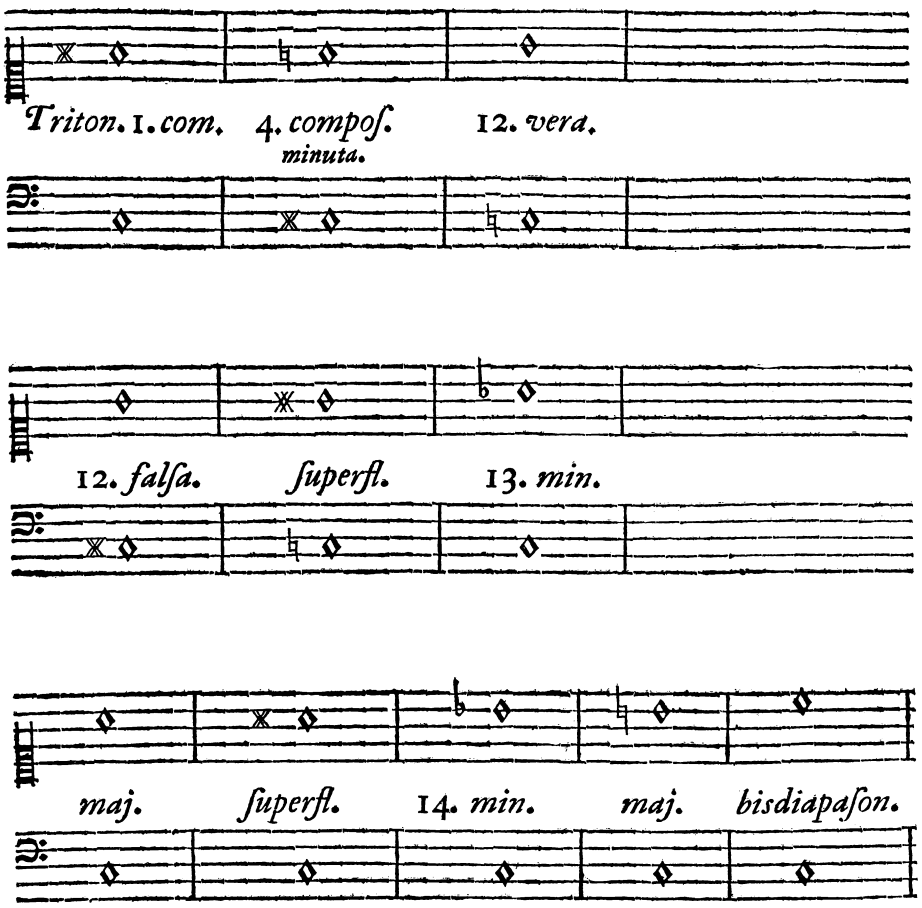


Omnia intervalla ſimplicia unâ ſerie collocata.

Eadem intervalla primò compoſita.

All the simple intervals arranged in one series.

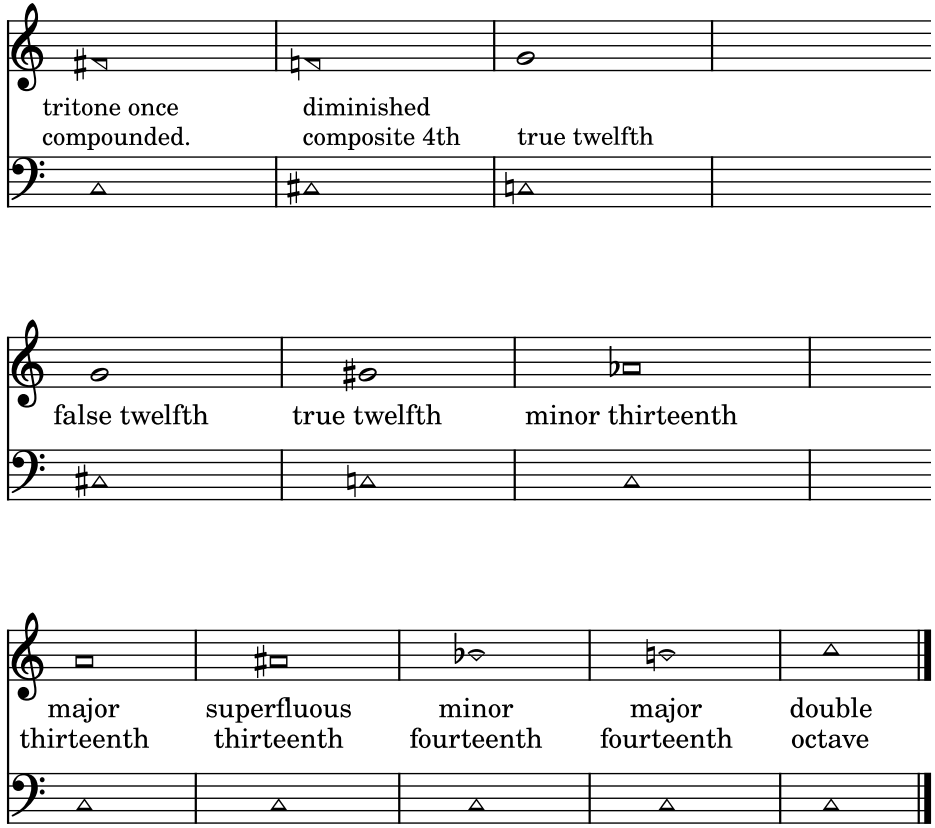


The same intervals compounded once.



Idem judicium eſto de bis compoſitis incipiendo à ſecunda Octava.

Hucuſque conſonantia intervalla cùm diſſonantibus promiſcuè deſignata ſunt; ſupereſt, ut eadem ſeorſim deſcribantur; ut quænam ſint, & qualia, meliùs internoſcantur. Quidnam rei conſonantia ſit, & diſſonantia, haud verboſiùs exponendum eſt, cùm ex ipſis nominibus conſtet, conſonantiam eſſe, quæ blandè ſinè ulla diſcrepantia aures afficit, animumque mulcet. Contrà verò diſſonantiam, quæ durum quid, & aſperum auribus obtrudens, offenſionem magìs quàm delectationem affert.



The same judgment applies to those compounded twice, beginning from the second octave.

Up to this point consonant intervals have been designated together with dissonant ones; it remains to describe them separately, so that what they are and of what sort they are may be better understood. What consonance and dissonance are needs no verbose explanation, since from the very names it is clear that consonance is that which gently, without any discrepancy, affects the ears and soothes the mind; whereas dissonance, thrusting something harsh and rough upon the ears, brings offense rather than delight.

Con-

Conſonantiæ ſunt

1. 3. 5. 6. 8. cum ſuis compoſitis.

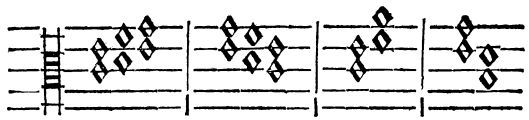
Ex his quædam *perfectæ* ſunt, quædam *imperfectæ*. Perfectæ ſunt, Uniſonus, Quinta, & Octava. Imperfectæ, Sexta, & Tertia. Reliquæ, ùt Secunda, Quarta, Quinta falſa, Tritonus, Septima cum ſuis compoſitis diſſonantiæ ſunt.

Atque hæc ſunt elementa, quibus univerſus Muſicæ concentus perficitur. Hujus finis eſt oblectare: oblectationem ſonorum varietas parit. Varietas tranſgrediendo de uno intervallorum ad aliud generatur. Tranſgreſſio motu perficitur; ideo quid rei motus ſit, veſtigandum ſupereſt.

Motus *muſicus* eſt ille ambitus, quo de uno intervallo ad aliud versùs partem auctam, vel gravem fit progreſſio, quæ progreſſio cùm triplici modo fieri ſoleat, triplex item ſtatuendus eſt Motus *rectus*, *contrarius*, & *obliquus*.

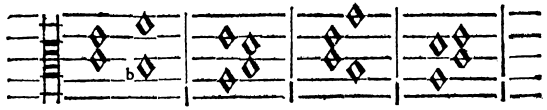
Motus *rectus* fit, quando duæ, vel plures partes aſcendendo, vel deſcendendo rectâ ſe viâ ſequuntur, ſive id fiat per gradum, vel per ſaltum, ùt in exemplo patet:

Motus rectus.

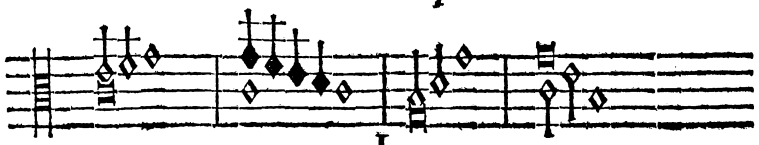


Motus *contrarius* eſt, quando pars una, deſcendente alterâ, & viciſſim, per ſaltum, vel gradatim aſcendit. E. G.

Motus contrarius.



Motus *obliquus* eſt, quando una pars per ſaltum, vel gradum ſe movet, alterâ immobili permanente, ùt videre eſt in exemplis:

Motus obliquus.

The consonances are:

1, 3, 5, 6, 8, with their compounds.

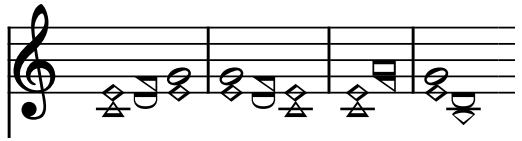
Of these, some are perfect, some imperfect. Perfect are the unison, the fifth, and the octave. Imperfect are the sixth and the third. The rest—such as the second, the fourth, the false fifth, the tritone, the seventh with its compounds—are dissonances.

And these are the elements by which the whole harmony of music is completed. Its end is delight; delight is produced by variety of sounds. Variety is generated by passing from one interval to another. This passage is accomplished by motion; therefore we must inquire what musical motion is.

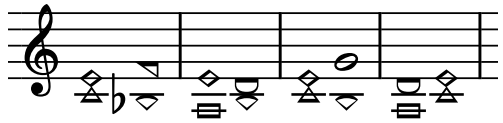
Musical motion is that course by which the progression is made from one interval to another, either upward or downward. Since this progression may occur in three ways, three kinds of motion must be established:

Direct, contrary, and oblique motion.

Direct motion occurs when two or more parts ascend or descend together in a straight course, whether by step or by leap, as is shown in the example:

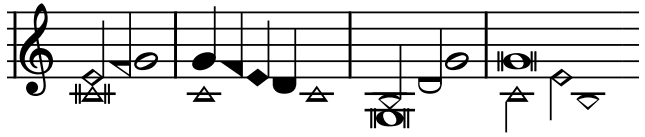
Direct motion.

Contrary motion occurs when one part ascends while the other descends, and vice versa, whether by leap or by step. For example:

Contrary motion



Oblique motion occurs when one part moves by leap or step, while the other remains stationary, as is seen in the examples:

Oblique motion.



Cognito triplici hoc motu, videndum eſt, qua ratione in uſu practico adhibendus ſit. Quæ doctrina ſequentibus quatuor cardinalibus veluti *Regulis* continetur.

Regula prima.

De Conſonantia perfecta ad perfectam proceditur per motum contrarium, vel obliquum.

Regula ſecunda.

De Conſonantia perfecta ad imperfectam per omnes tres motus.

Regula tertia.

De imperfecta ad perfectam per motum contrarium, vel obliquum.

Regula quarta.

De imperfecta, ad imperfectam per omnes tres motus.

Ubi animadvertes, motum obliquum in omnibus quatuor progreſſionibus eſſe licitum.

Ab hac triplicis motûs cognitione, uſuque recto pendet, (ùt dicere ſolemus) Lex, & Propheta.

Atque hæc erant, quæ ad Theoreticam Muſicam, intimosque artis receſſus pertinentia præſente Libro paucis declaranda duximus, nunc ad rem ipſam, partemque Operis ſecundam gradum promoveamus.

Having understood these three kinds of motion, we must see how they are to be used in practice. This doctrine is contained in the following four principal rules:

First Rule.

From a perfect consonance to a perfect consonance one proceeds by contrary or oblique motion.

Second Rule.

From a perfect consonance to an imperfect one by all three motions.

Third Rule.

From an imperfect consonance to a perfect one by contrary or oblique motion.

Fourth Rule.

From an imperfect consonance to an imperfect one by all three motions.

You will observe that oblique motion is allowed in all four progressions.

Upon the knowledge and right use of this threefold motion depends, as we are wont to say, "the Law and the Prophets."

And these were the things which, concerning Theoretical Music and the innermost recesses of the art, we judged should be briefly explained in this book. Now let us advance to the matter itself, and to the second part of the work.

LIBER

LIBER SECUNDUS.

DIALOGUS.

Joſephus. Te, venerande Magiſter, accedo, Muſicæ Præceptis, ac Inſtitutis imbuendus.

Aloyſius. Quid, muſicalem Compoſitionem addiſcere cupis?

Joſeph. Ità ſanè.

Aloyſ. An neſcis, Muſicæ Studium immenſum eſſe mare, neque Neſtoris annis terminandum? Verè rem difficilem, onusque Ætnâ gravius ſuſcipere intendis. Etenim ſi univerſo in genere vitæ deligendo res eſt plena difficultatis, utpote de qua electione aut rectè, aut ſiniſtrè ſuſceptâ, totius reliquæ vitæ dependet aut bonæ, aut malæ fortunæ curſus; quantâ præcautionis providentiâ utendum erit ei, qui hujus Diſciplinæ viam ingredi cogitat, antequam conſilium ſumere, & de ſe decernere audeat. Muſicus enim, & Poëta naſcuntur. Retro meminiſſe te oportet, an à tenera ætate, ſingulari quodam naturæ impetu impulſum te ad hoc ſtudium ſenſeris, Concentûsque oblectamento vehementer moveri tibi contingat?

Joſeph. Vehementiſſimè quidem. A quo tempore enim necdum bene rationis uſus mihi fuerat, neſcio, cujus ardoris vehementiâ raptus, omnes meas cogitationes, omnemque curam in Muſicam conjiciens, etiamnum mirabili planè ad eam addiſcendam cupiditate inflammatus, quaſi invitus rapior, dieque, noctuque aures dulci modulamine circumſonare videntur; ità ut de vocationis meæ veritate nullam prorsùs dubitandi cauſam habere mihi videar. Nec me abſtrahit laboris gravitas,

BOOK TWO.
DIALOGUE.

Joseph. I come to you, venerable Master, to be instructed and trained in the precepts of Music.

Aloysius. What, do you wish to learn musical composition?

Joseph. Indeed I do.

Aloys. Do you not know that the study of Music is a boundless sea, not to be crossed even in the years of Nestor? Truly you intend to undertake a difficult task, a burden heavier than Aetna. For if choosing one's way of life in general is full of difficulty—since on that choice, whether rightly or wrongly made, depends the whole course of one's future life, whether fortunate or unfortunate—how much foresight and caution must be used by the one who thinks of entering the path of this discipline, before he dares to take counsel and decide concerning himself. For a musician, like a poet, is born. You must look back and consider whether from your tenderest age you felt yourself driven to this study by some singular impulse of nature, and whether you have found yourself strongly moved by the delight of musical harmony.

Joseph. Most strongly indeed. For from the time when I had scarcely begun to use reason, I was seized—by I know not what vehemence of ardor—and cast all my thoughts and all my care into music. Even now, inflamed with a truly wondrous desire to learn it, I am drawn to it as if unwillingly, and day and night my ears seem to be surrounded by sweet melody, so that I seem to have no reason at all to doubt the truth of my vocation. Nor does the weight of labor deter me, for I trust that, with nature's help, it will not be difficult for me to overcome it. For I once heard an old man say that study is rather a pleasure than a toil.

vitas, quam ſuperare, naturâ adjuvante, non mihi difficilè futurum confido. Audivi enim à quodam Sene: Studium, voluptatem eſſe potiùs, quàm laborem.

Aloyſ. Mirabiliter delector, intelligens naturæ tuæ propenſionem. Unicam adhuc tibi movebo difficultatem, quâ ſublatâ, in numerum meorum te adſcribam.

Joſeph. Edic liberè, Magiſter obſervande. Sed certus ſum, neque iſto tuo, neque ullo alio motivo me deterritum iri, ut de concepto meo Conſilio deſiſtam.

Aloyſ. Forſan ſpe futurarum divitiarum, rerumque familiarium copiâ ad amplectendum hujus vitæ ſtatum titillaris? Si ità res ſe habet, crede mihi, muta conſilium. Non enim Plutus, ſed Apollo præeſt Parnaſſo. Aliâ viâ incedendum eſt iis, qui divitiarum cupiditate incenſi, aditum ad eas reperire volunt.

Joſeph. Minimè. Vellem tibi perſuadeas, alium voti mei ſcopum non eſſe, quàm ipſum Muſices amorem, abſtractum ab omni lucri deſiderio. Accedit, quod meminerim, perſæpe à Præceptore meo admonitum me, ut mediocri vivendi ratione contenti, virtutis, famæ, nominisque ſtudioſiores eſſe cuperemus, quàm facultatum, eò quòd virtus ſui ipſius ſit præmium.

Aloyſ. Incredibiliter lætor, reperiſſe me juvenem meo genio accommodatum; ſed notane tibi ſunt ea omnia, quæ de Intervallis, Conſonantiarum Diſſonantiarumque diviſione, de Motuum varietate, & de quatuor Regulis Libro præcedenti dicta ſunt.

Joſeph. Nîl illorum, quantum quidem credo, me latere exiſtimo.

Aloyſ. Ergo ad opus pergamus, exordium ſumendo ab ipſo Deo ter maximo, omnium Scientiarum fonte.

Joſeph. Antequam initium Lectionum ineamus, liceat mihi, venerande Magiſter, primò interrogare, quid nomine *Contrapuncti* intelligendum ſit, quod vocabulum adeò commune non ſolùm in ore peritorum, verùm etiam Muſicæ Artis expertium tam frequenter verſari audio?

Aloyſ. Rectè interrogas. Hoc enim erit noſtri Studii, noſtrique laboris princeps objectum. Scire oportet, loco Notularum modernarum antiquis temporibus puncta adhibita fuiſſe, ità ut Compoſitio punctis contra puncta poſitis *Contra-*

Aloys. I am wonderfully delighted, seeing the inclination of your nature. One difficulty only I will still raise for you; once removed, I shall enroll you among my pupils.

Joseph. Speak freely, revered Master. But I am certain that neither by this nor by any other motive will I be deterred from my chosen purpose.

Aloys. Perhaps you are drawn by the hope of future wealth and abundance of worldly goods to embrace this way of life for its usefulness? If this is the case, believe me, change your plan. For it is not Plutus but Apollo who presides over Parnassus. Those who burn with desire for riches must walk another path if they wish to find access to them.

Joseph. Not at all. I wish you to be persuaded that the aim of my desire is nothing other than the love of Music itself, apart from all desire for gain. Moreover, I often remember being warned by my teacher that we should be content with a modest way of life, and flourish in the pursuit of virtue, reputation, and honor rather than wealth; since virtue is its own reward.

Aloys. I am incredibly glad to have found a young man suited to my own disposition. But tell me—are you familiar with all that was said in the previous book concerning intervals, the division of consonances and dissonances, the variety of motions, and the four rules?

Joseph. I believe that none of these things are unknown to me.

Aloys. Then let us proceed to the work, beginning from God Himself, thrice-great, the source of all knowledge.

Joseph. Before we begin the Lessons, allow me, venerable Master, first to ask what is meant by the term Counterpoint; for I hear this word so commonly used, not only by experts, but even by those unskilled in the Art of Music.

Aloys. You ask rightly. For this will be the chief object of our Study and our labor. You must know that in place of modern notes, the ancients used points, so that a composition made by placing point against point came to be called Counterpoint.

trapunctum appellari consueverit ; quod etiamnum hodie non obstantibus aliarum Notularum figuris , usuvenit ; & nomine Contrapuncti Compositio intelligitur ad Artis Regulas elaborata. Complures Contrapunctum tanquam Genus continet Species , quas succesſivè videbimus. Interim fiat principium à Specie simpliciſſima , nempe :

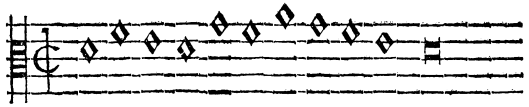
EXERCITII I.

LECTIO PRIMA,

De Nota contra Notam.

Joſeph. Bonitate tuâ satisfactum est primæ Quæstioni meæ. Nunc docere ne graveris, quid sit prima ista Species Contrapuncti, *Nota contra Notam.*

Aloyſ. Dicam. Est duarum, vel plurium vocum, Notarumque valore æqualium simpliciſſima Compoſitio, meris Consonantiis constans. Nec interest, cujus qualitatis Notæ, modò omnes sint ejusdem valoris. Quia tamen Nota semibrevis commodiorem intelligendi præbet viam, eâ in his Exercitationibus nobis utendum esſe, consultum existimavi. Fiat ergo principium, Auspice Deo, à Compoſitione duarum vocum, ponendo quendam Cantum firmum , vel à proprio judicio inventum, vel ex aliquo Libro Chorali assumptum pro fundamento. E. G.



Cuilibet harum Notarum superiùs in Cantu detur sua particularis consonantia , obsservatâ motuum , regularumque ratione , in fine præcedentis Libri expreſsâ, adhibendo ùt plurimum, motum contrarium, vel obliquum , quorum usu non facilè in errorem incidere possumus. Majori præcautione opus erit in progreſsione de una Nota ad alteram per motum rectum , ubi, quia majus errandi periculum , major etiam ad Regulas attentio adhibenda est.

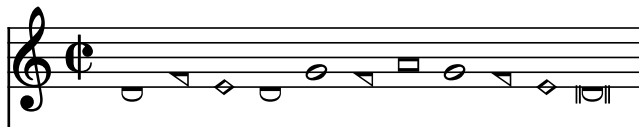
Joſeph. Ex motuum , regularumque perspicientia, omnia, quæ modò dixisti, mihi nota esſe videntur : verùm memini, te distinctionem feciſſe , Consonantiarum quasdam esſe perfe-

And even today, despite the various shapes of modern notes, the usage remains; and by the name Counterpoint is understood a composition worked out according to the Rules of the Art. Counterpoint, as a genus, contains several species, which we shall examine in order. Meanwhile, let us begin with the simplest species, namely:

EXERCISE I
FIRST LESSON,
On Note against Note.

Joseph. By your kindness my first question has been answered. Now do not hesitate to teach me what that first Species of Counterpoint is, Note against Note.

Aloys. I shall tell you. It is the simplest composition of two or more voices whose notes are of equal value, consisting solely of consonances. Nor does it matter of what quality the notes are, provided they are all of the same value. Yet because the whole note offers a more convenient way of understanding, I have judged it best that we use it in these exercises. Let us therefore begin, with God's favor, from a composition of two voices, placing a certain cantus firmus—either invented by your own judgment or taken from some choral book—as the foundation. For example:



To each of these notes in the upper voice give its own particular consonance, observing the rules of motion set forth at the end of the preceding book, using for the most part contrary or oblique motion, by which we are not easily led into error. Greater caution will be needed in progressing from one note to another by similar (direct) motion, where, because the danger of error is greater, greater attention must be given to the rules.

Joseph. From my understanding of the motions and rules, all that you have just said seems known to me; yet I remember that you made a distinction among consonances: some are perfect, others imperfect. It seems necessary to know whether some difference must also be made in using them.

ctas, quaſdam imperfectas, ſcire, ùt mihi videtur, neceſſe eſt, an etiam in iis adhibendis quædam differentia ſit facienda?

Aloyſ. Patientiam habe, omnia dicam. Magnam eſſe quidem diſparitatem inter Conſonantias perfectas, & imperfectas, quod alibi dicemus; hoc loco autem præter motûs rationem, & quòd plures imperfectæ, quàm perfectæ adhibendæ ſint, planè æqualis earum uſus eſt: excepto principio, & fine, quorum utrumque Conſonantiâ perfectâ conſtare debet.

Joſeph. Non diſpliceat tibi care Magiſter, explanare rationes, cur plures Conſonantiæ imperfectæ, quàm perfectæ hoc loco ponendæ ſint, & cur principium, & finis Conſonantiâ perfectâ conſtare debeant?

Aloyſ. Tuâ aviditate, quam tamen laudo, fit, ut vix non quædam mihi præpoſterè dicenda ſint. Dicam tamen, ſed non totum, ne intellectus principio tot rerum varietate obrutus confundatur. Scito ergo. Conſonantias imperfectas, perfectis magìs harmonicas eſſe, ob rationem alibi dicendam. Quapropter, ſi hujus Speciei, & duarum tantùm partium, & aliàs ſimpliciſſima Compoſitio perfectis Conſonantiis impleretur, inanem, & omni harmoniâ vacantem exiſtere, neceſſe foret. Quod attinet ad principium, & finem, cape hanc rationem. Principium perfectionis, & finis requietis index eſt. Cùm autem Conſonantiæ imperfectæ, & perfectionis expertes ſint, & finem concludere non poſſint, ideo principium, & finis Conſonantiis perfectis conſtare debent. Poſtremò notandum eſt, quòd ſi Cantus firmus in parte inferiori reperiatur, penultimæ Notæ Sexta major; ſi in ſuperiore, Tertia minor danda ſit.

Joſeph. Suntne iſta omnia, quæ ad ſpeciem hujus Contrapuncti requiruntur?

Aloyſ. Non omnia quidem, ſed quæ ſufficiunt ad fundamenta jacienda. Cætera ex correctione patebunt. Ad opus ergo te accingas, & poſito Cantu firmo pro fundamento, Contrapunctum in clave Cantûs, dictâ hucuſque ratione ſuperædificare tenta.

Joſeph. Faciam, ùt poſſum.

Con-

Aloys. Have patience; I shall explain everything. There is indeed a great disparity between perfect and imperfect consonances, as we shall say elsewhere; but here, apart from the rules of motion and the fact that more imperfect than perfect consonances must be used, their use is entirely equal—except at the beginning and the end, both of which must consist of a perfect consonance.

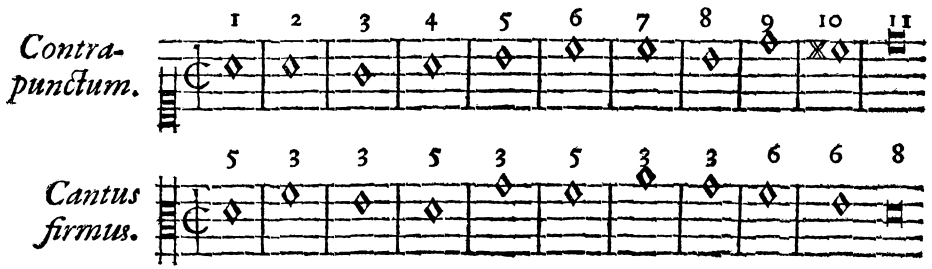
Joseph. Do not be displeased, dear master, to explain the reasons why more imperfect than perfect consonances must be placed here, and why the beginning and end must consist of a perfect consonance.

Aloys. Your eagerness—though I praise it—forces me to say some things out of order. I shall speak, but not everything, lest your understanding be overwhelmed at the outset by too much variety. Know, then, that imperfect consonances are more harmonious than perfect ones, for a reason to be explained later. Therefore, if a composition of this species—of only two parts and otherwise most simple—were filled with perfect consonances, it would necessarily be empty and devoid of all harmony. As for the beginning and end, take this reason: the beginning is a sign of perfection, and the end a sign of repose. But since imperfect consonances lack perfection and cannot conclude an ending, therefore the beginning and end must consist of perfect consonances. Finally, note that if the cantus firmus is in the lower part, the penultimate note must be a major sixth; if in the upper part, a minor third.

Joseph. Are these all the requirements for this species of counterpoint?

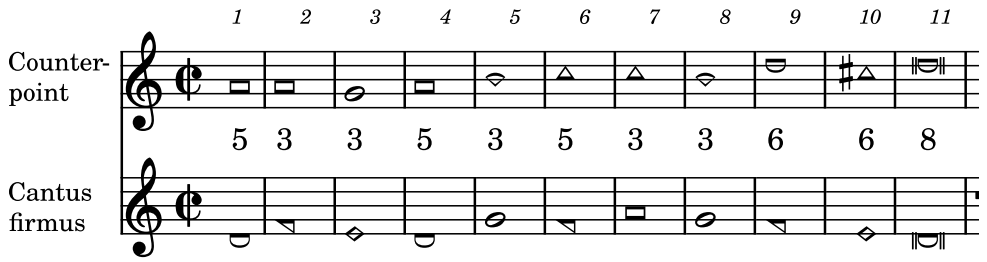
Aloys. Not all, but enough to lay the foundations. The rest will become clear through correction. Therefore prepare yourself for the work, and having set the cantus firmus as the foundation, try to build the counterpoint above it in the same key, according to what has been said.

Joseph. I shall do what I can.



Aloyſ. Rem acu tetigiſti. Miror, & perſpicacitatem, & attentionem tuam: ſed quem in finem numeros appoſuiſti, & ſuper Cantum, & ſuper Altum ?

Joſeph. Numeris ſuper Altum poſitis Conſonantias, quas adhibui, deſignare volui, ut habitis ante oculos de una Conſonantia ad alteram motibus minùs a recta progrediendi ratione deflecterem. Poſiti autem ſuper Cantum. E. G. 1. 2. 3. 4. 5. 6. 7. &c. nîl aliud indicant, quàm Notarum numerationem, ut tibi, venerande Magiſter, monſtrarem, ſi rem tetigi, non caſu, ſed ratione id accidiſſe. Juſſiſti principium fieri à Conſonantia perfecta, quod feci, poſitâ Quintâ. De prima Nota ad ſecundam, id eſt: de Quinta ad Tertiam, ſive de perfecta ad imperfectam progreſſus ſum motu obliquo, licèt id fieri poſſet per omnes tres motus. De ſecunda nota ad tertiam, nempe de Tertia ad Tertiam, ſive de imperfecta ad imperfectam adhibui motum rectum, ſecundùm Regulam, quæ dicit: de imperfecta ad imperfectam per omnes tres motus. De tertia Nota ad quartam, ſive de Tertia imperfecta Conſonantia ad Quintam perfectam, inceſſi per motum contrarium, juxta regulam dicentem: de imperfecta ad perfectam per motum contrarium. De quarta Nota ad quintam, ſeu de perfecta ad imperfectam progreſſus ſum regulâ permittente per motum rectum. De quinta ad ſextam Notam, id eſt de imperfecta ad perfectam, regulâ ita jubente adhibui motum contrarium. De ſexta Nota ad ſeptimam itum eſt per motum obliquum nulli errori ſubjectum. De Septima ad Octavam, ſive de imperfecta ad imperfectam, ùt regula permittit, motu recto perrexi. De octava Nota ad nonam, utpote de imperfecta ad imperfectam, motu quocunque. De nona ad decimam Notam eadem rectitudinis eſt ratio, ubi decima Nota, id eſt penultima, ùt juſſiſti, ſextam majorem efficit, quia Cantus firmus in inferiori



Aloys. You have hit the mark exactly. I admire both your insight and your attention. But why have you placed numbers both above the cantus and above the upper voice?

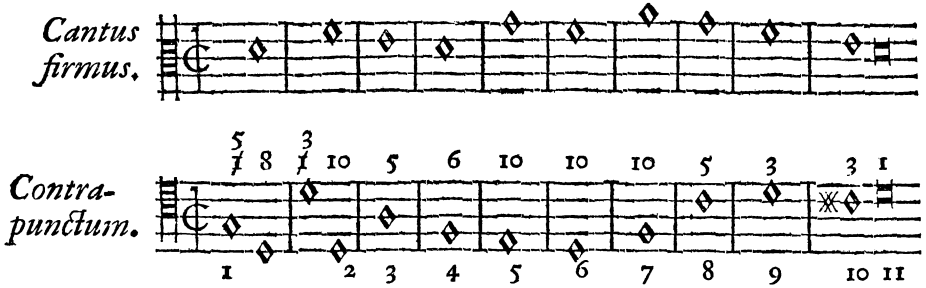
Joseph. By the numbers placed above the upper voice I wished to indicate the consonances I used, so that, having them before my eyes, I might not deviate from the proper manner of progressing from one consonance to another. The numbers placed above the cantus—1, 2, 3, 4, 5, 6, 7, etc.—indicate nothing more than the numbering of the notes, so that I might show you, venerable master, that if I have hit the mark, it happened not by chance but by reason. You ordered that the beginning be made with a perfect consonance, which I did by placing a fifth. From the first note to the second—that is, from the fifth to the third, or from a perfect to an imperfect consonance—I progressed by oblique motion, though it could have been done by any of the three motions. From the second note to the third—that is, from a third to a third, or from an imperfect to an imperfect consonance—I used similar motion, according to the rule that says: from imperfect to imperfect by any of the three motions. From the third note to the fourth—that is, from the imperfect third to the perfect fifth—I proceeded by contrary motion, according to the rule that says: from imperfect to perfect by contrary motion. From the fourth to the fifth note—that is, from perfect to imperfect—I progressed, as the rule permits, by similar motion. From the fifth to the sixth note—that is, from imperfect to perfect—I used contrary motion, as the rule requires. From the sixth to the seventh note I went by oblique motion, subject to no error. From the seventh to the eighth—that is, from imperfect to imperfect—I proceeded by similar motion, as the rule allows. From the eighth to the ninth note, being from imperfect to imperfect, by any motion. From the ninth to the tenth note the same reasoning applies; and the tenth note, being the penultimate, as you ordered, forms a major sixth, because the cantus firmus is in the lower part.

eſt parte. De decima Nota ad undecimam viam tenui ſecundùm regulam, quæ præcipit, ut, ſi quis de imperfecta ad perfectam pergere velit, id motu contrario fiat. Undecima Nota, utpote finalis, ſicut præcepiſti, perfecta Conſonantia eſt.

Aloyſ. Optimum in hac re ratiocinatorem egiſti. Quapropter te ſperare jubeo; ſi triplices illos motus, differentesque quatuor regulas ità memoriæ infixas habueris, ut etiam levi ad eas reflexione factâ, non aberrare tibi contingat, facilem jam ad ulteriores progreſſus viam tibi ſtratam eſſe. Jam perge, relicto Cantu firmo, ùt eſt in Alto, infra pone Tenorem pro parte Contrapuncti, eâ tamen differentiâ, ut ſicut priori in exemplo Conſonantiarum rationem de Cantu firmo aſcendendo dimenſus es, ita nunc eam de ipſo Cantu firmo deſcendendo ad fundamentum dimetiaris.

Joſeph. Hoc difficiliùs factu mihi videtur.

Aloyſ. Apparet. Quam difficultatem aliis etiam in Scholaribus me obſervâſſe recordor: ſed non tanti eſt, ſi, ùt dixi, animadverteris rationem Conſonantiarum, de Cantu firmo ad fundamentum numerando, inquirendam eſſe.



Joſeph. Quare, Magiſter venerande, primam, & ſecundam Notam ſigno erroris notâſti? Nonne in Quinta, Conſonantiâ perfectâ incepi? de illa quidem in ſecundam Notam, nempe Tertiam, motu recto perrexi, ſed regulâ permittente, quæ dicit: de perfecta ad imperfectam, motu quocunque; libera me, te quæſo, confeſtim hoc meo, quo premor angore, totus namque erubeſco.

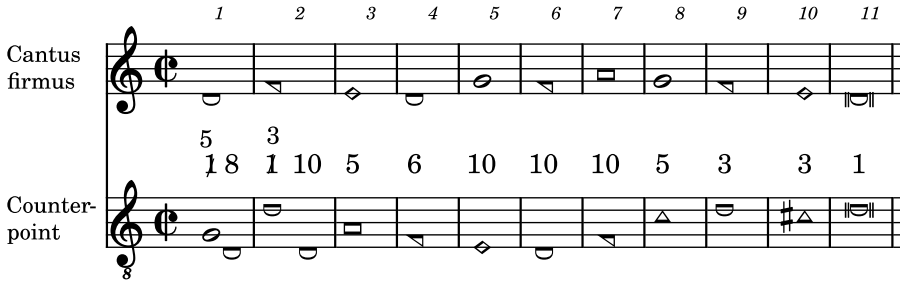
Aloyſ. Mitte angorem care fili; primus enim error non tuâ culpâ accidit, quia nondum Præceptum habuiſti, quòd Contrapunctum eidem tono, quo Cantus firmus eſt, ſit accommodandum, quod tibi modò dictum volo. Cùm enim præcedentis exempli Cantus firmus exiſtentiam ſuam habeat in *D, la, ſol, re*, ùt principium, & finis monſtrat, tu autem in *G, ſol,*

From the tenth to the eleventh note I followed the rule that commands that if one wishes to go from imperfect to perfect, it must be done by contrary motion. The eleventh note, being the final, is, as you prescribed, a perfect consonance.

Aloys. You have reasoned excellently in this matter. Therefore I bid you hope; if you have fixed in memory those three motions and the four different rules so firmly that even with slight reflection you do not stray from them, the way to further progress is already made easy for you. Now continue: leaving the cantus firmus as it is in the upper voice, place the Tenor below as the counterpoint, with this difference—that just as in the previous example you measured the consonances by ascending from the cantus firmus, so now you must measure them by descending from the cantus firmus to the foundation.

Joseph. This seems more difficult for me to do.

Aloys. It appears so. I recall observing this difficulty in other students as well; but it is not so great, if, as I said, you consider that the consonances must be reckoned from the cantus firmus down to the foundation.



Joseph. Why, venerable Master, have you marked the first and second notes with a sign of error? Did I not begin with a Fifth, a perfect consonance? From that I indeed proceeded to the second note, namely the Third, by similar motion, but the rule permits this, which says: from a perfect to an imperfect consonance, by any motion. Free me, I beg you, at once from this anxiety that weighs on me, for I am entirely ashamed.

Aloys. Set aside your anxiety, dear son; for the first error did not occur through your fault, since you had not yet received the precept that the Counterpoint must be adapted to the same mode as the cantus firmus—a point I now wish to tell you. For since the cantus firmus of the preceding example lies in D la sol re, as the beginning and end show, and you began in G sol re ut, it is clear that you began outside the mode.

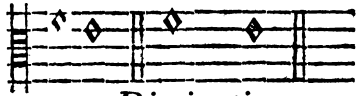
G, *ſol*, *re*, *ut*, inceperis, extra tonum te initium duxiſſe, clarum eſt; ideo in correctione locò Quintæ Octavam poſui, quæ eſt cum cantu firmo ejuſdem toni.

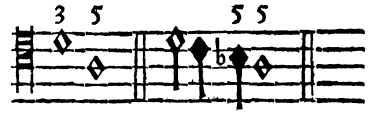
Joſeph. Gaudeo, hunc errorem, quem jam memoriâ teneo, inſcitiâ, non incuriâ meâ accidiſſe; ſed quæ ratio eſt ſecundi erroris in ſecunda Nota ſignati?

Aloyſ. Non eſt error ſecundæ Notæ reſpectu primæ, ſed ſecundæ Notæ reſpectu tertiæ: progreſſus enim es de Tertia ad Quintam motu recto, contra regulam, quæ dicit: de imperfecta ad perfectam, per motum contrarium; qui error facilis eſt correctionis, Notâ ſecundâ per motum obliquum in *D*, *la*, *ſol*, *re*, inferiùs permanente, & Decimam efficiente, quo caſu de ſecunda Nota ad tertiam, nempe de Decima ad Quintam, ſive de imperfecta ad perfectam, ùt regula præcipit, per motum contrarium progredi contigit. Nec te affligi volo parvum ob hunc errorem; vix enim poſſibile eſt incipienti, tantæ eſſe attentionis, ut nullum prorsùs in errorem incidat. Uſus eſt rerum magiſter. Interim contentum eſſe te jubeo, quòd reliquum rectè feceris; maximè, quòd penultimæ Notæ, Cantu firmo in parte ſuperiori exiſtente, ùt paulò antè monui, Tertiam minorem dederis.

Joſeph. Ut deinceps majori attentione huic regulæ incumbam, eamque firmiùs memoriæ mandem, non diſpliceat tibi rationem explanare, cur ab imperfecta Conſonantia in perfectam ire, per motum rectum non liceat? non enim videtur res tanti momenti.

Aloyſ. Dicam. Quia eo caſu duæ Quintæ continenter ſe ſequuntur, quarum una expreſſa, ſeu aperta eſt, altera verò coperta, ſeu abſcondita eſt, quam ultimam per diminutionem palàm fieri, nunc monſtrabo in exemplo.



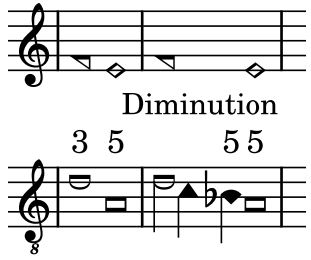
Diminutio.

Quæ diminutionem faciendi facultas perito Cantori adimi non poteſt, maximè voce ſolâ cantanti; quod idem intelligendum eſt de progreſſu Octavæ ad Quintam per motum rectum,

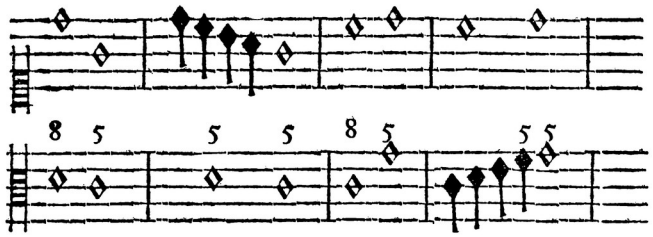
Therefore in the correction I placed an octave instead of the fifth, which is of the same mode as the cantus firmus.

Joseph. I am glad that this error—which I now remember—occurred through ignorance, not negligence. But what is the reason for the second error marked on the second note?

Aloys. It is not an error of the second note with respect to the first, but of the second with respect to the third. For you progressed from the Third to the Fifth by similar motion, against the rule which says: from an imperfect to a perfect consonance, by contrary motion. This error is easily corrected: let the second note remain lower, in D la sol re, by oblique motion, forming a Tenth; in which case, from the second note to the third—that is, from the Tenth to the Fifth, or from an imperfect to a perfect consonance—you proceed, as the rule prescribes, by contrary motion. Nor do I wish you to be distressed by this small error; for it is scarcely possible for a beginner to be so attentive as to fall into no error at all. Practice is the teacher of things. Meanwhile, I bid you be content that you have done the rest correctly—especially that, since the cantus firmus is in the upper part, you gave the penultimate note a minor third, as I mentioned earlier.

Joseph. So that I may henceforth apply greater attention to this rule, and fix it more firmly in memory, do not be displeased to explain the reason why it is not permitted to go from an imperfect consonance to a perfect one by similar motion. It does not seem to be a matter of great importance.

Aloys. I shall explain. Because in that case two Fifths follow one another consecutively: one expressed, or open, the other hidden, or concealed, which latter can be made evident by diminution, as I shall now show in an example.

The ability to diminish in this way cannot be taken from a skilled singer, especially one singing alone; and the same is to be understood of the progression from an octave to a fifth by similar motion,

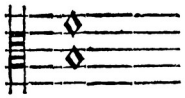
ctum, quo casu ob eandem rationem duas Quintas immediatè se sequi contingeret, ùt sequens exemplum monstrat.

Vides ergo, quomodo diminuendo Quintæ saltum, duæ Quintæ, quarum una antè per diminutionem abscondita erat, patefiant. Ex quo colligi potest, à Legislatoribus cujuscunque Facultatis nihil frustrà, nihil, quod ratione non fundatum sit, institutum fuisse.

Joseph. Video, & miror.

Aloys. Nunc perge, & eandem lectionem faciendo, percurre omnes tonos intrà hanc Octavam contentos, & naturali ordine gradatim sequentes. In D. incepisti; sequuntur modò: E. F. G. A. C.

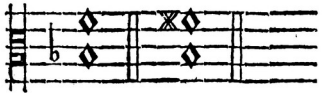
Joseph. Quare B. omisisti, situm inter A. & C.?

Aloys. Quia quintam Consonantiam non habet, nec adeò tonum constituere potest, ùt suo loco fusiùs dicemus; videamus exemplum.



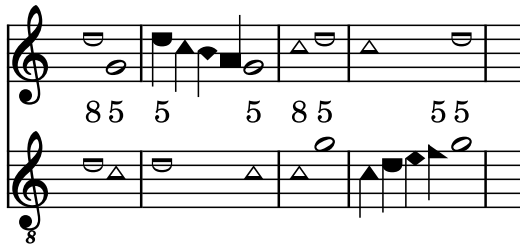
Quæ utpote duobus tantùm tonis, & tot semitoniis constans, Quinta falsa, & dissonans est; cùm Quinta vera, sive consonans constare debeat tribus tonis, & semitonio, ùt Libro primo dictum est.

Joseph. Non posset ex ista Quinta falsa effici Quinta consonans, adjungendo vel Notæ inferiori b., vel superiori Diesin 𝄪. ùt in hoc exemplo?



Aloys. Potest, sed hoc casu, utpote Quintâ hâc extra genus Diatonicum existente, non ampliùs tonus naturalis, de quibus

in which case, for the same reason, two Fifths would immediately follow one another, as the following example shows.

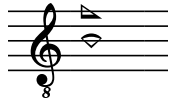


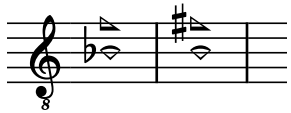
You see, therefore, how by diminishing the leap of a Fifth, two Fifths—one previously hidden by diminution—become exposed. From this it may be gathered that the legislators of any discipline have instituted nothing in vain, nothing not founded on reason.

Joseph. I see, and I marvel.

Aloys. Now continue, and by doing the same exercise, run through all the modes contained within this octave, following them step by step in natural order. You began in D; now follow: E, F, G, A, C.

Joseph. Why have you omitted B, which lies between A and C?

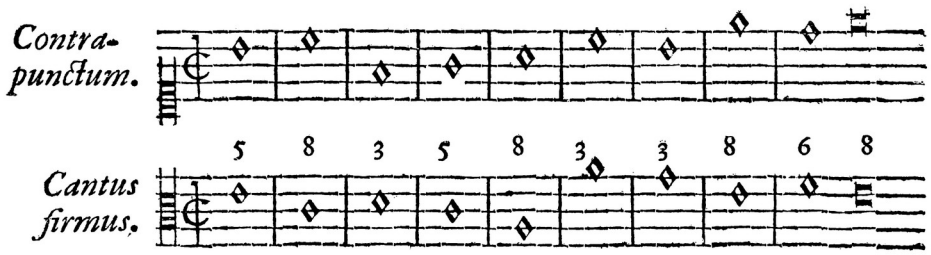
Aloys. Because it does not have a perfect fifth, and therefore cannot constitute a mode, as we shall say more fully in its proper place. Let us look at the example.

Since it consists of only two whole tones and as many semitones, it is a false and dissonant Fifth; whereas a true or consonant Fifth must consist of three whole tones and a semitone, as was said in the first book.

Joseph. Could not this false Fifth be made a consonant Fifth by adding a flat to the lower note or a sharp to the upper note, as in this example?

Aloys. It could; but in that case, since such a Fifth lies outside the diatonic genus, it would no longer be a natural mode—of which we are speaking now—but a transposed mode, of which we shall speak in its place.

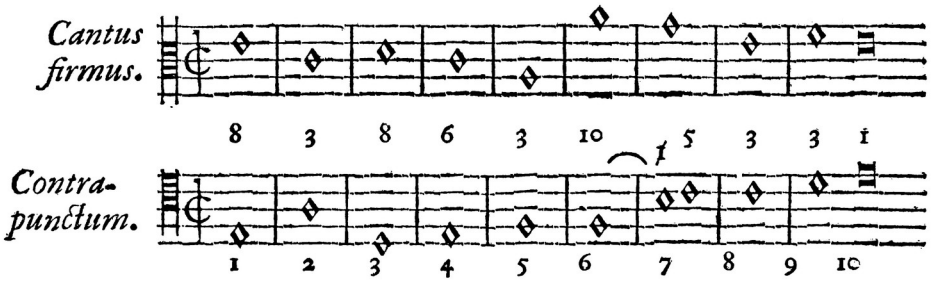
quibus ſolùm nunc tractandum eſt, ſed tonus tranſpoſitus eſſet, de quibus ſuo loco.

Joſeph. Eſtne quædam diſparitas inter hos tonos?

Aloyſ. Eſt vel maximè. Etenim propter varium ſitum ſemitoniorum cujuſvis Octavæ diſpar etiam oritur modulandi ratio, ſed quæ nunc ad te non attinet. Age ergo, tuumque exercitium reſumendo, conſtruito Contrapunctum ſuper Cantum firmum, quem tibi modò in E. præſcripturus ſum.



Ad amuſſim feciſti. Nunc poſito Cantu firmo in parte ſuperiori, infrà in Tenore confice Contrapunctum.

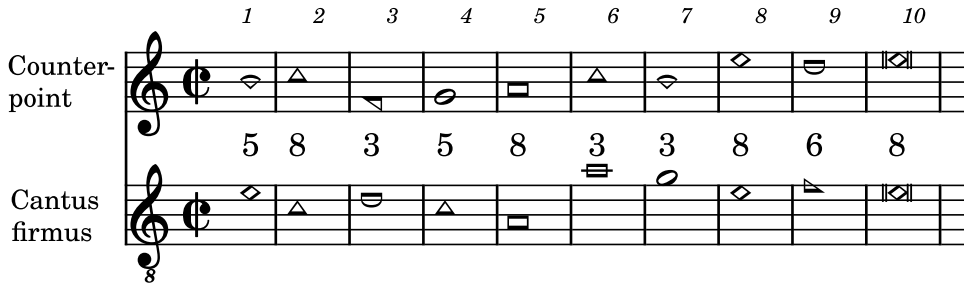


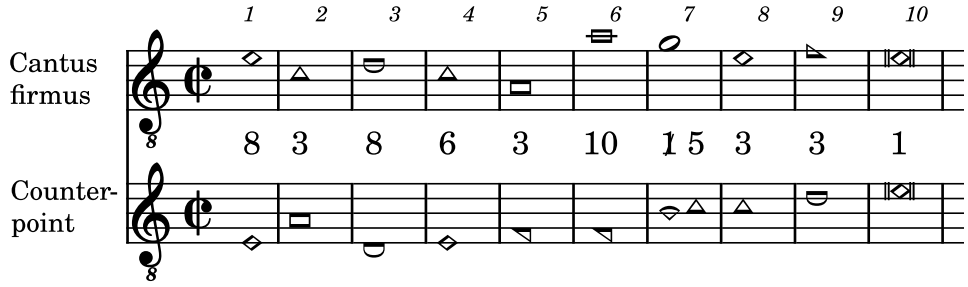
Joſeph. Ergo denuò in errorem incidi? Si hoc in Bicinio, inque Specie ſimpliciſſima mihi contigit, quid fiet in Triciniis, Quatriciniis, & plurium vocum Compoſitionibus? Quæſo dic mihi, qualem errorem indicet virgula ducta de ſexta ad ſeptimam Notam cum cruce ſuprà poſita?

Aloyſ. Noli moleſtè ferre errorem, quem, quia nondum admonitus fuiſti, evitare non poteras: nec anticipatò cruciari velis timore de plurium vocum compoſitione; quia uſus paulatim multò te circumſpectiorem, facilioremque reddet. Non dubito, quin ſæpiùs audiveris tritum illud ſermone proverbium: *mi* contra *fa*, eſt diabolus in Muſica, quod feciſti procedendo de ſexta Nota *fa*, ad ſeptimam *mi* per ſaltum Quartæ majoris, ſive Tritoni, quem cantu difficilem, atque

 malè

Joseph. Is there some difference between these modes?

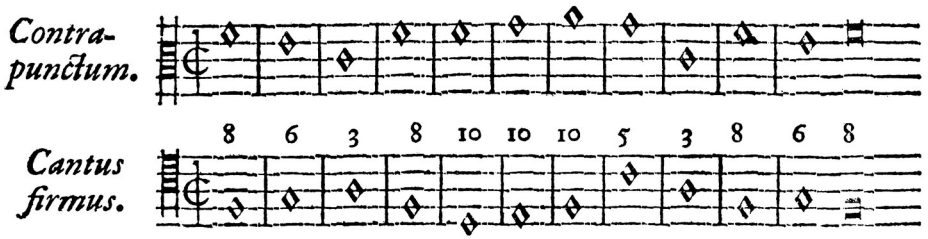
Aloys. Very much so. For because of the varying placement of semitones in each octave, a different manner of modulation arises; but this does not concern you yet. Now resume your exercise and construct a Counterpoint over the cantus firmus which I shall now give you in E.

You have done it exactly. Now, with the cantus firmus placed in the upper part, compose the Counterpoint below in the Tenor.



Joseph. So I have fallen into error again? If this happens to me in a bicinium and in the simplest species, what will happen in tricinia, quadricinia, and compositions of many voices? Tell me, I beg, what error is indicated by the stroke drawn from the sixth to the seventh note with a cross placed above it?

Aloys. Do not take the error badly, since you could not avoid it, not having yet been warned. Nor should you torment yourself beforehand with fear of multi-voice composition; for practice will gradually make you much more cautious and more skillful. I do not doubt that you have often heard the well-known proverb: mi contra fa est diabolus in Musica—which you have committed by proceeding from the sixth note fa to the seventh mi by the leap of a major fourth, or tritone, which is difficult to sing and

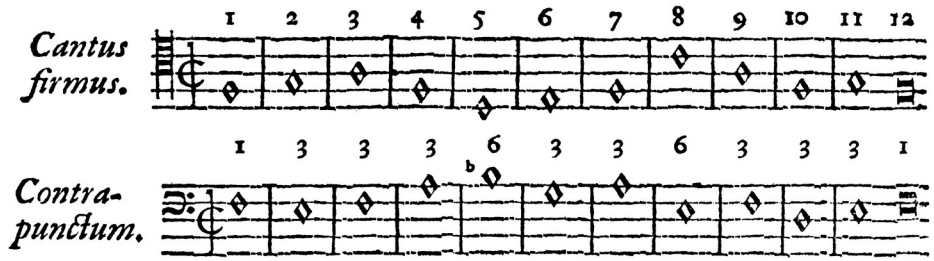
malè ſonantem in Contrapuncto adhibere nefas eſt. Fac bono animo ſis, & de tono E. procede ad tonum F.



De principio uſque ad finem bene.

Joſeph. Hac vice Cantum firmum in Tenore mihi præſcripſiſti, intereſtne aliquid?

Aloyſ. Nihil quidem aliud, quàm ut differentes claves ſemper tibi familiariùs innoteſcant. Ubi tamen animadvertendum eſt, proximas claves ſemper conjungendas eſſe, ut ratio Conſonantiarum ſimplicium à compoſitis faciliùs diſcerni poſſit. Nunc poſito Baſſo infrà Cantum firmum, in eo formetur Contrapunctum.

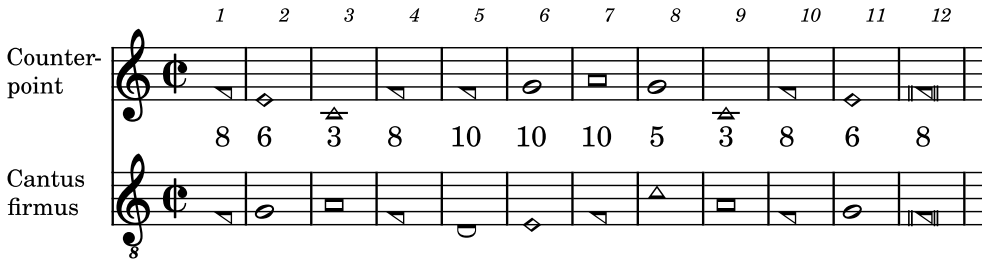


Bene quidem; ſed quâ ratione de quarta Nota uſque ad ſeptimam incluſivè Cantum firmum Contrapuncto tranſcendiſti?

Joſeph. Quia aliàs illuc uſque ſemper per motum rectum eundum mihi fuiſſet cum cantandi ratione minùs concinnâ.

Aloyſ. Optimâ obſervantiâ id feciſti: maximè quòd Cantum firmum hoc caſu tanquam partem inferiorem pro fundamento ſtatuens, ſuper eo rationem Conſonantiarum dimenſus fueris. Pergamus ad tonum G.

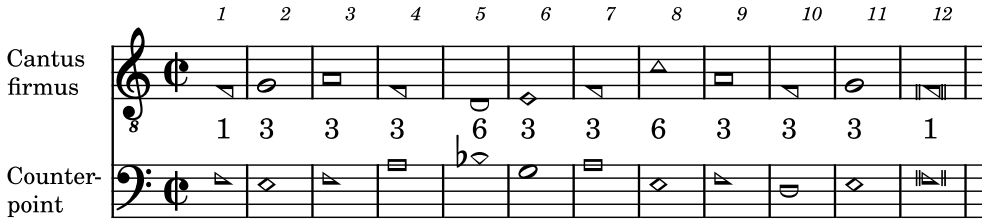
forbidden to use in Counterpoint because of its harsh sound. Take heart, and proceed from the mode of E to the mode of F.



From beginning to end, well done.

Joseph. This time you assigned the cantus firmus to the Tenor. Does that make any difference?

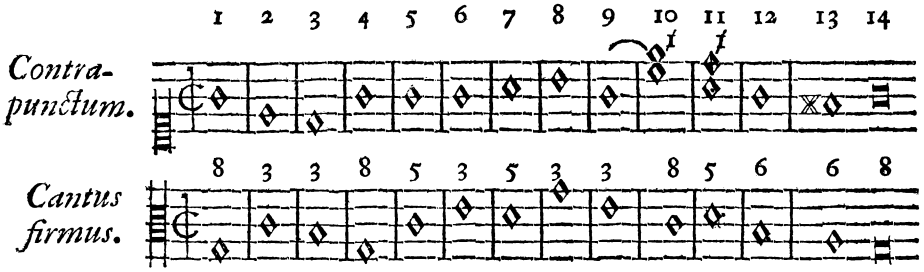
Aloys. Nothing except that you may become more familiar with different clefs. Yet it must be noted that adjacent clefs should always be paired, so that the relation of simple consonances to compound ones may be more easily discerned. Now, with the Bass placed below the cantus firmus, form the Counterpoint upon it.



Quite good; but on what grounds did you carry the counterpoint above the cantus firmus from the fourth note all the way to the seventh inclusive?

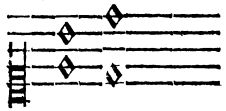
Joseph. Because otherwise I would have had to proceed all the way there by similar motion, which did not suit the manner of singing that seemed natural to me.

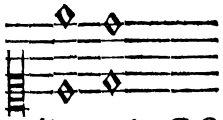
Aloys. You acted with excellent discernment—especially in treating the cantus firmus in this case as the lower part and foundation, and measuring the consonances above it accordingly. Let us proceed to the mode G.



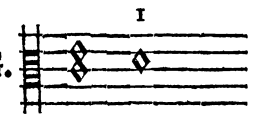
Joſeph. Maximâ, quâ potui attentione Contrapunctum hoc perfeci, nihilominùs de novo duo ſigna, errorum indices conſpicio, nempe de Nota nona ad decimam, & de decima ad undecimam.

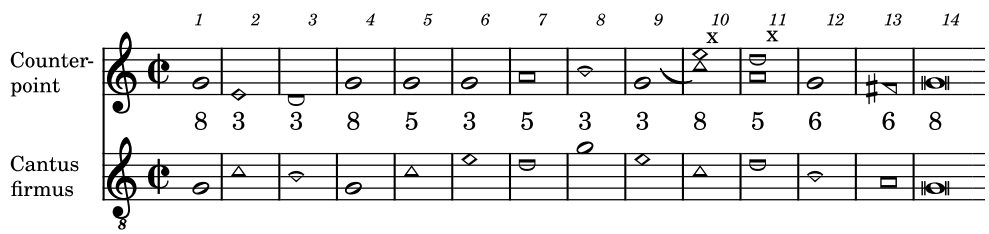
Aloyſ. Impatiens mihi videris; ſummopere tamen delector hoc tuo à regulis non declinandi ardore. Quomodo autem evitare cupis illas minutias, de quibus præceptum nondum habuiſti? De Nota nona ad decimam adhibuiſti ſaltum Sextæ majoris, qui in Contrapuncto, in quo Compoſitionis genere omnia cantu facillima eſſe debent, prohibitus eſt. Deinde de decima Nota ad undecimam progreſſus es de Decima in Octavam conjunctis partibus, id eſt, parte inferiori aſcendendo, & parte ſuperiori deſcendendo gradatim ſe movente, quæ Octava ab Italis Octava *Battuta*, à Græcis *Theſis* nuncupata, quia in principio tactûs venit, prohibetur. Verùm ego hujus rei causâ diu multumque ſcrutatus, neque vitii rationem, neque diſparitatem reperire potui, cur hujus

exempli Octava  approbanda,

ſequentis autem  reprobanda ſit?

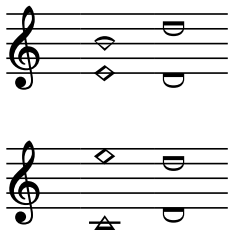
cùm utriusque paradigmatis Octava motu contrario perficiatur. Alia ratio habenda eſt Uniſoni, eodem modo producti,

nempe de Tertia in Uniſonum, E.G.  quo caſu Uniſonus, utpote in proportione æqualitatis conſiſtens admodum parùm auditur, & quaſi abſorptus & perditus videtur; ob quam rationem in hac ſpecie Contrapuncti excepto

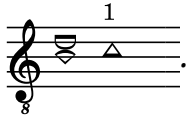


Joseph. With the greatest attention I could muster I completed this counterpoint; nevertheless I again see two signs marking errors—namely from the ninth note to the tenth, and from the tenth to the eleventh.

Aloys. You seem impatient; yet I am greatly delighted by your zeal not to deviate from the rules. But how do you expect to avoid those subtleties about which you have not yet received instruction? From the ninth to the tenth note you used a leap of a major sixth, which is forbidden in counterpoint, where everything must be as easy as possible to sing. Then from the tenth to the eleventh note you proceeded from a Tenth to an Octave with the parts moving in similar motion—i.e., the lower part ascending and the upper part descending stepwise. This octave, called by the Italians the octava battuta and by the Greeks thesis (because it falls on the beginning of the beat), is prohibited. Yet I have long examined this matter and could find neither the reason for the fault nor the difference why the first octave in this example:

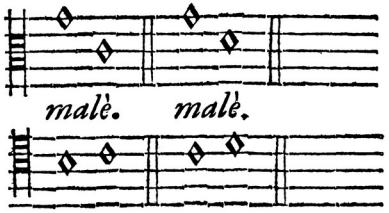


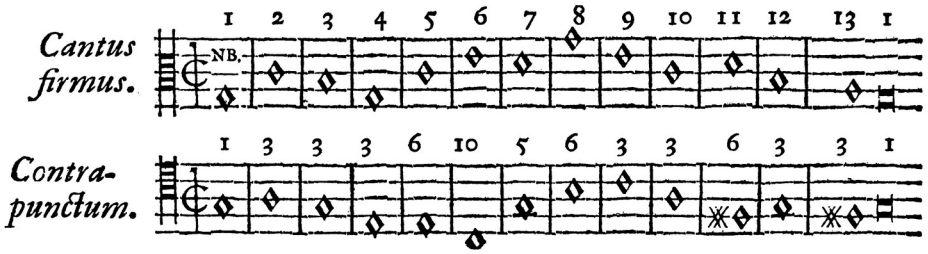
should be approved, while the following one should be rejected, even though in both examples the octave is formed by contrary motion.

Another consideration must be given to the unison produced in the same way—for example, from a Third to a Unison, e.g.  .

In such a case the unison, being a proportion of equality, is scarcely heard at all and seems absorbed and lost. For this reason, in this species of counterpoint, except at the beginning and end, it must never be used.

principio, & fine nunquam ponendus eſt. Ut autem revertar ad Octavam ſuperiùs allatam, quæ battuta dicitur, liberum arbitrium vel evitandi vel adhibendi tibi relinquo; parùm enim intereſt. Quòd ſi autem de remotiore quadam Conſonantiâ per ſaltum lapſus conjunctim in Octavam fiat, nec in Compoſitione plurium vocum tolerandum puto. E. G.

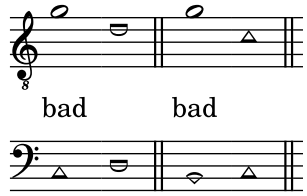
Quod à potiore de uniſono quoque intelligendum eſſe cupio. E. G.

In Compoſitione octo vocum ſimiles ſaltus maximè in Baſſis, inque partibus, quæ eorum vices funguntur, vix evitari poterunt, ùt ſuo loco dicetur. Reſtat adhuc ultimi exempli Contrapunctum in parte inferiori ponendum.



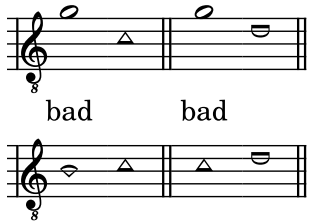
Joſeph. Quid ſibi vult illud in prima Contrapuncti Nota poſitum NB. ?

Aloyſ. Indicat, quòd aliàs malè fiat motus de Uniſono ad aliam Conſonantiam per ſaltum, neque in Uniſonum eodem modo, ùt paulò antè dictum eſt. Verùm cùm ille ſaltus

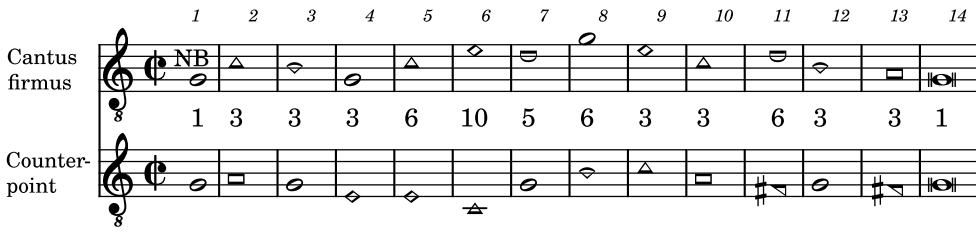
But to return to the octave mentioned above, which is called battuta, I leave it to your free judgment whether to avoid it or use it; it matters little. However, if by a leap from some more distant consonance the parts arrive together at an octave, I do not think it should be tolerated in multi-voice composition. For example:



The same must be understood regarding the unison. For example:



In an eight-voice composition such leaps—especially in the basses and in parts that take their role—can scarcely be avoided, as will be said in its place. Now the counterpoint for the last example must still be placed in the lower part.

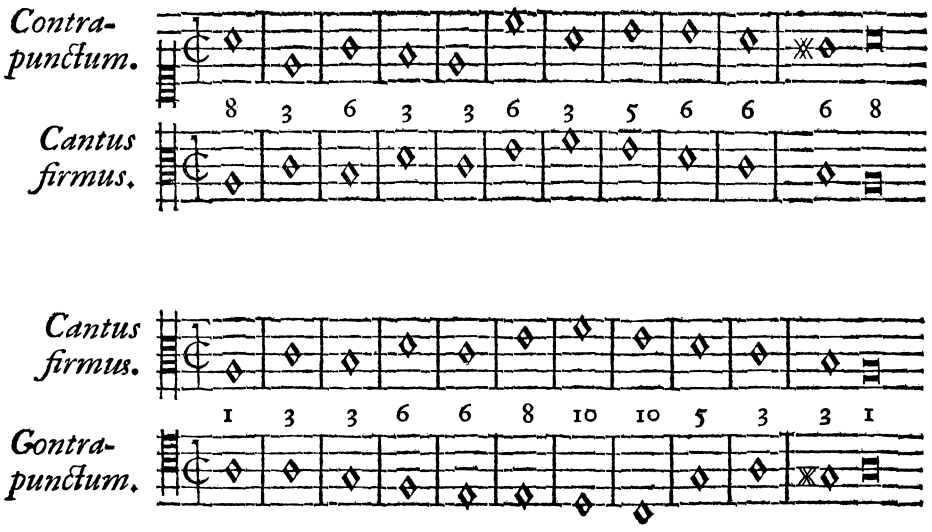


Joseph. What does that "NB." placed on the first note of the counterpoint mean?

Aloys. It indicates that ordinarily it is wrong to move from a unison to another consonance by a leap, or into a unison in the same manner, as was said a little earlier. But since that leap is in the cantus firmus, which cannot be altered, it must be tolerated here. It would be

tus ex parte Cantûs firmi ſit, qui immutari non poteſt, ideo hîc loci tolerandus eſt. Alia ratio eſſet, ſi nulla Cantûs choralis obligatione obſtricto, liberum arbitrium faciendi, quid velis, tibi foret. Sed quare in undecima Nota appoſuiſti Dieſin? utpote quæ in genere Diatonico adhiberi non admodum ſolita eſt.

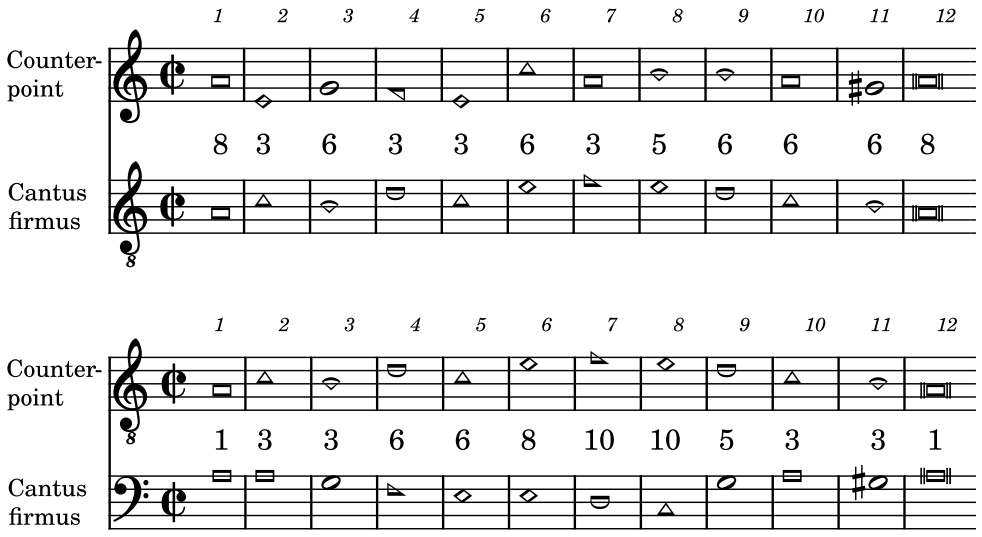
Joſeph. Sextam eo loco ponere mihi viſum eſt. Ego autem audivi, cùm artem canendi addiſcerem, quòd *fa* ad deſcenſum, *mi* verò ad aſcenſum inclinet; cùm autem de illa Sexta in Tertiam aſcendendo fiat progreſſus, ideo Dieſin adjunxi, ut illa aſcendendi ratio magis autentica efficiatur. Accedit, quòd illud F. undecimæ Notæ abſque Dieſi poſitum, cum F. decimæ tertiæ Notæ Dieſi adjunctâ nonnihil adverſam relationem efficeret.

Aloyſ. Optimam adhibuiſti obſervationem. Nunc omnem offenſionis lapidem ſublatum eſſe arbitror. Perge ergo ad A. & C. duos ſuperſtites Tonos.

Ex duobus ultimis exemplis apparet, omnia ſcitu ad hanc ſpeciem neceſſaria tibi jam nota eſſe. Nunc ſecunda ſpecies perſequenda eſt.

otherwise if, free from the obligation of a choral melody, you had complete liberty to write whatever you wished. But why did you place a sharp on the eleventh note, since such an accidental is not commonly used in the diatonic genus?

Joseph. It seemed to me that a sixth should be placed there. And I had heard, when learning to sing, that fa tends downward and mi upward. Since from that sixth the progression ascends to a third, I added the sharp so that the upward motion might be more authentic. Moreover, that F on the eleventh note, if left without a sharp, would have produced a somewhat awkward relation with the F on the thirteenth note, which has a sharp.

Aloys. You have made an excellent observation. Now I think every stumbling block has been removed. Proceed, then, to A and C, the two remaining modes.

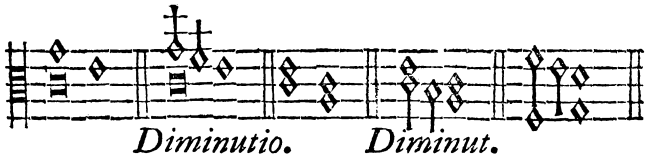


From these last two examples it is clear that everything necessary to know for this species is now familiar to you. We must now pursue the second species.

EXERCITII I.

LECTIO SECUNDA,

Sive ſecunda Contrapuncti Species.

ANtequam hanc ſecundam Contrapuncti Speciem explicandam aggrediar, præſciendum eſt, nos in præſentiarum in Tempore binario verſari, cujus tactus ſive menſura duabus partibus æqualibus conſtat, quarum una in depreſſione, altera in elevatione manûs conſiſtit. Depreſſio manûs græcè *Theſis*, elevatio autem *Arhſis* appellatur, quibus duobus vocabulis pro majori commoditate hoc in Exercitio deinceps nobis utendum duxi. Species iſta ſecunda fit, quando duæ minimæ contra unam ſemibrevem ponuntur, quarum una in Theſi veniens, ſemper conſonans ſit, neceſſe eſt, altera verò in Arhſi, ſi de Nota præcedenti in ſequentem gradatim moveatur, etiam diſſonans eſſe poterit; ſi autem per ſaltum labatur, neceſſariò Conſonans eſſe debet. Poteſt autem hac in Contropuncti Specie diſſonantia non aliter habere locum, quàm per diminutionem, id eſt, implendo illud ſpatium intermedium, quod intercedit duas Notas per ſaltum Tertiæ diſtantes. E. G.

Diminutio. *Diminut.*

Nec ratio habenda eſt illius Notæ diminutæ, an conſonans, vel diſſonans ſit, ſufficit implere illud ſpatium vacuum inter duas Notas relictum per ſaltum Tertiæ diſtantes.

Joſeph. Suntne præterea illa, quæ de prima Contrapuncti Specie ratione motûs, & progreſſuum præcepta ſunt, etiam hîc obſervanda?

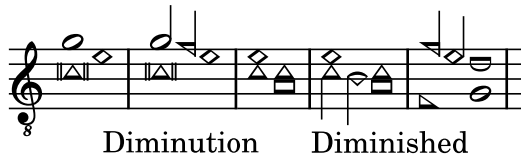
Aloyſ. Vel maximè; præter penultimum tactum, cujus in hac Specie Nota prima Quintam, ſecunda Sextam majorem habere debet, ſi Cantus firmus, ſive Choralis in parte inferiori exiſtat: ſi verò Cantus firmus in parte ſuperiori fuerit, pri-

EXERCISE I.
SECOND LESSON,
Or the Second Species of Counterpoint

Before I proceed to explain this second species of counterpoint, it must first be known that we are presently dealing with duple time, whose beat or measure consists of two equal parts, one of which is the downbeat, the other the upbeat. The downbeat is called Thesis in Greek, and the upbeat Arsis. For greater convenience, I have decided that in this exercise we shall use these two terms from now on.

This second species occurs when two minims (half-notes) are set against one semibreve (whole note). Of these two minims, the one that falls on the Thesis must always be consonant; the other, on the Arsis, may be dissonant if it moves stepwise from the preceding note to the following one. But if it moves by a leap, it must necessarily be consonant.

In this species of counterpoint, dissonance can occur only through diminution, that is, by filling in the intermediate space between two notes a third apart that are approached by leap. For example:



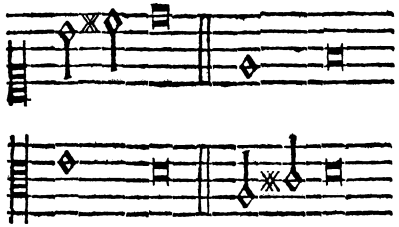
Diminution Diminished

Nor must any account be taken of whether that diminished note is consonant or dissonant; it is enough that it fills the empty space left between two notes a third apart.

Joseph. Are the rules concerning motion and progressions that were prescribed for the first species of counterpoint also to be observed here?

Aloys. Very much so — except for the penultimate beat, whose first note in this species must have a fifth, and the second a major sixth, if the cantus firmus or chorale is in the lower part. But if the cantus firmus is in the upper part, the first note of the penultimate beat must have a fifth, and the second a minor third. An example will make the matter clear.

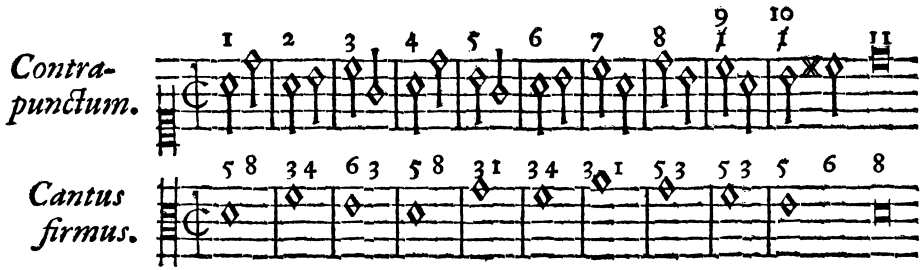
prima Nota penultimi tactûs Quintam, ſecunda verò Tertiam minorem habeat. Exemplum rem claram exhibebit.



Pro majori facilitate juvabit habere rationem finis, antequam ſcribere incipias. Nunc ad opus progrediamur de manu ad manum, ſuperiores Cantus firmos adhibendo.

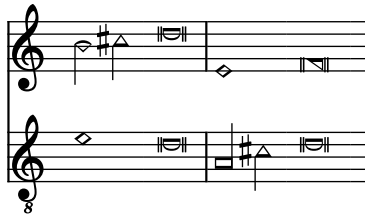
Joſeph. Faciam. Sed patientiam tuam imploro, ſi in errores incidere mihi contingat ; non enim habeo hujus rei, nisì valde confuſam cognitionem.

Aloyſ. Fac ùt poteris ; nihil moleſtè feram ; correctio obſcuritatis tenebras facilè diſſipabit.



Joſeph. Haud inanis fuit hic meus errandi timor ; duo enim conſpicio reprehenſionis ſigna : primum in prima tactûs noni figura, ſecundum in decimi tactûs prima Nota, quorum neutrum intelligo ; utrobique enim de Conſonantia imperfecta ad imperfectam motu contrario progreſſus ſum.

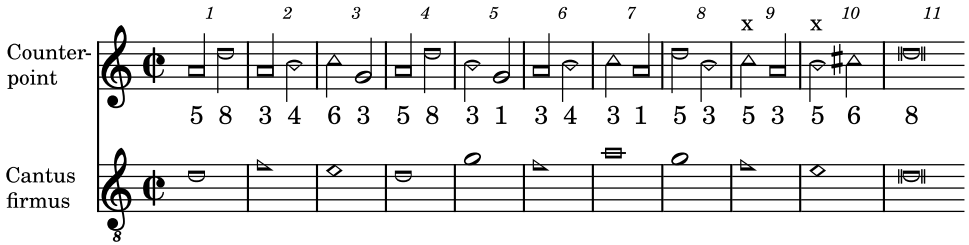
Aloyſ. Rectè ratiocinaris, ſunt enim duo errores, quorum una, & eadem eſt ratio, quam ignorare te, nondum ſcilicet edoctum, haud miror. Notandum ergo eſt, quòd ſaltus Tertiæ neque duas Quintas, neque duas Octavas ſalvare poſſit ; quia nota illa intermedia in Arhſi veniens, quaſi pro non adjecta habetur, eò quòd ob exiguitatem valoris, parvitatemque ſpatii ſonum ità temperare non valeat, quin duarum Quintarum,



For greater ease, it will help to consider the ending before you begin to write. Now let us proceed to the work, hand to hand, using the upper cantus firmi.

Joseph. I will. But I beg your patience if I happen to fall into errors; for I have only a very confused understanding of this matter.

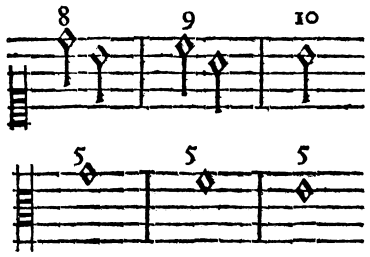
Aloys. Do what you can; I shall take nothing amiss. Correction will easily dispel the darkness of obscurity.



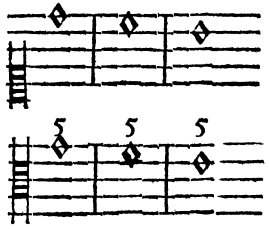
Joseph. My fear of making mistakes was not unfounded; for I see two marks of censure: the first in the first note of the ninth beat, the second in the first note of the tenth beat. I do not understand either; for in both cases I proceeded from an imperfect consonance to an imperfect consonance by contrary motion.

Aloys. Your reasoning is correct; for there are indeed two errors, both arising from the same cause — which I do not wonder you are unaware of, since you have not yet been taught it. It must therefore be noted that a leap of a third cannot "save" (i.e., avoid) two fifths or two octaves; because the intermediate note, falling on the Arsis, is considered as if it were not added, since by reason of its small value and the shortness of the space it cannot sufficiently temper the sound so that the succession of two fifths or two octaves is not perceived by the ear.

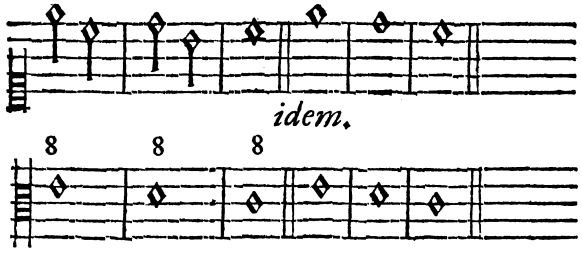
tarum, vel Octavarum ratio auribus percipiatur. Videamus priora exempla à tactu octavo incipiendo.

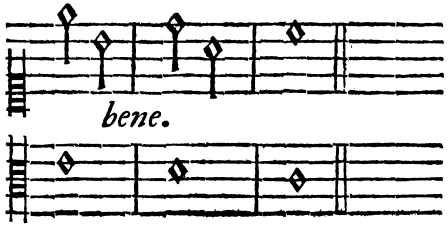


Etenim ſi Nota illa intermedia, vel in Arhſi exiſtens, pro non adjecta habetur, Hypotheſes illas apertè ſic ſe habere perſpicuum eſt.

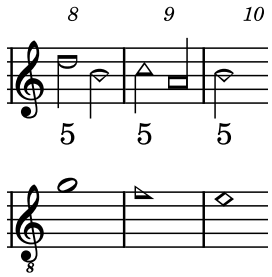


Idem Judicium ferendum eſt de Octavis.

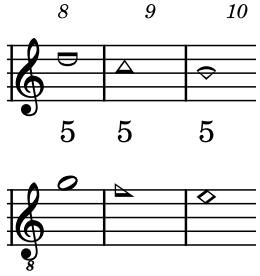


Alia ratio habenda eſt ſaltûs majori ſpatio conſtantis. E. G. Quartæ, Quintæ, Sextæ: quo caſu diſtantia de prima Nota ad ſecundam quaſi oblivionem auribus afferre videtur ſoni Notæ primæ in Theſi, ad Notam ſecundam pari modo in Theſi exiſtentem. Videamus ſuperiùs exemplum de Octavis, quarum conſecutio ſaltu Quartæ intervenientis ſalvatur.

Qui

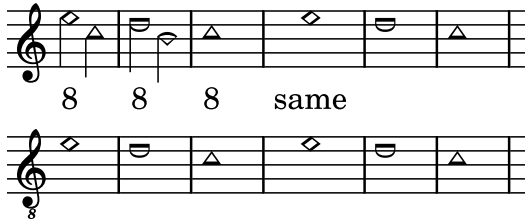
Let us look at the earlier examples beginning with the eighth beat.



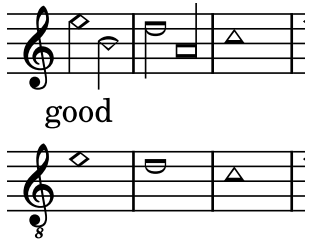
For that intermediate note, being on the Arsis, is considered as if not added; it is clear that the hypotheses stand thus.



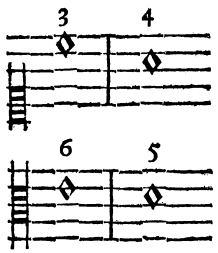
The same judgment must be made concerning octaves.

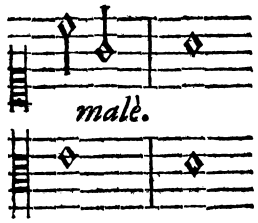


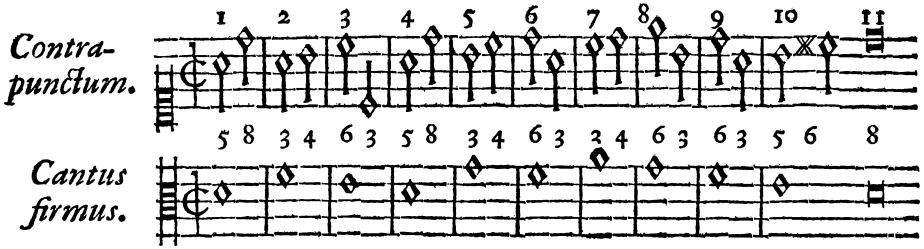
A different rule applies to leaps of greater interval — e.g., a fourth, fifth, or sixth — in which case the distance from the first note to the second seems to bring about a kind of forgetfulness in the ear of the sound of the first note on the Thesis, when the second likewise stands on the Thesis. Let us look at the above example of octaves, whose succession is "saved" by the intervening leap of a fourth.



Qui ſaltus Quartæ pari ratione cauſa eſt, cur præcedentis Lectionis tuæ tertium tactum ad quartum nullâ erroris notâ ſignaverim; quia non habitâ ratione illius Notæ intermediæ Hypotheſis ſic ſe haberet.

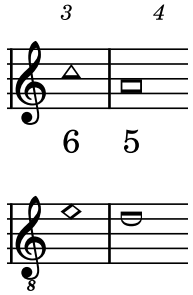


Qui progreſſus contra regulam eſt, quæ dicit: de imperfecta ad perfectam per motum contrarium; quem errorem aufert ſaltus Quartæ hac ratione:

Corrigatur modò præcedens tua Lectio.

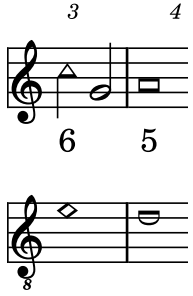


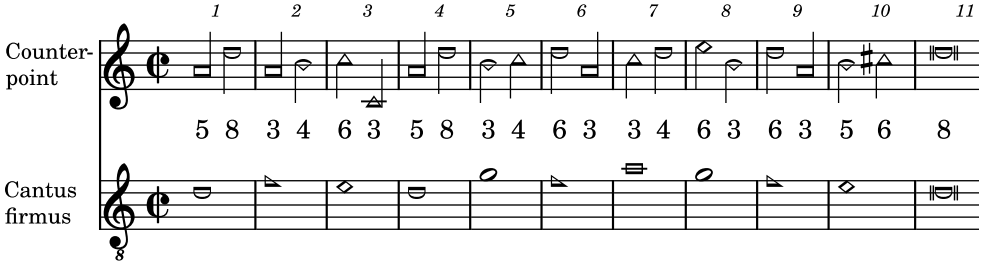
Jam video, hucuſque dicta ſatis memoriâ te tenere. Antequàm autem hanc ipſam Lectionem cum Contrapuncto in inferiori parte componendam aggrediaris, quædam animadvertenda pro majori facilitate, ſcitu perquàm utilia tibi proponam. Primum eſt, quòd locò primæ Notæ pauſa medii tactûs valoris poni poſſit. Secundum, ſi ambas partes ita vicinè, itaque conjunctim reperiri contingat, ut vix ſciatur, quorſum eundum, nec facultas tibi ſit, per motum contrarium

This leap of a fourth is also the reason why I marked no error in the progression from the third to the fourth beat of your previous lesson; for without considering the intermediate note, the hypothesis would stand thus:

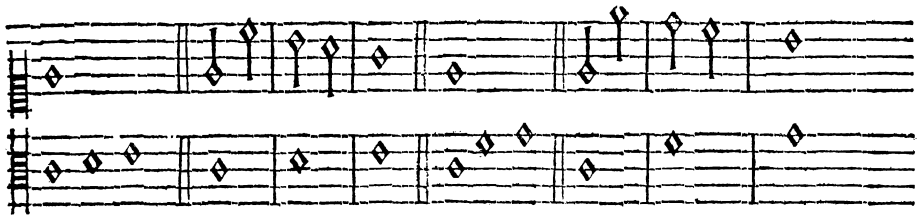


Which is a progression contrary to the rule that says: "from an imperfect to a perfect consonance by contrary motion." This error is removed by the leap of a fourth, as follows:

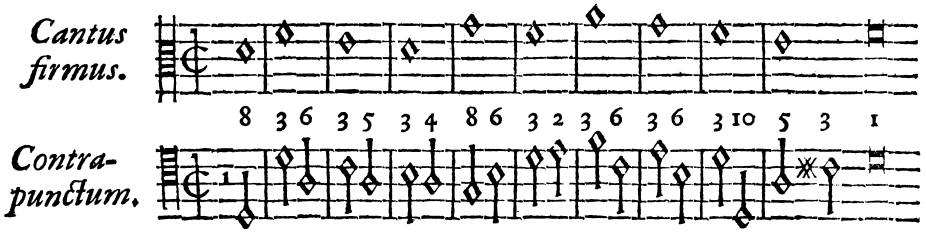


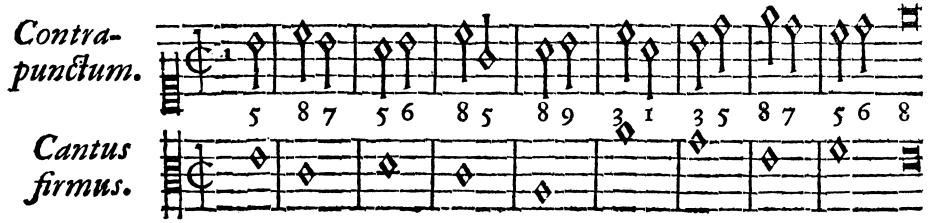
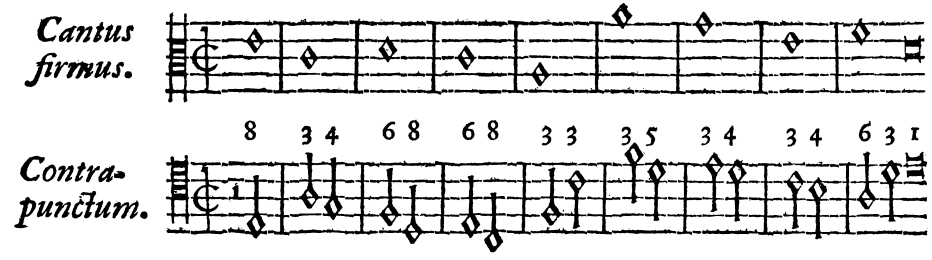
Now let your previous lesson be corrected.

I see that you have retained well enough what has been said so far. But before you attempt to compose this same lesson with the counterpoint in the lower part, I will point out a few things for your greater ease, very useful to know. First, in place of the first note, a rest of half-beat value may be set. Second, if both parts happen to be so close and so joined that one scarcely knows where to go, and you have no possibility of proceeding by contrary motion,

rium progrediendi, id effici posse per saltum Sextæ minoris, (qui licitus est) vel Octavæ, ùt in exemplis sequentibus :



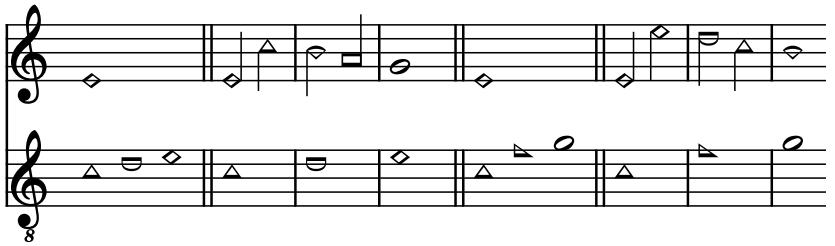
Jam perge, & præcedentis Lectionis Contrapunctum in parte inferiori construe.

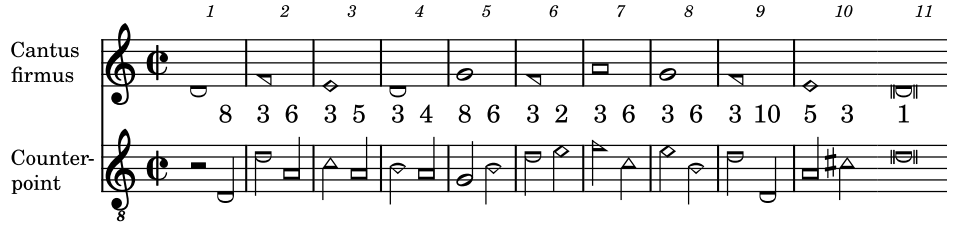


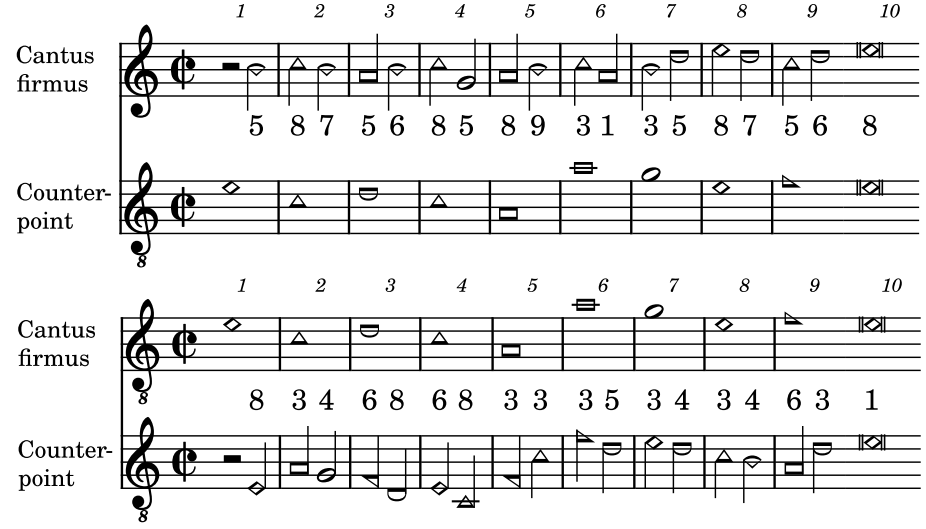
Nunc resumendo omnes Cantus firmos in prima Contrapuncti Specie præscriptos, percurre reliquos quinque tonos, superiùs, inferiùsque Contrapunctum fabricando.



Joseph. Memini te paulò ante præcepisse, hujus Speciei Contrapuncto in inferiori parte existente, penultimi tactûs primæ Notæ

this may be achieved by a leap of a minor sixth (which is permitted) or of an octave, as in the following examples:



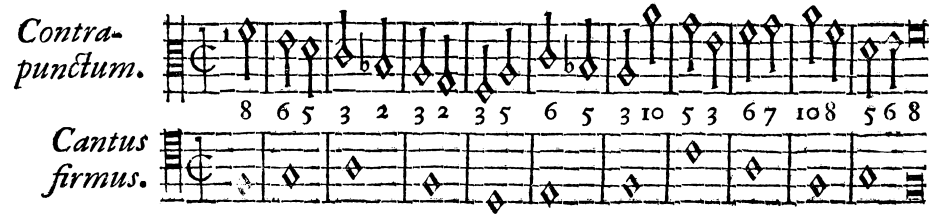
Now continue, and construct the counterpoint of the previous lesson in the lower part.

Now, taking up again all the cantus firmi prescribed in the first species of counterpoint, run through the remaining five tones, composing counterpoint above and below.

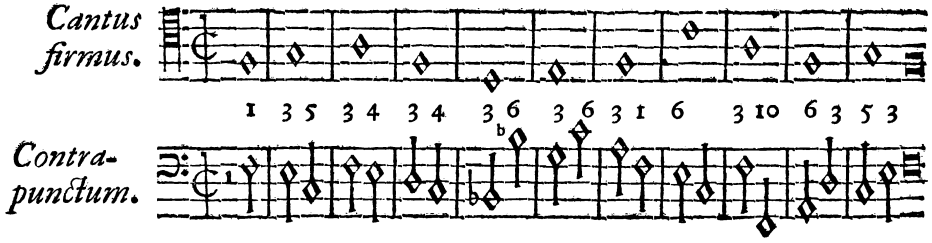


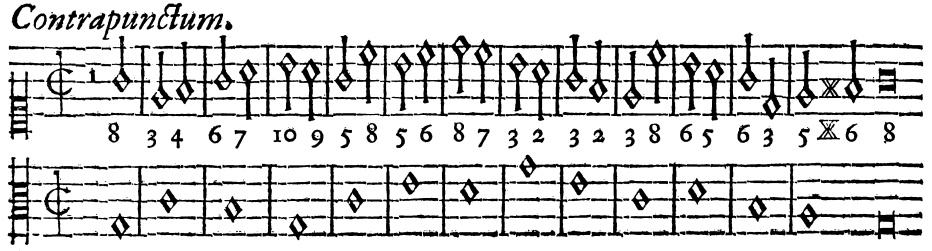
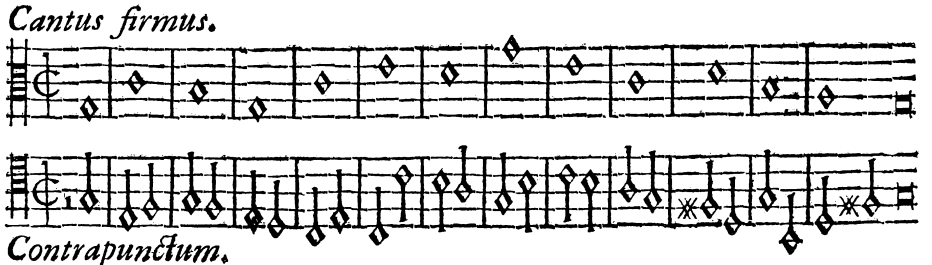
Joseph. I remember that you instructed earlier that when the counterpoint of this species is in the lower part, the first note of the penultimate beat

Notæ Quintam dari oportere: verùm patet, in illa hujus toni Hypotheſi Quintæ, utpote diſſonantiæ locum non eſſe, ob rationem *mi* contra *fa*; quapropter locò Quintæ Sextam adhibendam eſſe judicavi.

Aloyſ. Summopere mihi placet hæc tua obſervatio. Perge nunc, reliquosque quatuor tonos pari exercitio proſequere.

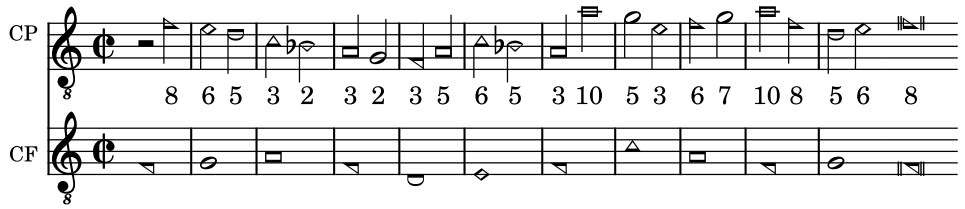


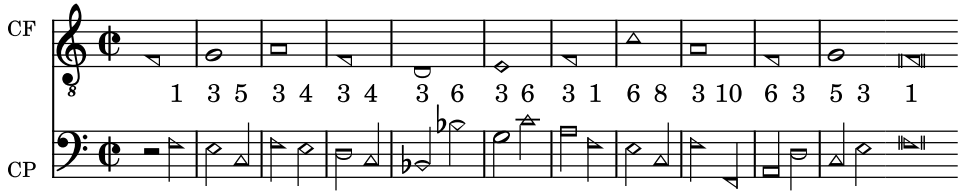
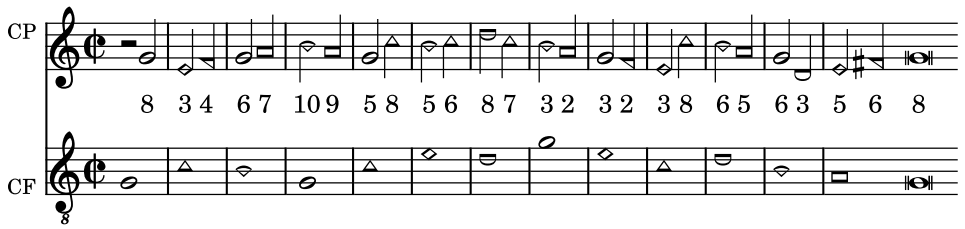
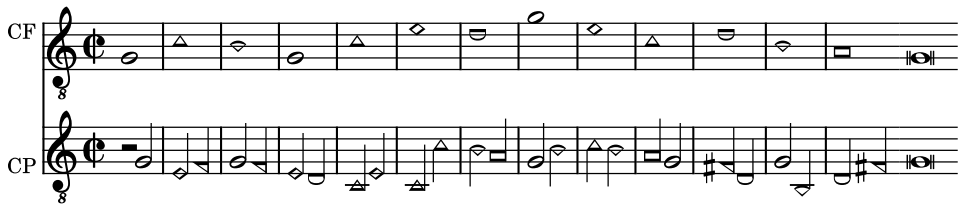


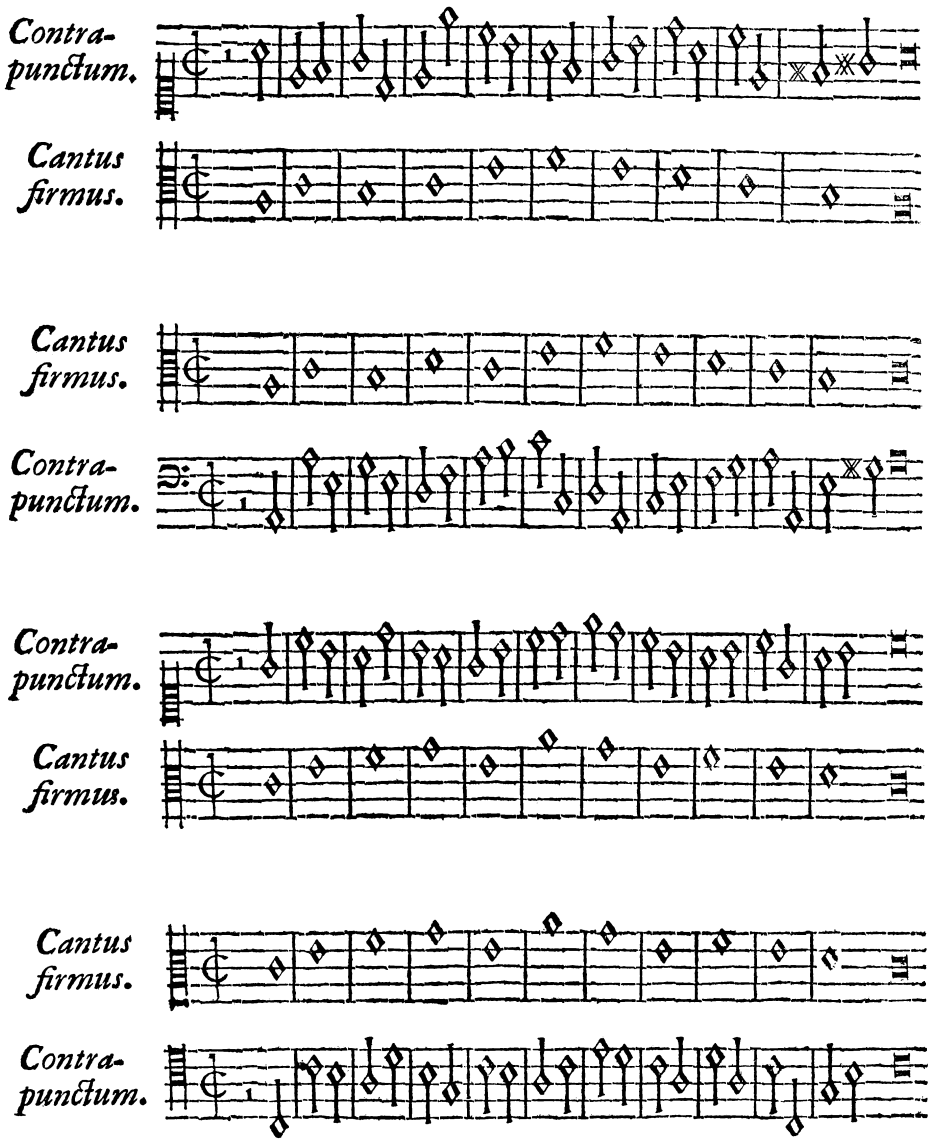
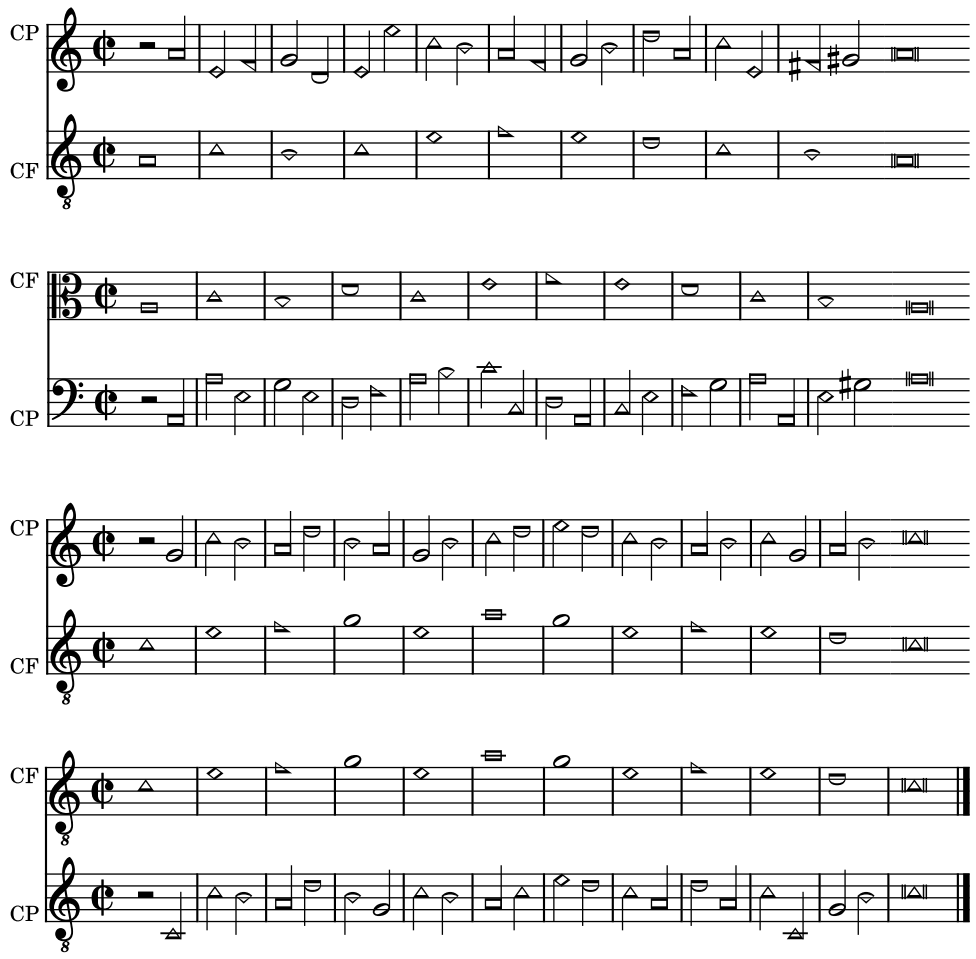


must take a fifth. But it is clear that in this tone the fifth cannot occur, since it would be a dissonance because of the mi against fa; therefore I judged that a sixth should be used instead of the fifth.

Aloys. Your observation pleases me greatly. Continue now, and pursue the remaining four tones with the same exercise.





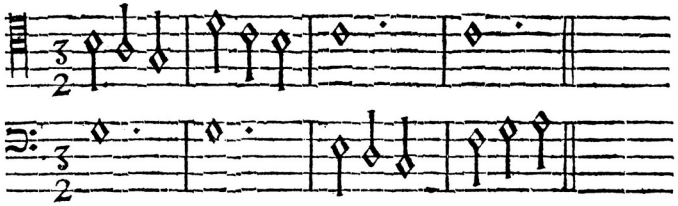


Satis commodè rem inſtituiſti. Sic Dii laboribus omnia vendunt; oportet te etiam memorem eſſe illius: Gutta cavat lapidem, non vi, ſed ſæpè cadendo: quô docemur, indefeſſum Studii laborem ad Scientiarum adeptionem requiri, ita ut nullum diem ſinè linea (ùt dicitur) præterlabi ſinamus. Præterea te hoc loco admonendum etiam duxi, ut non ſolùm illius tactûs

You have handled the matter quite well. Thus the gods sell all things for labor; you must also remember this: "The drop hollows the stone, not by force, but by falling often." By which we are taught that unwearied labor of study is required for the attainment of knowledge, so that we allow no day to pass without a line, as the saying goes. Moreover, I must also warn you here that you should consider not only the beat on which you are working, but also the ones that follow.

tactûs, in quo laboras, rationem habeas, verùm etiam & ſequentium.

Joſeph. Ità ſanè eſt venerande Magiſter. Etenim in componendis præcedentibus Contrapuncti exercitationibus vix ſciviſſem, quid faciendum, nisì, antequam ſcribere decreverim, unum, vel alterum tactum conſiderando, quid ibi quadraret, anticipaviſſem.

Aloyſ. Mirum in modum lætor qui te tam ſagacem videam, teque etiam, atque etiam hortor, ut incepto, & attentionis, & conſtantiæ ſtudio difficultatis gravitatem ſuperare nitaris, neque te oneris aſperitate opprimi, neque blanditiis, velut jam multa calleas, ab aſſiduitate Studii abſtrahi patiare. Quo facto paulatim in tenebris lumen exoriri, & aliqua ex parte caliginis velum tibi ſublatum iri ſenties, ac gaudebis. Hîc quoque dicendum eſſet de Tempore Ternario, ubi tres Notæ contra unam ponuntur. Quia autem ob rei facilitatem parvi eſt momenti res, operæ pretium non arbitror, peculiarem eâ causâ Lectionem ſtatuere. Pauca exempla ad hujus rei intelligentiam ſufficere modò compertum erit.

Ubi Nota intermedia, quia omnes tres gradatim moventur, diſſonantia eſſe poteſt; ſecùs eſſet, ſi una vel altera ſaltaret ex his tribus, quo caſu omnes tres conſonantes eſſe oportere, ex jam dictis, conſtat.

EXERCITII I.

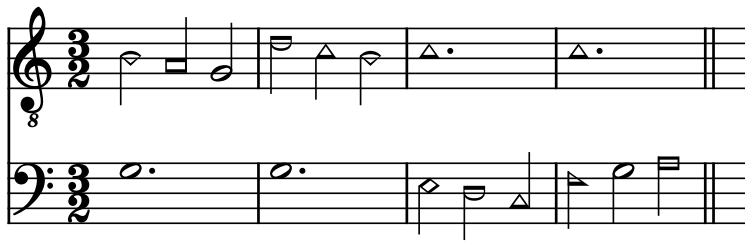
LECTIO TERTIA,

De tertia Contrapuncti Specie.

TErtia Contrapuncti Species eſt quatuor ſemiminimarum contra unam ſemibrevem Compoſitio. Ubi principiò animadvertendum eſt, quòd, ſi quinque ſemiminimas vel aſcendendo, vel deſcendendo continuò gradatim ſe ſequi contingat,

Joseph. Indeed, venerable Master, it is so. For in composing the previous counterpoint exercises, I would scarcely have known what to do unless, before deciding to write, I had anticipated by considering one or two beats ahead what would fit there.

Aloys. I am exceedingly glad to see you so perceptive, and I urge you again and again to strive, by diligence and constancy, to overcome the weight of difficulty, and not to let yourself be crushed by the burden, nor to be drawn away from steady study by the flattering notion that you already know much. By doing so, you will gradually see light arise in the darkness, and you will feel the veil of obscurity lifted in part, and you will rejoice.

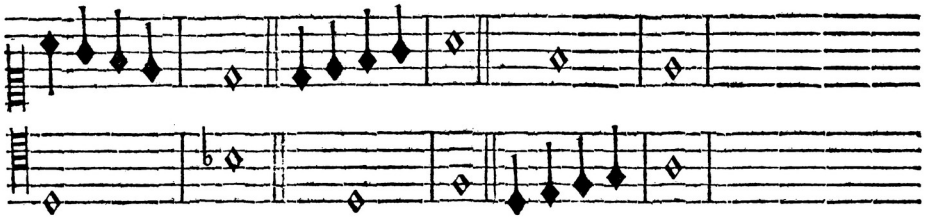
Something should also be said here about triple time, where three notes are set against one. But because the matter is easy and of little importance, I do not think it worthwhile to establish a separate lesson for it. A few examples will suffice for understanding.

Where the intermediate note, because all three move stepwise, may be dissonant; otherwise, if one or another of the three were to leap, then all three must be consonant, as is clear from what has already been said.

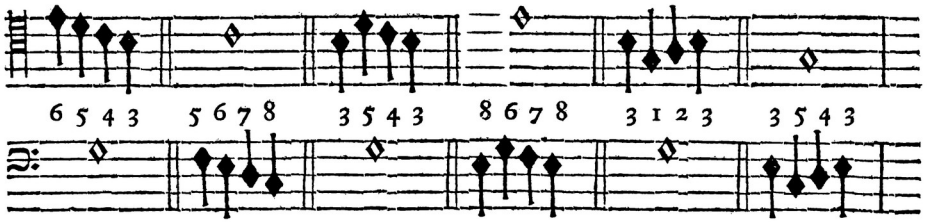
EXERCISE I.

LESSON THREE,

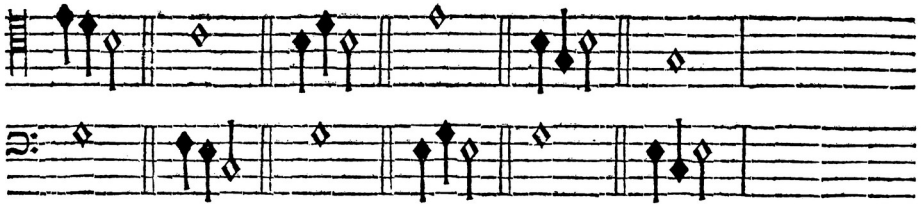
On the Third Species of Counterpoint.

The third species of counterpoint is the composition of four quarter notes against one whole note. First it must be observed that if five quarter notes follow one another stepwise, either ascending or descending, the first must be a consonance, the second may be a

tingat, prima Conſonans eſſe debeat, ſecunda diſſonans eſſe poſſit. Tertia denuò Conſonans ſit, neceſſe eſt. Quarta diſſonans eſſe poterit, ſi quinta Conſonantia fuerit; ùt in exemplis:

Fallit autem primò in Hypotheſi, ubi ſecundam & quartam Notam Conſonantias eſſe contingit, quo in caſu tertia Nota diſſonantia eſſe poterit; ùt in ſequentibus exemplis:

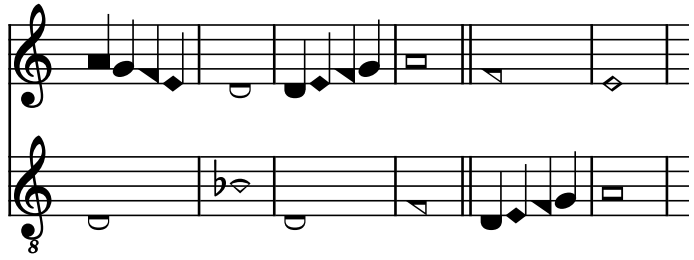


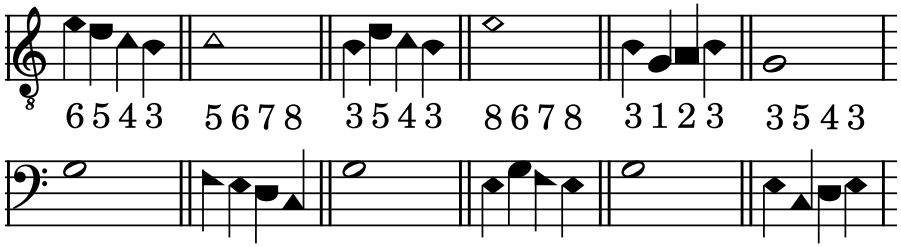
Ubi omnium Paradigmatum tertia Nota Conſonans eſt, & diminutio ſaltûs tertiæ vocatur. Ut autem hujus rei veritas pateſiat, exempla hæc ad ſubſtantiam revocanda ſunt, hac ratione:



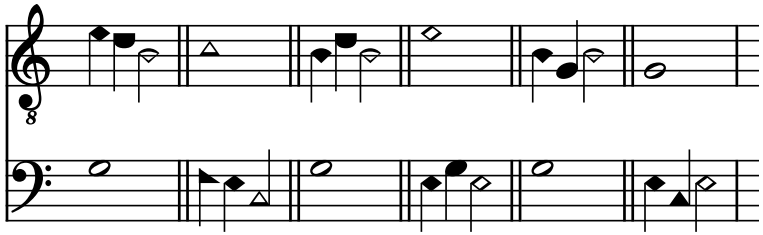
Ex quibus clarè videtur, tertiam illam Notam, nempe diſſonantiam aliud non eſſe, quàm diminutionem ſaltûs tertiæ, quâ impletur ſpatium intermedium de Nota ſecunda ad tertiam repertum, quod ſpatium nunquam diminutionem, ſive impleri Notâ intermediâ recuſat.

Secunda Hypotheſis eſt, ubi à cõmuni regula deflectitur, Nota cambita, ab Italis *Cambiata* nuncupata, ubi de Nota ſecunda diſſonan-

dissonance, the third must again be a consonance, and the fourth may be a dissonance if the fifth is a consonance; as in the examples:

There is, however, an exception in the first hypothesis, where it happens that the second and fourth notes are consonances, in which case the third note may be a dissonance; as in the following examples:



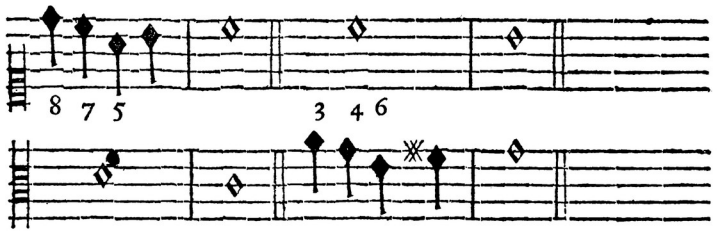
In all these examples the third note is a consonance, and this is called the diminution of a third-leap. But in order that the truth of this matter may be made clear, these examples must be reduced to their essential form, in this manner:

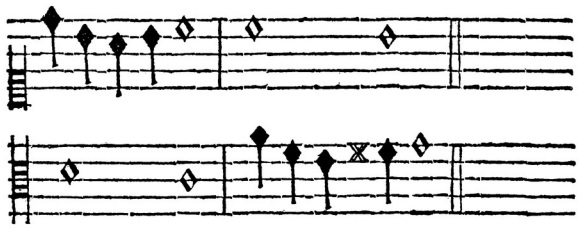


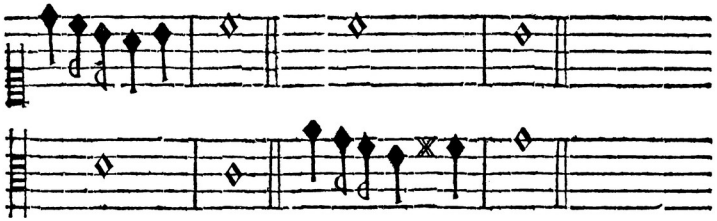
From these it is clearly seen that that third note—namely the dissonance—is nothing other than the diminution of a leap of a third, by which the intermediate space between the second and third notes is filled in, a space which never refuses diminution, that is, the filling-in of the intermediate note.

The second hypothesis is where one departs from the common rule: the nota cambiata (changed note), called cambiata by the Italians (changed note), where from the second note, being a dissonance, one

ſonante per ſaltum fit progreſſus in Conſonantiam, ùt in ſequentibus exemplis videndum eſt.



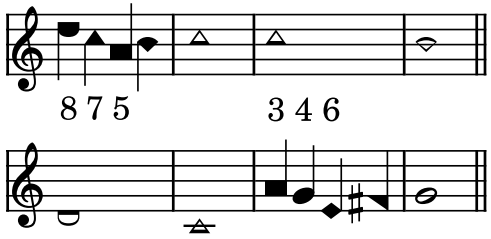
Saltus ille tertiæ de ſecunda Nota in tertiam eſſe deberet de rigore de prima Nota ad ſecundam ; in quo caſu ſecunda Nota ſextam Conſonantiam efficeret, hac ratione :

In qua Hypotheſi, ſi de prima Nota ad ſecundam diminutio fieret, res ità ſe haberet :



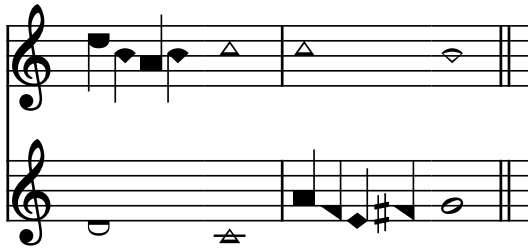
Cùm autem hac in Contrapuncti Specie Fuſis locus non ſit, ideo placuit gravibus authoritate Viris primum exemplum, ubi ſecunda Nota ſeptimam habet, fortaſſis propter majorem Cantûs concinnitatem ratum habere.

Ultimùm reſtat monſtrare, quomodo penultimus tactus, qui, ùt plurimum cæteris plus difficultatis habet, conficiendus ſit ; qui, ſi Cantus firmus in inferiori ſit parte, hac ratione conſtitutus eſſe debet.

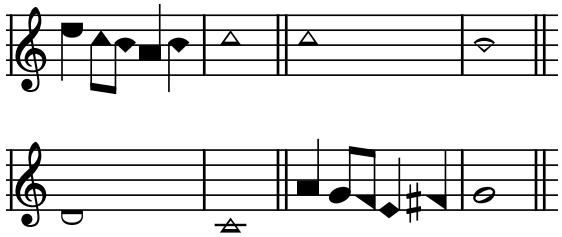
proceeds by a leap into a consonance, as may be seen in the following examples:



That leap of a third from the second note to the third ought strictly to be from the first note to the second; in which case the second note would form a sixth, in this manner:

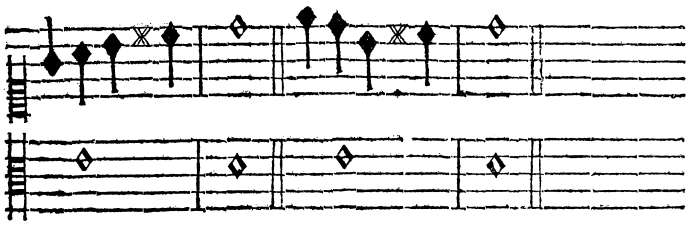


In which hypothesis, if a diminution were made from the first note to the second, the matter would stand thus:

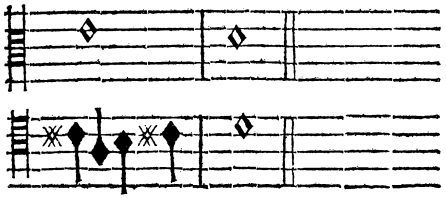


But since in this species of counterpoint there is no place for dotted notes, it has therefore pleased men of great authority to approve the first example—where the second note forms a seventh—perhaps because of the greater smoothness of the melody.

It remains lastly to show how the penultimate beat—which for the most part has more difficulty than the others—is to be formed; which, if the cantus firmus (fixed melody) is in the lower part, must be constructed in this manner:



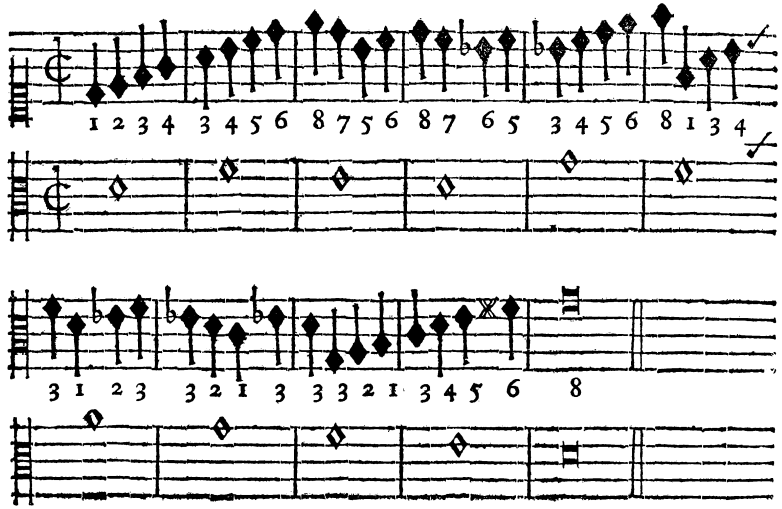
Si autem Cantus firmus in ſuperiori ſit parte, ſequenti modo faciendus erit.



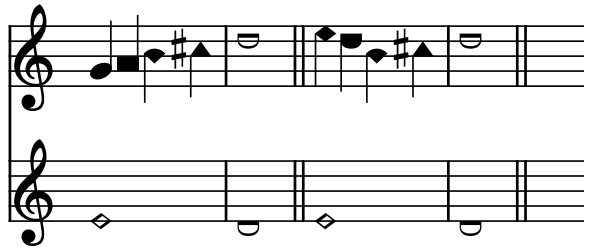
His ergo cognitis, & iis, quæ de aliis Speciebus jam dicta ſunt, facilem tibi futuram ſpero hujus Speciei Compoſitionem : ſed iterum, atque iterum admonitum te volo, ut ſequentis tactûs vel maximè rationem habeas, ſi nullum in progrediendo obicem offendere cupis. Ergo Opus aggrediare, omnes in prima Lectione præſcriptos Cantus firmos eo ordine perſequendo.

Contra-punctum.

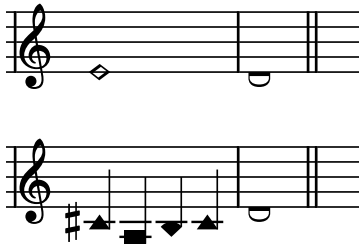
Cantus firmus.



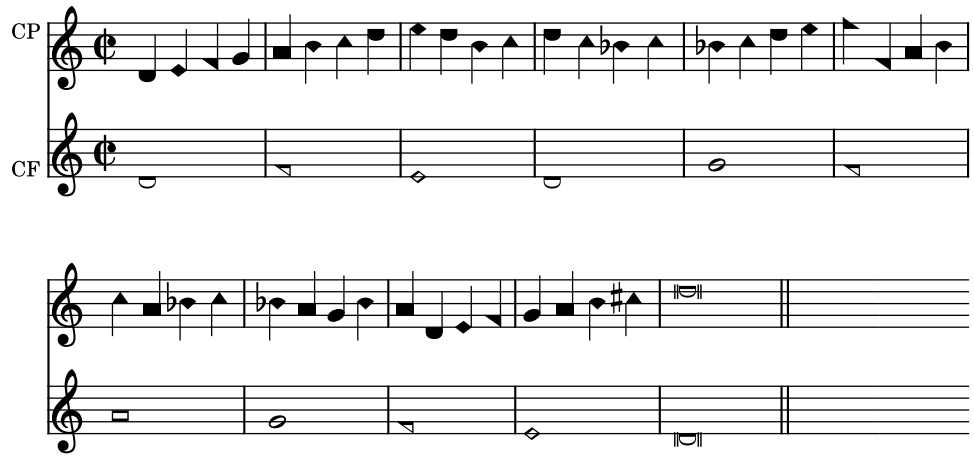
Can-

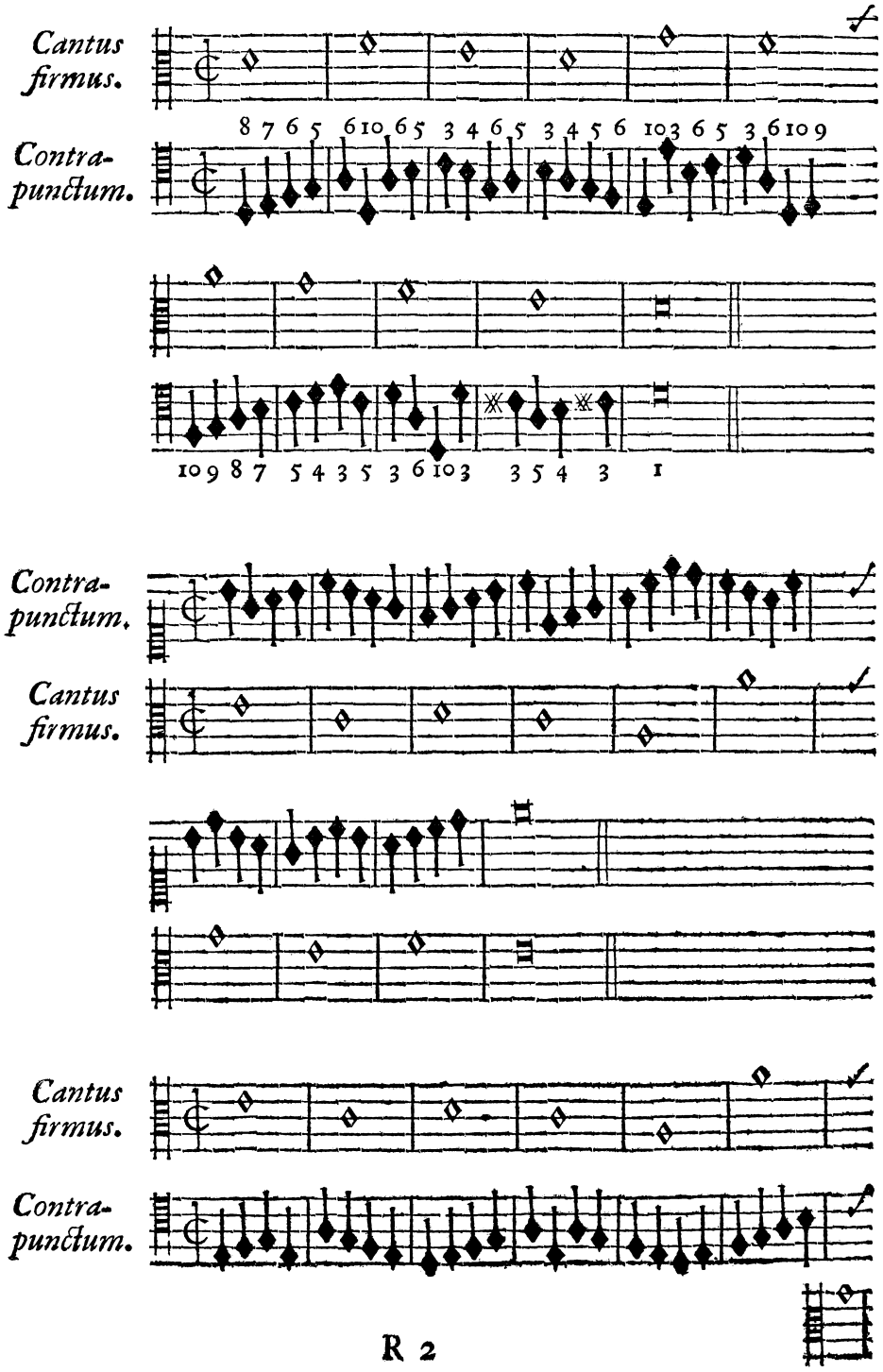
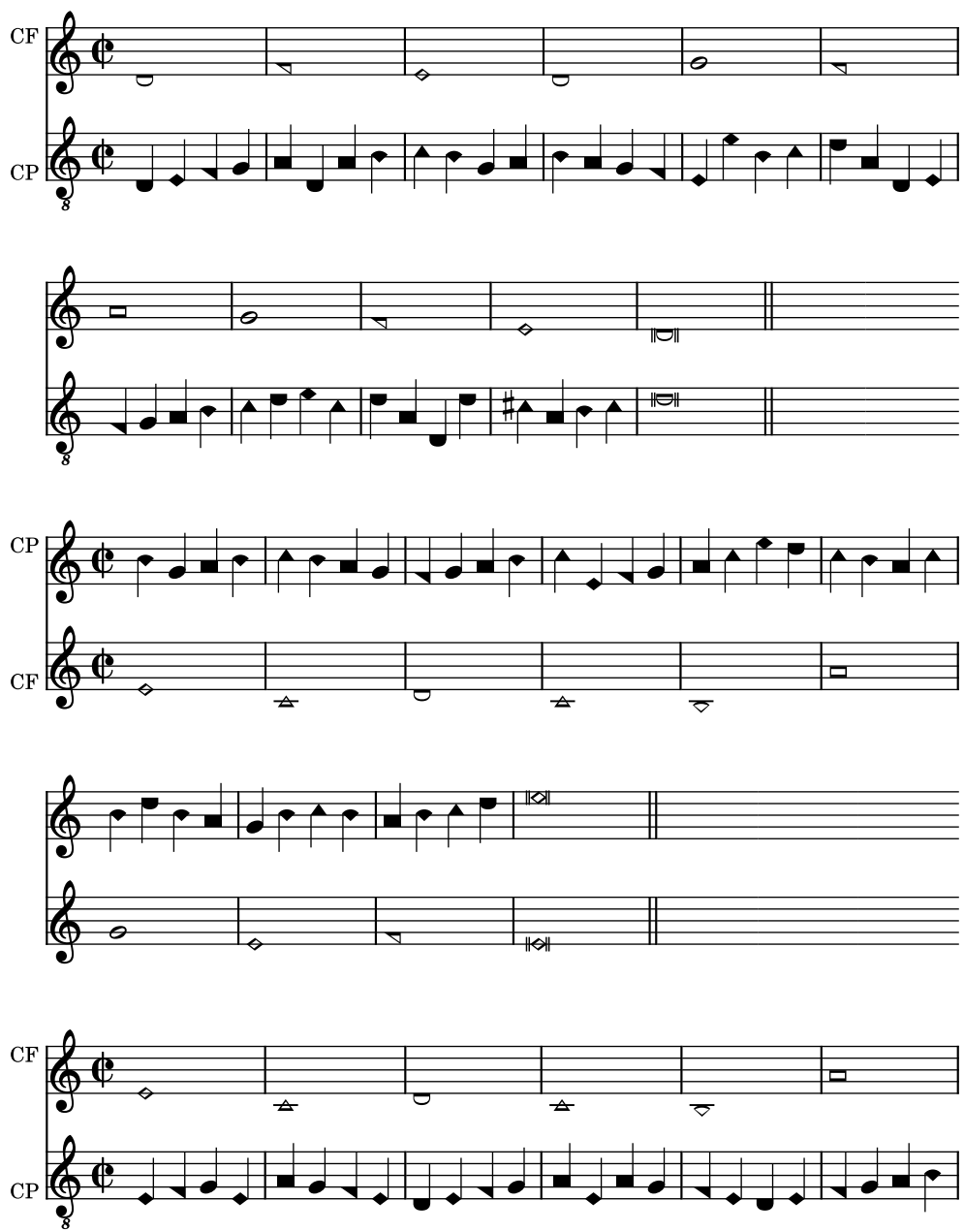


But if the cantus firmus is in the upper part, it must be done in the following way:



These things being known, and those already said concerning the other species, I hope the composition of this species will be easy for you; but again and again I wish to warn you to pay the greatest attention to the following beat, if you wish to avoid encountering any obstacle in your progress. Therefore set to work, going through all the cantus firmi prescribed in the first lesson in the same order.




Cantus firmus.
Contra-punctum.
8 7 6 5 6 10 6 5 3 4 6 5 3 4 5 6 10 3 6 5 3 6 10 9
10 9 8 7 5 4 3 5 3 6 10 3 3 5 4 3 1
Contra-punctum.
Cantus firmus.
Cantus firmus.
Contra-punctum.

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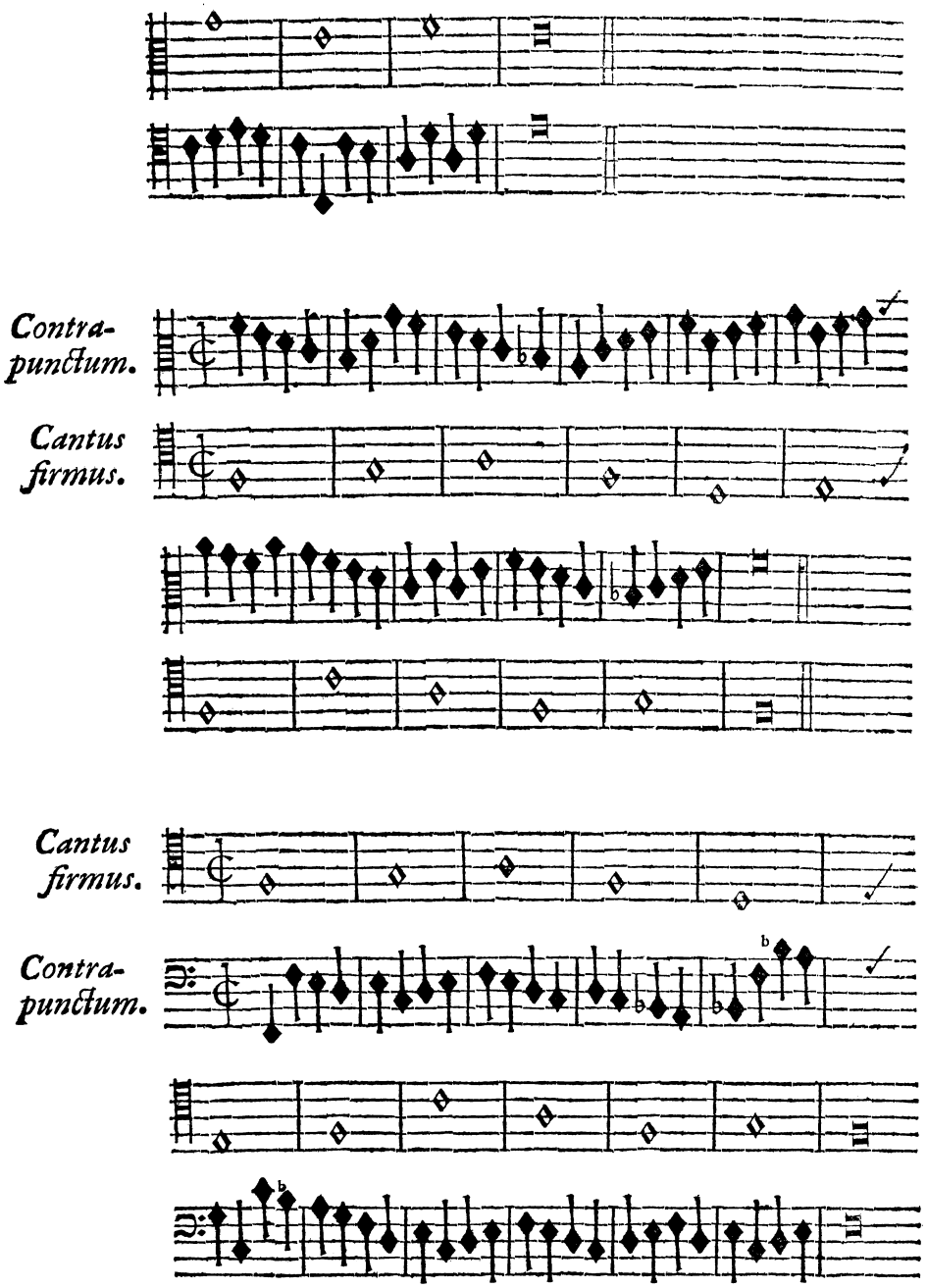
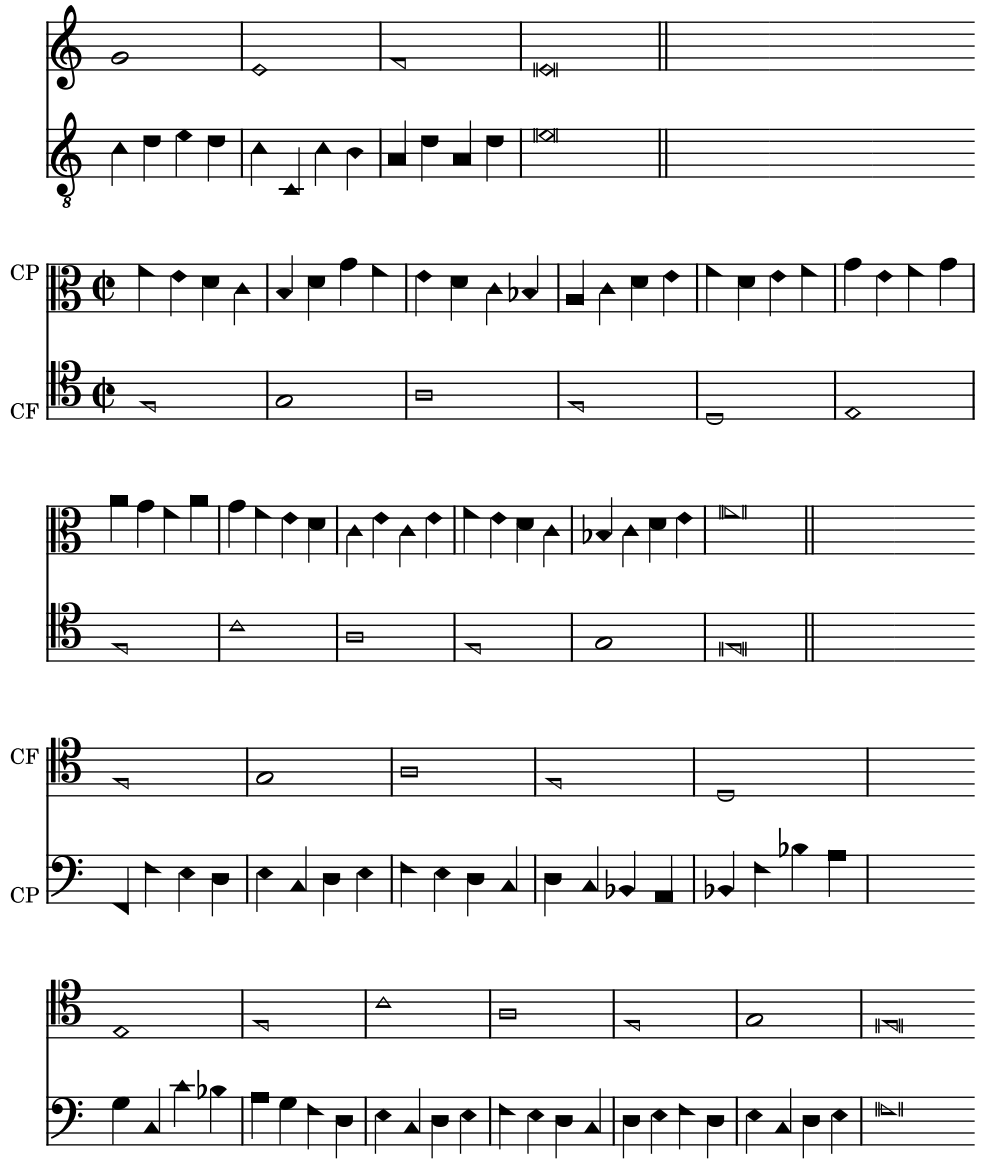


Aloyſ. Quare nonnullis in locis b. mollia adhibuiſti, ſignum in genere Diatonico, in quo verſamur, haud uſitatum? *Jo-*

Aloys. Why have you used b mollia (soft B, i.e., B-flat) in several places, a sign not commonly used in the diatonic genus in which we are working?

Joſeph. Viſum mihi eſt, aliàs aſperam ex *mi* contra *fa* exorituram relationem. Nec obeſſe arbitror generi Diatonico, cùm b. mollia illa non ſubſtantialiter, ſed accidentaliter ex neceſſitate immixta eſſe conſtet.

Aloyſ. Optimè animadvertiſti; ob eandem enim rationem nonnunquam etiam Dieſes adhibendæ ſunt: ſed ubi, quando, perſpicaci ponderandum eſt judicio. Ex præcedentibus exemplis ſatìs memoriâ tenere videris ea, quæ ad hanc Speciem requiruntur. Reliquos tres adhuc ſuperſtites tonos: G. A. C., ne nimis prolixi ſimus, privato ſtudio perſequendos tibi relinquo. Sit igitur

EXERCITII I.

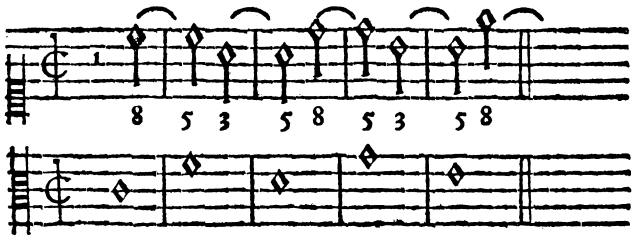
LECTIO QUARTA,

De quarta Contrapuncti Specie.

SPecies hæc conſtat duabus minimis contra ſemibrevem in uno eodemque loco poſitis, virgulâque curvatâ ſubductis, quarum prima in Arhſi, ſecunda in Theſi veniat, neceſſe eſt. Quæ Species *Ligatura*, vel *Sincope* appellari conſuevit; eſtque duplex, Conſoantiæ, & Diſſonantiæ.

Ligatura Conſonantiæ eſt illa, cujus utraque minima, & in Arhſi, & in Theſi Conſonantia eſt. Exemplis res clara fiet.

Ligaturæ Conſonantiæ.



Ligaturæ Diſſonantiæ, cujus prima Nota, nempe in Arhſi quidem Conſonans, (quod ſemper eſſe debet) ſecunda verò in Theſi diſſonans eſt, ùt in ſequenti exemplo videre licet.

Joseph. It seemed to me that otherwise a harsh relation would arise from mi contra fa (mi against fa). Nor do I think it harms the diatonic genus, since those b mollia are introduced not substantially but accidentally, out of necessity.

Aloys. You have observed very well; for the same reason sharps must sometimes be used also; but where and when must be weighed with perceptive judgment. From the preceding examples you seem to remember well enough the things required for this species. The remaining three modes—G, A, and C—I leave for you to pursue in private study, lest we be too lengthy. Therefore let it be

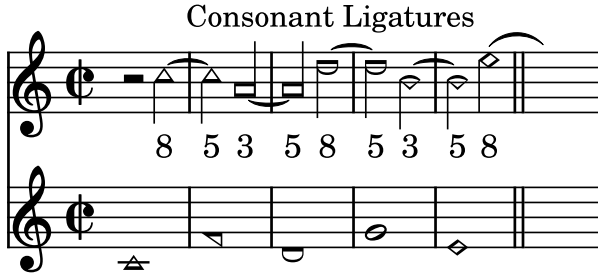
EXERCISE I.

LESSON FOUR,

On the Fourth Species of Counterpoint.

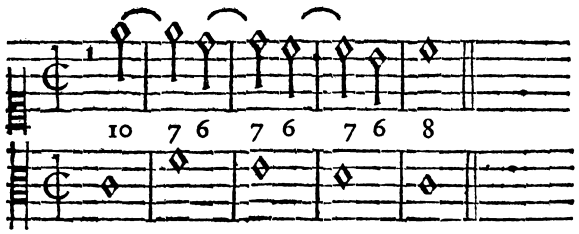
This species consists of two half notes set against a whole note in the same place, joined by a curved stroke, of which the first must fall on the Arsis (up-beat) and the second on the Thesis (down-beat). This species is commonly called the Ligature (tie) or Syncopation, and it is twofold: of consonance and of dissonance.

A ligature of consonance is that in which both half notes, in both Arsis and Thesis, are consonances. The matter will be clear from examples.



A ligature of dissonance is that in which the first note, namely in the Arsis, is indeed a consonance (as it always must be), but the second, in the Thesis, is a dissonance; as may be seen in the following example.

Ligaturæ Diſſonantiæ.

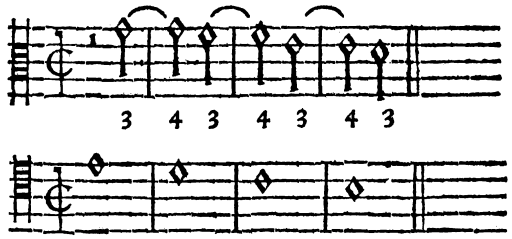


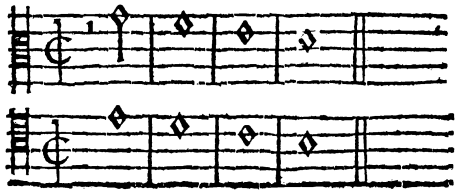
Porrò cùm Diſſonantiæ hoc loco non accidentaliter, vel per diminutionem, quemadmodum in præcedentibus Speciebus, ſed ſubſtantialiter, & in Theſi veniant, neque ex ſe, utpote auribus inimicæ quidquam grati exhibere valeant, amœnitatemque ſuam à Conſonantia continuò ſequente, in quam reſolvuntur, participare neceſſe ſit ; quapropter nunc dicendum eſt

De

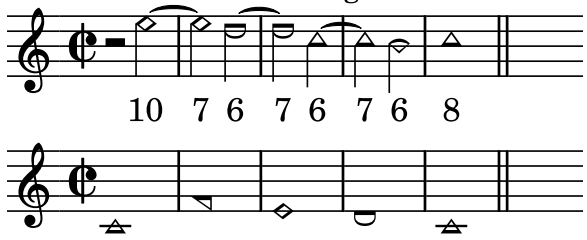
Diſſonantiarum Reſolutione.

Antequam dicere aggrediar, quô modô Diſſonantiæ reſolvendæ ſint, juvat ſcire, Notas ligatas quaſi vinculis conſtrictas haud aliud eſſe, quàm Notæ ſequentis retardationem, quæ poſteà tanquam ſervitute ſolutæ in libertatem exire videntur. Quam ob rem diſſonantiæ ſemper in proximam Conſonantiam gradatim deſcendendo ſe moventem reſolvendæ ſunt ; ùt in ſequenti exemplo clarè videtur :



Quod Paradigma, ſublatâ retardatione ſic ſe habet :

Ex

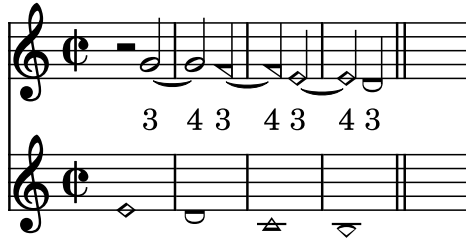
Dissonant Ligatures



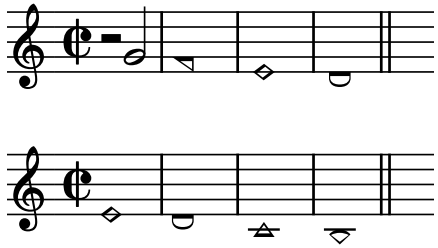
Moreover, since dissonances in this place arise not accidentally or by diminution, as in the preceding species, but substantially and in the Thesis, and since of themselves, being hostile to the ear, they can offer nothing pleasing, it is necessary that they receive their agreeableness from the consonance that immediately follows, into which they resolve. Therefore it is now time to speak

On The Resolution of Dissonances.

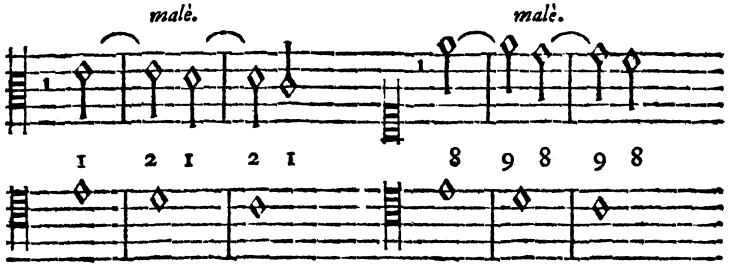
Before I begin to explain how dissonances are to be resolved, it is helpful to know that tied notes, as if bound with chains, are nothing other than a retardation of the following note, which afterwards, as though freed from servitude, seems to pass into liberty. For this reason dissonances must always be resolved into the nearest consonance by stepwise descent, as is clearly seen in the following example:



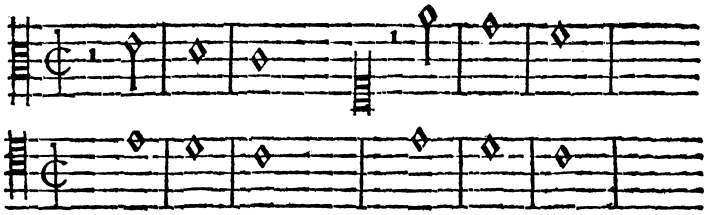
This pattern, with the retardation removed, stands thus:

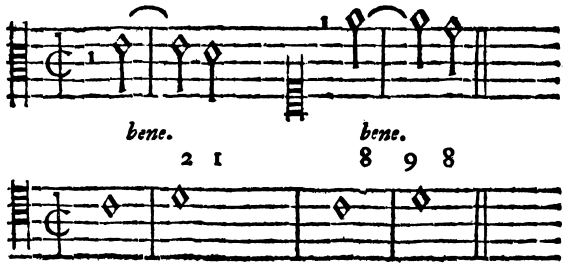


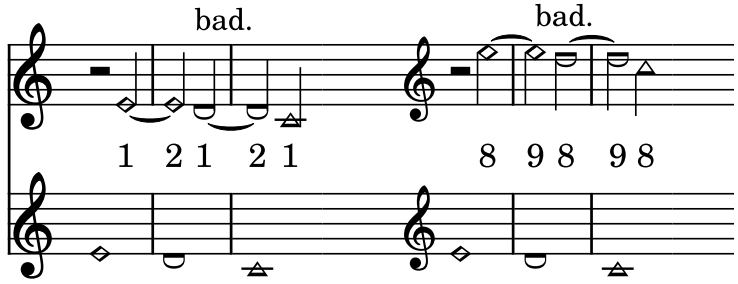
Ex quo colligitur, rem ſcitu facilem fore, in qualem Conſonantiam quælibet diſſonantia reſolvenda ſit, videlicet in illam, quæ, ſublata retardatione, immediatè in ſequentis tactûs Theſi reperitur. Inde fit, quòd Cantu firmo in inferiori parte exiſtente, Secunda in Uniſonum, Quarta in Tertiam, Septima in Sextam, Nona in Octavam reſolvi debeant. Quæ ratio ità veritate nititur, ut hanc ob cauſam neque de Uniſono in Secundam, neque de Octava in Nonam per Ligaturam progredi liceat; quemadmodum in ſequentibus exemplis videndum eſt.



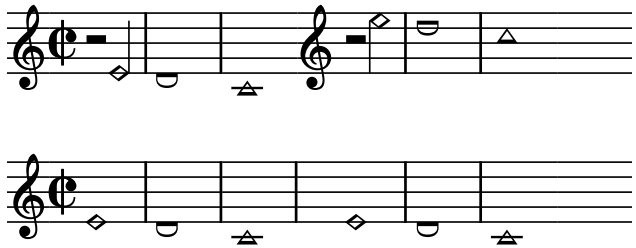
Quâ retardatione ſublatâ immediatam duorum Uniſonorum in primo exemplo, & duarum Octavarum conſecutionem in ſecundo videbis, hac ratione:



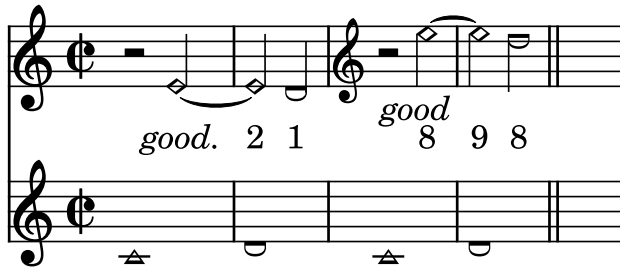
Contra fit, ſi de Tertia in Secundam, & de Decima in Nonam fit progreſſus, hoc modo:

From this it is gathered that it will be easy to know into which consonance any dissonance must be resolved—namely, into that which, the retardation removed, lies immediately on the Thesis of the following beat. Hence it follows that, when the cantus firmus (fixed melody) is in the lower part, the Second must resolve to the Unison, the Fourth to the Third, the Seventh to the Sixth, and the Ninth to the Octave. This principle is so firmly grounded in truth that for this reason it is not permitted to proceed by ligature from the Unison to the Second, nor from the Octave to the Ninth, as may be seen in the following examples:

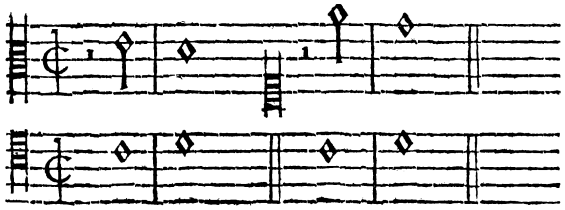


With the retardation removed, you will see the immediate succession of two Unisons in the first example, and of two Octaves in the second, in this manner:

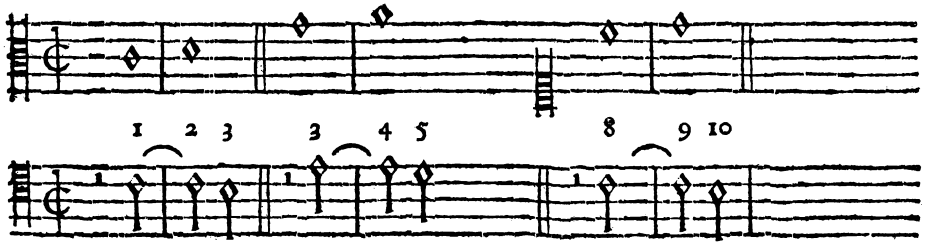


On the other hand, if the progression is from the Third to the Second, or from the Tenth to the Ninth, it is valid, as follows:



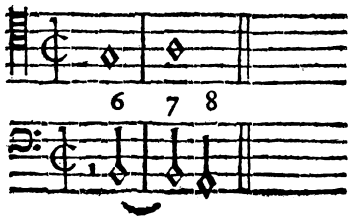
Quæ Hypotheſes rectè conſiſtunt, quia etiam retardatione, ſive ligaturâ ſublatâ valerent, ſequenti nempe ratione:

Poſtquam monſtratum eſt, quales diſſonantiæ adhiberi poſſint, quôque modo reſolvi debeant, ſi Cantus firmus in inferiori exiſtat parte, jam dicendum reſtat, quibus diſſonantiis locus ſit, qua ratione reſolvendæ, Cantu firmo in ſuperiori parte exiſtente.

Dico igitur, quòd hoc loco Secunda reſoluta in Tertiam; Quarta reſoluta in Quintam, Nona reſoluta in Decimam adhiberi poſſint. E. G.



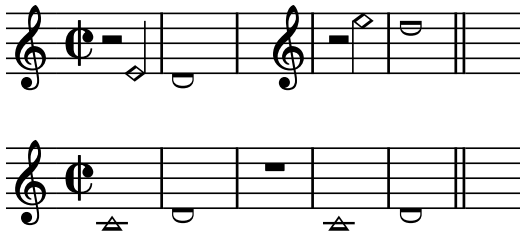
Joſeph. Quare hoc loco omittis Septimam? ergo Cantu firmo in parte ſuperiore exiſtente adhiberi non poteſt? non moleſtè feras mihi dicere rationem.

Aloyſ. Septimam conſultò hoc loco à me omiſſam eſſe fateor. Ratio autem vix alia afferri poterit, quàm Authorum Claſſicorum gravis Authoritas, cui in praxi plurimum tribuendum eſt, quorum vix ullus reperitur, qui Septimâ hac ratione in Octavam reſolutâ uſus fuerit.

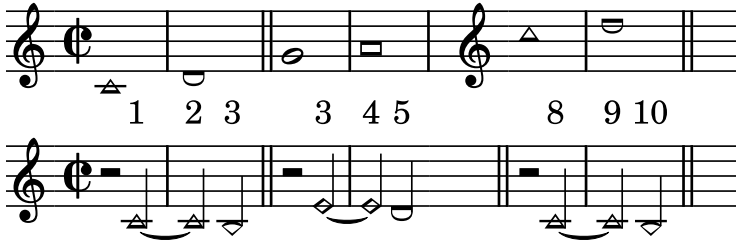


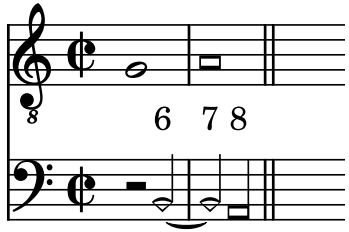
Poſſet

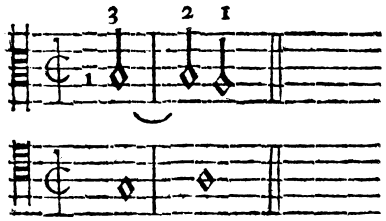
These hypotheses stand correctly, because even with the retardation or ligature removed they would still be valid, namely in the following manner:



After showing what dissonances may be used and how they must be resolved when the cantus firmus is in the lower part, it remains to say what dissonances may be used and how they must be resolved when the cantus firmus is in the upper part.

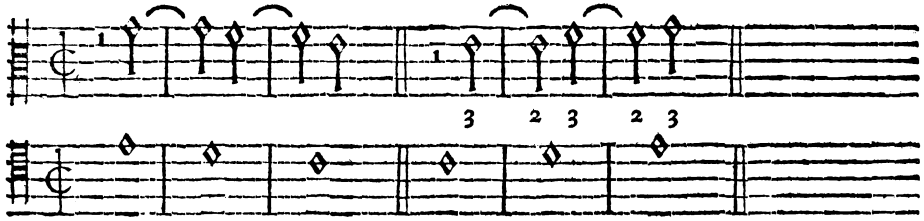
I say, therefore, that in this case the Second resolved to the Third, the Fourth resolved to the Fifth, and the Ninth resolved to the Tenth may be used. For example:

Joseph. Why do you omit the Seventh here? Can it not be used when the cantus firmus is in the upper part? Do not be unwilling to tell me the reason.

Aloys. I confess that I have deliberately omitted the Seventh here. The reason can scarcely be anything other than the weighty authority of the classical authors, to whom in practice much must be granted; scarcely any of them is found to have used the Seventh resolved to the Octave in this manner.

Poſſet fortaſſe dici, ideo Septimam hac ratione reſolutam non valere, quia in Conſonantiam perfectam, nempe Octavam, à qua parùm admodum harmoniæ habere poteſt, reſolvitur, nisì eosdem apud Authores Secunda, quæ Septima inverſa eſt, in Uniſonum reſoluta frequenter reperiretur, à quo, tanquam Conſonantiarum perfectiſſimo multò minùs harmoniæ, diſſonans participare poteſt. Uſui hac in parte, gravibus ab Authoribus introducto conſulendum puto. Videamus exemplum Septimæ in Secundam inverſæ, & in Uniſonum reſolutæ :

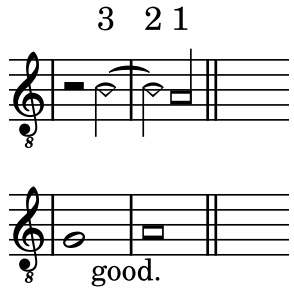


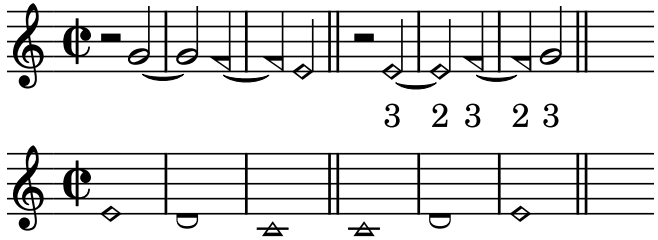
bene.

Joſeph. Antequam ad Lectionem progrediar, liceat mihi pace tuâ quærere, an retardatio, ſive Ligatura in Diſſonantia etiam in aſcendendo locum habeat ? Videtur enim par ratio ſequentium exemplorum :



Aloyſ. Quæſtionem moves, nodo Gordio ſolutu difficiliorem, quam tu etiamnum in diſciplinæ hujus limite hærens capere non potes, ideo alio loco ſolvendam. Quamvis ſi retardationem ſuſtuleris, par ratio Tertiarum videtur & aſcensûs, & deſcensûs, quandam tamen differentiam ſubeſſe, ſuo loco, ùt dixi, explanabitur. Interim omnes diſſonantias in proximam Conſonantiam deſcendendo reſolvendas eſſe jure Magiſtri tibi dico. Cæterùm hujus Speciei penultimus tactus Septimâ in Sextam reſolutâ, ſi Cantus firmus in inferiori ſit parte, concludendus eſt : ſin autem Cantus firmus in ſuperiori

One might perhaps say that the Seventh so resolved is invalid because it resolves into a perfect consonance, namely the Octave, from which a dissonance can receive very little harmony—were it not that the same authors frequently use the Second (which is the inversion of the Seventh) resolved into the Unison, from which, being the most perfect of consonances, a dissonance can receive even less harmony. In this matter I think we must follow the usage introduced by authors of great authority. Let us look at an example of a Seventh inverted into a Second and resolved into the Unison:

Joseph. Before I proceed to the lesson, allow me, with your permission, to ask whether retardation or ligature may also occur in an ascending dissonance? For the following examples seem to show equal reason:

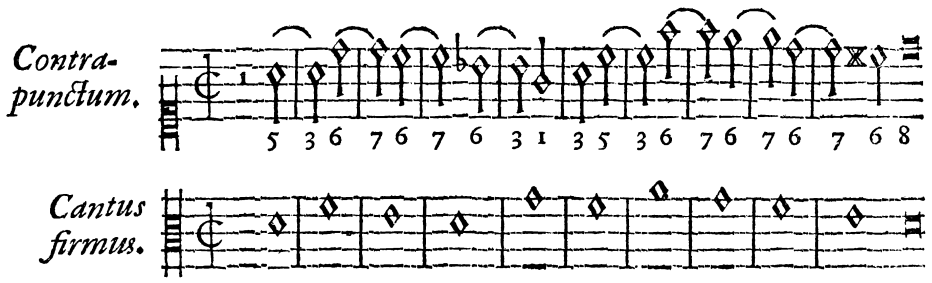


Aloys. You raise a question wrapped in a Gordian knot, more difficult than you, still standing on the threshold of this discipline, can yet grasp; therefore it must be solved elsewhere. Although, if you maintain the retardation, the reasoning of thirds seems equal in ascent and descent, yet some difference lies beneath, which will be explained in its proper place, as I said. Meanwhile, as your teacher, I tell you that all dissonances must be resolved into the nearest consonance by descending step. Furthermore, the penultimate beat of this species must end with a Seventh resolved to a Sixth if the cantus firmus is in the lower part; but if the cantus firmus is in the upper part,

riori parte exiſtat, cum Secunda in Tertiam minorem reſolutâ, inde in Uniſonum procedendo finiendus erit.

Joſeph. Eritne ſingulis tactibus danda ſua Ligatura?

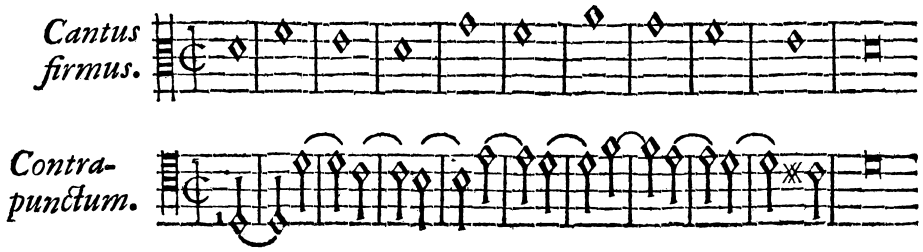
Aloyſ. Omninò, ubi eſſe poterit. Offendes autem nonnunquam tactum, cui Ligaturæ locus non erit, quo caſu tactus ille Minimis ſolutis implendus erit, donec facultas faciendæ Ligaturæ ſe præſentet. Age, Ligaturisque incumbe.



Rectè quidem feciſti; ſed quare in quinto tactu omiſiſti Ligaturam? poterat enim eſſe, ſi poſt Tertiam illam, Quintam adhibuiſſes; quæ prima Ligaturæ Nota fuiſſet, & manendo in eadem linea, in ſequentis tactûs Theſi, nempe Sextâ alteram Ligaturam habuiſſes. Dixi enim, nullam occaſionem prætereundam eſſe.

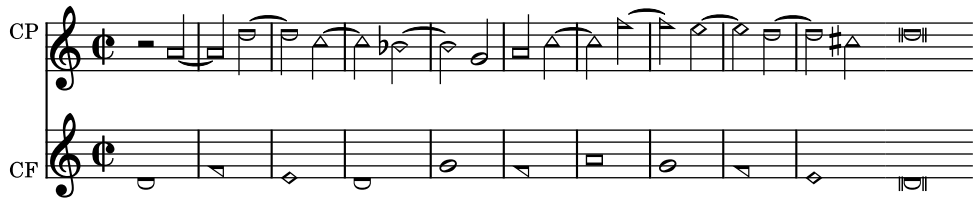
Joſeph. Poteram certè; ſed conſultò omiſi, ne in odioſam inciderem repetitionem, continuò enim antè in tertio & quarto tactu easdem poſui Ligaturas.

Aloyſ. Satis providâ uſus es obſervatione, quia & canendi rationi, & progreſsûs concinnitati haud parùm concedendum eſt. Perge.



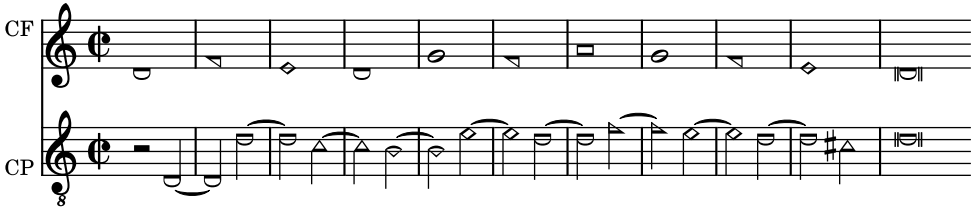
it must end with a Second resolved to a minor Third, and thence proceed to the Unison.

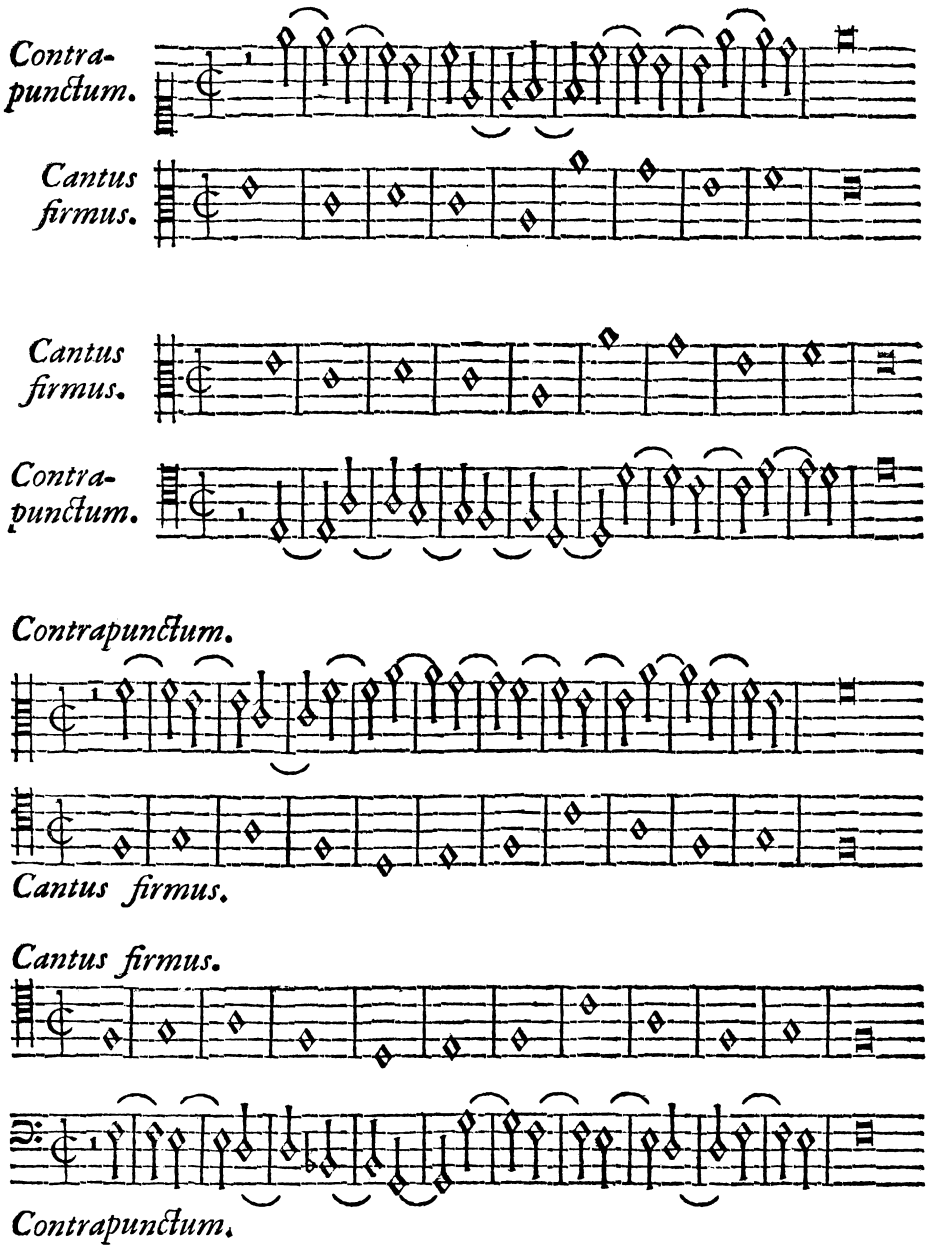
Joseph. Must each beat have its own ligature?

Aloys. Certainly, wherever it can be. Yet you will sometimes encounter a beat in which no ligature is possible, in which case that beat must be filled with free half notes until the opportunity for a ligature presents itself. Come, apply yourself to the ligatures.

You have done well; but why did you omit the ligature in the fifth beat? It could have been there, if after that Third you had used a Fifth, which would have been the first note of the ligature, and by remaining on the same line, on the Thesis of the next beat—namely the Sixth—you would have had the second note of the ligature. I said that no opportunity must be neglected.

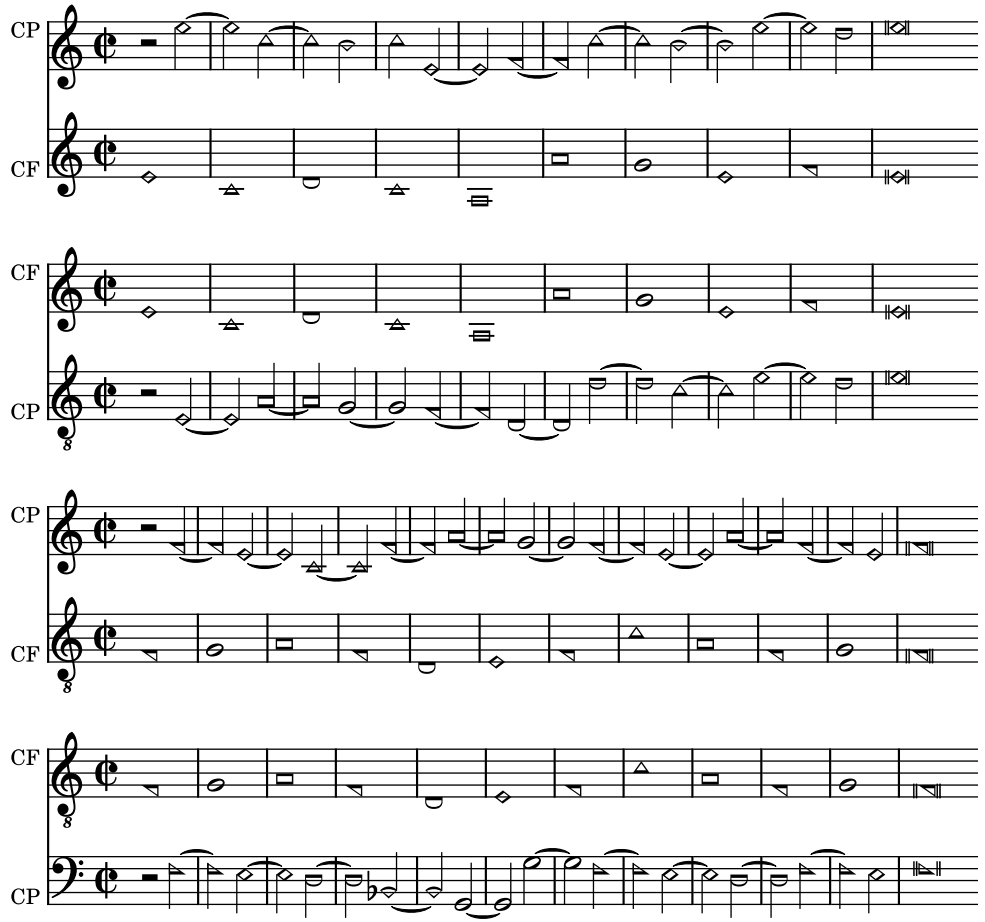
Joseph. I certainly could have; but I omitted it deliberately, lest I fall into a tedious repetition, for I had already placed the same ligatures in the third and fourth beats.

Aloys. You have used a sufficiently prudent observation, for something must be granted both to the manner of singing and to the smoothness of the progression. Continue.



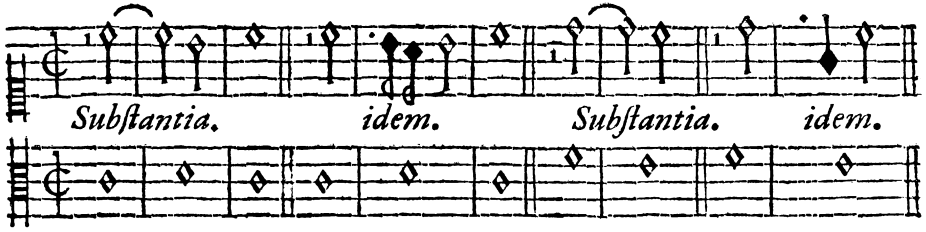
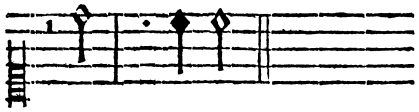
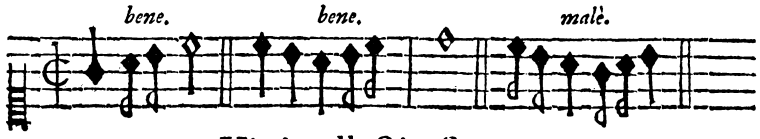


Aloyſ. Hæc exempla pro nunc ſufficiant. Cùm autem Ligaturis haud mediocre condimentum Muſicali Concentui accedat, ideò non ſolùm reliquorum trium tonorum Cantus firmos hac ratione perſequendos tibi commendo, verùm etiam,



Aloys. These examples are enough for now. Since ligatures add no small seasoning to musical harmony, I therefore commend to you not only the remaining three modes to be pursued in this manner, but also that, having found others in this species, you exercise yourself greatly in them; nor will you ever exercise yourself enough.

ut repertis aliis in hac Specie plurimùm te exerceas, nec unquam ſatìs exercebis.

Pro concinnitate ſequentis Speciei hîc præmitto, Ligaturas hucuſque dictas etiam alio modo fieri poſſe, quæ quidem ſubſtantiam minimè evertunt, afferunt tamen concentui celerioris motûs rationem. E. G.

Ex quibus clarè colligitur, primam & tertiam Hypotheſin eſſe ſubſtantiam; ſequentes, ubi *idem* adjunctum eſt, eſſe variationes, vel cantûs, vel motûs gratiâ adhibitas. Rumpi etiam ſolet Ligatura modo ſequenti:

Prætereà nonnunquam duæ Fuſæ ſequenti Speciei immiſceri poſſunt; quæ quidem in ſecunda, & quarta tactûs parte, nunquam verò in prima & tertia ponendæ ſunt.

His intellectis ſit

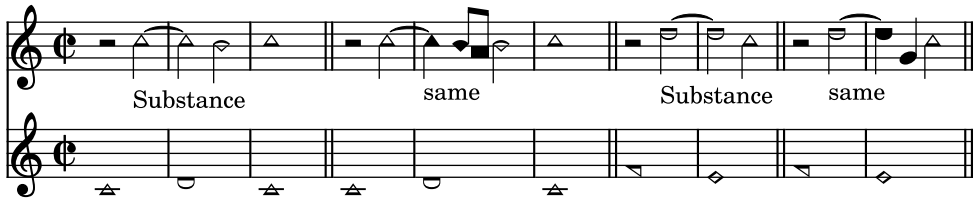
EXERCITII I.

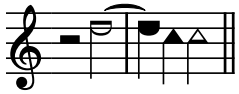
LECTIO QUINTA,

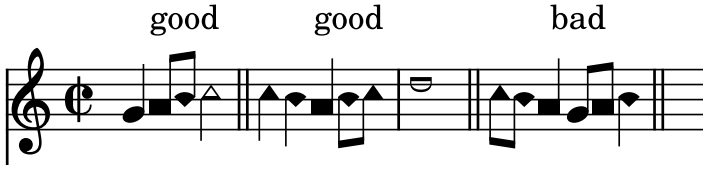
De quinta Specie Contrapuncti.

SPecies iſta Contrapunctum floridum appellatur, ſic dictum, quia omnis generis ornatu, canendi gratiâ, flexibili motuum facilitate, concinnâ figurarum varietate, ùt hortus floſculis florere debet. Quemadmodum enim omnibus reliquis Arith-

For the sake of the elegance of the next species, I here premise that the ligatures described thus far may also be made in another way, which does not overturn their substance, but gives the harmony a sense of quicker motion. For example:



From these it is clearly gathered that the first and third hypotheses are the substance; the following ones, where the same structure is present, are variations introduced for the sake of melody or motion. The ligature is also sometimes broken in the following manner:

Moreover, two fusae (eighth-notes) may sometimes be inserted into the next species; these must be placed in the second and fourth parts of the beat, never in the first or third.



These things understood, let it be

EXERCISE I.
LESSON FIVE,

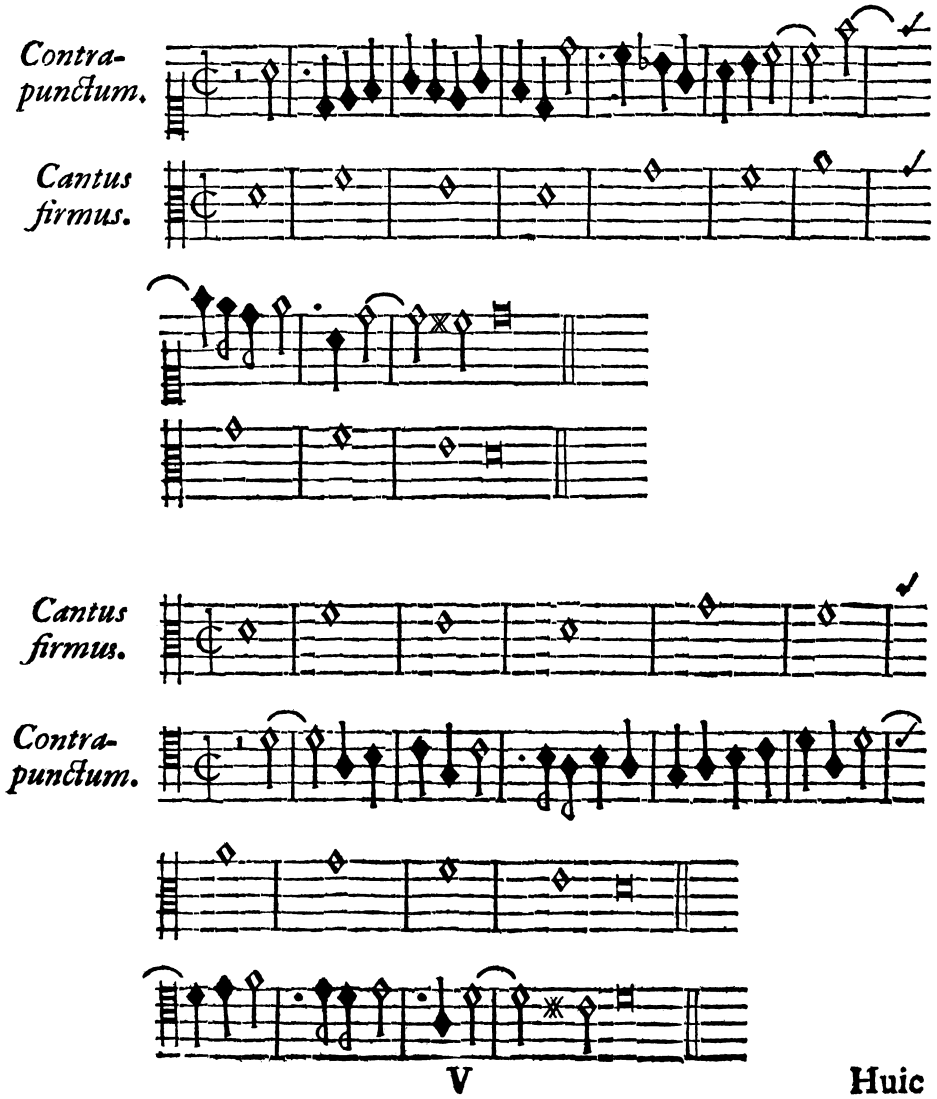
On the Fifth Species of Counterpoint.

This species is called contrapunctum floridum (florid counterpoint), so named because, with ornament of every kind, for the sake of singing, with flexible ease of motion and a graceful variety of figures, it ought to bloom like a garden with little flowers. For just as in all the remaining species of common arithmetic—such as numeration, addition,

Arithmeticæ vulgaris Speciebus, ùt Numeratione, Additione, Subtractione, & Multiplicatione in Diviſione utimur, ità Species hæc nihil aliud eſt, quàm præcedentium Contrapuncti Specierum Congeries & Compoſitio, nec quidquam novi hîc monendum occurrit, præter Cantûs concinnitatem, cui plurimùm curæ, atque operæ dandum eſſe, nunquam è memoria tibi excidat, hortor, atque obſecro.

Joſeph. Dabo operam pro viribus meis, ſed vix mihi fido, nullo exemplo ante oculos habito, calamum prehendere.

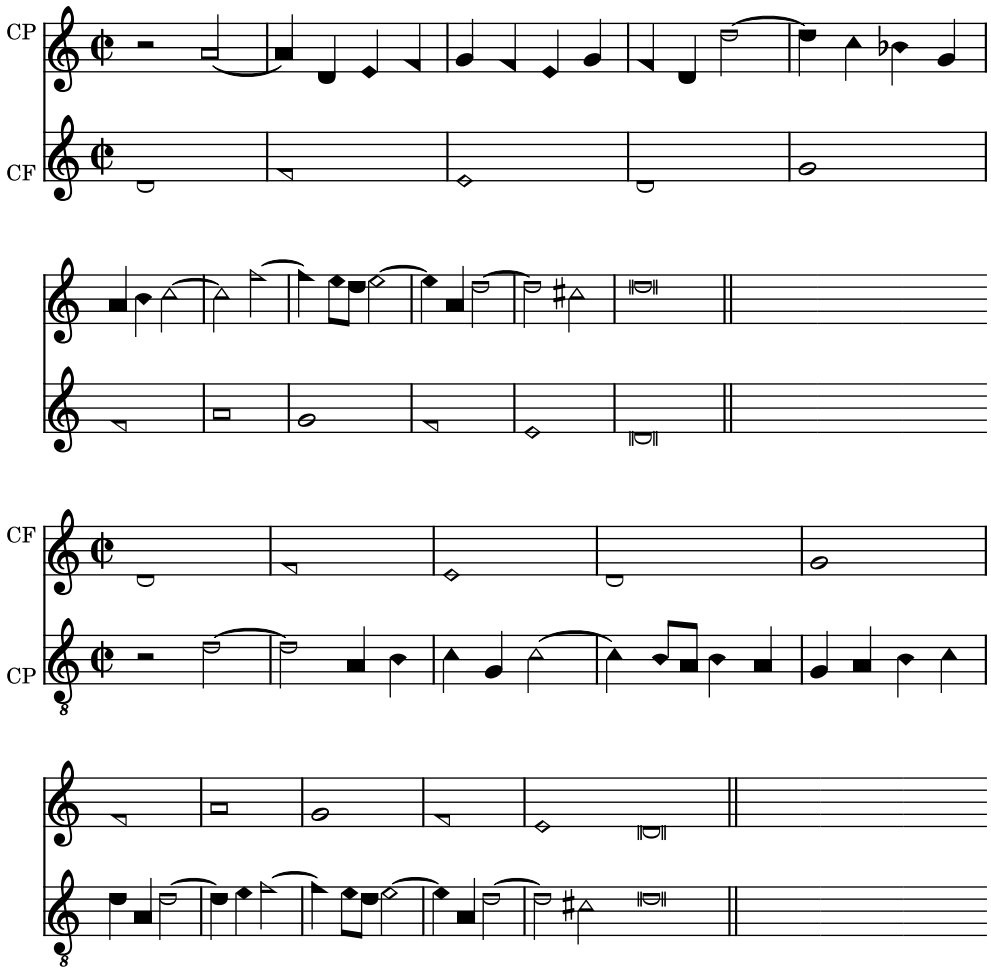
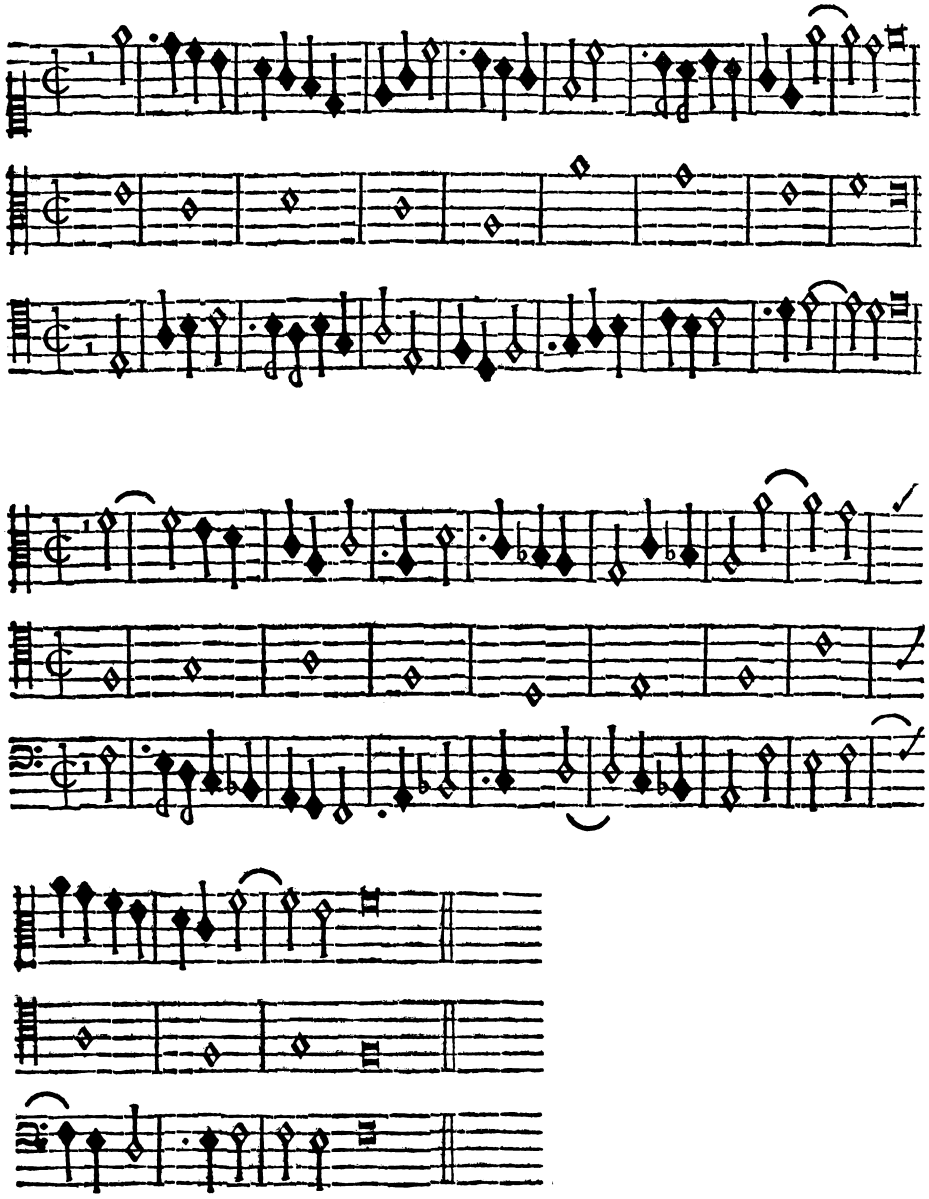
Aloyſ. Alacri animo ſis, formabo tibi primum exemplum.

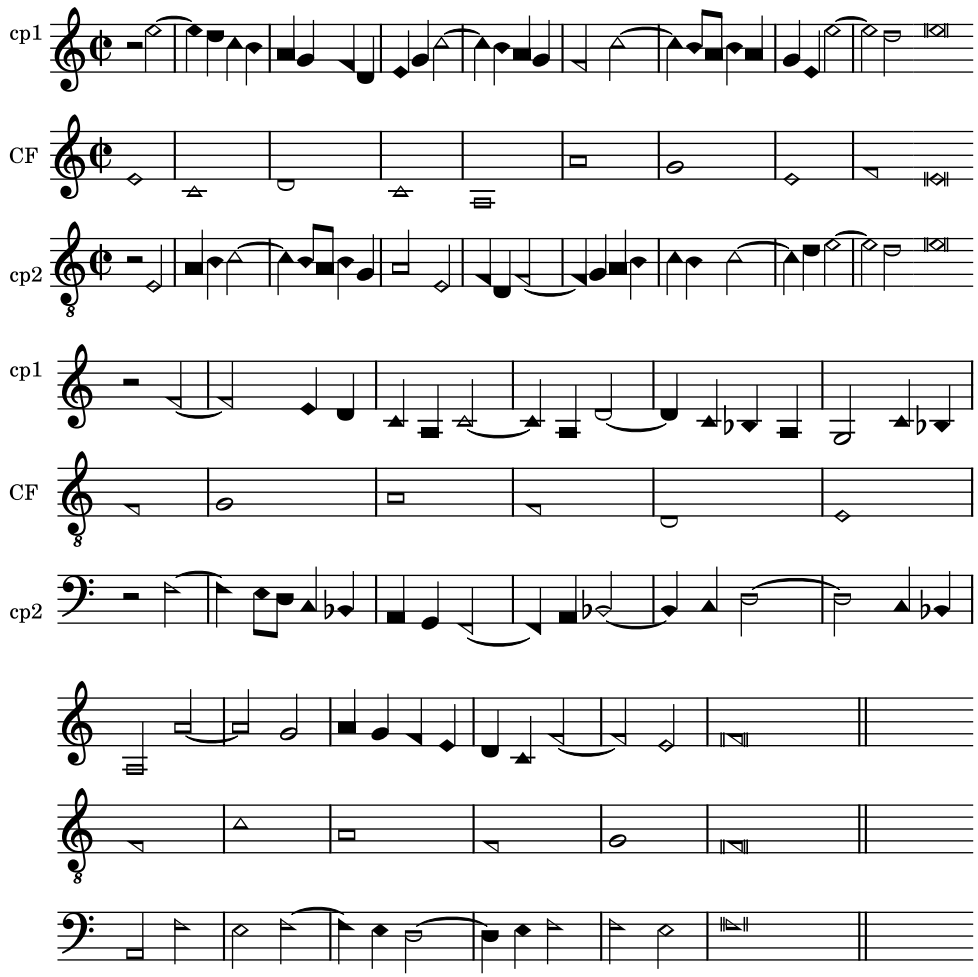


Huic

such as numeration, addition, subtraction, and multiplication—we make use of division, so this species is nothing other than a heap and combination of the preceding species of counterpoint; nor does anything new occur here that must be taught, except the smoothness of the melody, to which you must never forget to give the greatest care and effort, as I urge and entreat you.

Joseph. I will do my best, but I scarcely trust myself to take up the pen without having an example before my eyes.

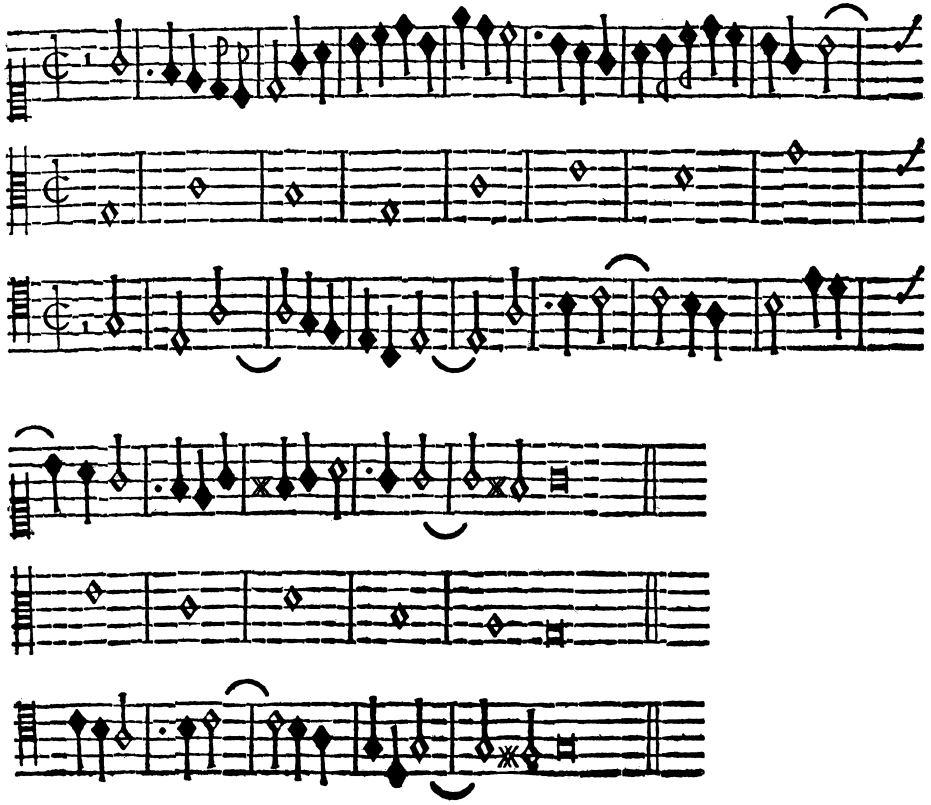
Aloys. Take heart; I will form the first example for you.

Huic igitur formulæ reliquorum Cantuum firmorum Contrapuncta tibi accommodanda reſtant.

Aloyſ. Satìs induſtriosè operatus es; & quod mihi non mediocriter placet, eſt, te non parùm attentionis canendi rationi tribuiſſe, quodque in introitu in Theſin plerumque motum

To this pattern, therefore, you must adapt the counterpoints for the remaining cantus firmi.

You have worked with sufficient industry; and what pleases me not a little is that you have given no small attention to the manner of singing, and that in entering upon the Thesis you have for the most part used

tum obliquum, vel Syncopen adhibueris, quam induſtriam, quia plurimùm Contrapuncto affert gratiæ, ulteriùs tibi commendatam volo.

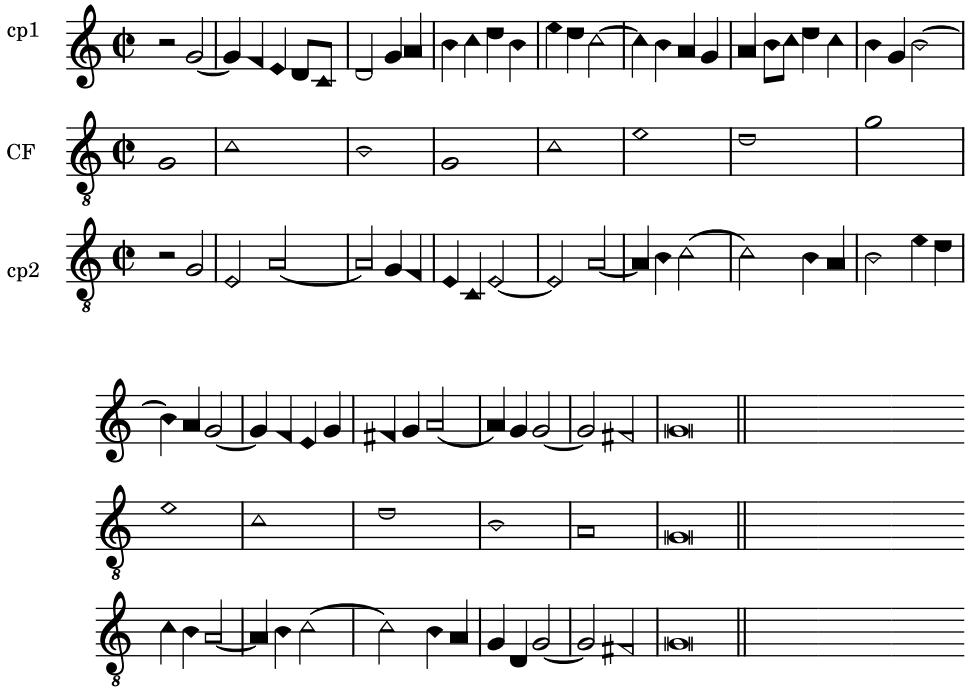
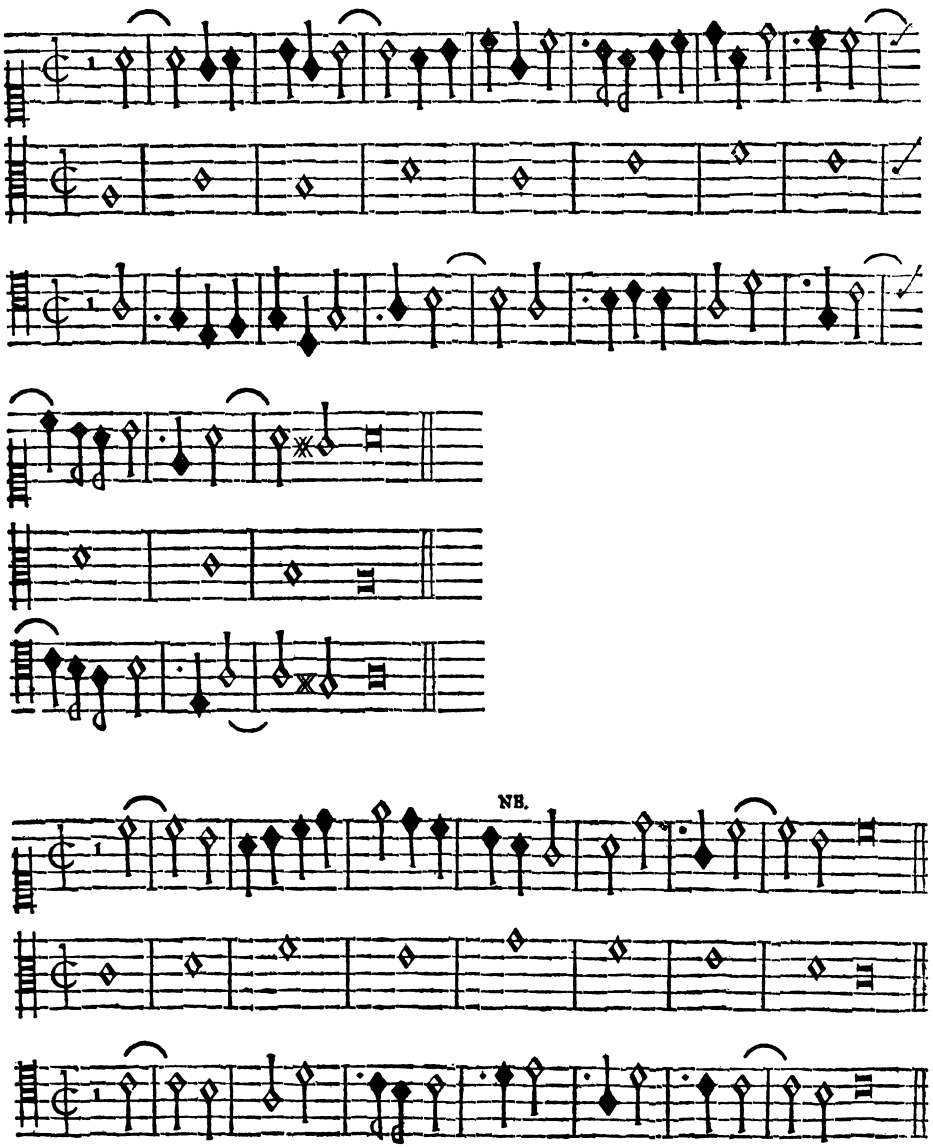
Joſeph. Incredibili afficior lætitiâ, quòd intelligam, non in totum tibi diſplicere & ſtudii, & conatûs mei rationem, quo blandimento animatus, non mediocrem fructum brevi tempore facturus mihi videor. In præſentia tua, vel privato ſtudio reſtantes tres toni mihi perſequendine ſunt?

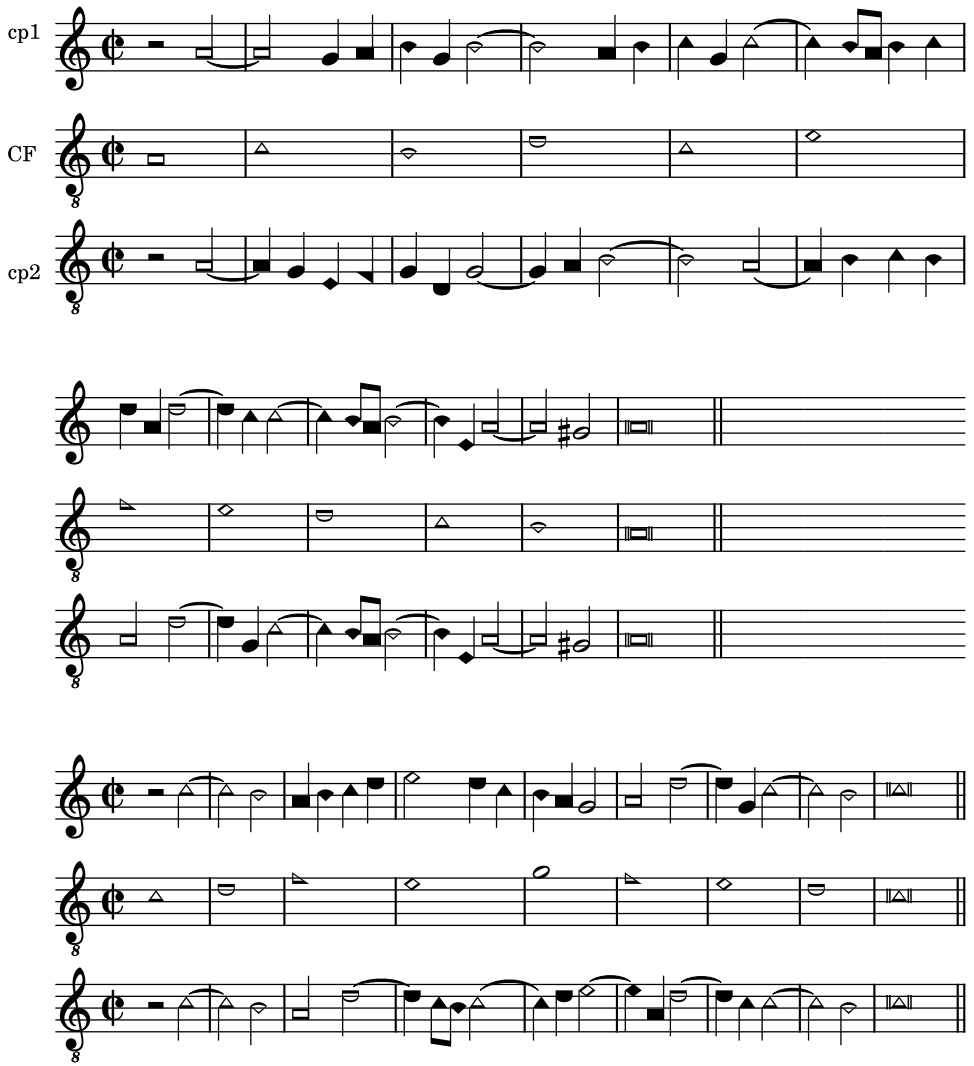
Aloyſ. Cùm Species hæc tantæ ſit utilitatis, ut vix dicere queam, ideo abſolves tu quidem prædictos tres tonos me præſente; verùm te admonitum volo, ut hoc in genere Compoſitionis ſemper magìs magisque te exercere nunquam deliſtas.

Joſeph. Conſilium tuum ſemper ùt Legem obſervabo.

oblique motion or syncopation—an effort which, because it brings much grace to counterpoint, I wish to commend to you further.

Joseph. I am filled with incredible joy to understand that the method of my study and effort does not entirely displease you; encouraged by this kindness, I seem to myself likely to make no small progress in a short time. Must I pursue the remaining three modes in your presence, or in private study?

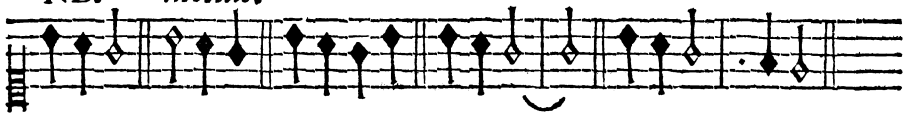
Aloys. Since this species is of such great usefulness that I can scarcely express it, you shall indeed complete the aforesaid three modes in my presence; but I wish to warn you that in this kind of composition you must never cease to exercise yourself more and more.

Joseph. I will always observe your advice as a law.

Joſeph. Quid indicat illud nota bene NB. in quinto tactu ultimi exempli in Cantu poſitum?

Aloyſ. Ne te afflixeris. Nihil enim de hoc ex me adhuc intellexiſti; nunc tibi non quidem Præcepti, ſed Conſilii causâ dico, quòd poſitis duabus Semiminimis in principio tactûs, abſque Ligatura continuò ſequente, ratio cantandi clau-

Joseph. What does that nota bene "NB." placed in the fifth beat of the last example in the melody indicate?

Aloys. Do not trouble yourself. You have understood nothing of this from me yet; now I tell you, not as a precept but as advice, that when two quarter notes are placed at the beginning of a beat without a ligature immediately following, the manner of singing seems to limp.

claudicare videatur. Quapropter conſultiùs erit, ſi duas tantùm Semiminimas in principio tactûs adhibere lubet, eas cum ſequenti Ligatura ſtringere, vel cum duabus adhuc aliis Semiminimis curſum facilitare; ùt in ſequenti exemplo:

NB. *meliùs.*

Jam Bicinium, omnes quinque Species percurrendo, ſuppoſito Cantu firmo, Deo ſint Laudes, abſolvimus. Nunc redeundum eſt ad principium, nempe Notam contra Notam in Tricinio, videndumque, quid in quavis Specie obſervandum, quâque ratione trium partium Compoſitio inſtitui debeat.

EXERCITII II.

LECTIO PRIMA.

De Nota contra Notam in Tricinio.

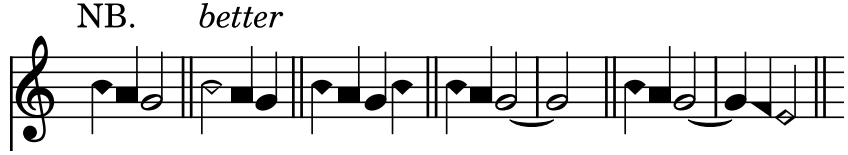
TRicinii Compoſitionem omnium perfectiſſimam eſſe, vel ex eo conſtat, quòd in illa ſinè adjumento alterius partis perfecta harmoniæ Trias haberi poſſit, cùm tertiæ, ſive plurium partium acceſſio nîl aliud quali ſit, quàm cujusdam partis in Triade harmonica jam exiſtentis repetitio; unde quaſi in Proverbium abivit, cui Tricinii rectè efficiendi poteſtas eſſet, ei viam ad plurium partium compoſitionem, quàm latiſſimè patere.

Joſeph. Incredibili animus meus exardeſcit ſciendi deſiderio, qua ratione Compoſitio hæc conficienda ſit; verùm haud modicum metuo, ne rei gravitas tenuitatem opprimens, votum meum hæſitare faciat.

Aloyſ. Non eſt, quod metuas; etenim quia in conficiendis Bicinii Speciebus non magnoperè laborâſti, ſpe fretus eſto, intellectis primùm iis, quæ tibi dicturus ſum, & ſumpto principio à Specie ſimpliciſſima, Notâ nempe contra Notam, eoque ordine, quem in Bicinio obſervavimus, progrediendo, non adeò difficilem tibi futurum Tricinii conficiendi laborem.

Eſt igitur Species hæc trium partium ſimpliciſſima Compoſitio

Therefore it will be more prudent, if you wish to use only two quarter notes at the beginning of a beat, to bind them with the following ligature, or to ease the motion by adding two more quarter notes, as in the following example:

NB. *better*

Now, having gone through all five species in a bicinium, with the cantus firmus assumed—thanks be to God—we have completed it. Now we must return to the beginning, namely Note against Note in a tricinio, and see what must be observed in each species, and in what manner the composition of three parts must be established.

EXERCISE II.
LESSON ONE.
On Note against Note in a Tricinio.

That the composition of a tricinio (three-voice texture) is the most perfect of all is clear even from this: that in it, without the aid of any other part, a perfect Trias harmonica (harmonic triad) can be obtained; for the addition of a third or more parts is almost nothing other than a repetition of some part already present in the harmonic triad. Hence it has almost passed into a proverb: whoever has the ability to construct a tricinio correctly has the widest possible path open to the composition of many-voiced works.

Joseph. My mind burns with incredible desire to know by what method this composition is to be made; yet I greatly fear that the weight of the matter, overwhelming my weakness, may cause my intention to falter.

Aloys. There is no reason to fear; for since you did not labor greatly in constructing the species of the bicinium, be confident in hope that, once you understand what I am about to tell you, and begin from the simplest species—Note against Note—and proceed in the same order we observed in the bicinium, the labor of constructing a tricinio will not be so difficult.

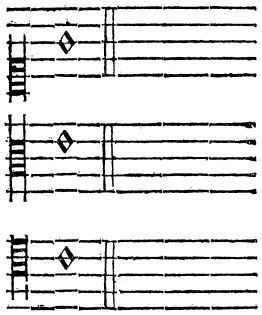
This species, then, is the simplest composition of three parts, with equal notes—that is, three whole notes—and consisting solely of consonances.

poſitio Notis æqualibus, ſive tribus ſemibrevibus, merisque Conſonantiis conſtans.

Ubi primùm notandum, in omni Hypotheſi, ſive tactu, nisì alia ratio impediat, Triadem harmonicam adhibendam eſſe.

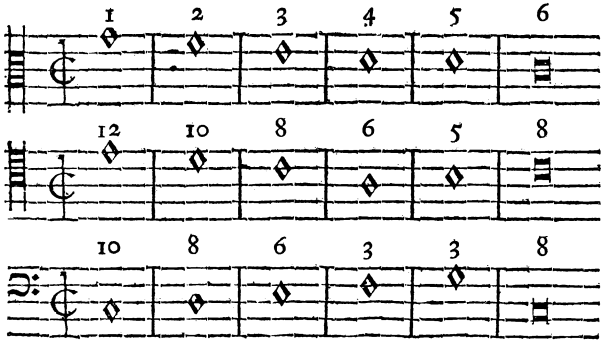
Joſeph. Quid eſt Trias harmonica ?

Aloyſ. Eſt Siſtema Tertiâ & Quintâ conſtans. E. G.



Joſeph. Sed quæ ſunt illæ rationes, quæ hanc Triadem faciendi facultatem nonnunquam adimunt ?

Aloyſ. Eſt ſubinde concinnioris Cantûs ratio, cujus gratia nonnunquam locò Triadis alia Conſonantia, vel Sexta, vel Octava adhiberi ſolet : ſæpiùs etiam immediatam duàrum perfectarum conſecutionem evitandi neceſſitas poſtulat, ut relictâ Triade, aut Sextâ locò Quintæ, aut Octavâ, aut utrâque utendum ſit ; quemadmodum ſequenti exemplo monſtrabo :

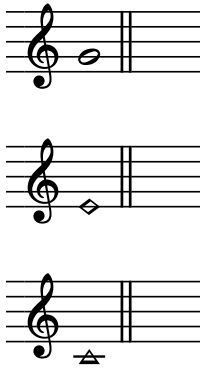


Joſeph. Videtur mihi, pace tuâ, Magiſter, dixerim, in ſecundo tactu Triadem abſque ratione prætermiſſam eſſe ; poterat enim, meo quidem judicio, Tenori Quinta dari, quæ Triadem conſtituiſſet, & poſteà tertio in tactu eidem Tenori

First it must be noted that in every hypothesis, or beat, unless some other reason prevents it, the harmonic triad must be used.

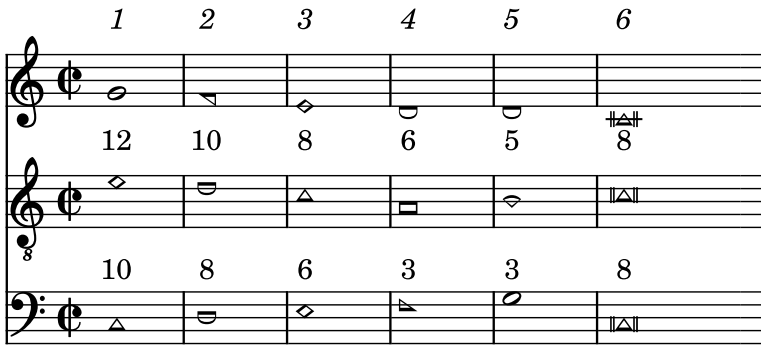
Joseph. What is the harmonic triad?

Aloys. It is a system consisting of a third and a fifth. For example:



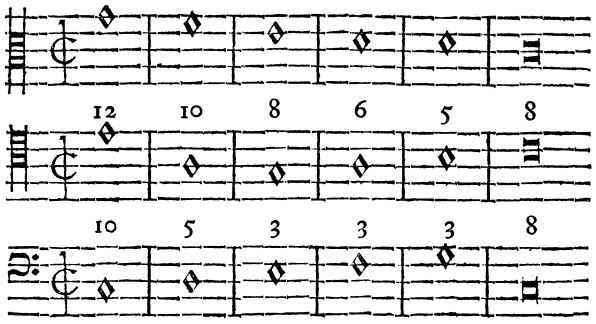
Joseph. But what are the reasons that sometimes take away the possibility of forming this triad?

Aloys. Sometimes it is the smoother flow of the melody, for the sake of which another consonance—either a sixth or an octave—is used in place of the triad. More often the necessity of avoiding the immediate succession of two perfect consonances requires that, leaving aside the triad, one must use either a sixth in place of the fifth, or an octave, or both; as I will show in the following example:



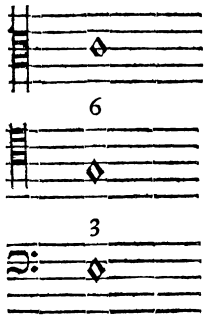
Joseph. It seems to me—if I may say so with your permission, Master—that in the second beat the triad was omitted without reason; for in my judgment the Tenor could have been given the fifth, which would have formed the triad, and then in the third beat the same Tenor could have

ri Tertia tribui, relictis cæteiis Hypotheſibus, ùt à te poſitæ ſunt. E. G.

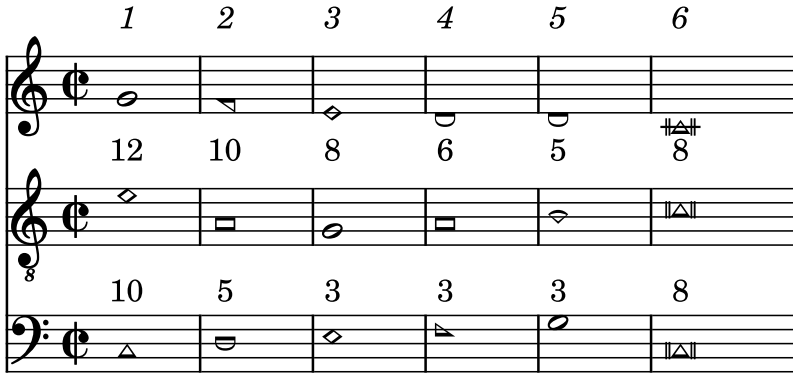


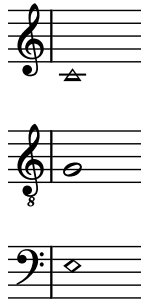
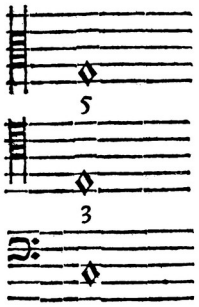
Ambitus iſti neque progreſſuum rationi, neque concinnitati Cantûs obeſſe videntur.

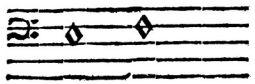
Aloyſ. Non parùm mihi placet hæc tua reflexio, neque reprobari poteſt hoc tuum Paradigma. Verùm primum, nempe meum, & naturæ, & ordini, varietatique magìs conſentaneum eſſe, quis non perſpicit? Naturæ, & ordini, quia in illo pars Tenoris pulchrè gradatim abſque ſaltu deſcendendo movetur uſque ad tertium tactum incluſivè, in quo Sexta reperitur, quam Nota *mi* in parte fundamenti poſita, præ aliis Conſonantiis amat, ùt jam alibi dictum eſt, & iteratò hujus rei rationem explanatiùs dicturus ſum. Ponatur primò illa ſexta Hypotheſis:



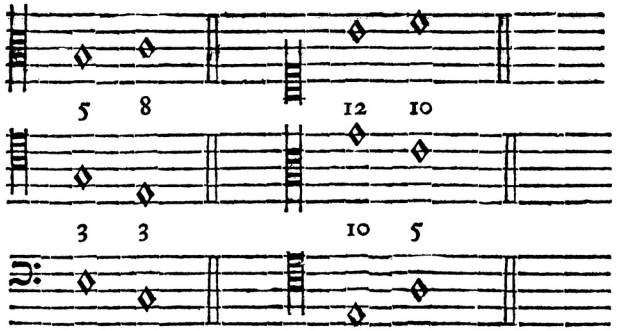
Ubi Nota Sextam efficiens quaſi de loco proprio ad improprium translata tenenda eſt, & in loco proprio ſic ſe haberet:

been given the fifth, which would have formed the triad, and then in the third beat the same Tenor could have been given the third, leaving the other hypotheses as you placed them. For example:

This range seems to hinder neither the logic of the progressions nor the smoothness of the melody.

Aloys. Your reflection pleases me not a little, nor can your example be rejected. But who does not see that the first, that is, mine, is more in accord with nature, order, and variety? With nature and order, because in it the Tenor part moves beautifully downward stepwise without a leap up to the third beat inclusive, in which the sixth appears —a note which mi, when placed in the fundamental part, prefers above other consonances, as has already been said elsewhere, and of which I will again explain the reason more clearly. Let that sixth hypothesis be set forth first:

Here the note forming the sixth must be regarded as transferred from its proper place to an improper one; and in its proper place it would stand thus:

Illud C. ergo in loco proprio poſitum Triadem harmonicam conſtituit, quod translatum per Octavam in acutum, cæteris partibus reſtantibus, Sextam efficiat neceſſe eſt. Quod verò tantùm intelligendum eſt, quando poſt illud *mi*, *fa* ſequitur, ùt in exemplo:

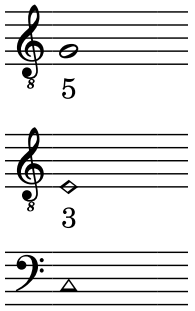


Si autem illud *mi* aliò ſaltare contingat, magìs Quintam, quàm Sextam deſiderat, ùt in ſequentibus exemplis:

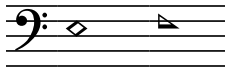


Nunc eò, quò nonnihil receſſimus, redeundum eſt; nempe ad rationem, cur Paradigma meum etiam varietati accommodatius ſit, quia in illo, A. in Tenoris parte tantùm ſemel, in tuo verò exemplo bis reperitur, ùt videre licet:

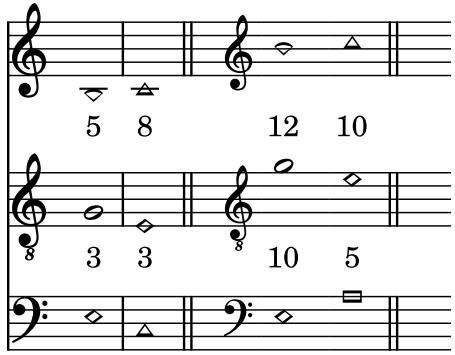
Cui



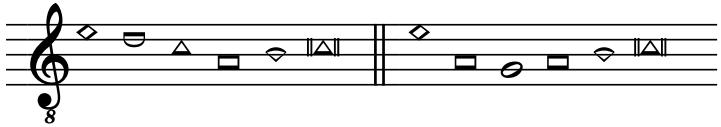
That C, therefore, placed in its proper position, forms the harmonic triad; but transferred by the octave into the upper register, with the other parts remaining, it necessarily forms a sixth. But this is to be understood only when fa follows that mi, as in the example:



But if that mi happens to leap elsewhere, it desires a fifth rather than a sixth, as in the following examples:



Now we must return to the point from which we somewhat digressed—namely, the reason why my example is also more suited to variety: because in it the A in the Tenor part appears only once, whereas in your example it appears twice, as may be seen:

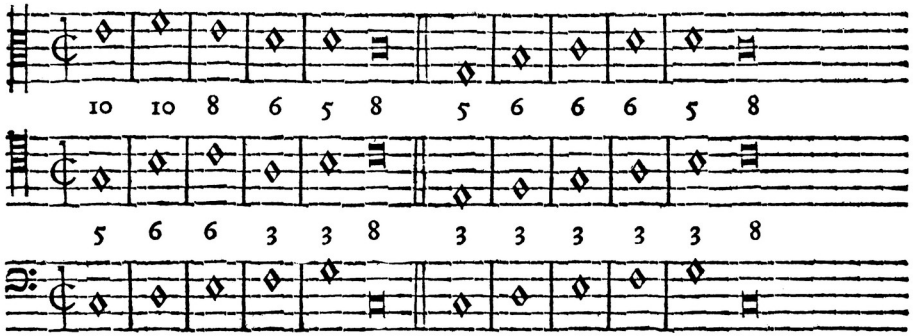
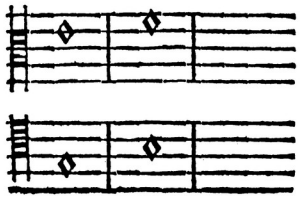


Cujus varietatis magnam rationem habendam eſſe, pro nunc & ſemper tibi dictum volo.

Joſeph. Ne moleſtè tuleris, ſi quæram, cur in tuo paulò ante allato Paradigmate à partibus tam longè ſejunctis principium ſumſeris? Videtur enim mihi, alio quoque modo fieri potuiſſe.

Aloyſ. Minimum moleſtiæ, ſed plurimùm oblectamenti mihi affert ſpe plena curioſitas tua. Nonne vidiſti, Baſſum illius exempli continuò gradatim aſcendendo moveri? Quam ob rem, ut partes, motu contrario ſibi obviam eundi campum habeant, partibus à parte fundamentali remotis initium fieri neceſſe eſt.

Sed quibus exemplis Paradigma illud alio modo fieri potuiſſe demonſtras?

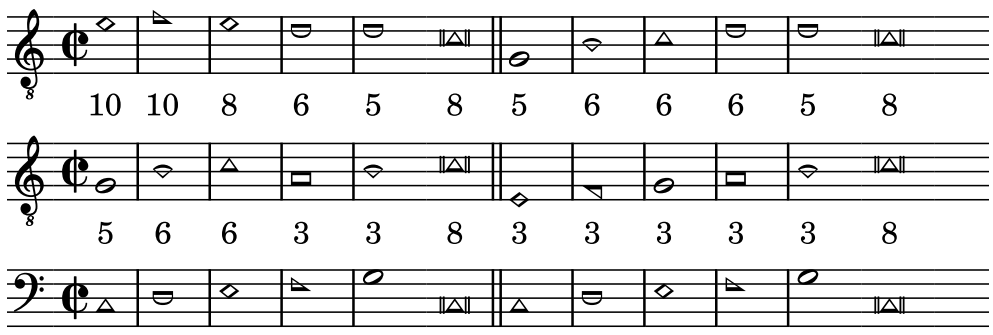
Joſeph. Duo exempla, non demonſtrandi gratiâ, ſed, ut docear, adducam.

Aloyſ. Non admodum improbo exempla hæc tua. Verùm vides, in primo exemplo, de primo tactu ad ſecundum, omnes tres partes aſcendendo, partim per gradum, partim per ſaltum moveri, quod abſque quadam inconvenientia vix contingere poteſt. Quod hîc accidit in parte Tenoris reſpectu Alti de primo ad ſecundum tactum, qui ſic ſe habent:

Of which variety you must take great account, now and always, as I tell you.

Joseph. Do not be displeased if I ask why, in your example just given, you began the parts at such distant intervals? It seems to me it could have been done in another way.

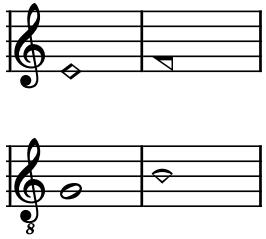
Aloys. Your curiosity, full of hope, brings me not the least annoyance but the greatest delight. Did you not see that the Bass of that example moves continuously upward stepwise? Therefore, so that the parts may have room to meet each other by contrary motion, it is necessary to begin the parts far from the fundamental part.

But by what examples do you show that the pattern could have been made in another way?

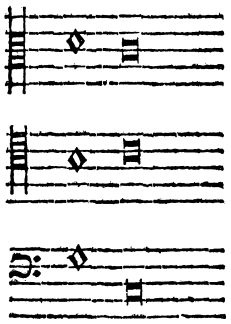
Joseph. I will bring two examples, not to prove anything, but so that I may be taught.



Aloys. I do not greatly disapprove of your examples. But you see that in the first example, from the first beat to the second, all three parts move upward, partly by step and partly by leap, which can scarcely occur without some inconvenience. This happens here in the Tenor with respect to the Alto from the first to the second beat, which stand thus:



Ex quo perſpicuum eſt, ſublatâ parte fundamentali progreſſum illum vitioſum eſſe, quia non ſolùm de imperfecta ad perfectam fit ambitus, ſic loquendo, verùm, quod pejus eſt, nec illa Quinta Conſonans eſt, ſed Quinta falſa. Regularum enim ratio non ſolùm reſpectu Baſſi, verùm etiam aliarum partium inter ſe, ubi fieri poteſt, quamvìs in plurium partium compoſitione non ità rigorosè obſervanda eſt: ità ut etiam in Tricinio partium reſpectu Baſſi ponderoſas ob rationes nonnunquam ab eo rigore recedere liceat, quod videndum eſt in prioris exempli penultimo tactu, ubi de Quinta in Octavam finalem, nempe de perfecta ad perfectam motu recto fit progreſſus, quia aliter fieri non poteſt. E. G.



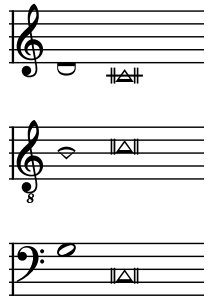
Joſeph. Non poſſet illud inconveniens poſitâ Decimâ locò Octavæ evitari?

Aloyſ. Poſſet quidem, ſed videtur per illam imperfectam non ſatìs ſtabiliri ratio & perfectionis, & quietis in Nota finali requiſita. Alia ratio eſt Quadricinii, ubi accedente Quintâ, non obſtante Tertiâ illis rationibus ſatisfieri videtur.

Joſeph. Sed quid dignum reprehenſione reperis in ſecundo meo exemplo?

Aloyſ. Non quidem aliud, quàm, quòd Sextæ in aſcendendo in Theſi nonnihil aſperitatis habeant. Sextæ in Arhſi venientes (quibus in hac Specie locus non eſt) utpote minùs roboris habentes tolerabiliores ſunt, ùt alibi latiùs dicetur.

Jam ad exercitium pergamus. Ut autem intuitu cujusdam formulæ Speciei hujus confectio facilior tibi contingat, componam ego tibi primum exemplum, quod pro trium partium ratione etiam triplex erit; primò ut Cantus firmus in parte ſuperiori: ſecundo in parte media: tertiò in parte fundamen-

From this it is clear that, with the fundamental part removed, that progression is faulty, because not only is the motion from an imperfect to a perfect consonance—so to speak—but what is worse, that fifth is not a consonant fifth but a false fifth. For the rules apply not only with respect to the Bass but also, where possible, among the other parts themselves, although in multi-voice composition they are not to be observed so rigidly. Thus even in a tricinio, for weighty reasons, one may sometimes depart from that strictness with respect to the Bass, as is seen in the penultimate beat of the earlier example, where the progression from the fifth to the final octave—that is, from perfect to perfect—occurs by similar motion, because it cannot be done otherwise. For example:

Joseph. Could not that inconvenience be avoided by placing a tenth instead of the octave?

Aloys. It could indeed; but by that imperfect consonance the stability of the perfection and repose required in the final note would not be sufficiently established. The case is different in a quadricinium, where, with the addition of the fifth, despite the third, those requirements seem to be satisfied.

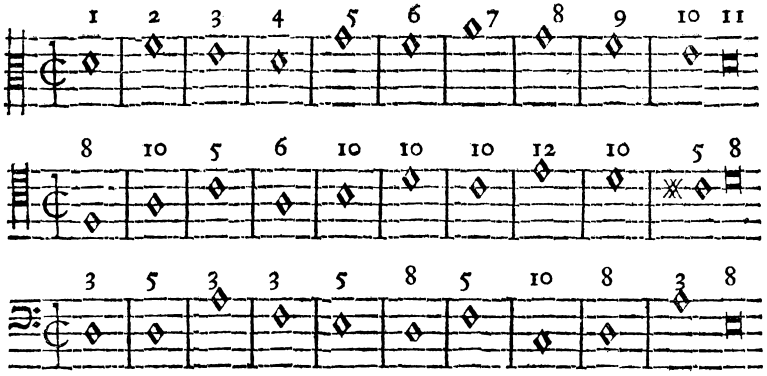
Joseph. But what do you find worthy of blame in my second example?

Aloys. Nothing except that sixths ascending on the Thesis have a certain harshness. Sixths occurring on the Arsis (which have no place in this species), being of less strength, are more tolerable, as will be said more fully elsewhere.

Now let us proceed to the exercise. So that the construction of this species may be easier for you by looking at a certain pattern, I will compose the first example for you, which, according to the nature of three parts, will also be triple: first with the cantus firmus in the upper part; second in the middle part; third in the fundamental part.

damentali ſtatuatur, quod & tibi deinceps obſervandum erit, aſſumendo ſuperiùs ordine præſcriptos Cantus firmos.

Cantus firmus.

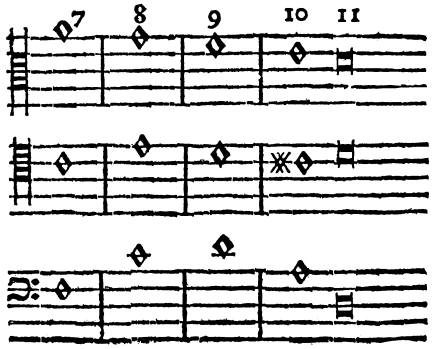


In quo exemplo videri poteſt, in omni tactu, ubi nulla ratio alia vetuit, Triadem harmonicam adhibitam fuiſſe, habitâ aliâs progreſſuum, motuumque ratione ſæpiùs jam repetitâ.

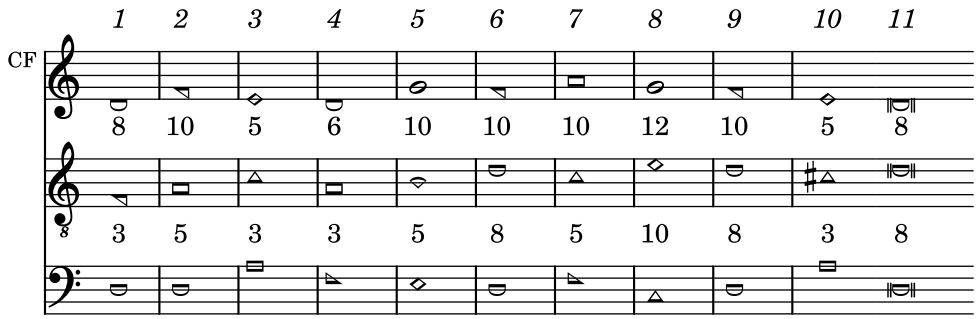
Joſeph. Videtur tamen de ſeptimo tactu in octavum in parte Alti reſpectu Baſſi te non obſervâſſe regulam, quæ dicit: de imperfecta ad perfectam per motum contrarium. Tu autem per rectum progreſſus es.

Aloyſ. Rectè obſervas quidem; ſed meminiſſe oportet, paulò ante me dixiſſe, quòd ad evitandum majus inconveniens in Tricinio, ubi aliter fieri non poſſet, nonnunquam de eo rigore recedere liceat.

Joſeph. Memini quidem; ſed videtur mihi, etiam obſervatâ illâ regulâ rectè fieri potuiſſe, ſi de F. in Baſſo in C. acutum, per motum contrarium progreſſus inſtitutus fuiſſet, hac ratione:



This you must also observe henceforth, using the cantus firmi prescribed above in order.

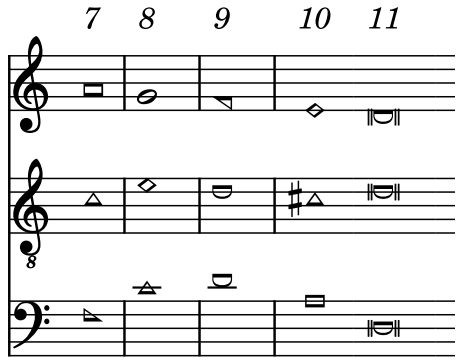


In this example it may be seen that in every beat, where no other reason forbade it, the harmonic triad has been used, with due regard to the logic of the progressions and motions, as has often been repeated.

Joseph. Yet it seems that from the seventh beat to the eighth, in the Alto with respect to the Bass, you did not observe the rule that says: from an imperfect to a perfect consonance by contrary motion. But you proceeded by similar motion.

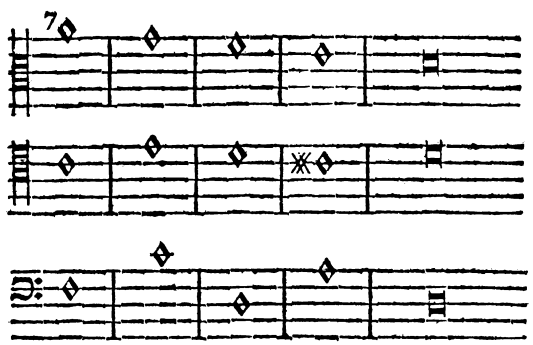
Aloys. You observe correctly; but you must remember that I said a little earlier that, to avoid a greater inconvenience in a tricinio, where it cannot be done otherwise, one may sometimes depart from that strictness.

Joseph. I do remember; but it seems to me that even observing that rule it could have been done correctly, if from F in the Bass to high C the progression had been made by contrary motion, in this manner:

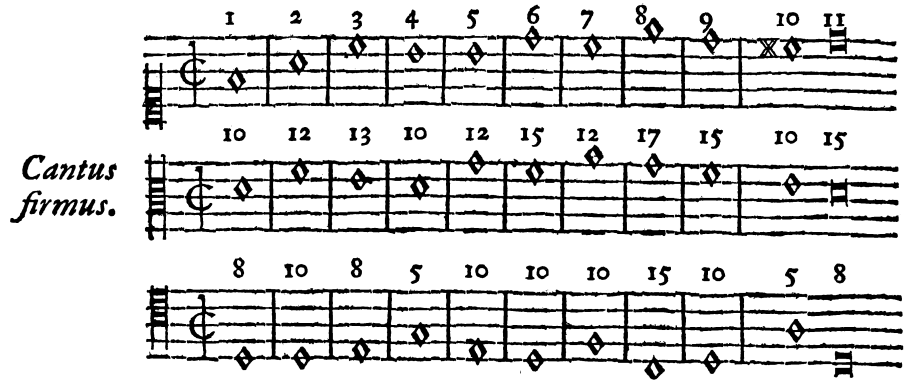


Aloyſ. Eſſet quidem proviſum illi inconvenienti per motum contrarium ; verùm non vides tali ratione duas ſimiles transgreſſiones immediatè ſecuturas eſſe ? de nona Nota in decimam, & de decima in ultimam ? Deinde in nono tactu pars Tenoris cum Baſſo, ùt vides, faceret uniſonum, qui minùs confert harmoniæ, quàm Octava, præterquàm quòd in hoc genere Compoſitionis abſque gravi neceſſitate non liceat tranſcendere limites quinque linearum.

Joſeph. In totum convictus ſum rationibus adductis, ſed fortaſſis ſic fieri poſſet ?



Aloyſ. Minùs. Nonne meminiſti, ſaltum Sextæ majoris prohibitum eſſe? Quid judicandum de ſaltu Septimæ? quia naturali cantûs facilitati conſulendum eſt. Jam Cantu firmo in media parte poſito exemplum faciam.

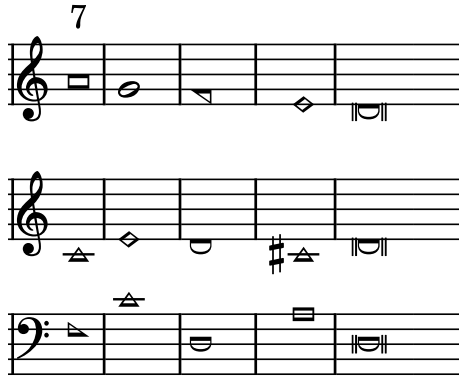


Joſeph. Quare in nono tactu Octavam adhibuiſti ? mihi enim videtur, abſque impedimento, Quintæ, ſive Triadi harmonicæ locum ibi fuiſſe.

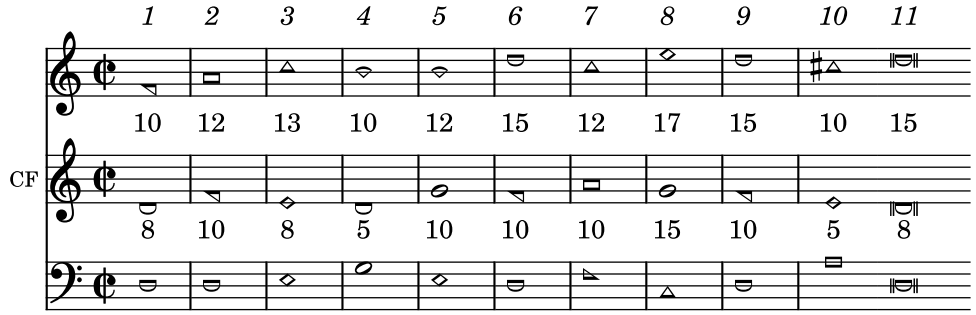
Aloyſ.

Aloys. That inconvenience would indeed have been avoided by contrary motion; but do you not see that by that method two similar violations would immediately follow—from the ninth note to the tenth, and from the tenth to the last? Then in the ninth beat the Tenor with the Bass, as you see, would form a unison, which contributes less to harmony than the octave; besides, in this kind of composition it is not permitted, without grave necessity, to exceed the limits of the five-line staff.

Joseph. I am entirely convinced by the reasons you have given; but perhaps it could be done thus?

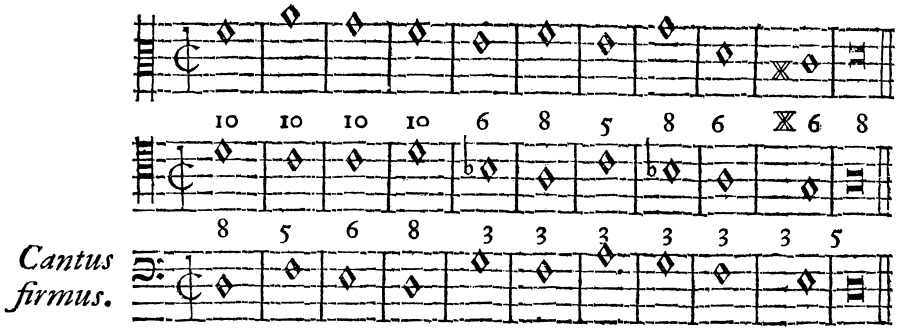


Aloys. Less so. Do you not remember that the leap of a major sixth is forbidden? What then shall we say of the leap of a seventh? For the natural ease of singing must be considered. Now I will make an example with the cantus firmus placed in the middle part.



Joseph. Why did you use the octave in the ninth beat? It seems to me that the fifth, or the harmonic triad, could have been used there without impediment.

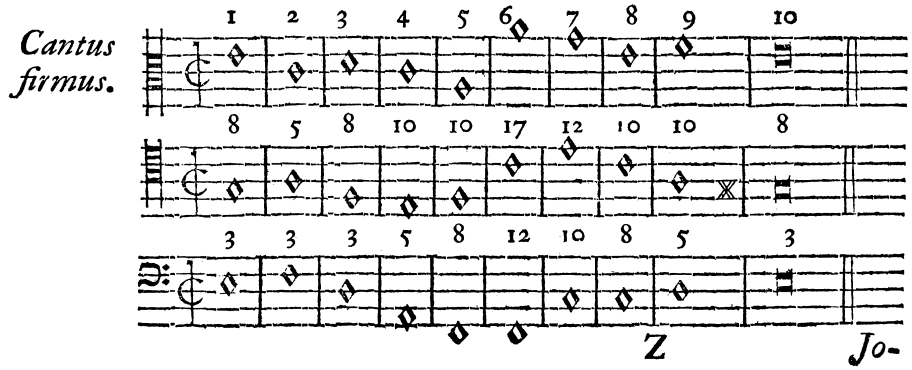
Aloyſ. Ità eſt, potuiſſet. Verùm paulò accuratiùs inſpiciens, comperies Octavam meliùs inſervire Cantûs rationi, cui, ùt jam ſæpè dixi, & ſæpiùs dicendum erit, multum tribuendum eſt. Reſtat adhuc exemplum perficiendum, in quo Cantus firmus in parte fundamentali exiſtat.



Joſeph. Nihil dubii hoc in Paradigmate reperio, præter tactum finalem; videtur enim, poſitâ locò Quintæ Tertiâ plùs harmoniæ Hypotheſi illi acceſſurum fuiſſe.

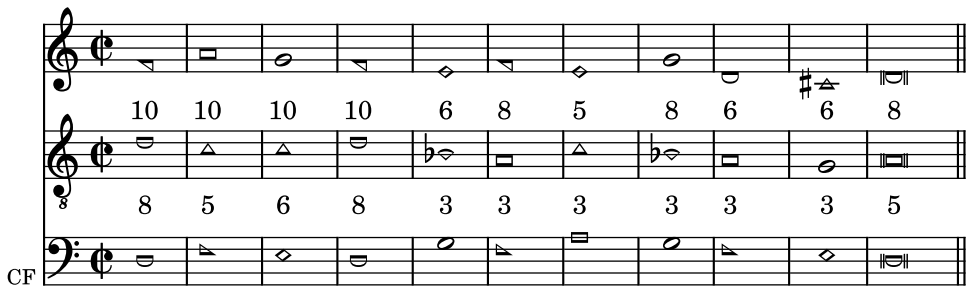
Aloyſ. Rectè quidem judicas. Sed qualem Tertiam ponendam fuiſſe putas, majorem, vel minorem? Si minorem, nonne ſentis, ad concludendum finem non valere? Si majorem, æquè nihil ad rem: cùm enim tonus iſte conſiſtentiam ſuam habeat cum Tertia minore, nempe F. abſque Dieſi, quâ modulatione per totum Cantûs firmi curſum aures aſvetæ, neſcio quid offenſionis ex auditu Dieſis in fine percipiant. Ideò conſultiùs erit Tertiam omnino omittere.

Nunc ad reliqua quinque ſuperſtitum tonorum exempla componenda progredere, eâ ratione, ut Cantus firmus in omnibus tribus partibus, ùt jam dixi, & feci, ſtatuatur; obſervatis hucuſque dictis.



Z Jo-

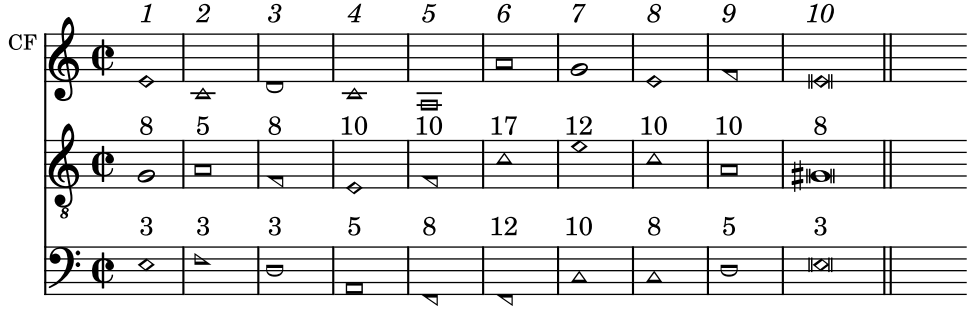
Aloys. So it is—it could have been. But looking a little more carefully, you will find that the octave serves the melodic line better, to which, as I have often said and must often say, much must be granted. One example still remains to be completed, in which the cantus firmus is in the fundamental part.



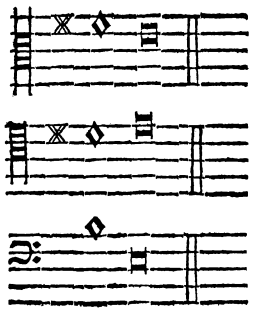
Joseph. I find nothing doubtful in this pattern except the final beat; for it seems that, if a third had been placed instead of the fifth, more harmony would have been added to that hypothesis.

Aloys. You judge rightly. But what kind of third do you think should have been placed—major or minor? If minor, would it not be too soft to conclude the ending? If major, it is equally unsuitable; for since this mode has its consistency with the minor third—namely F without a sharp—the ears, accustomed to that modulation throughout the course of the cantus firmus, would perceive something offensive in hearing the sharp at the end. Therefore it is more prudent to omit the third altogether.

Now proceed to compose the examples for the remaining five modes, in such a way that the cantus firmus is placed in all three parts, as I have said and done, observing all that has been stated thus far.

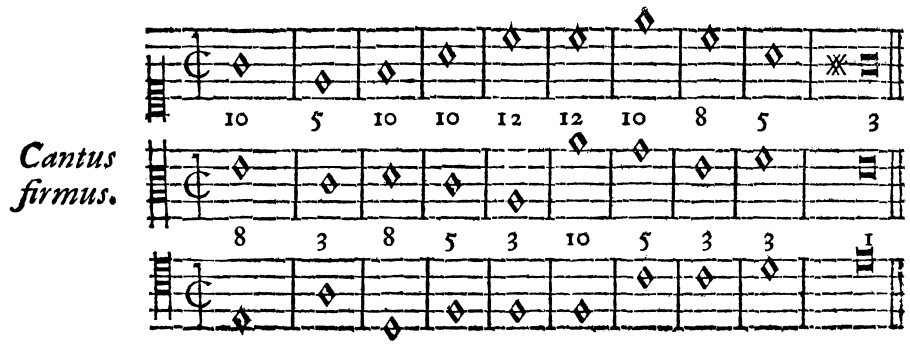


Joſeph. Non parùm laboravi in conficienda clauſula finali. Cùm enim in Hypotheſi penultimi tactûs clauſulæ formali; E. G.

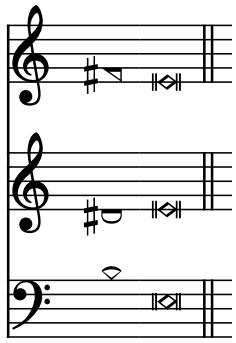


locus non ſit, neque tonus hic naturali ratione Quintam perfectam eo loco habeat, neque Dieſin admittat, cujus beneficio clauſula formalis formari poſſit, ideo viſum mihi fuit, non alio modo, quàm quo feci, finem poſſe concludi. Nonnihil tamen dubius hæreo propter Tertiam majorem in Nota finali poſitam. Etenim recordor, te ſuprà dixiſſe, in ſimilibus tonis Tertiam omnino omittendam, ejuſque loco Quintam eſſe ponendam.

Aloyſ. Valde prudenti uſus es judicio de clauſula finali, nempe propter peculiarem Semitoni ſitum clauſulæ formali ibi locum non eſſe, ideoque peculiarem etiam clauſulam formandam fuiſſe; neque ſcrupulum tibi moveat Tertia major in Nota finali poſita, quod enim dixi de omittenda Tertia, intelligendum eſt, ubi fieri poteſt. Rectè ergo judicâſti, in illius exempli Nota finali abſque immediata duarum Quintarum conſecutione Quintam poni non potuiſſe, ergo Tertiam majorem ejus loco adhibendam fuiſſe, quia minor propter majoris imperfectionis rationem minùs apta eſt ad concludendum finem. Perge.

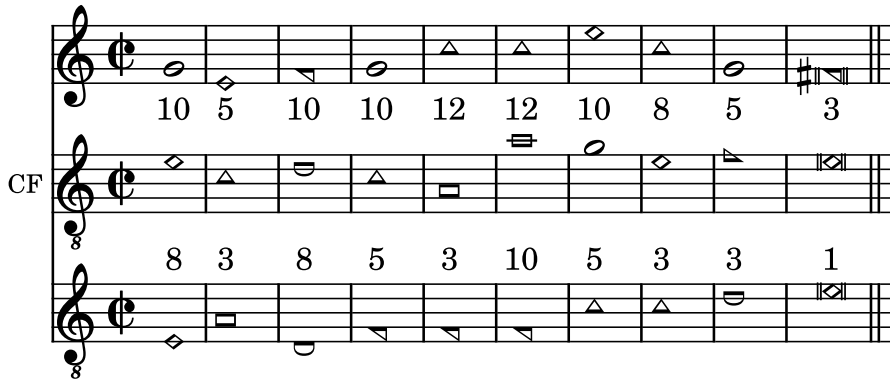


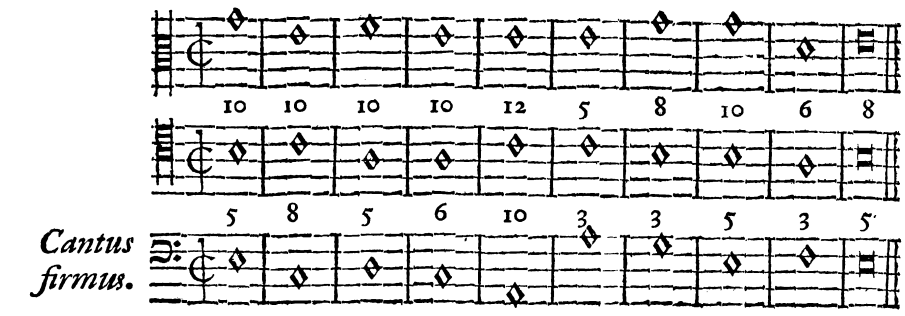
Joseph. I labored not a little in forming the final cadence. For since, in the hypothesis of the penultimate beat, there is no place for a formal cadence,

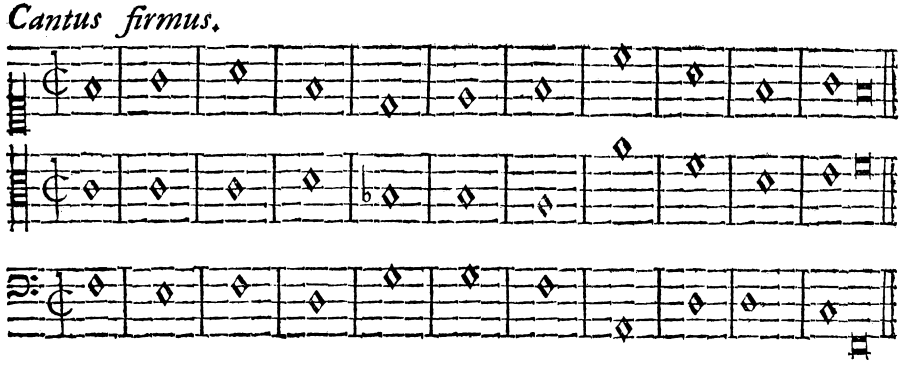


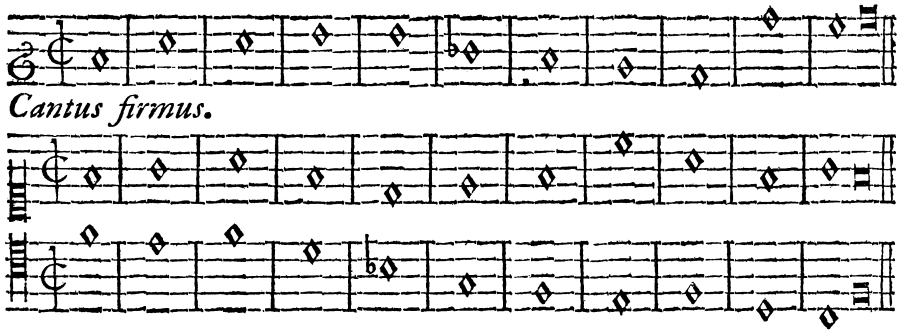
and since this mode does not naturally have a perfect fifth in that position, nor does it admit a sharp by whose help a formal cadence could be made, it therefore seemed to me that the ending could be concluded in no other way than the one I used. Yet I remain somewhat doubtful because of the major third placed on the final note. For I recall that you said above that in similar modes the third must be entirely omitted, and the fifth put in its place.

Aloys. You have used very prudent judgment regarding the final cadence—namely, that because of the peculiar placement of the semitone, there is no place for a formal cadence there, and therefore a special cadence had to be formed. Nor should the major third placed on the final note trouble you; for what I said about omitting the third is to be understood only where it can be done. You judged rightly that in the final note of that example the fifth could not be placed without the immediate succession of two fifths; therefore the major third had to be used in its place, because the minor, on account of its greater imperfection, is less suitable for concluding the end. Continue.




10 10 10 10 12 5 8 10 6 8
5 8 5 6 10 3 3 5 3 5
Cantus firmus.


Cantus firmus.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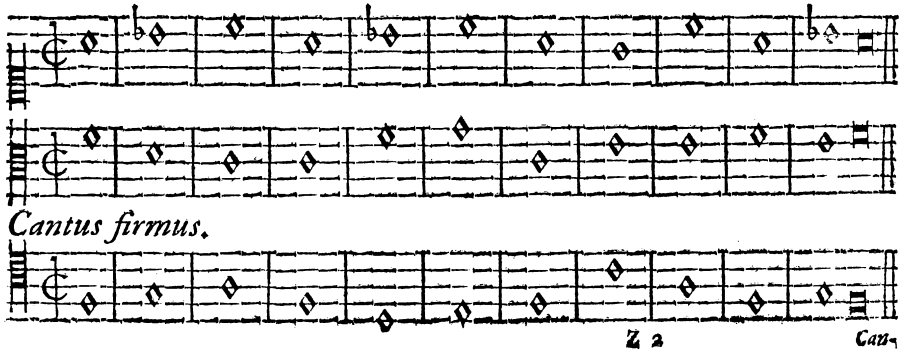
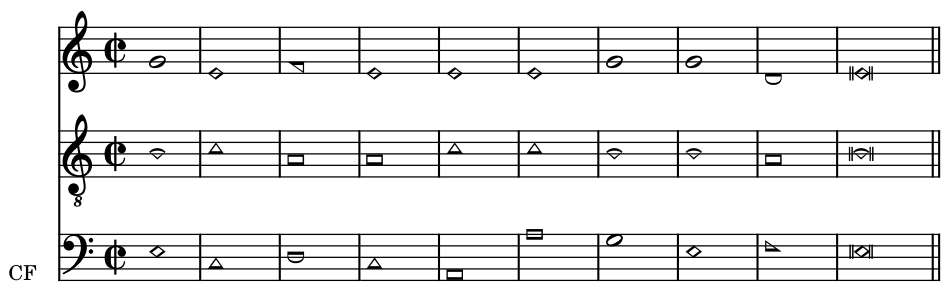

Cantus firmus.

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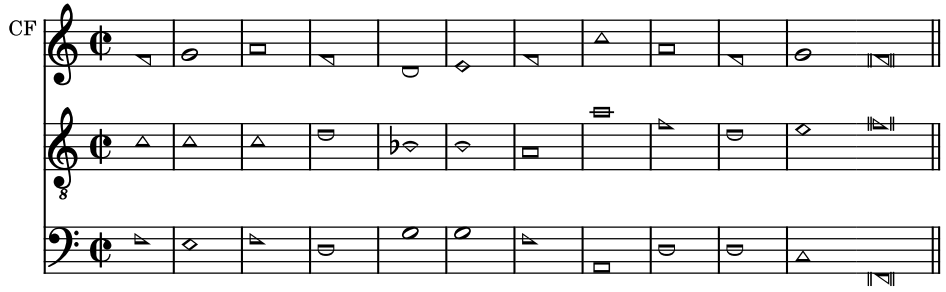
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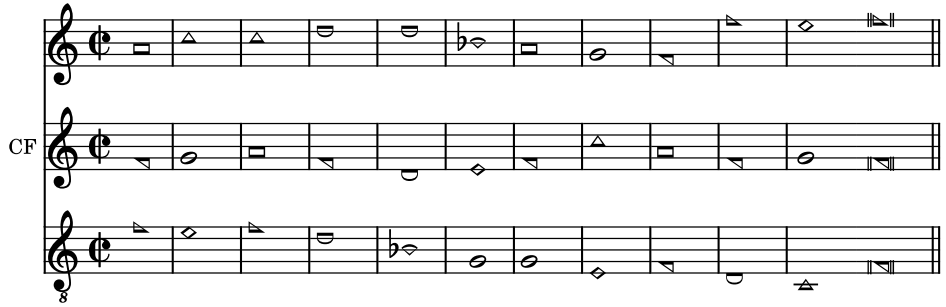
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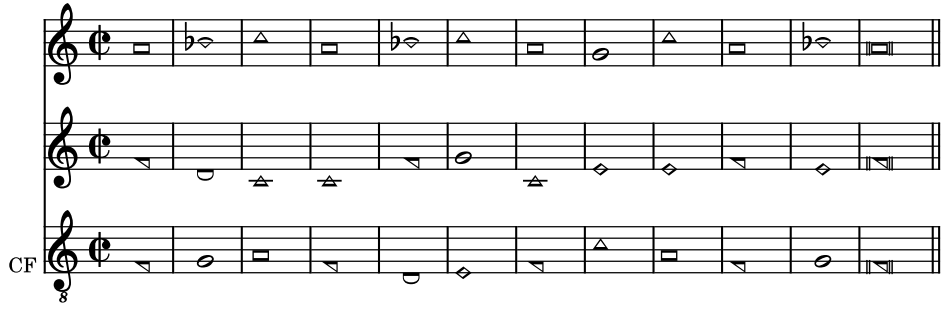
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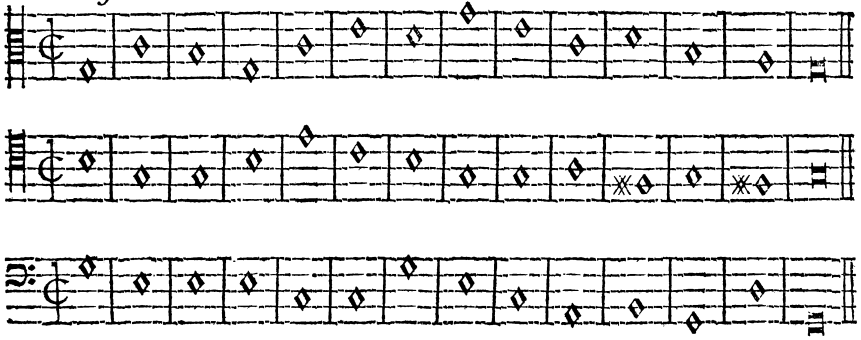
Cantus firmus.



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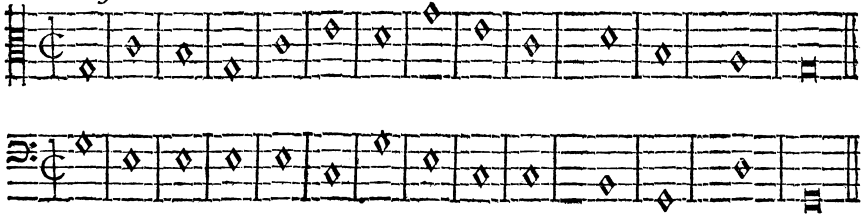

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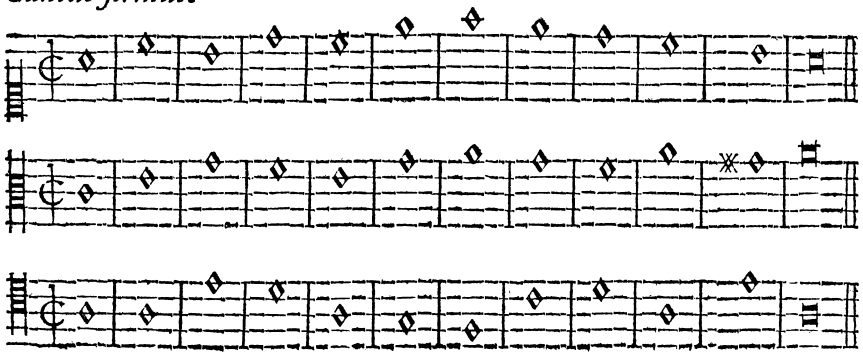
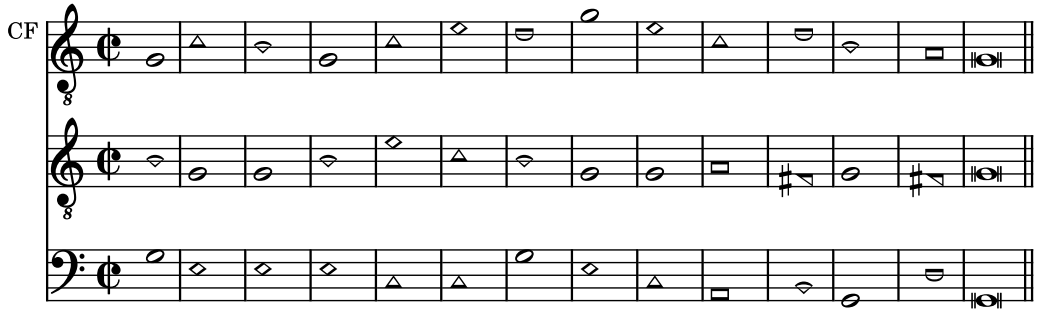
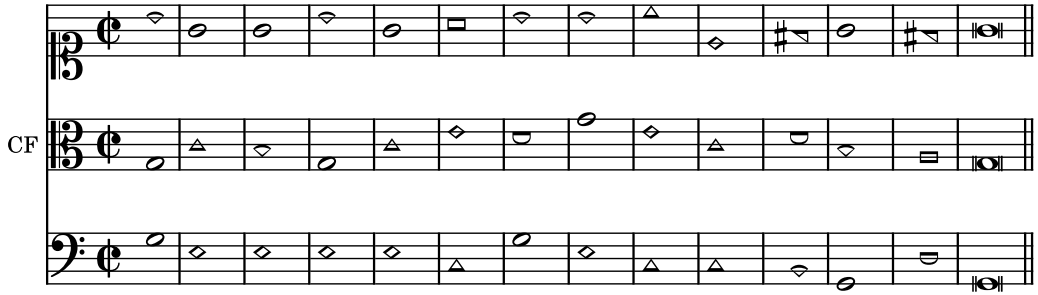
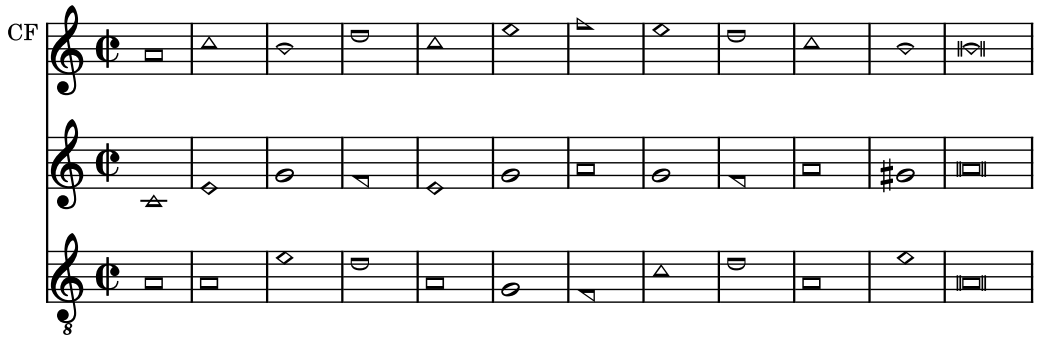
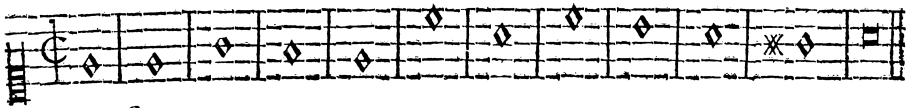
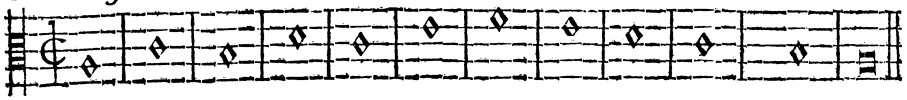
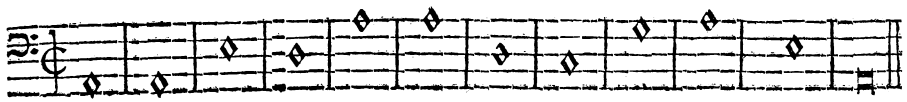
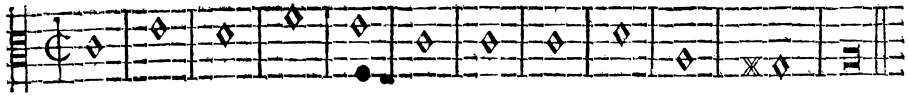


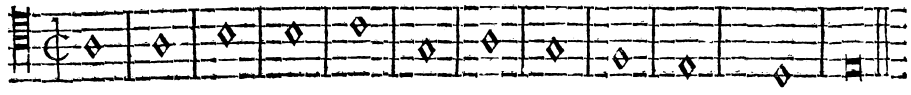




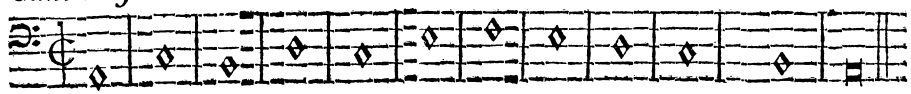




Cantus firmus.





Cantus firmus.



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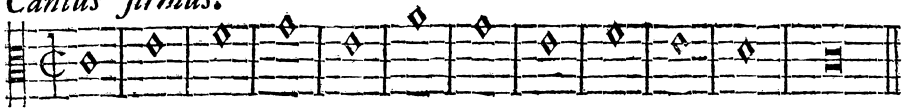
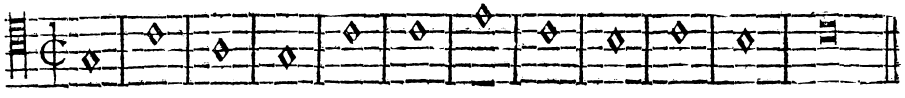
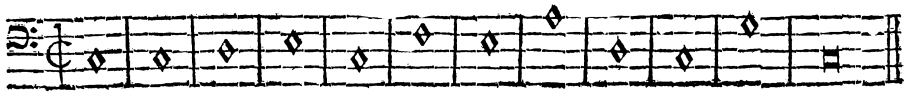
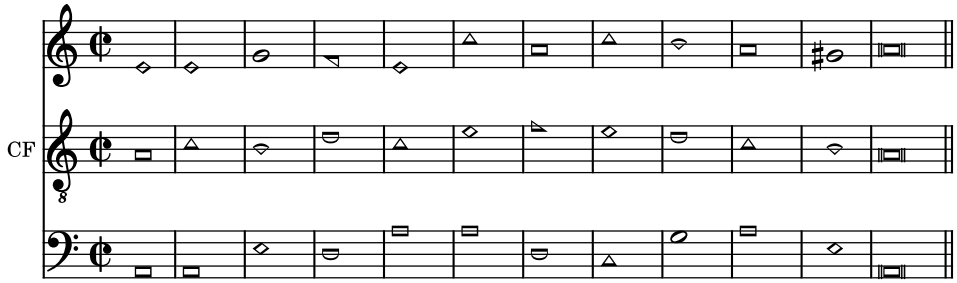
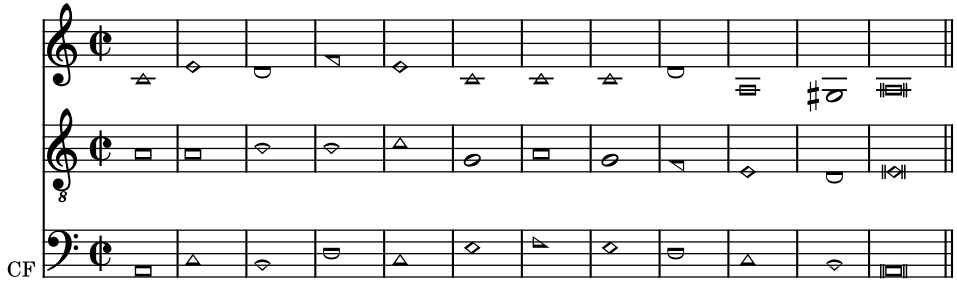
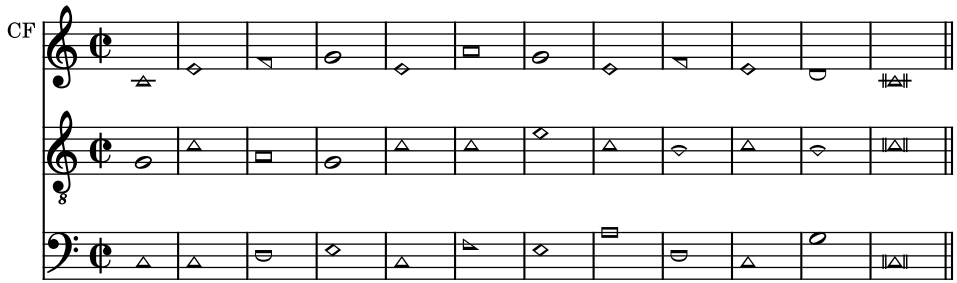


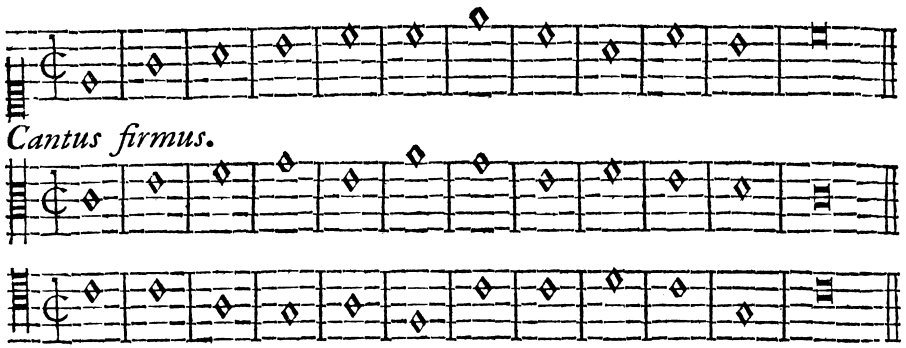












Joſeph. Ecce reliqua tonorum exempla, meo quidem judicio eatenus bene facta, quatenus Cantûs firmi reſtrictio non impedit. Verùm nisì Cantûs firmi neceſſitas componendi libertatem reſtrinxiſſet, multis in locis compoſitionem Triade harmonicâ locupletiorem effici potuiſſe, exiſtimo.

Aloyſ. Rectè ſentis, atque ità ſuo tempore in Compoſitione libera, nullâ Cantûs firmi obligatione reſtrictus oneraberis: ſed incredibile eſt, quantum utilitatis hæc exercitatio, nempe ſuper Cantum firmum ædificando hujus Diſciplinæ ſtudioſo afferat, quâ ſolâ, & uſu, & ratione inſtructus, ad veram artis hujus cognitionem perduci poteſt, quam exercitationem iterum, atque iterum, (ſæpiùs enim inculcanda eſt,) ſummoperè tibi commendo.

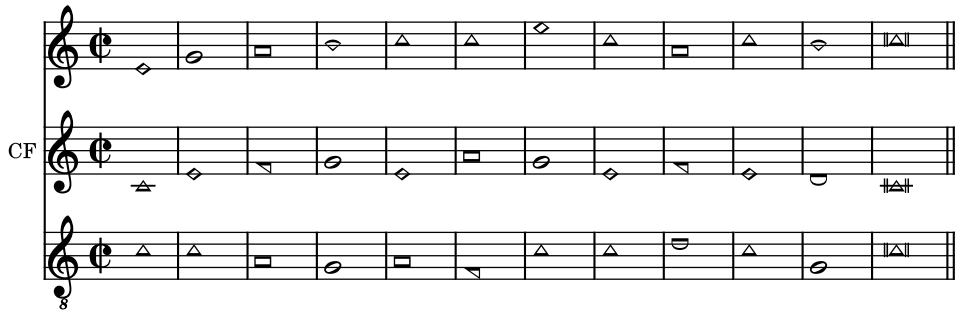
Abſolutis primæ Speciei exemplis, ad ſecundam Speciem modò progrediendum eſt.

EXERCITII II.

LECTIO SECUNDA,

De Poſitione Minimæ contra Semibrevem in Tricinio.

Hîc loci in memoriam revocanda, obſervandaque ſunt, quæ de hac Specie in Bicinio præcepimus, quæque ratione Triadis harmonicæ de Semibrevi contra Semibrevem in Tricinio diximus; cum hac tamen relaxatione, quòd in hac Tricinii Specie in gratiam Triadis harmonicæ Minima per ſaltum Tertiæ nonnunquam duas Quintas ſalvare queat; E. G.

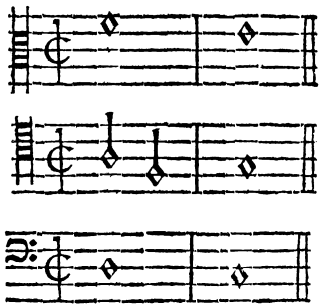


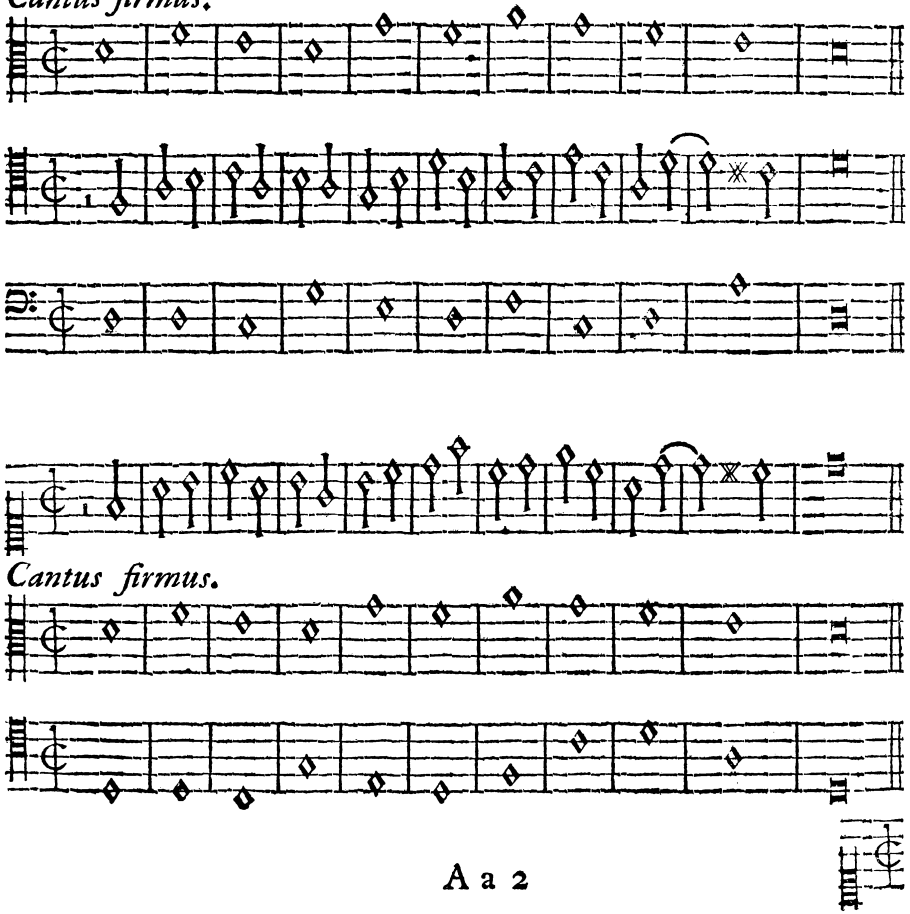
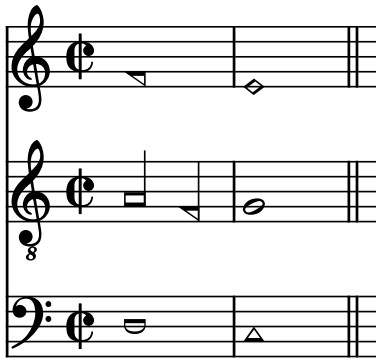
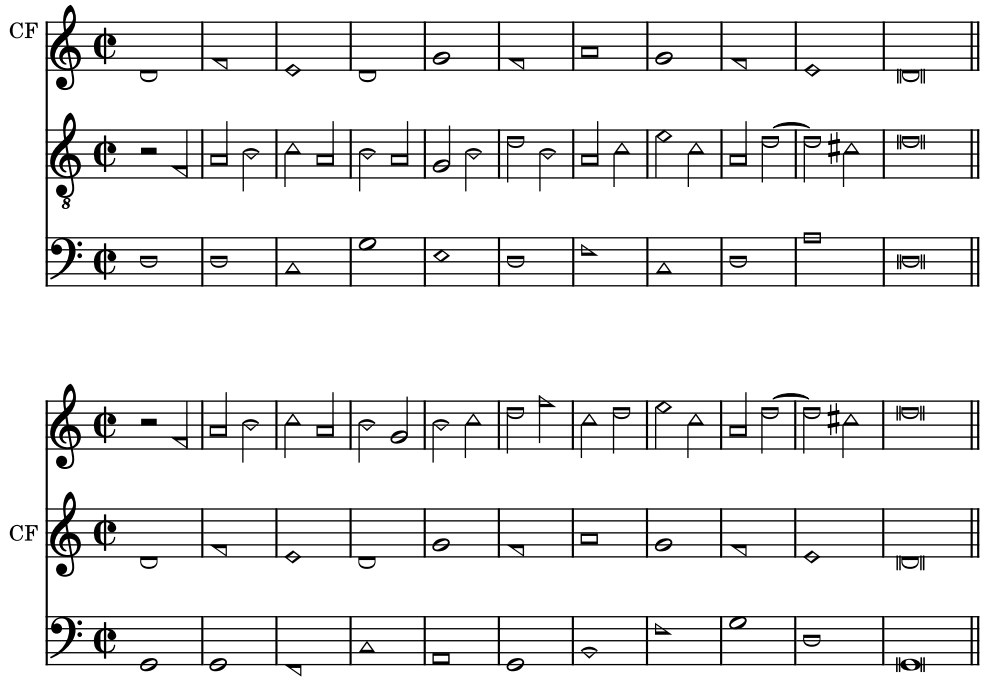
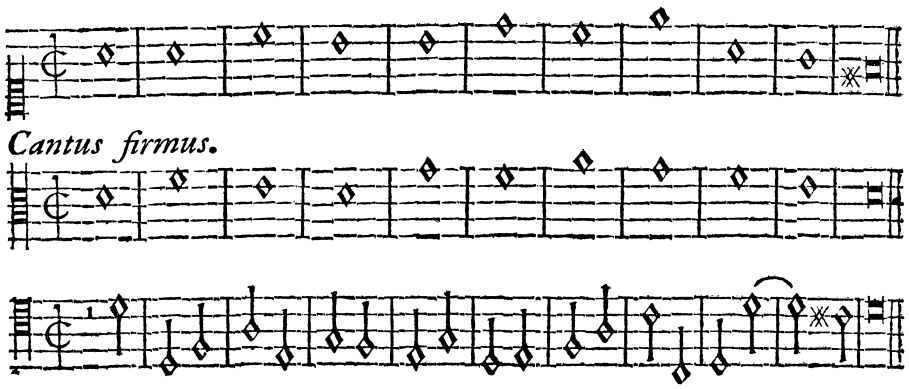
Joseph. Here are the remaining examples of the modes, in my judgment well made insofar as the restriction of the cantus firmus (fixed melody) does not hinder them. But unless the necessity of the cantus firmus had restricted the freedom of composition, I think the composition could in many places have been enriched with the harmonic triad.

Aloys. You judge rightly, and so in due time, in free composition, when you are no longer restricted by the obligation of a cantus firmus, you will be unburdened. But it is incredible how much usefulness this exercise—namely, building upon a cantus firmus—brings to the student of this discipline; for by this alone, instructed both by practice and by reason, one can be led to the true understanding of this art. This exercise, again and again (for it must often be repeated), I strongly commend to you.

Having completed the examples of the first species, we must now proceed to the second species.

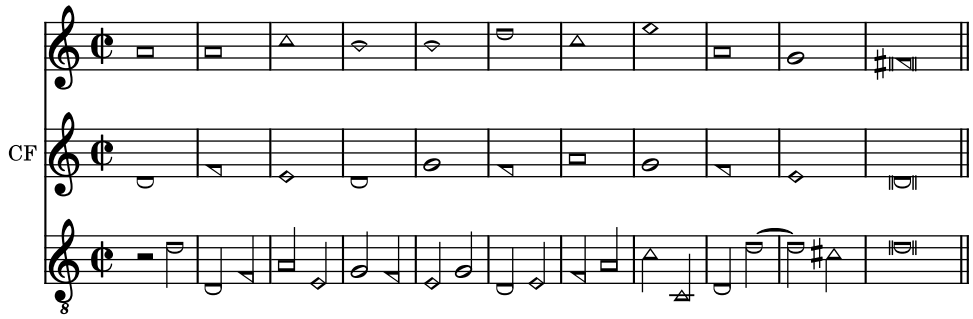
EXERCISE II.
LESSON TWO,
On the Placement of the half note against the whole note in the Tricinio.

Here you must recall and observe what we taught about this species in the bicinium, and what we said concerning the harmonic triad in the case of whole note against whole note in the tricinio; with this relaxation, however: that in this species of the tricinio, for the sake of the harmonic triad, a half note by a leap of a third may sometimes save two fifths. For example:

Quæ Hypotheſis in Bicinio prohibita, in Tricinio in gratiam Triadis harmonicæ, ùt dixi, toleratur. Pro formula tria tibi exhibeo exempla, ut reliquorum confectio facilior tibi eveniat.

Cantus firmus.

This hypothesis, forbidden in the bicinium, is tolerated in the tricinio for the sake of the harmonic triad, as I said. As a model I give you three examples, so that the construction of the remaining ones may be easier for you.

Joſeph. Memini te ſuperiùs dixiſſe, ubi de hac Specie in Bicinio præcepta tradidiſti, nunquam duas Minimas in una eademque linea ponendas; proinde Ligaturæ hîc loci locum non eſſe: verùm in omni trium horum Paradigmatum clauſula finali, non ſolùm Ligaturam, ſed etiam ultimi exempli Notam finalem Tertiâ majore, quàm non minùs prohibitam fuiſſe recordor, terminatam conſpicio.

Aloyſius. Probè quidem meminiſti. Cùm verò nulla ferè ſit ſinè exceptione Regula, intelligendum puto, ubi ità faciendi facultas reperitur, quod in Bicinio ſemper, non item in Tricinio uſuvenire poteſt, ùt in præcedentibus exemplis haud obſcurè videre eſt, ubi in Ligaturæ Hypotheſi poſitis duabus Minimis ſolutis, aut Uniſonum vitioſum, aut octavam harmoniâ vacuam exiſtere neceſſè eſſet. Tertiam majorem in tertii exempli Nota finali poſitam neceſſitas excuſat, eò quòd ibidem in parte Cantûs Quintæ locus non ſit; quia poſitâ Quintâ, duæ Quintæ immediatè ſe ſe ſequentes exorituræ eſſent.

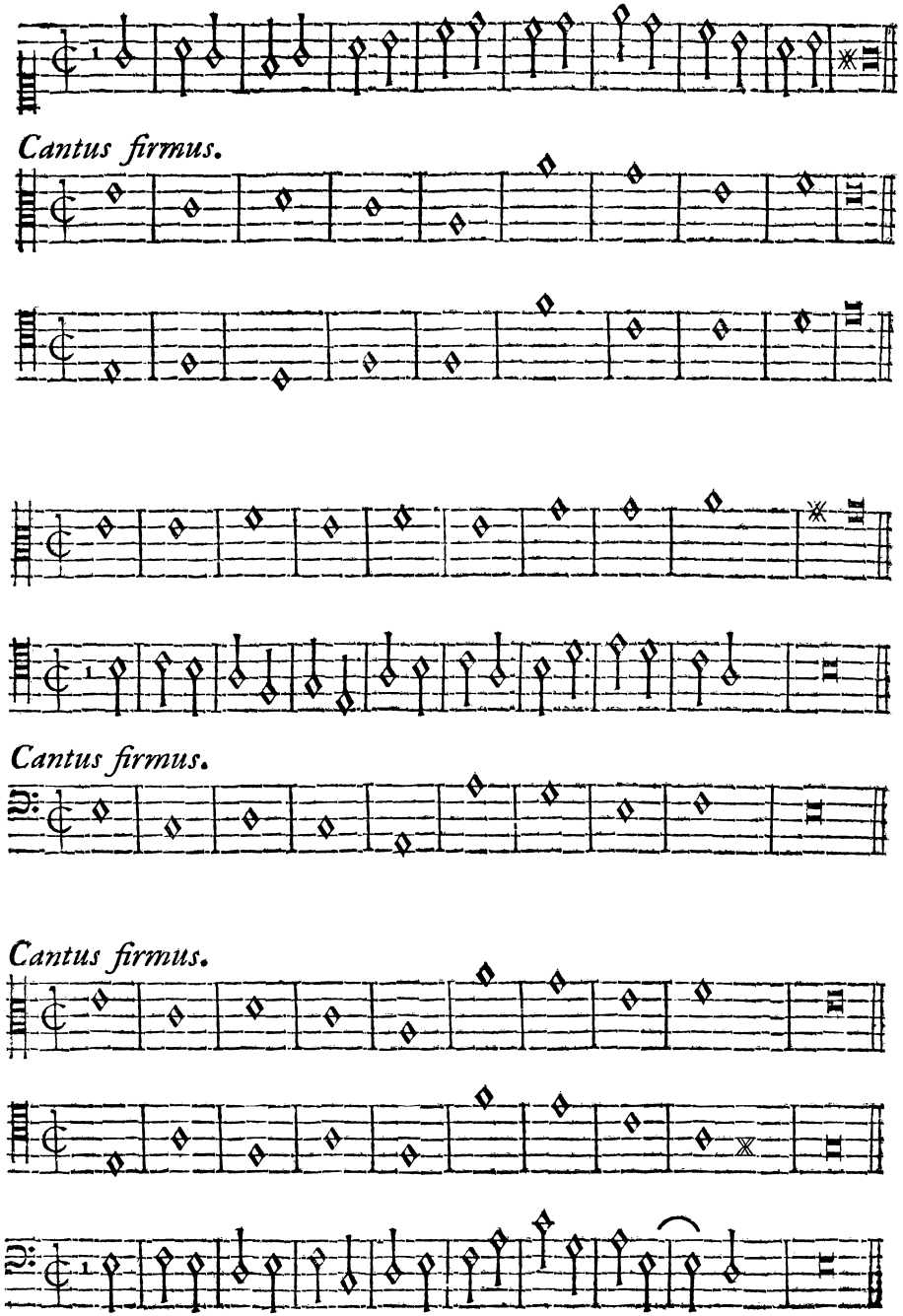
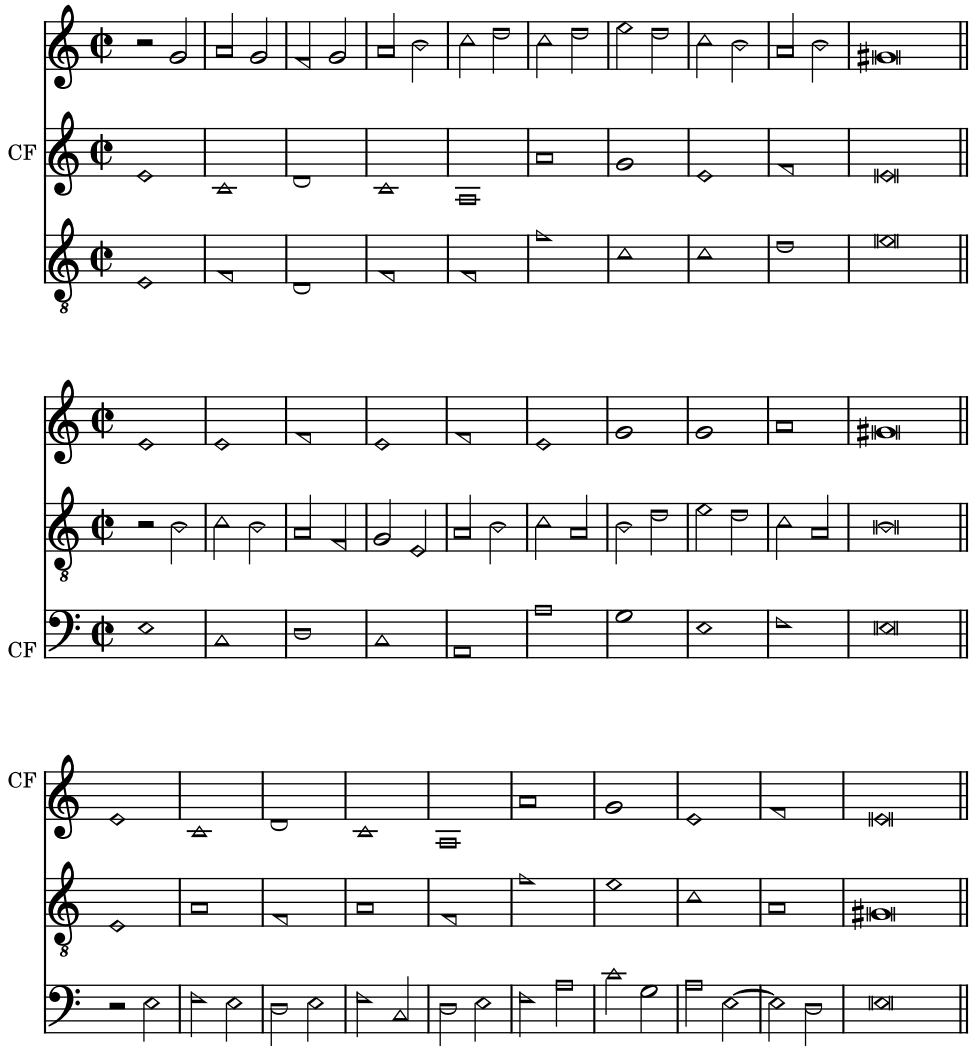
Ex iſtis exemplis conſtat in una partium per totum Cantûs firmi curſum Minimas, in reliquis autem duabus meras Semibreves conſtituendas eſſe, ità ut Minimæ cum duabus Semibrevibus ritè ſemper concordent, habitâ aliàs motuum, regularumque ratione, quoad fieri poteſt.

Age nunc ſis reliquorum tenorum confice exempla, obſervatâ triplici mutatione à me monſtratâ.

Joseph. I remember that you said earlier, when giving the rules for this species in the bicinium, that two half notes must never be placed on the same line, and therefore that there is no place here for a ligature. Yet in the final cadence of all three of these patterns I see not only a ligature, but also in the last example the final note ending with a major third, which I also recall being forbidden.

Aloys. You remember well. But since scarcely any rule is without exception, I think it must be understood that where the possibility of doing so is found—which can always be found in the bicinium, but not in the tricinio—it may be done, as is not obscure in the preceding examples. For in the hypothesis of the ligature, if the two half notes were left free, either a faulty unison or an octave empty of harmony would necessarily result. The major third placed on the final note of the third example is excused by necessity, because in that place in the melody there is no room for the fifth; for if the fifth were placed, two fifths would immediately follow one another.

From these examples it is clear that in one of the parts throughout the entire course of the cantus firmus half notes must be used, and in the other two parts pure whole notes, so that the half note may always agree properly with the two whole notes, with due regard to the rules of motion as far as possible.

Now go on and compose examples for the remaining modes, observing the threefold alternation I have shown you.


Cantus firmus.
Cantus firmus.
Cantus firmus.

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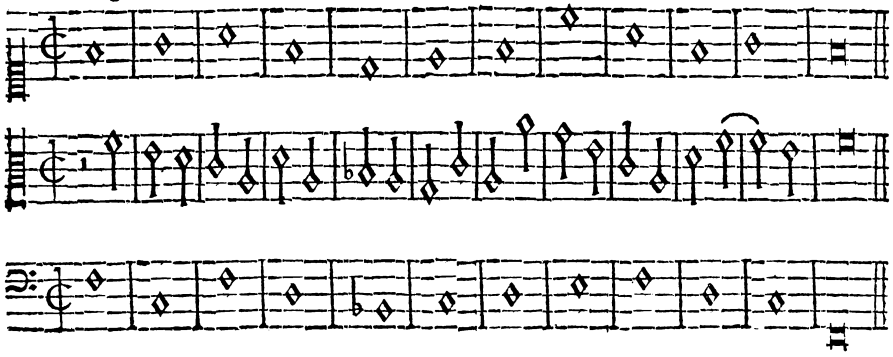
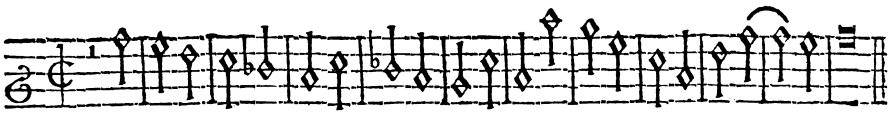
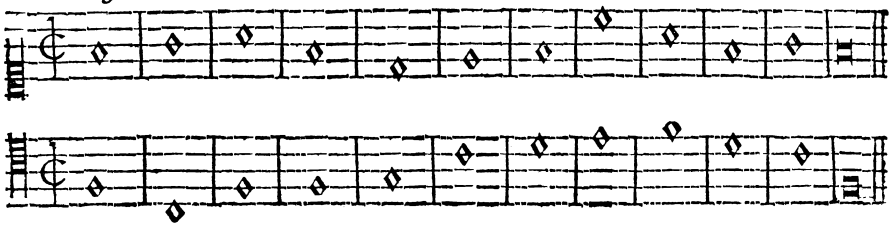
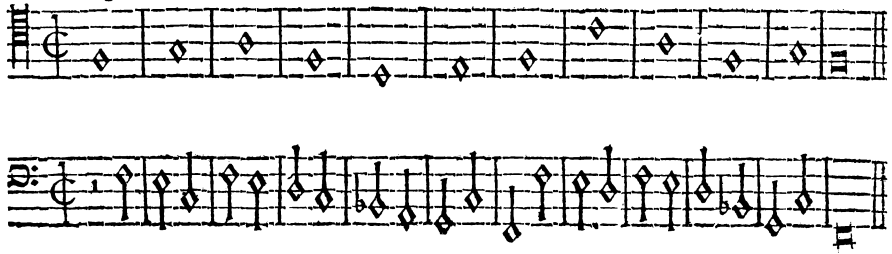
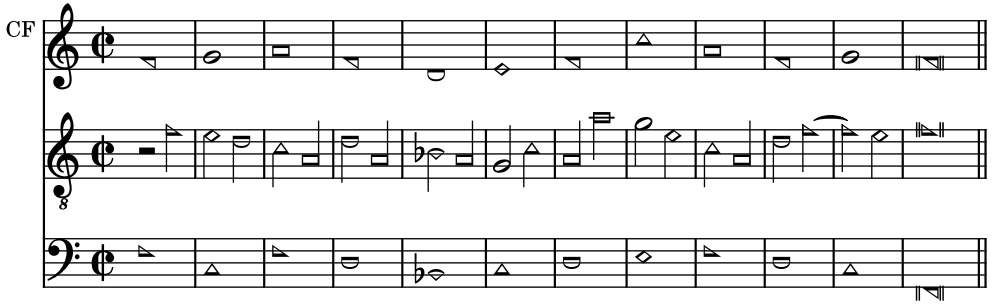
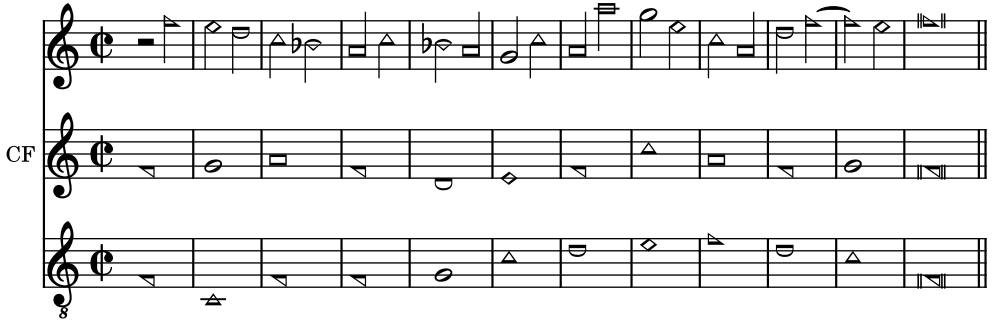
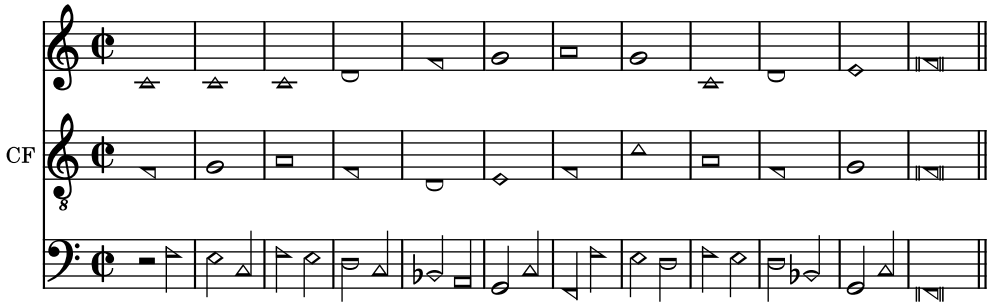
Cantus firmus.



Cantus firmus.

Cantus firmus.

Aloys.

Aloys. Hæc exempla in præsenti sufficiant, reliquos tres tonos privato studio pari ratione persequendos tibi relinquo.

Joseph. Plenum difficaltatis hoc Studium reperio, ità ut nonnunquam ferè impossibilis mihi videatur progressio de uno tactu ad alterum.

Aloys. Fateor, non adeò facilem fore harum Specierum confectionem, in quibus necesse est, duas minimas cum Semibrevibus in duabus partibus positis, observatisque observandis ritè consonare; quod factu difficile, vixque possibile est, nisì unum vel alterum tactum sequentem, antequàm scribere decreverim, anticipatò jam mente conceperim, quemadmodum jam sæpiùs dixi : sed vix exprimi verbis potest, quantam utilitatem, quantamque scribendi facilitatem Studiosis afferat hæc exercisatio, in qua bene versati sublato postmodum Cantûs firmi rigore, tanquam vinculis laxati, liberiores Compositiones quasi ludibundi conficere se posse exultantes comperient.

EXERCITII II.

LECTIO TERTIA.

Cùm idem progrediendi ordo in hac Tricinii Compositione nobis propositus sit, quo in dandis Bicinii Lectionibus usi sumus, per se patet, hanc Lectionem esse Compositionem, quatuor Semiminimis contra Semibrevem positis constantem; cum hac tamen differentiali observatione, ut quemadmodum Semiminimæ in Bicinii compositione cum unius duntaxat partis Semibrevi, ità in hac Tricinii compositione cum duarum partium Semibrevibus consonare debeant : observatis cæterùm iis, non solùm quæ de Specie in Bicinii Lectione dicta, verùm etiam iis, quæ de Tricinii Speciebus hucusque præcepta sunt.

Joseph. Nihilne aliud hìc loci observandum occurrit?

Aloys. Nihil, nisì quòd sicut in omnibus Contrapuncti Speciebus, ità etiam in hac, maxima ratio habenda sit illarum Notarum, quæ in Thesi consistunt.

Joseph. Tentabo ergo, an absque proposito exemplo hanc Speciem conficiendi facultas sit.

Aloys. Placet. Sed attende, ut si primâ Arhsis Semimi-

Aloys. These examples are enough for the present; I leave the remaining three modes for you to pursue in private study in the same manner.

Joseph. I find this study full of difficulty, so that sometimes the progression from one beat to the next seems almost impossible to me.

Aloys. I admit that the construction of these species will not be very easy, since it is necessary that two half notes agree with whole notes placed in two parts, and with all the required conditions observed —something difficult to do and scarcely possible unless one or two of the following beats are already conceived in the mind before I decide to write, as I have often said. But it can scarcely be expressed in words how much usefulness and ease in writing this exercise brings to students; for those who become well-versed in it, once the strictness of the cantus firmus is removed, will find themselves, as if freed from chains, able to compose freer works almost playfully, rejoicing in their ability.

EXERCISE II.
LESSON THREE.

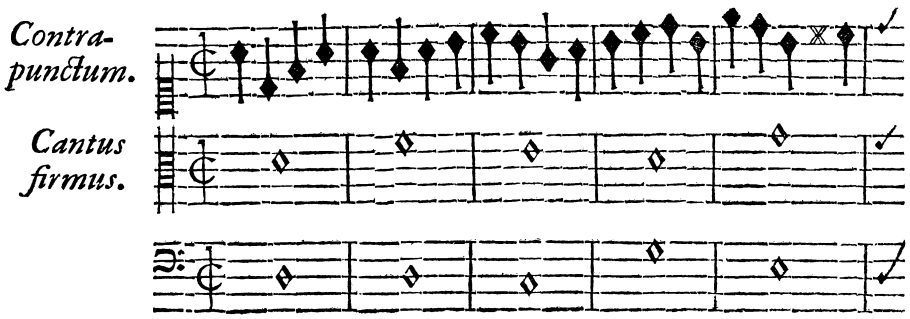
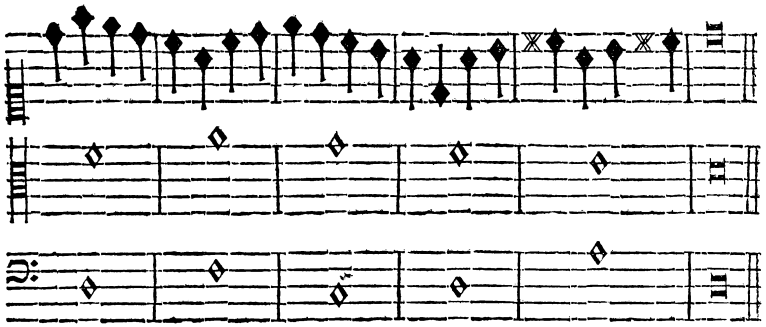
Since the same order of progression has been proposed for us in this tricinio composition as we used in giving the lessons on the bicinium, it is clear in itself that this lesson consists of the composition of four quarter notes set against a whole note; with this additional observation, however: that just as the quarter notes in the bicinium must agree with the whole note of only one part, so in this tricinio they must agree with the whole notes of two parts; and with all the other things observed, not only those said about this species in the bicinium lesson, but also those prescribed thus far concerning the species of the tricinio.

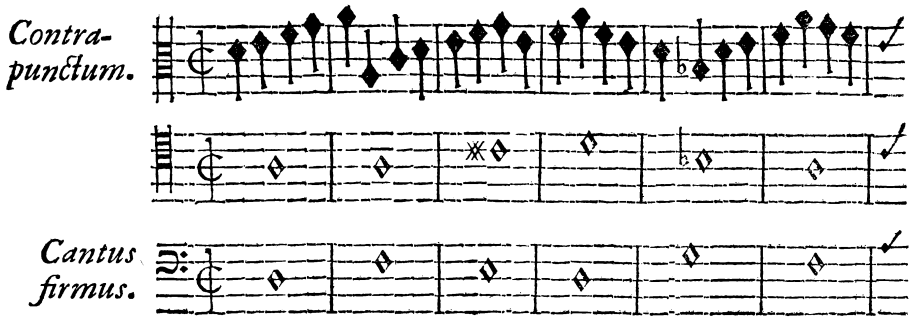
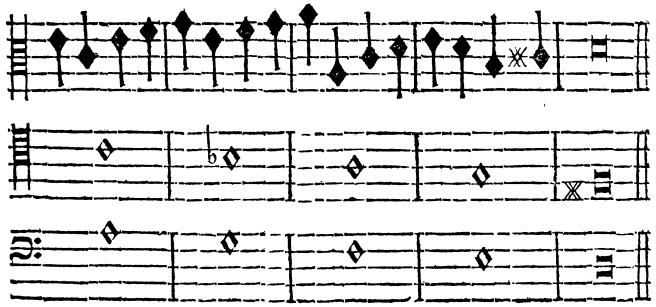
Joseph. Is there nothing else to be observed here?

Aloys. Nothing, except that, as in all species of counterpoint, so also in this one, the greatest attention must be given to those notes which fall on the Thesis (down-beat).

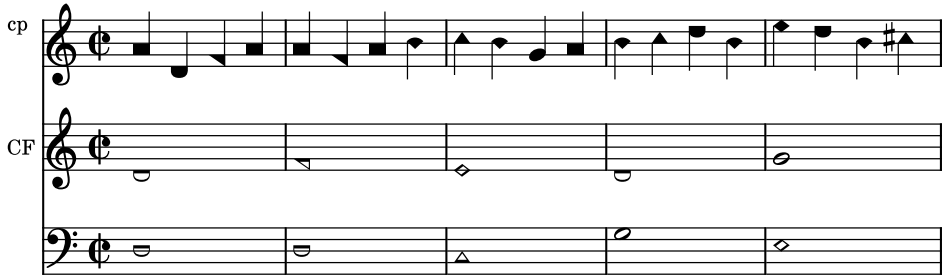
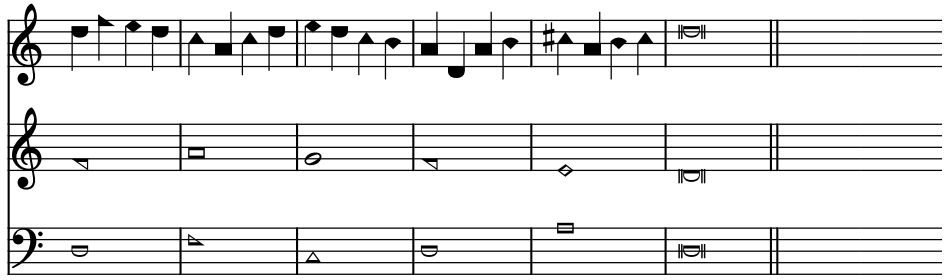
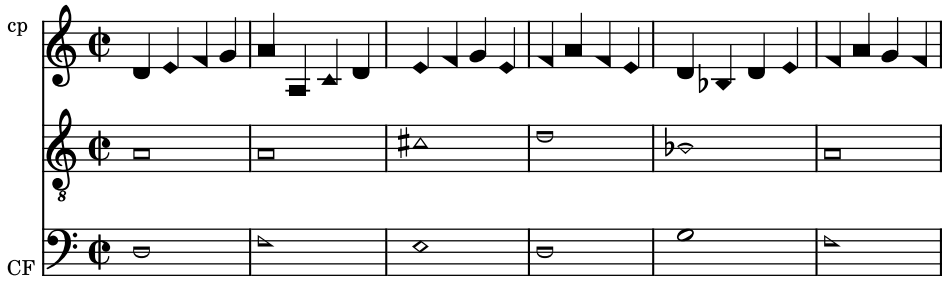
Joseph. I will try, then, whether I can construct this species without a proposed example.

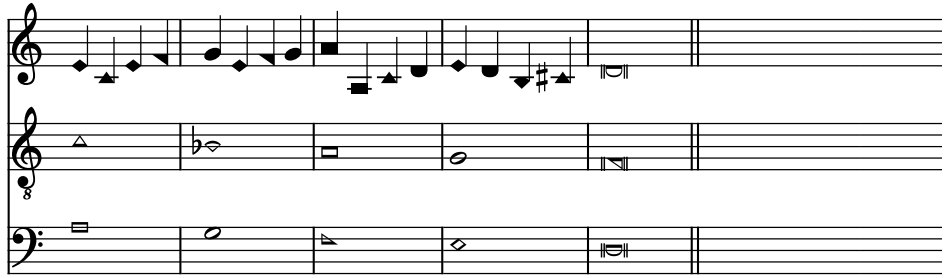
Aloys. Good. But take care that if the first Arsis (up-beat) quarter note cannot form the

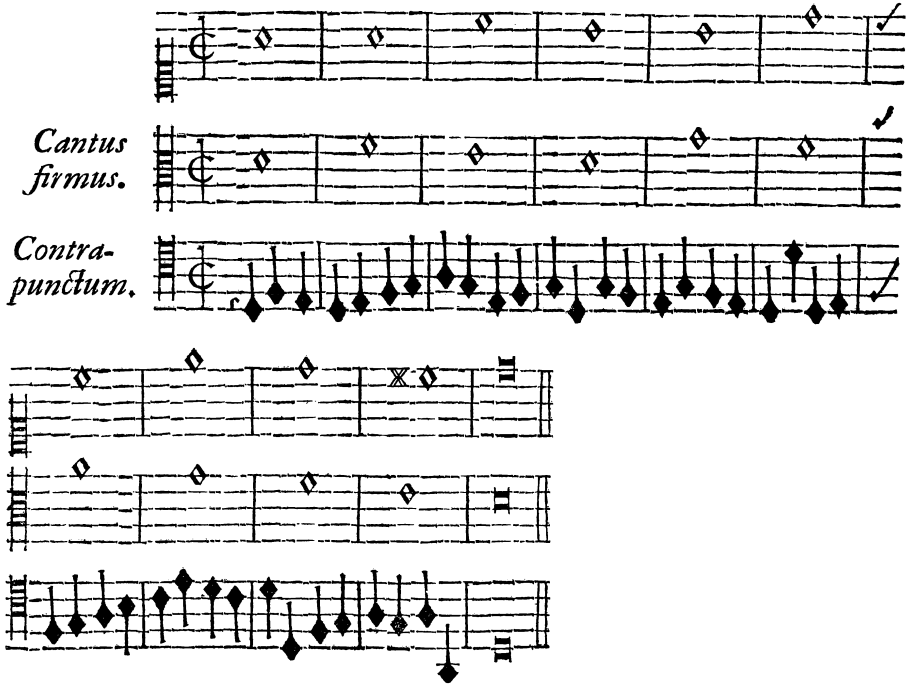
nimâ Triadem harmonicam conſtituere non poſſis, id ſecundâ vel tertiâ efficere procures.



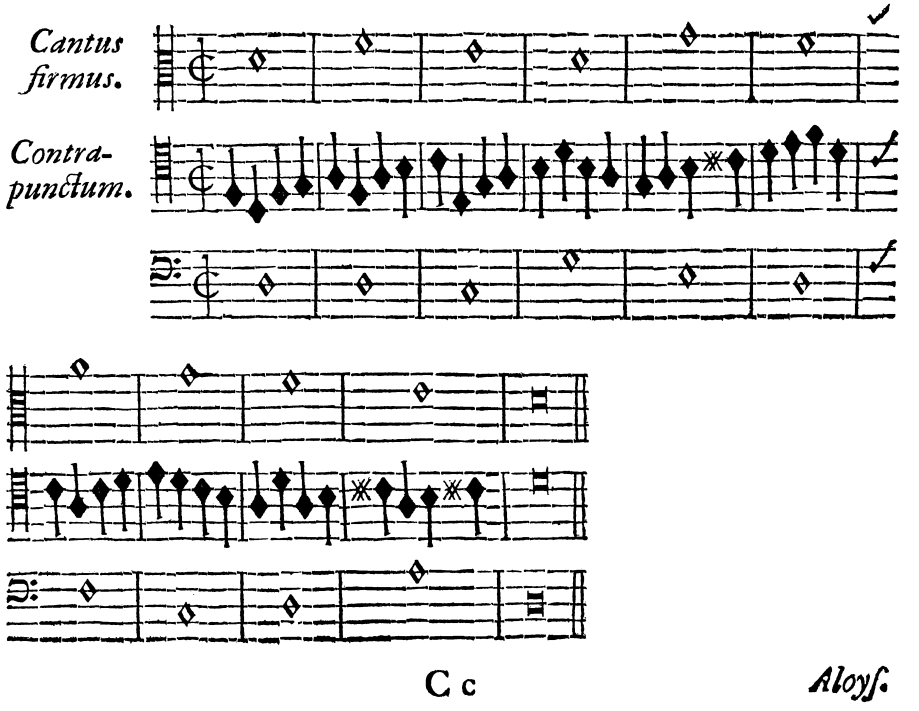


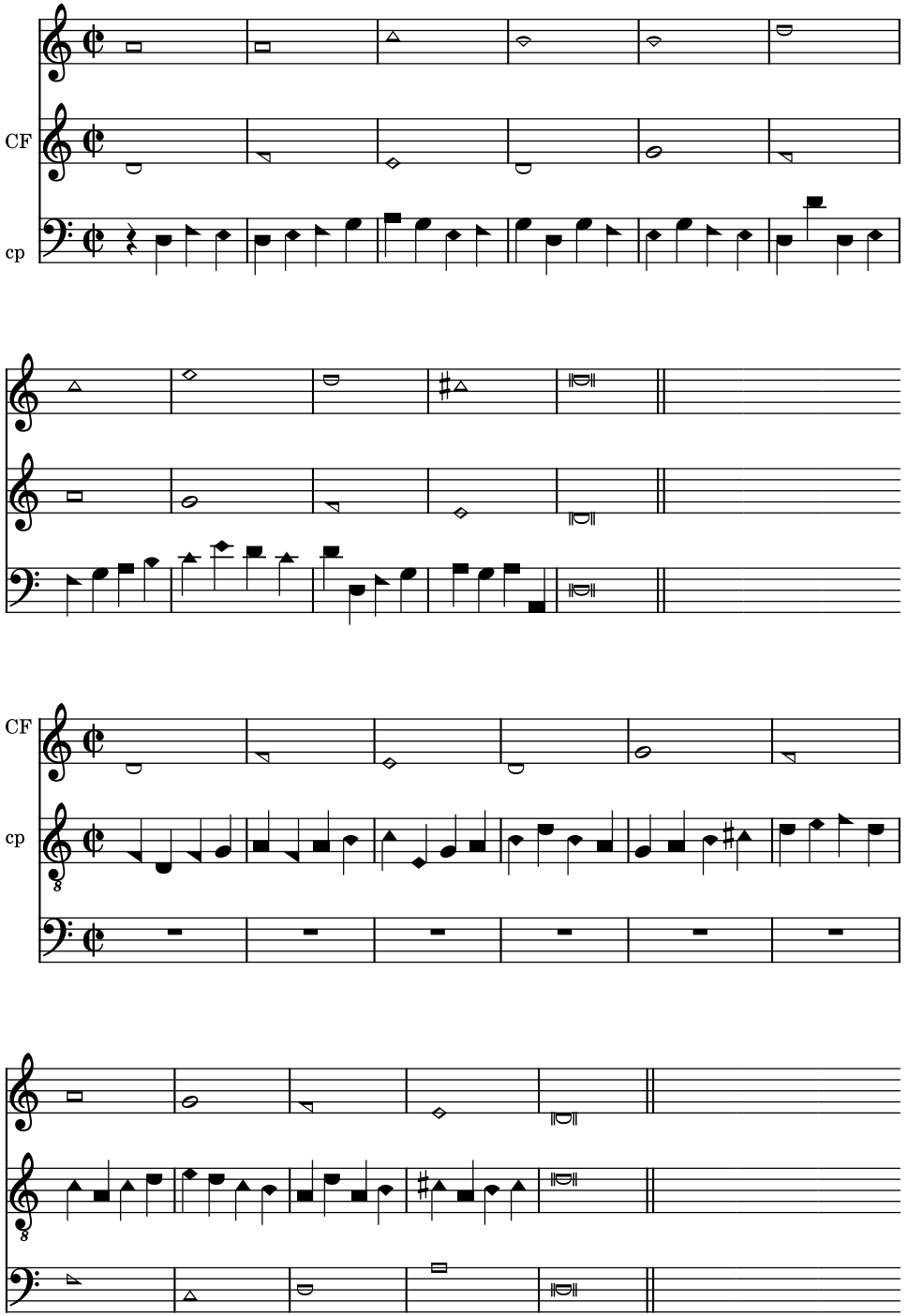
harmonic triad, you try to have the second or third do so.






Cantus firmus.
Contra-punctum.


Cantus firmus.
Contra-punctum.

Aloys.

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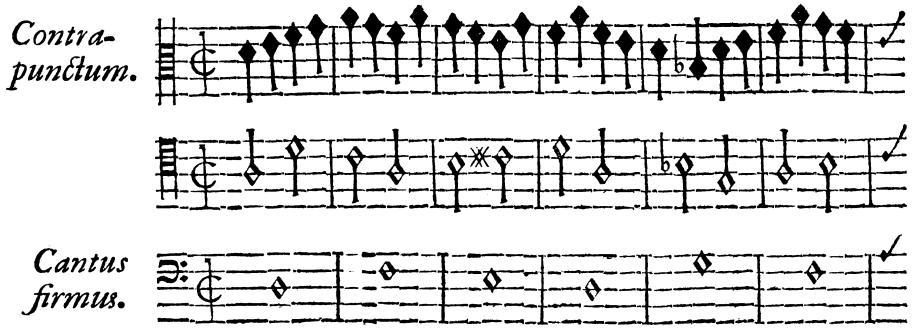
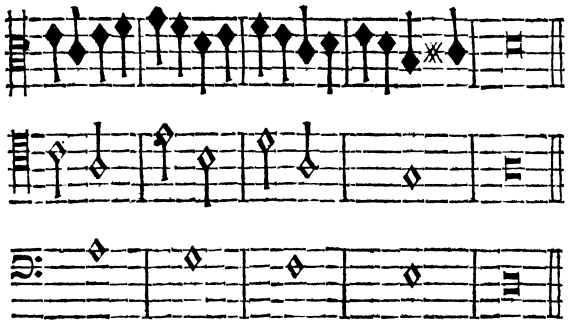
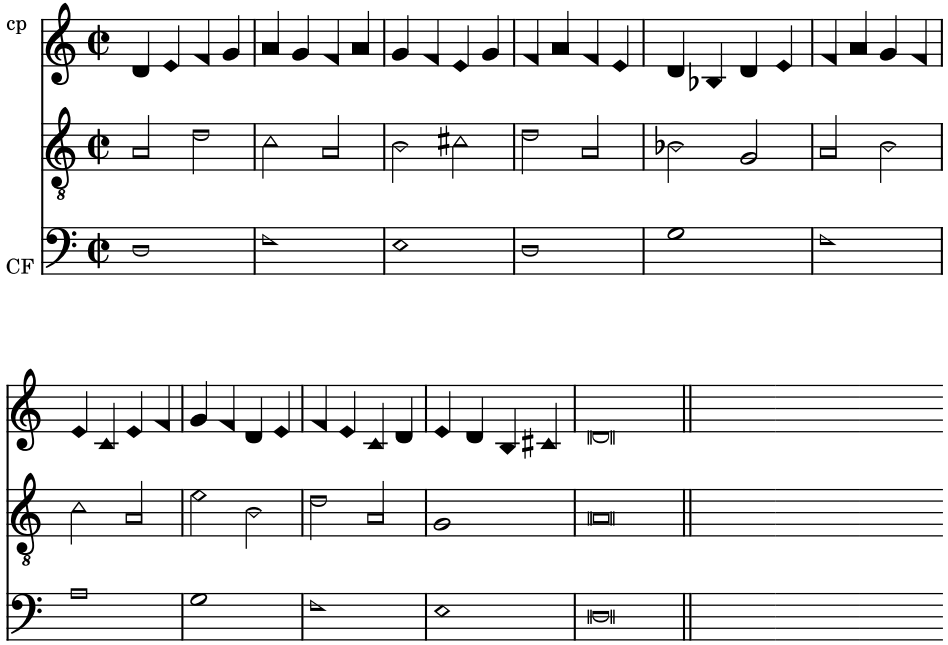
Aloys. Ex his haud malè factis exemplis intelligo, te jam haud mediocri quadam cognitione imbutum esse. Unde cæterorum tonorum Cantuumque firmorum exempla privato studio domi tuæ persolvenda tibi commendo. Quibus peractis si lubet, omnes sex tonorum Cantus firmi denuò resumi, atque perfici poterunt, ità ut in una partium Semiminimæ, in altera Minimæ, tertiâ Semibreves ponantur ad sequentis exempli tenorem.

Quantum ornamenti, præstantiæque triplex hæc figurarum distinctio Compositioni Musicæ afferat, vix dici potest. Quapropter hanc exercitationem cum ea triplici, vel quadruplici mutatione, quâ hucusque usi sumus, maximæ tibi curæ esse, magnopere cuperem.

Joseph. Faciam, maximâ quâ potero virium contentione. Consilium enim tuum semper pro lege mihi tenendum est.

Aloys. Nunc persolvenda nobis restat

EXER-

Aloys. From these examples, which are not badly done, I understand that you are already endowed with no small degree of understanding. Therefore I commend to you the completion at home, in private study, of the examples for the remaining modes and cantus firmi. When these are finished, if you wish, all six cantus firmi of the modes may be taken up again and completed in such a way that in one part quarter notes, in another half notes, and in the third whole notes are placed, according to the pattern of the following example:

How much ornament and excellence this threefold distinction of note-values brings to musical composition can scarcely be expressed. Therefore I greatly desire that this exercise, with the triple or quadruple alternation we have used thus far, be of the greatest care to you.

Joseph. I will do so with all the strength I can. For your advice must always be held by me as a law.

Aloys. Now what remains for us to complete is

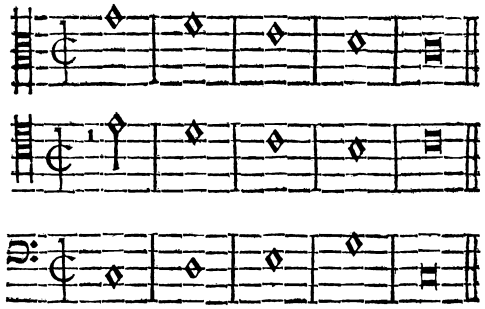
EXERCITII II.

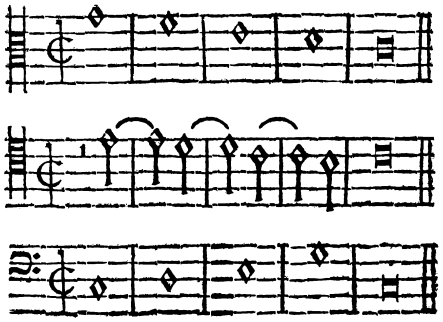
LECTIO QUARTA,

De Ligatura.

Hîc in mentem revocanda ſunt, quæ de Ligaturis in Bicinio dicta ſunt, quarum efficiendi ratio in Tricinio minimè mutanda, ſed ad amuſſim obſervanda eſt. Ad rem pertinebit modô docere, quemadmodum accedente parte tertiâ ſocialis illius concordantia inſtituenda ſit. Refert quoque hîc meminiſſe, quod ſuprà dictum eſt; Ligaturam nempe aliud non eſſe, quàm retardationem Notæ ſequentis. Unde Paradoxum ſit: eandem tertiæ parti tribui oportere concordantiam, quæ omiſsâ Ligaturâ adhibenda fuiſſet; quam veritatem ſequentia Paradigmata perſpicuam reddent.

Sinè Ligatura.



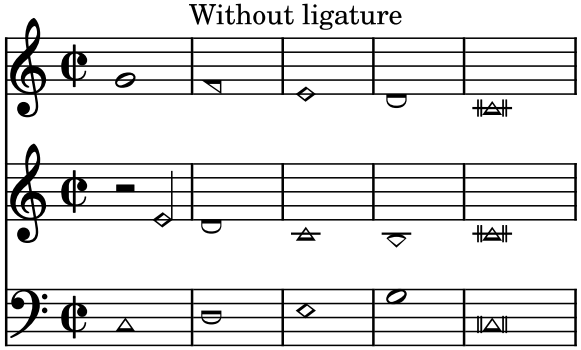
Cum Ligatura.

EXERCISE II.

LESSON FOUR,

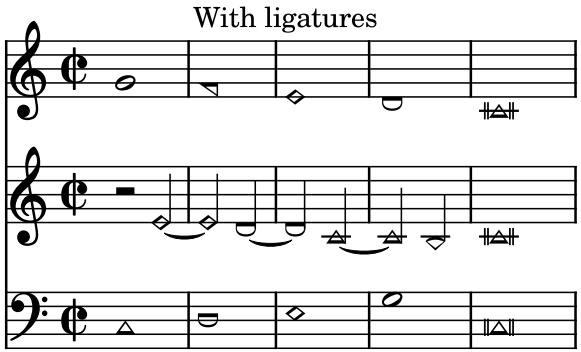
On the Ligature.

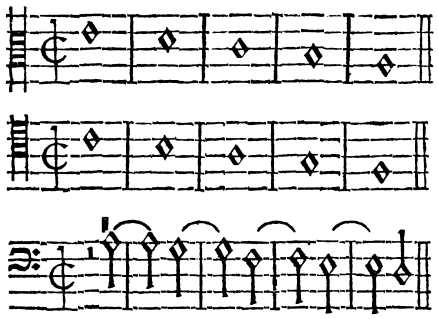
Here you must recall what was said about ligatures in the bicinium, whose method of construction must not be changed in the tricinio, but observed exactly. What pertains to the matter now is to show how, with the addition of the third part, the special consonance belonging to it must be established. It is also important here to remember what was said above, namely that the ligature is nothing other than the retardation of the following note. Whence it follows as a paradox that the same consonance must be given to the third part which would have been used had the ligature been omitted. The following patterns will make this truth clear.

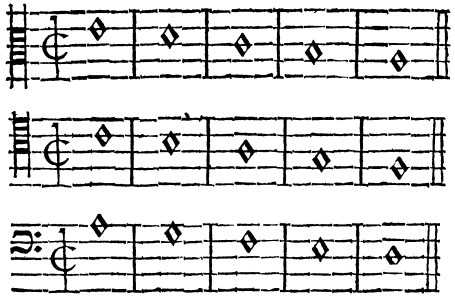
Without ligature



With ligatures



Ubi clarè videtur, utroque in exemplo eandem tertiæ partis eſſe concordantiam nihil impediente ligaturâ; quod idem intelligendum puto de Ligaturis in parte gravi, ſive fundamentali collocatis. E. G.

Hæ Hypotheſes, ſublatis Ligaturis, ob Triadis harmonicæ rationem, nisì alia ratio, nempe immediata plurium Quintarum conſecutio impediret, ſic ſe haberet:



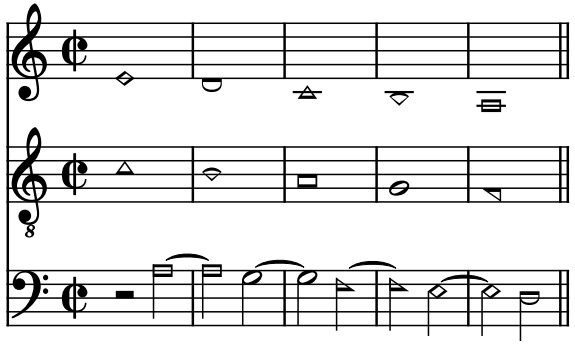
Quod vitioſum exemplum, non alium in finem oculis, Joſephe, ſubjeci, quàm ut in cognitionem venires, concordantiarum rationem à Ligaturis minimè immutari; ſed prorſus eandem permanere.

Joſeph. Ex hac tua aſſertione Magiſter, ſcrupulus mihi oritur, quem, nî moleſtè fers, aperiam.

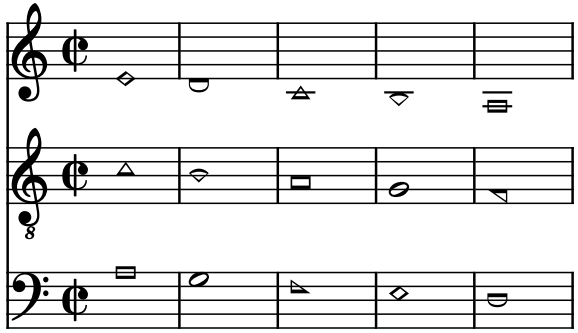
Aloyſ. Edic audacter. Longum etenim tuum hucuſque ſilentium vix non in dubitationem me induxit, an, quæ nihil interpellatus dixi, intelligeres, nec ne?

Joſeph. Si Ligaturæ, ùt dictum eſt, nihil immutant, æquè primum Ligaturis conſtans exemplum paulò ante allatum, quàm ſecundum vitioſum erit. Proinde, ſicut in ſecundo ſemotis

There it is clearly seen that in both examples the consonance of the third part is the same, the ligature causing no hindrance; and I think the same must be understood of ligatures placed in the lower, or fundamental, part. For example:



These hypotheses, with the ligatures removed, would stand thus for the sake of the harmonic triad, unless another reason—namely the immediate succession of several fifths—prevented it:



This faulty example I placed before your eyes for no other purpose than that you might come to understand that the nature of the consonances is not changed by the ligatures, but remains entirely the same.

Joseph. From this assertion of yours, Master, a difficulty arises for me, which, if you do not mind, I will reveal.

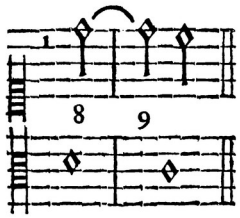
Aloys. Speak boldly. Your long silence until now had almost led me to doubt whether you understood what I said without interruption.

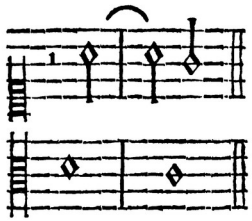
Joseph. If ligatures, as you said, change nothing, then the first example, which contains ligatures, will be just as faulty as the second. For just as in the second example, with the ligatures removed, the

motis Ligaturis exemplo immediata apparet plurium Quintarum conſecutio ; ob eandem rationem, ſi Ligaturæ pro non adjectis habentur, nec primnm Ligaturis inſtructum exemplum eodem vitio vacabit.

Aloyſ. Valde mihi placet hoc tuum ſagax argumentum, magnæ attentionis tuæ indicium. Verùm, præterquam, quod Clarorum in arte peritorum auctoritati permultum tribuendum ſit, qui primum exemplum approbant, ſecundum improbant, ſcire te oportet, dum dicerem, *Ligaturas nihil immutare*, duntaxat de concordantiarum natura & eſſentia, quæ in utroque exemplo æqualis eſt, intelligendum eſſe. Cæterùm maximam Ligaturis eſſe vim, poteſtatemque ſalvandi, & immutandi quis ambigere poteſt ?

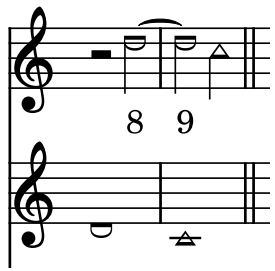
Joſeph. Solutum eſſet hac tuâ diſtinctione argumentum meum, nisì obſtaret Ligaturæ exemplum ſuprà in Bicinio adductum, quod ùt impotens ſalvandi duas Octavas, ideo tanquam vitioſum rejeciſti. E. G.

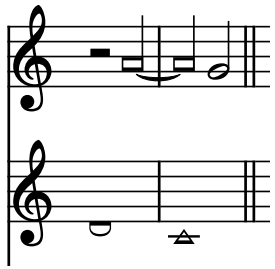


Sicut Ligatura in hoc exemplo non valet efficere, quò minùs duæ vitioſæ Octavæ eveniant ; itâ eadem prorſus ratione in ſequenti non valebit ſalvare duas Quintas :

Aloyſ. Ad hanc tuam haud exilem objectionem ſolvendam non ignorare expedit, multa in parte acuta, quia perceptibiliore, magisque auribus perviâ prohiberi, quæ in parte gravi, per gravitatem non nihil offuſcata, nec admodum auditum ferientia, tolerantur. Acumen enim diſtinguit ; gra-

immediate succession of several fifths appears, so for the same reason, if the ligatures are considered as not added, the first example, furnished with ligatures, will not be free from the same fault.

Aloys. I am greatly pleased by this keen argument of yours, a sign of your great attentiveness. But besides the fact that much must be granted to the authority of distinguished experts in the art, who approve the first example and disapprove the second, you must know that when I said ligatures change nothing, it was to be understood only of the nature and essence of the consonances, which is the same in both examples. As for the rest, who can doubt that ligatures have great force and power to save and to alter?

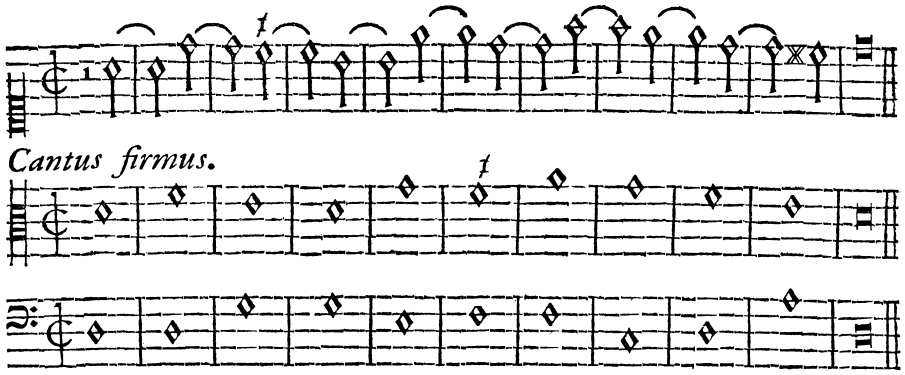
Joseph. My argument would be solved by this distinction of yours, were it not for the example of the ligature brought above in the bicinium, which is unable to save two octaves, and is therefore rejected as faulty. For example:

Just as the ligature in this example does not suffice to prevent two faulty octaves from occurring, so by exactly the same reasoning it will not be able to save two fifths in the following example:



Aloys. To resolve this objection of yours—which is by no means slight—it is useful to know that many things are forbidden in the upper part, because it is more perceptible and more exposed to the ear, which in the lower part, being somewhat obscured by its gravity and not striking the ear so strongly, are tolerated. Sharpness distinguishes; gravity

vitas ſupprimit. Deinde ut ad fundamentaliorem perveniamus rationem, diſparitatemque, in memoriam revocentur, quæ alibi de Conſonantiarum perfectione diximus: Quintam nempe perfectam; Octavam perfectiorem; Uniſonum perfectiſſimam eſſe Conſonantiam; & quò perfectiores, eò minùs Harmoniæ habere. Porrò compertum eſt, Diſſonantes ex ſe omni Harmoniæ amœnitate, gratiâque vacuas eſſe; ſed quidquid grati, blandique auribus exhibent, id totum beneficio ſequentis Conſonantiæ, in quam reſolvuntur, attribuendum eſſe. Ex quo perſpicuum eſt, diſſonantiam in Quintam reſolutam tolerabiliorem eſſe, quàm ea, quæ in Octavam reſolvi contingit. Unde mirum non eſt, primum exemplum ùt vitioſum à bonis Authoribus rejici, ſecundum verò Contrapuncti præceptis accommodatum probari. Denique quò magìs Conſonantia perfecta, in quam Diſſonantia reſolvitur, ad Conſonantiæ imperfectæ accedit naturam, eò faciliore excuſatione, indulgentiaque proſequenda erit reſolutio. Nunc ſi tibi ſatisfactum eſt, ad hujus Contrapuncti Speciei exercitium te accinge.

Joſeph. Faciam, ùt jubes.

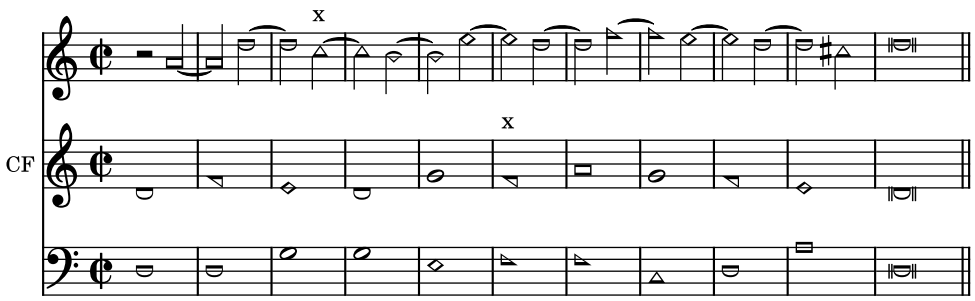


Aloyſ. Cur in tertii tactûs Syſtemate in parte cantûs, erroris, vel potiùs dubii ſignum poſuiſti?

Joſeph. Nondum memoria mihi excidit, primam Ligaturæ partem Conſonantiam eſſe debere: ego autem eo loco diſſonantiam, nempe Quartam adhibui; partim quia ob neceſſitatem duarum Semibrevium ponendarum, aliter fieri poſſe, non reperiebam: partim quia ejuſmodi exempla bonis in Authoribus me conſpexiſſe recordor.

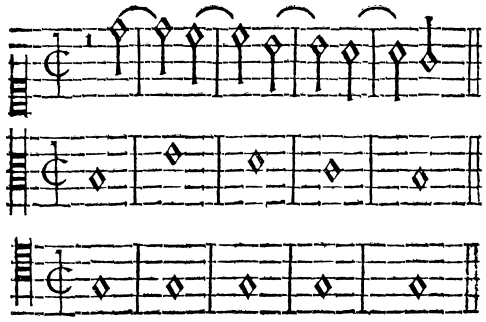
Aloyſ.

suppresses. Then, to come to a more fundamental reason, let us recall what we said about the perfection of consonances: that the fifth is perfect, the octave more perfect, the unison most perfect; and the more perfect they are, the less harmony they contain. Moreover, it is well known that dissonances, of themselves, are empty of all pleasantness and charm; but whatever grace or sweetness they offer to the ear must be attributed entirely to the consonance that follows, into which they resolve. From this it is clear that a dissonance resolved into a fifth is more tolerable than one resolved into an octave. Hence it is no wonder that the first example is rejected as faulty by good authors, while the second is approved as conforming to the rules of counterpoint. Finally, the more the perfect consonance into which the dissonance resolves approaches the nature of an imperfect consonance, the more easily and indulgently the resolution is to be excused. Now, if you are satisfied, prepare yourself for the exercise of this species of counterpoint.

Joseph. I will do as you command.

Aloys. Why did you place a sign of error—or rather of doubt—on the melody part in the system of the third beat?

Joseph. I have not yet forgotten that the first part of the ligature must be a consonance; but in that place I used a dissonance, namely the fourth—partly because, owing to the necessity of placing two whole notes, I found no other way to proceed, and partly because I recall having seen such examples in good authors.

Aloyſ. Valde laudabilis eſt ſcrupulus iſte tuus, quô magnæ attentionis tuæ prodis teſtimonium. Verùm nihil refert, quò minùs Hypotheſis illa regularum rigori ſit conſentanea. Etenim quod dictum eſt, primam Ligaturæ partem Conſonantiam eſſe debere, intelligendum eſt de illis Hypotheſibus, in quibus pars fundamentalis ſingulis tactibus movetur; non de illis, quarum baſis in manſione, ùt dicere ſolemus, & in eodem ſitu detinetur: quo caſu ligatura meris diſſonantiis conſtans, non ſolùm non vitioſa eſt, verùm etiam elegantiſſimè adhibetur; quemadmodum ſequenti Paradigmate demonſtratur.

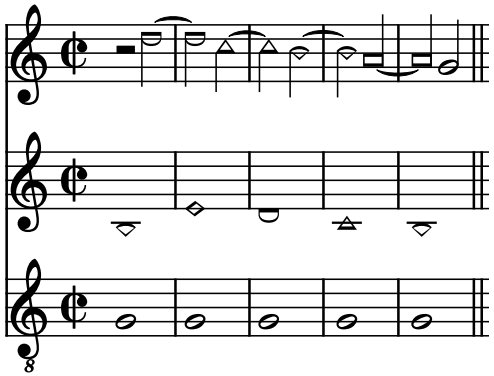


Quid alterum dubitationis ſignum in ſexto tactu exiſtens indicat?

Joſeph. Non ignoro Septimam ſibi Tertiam comitem poſtulare; ego autem ibidem propter Cantûs firmi neceſſitatem Octavam illi aſſociavi.

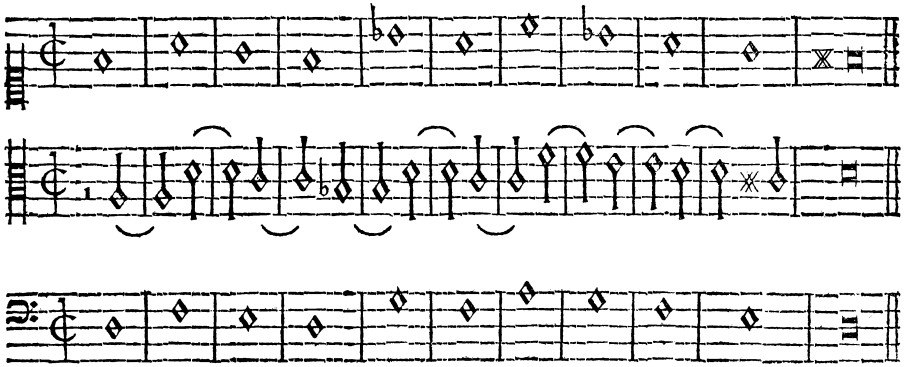
Aloyſ. Meminiſſe te oportet, hîc loci nos in lectionibus verſari, & in ſtudio, ſingulis tactibus Ligaturam inſerendi, non admodum rigori cæterarum concordantiarum inhærentes, de quibus alibi & diximus, & dicturi ſumus. Aliud judicium ferendum eſt de libero componendi genere, ubi nulla neceſſitas impedit, quò minùs cuilibet diſſonantiæ propria ſua co cordantia tribui poſſit: ideo Septima illa cum Octavâ aſſociata tolerari debet. Nunc ad reliquas exercitationes.

Aloys. That scruple of yours is very praiseworthy, showing great attentiveness. But it does not matter that this hypothesis does not conform to the strictness of the rules. For what was said—that the first part of the ligature must be a consonance—must be understood of those hypotheses in which the fundamental part moves in every beat; not of those whose foundation remains stationary, as we say, and stays on the same pitch; in which case a ligature consisting entirely of dissonances is not only not faulty, but is used most elegantly, as the following pattern shows:

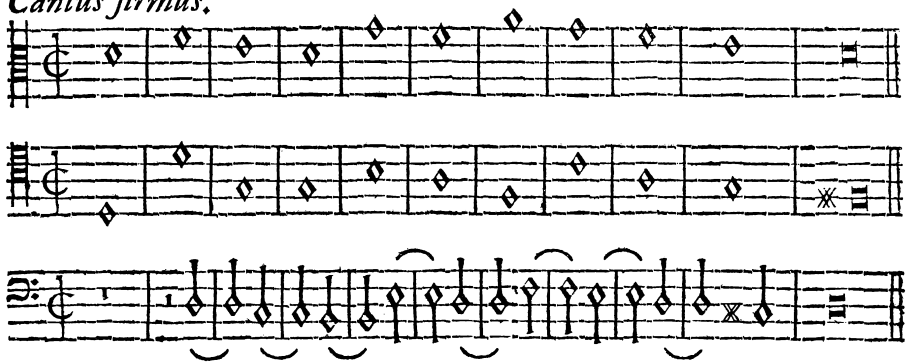


What does the second sign of doubt in the sixth beat indicate?

Joseph. I know that the seventh requires the third as its companion; but in that place, because of the necessity of the cantus firmus, I associated the octave with it.

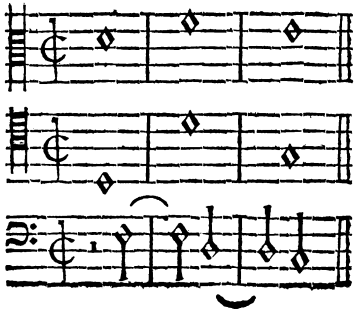
Aloys. You must remember that here we are dealing with lessons, and with the study of inserting a ligature in every beat, not adhering too strictly to the other consonances, about which we have spoken and shall speak. A different judgment must be made in free composition, where no necessity prevents each dissonance from receiving its proper consonance; therefore that seventh associated with the octave must be tolerated. Now to the remaining exercises.

Cantus firmus.

Cantus firmus.



Aloyſ. Quare in ultimi exempli parte fundamentali ab initio unius tactûs pauſam adhibuiſti ?

Joſeph. Quia Ligaturam ibidem adhiberi poſſe, non inveni; neque aliâ Contrapuncti Specie illud ſpatium implendum exiſtimavi ; ideo pauſæ beneficio me juvare tentavi.

Aloyſ. Non mihi diſplicet induſtria iſta tua. Interim tamen ſequenti modo fieri potuiſſet.

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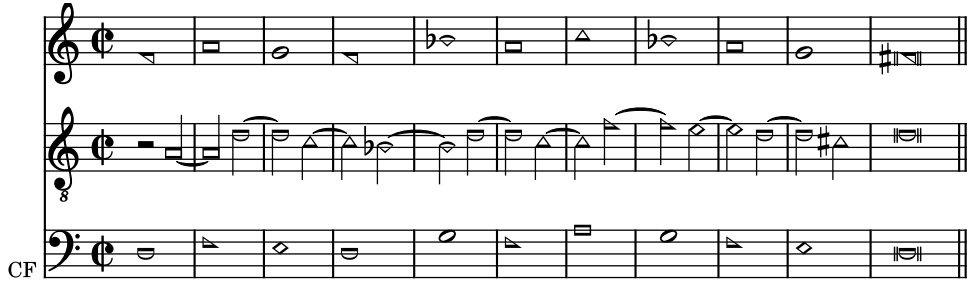
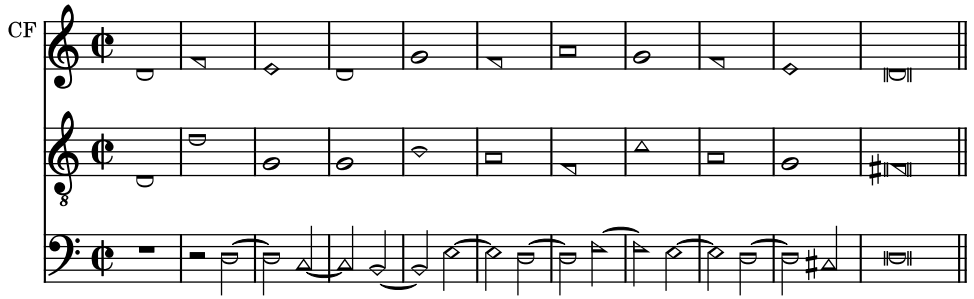
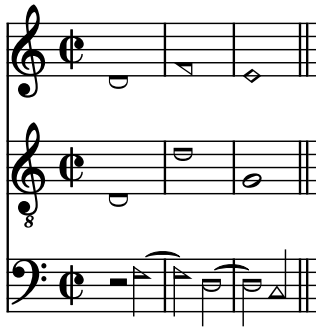
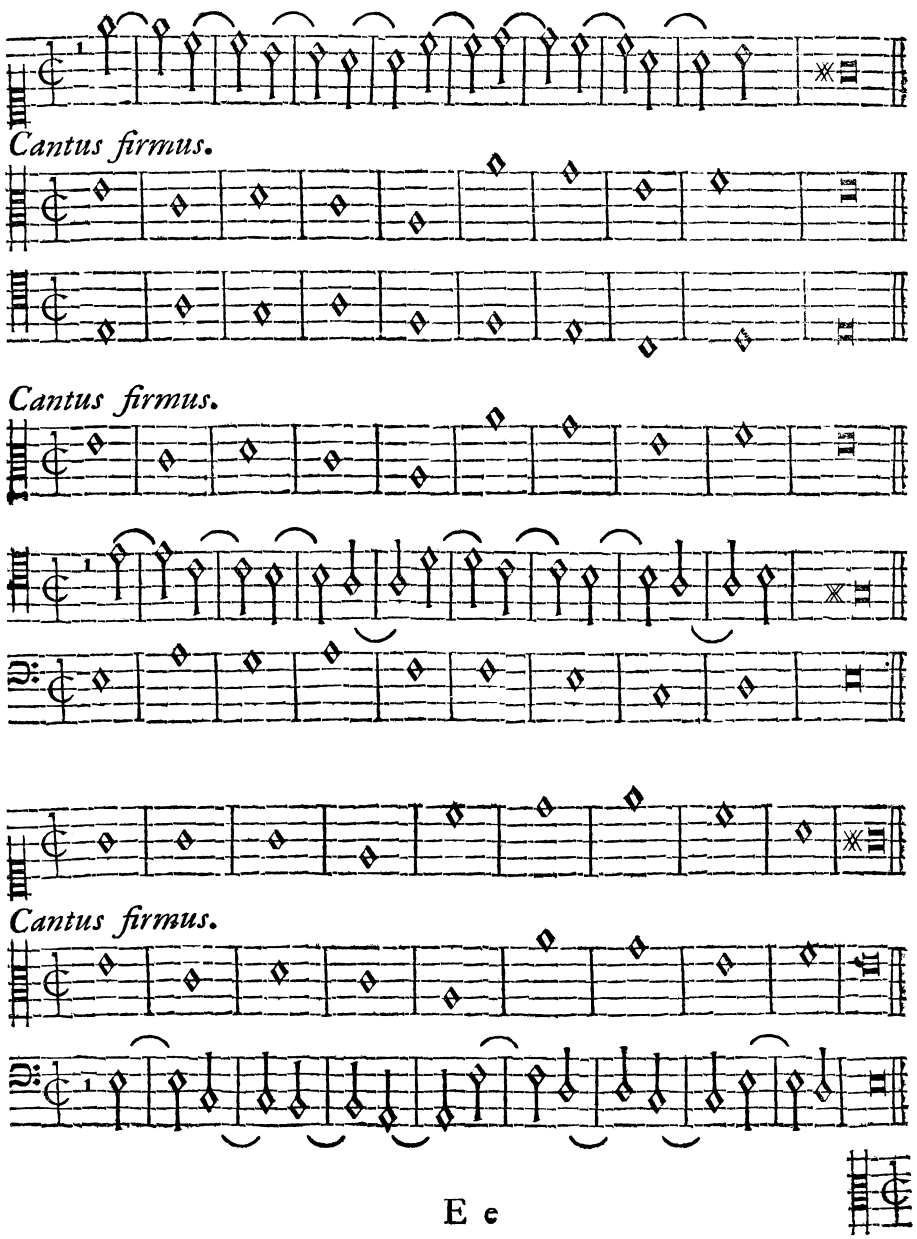
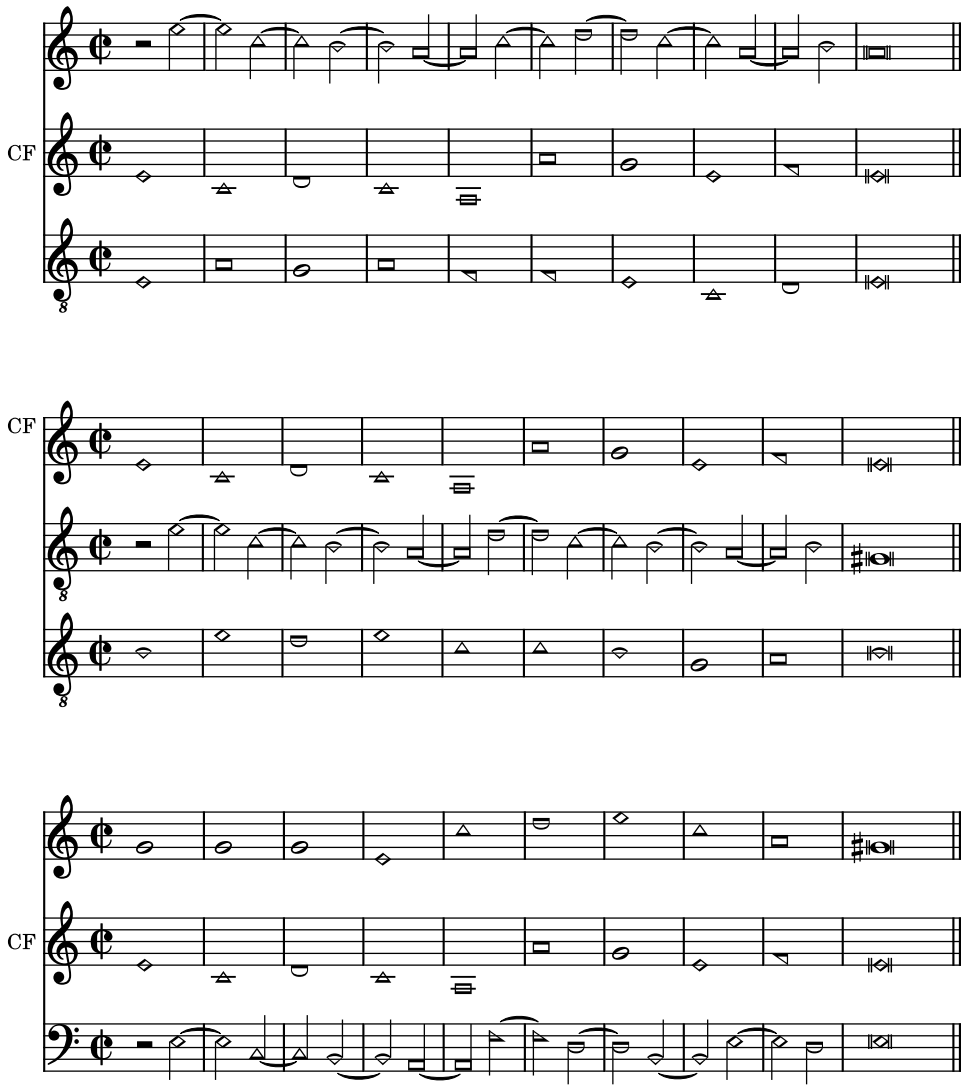
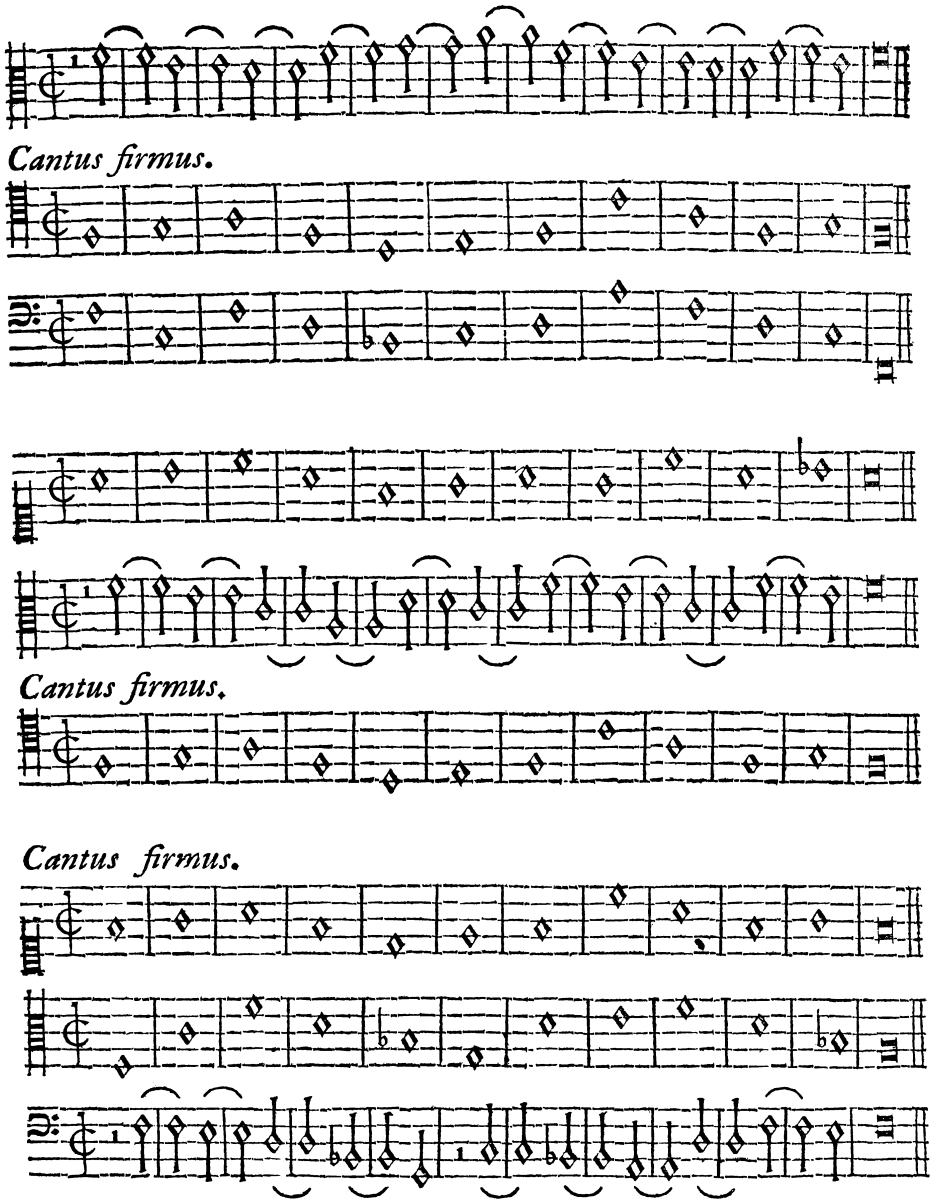
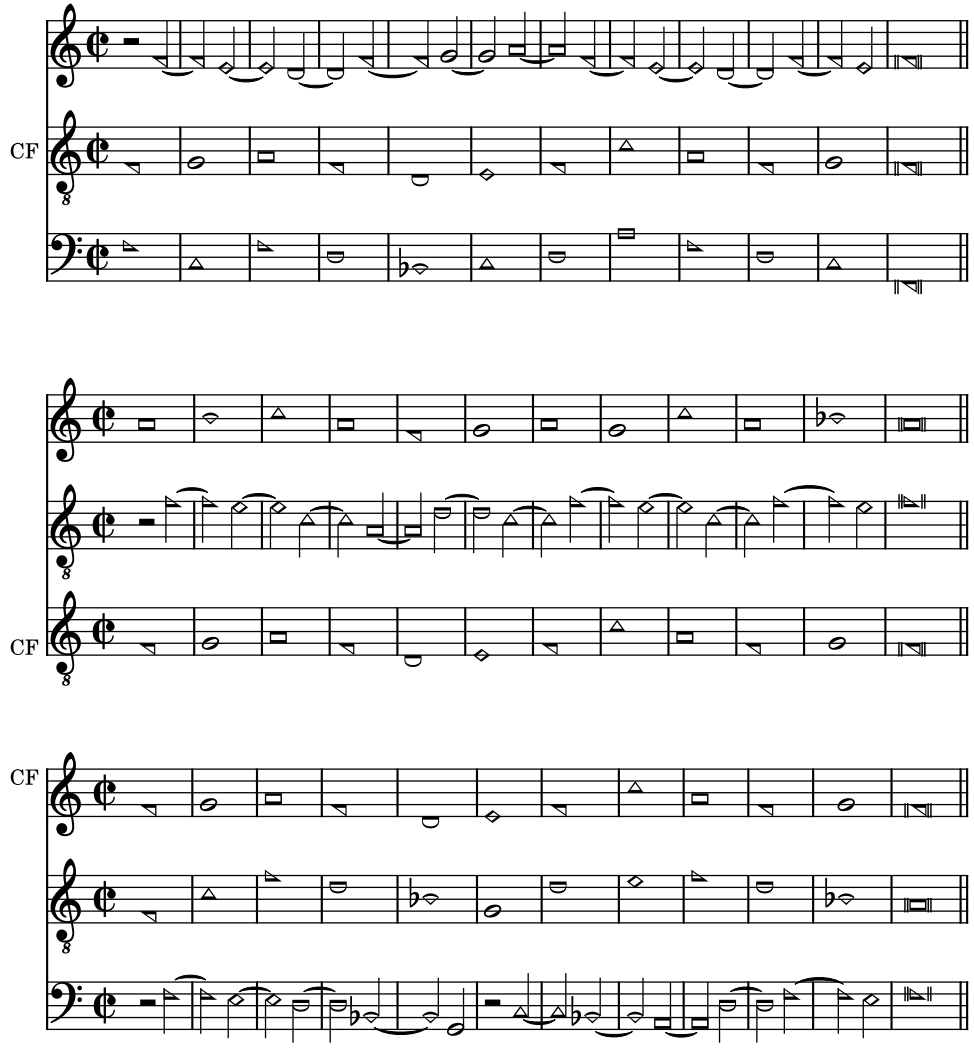

Aloys. Why did you place a rest at the beginning of one beat in the fundamental part of the last example?

Joseph. Because I could not find a way to use a ligature there, nor did I think that space should be filled with another species of counterpoint; therefore I attempted to help myself by means of a rest.

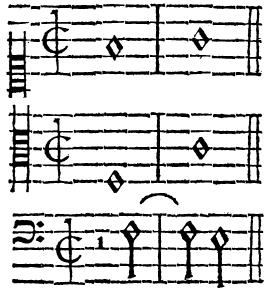
Aloys. That industry of yours does not displease me. Yet it could have been done in the following way:

Ubi Tenor in primo tactu basis officio fungitur : id quod non solùm Tenori, verùm etiam Alto ; imò si possibile esset, Cantui usu venit ; ita ut pars gravissima qualiscunque illa pro fundamento statuenda, & super eandem ædificandum sit. Jam ad duorum adhuc tenorum naturali ordine se sequentium exempla conficienda progredere.



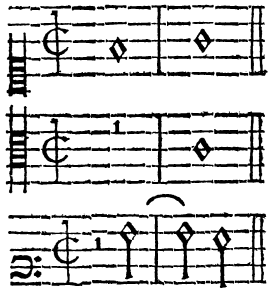
Here the Tenor in the first beat performs the office of the foundation; something that not only the Tenor but also the Alto—and indeed, if it were possible, the melody—may do; so that the lowest part, whatever it may be, must be set as the foundation, and the rest built upon it. Now proceed to compose examples for the two remaining modes that follow in natural order.



Aloyſ. In tantum exercitia hæc ad Harmoniæ rationem elaborata ſunt, quantum ligaturæ cum duabus ſemibrevibus in omnibus tactibus poſitæ permittunt : neque enim hujus ſpeciei reſtrictio Harmoniam admittit, omnibus numeris abſolutam. Præterquam, quòd hîc potior ratio ligaturarum, ut jam ſæpius diximus, habeatur, quæ tot diverſis modis hoc in exer-

Aloys. These exercises have been worked out as far as the nature of harmony allows, insofar as the ligatures with two whole notes can be placed in every beat; for the restriction of this species does not admit harmony fully complete in every respect. Besides, here the chief consideration is the ligatures, as we have often said, which, appearing in so many different ways in this exercise, provide a perfect understanding of their construction. Therefore I strongly commend this exercise to you

exercitio reſultantes perfectam ſui conficiendi cognitionem exhibent. Quapropter exercitium hoc præcipuum Compoſitionis ornamentum, ſummopere tibi commendo. Id, quod meliùs fieri potuiſſet, eſt primus exercitii tactus, qui ſic ſe habet:



Ubi inter Altum, & Cantum occulta duarum Quintarum oritur conſecutio, quæ facilis auribus perceptu in Tricinio evitari debet, quod fiet unius pauſæ poſitione in Alto, hoc modo:



Joſeph. Simili juvamine ſum uſus in ejusdem exercitii Baſſi ſexto tactu; ubi ſecùs ulteriùs per ligaturarum ſeriem progrediendi poteſtatem non inveni.

Aloyſ. Induſtriosè factum eſt. Præceptum enim de ligaturis in omnibus tactibus faciendis, intelligi non aliter debet; quàm ubi facultas adeſt. Reliquos tres tonos pari exercitio percurres.

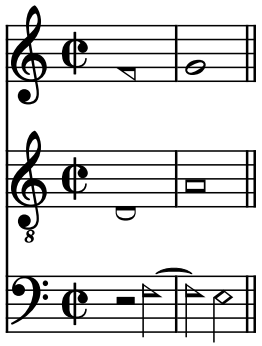
EXERCITII II.

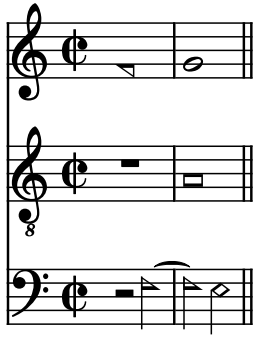
LECTIO QUINTA.

De Contrapuncto florido.

Aloyſ. Quid, & qualis hæc ſpecies ſit, ex iis, quæ de ea in Bicinio diximus, conſtare tibi ſuppono: eſſe nempè Compoſi-

as a principal ornament of composition. What could have been done better is the first beat of the exercise, which stands thus:



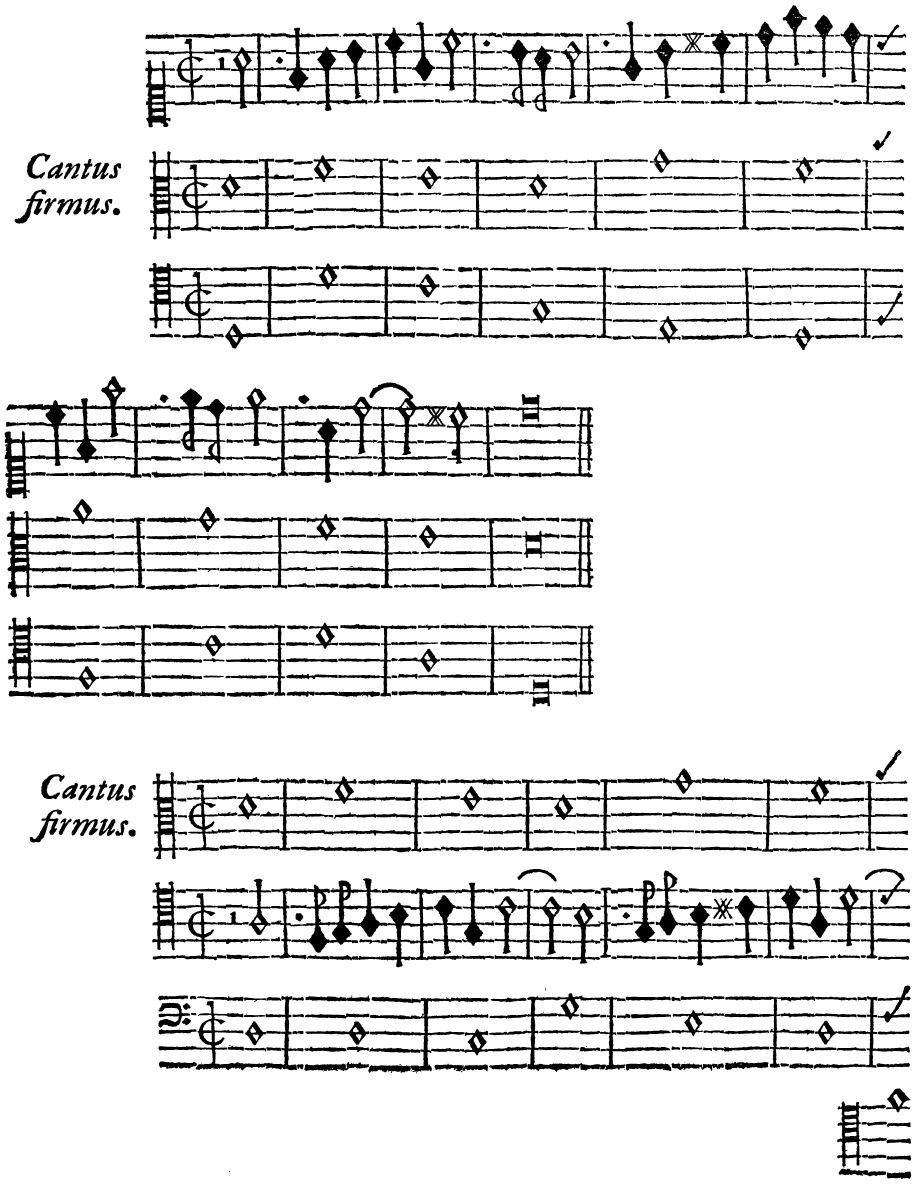
Here between the Alto and the Melody there arises a hidden succession of two fifths, which, being easily perceived by the ear in a tricinio, must be avoided; which will be done by placing a rest in the Alto, in this manner:

Joseph. I used a similar aid in the sixth beat of the Bass in the same exercise, where I found no further possibility of proceeding through a series of ligatures.

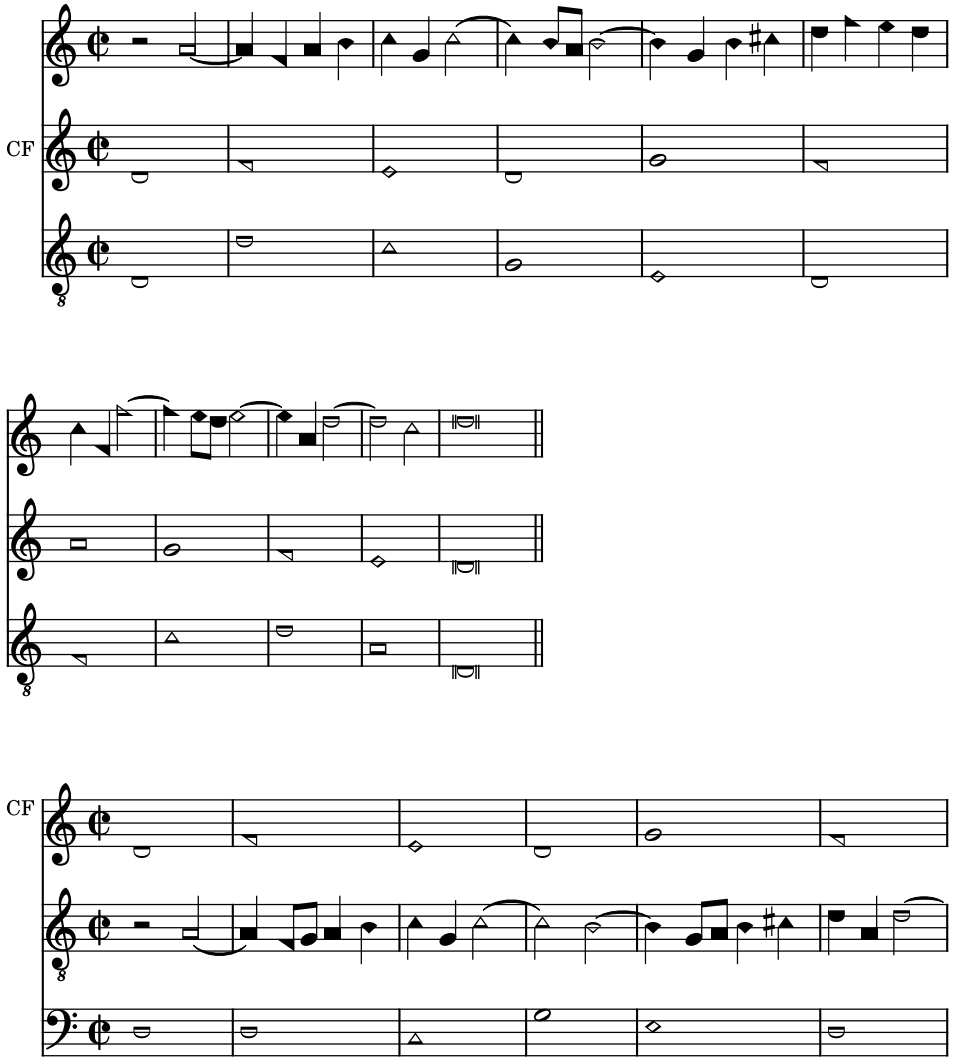
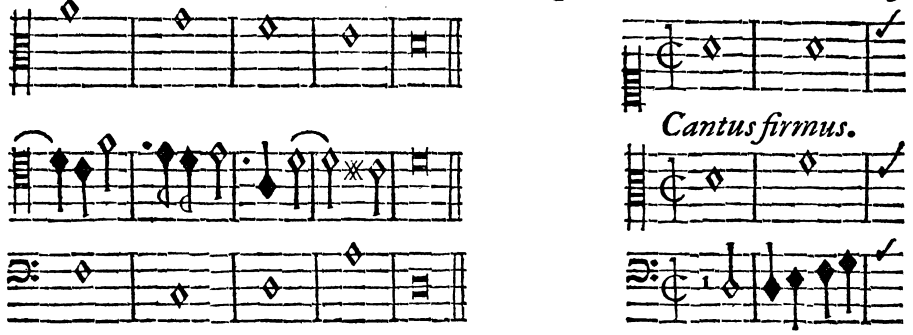
Aloys. It was done industriously. For the precept about making ligatures in every beat must be understood only where the possibility exists. You will go through the remaining three modes with equal exercise.

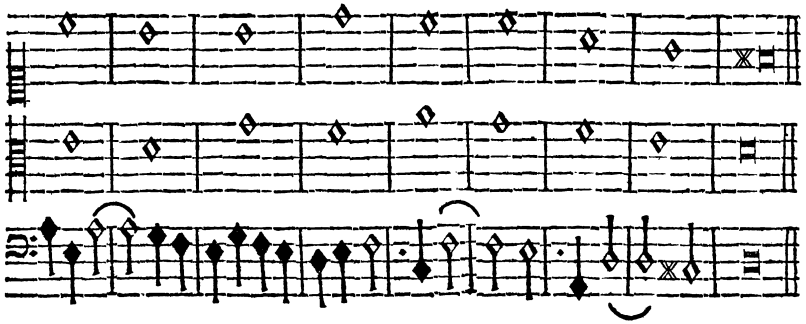
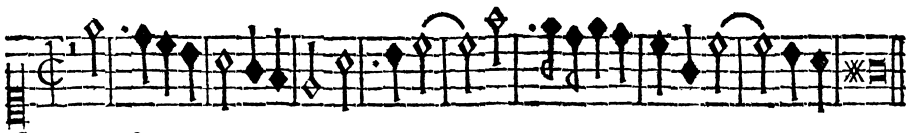
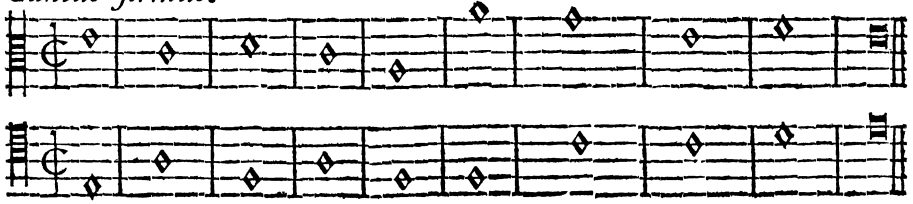
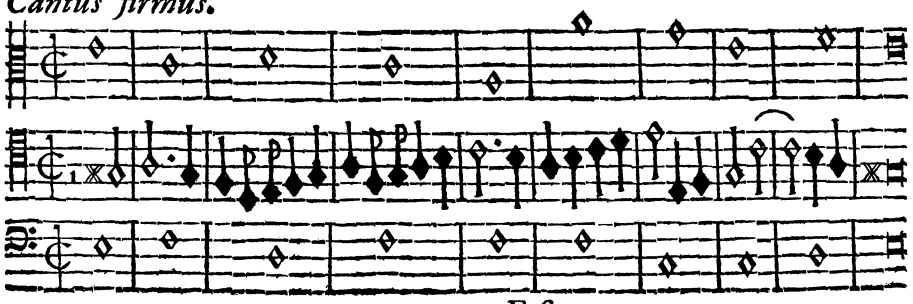
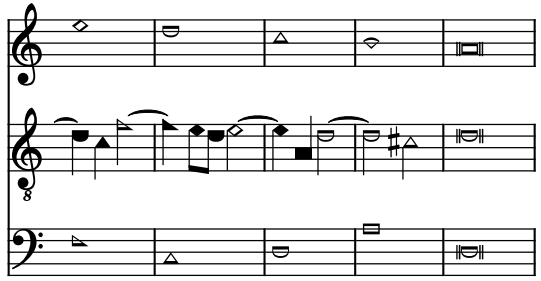
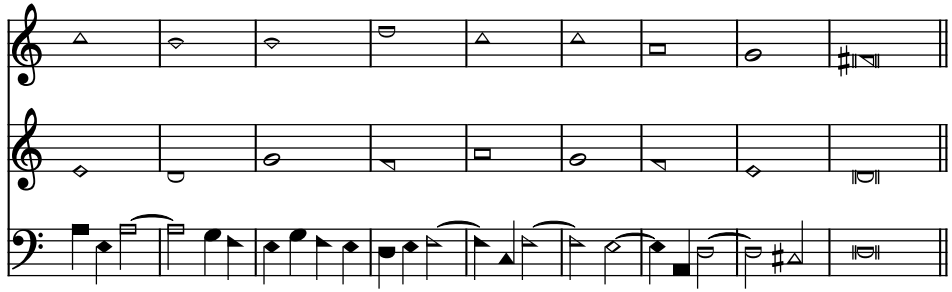
EXERCISE II
LESSON FIVE.
On florid counterpoint.

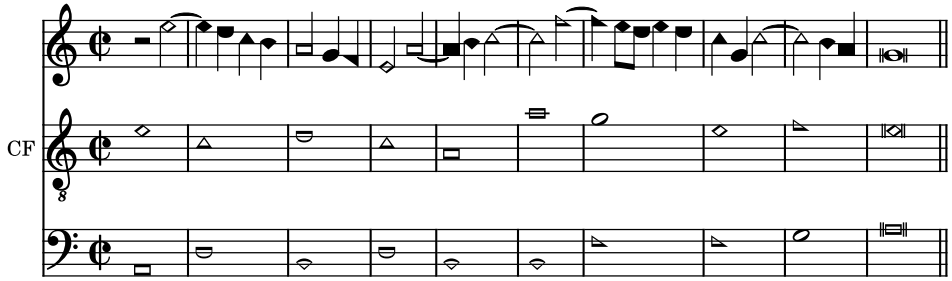
Aloys. What this species is, and of what sort, I assume is clear to you from what we said about it in the bicinium: namely, that it is the composition

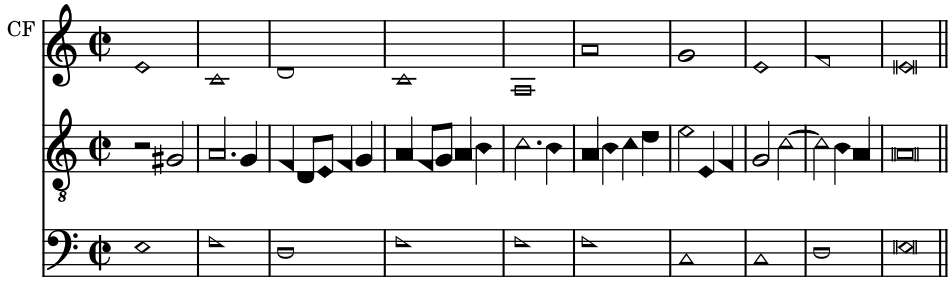
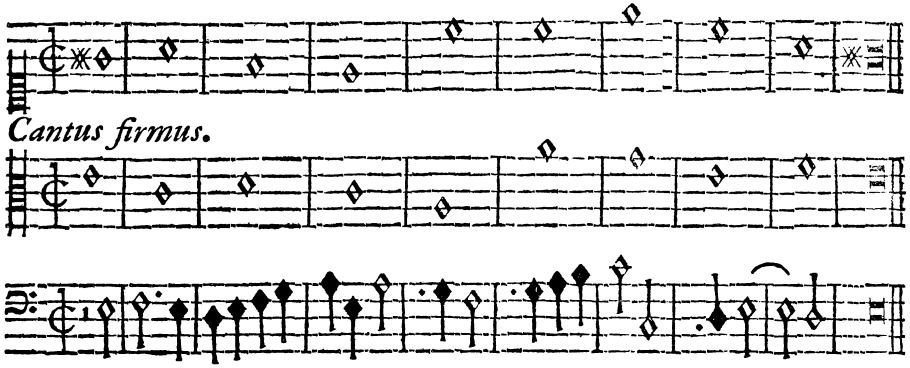
poſitionem omnium quinque ſpecierum, pulchrâ concinnaque canendi ratione ordinatam. Quomodo autem reliquarum duarum partium ſemibrevibus conſtantium, inſtituendæ ſint concordantiæ, ex Tricinii exercitationibus hucuſque ventilatis tibi jam innoteſcere minimè dubito : adeò ut hìs ulteriùs explicandis hîc immorari ſupervacaneum exiſtimem. Unde ſinè mora ad exempla procedamus.



of all five species, arranged in a beautiful and graceful manner of singing. How the consonances of the other two parts, consisting of whole notes, are to be established, I do not at all doubt that you now understand from the tricinio exercises discussed thus far; so that I consider it unnecessary to dwell further on explaining them here. Therefore let us proceed without delay to the examples.



Cantus firmus.

Cantus firmus.

Cantus firmus.







Aloyſ. Sufficiant pro nunc hæc exempla : ſi reliquorum quatuor tonorum exercitationes privato ſtudio, & pari diligentiâ proſequêris, perfectam hujus ſpeciei conficiendæ cognitionem facili ratione adipiſceris. Ubi animadvertendum, motum obliquum ut plurimùm in quovis tactu magnam labori huic addere facilitatem. Nunc ad Quatricinium.

EXERCITII III.

LECTIO PRIMA.

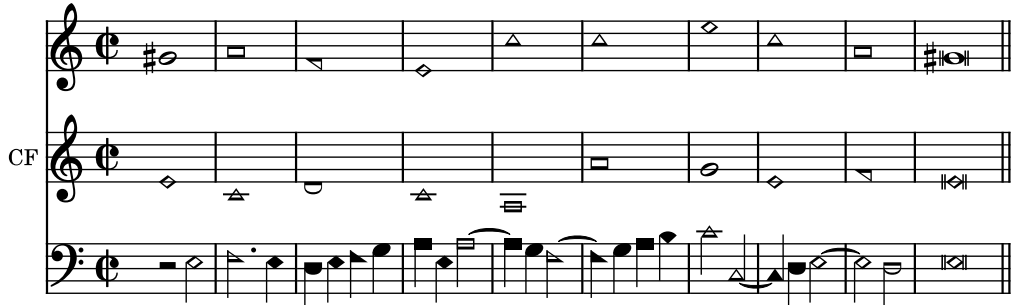
De Quatricinio,

Sive

Quatuor partium Compoſitione.

PErfectam Triadis harmonicæ rationem Tricinio, ſive trium partium Compoſitione contineri jam ſuprà dictum eſt. Inde patet, accedenti parti quartæ locum aliter non eſſe, nisì per duplicationem cujusdam Conſonantiæ in Tricinio jam poſitæ exceptis quibusdam diſſonantiarum Hypotheſibus, alio loco demonſtrandis. Tametſi ratione intervalli, ſitusque inter Uniſonum, & Octavam maxima ſit diſparitas ; reſpectu tamen vulgaris denominationis omnino nulla eſt : æquè enim Uniſonus, quàm Octava v. g. C. appellari, & Octava ferè Uniſonus repetitus æſtimari conſuevit.

Itaque Quatricinii ſiſtema præcipuè Tertià, Quintâ, atque Octavâ conſtabit. Ubi propter vitioſas progreſſiones (quod crebriùs contingit) Octava haberi non poterit, illius loco Tertia frequentiùs, Sexta rariùs duplicanda erit. Cæterùm progreſſionum, motuumque Regulæ Libro primo præſcriptæ, quoad fieri



Aloys. Let these examples suffice for now; if you pursue the exercises of the remaining four modes in private study and with equal diligence, you will easily acquire a perfect understanding of how to construct this species. Note that oblique motion in almost every beat adds great ease to this work. Now to the quatricinium (four-voice composition).

EXERCISE III.

LESSON ONE.

On the Quatricinium,

or

The Composition of Four Parts.

It has already been stated above that the full nature of the harmonic triad is contained in the tricinio, that is, in three-part composition. From this it is clear that when a fourth part is added, it can have no other place than by doubling some consonance already present in the tricinio, except for certain hypotheses of dissonances to be shown elsewhere. Although, as an interval and in its placement between unison and octave, there is a great disparity, yet with respect to ordinary naming there is none at all: for both the unison and the octave—for example, G—are called by the same name, and the octave is regarded almost as a repeated unison.

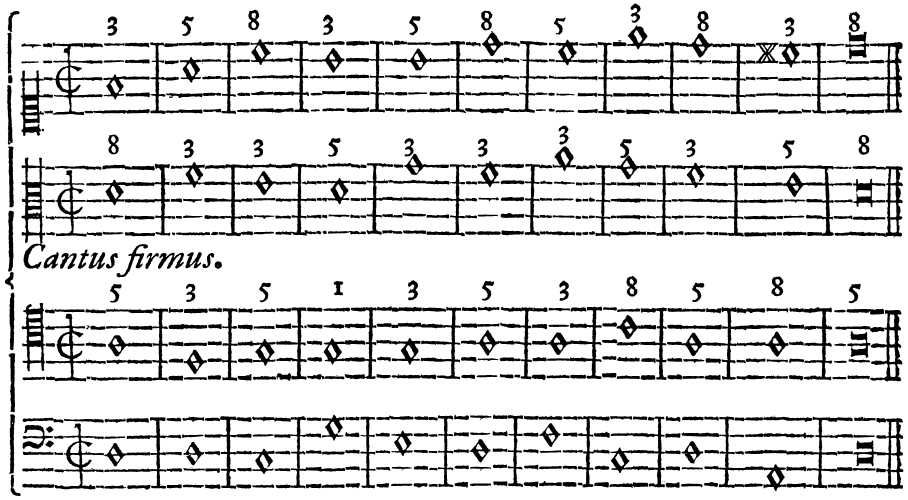
Thus the system of the quatricinium (four-voice texture) will chiefly consist of the third, the fifth, and the octave. When, because of faulty progressions (which frequently happens), the octave cannot be used, the third must more often, and the sixth more rarely, be doubled in its place. Otherwise, the rules of progressions and motions prescribed in the first book are to be observed as far as possible,

fieri poterit, obſerventur ; habito reſpectu non ſolùm partium ad fundamentum, verùm etiam partium ad partes inter ſe. Dixi: quoad fieri poterit, quia nonnunquam neceſſitas feret, aut ratio canendi, aut imitatio, aut ſubjecti rigor, ut Quintarum, vel Octavarum cooperta conſecutio toleranda ſit.

Interim quò minùs à communibus regulis aberrare contigerit, eò perfectior reſultabit Quatricinii Compoſitio.

Joſeph. Per caliginem hanc orationem tuam perſpicere mihi videor. Unde ad rem clariùs intelligendam, uno, vel altero exemplo mihi opus eſt.

Aloyſ. Sequenti Paradigmate mentis tuæ caligo facilè diſſipabitur.



Ne mirare, Conſonantias, vel intervalla compoſita contra morem hucusque uſurpatum numeris ſimplicibus notata ; id enim haud alium in finem faciendum putavi, quàm ut Conſonantiarum duplicatio magìs perſpicua ſe oculis exhiberet. Hanc ergo formulam imitatione perſequere ; & ſi quid adhuc dubii tibi ſupereſt, indica : ſi nihil, hunc ipſum Cantum firmum, ſingulis in partibus ſucceſſivè exiſtentem ſolitâ mutatione pertractandum aſſume.

Joſeph. Intereſt ne, quænam Conſonantiarum cuilibet tribuatur parti?

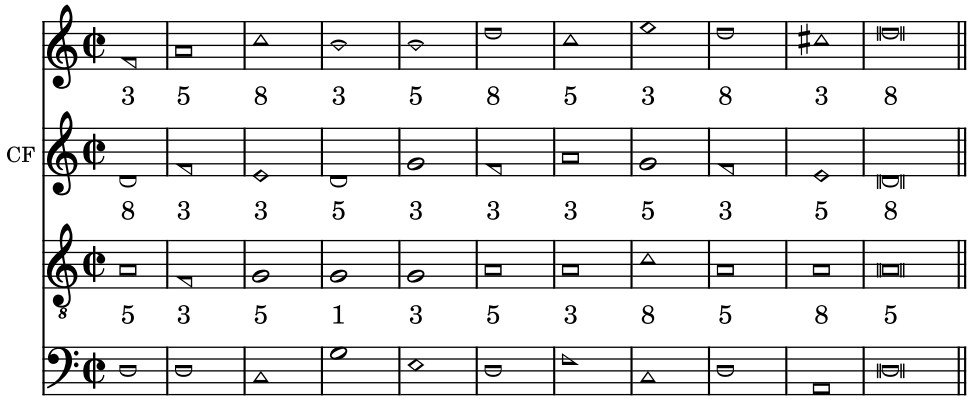
Aloyſ. Intereſt vel maximè ; quod tibi ex Tricinii exercitationibus, & dictis jam innotuiſſe exiſtimo. Præterquam enim, quòd ordine naturali, ubi fieri poteſt, quælibet Conſo-

with regard not only to the parts in relation to the foundation, but also to the parts in relation to one another. I said "as far as possible," because necessity will sometimes demand—be it the nature of the melody, or imitation, or the strictness of the subject—that a hidden succession of fifths or octaves be tolerated.

Meanwhile, the less you are forced to depart from the common rules, the more perfect the composition of the quatricinium will be.

Joseph. I seem to see your discourse, though as through a mist. Therefore, to understand the matter more clearly, I need one or two examples.

Aloys. The following pattern will easily disperse the mist in your mind.



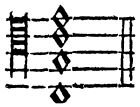
Do not be surprised that compound consonances or intervals, contrary to the custom observed thus far, are here marked with simple numbers; I thought this should be done for no other reason than that the duplication of the consonances might present itself more clearly to the eye. Imitate this pattern, then; and if any doubt still remains, point it out. If not, take up this same cantus firmus (fixed melody), treating it successively in each part with the usual alternation.

Joseph. Does it matter which consonances are assigned to which part?

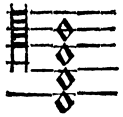
Aloys. It matters greatly; and I think this is already known to you from what has been said and practiced in the tricinio. For besides the fact that, in the natural order, each consonance claims its own proper place where this can be done, it is very important to foresee whether, from the system of the first beat disposed in such a way, the progression to the second, third, or even fourth beat can rightly be made.

nantia proprium ſibi locum vindicet, plurimùm refert circumſpicere, an à primi tactûs ſiſtemate tali modo diſpoſito, progreſſio ad ſecundum, tertium, vel etiam quartum tactum inſtitui ritè poſſit: ſi minùs, mutanda erit primi tactûs ſtructura; atque ita collocandæ erunt Conſonantiæ, ut ad ſequentes tactus commoda, rectaque fiat progrediendi facultas.

Joſeph. Qualis eſt autem, de quo dixiſti, proprius ille locus, quem Conſonantiæ ſibi vendicant?

Aloyſ. Eſt ordo ex diviſione Octavæ harmonicè inſtitutâ productus. Patet ergo ex tali diviſione primò generari Quintam, ex ulteriori Quintæ diviſione Tertiam produci: qui ordo in Conſonantiis collocandis tenendus eſt; niſi alia, nempe in ſequentes tactus progrediendi ratio, quod frequentiſſimè fiet, impediat. Subjicio exemplum, naturalem Conſonantiarum ordinem exhibens:

Vides primo loco Quintam poſitam, generatam ex Octavæ diviſione: Secundò Octavam per ſe exiſtentem: tertiò Tertiam, ſive Decimam per Quintæ diviſionem productam.

Joſeph. Secundùm ſiſtema hodiernæ noſtræ Claviaturæ videtur Tertia primo loco, atque ante Quintam eſſe ponenda, proinde Quatricinii ſiſtema ſequenti ordine collocandum.

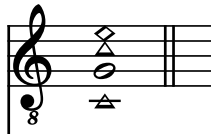


Aloyſ. Videtur quidem; ſed re ipsâ non eſt ita; Nam præterquam, quòd ordinis judicium à Diviſione, non à Claviaturæ diſpoſitione ſumendum ſit, accedit, quòd Tertia in loco gravi, proximoque parti fundamentali poſita confuſum, & obfuſcatum edat ſonum; atque quò majores fuerint termini, ſive numeri intervallum aliquod conſtituentes, eò acutiorem ejusdem intervalli fore ſonum: inde quoque acutiorem ſibi vendicare locum. Porrò termini conſtitutivi Quintæ ſunt, 2. 3. qui per Additionem uniti dant. 5. Termini verò Tertiæ ſunt 4. 5. uniti faciunt. 9. Inde naturali ratione Quintam in graviori, Tertiam in acutiori loco ponendam eſſe conſtat. Per-

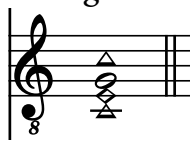
If not, the structure of the first beat must be changed, and the consonances so placed that a convenient and correct possibility of proceeding to the following beats is obtained.

Joseph. And what is that proper place which, as you said, the consonances claim for themselves?

Aloys. It is the order produced from the division of the octave structured harmonically. From such a division it is clear that the fifth is generated first; and from the further division of the fifth, the third is produced. This order is to be maintained in placing the consonances, unless another consideration—namely the way of progressing to the later beats, which will very often happen—prevents it. I subjoin an example showing the natural order of the consonances:

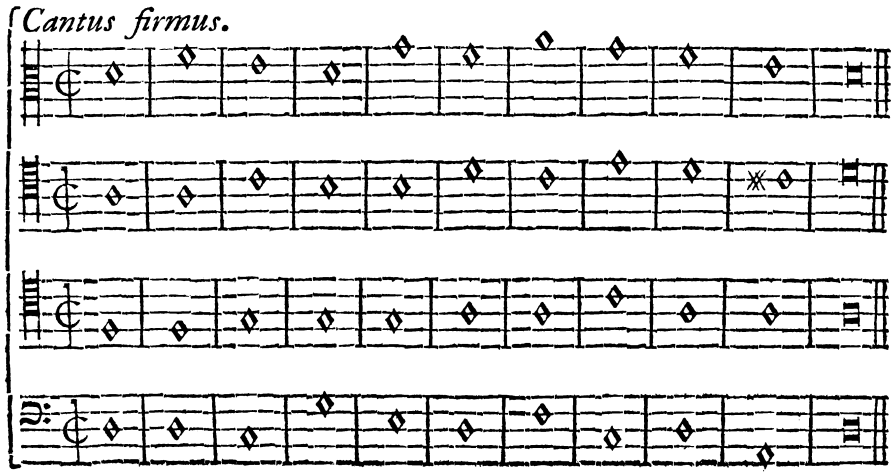


You see that in the first place stands the fifth, generated from the division of the octave; in the second, the octave existing by itself; in the third, the third or the tenth, produced by the division of the fifth.

Joseph. According to the system of our modern keyboard, the third seems to be placed first and before the fifth; therefore the system of the quatricinium should be arranged in the following order.

Aloys. It may seem so, but in reality it is not so. For besides the fact that judgment about order must be taken from division, not from the arrangement of the keyboard, it is also the case that the third, when placed in the low register near the fundamental part, produces a confused and darkened sound; and the larger the terms, that is, the numbers constituting an interval, the higher will be the pitch of that interval, and thus it will claim a higher place. Now the constitutive terms of the fifth are 2 and 3, which joined by addition give 5. The terms of the third are 4 and 5, which joined give 9. Hence, by natural reasoning, the fifth must be placed in the lower, the third in the higher position.

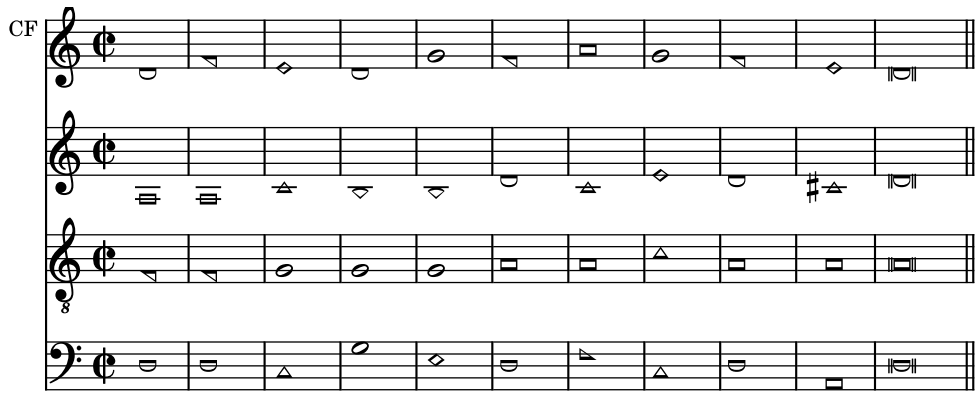
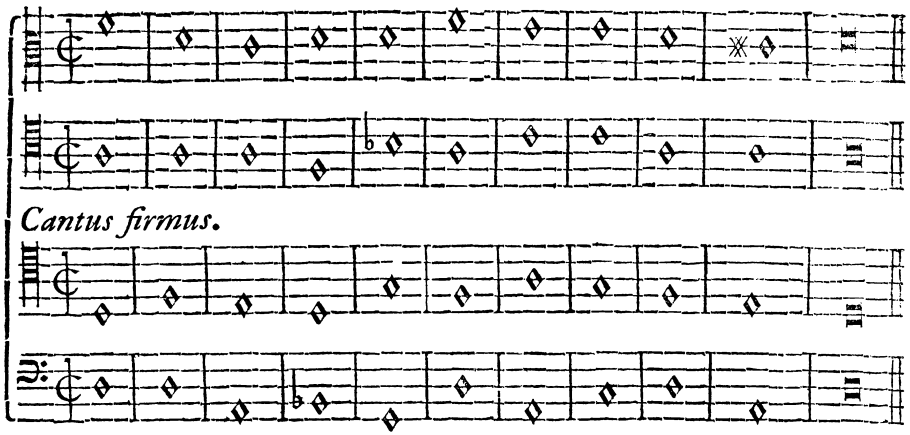
Perge nunc, ſi nihil dubii tibi reſtat, ad reſtantes hujus, & reliquorum tonorum lucubrationes, omnium tactuum ergaſtula circumſpiciendo, comparandoque partes ſingulas ſingulis, ne quid regulis adverſum obrepere contingat: ad quod perficiendum magnâ circumſpectione opus eſt; ubi animadvertendum, ut non ſolùm partes cum Baſſo, ſive parte fundamentali, verùm etiam partes mediæ inter ſe ad regularum tenorem ritè concordent.

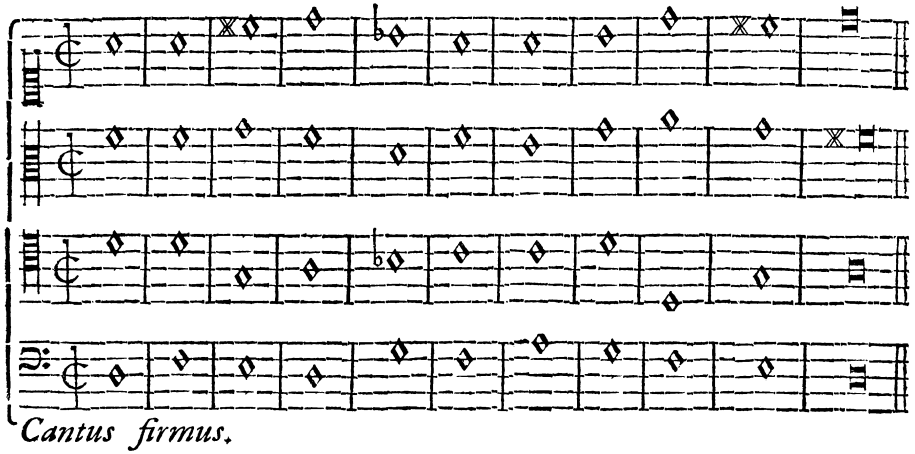
Joſeph. Ex præcedenti Quatricinii exemplo comperi, inter partes medias nonnunquam Quartam evenire, quam diſſonantiam eſſe ſtatuiſti, quamque prohibitam in Compoſitione meris ſemibrevibus conſtante affirmâſti; quod absque ſcrupulo præterire non potui.

Aloyſ. Rectè quidem. Verùm recordandum eſt, naturam Conſonantiarum, diſſonatiarumque habito reſpectu ad partem fundamentalem metiendam eſſe; quidquid inter partes medias inde eveniat: modò alia abſurda, ut conſecutio duarum Quintarum, aut Octavarum evitetur. Cæterùm attendendum eſt, quò contractiores in anguſtum partes, eò perfectiorem reſultare Harmoniam, juxta illud: Vis unita fortior. Si quid ardui tibi objicietur, ùt fieri ſecùs haud poteſt; cogita ad virtutem per aſpera aſcendendum eſſe: neque Athletas absque adverſario ad Olympia deſcendere; Indefeſſo ſtudio, conſtantiâ, ſcutoque patientiæ in certamine opus eſſe.

Now, if no doubt remains for you, proceed to the remaining studies of this and the other modes, inspecting carefully the "workshops" of all the beats and comparing each part with each, lest anything contrary to the rules creep in. To achieve this, great circumspection is needed; you must observe that not only the parts with the bass, or fundamental part, but also the middle parts among themselves must agree rightly according to the tenor of the rules.

Joseph. From the preceding example of the quatricinium I have learned that between the middle parts a fourth sometimes occurs—a dissonance which you have declared, and which you affirmed was forbidden in a composition consisting entirely of whole notes; something I could not pass over without scruple.

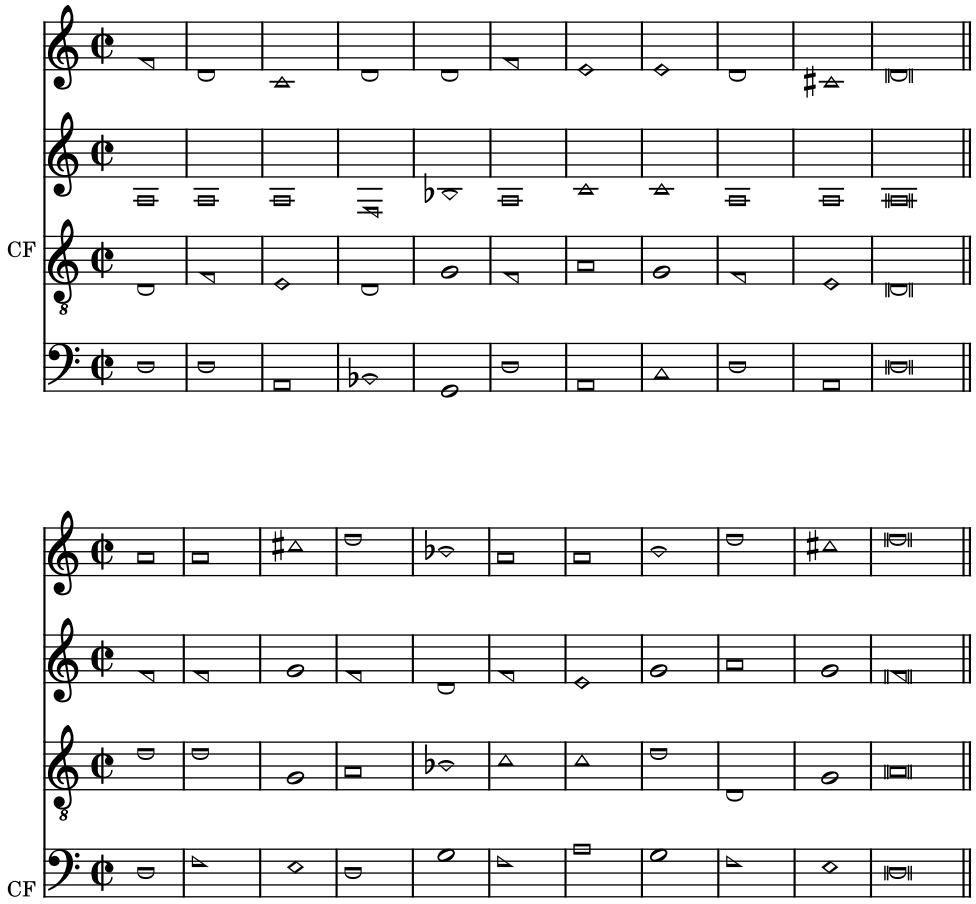
Aloys. Quite right. But it must be remembered that the nature of consonances and dissonances must be measured with respect to the fundamental part; whatever happens between the middle parts is of no consequence, provided other absurdities—such as the succession of two fifths or two octaves—are avoided. Moreover, it must be noted that the more closely the parts are drawn together into a narrow compass, the more perfect the resulting harmony, according to the saying Vis unita fortior ("United strength is stronger"). If anything difficult presents itself to you—as indeed cannot fail to happen—remember that one must ascend to virtue through hardships, and that athletes do not go down to the Olympic games without an adversary. Unwearied study, constancy, and the shield of patience are needed in the contest.





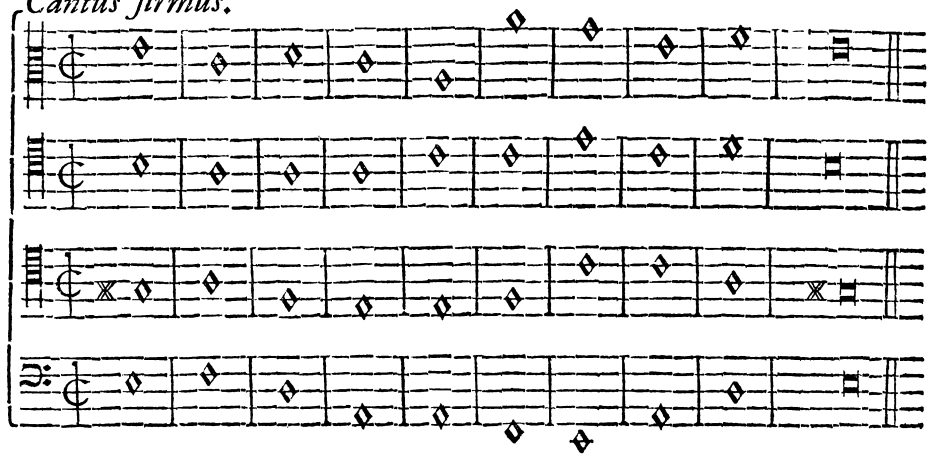
Joſeph. Non ſinè hæſitatione in primis duobus exemplis partem Tenoris contra admonitionem tuam ita Baſſo appropinquavi, ut Tertiis in gravi conſiſtere plerumque contingat; quod feci, quia aliter fieri poſſe, non inveni, propter neceſſitatem Cantûs firmi in omnibus quatuor partibus collocandi. Quapropter ea cenſuræ tuæ, correctionique ſubmitto.

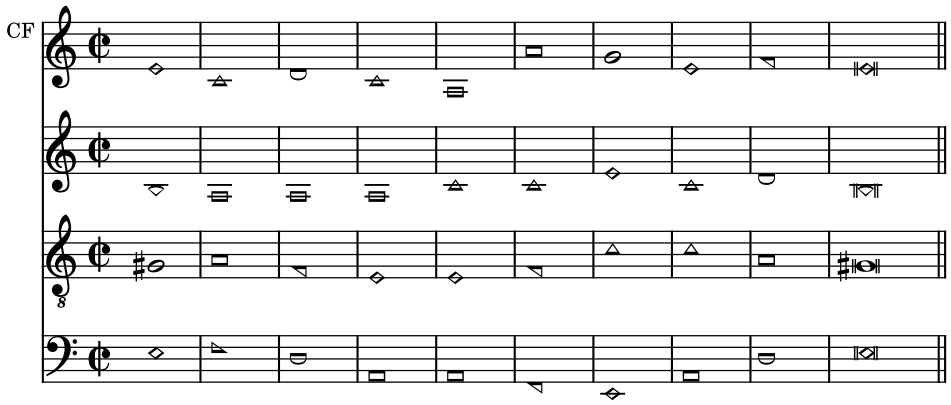
Aloyſ. Verum eſt, ſtante Cantûs firmi neceſſitate, Paradigmata illa exercitii gratiâ tantum inſtituta fieri non potuiſſe, ſecùs res ſe habebit, ubi inventio à proprio arbitrio tuo dependet. Quantum autem utilitatis exercitia hæc Studioſis afferant, ſucceſſu temporis admirabundus comperies. Duorum adhuc tonorum exempla ſolitâ quadruplici Cantûs firmi mutatione perficienda reſtant. Perge.

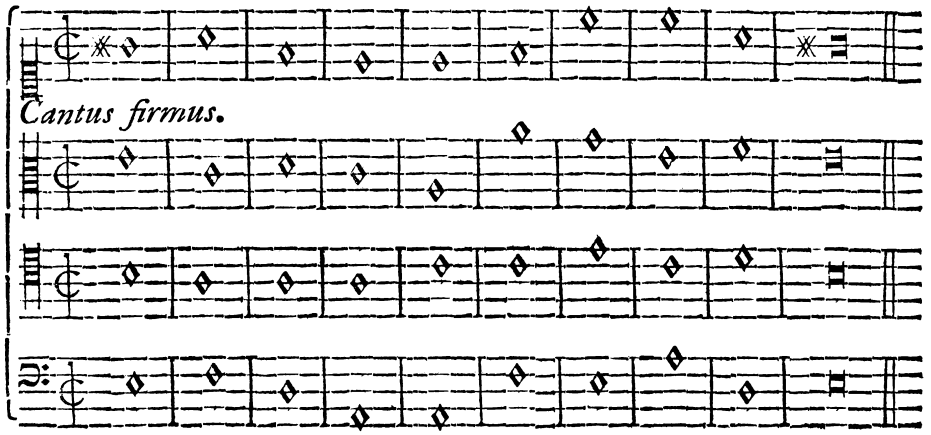
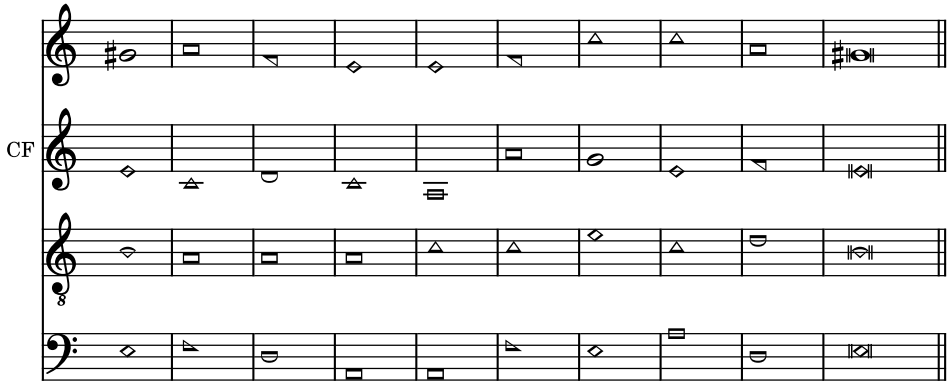
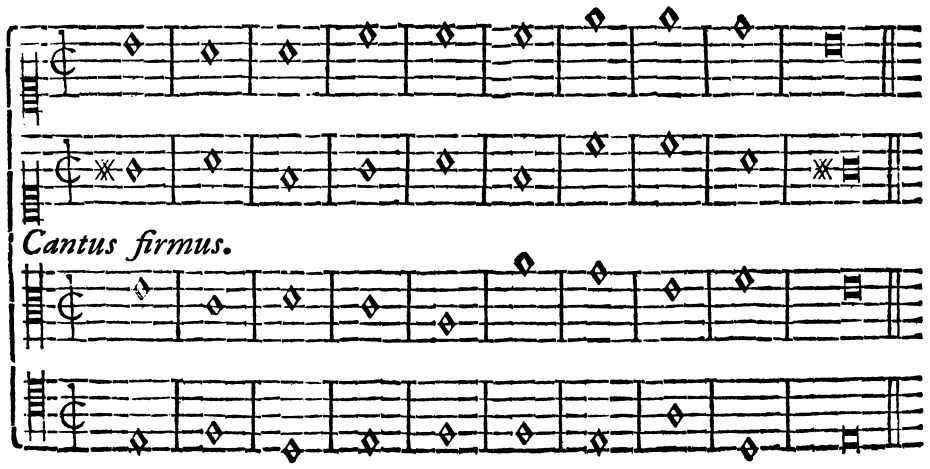
Can-

Joseph. Not without hesitation, in the first two examples, I brought the Tenor so near the Bass—contrary to your warning—that it often happens that they stand in thirds in the low register; I did this because I found no other way, owing to the necessity of placing the cantus firmus (fixed melody) in all four parts. Therefore I submit these to your judgment and correction.

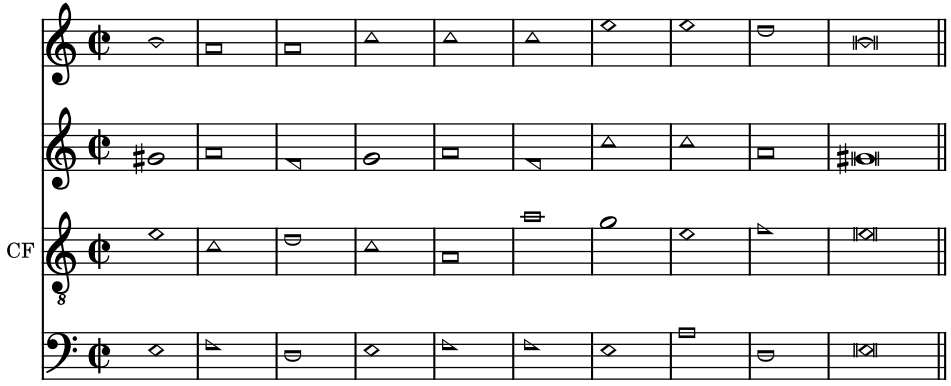
Aloys. It is true that, given the necessity of the cantus firmus, those patterns could not have been constructed otherwise for the sake of exercise; the case will be different when the invention depends on your own discretion. But you will discover with admiration, as time goes on, how much usefulness these exercises bring to students. Two examples of the remaining modes still remain to be completed with the usual fourfold alternation of the cantus firmus. Continue.

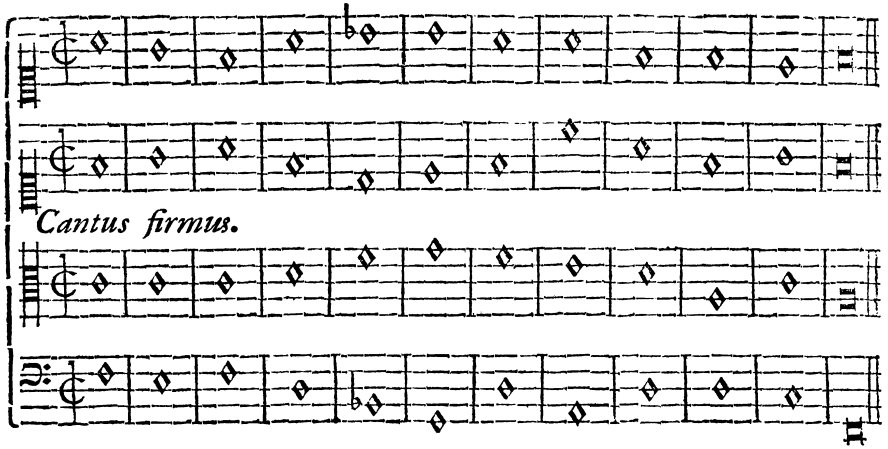
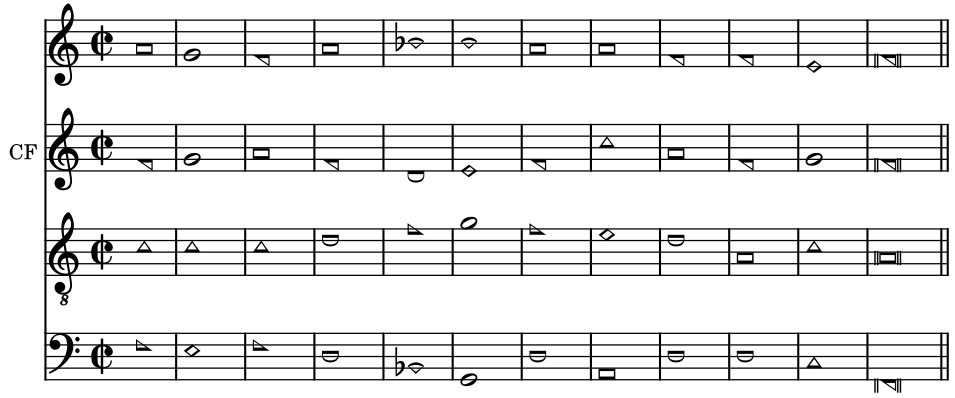
Cantus firmus.










Cantus firmus.

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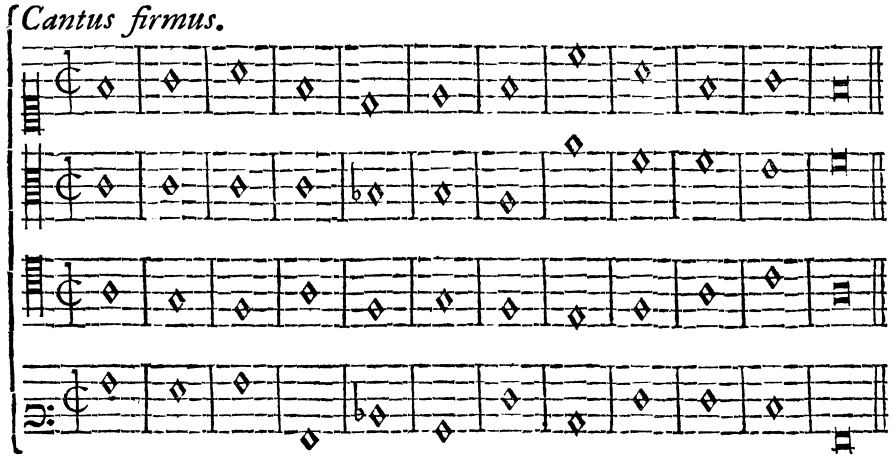
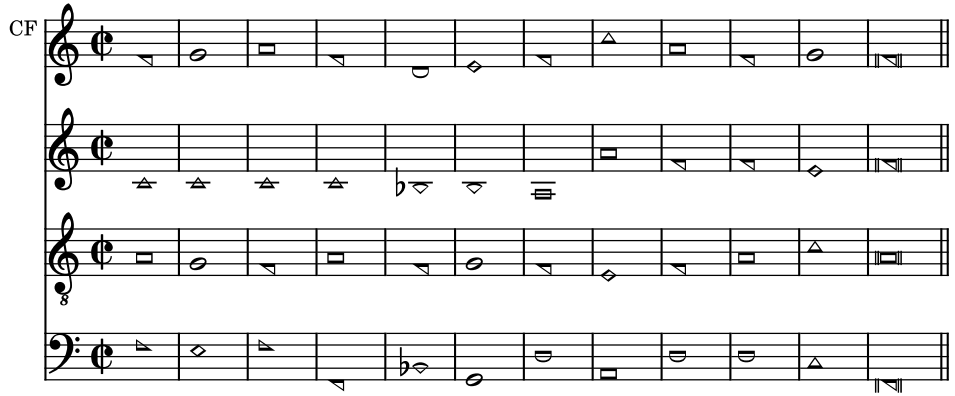

Cantus firmus.

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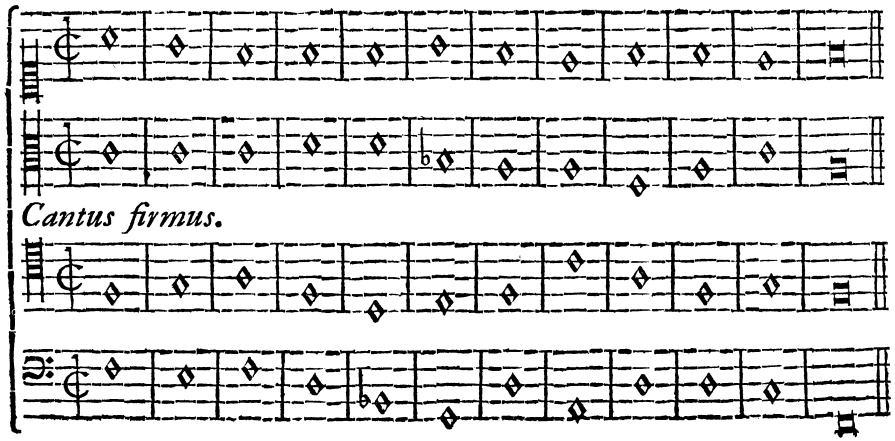
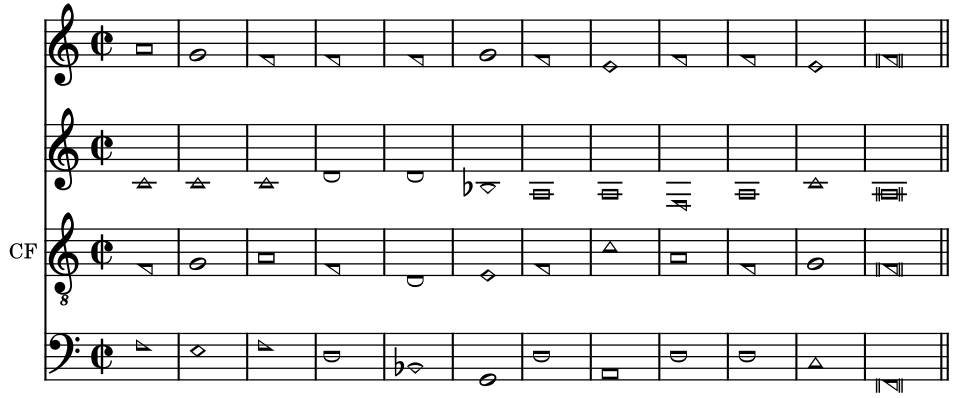

Cantus firmus.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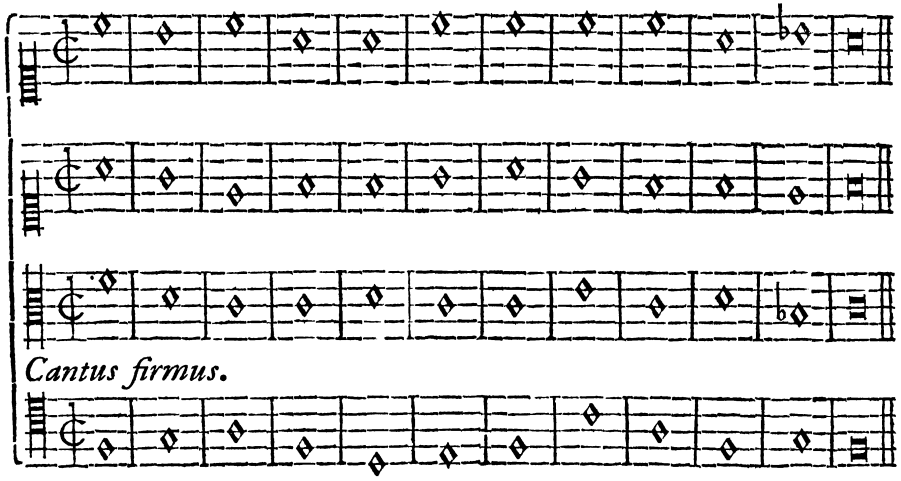
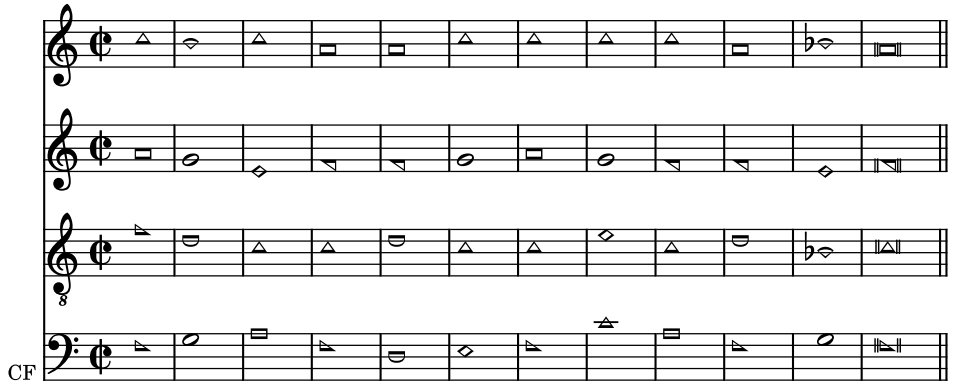


Hæc exempla utcunque bene facta ſunt : neque immutabilis Cantûs firmi neceſſitas admittit , ut compoſitio talis ad motuum, aliarumque regularum rigorem ad amuſſim elaborari poſſit , ùt facilè fit in compoſitione libera. Reliquorum trium tonorum exempla pari ratione quadruplicique Cantûs firmi poſitione perficienda ſunt.

EXERCITII III.

LECTIO SECUNDA,

De Minimis contra Semibrevem.

Hîc in memoriam revocare neceſſe eſt , quæ ſuprà de hac Contrapuncti ſpecie in Tricinio diximus ; quæ omnia in hac Quatricinii compoſitione quoque tenenda ſunt : neque alia differentia eſt, quàm quòd ibi duæ Minimæ cum duabus ſemibrevibus , hîc verò cum tribus concordare debeant : obſervatis de cætero , quæ de Quatricinio Notæ contra Notam dicta ſunt : quantum tamen reſtrictum hujus ſpeciei ligamen permittet. Ad exercitium exemplorum perge.

These examples are fairly well done; and the unchangeable necessity of the cantus firmus does not allow such a composition to be worked out with exact rigor according to the rules of motion and other regulations, as is easy in free composition. The examples of the remaining three modes must be completed in the same manner, with the cantus firmus placed in four positions.

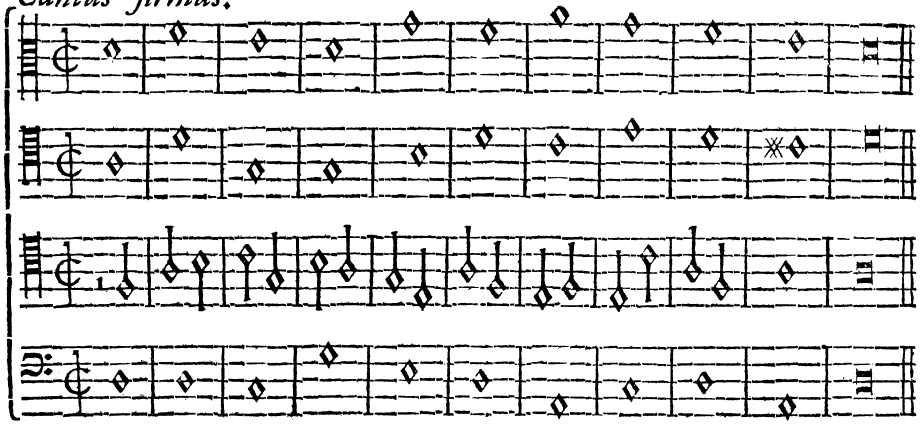
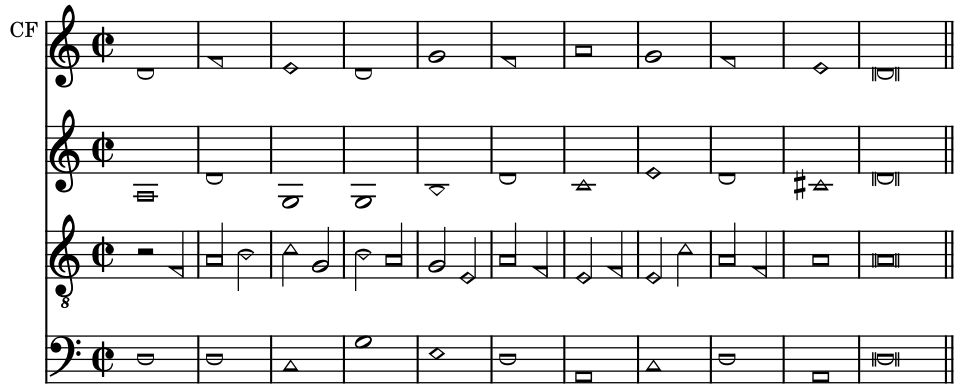
EXERCISE III

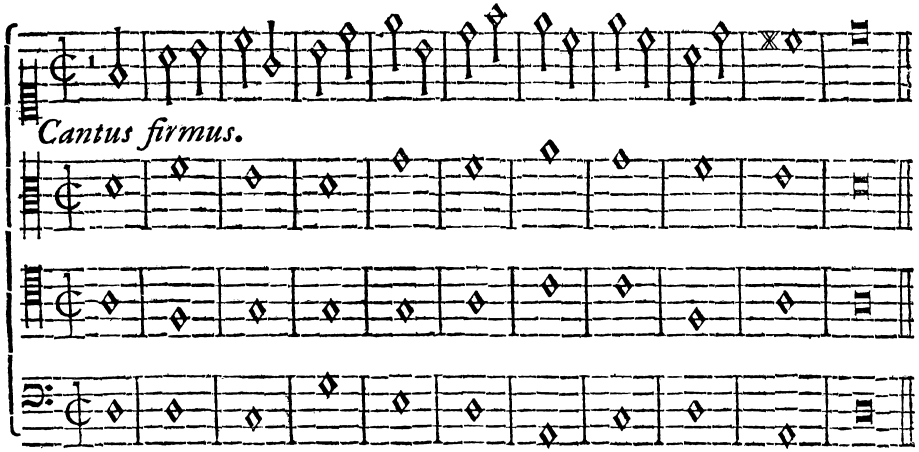
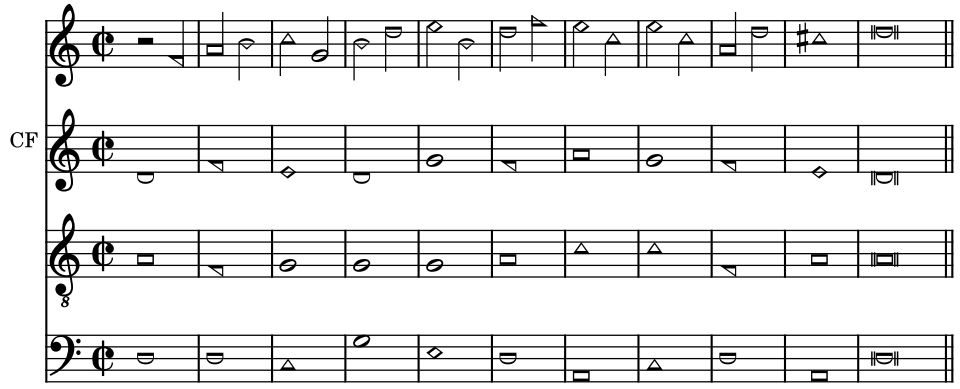
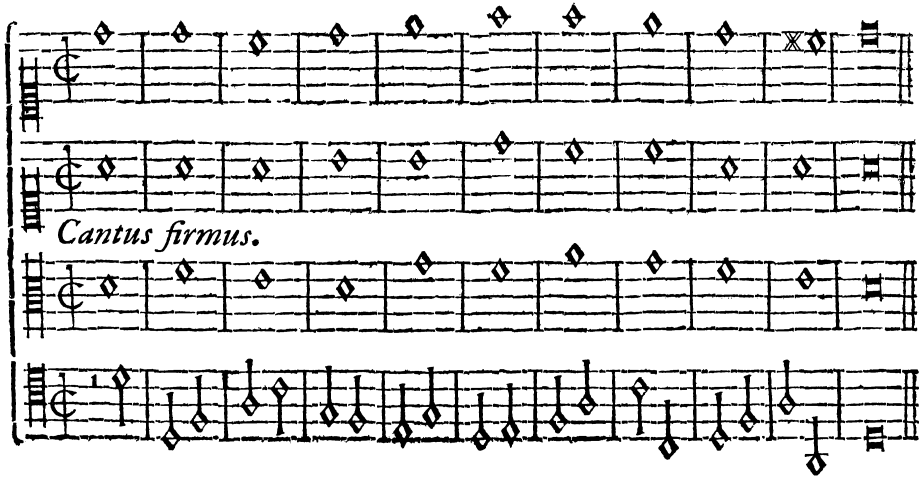
LESSON TWO,

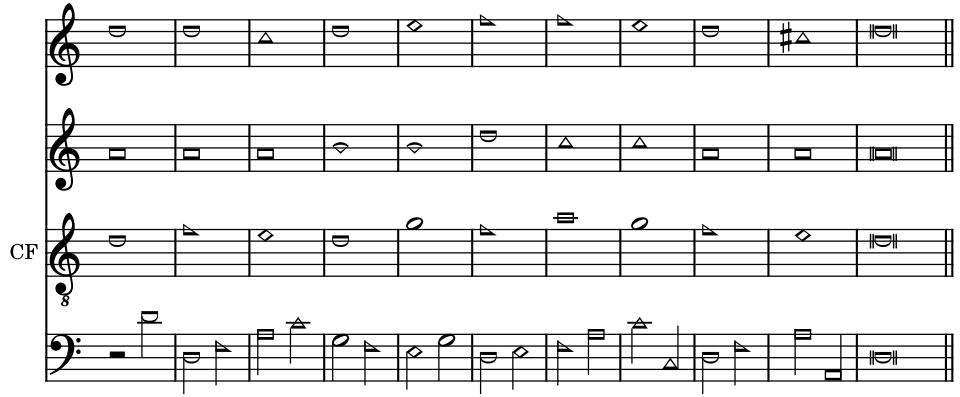
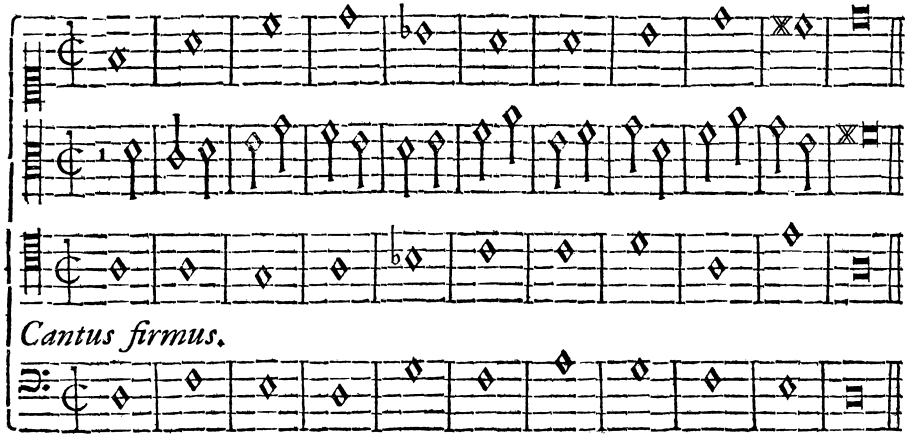
On half notes against the whole note.

Here it is necessary to recall what we said above about this species of counterpoint in the tricinio; all of which must also be observed in this composition of the quatricinium. The only difference is that there two half notes must agree with two whole notes, but here with three; and otherwise, what was said about Note against Note in the quatricinium must be observed, as far as the restricted nature of this species allows. Proceed to the exercise of examples.

Cantus firmus.







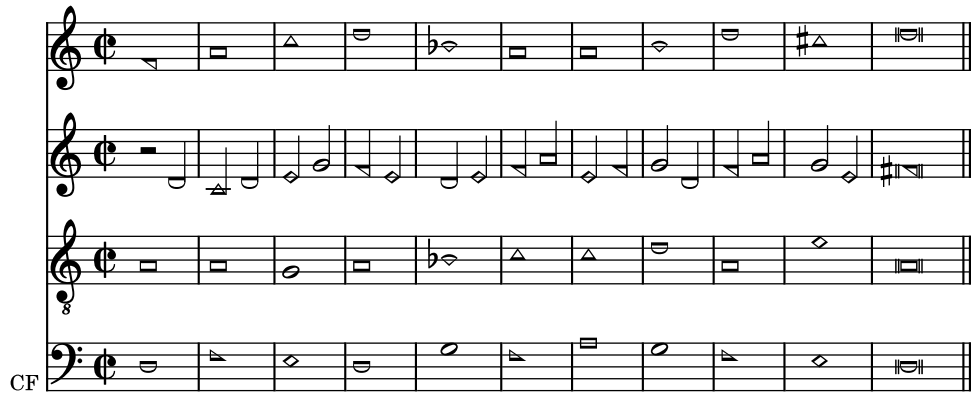
Joſeph. Difficultate plenam experior hanc Contrapuncti Speciem, nec ulla difficilior hucuſque mihi viſa fuit, ità ut penultimam eâ lege nullo modo proſequi potuerim.

Aloyſius. Ità eſt. Verùm hujus rei gravitas oritur ex obligatione ponendi duas Minimas contra tres Semibreves: non alium in finem inſtituta, quàm ut concordandi ſcientiam adipiſcaris, teque cautum, circumſpectum, & attentum in ſtruendis Concordantiis reddat. Inde mirum non eſt, in quibusdam Hypotheſibus proſequendi facultatem non reperiri: neque in ullo libero componendi genere uſuveniet, ut tanta tactuum ſeries hac lege protrahenda ſit. Non enim usûs, ſed exercitii gratiâ lectiones hæ adinventæ ſunt: quemadmodum Syllabizzatio à legere ſciente non ampliùs practicatur; ità quoque Contrapuncti Species ad addiſcendum ſolummodo præſcribuntur. Cætera tonorum exempla domeſtico ſtudio tuo committo.

EXERCITII III.

LECTIO TERTIA,

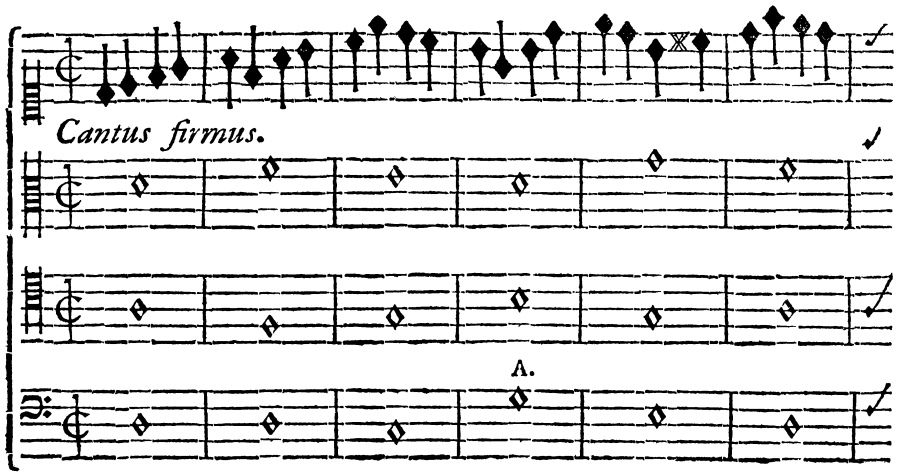
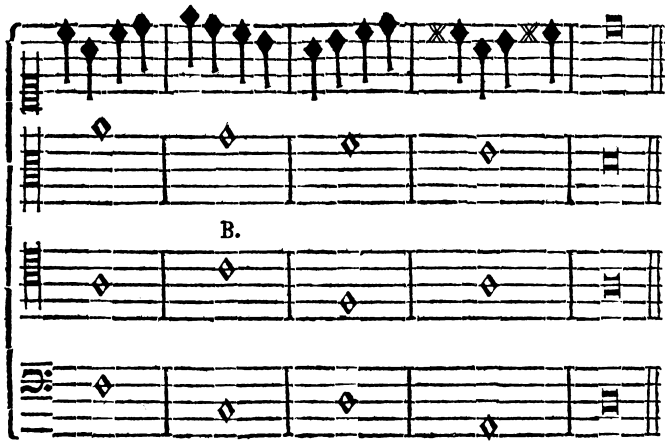
De Semiminimis contra Semibrevem.

QUæ hujus Speciei requiſita ſint, ex iis, quæ de hac in compoſitione duarum, triumque partium, dicta ſunt, petantur: adeò ut hic nihil aliud addendum ſit, præter Concordantiæ diſparitatem. Quemadmodum enim in Bicinio Concordantia cum una Semibrevi, in Tricinio cum duabus inſti-

Joseph. I find this species of counterpoint full of difficulty, and none has seemed harder to me thus far, so that I could not in any way carry out the penultimate beat according to its rule.

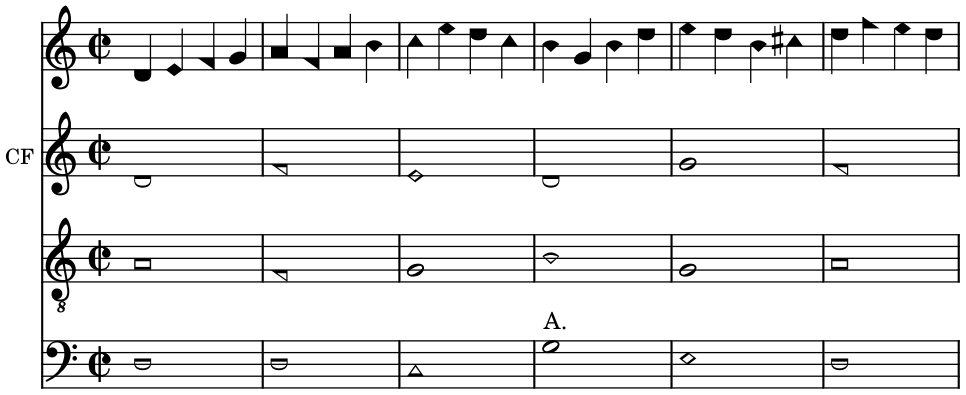
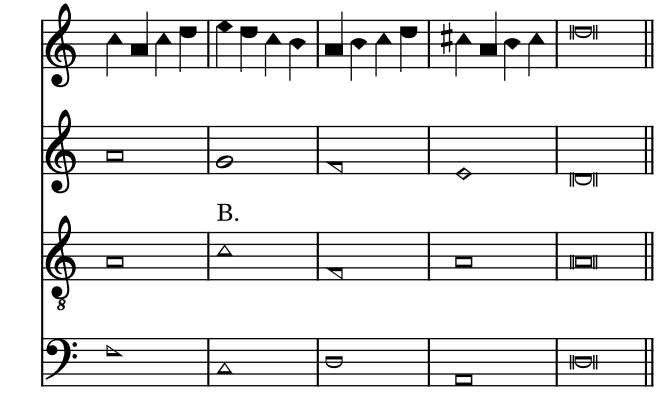
Aloysius. So it is. But the difficulty of this matter arises from the obligation of placing two half notes against three whole notes: instituted for no other purpose than that you may acquire the knowledge of concord, and become cautious, circumspect, and attentive in forming consonances. Hence it is no wonder that in certain hypotheses the possibility of continuing cannot be found; nor will it ever occur in any kind of free composition that such a long series of beats must be carried out under this rule. For these lessons were invented not for use, but for exercise; just as syllabification is no longer practiced by one who knows how to read, so the species of counterpoint are prescribed solely for learning. I commit the remaining examples of the modes to your private study.

EXERCISE III
LESSON THREE,
On quarter notes against the whole note.

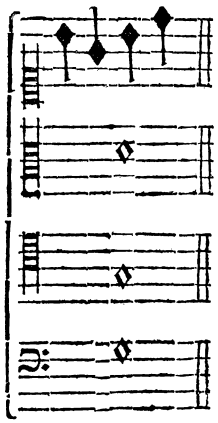
What is required for this species must be gathered from what has been said about it in the composition of two and three parts; so that nothing else need be added here except the difference in concord. For just as in the bicinium the concord must be made with one whole note, and in the tricinio with two,

tuenda eſt : ita in hac Quatricinii Compoſitione quatuor Semiminimæ cum tribus Semibrevibus juxta Harmoniæ regulas aliunde tibi notas concordandæ ſunt. Exemplis res clarior apparebit.



Joſeph. Quare (pace tuâ Magiſter interrogo) in quarto tactu Tertiam duplicâſti? Videtur enim poſito, loco unius Tertiæ Uniſono in parte Tenoris ſequenti modo fieri potuiſſe.

so in this composition of the quatricinium four quarter notes must be made to agree with three whole notes according to the rules of harmony already known to you. The matter will appear clearer from examples.



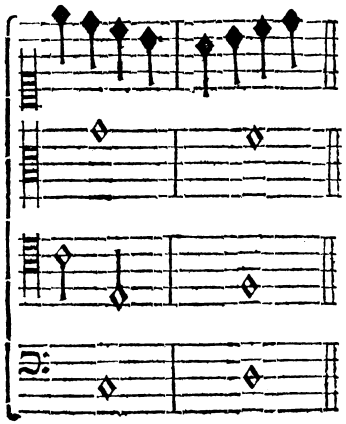
Joseph. Why (with your permission, Master, I ask) did you double the third in the fourth beat? For it seems that, instead of one of the thirds, a unison in the Tenor part could have been used in the following manner:

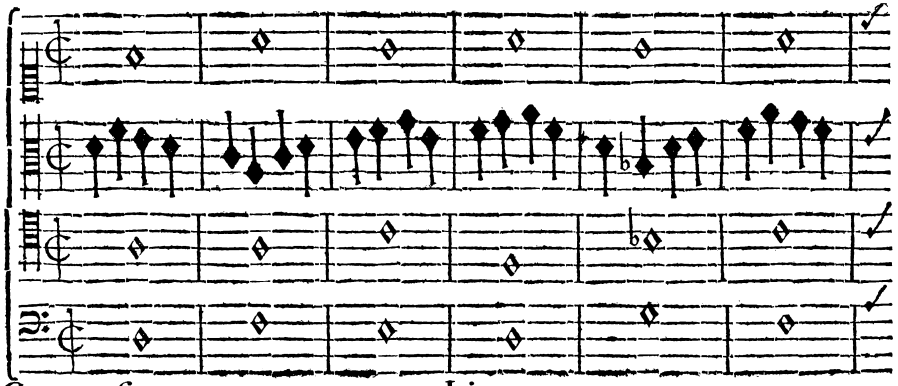


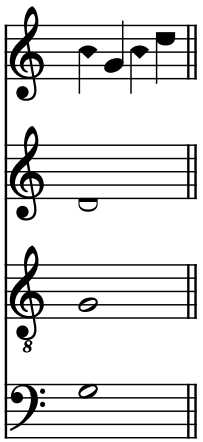
Aloyſ. Reſpondeo, potuiſſe quidem ità fieri; verùm præterquàm, quòd Uniſono in Theſi poſito Compoſitionis plenitudini haud parùm dematur; accedit, quòd Tertia, ſeu Decima in parte Cantûs utpote vagabunda, manca foret; quia non continuò audiretur. Quid ſecundum dubii ſignum B. indicat?

Joſeph. Videtur progreſſio illa inter Altum, & Tenorem vitioſa, quia de perfecta ad perfectam motu recto proceditur.

Aloyſ. Dico, progreſſionem illam ob Semibrevium neceſſitatem aliter fieri non poſſe, ideoque tolerandam eſſe, quæ facilis emendationis foret, ſi Tenoris Semibrevis dividi ſequenti modo poſſet.

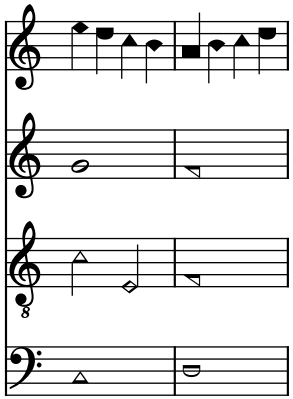


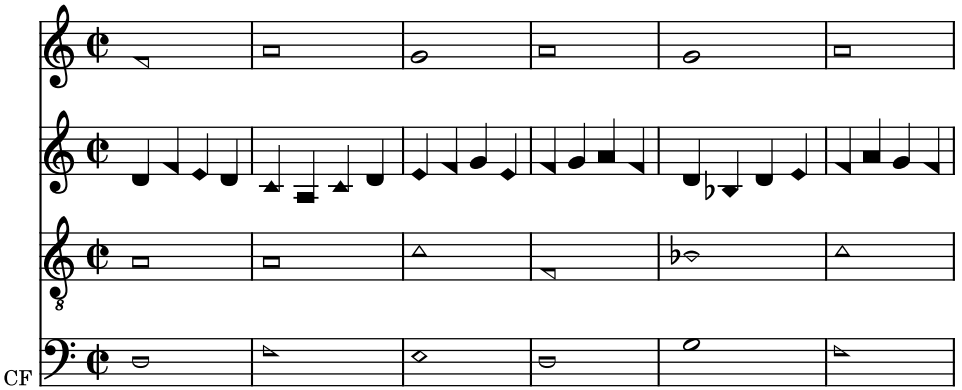
Quod etiam de priorum Specierum exemplis intelligi volo, in quibus multa reperiuntur, quæ abſque Semibrevium neceſſitate tanquam vitioſa rejicienda forent. Nunc ad reliqua exemplaria variatâ Cantûs firmi collocatione progredere.

Cantus firmus.

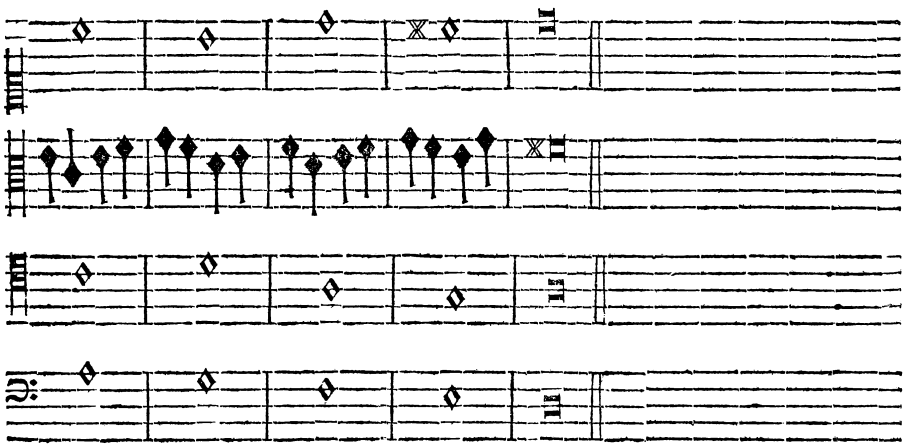


Aloys. I answer that it could indeed have been done so; but besides the fact that a unison placed on the Thesis detracts not a little from the fullness of the composition, it also follows that the third—or tenth—in the melody part, being as it were wandering, would be incomplete, because it would not be heard continuously. What does the second sign of doubt, the letter B, indicate?

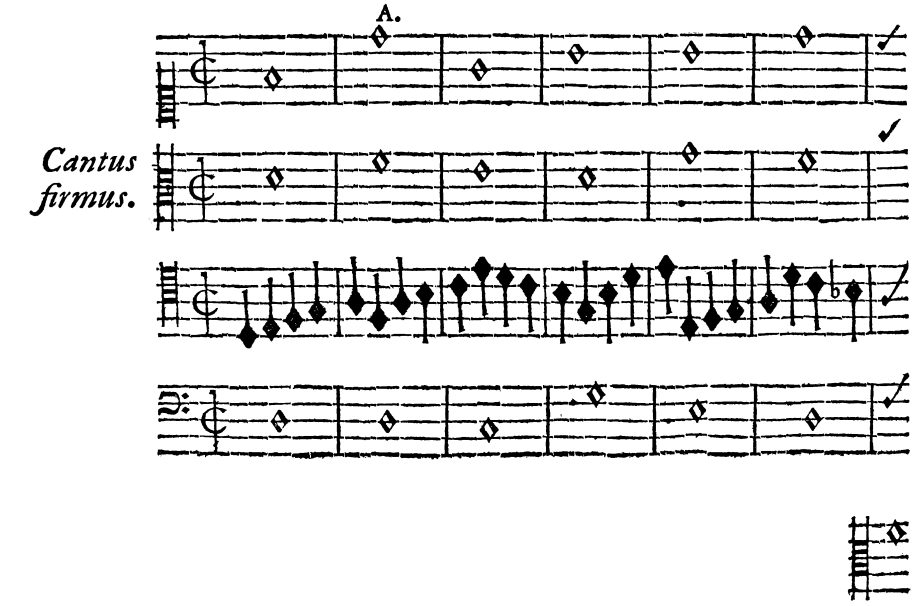
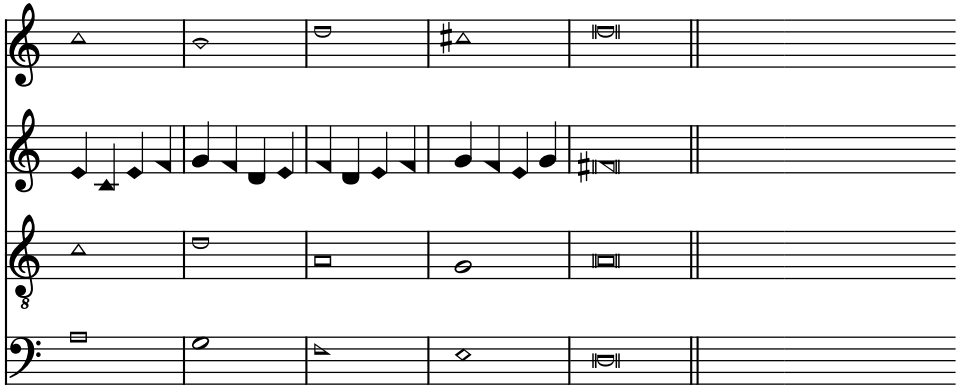
Joseph. It seems that the progression between the Alto and the Tenor is faulty, because it proceeds from perfect to perfect by similar motion.

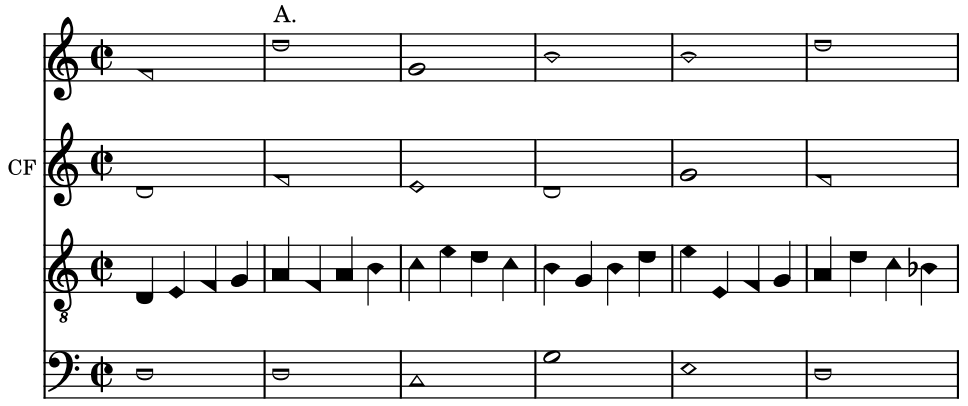
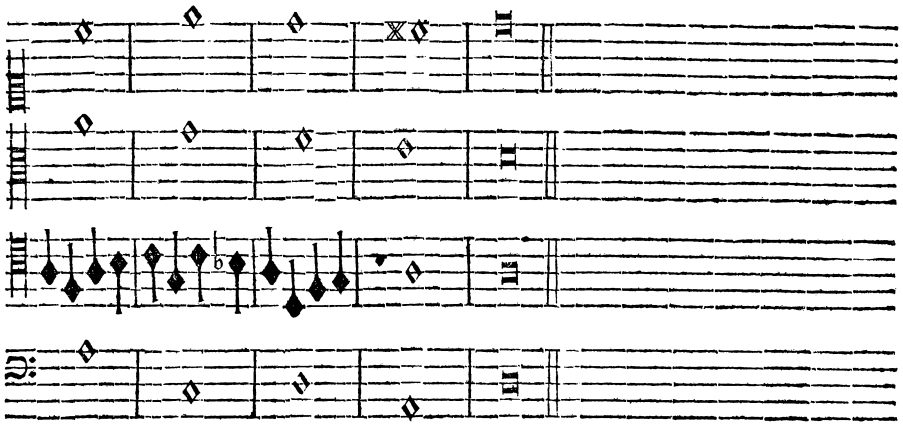
Aloys. I say that this progression, owing to the necessity of the whole notes, cannot be done otherwise, and therefore must be tolerated; it would be easily corrected if the Tenor's whole note could be divided in the following way.

This I also wish to be understood of the examples of the preceding species, in which many things are found that, without the necessity of the whole notes, would have to be rejected as faulty. Now proceed to the remaining examples with varied placement of the cantus firmus.



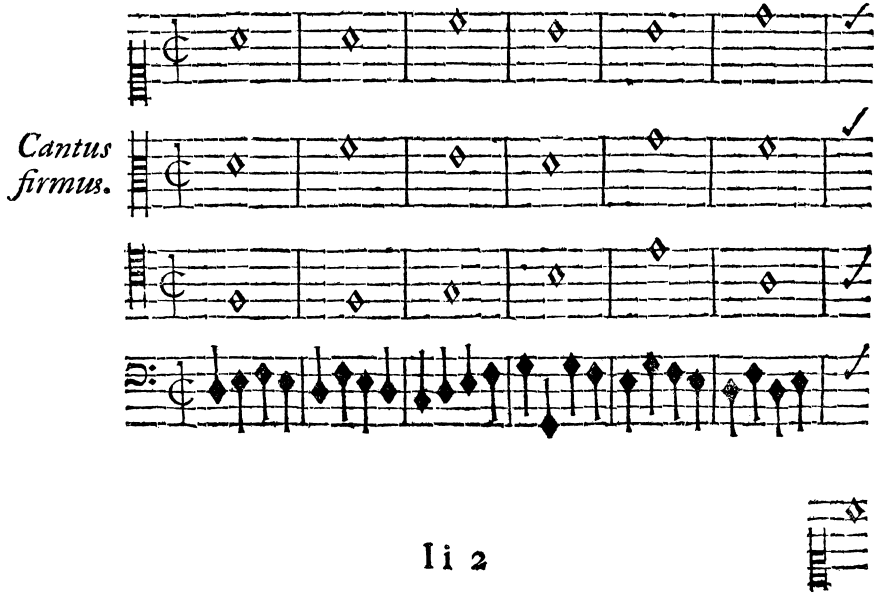


Aloyſ. Egregium, Joſephe, hoc exemplo profectûs tui ſpecimen dediſti: quô prodis, nihil ex his, quæ per tot Lectionum curſum dicta ſunt, memoriâ tibi excidiſſe; quia primâ Semiminimâ ad Harmoniæ completæ rationem poſitâ, cum reliquis tribus hinc inde vagando planâ viâ in ſequentem tactum ingreſſus es, cæteraque Harmoniæ, ſpecierumque præcepta ritè obſervâſti: quod fit, quum Theſis, ſive cujusvis tactûs principium completâ Harmoniâ, id eſt, Tertia, & Quintâ proviſum eſt. Perge.

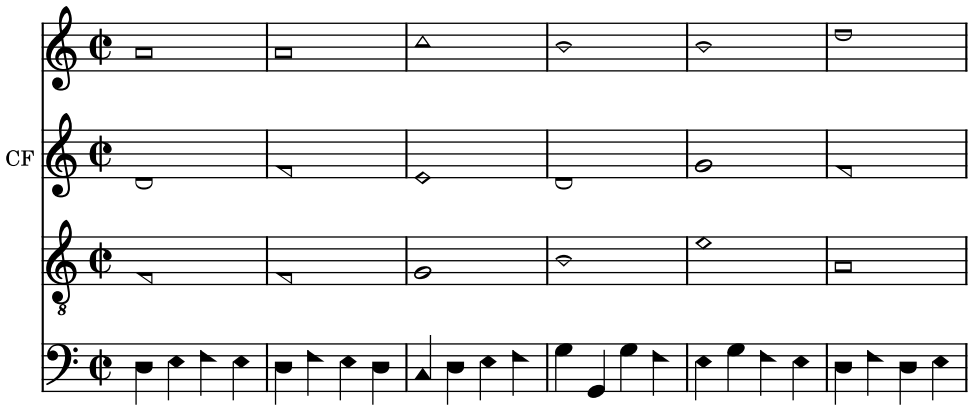
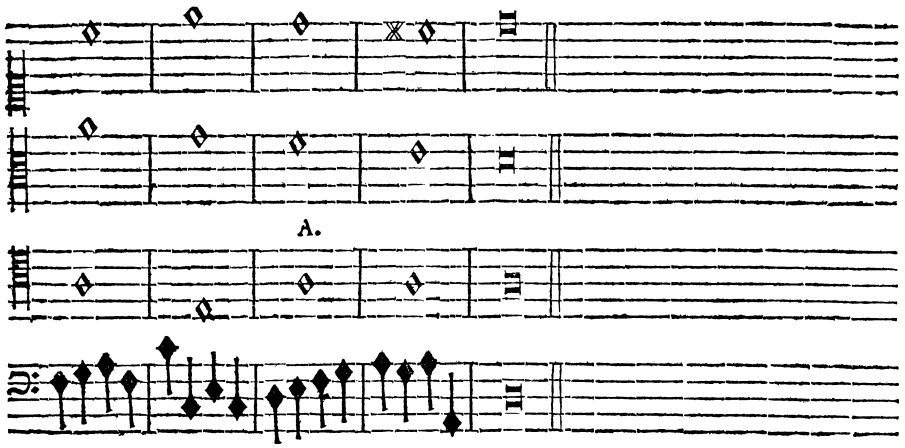


Aloys. Excellent, Joseph—you have given in this example a clear proof of your progress. By it you show that nothing said throughout so many lessons has slipped from your memory; for since you placed the first quarter note according to the requirement of complete harmony, and with the remaining three, wandering here and there, entered smoothly into the next beat, you have rightly observed the other rules of harmony and of the species. This happens when the Thesis (down-beat), that is, the beginning of any beat, is provided with complete harmony—that is, with the third and the fifth. Continue.

Aloyſ. Quid ſibi vult illa Litera A. in cantûs ſecundo tactu poſita ?

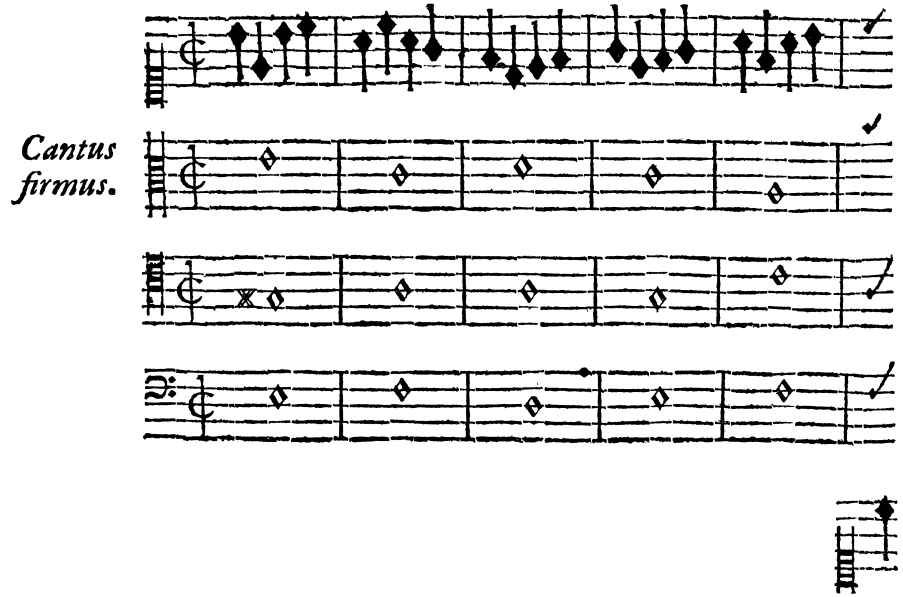
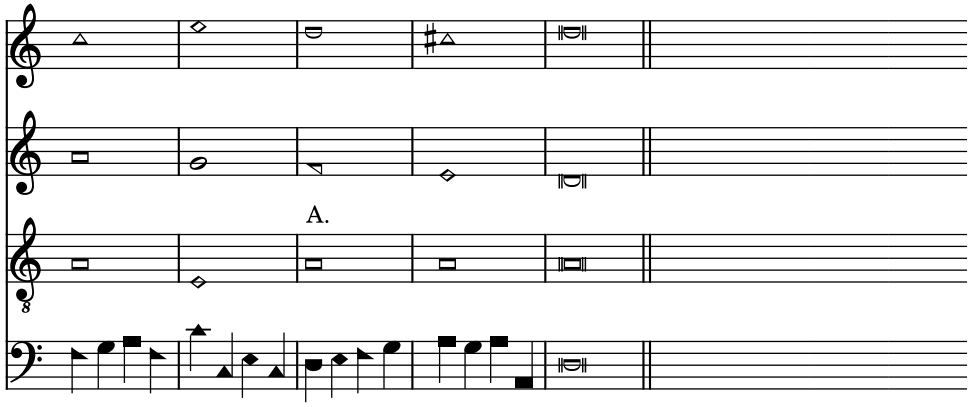
Joſeph. Non nihil ſcrupuli mihi movet progreſſio illa de Conſonantia perfecta ad perfectam motu recto, id eſt de Quinta ad Octavam : eò magìs, quòd partibus extremis evenire contingat.

Aloyſ. Jam paulò ante dixi, iterumque repeto, ſimiles progreſſiones ſubinde neceſſitati Semibrevium condonandas eſſe : quippe quum ne in libero quidem componendi genere ſemper evitari poſſint : quæ tamen in partibus mediis, quàm in extremis, ùt rectè animadvertiſti, tolerabiliores erunt.

Aloys. What does that letter A placed in the melody in the second beat mean?

Joseph. That progression from perfect consonance to perfect consonance by similar motion—that is, from the fifth to the octave—causes me some scruple, all the more because it occurs in the outer parts.

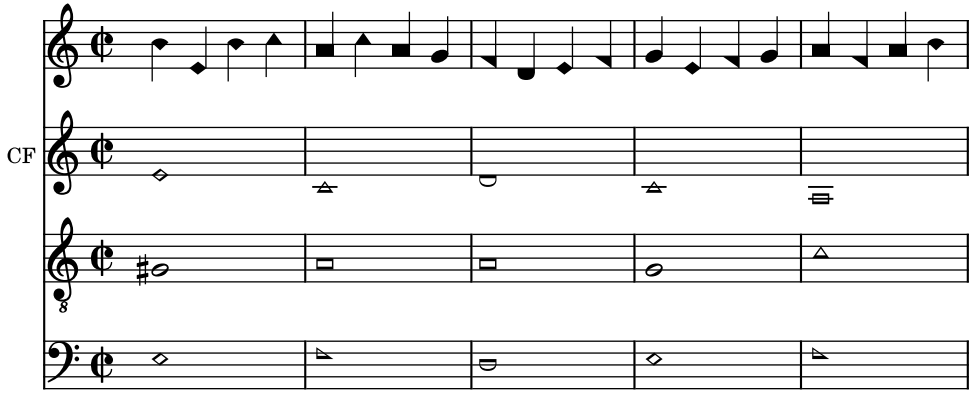
Aloys. I said a little earlier, and I repeat, that such progressions must sometimes be pardoned because of the necessity of the whole notes; for even in free composition they cannot always be avoided. Yet, as you rightly observed, they are more tolerable in the middle parts than in the outer ones.



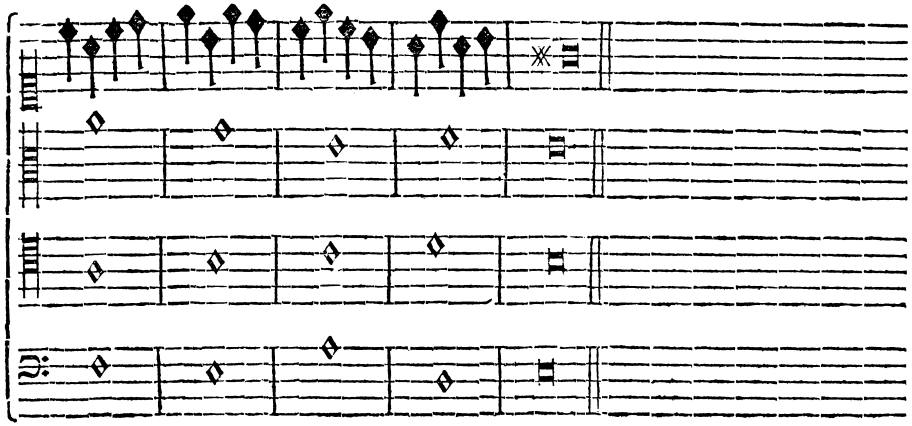
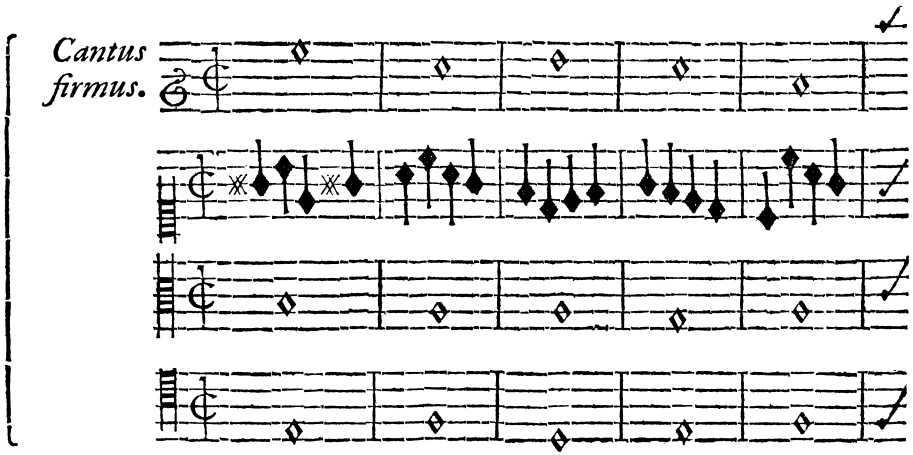
Joſeph. Litera A. in Tenoris parte poſita denotat, ob eandem rationem antè dictam à communi regula aberratum eſſe; quia tranſitus fit de Conſonantia imperfecta ad perfectam, nempe de Tertia ad Quintam motu recto, inde etiam ob ejusdem neceſſitatis cauſam tolerari poſſe exiſtimo.

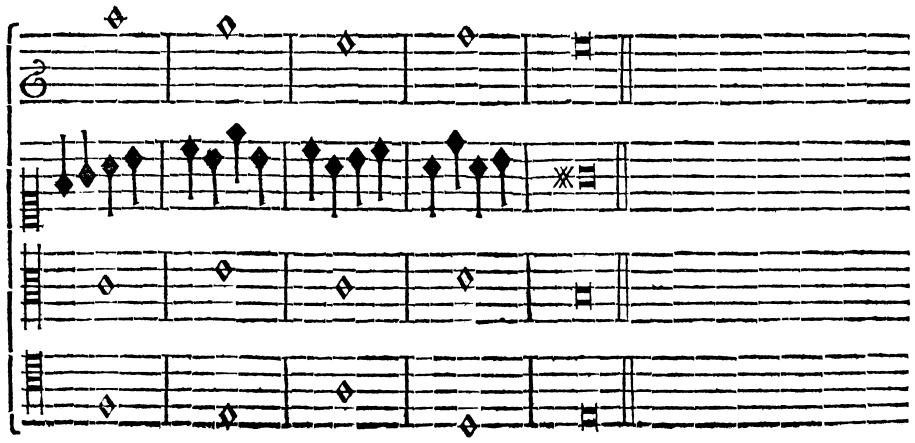
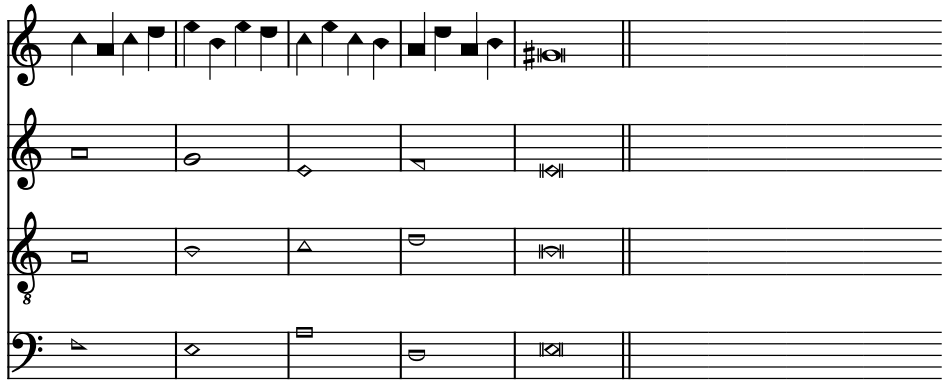
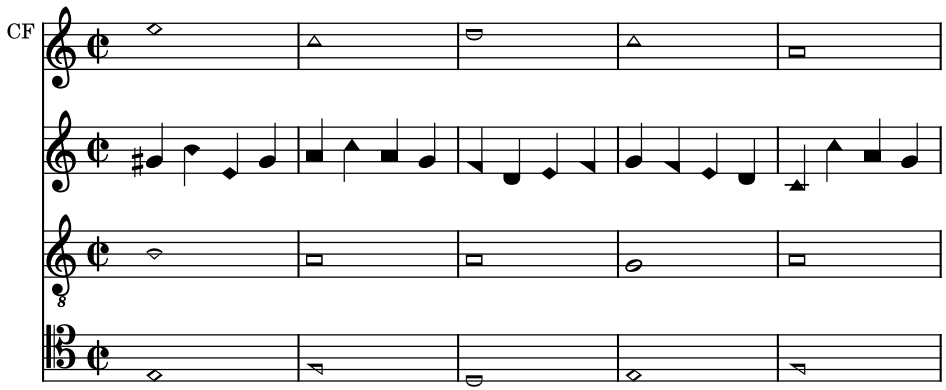
Aloyſ. Rectè exiſtimas. Perſpicuum enim eſt, ſtante hujus Speciei rigore, meliùs fieri non potuiſſe: defectumque illum eò difficiliorem eſſe perceptu, quia in parte media exiſtit. Nunc alterum tonum, *E, la, mi*, ob Quintæ defectum omnium difficillimum factu, me præſente pari ratione perficias. Cæteros domi tuæ lucubrandos tibi committo.



Joseph. The letter A placed in the Tenor part indicates that, for the same reason just mentioned, the common rule has been departed from; because the progression is from imperfect to perfect consonance, namely from the third to the fifth by similar motion. Therefore I think it can be tolerated for the same reason of necessity.

Aloys. You judge rightly. It is clear that, given the strictness of this species, it could not have been done better; and that defect is all the harder to perceive because it occurs in the middle part. Now complete the next mode, E–la–mi (the Phrygian mode), the most difficult of all because of the lack of a fifth, in my presence and in the same manner. The others I leave for you to work out at home.





Cantus
firmus.



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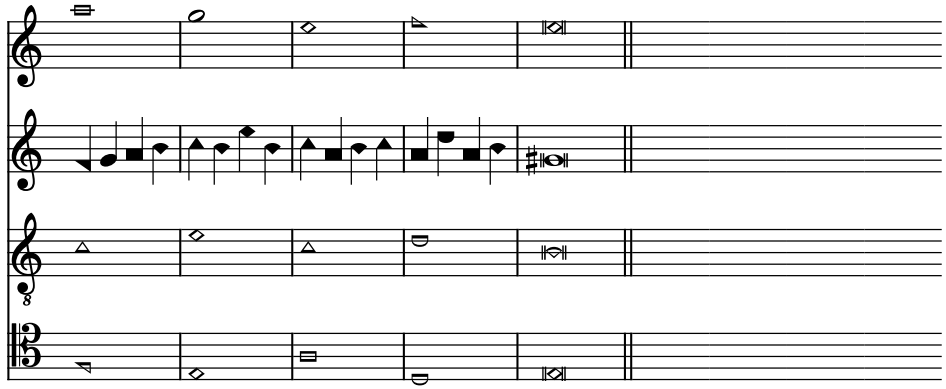
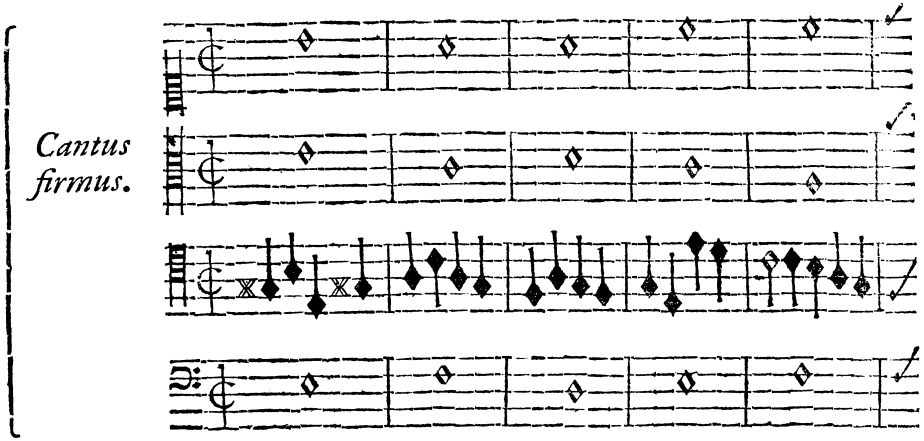
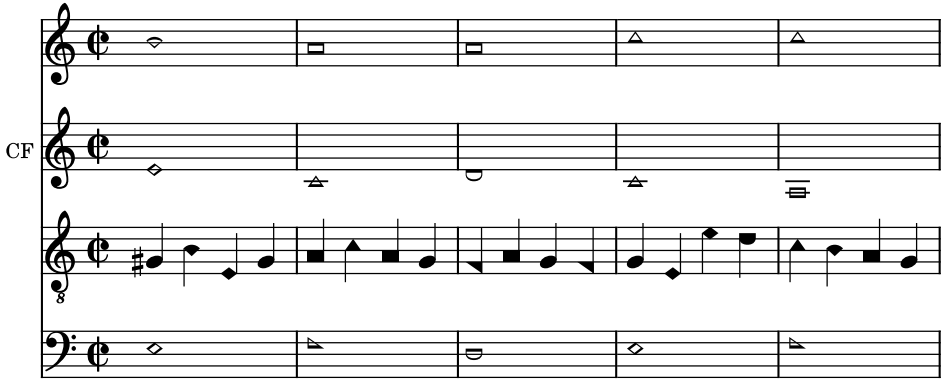
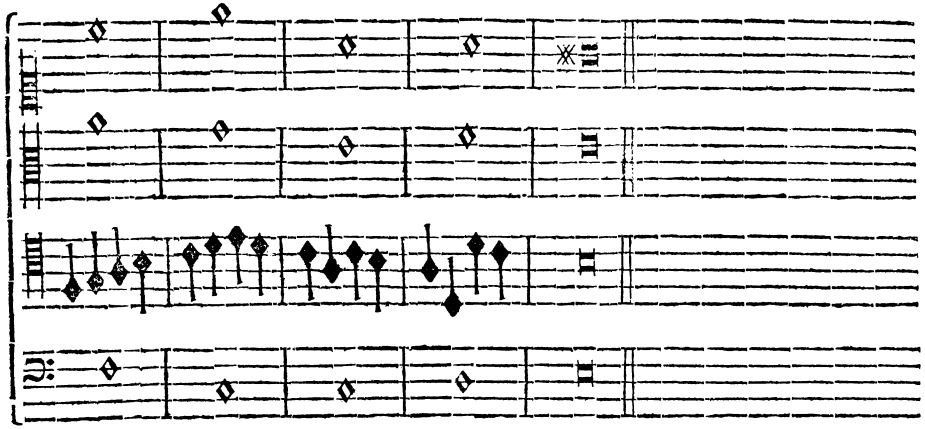
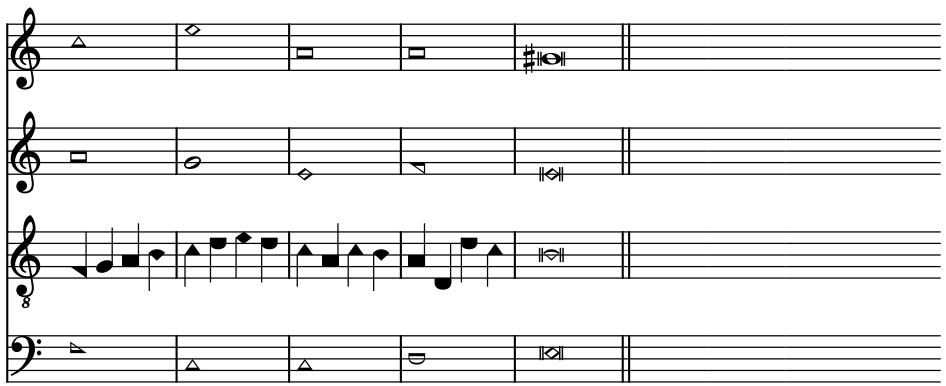
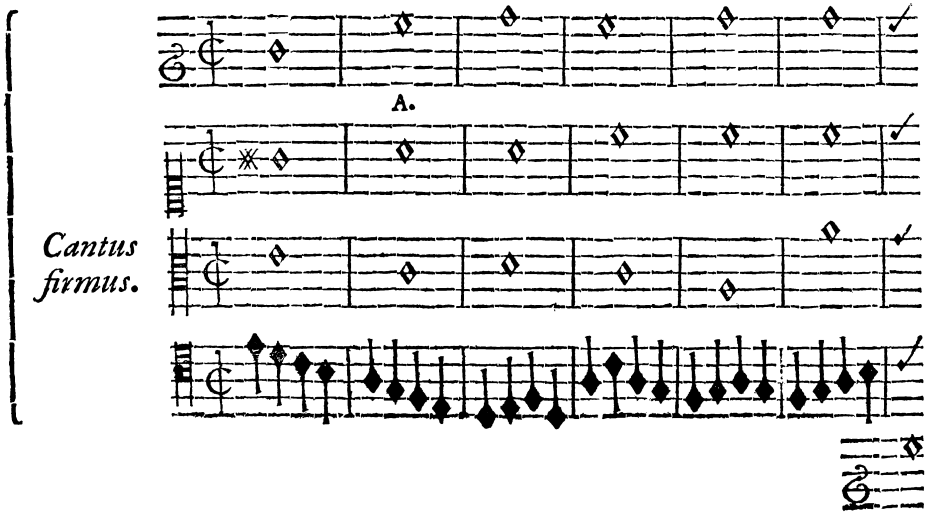
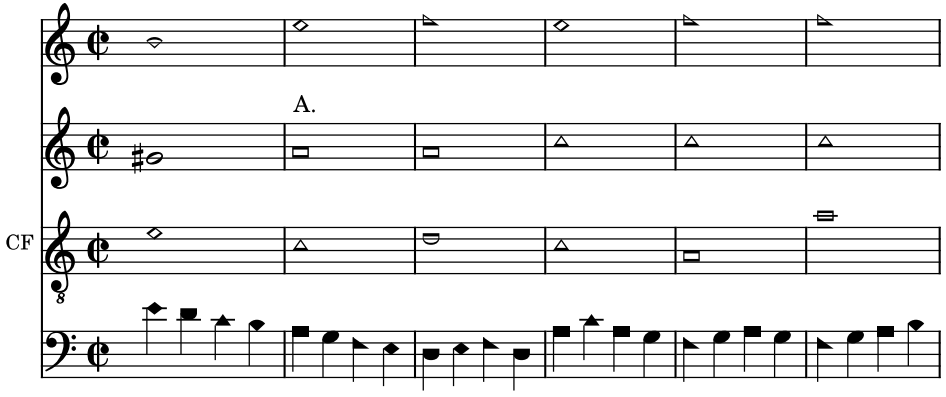
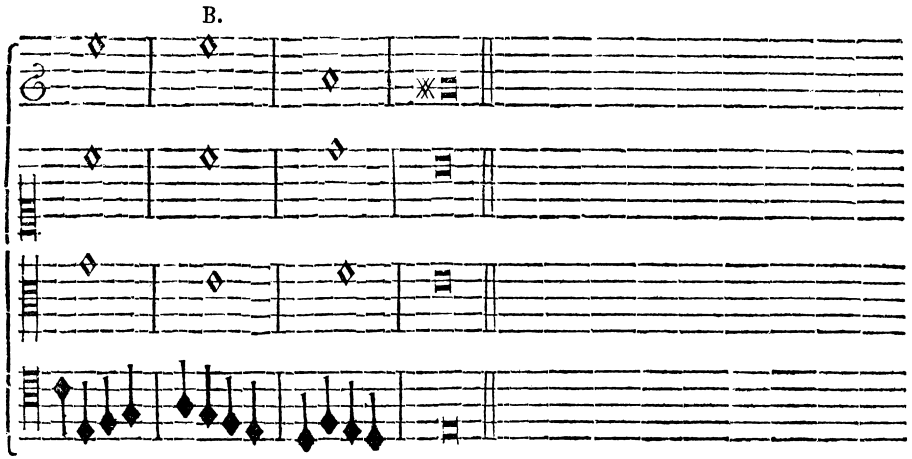


Cantus firmus.





Cantus firmus.



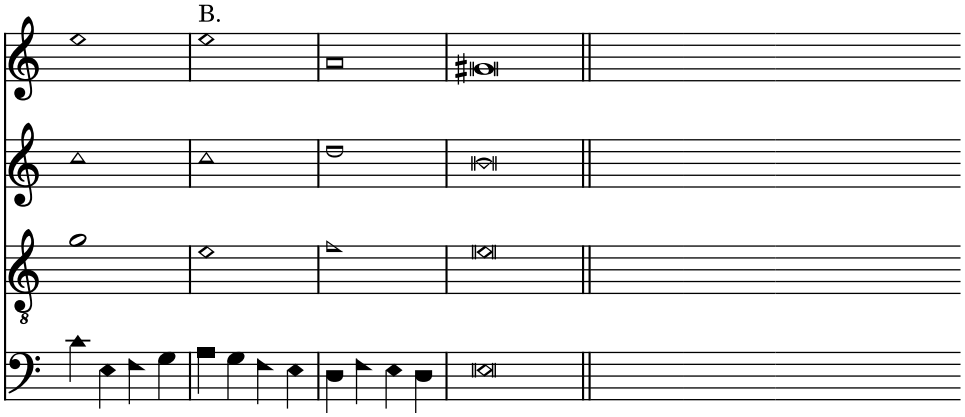
Joſeph. Inter partes Cantûs, & Violini de prima ad ſecundam Notam ambitus reperitur de Tertia ad Quintam, contra communem regulam, quem ob prædictas rationes emendare nolui, nec inveni.

Aloyſ. Hîc loci multa neceſſitati concedenda fore, quæ aliàs in libero componendi genere vitanda ſunt, perſæpe jam inculcavi : quemadmodum etiam tranſitus ille B. de Octava ad Quintam reſpectu partis infimæ motu recto, vitio notandus non eſt, propter hujus Speciei difficultatem. Jam reliquis tonis domi tuæ abſolvendis ad aliam Speciem pergamus.

EXERCITII III.

LECTIO QUARTA,

ABſolutâ Semiminimarum Specie, Ligaturas ſequi, e Bicinii, Triciniique Lectionibus compertum eſt : & quidnam eæ ſint, pari ratione jam perſpectum eſſe præſuppono. Nunc explanandum reſtat, quaſnam Concordantiarum comites in Quatricinio ligaturæ ſibi adſciſcant : de quo nonnihil dictum eſt in Tricinii compoſitione ; eas nempe concordantias, quas ſublatâ ligaturâ petunt, ſibi poſtulare ; ob rationem ibi dictam, quòd ligatura nîl aliud ſit, quàm retardatio ſequentis Notæ, quæ quoad concordantias nihil immutat. Exempla rem manifeſtabunt.

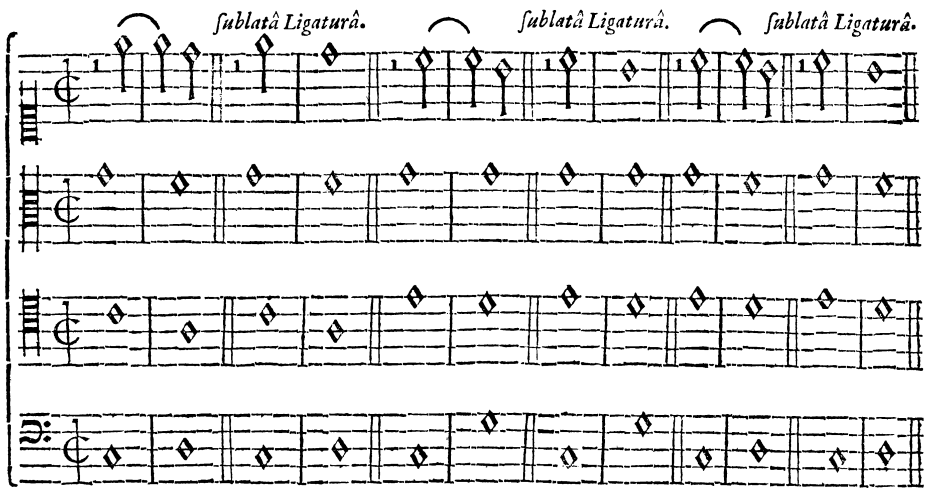


Joseph. Between the melody and the violin parts, from the first to the second note, there is a progression from the third to the fifth, contrary to the common rule; which, for the reasons already stated, I did not wish to correct, nor did I find a way.

Aloys. I have often insisted that in this place many things must be granted to necessity which in free composition must be avoided; likewise that the progression marked B—from the octave to the fifth in the lowest part by similar motion—is not to be marked as faulty because of the difficulty of this species. Now, having finished the remaining modes at home, let us proceed to another species.

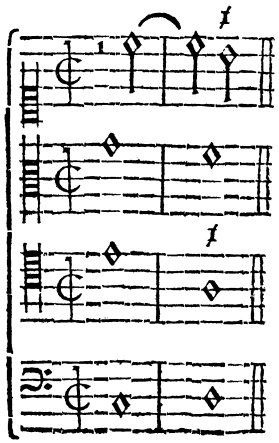
EXERCISE III.

LESSON FOUR.

Having completed the species of quarter notes, it is known from the lessons on the bicinium and tricinio that ligatures follow; and what they are, I assume is now equally clear. What remains is to explain what consonances ligatures in the quatricinium take as their companions. Something of this was said in the tricinio: namely, that they demand the same consonances which, with the ligature removed, would be required —because, as was said there, the ligature is nothing other than the retardation of the following note, which changes nothing with respect to consonances. Examples will make the matter clear.

Quæ Paradigmata clarè demonſtrant, easdem concordantias, & Notis ligtis, & ſolutis eſſe.

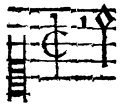
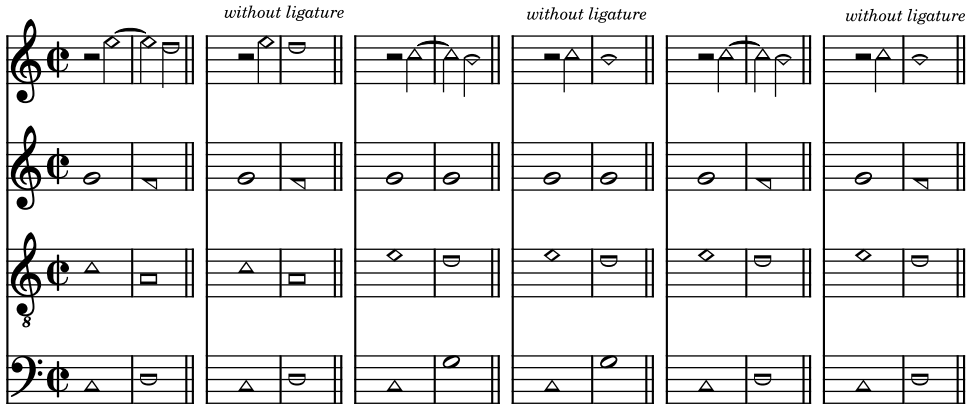
Joſeph. Nunquamne, venerande Magiſter, fallit iſta Regula?

Aloyſ. Fallit in quibusdam Hypotheſibus hujus Lectionis; ubi obligatio eſt ligaturas cum tribus Semibrevibus concordandi per integram tactûs menſuram: quod primò uſuvenire non poteſt in Hypotheſi, ubi Septimæ in ligatura Quinta aſſociatur. E. G.



Ubi reſolutio Ligaturæ cum Tenoris parte vitioſam diſſonantiam, & quàm maximè evitandam efficit.

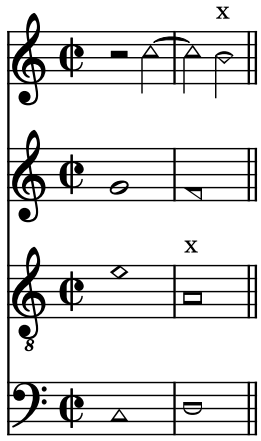
Joſeph. Quid remedii hoc in caſu?

Aloyſ. Semibrevis in parte Tenoris dividenda eſt hoc modo.



These patterns clearly show that the consonances are the same whether the notes are tied or free.

Joseph. Does this rule never fail, venerable Master?

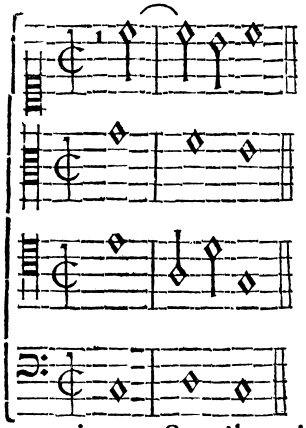
Aloys. It fails in certain hypotheses of this lesson—where the obligation is to make the ligatures agree with three whole notes for the entire measure. This cannot occur in the hypothesis where a seventh in the ligature is associated with a fifth, for example:



Here the resolution of the ligature produces with the Tenor part a faulty dissonance, and one that must be avoided as much as possible.

Joseph. What remedy is there in this case?

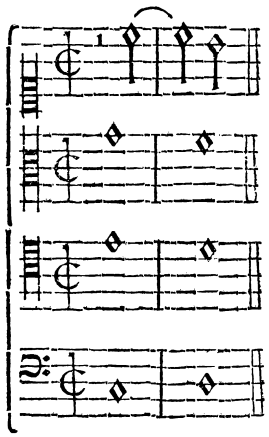
Aloys. The whole note in the Tenor part must be divided in this way:



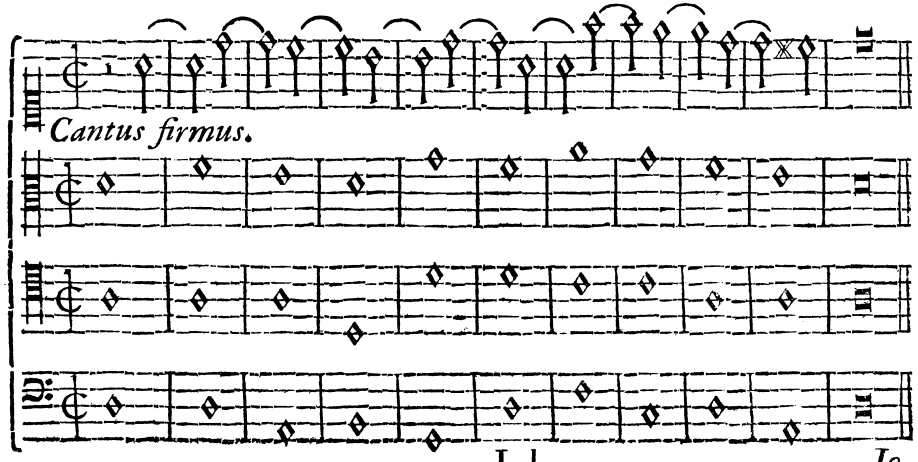
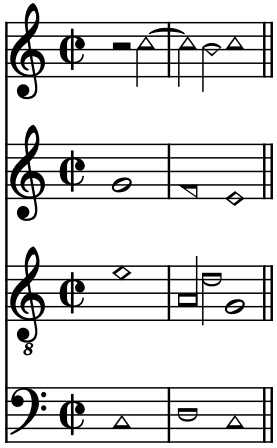
Joſeph. Sed hæc Species non admittit Semibrevis diviſionem.

Aloyſ. Ità eſt. Ubi fieri poteſt. Verùm plures caſus evenient, ùt exemplis mox ſubjiciendis edoceberis, ubi neceſſitate poſtulante, diviſione ſuperſedere nequaquam poteris. Quamobrem trium Semibrevium rigor hac in Specie non admodum ſtrictè teneri poteſt.

Joſeph. Septima cum Octava comitata, Semibrevis diviſioni obnoxia minimè foret: utì in priori exemplo tuo videndum eſt, hoc modo:



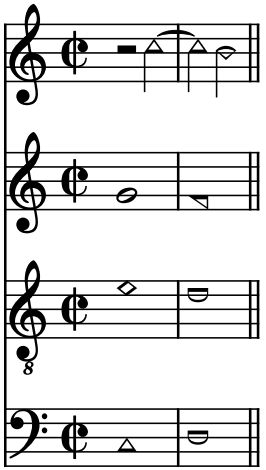
Aloyſ. Verum eſt in hoc caſu, ubi nihil impedit, quò minùs Octava locò Quintæ dari poſſit; ſed in Hypotheſibus, quarum plurimæ reperiuntur, ubi propter præcedentium Notarum, ſubſequentiumque ſeriem Octavæ locus non eſt, Quinta neceſſariò uſurpari debet: tum Semibrevis utì in multis aliis eventibus dividenda eſt; quemadmodum Paradigmata ſequentia docent.

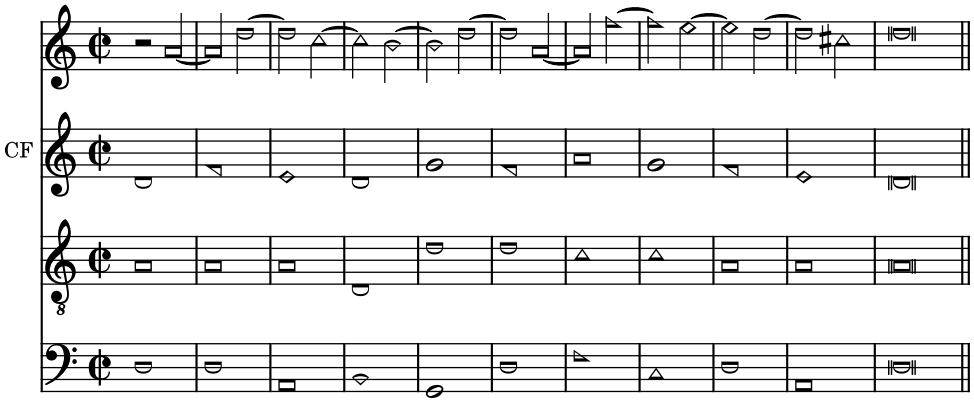


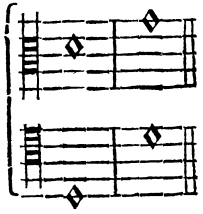
Joseph. But this species does not admit the division of a whole note.

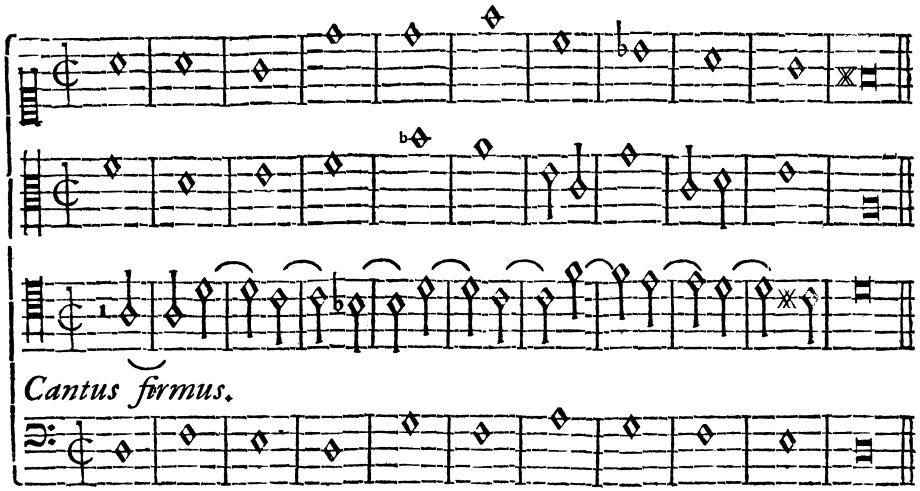
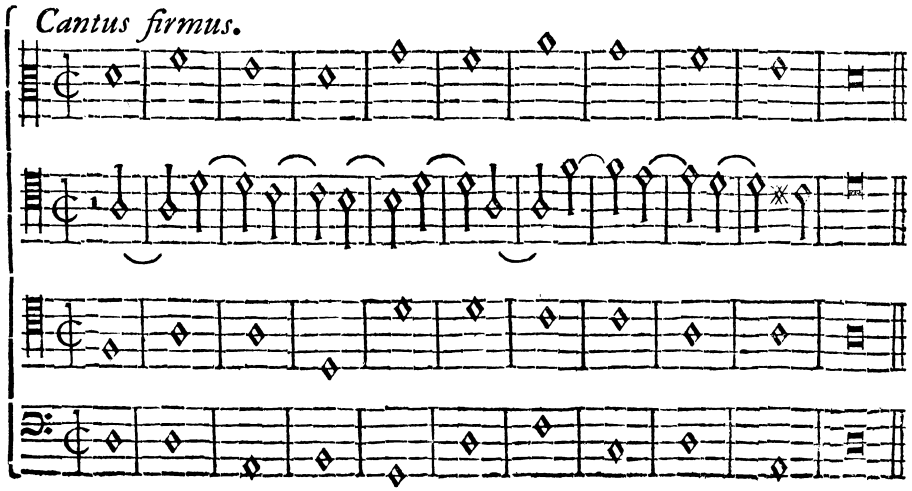
Aloys. True—where it can be avoided. But many cases will occur, as you will soon learn from the examples, where, necessity demanding it, you will by no means be able to refrain from dividing. Therefore the strict requirement of three whole notes in this species cannot always be held rigidly.

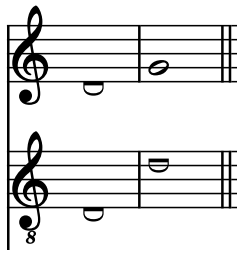
Joseph. A seventh accompanied by an octave would not be subject to the division of the whole note, as is seen in your earlier example, in this way:



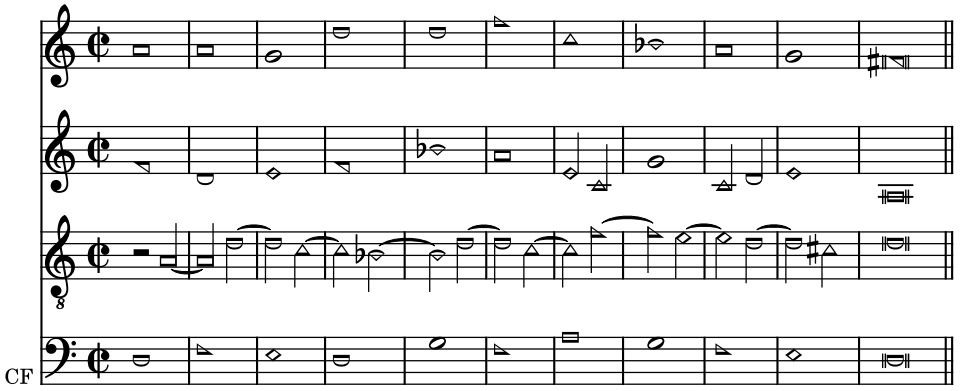
Aloys. True—in this case, where nothing prevents the octave from being given instead of the fifth. But in many hypotheses—of which there are many—where, because of the series of preceding and following notes, there is no place for the octave, the fifth must necessarily be used; then the whole note, as in many other situations, must be divided, as the following patterns show.

Joſeph. Nihil dubii in hoc exemplo reperio, præter progreſſionem de quarto in quintum tactum inter partes Tenoris & Alti.

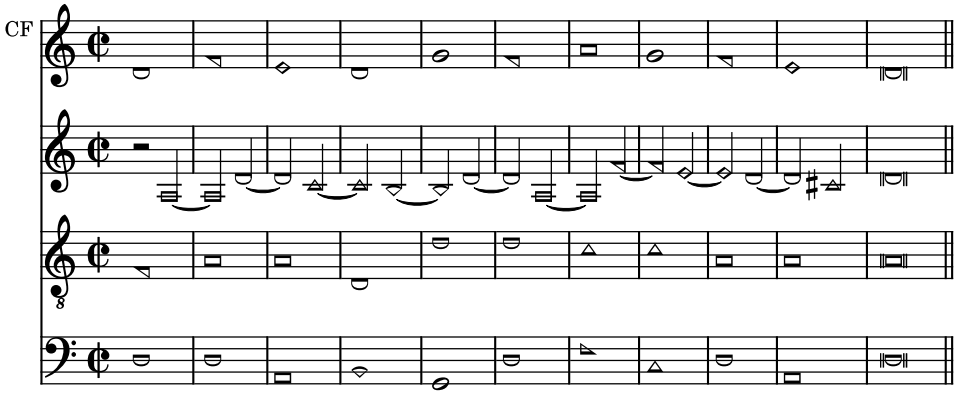
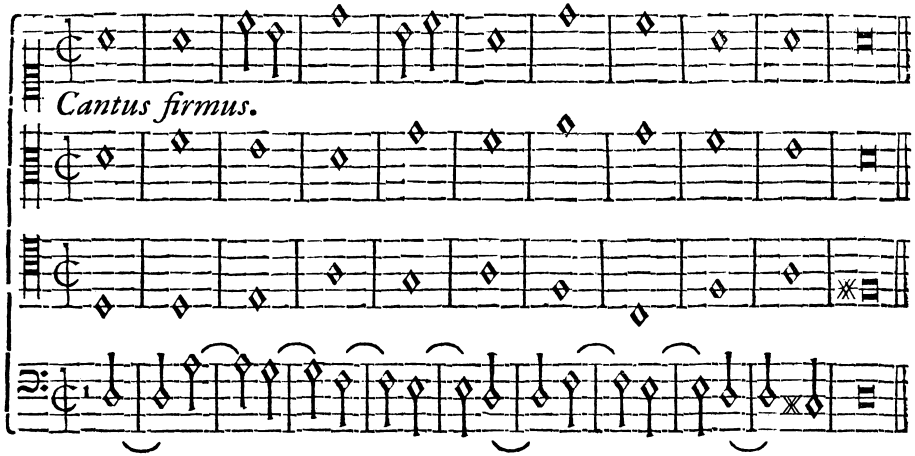


Aloyſ. Ne ſcrupulum tibi moveat hæc progreſſio, ſcire te oportet, Quartam inter medias partes, aut nullius momenti haberi, aut officio Conſonantiæ imperfectæ perfungi. Unde progreſſio illa perinde tenenda eſt, ac ſi de Conſonantia perfecta ad imperfectam motu recto procederes : quod bene notandum eſt. Nunc ad reliqua exempla.

Joseph. I find nothing doubtful in this example except the progression from the fourth to the fifth beat between the Tenor and Alto parts.



Aloys. Do not let that progression trouble you. You must know that the fourth between the middle parts is either of no importance or serves the office of an imperfect consonance. Therefore that progression must be regarded just as if you were proceeding from perfect to imperfect consonance by similar motion. This must be well noted. Now to the remaining examples.

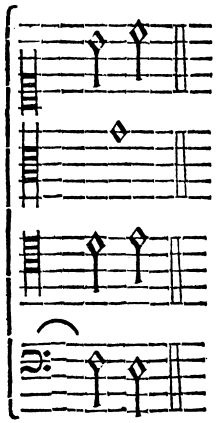




Quibus exemplis palàm fit, ligaturas in Quatricinio aut non ubique tribus integris Semibrevibus (ùt hæc Species requirit) ſtipari, aut ſi poſſunt, haud abſolutâ omnibus legibus Harmoniâ compleri poſſe.

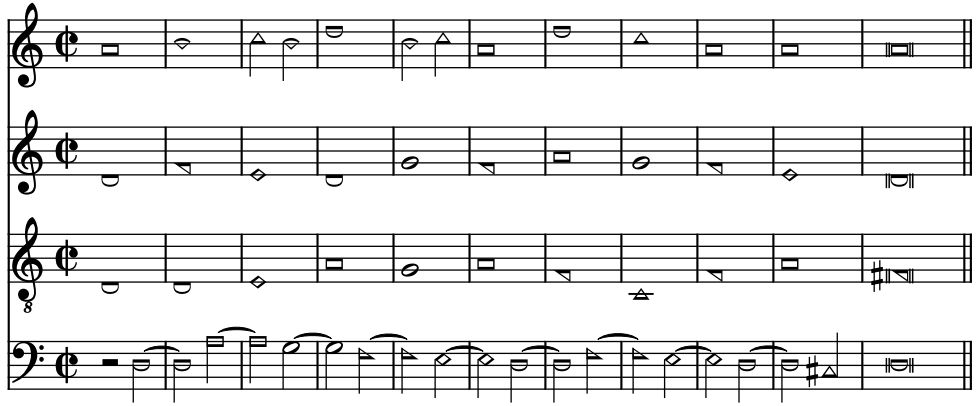
Joſeph. In quibus exemplis una, alterave Semibrevium diviſa ſit, & ob quam cauſam, conſpicio, verùm ubi deeſſe dicis aliquid Harmoniæ, non video.

Aloyſ. Non animadvertis, in exempli primi ſexto tactu Theſi deeſſe Quintam? quæ tamen ad completam Harmoniam ſummopere neceſſaria eſt. Deinde in ultimi exempli quinto tactu Secunda duplicata eſt, deficiente Sextâ, quæ ad integram Harmoniæ rationem deſideratur, ùt ſequens exemplum demonſtrat.



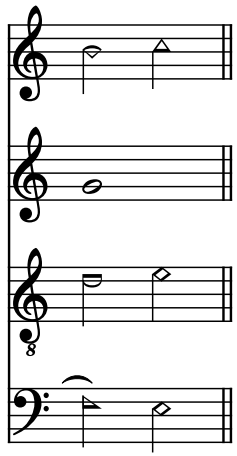
Poſtremò in ejuſdem exempli quinto tactu quarta duplicata eſt; cùm de rigore Secunda potiùs, quàm Quarta duplicanda ſit.

Joſeph. Quæ ratio eſt duplicandæ Secundæ potiùs, quàm Quartæ?

Aloyſ. Non tam ratione Secundæ, vel Quartæ, quàm ratione completæ Harmoniæ intereſt. Cùm enim Harmoniæ plenitudo conſiſtat in ſociatione Tertiæ, Quintæ, & Octavæ; in dicto autem exemplo locò Octavæ Quinta duplicata reperiatur, ibidem Harmoniam

From these examples it is clear that ligatures in the quatricinium either cannot everywhere be supported by three full whole notes (as this species requires), or, if they can, the harmony cannot be completed in all respects.

Joseph. I see in which examples one or two whole notes have been divided, and for what reason; but where you say that something is lacking in the harmony, I do not see it.

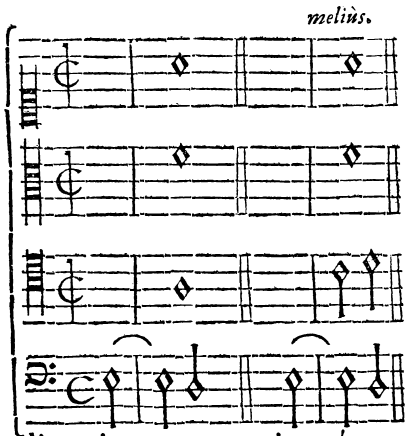
Aloys. Do you not notice that in the sixth beat of the first example the Thesis lacks the fifth?—which is most necessary for complete harmony. Then in the fifth beat of the last example the second is doubled, the sixth being absent, which is required for full harmony, as the following example shows:



Finally, in the same example's fifth beat the fourth is doubled, whereas strictly the second should be doubled rather than the fourth.

Joseph. What is the reason for doubling the second rather than the fourth?

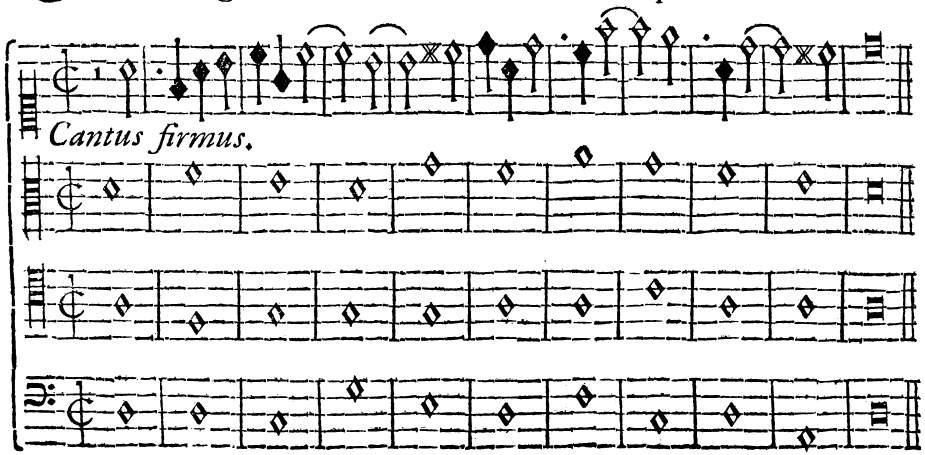
Aloys. It is not so much a matter of the second or the fourth themselves, as of complete harmony. For the fullness of harmony consists in the association of the third, fifth, and octave; but in the example in question, in place of the octave the fifth is doubled, and therefore the harmony is not complete in every respect.

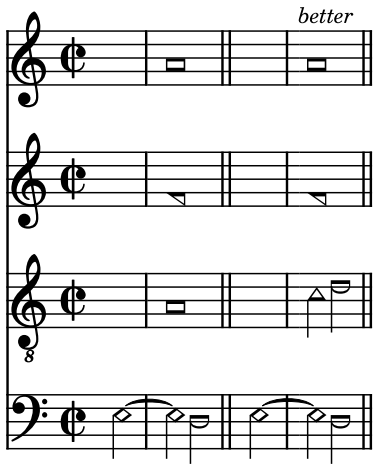
moniam non omnibus numeris abſolutam fore conſtat. Hîc autem non loquor de prima tactûs parte, ubi Secunda ineſt, quæ Octavam ſibi comitem haud admittit, ſed de ſecunda tactûs parte, in qua Octava deſideratur. E. G.

Similes haud magni momenti defectus hujus Speciei rigori concedendi ſunt, ob eximium, quod hæc exercitatio Studioſis affert emolumentum: quia non ſolùm rectam componendi rationem edocet; verùm etiam, quantum ſubinde, neceſſitate poſtulante à rigore regularum recedere permiſſum ſit. Formulam ante oculos tibi poſui, quam imitando reliquos quinque tonos æquali ratione, mutationeque pertractandos tibi relinquo. Nunc ad quintam hujus Studii Speciem progrediamur.

EXERCITII III.

LECTIO QUINTA.

SPeciem hanc, Joſephe, Contrapuncto florido contineri, non ignoras. Quid ſit, qua ratione fiat, ex præcedentibus, & exemplis, & dictis nota tibi fore ſuppono: maximè ex Tricinii ſpeciei hujus exercitio: ut adeò nihil hîc novi addendum ſit, præter quartam partem, Semibrevi conſtantem, juxtaque Quatricinii regulas intexendam. Ad exempla.

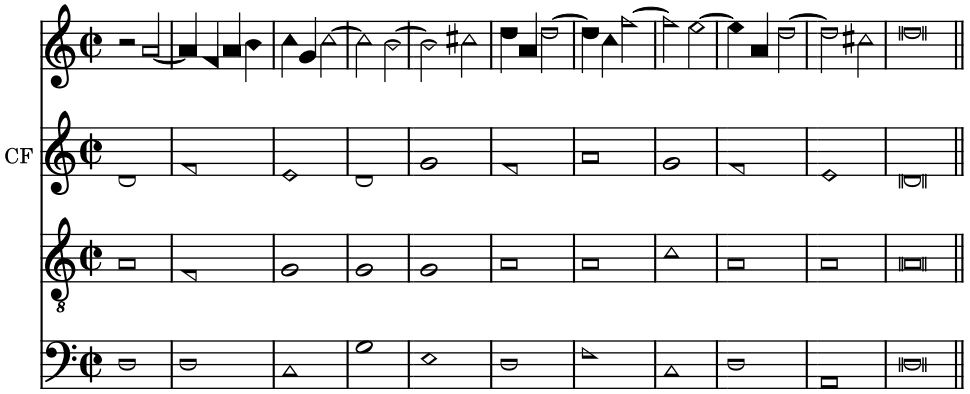


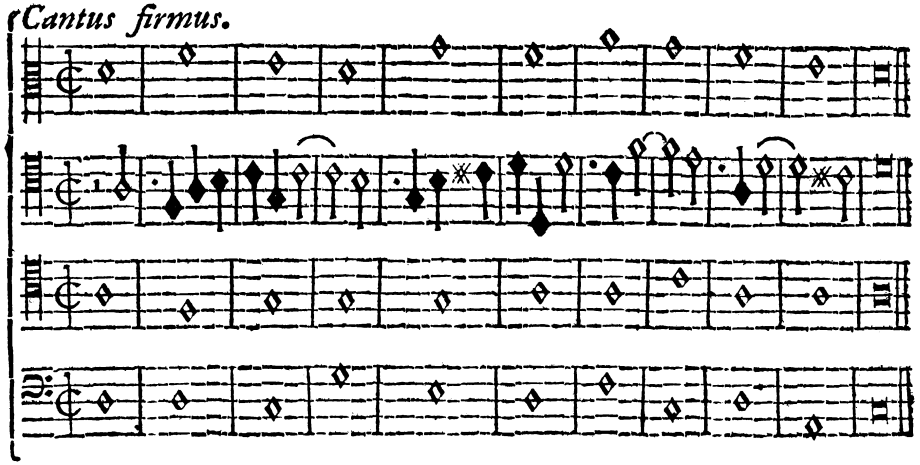
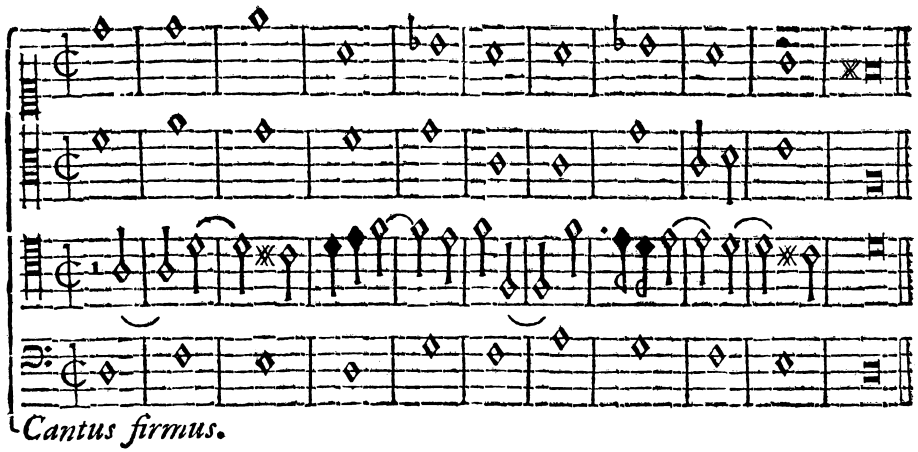
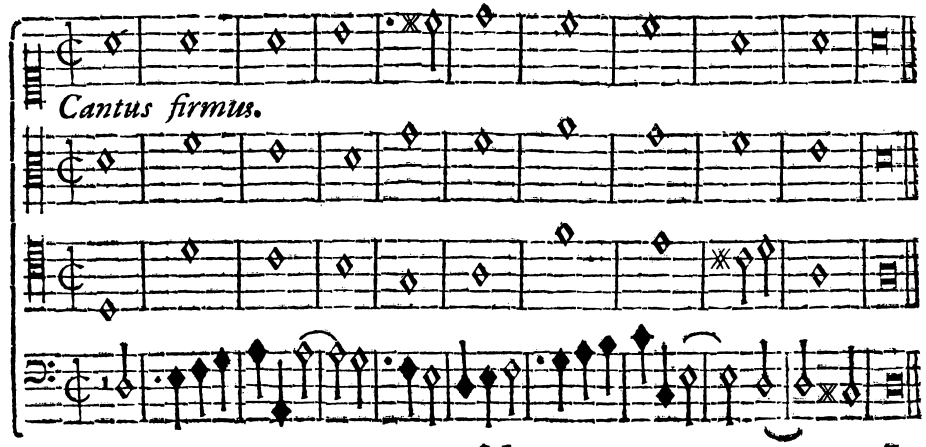
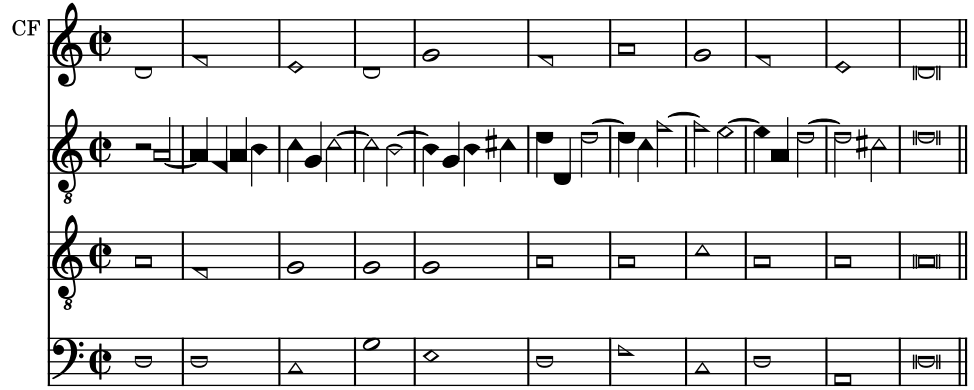
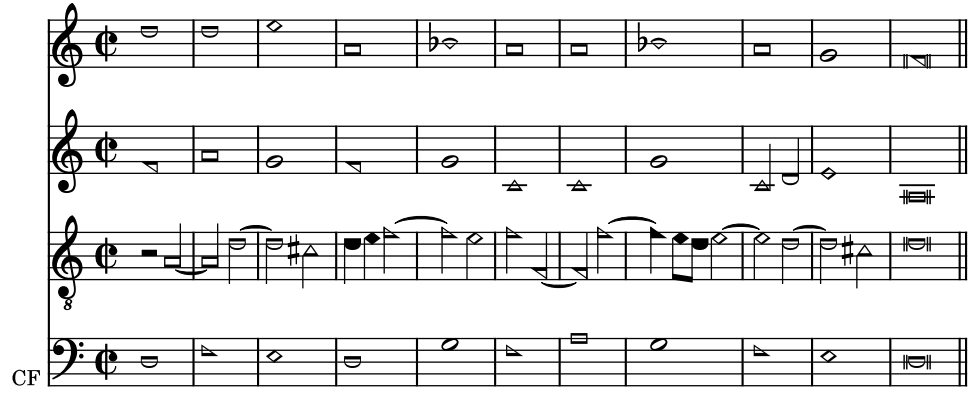
Here I do not speak of the first part of the beat, where the second appears—which does not admit the octave as its companion—but of the second part of the beat, in which the octave is required. For example:

Similar defects, of no great importance, must be allowed to the strictness of this species, because of the great benefit this exercise brings to students; for it not only teaches the correct method of composing, but also how far, when necessity demands, one may depart from the strictness of the rules. I have placed the pattern before your eyes; by imitating it, I leave to you the remaining five modes to be treated in the same manner and alternation. Now let us proceed to the fifth species of this study.

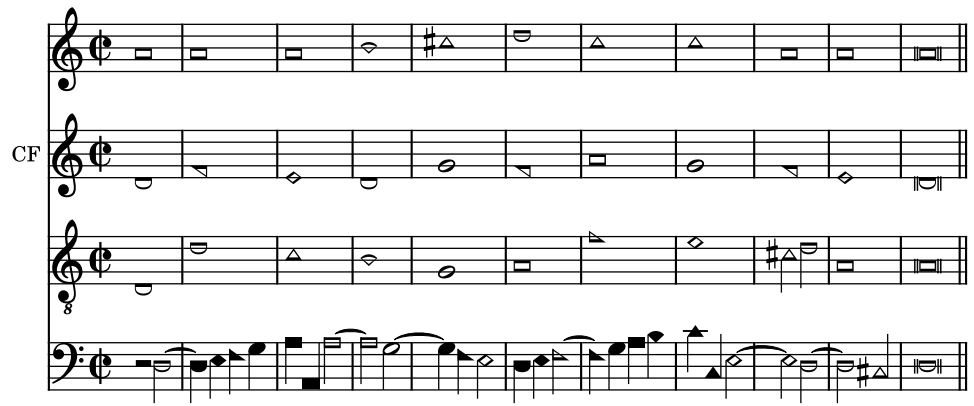
EXERCISE III.

LESSON FIVE.

You know, Joseph, that this species is contained within the contrapunctus floridus (florid counterpoint). What it is, and how it is made, I assume is already known to you from what has been said and shown in the preceding examples—especially from the exercises of this species in the tricinio—so that nothing new need be added here except the fourth part, consisting of whole notes, which must be woven in according to the rules of the quatricinium. To the examples.

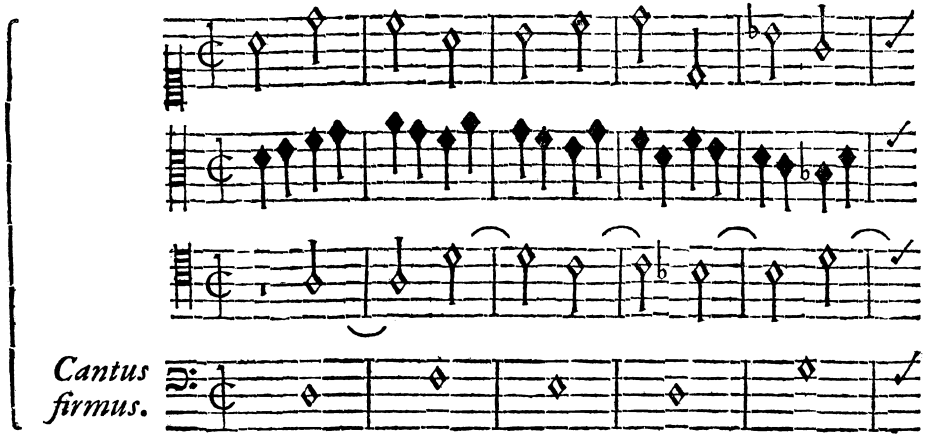
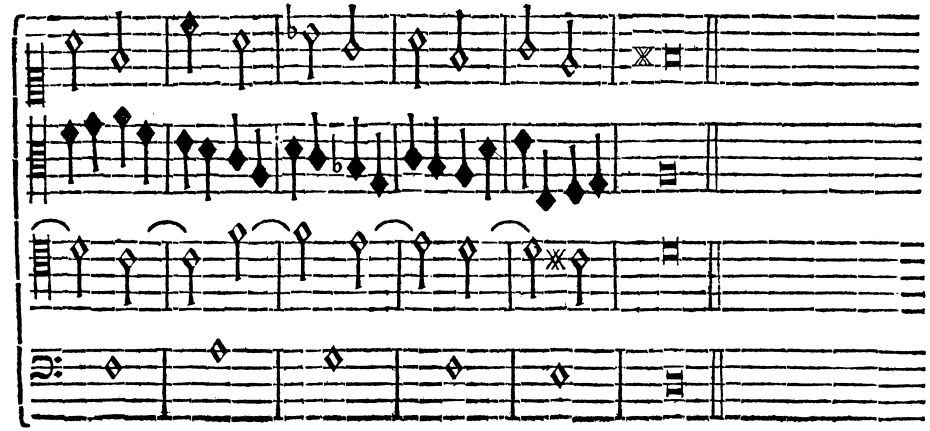


Cantus firmus.

Cantus firmus.

Cantus firmus.





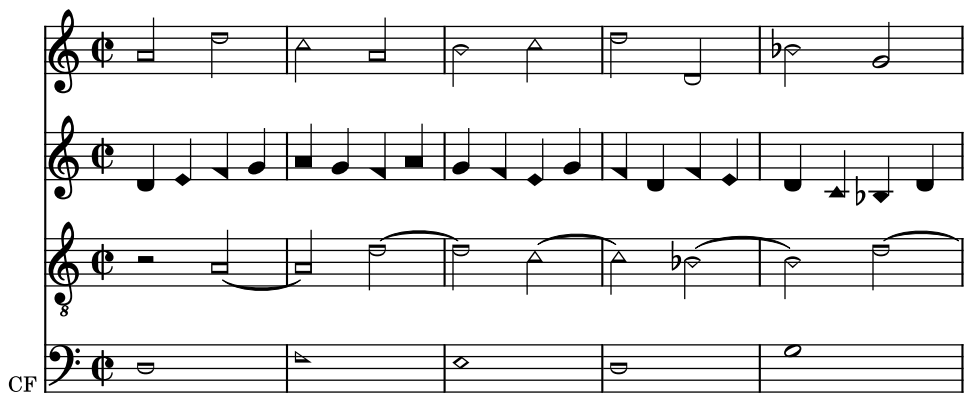
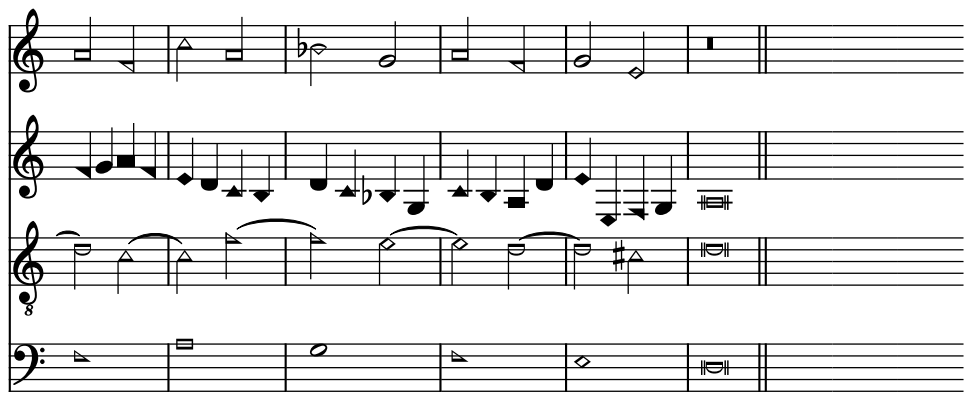
Joſeph. Video ſicut in ligaturarum ſpecie, ità quoque hîc unam alteramve Semibrevium diviſam eſſe.

Aloyſ. Quando dixi, quòd Semibreves integræ manere debeant, de poſſibilitate intelligi volo. Interim tamen advertes, ubi copia dabatur, id ubique ad amuſſim obſervatum eſſe. Reliqua quinque tonorum exercitia pari ſtudio proſequenda tibi committo.

Abſolutis jam quinque Speciebus, ubique ſeorſum exercitatis, id conjunctim nunc ut facias, quàm maximè te cohortor; ſervato eodem Cantu firmo, conjungas E. G. Minimas, Semiminimas, & Ligaturas; quo facto, quâlibet partium peculiarem motum nactâ, mira Compoſitioni varietas accedet. In exemplo.

Pari

Joseph. I see that, just as in the species of ligatures, so also here one or another whole note has been divided.

Aloys. When I said that whole notes ought to remain whole, I meant "where possible." Yet you will notice that wherever it could be done, it has been observed with exactness. I entrust to you the remaining exercises of the five modes, to be pursued with equal diligence.

Now that the five species have been completed, each practiced separately, I strongly urge you to combine them: keeping the same cantus firmus (fixed melody), join, for example, half notes, quarter notes, and ligatures. When this is done, each part having its own characteristic motion, a wonderful variety will be added to the composition. For example:

Pari ſtudio, & hunc, & cæteros quinque tonos quadruplici Cantûs firmi variatâ poſitione, ità ut in omnibus quatuor partibus exiſtere contingat, abſolvendos tibi impono: operamque dabis, ut quæ de qualibet Specie ſeparatim dicta ſunt, nunc cumulatè comprehendas: & quid muneris partium inter ſe exiſtat, (quod te ampliùs latere non puto) ad amuſſim obſerves. Quantum emolumenti hæc exercitia ritè inſtituta ſtudioſis afferant, haud facilè exprimi poteſt: quippe quum fermè nihil difficultatis oriturum ſit, quod his Speciebus exactis notum tibi futurum non ſit. Quapropter, ſi progreſſum hac in Diſciplina facere cupis, ſummâ contentione te exhortor, ut his quinque Speciebus exercendis haud exiguum tempus impendas; novoſque ſemper arbitrio tuo Cantus firmos eligendo, ad minimùm unius, alteriuſve anni ſpatium exercitio huic inſiſtendo conſumas. Neque præpoſterâ rapi te ſinas aviditate progrediendi ad Compoſitionem idealem, arbitrariamque; cujus dulcedine allectus hinc inde vagando tempus terens, haud unquam ad veri fundamenti adeptionem pervenies.

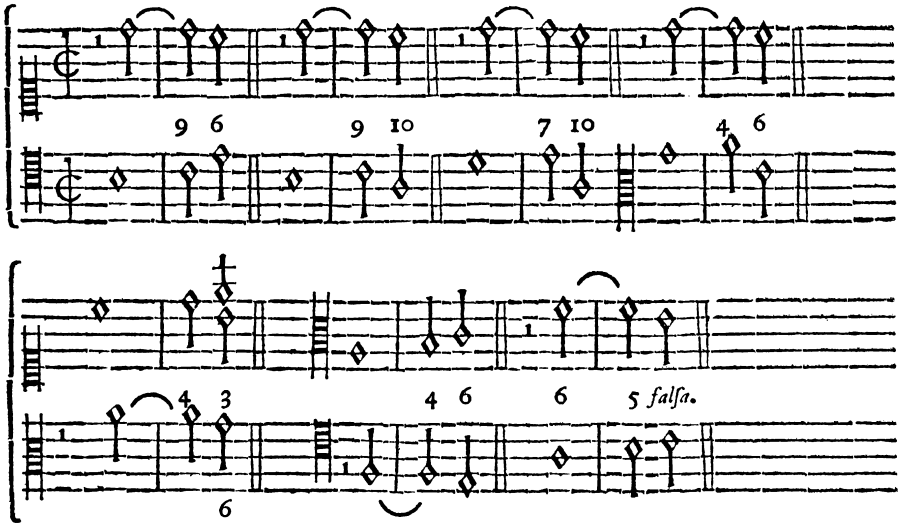
Joſeph. Per aſperam viam, Venerande Magiſter, valdeque ſpinoſam incedere jubes. Vix enim fieri poteſt, tanto tempore faſtidioſo huic labori vacando, ne tœdio afficiar.

Aloyſ. Ignoſco Joſephe, querellæ tuæ; tibique compatior. Verùm Muſæ montem incolere dicuntur, quò, niſi viâ præcipiti non pervenitur. Nullus eſt utcunque miſerabilis artis mechanicæ tyro, qui tyrocinio ad minimum tres annos non obſtringatur. Quid dicam de Muſica? quæ non has modò inertes, operariasque artes ingenio, difficultate, rerum copiâ longè antecellit, verùm nulli liberalium artium eſt ſecunda. Stimulet te futura ſudoris utilitas. Alliciat gloriæ ſpes. Incitet exinde ventura ſcribendi facilitas; deque, quæ ſcripturus es, rectâ ſcriptione certa fiducia. Quo facto, remoto ad tempus Cantu firmo, ad imitationum, fugarumque cognitionem te ducam. Priùs autem præmittendum eſt, extra Cantûs firmi reſtrictionem, quaſdam diſſonantiarum alio quoque modo reſolvi poſſe: V. G. Nonam in Sextam, & Decimam in Tertiam: Quartam in Sextam, & Tertiam infrà ſcripto modo. E. G.

With equal diligence, I assign to you the completion of this and the remaining five modes, with the cantus firmus placed in four different positions, so that it appears in all four parts. You will take care that what has been said separately about each species is now understood collectively; and that you observe exactly what the office of each part is in relation to the others (which I do not think is any longer unknown to you). How much benefit these exercises, rightly undertaken, bring to students can scarcely be expressed; for hardly any difficulty will arise that will not already be familiar to you from these species once mastered. Therefore, if you wish to make progress in this discipline, I urge you with the greatest insistence to devote no small amount of time to practicing these five species; and by continually choosing new cantus firmi at your own discretion, to spend at least a year or two in this exercise. Do not allow yourself to be carried away by a premature eagerness to advance to free, ideal composition; for if, enticed by its sweetness, you wander here and there wasting time, you will never reach the acquisition of true foundations.

Joseph. You command me, venerable Master, to walk a rough and very thorny path. It is scarcely possible to devote so much time to this tedious labor without being overcome by weariness.

Aloys. I forgive your complaint, Joseph, and sympathize with you. But the Muses are said to dwell on a mountain, which cannot be reached except by a steep path. There is no apprentice of even the most miserable mechanical craft who is not bound to at least three years of training. What shall I say of Music?—which not only surpasses these inert and manual arts in genius, difficulty, and abundance of material, but is second to none of the liberal arts. Let the future usefulness of your labor spur you on. Let the hope of glory entice you. Let the coming ease of writing encourage you, and the sure confidence that what you will write will be written correctly. When this is done, and the cantus firmus is set aside for a time, I will lead you to the understanding of imitation and fugue. But first it must be stated that outside the restriction of the cantus firmus certain dissonances may be resolved in other ways: for example, the ninth to the sixth, the tenth to the third; the fourth to the sixth, and the third in the manner written below. For example:



Joſeph. Quare hæ reſolutiones extra Cantum firmum uſuveniunt, & non etiam cum Cantu firmo?

Aloyſ. Non vides, utramque partem in reſolutione moveri? id autem cum Cantu firmo, qui immobilis eſt, fieri non poſſe? diſparitas itaque eſt: & patet his reſolutionibus, ubi motus obliquus indiſpenſabilis eſt, locum haud quaquam eſſe. Nunc ad Bicinii Compoſitionem, nullo Cantûs firmi vinculo reſtrictum, retrò te reducam.

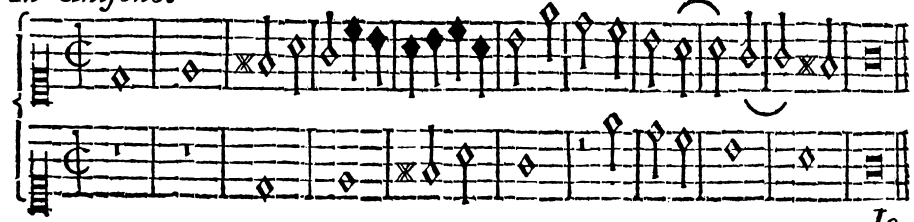
EXERCITII IV.

LECTIO UNICA,

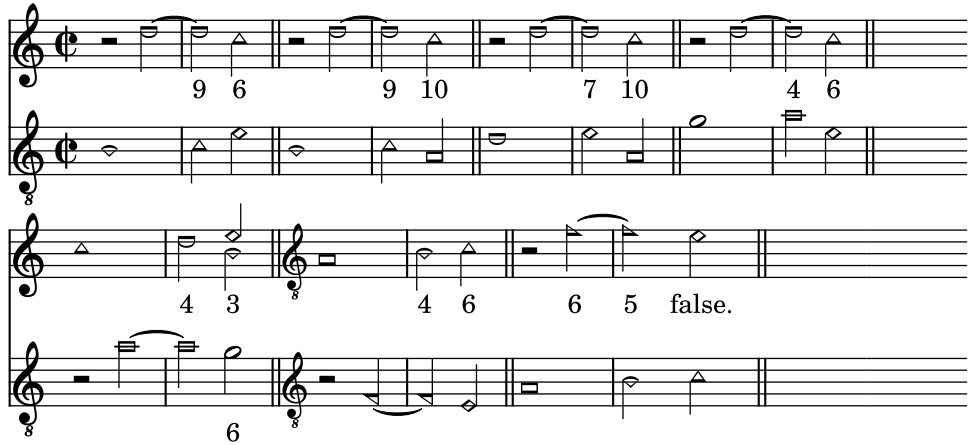
De Imitatione.

IMitatio fit, quando pars ſequens antecedentem ſequitur poſt aliquam pauſam ſervatis iiſdem intervallis, quibus antecedens inceſſit, nullâ Modi, Toni, Semitoniúque habitâ ratione; id quod fieri poteſt in Uniſono, Secundâ, Tertiâ, Quartâ, Quintâ, Sextâ, Septimâ, & Octavâ; quod exemplis clariùs patebit.

In Uniſono.



Jo-



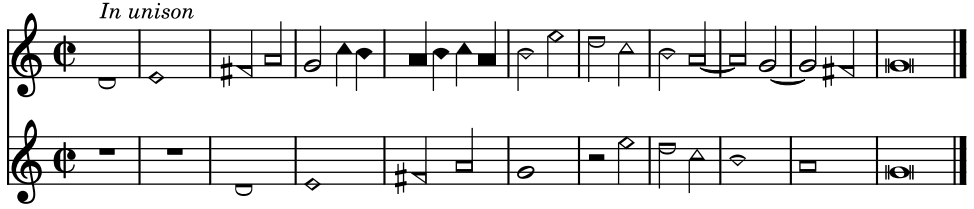
Joseph. Why do these resolutions occur outside the cantus firmus, and not also with the cantus firmus?

Aloys. Do you not see that in these resolutions both parts move? But this cannot be done with a cantus firmus, which is immovable. Therefore there is a difference: and it is clear that these resolutions, in which contrary motion is indispensable, can have no place. Now I will lead you back to the composition of the bicinium, unrestricted by any cantus firmus.

EXERCISE IV.

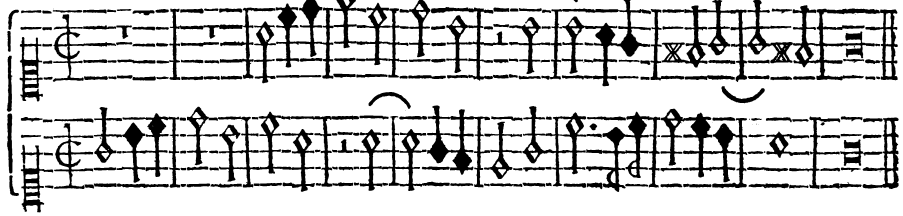
SOLE LESSON,

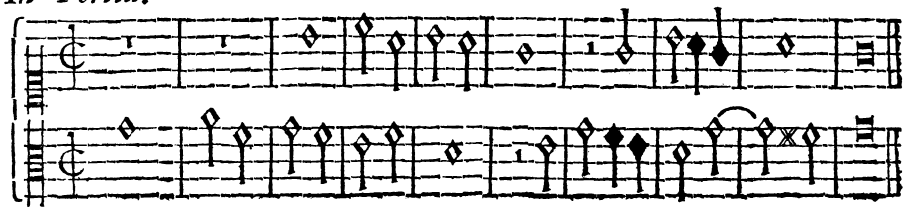
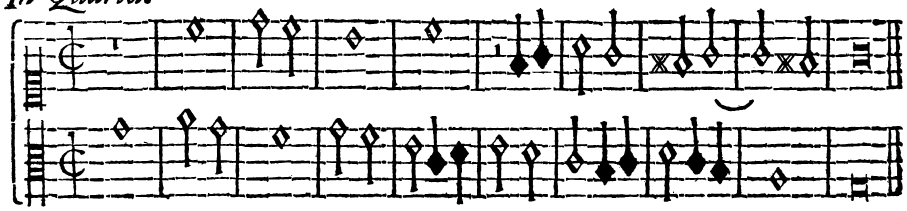
On Imitation.

Imitation occurs when the following part imitates the preceding one after some rest, preserving the same intervals in which the first part moved, without regard to mode, key, or semitone; which can be done at the unison, second, third, fourth, fifth, sixth, seventh, and octave, as will be clear from the examples.

Joſeph. Ex hoc Paradigmate colligo, non omnes antecedentis partis Notas in ſequente repetendas eſſe.

Aloyſ. Rectè collegiſti. Id enim muneris eſt Canonum, non imitationum, in quibus ſatìs eſt, quaſdam Notas proſequi.

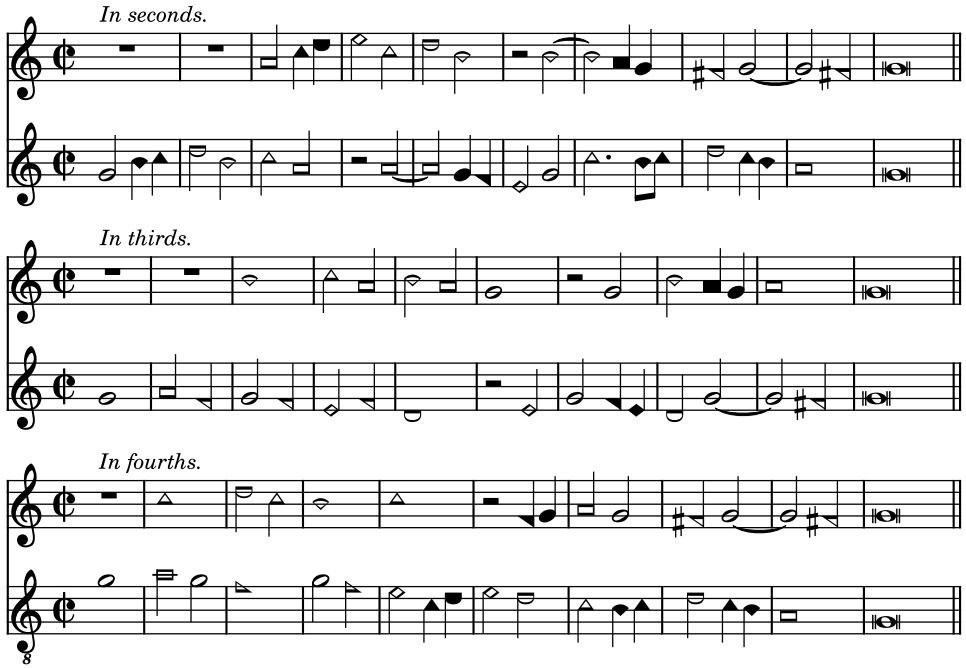
In Secunda.



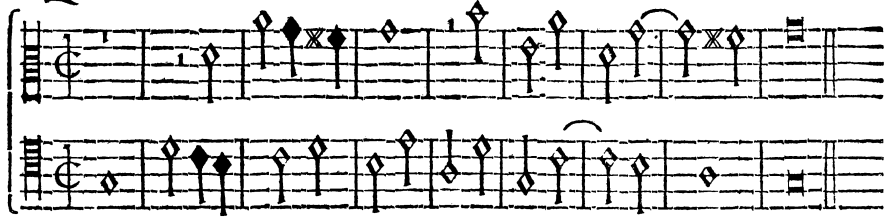
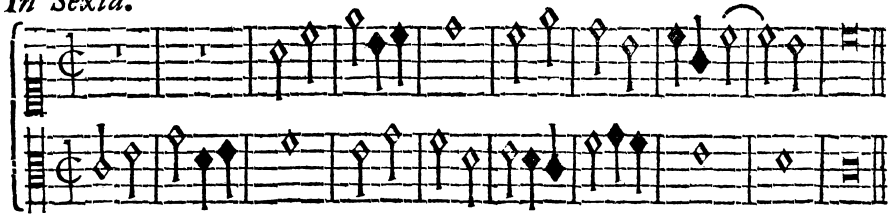
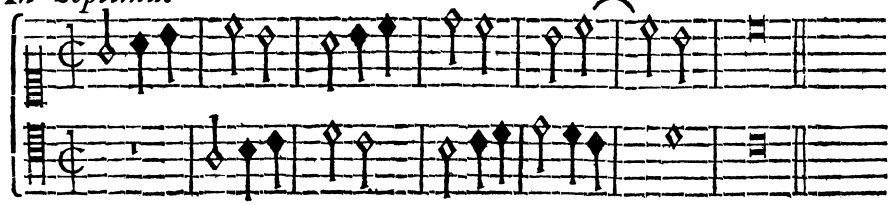
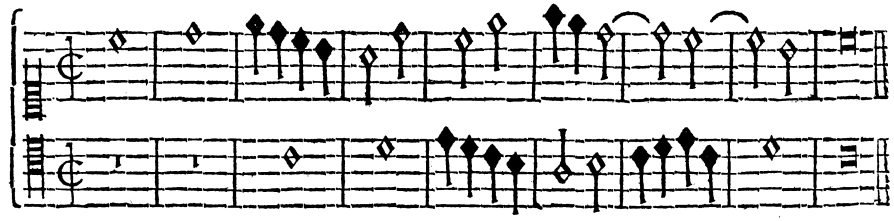
In Tertia.

In Quarta.

Joſeph. Videtur ultimum hoc exemplum incipere in C. & in G. finire.

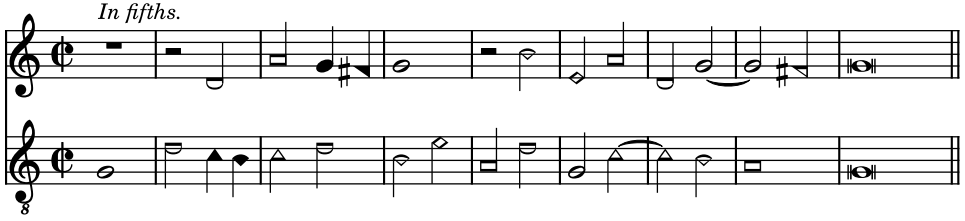
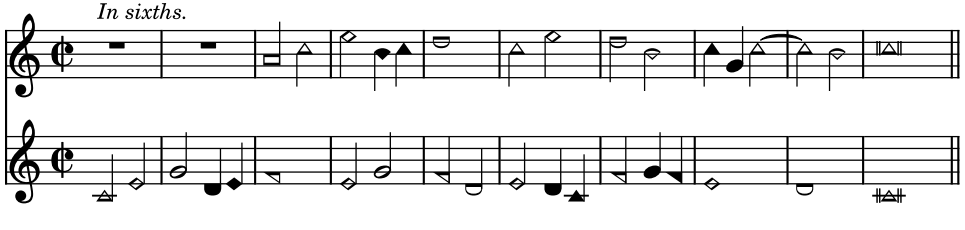
Aloyſ. Jam oblivio te cœpit illius, quod paulò antè dixi, imitationem nullo modi, vel toni vinculo obſtrictam, ſatìs eſſe, quocunque modo partem imitantem antecedentis veſtigia ſequatur? præterquam, quòd imitatio non tam in principio Concentûs, quàm in medio uſurpari ſoleat. Ubi rigor modi haud multùm attenditur.

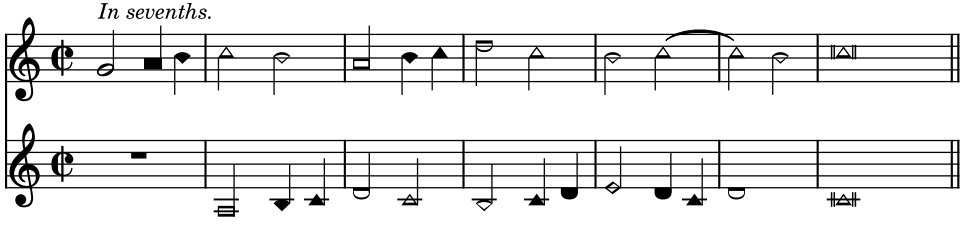
Joseph. From this pattern I gather that not all the notes of the leading part must be repeated in the following part.

Aloys. You have gathered rightly. That is the office of canons, not of imitations, in which it is enough to pursue some of the notes.

Joseph. It seems that this last example begins in C and ends in G.

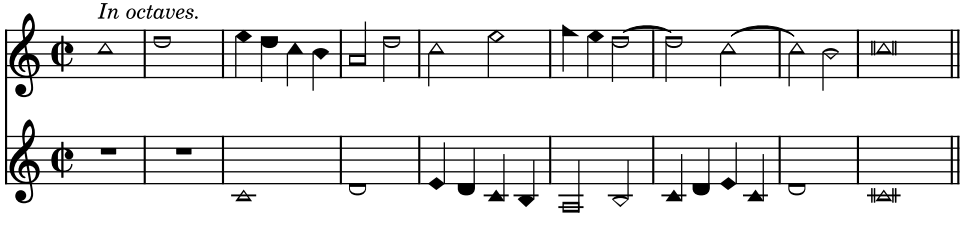
Aloys. Have you already forgotten what I said a moment ago—that imitation is bound by no constraint of mode or key, and that it is enough that the imitating part follow the footsteps of the preceding one in whatever way? Besides, imitation is used not so much at the beginning of a piece as in the middle, where the strictness of mode is not much regarded.

In Quinta.

In Sexta.

In Septima.

In Octava.

Aloyſ. Hæc exempla imitando non nihil temporis impendes; poſitis ad arbitrium aliquot Notis, quas de Nota ad Notam deſcriptas in quocunque intervallo libuerit, alteri parti inſeres, & ſupra quas variis Notarum figuris juxta Contrapuncti præcepta fabricando, habitâ concinnè canendi ratione Bicinium conficies.

In fifths.

In sixths.

In sevenths.



In octaves.



Aloys. By imitating these examples you will spend no small amount of time; having set down at your discretion some notes, written note-to-note in whatever interval you please, you will insert them into another part, and upon them, using various note-values according to the rules of counterpoint and with a graceful manner of singing, you will construct a bicinium.

EXER-

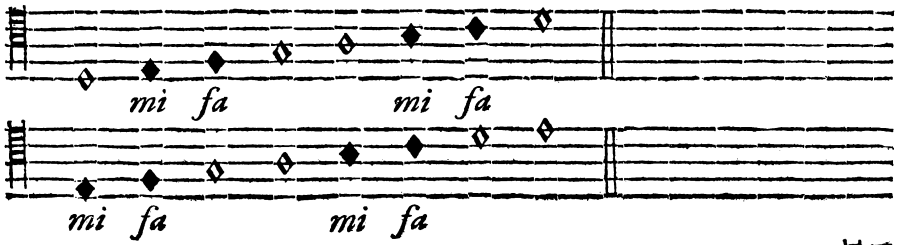
EXERCITII V.

LECTIO PRIMA,

De Fugis in genere.

FUgam à *fugere*, ac *fugare*, dictam complures haud aſpernandi Auctores affirmant ; ac ſi præcedens pars fugeret, fugata ab inſequente ; quod verum eſſet, nisì id ipſum, ùt dictum eſt, imitatio faceret. Quapropter alia definitio, quâ Fuga ab imitatione diſtinguatur, ſtatuenda eſt. Dico ergo : Fuga eſt quarundam Notarum in parte præcedenti poſitarum, ab ſequente repetitio, habitâ modi, ac plerumque toni, ſemitoniique ratione. Ad intelligentiam hujus definitionis, patet, neceſſe fore ſcire, quid ſit Modus. Per Modum intelligo id, quod vulgò tonus appellatur. Itaque dum dicitur : tonus primus, tonus ſecundus, &c. meliùs dicetur, Modus primus, Modus ſecundus, &c. ad tollendam æquivocationem intervalli ſeſquioctavi, 9. 8., atque ſeſquinoni, 10. 9. qui toni nomine quoque comprehenduntur. Cùm autem Modorum materia ſit fermè omnium intricatiſſima, nec incipienti facilis perceptu ; pro hîc, & nunc de eo tantum quantum ad noſtrum præſens propoſitum ſatìs eſt, explicâſſe contenti, plenam ejus explanationem ad finem Operis differendam exiſtimamus. Modus igitur eſt ſeries intervallorum infrà limites Octavæ contentorum, atque in vario, diſparique ſitu Semitoniorum collocatio. Huc pertinet illud Horatii :

Eſt modus in rebus, ſunt certi denique fines,
Quos ultrà, citraque nequit conſiſtere rectum.

Cùm varietas ſitûs Semitoniorum ſextuplex reperiatur, ſex Modi quoque ſtatuendi ſunt : qui ſequentibus Octavarum Siſtematibus exhibentur, nempe : D. E. F. G. A. C.

EXERCISE V.
LESSON ONE,
On Fugues in General.

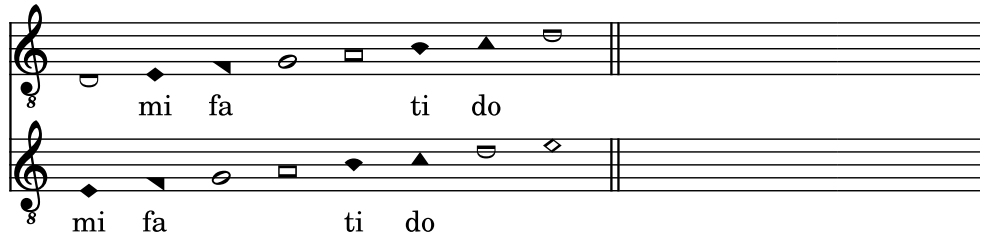
Many respectable authors affirm that the fugue is so called from fugere and fugare ("to flee" and "to chase"), as if the preceding part fled and were pursued by the following; which would be true, were it not that imitation itself does the same thing. Therefore another definition must be established by which the fugue is distinguished from imitation. I say, then: A fugue is the repetition by a following part of certain notes placed in the preceding part, with regard to the mode, and generally the key and semitone. To understand this definition, it is necessary to know what a modus (mode) is.

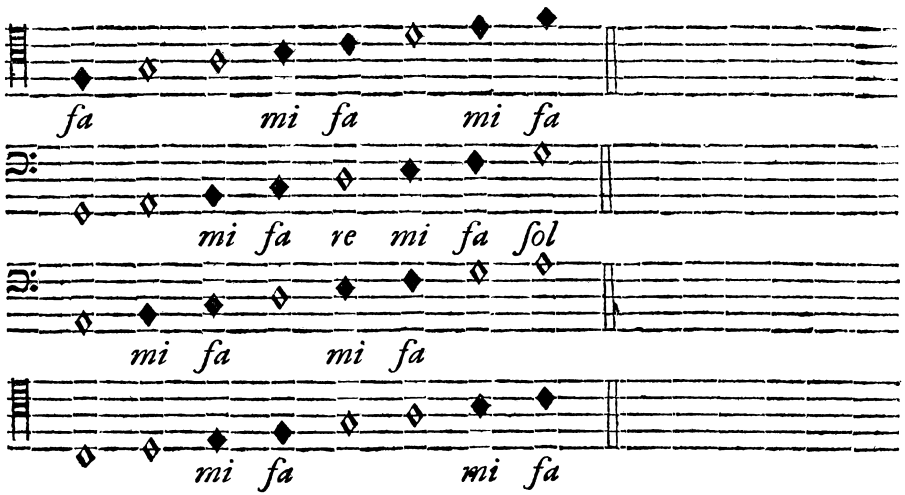
By modus I mean what is commonly called a tonus (tone). Thus, when one says "first tone," "second tone," etc., it is better to say "first mode," "second mode," etc., to avoid the ambiguity of the intervals sesquioctave (9:8) and sesquinone (10:9), which are also included under the name "tone." But since the subject of the modes is among the most intricate of all, and not easily grasped by a beginner, we are content here and now to explain only as much as is necessary for our present purpose, leaving a full explanation to the end of the work.

A mode is therefore a series of intervals contained within the limits of the octave, and the placement of semitones in various and differing positions. To this pertains the saying of Horace:

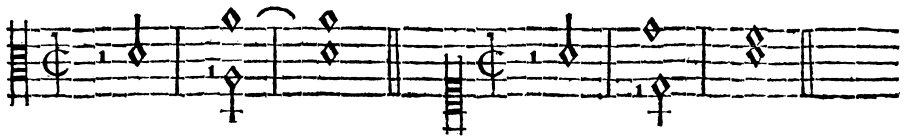
("There is a measure in things; there are fixed boundaries
beyond which right cannot stand.")

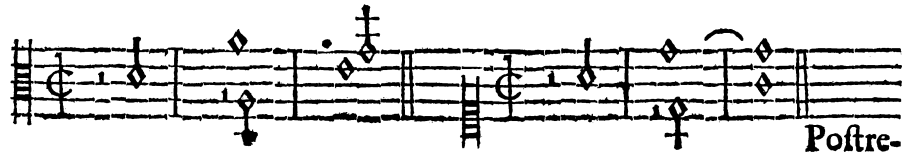
Since the placement of semitones is found in six different arrangements, six modes must also be established; which are shown in the following octave-systems, namely: D, E, F, G, A, C.

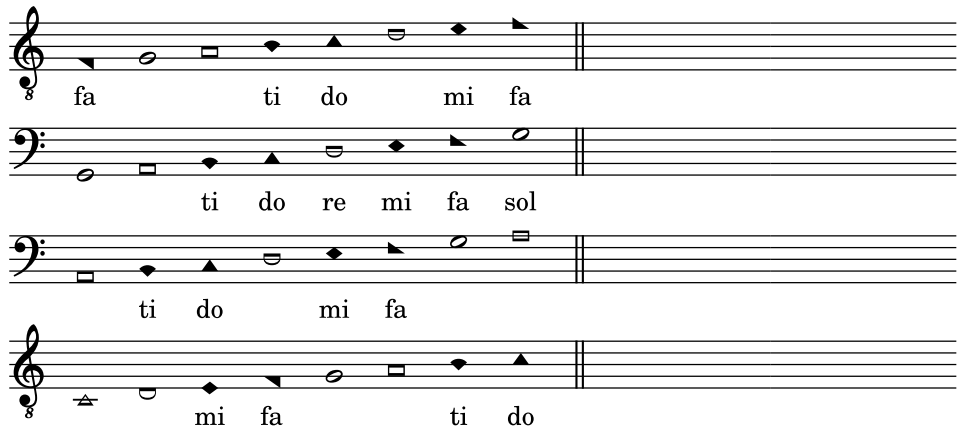


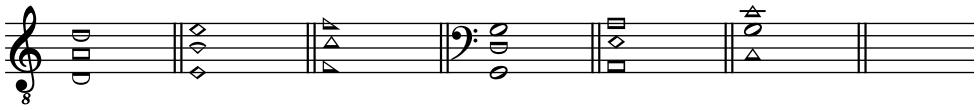
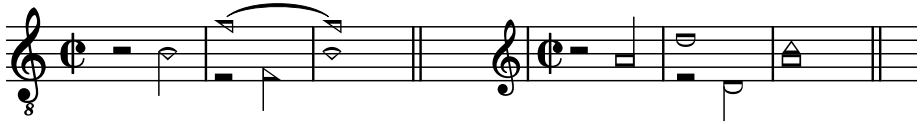
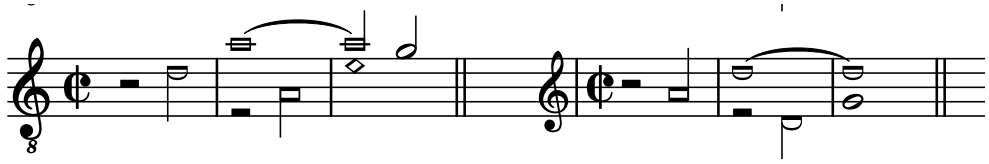


Vides in ſex diverſis locis Semitoniorum exiſtentiam ; numerando de cujuſvis Modi prima Nota : quæ diverſitas nigredine Notarum indicatur ; qui horum autem primus, ſecundus, tertiuſve ſit &c. in præſentia tranſibimus ; interim eo, quo hîc poſiti ſunt ordine, eos numerantes. Porrò Modus Quartæ, & Quintæ intrà Octavam exiſtentium limitibus terminatur : juxta quos limites Fugarum themata regulanda ſunt. E. C.

Adeò, ſi pars incipiens, Quintæ ſaltum efficiat, ne pars ſequens Modi, vel Octavæ limitem excedat, Quartæ ſaltum faciat, neceſſe ſit, & vice versâ. E. G.

Ad quod imitatio non obſtringitur ; cui ſufficit, eosdèm gradus, ac ſaltus ſequi. E. G.



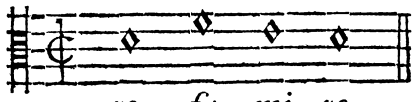
Poſtre-

You see the semitones in six different places, counting from the first note of each mode; this difference is indicated by the blackened notes. Which of these is first, second, third, etc., we pass over for now; meanwhile we number them in the order in which they are placed here.

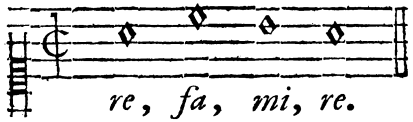
Furthermore, the mode is bounded by the limits of the fourth and fifth contained within the octave; according to these limits the subjects of fugues must be regulated. For example:

Therefore, if the opening part makes a leap of a fifth, the following part must make a leap of a fourth, lest it exceed the limit of the mode or the octave—and vice versa. For example:

Imitation, however, is not bound by this; it is enough for it to follow the same steps and leaps. For example:

Poſtremò, Fuga, extra intervalla Modum conſtituentia, v. g. extra Uniſonum, Octavam, & Quintam inſtitui non poteſt: Imitatio autem, ùt dictum eſt, in omnibus intervallis.

Joſeph. Hucuſque quæ de Modis, Imitatione, Fugaque dicta ſunt, percepiſſe mihi videor. Jam, ut edoceas, quæſo, quid in eligendis ſubjectis, & quo modo Fugæ inſtituendæ, atque ducendæ ſint: perſæpe enim audivi, rem Fugarum non omnibus hujus diſciplinæ Profeſſoribus notam eſſe.

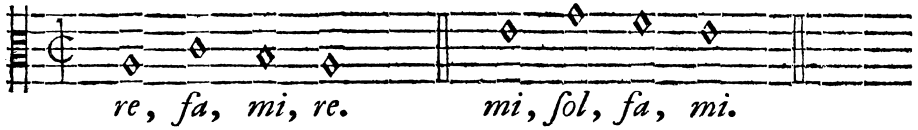
Aloyſ. Ità eſt, frequentior de illis auditur ſermo, quàm ſcientia. Ut itaque nihil earum te lateat, quod quæsîſti, exponam. Sex differentes modos dari, ex antè dicta demonſtratione conſtat. Ideoque cuivis Modorum etiam proprium ſuum ſubjectum aptandum eſſe. Ex quo infertur, thema primo Modo proprium, ſecundo, tertio, quarto, quinto, ſexto minimè convenire, propter tonorum, ſemitoniorumque ſitum, in quolibet Modo differenter repertum, ùt exempla monſtrabunt.

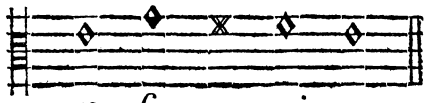
Subjectum primi Modi proprium.





Hoc thema, à parte ſequente in Quinta reaſſumptum eosdem tonos, & ſemitonia ſervat. E. G.

Id, quod in nullo cæterorum Modorum huic ſubjecto uſuvenire poteſt. Videamus Modum *A, la, mi, re,* qui ſolus Modo *D, la, ſol, re,* æqualitate proximus eſt:

Ubi in partis ſequentis penultima Nota locò *mi, fa* venit, & toni, ſemitoniique ſitus invertitur; nec ſubjecti ſiſtema in prima parte propoſitum, aliter haberi poteſt, niſi Dieſis penultimæ Notæ præpoſitæ adminiculo. E. G.

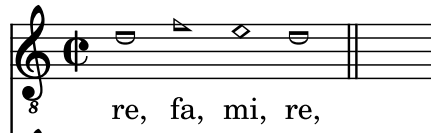
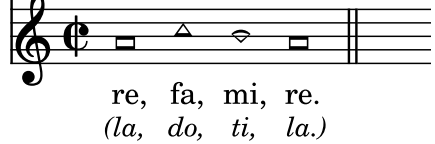


Id quod genus Diatonicum, in quo nunc verſamur, haud admittit; ſed omni

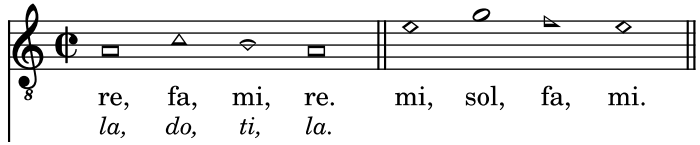
Finally, a fugue cannot be constructed outside the intervals that constitute the mode—for example, outside the unison, octave, and fifth; but imitation, as has been said, can be made in all intervals.

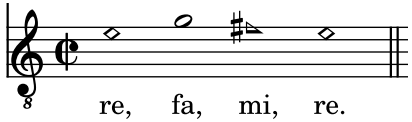
Joseph. Up to this point I think I have understood what has been said about modes, imitation, and fugue. Now, I beg you, teach me what must be considered in choosing subjects, and how fugues are to be begun and carried through; for I have often heard that the matter of fugues is not known to all professors of this discipline.

Aloys. So it is; one hears talk about these things far more often than one finds real knowledge. So that nothing you have asked about may remain hidden from you, I will explain. It is clear from what has already been shown that there are six different modes. Therefore each mode must also have its own proper subject. From this it follows that a theme proper to the first mode is by no means suitable for the second, third, fourth, fifth, or sixth, because the arrangement of tones and semitones is different in each mode, as the examples will show.

A subject proper to the first mode:

This theme, when taken up again by the following part at the fifth, preserves the same tones and semitones. For example:

This cannot occur in any of the other modes with this subject. Let us examine Mode A–la–mi–re, which alone is closest in structure to Mode D–la–sol–re:



Here, in the penultimate note of the following part, fa appears in place of mi, and the placement of tones and semitones is inverted; nor can the system of the subject proposed in the first part be preserved in any other way unless a sharp is placed before the penultimate note. For example:

But this the genus diatonicum (diatonic genus), in which we are now working, does not admit; one must abstain entirely from sharps

Dieſi, & b. molli in thematibus abſtinere debet : aliàs enim nunquam genuinam Modorum naturam compertam haberemus.

EXERCITII V.

LECTIO SECUNDA,

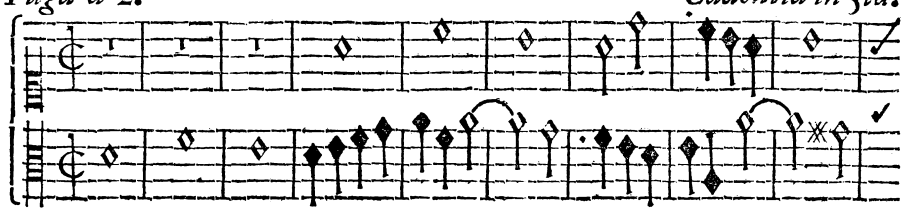
De Fuga duarum Partium

NUnc Fugas duarum partium, planâ primò, ac ſimpliciſſimâ methodo conficiendi artem te docebo. Accipe ſubjectum aliquot Notarum, Modo, in quo laborare intendis, appropriatum : illasque Notas inſcribe parti, quâ incipere animo concepiſti : quibus finitis, ſi modo nihil derogetur, ſecundæ parti, ſive ſequenti, in Quarta, vel Quinta intrando easdem Notas dabis : interim cum parte, quâ thema incepiſti, variis Notarum figuris modulando, ut Contrapuncto florido edoctus es, Contrapunctum ſuperinſtrues. Quo facto, poſt aliquantulam modulationem partes eò diſponendo primam clauſulam in quinta Modi conficies. Poſt hæc, poſitâ integri tactûs, aut dimidii pausâ, vel intercedente magno ſaltu, etiam absque pauſa, in parte fermè, quâ incepiſti, thema reaſſumes ; ſed alio, quàm inchoatum eſt, intervallo : pars altera, factâ aliquâ pausâ, priùs, quàm ſubjectum in altera parte finitum ſit, intrare tentabit ; deinde nonnihil modulando ſecundam clauſulam in Tertia Modi conſtrues. Poſtremò, poſito in utra partium ſubjecto, alteram partem in ſecundo protinus ſubjecti tactu, ſi res patitur, cum ſubjecto intrare facies ; & ita, conjunctis arctiſſimo vinculo partibus, clauſulâ finali Fugæ finem impones.

Joſeph. Memini, te ſuprà dixiſſe - - - -

Aloyſ. Differre dubium tuum nunc expedit, donec res obſcura Paradigmate clareſcat. Accipiam itaque ſubjectum, quò paulò ante Modi natura demonſtrata eſt, præſcriptoque modo id ducendo, Fugam efficiam ; quâ normâ facilior, & expeditior via tibi ſtrata erit. E. G.

Fuga à 2. *Cadentia in 5ta.*



and flats in subjects, otherwise we would never truly understand the nature of the modes.

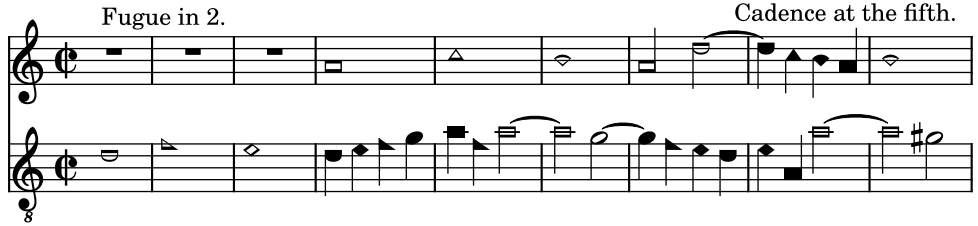
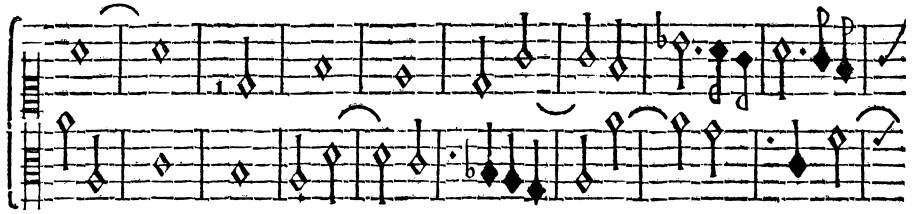
EXERCISE V.
LESSON TWO,
On the Fugue for Two Parts

Now I will teach you the art of composing two-part fugues in a clear and very simple method. Take a subject of several notes appropriate to the mode in which you intend to work, and write those notes in the part with which you have decided to begin. When these are finished, if nothing prevents it, give the same notes to the second part—the follower—entering at the fourth or fifth. Meanwhile, with the part in which you began the theme, you will weave a counterpoint, using various note-values as you have learned in contrapunctus floridus (florid counterpoint). When this is done, after a little modulation, arrange the parts so that the first cadence is made on the fifth of the mode. After this, with a rest of a whole or half measure, or by means of a large leap—even without a rest—you will reintroduce the theme in the part with which you began, but at an interval different from the opening. The other part, after a brief rest, will attempt to enter before the subject in the first part has finished; then, after some modulation, you will form the second cadence on the third of the mode. Finally, when the subject has been placed in either part, you will cause the other part to enter with the subject at the second beat, if possible; and thus, the parts being bound together in the closest union, you will end the fugue with the final cadence.

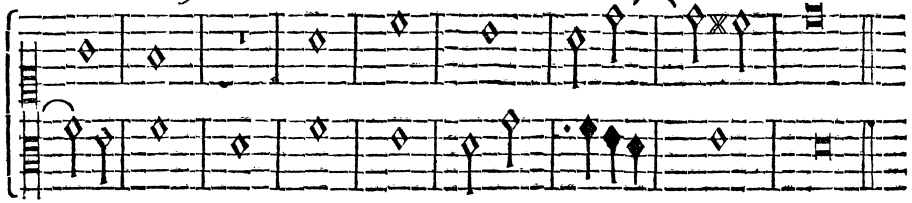
Joseph. I remember that you said above — — —

Aloys. It is better to postpone your question until the matter becomes clear through an example. I will therefore take the subject by which the nature of the modes was shown a little earlier, and by treating it according to the prescribed method, I will compose a fugue; by this pattern an easier and more direct path will be laid out for you. For example:

Fugue in 2. Cadence at the fifth.

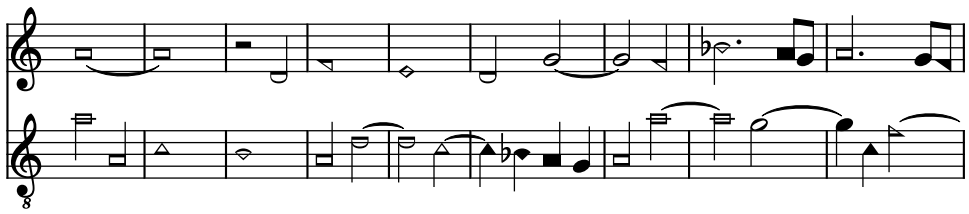


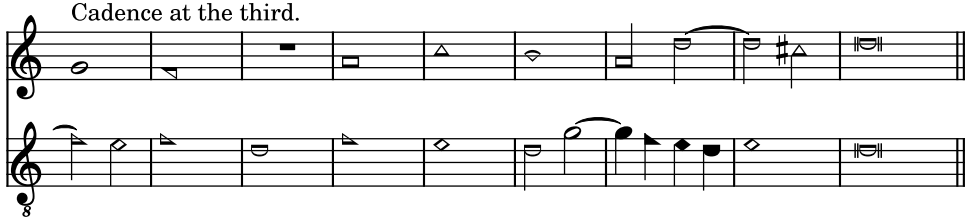
Cadentia in 3tia.



En finitum in Alto ſubjectum, à Cantu idque in Quinta Modi aſſumi; interim Altum Contrapuncti floridi ſpecie vagari; uſquedum ſubjectum in Cantu finitum ſit. *Poſt* factâ brevi, utraque in parte, modulatione, clauſulam in Quinta modi confici. *Quo facto*, vides ab Alto repeti ſubjectum in Quinta Modi inferiùs, absque pauſa quidem, quia per ſaltum: deinde poſitâ pausâ intrare partem Cantûs in Octava Modi; ſed maturiùs quàm in principio, nempe in tertio ſubjecti tactu, modulante interim Alto, finitoque denique ſubjecto in Cantu poſt brevem utriusque partis modulationem conſpicis ſecundam clauſulam in Tertia Modi conſtructam. *Poſtremò* comperies ſubjecto in Alto aſſumpto, ab Cantu extemplò in ſecundo tactu idem ſubjectum intextum eſſe, indeque clauſulâ finali finem imponi. Hac itaque ſimpliciſſimâ methodo uſus, ac cæterorum exercitatione Modorum gradatim pleniorem Fugarum ſcientiam adepturum te, nullum habe dubium.

Joſeph. Video in hac Fuga aliquibus in locis Dieſes, & b mollia poſita eſſe, contra præceptum de iisdem in Genere Diatonico non adhibendis.

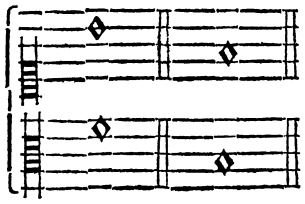
Aloyſ. Intelligendum puto de ſubjectis, ubi ad noſcendam Modorum naturam differentiamque, Semitonia naturali ratione venire debent, absque Dieſis, vel b mollis adminiculo. Alia ratio eſt in Modulatione, ubi Dieſis, atque b mollis uſus non ſolùm non prohibitus, verùm etiam ad evitandam aſperam *mi* contra *fa* relationem nonunquam neceſſarius eſt. Oriuntur autem dictæ relationes ex ſequenti fermè *mi* contra *fa* oppoſitione.

Cadence at the third.



Here you see the subject completed in the Alto, and taken up by the Cantus at the fifth of the mode, while the Alto wanders in the style of florid counterpoint until the subject in the Cantus is finished. After a brief modulation, you see a cadence formed in both parts on the fifth of the mode. Then you see the Alto repeat the subject a fifth lower, without a rest, because it enters by a leap; then, after a rest, the Cantus enters at the octave of the mode, but earlier than at the beginning—namely, at the third beat of the subject—while the Alto continues to modulate. When the subject in the Cantus is finished, after a short modulation in both parts, you see the second cadence formed on the third of the mode. Finally, when the subject is taken up in the Alto, you see the Cantus immediately enter with the same subject at the second beat, and the fugue is brought to an end with the final cadence. Using this simplest method, and by practicing the remaining modes, you will gradually acquire fuller knowledge of fugues—of this have no doubt.

Joseph. I see that in this fugue sharps and flats appear in several places, contrary to the rule that they are not to be used in the diatonic genus.

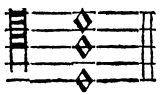
Aloys. You must understand this rule as applying to subjects, where the natural placement of semitones must be preserved in order to know the nature and distinction of the modes, without the aid of sharps or flats. It is otherwise in modulation, where the use of sharps and flats is not only permitted but sometimes necessary to avoid the harsh relation of mi contra fa ("mi against fa"). These relations arise from the following opposition of mi and fa:



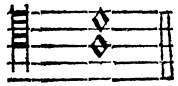
Jam aperi dubium tuum, quod ſuprà proponere in animo tibi fucrat.

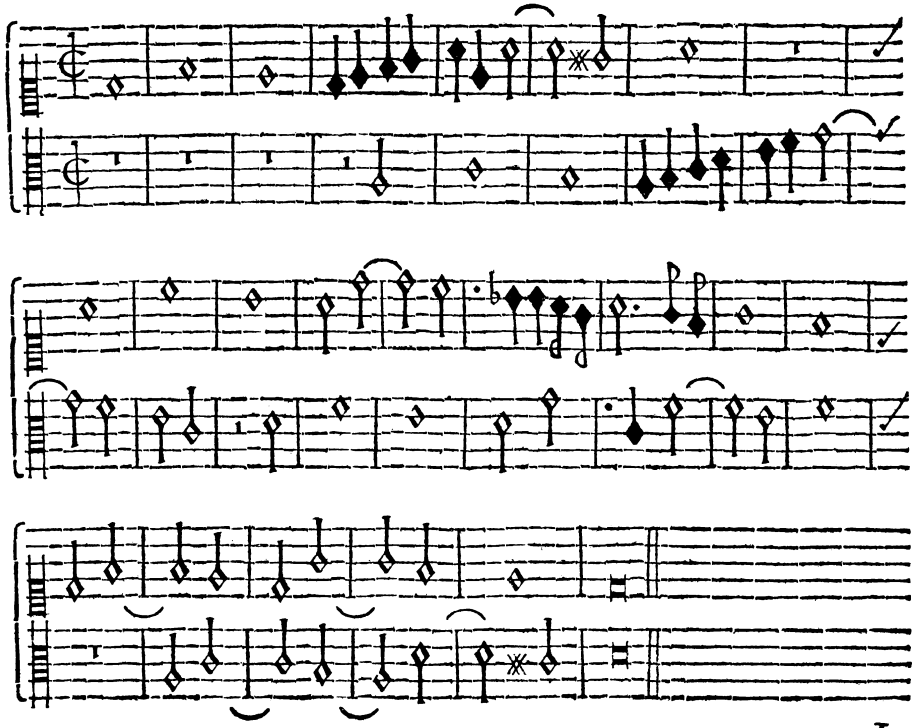
Joſeph. In diſcrimine inter Fugam, Imitationemque ſuprà dato aſſeruiſti, in Fuga parti ſequenti non nisì in Quinta, Octava, & Uniſono facultatem intrandi fore; paulò verò poſt in dogmate conficiendæ Fugæ de Quarta quoque mentionem feciſti.

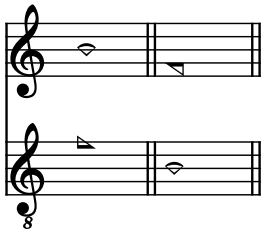
Aloyſ. Primo loco reſpectus habitus eſt ad Notam fundamentalem, ſequensque ſiſtema:



Cui quarta utique non ineſt: ſecundo verò loco ommiſsâ Notâ fundamentali, habitoque reſpectu partis ſuperioris ad ſequens ſiſtema reſpexi:



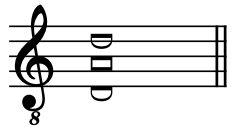
Quanquam etiam hoc in caſu A. illud Quinta dici poſſit, quia Nota fundamentalis D. ſubintelligi debet; unde ſequentis exempli pars ſecunda in Quinta ſequi cenſenda eſt. E. G.

Jo-



Now state the question you intended to ask earlier.

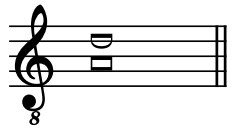
Joseph. In the distinction you gave above between fugue and imitation, you asserted that in a fugue the following part may enter only at the fifth, octave, or unison; yet a little later, in the rule for composing a fugue, you also mentioned the fourth.

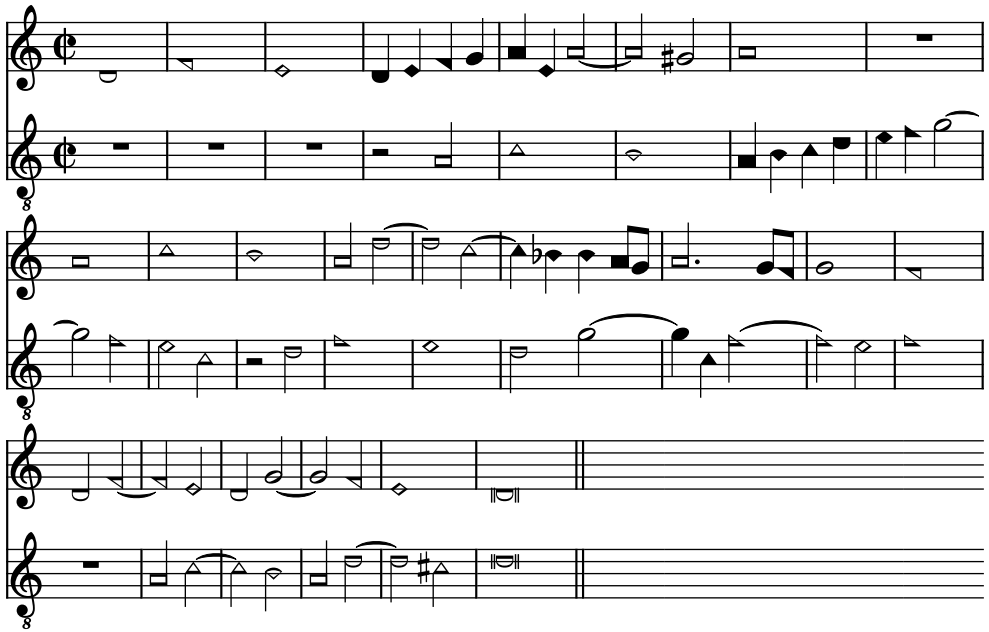
Aloys. In the first place, I was speaking with reference to the fundamental note and the system that follows it:

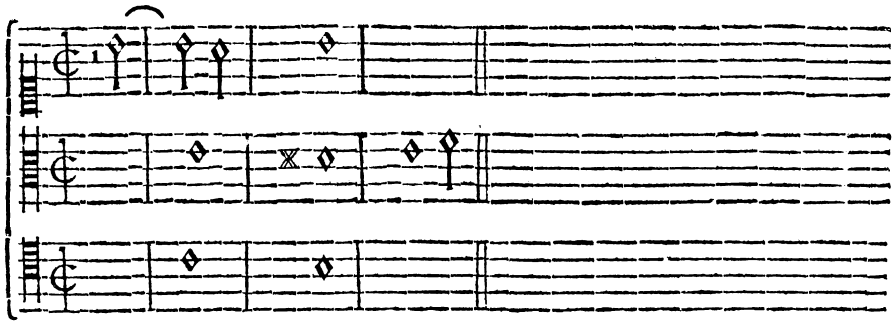


In which the fourth is not included.

But in the second place, leaving aside the fundamental note, I was speaking with reference to the upper part and the system that follows it:



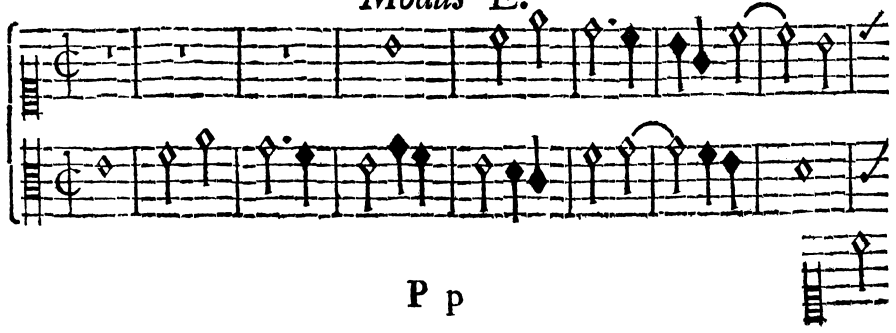
Although even in this case that A may be called a fifth, because the fundamental note D must be understood; therefore the second part in the following example is to be considered as entering at the fifth. For example:

Joſeph. Quid eſt, quòd ad finem Fugæ, idque in ſubjecto Notarum figuras nonnihil immutatas conſpicio ?

Aloyſ. Dico, non ſolùm permiſſum eſſe ſubinde ſolutas Notas ligari, verùm etiam, neſcio quid grati per hoc accedere Compoſitioni: imò perſæpe, ubi ſubjecta in anguſtum contrahi aliter non poſſunt, hâc figurarum fractione opùs eſſe. Nunc tu ipſe alterum modum E. ad manus accipe, & electo ſubjecto, nulli alteri Modorum conveniente, ad prioris exempli normam Fugam conficere tenta: hoc tamen diſcrimine, quòd, ſicut & in priori, & omnibus aliis Modis prima Clauſula in Quinta fit, ita in hoc in Sexta facienda ſit; ob rationem, quod accedente parte tertiâ, Tertiam majorem exigente, nimis aliena foret ab hoc Modo modulatio; peſſimaque ex *mi* contra *fa* oriunda modulatio propter mox ſecuturum F. aures offenderet. E. G.

Nunc intellectis omnibus, ùt puto, ad hoc neceſſariis, ad operam te accinge.

Joſeph. Si primum tentamen non bene mihi ceſſerit, ne ægrè feras, te Magiſter etiam, etiamque rogo.

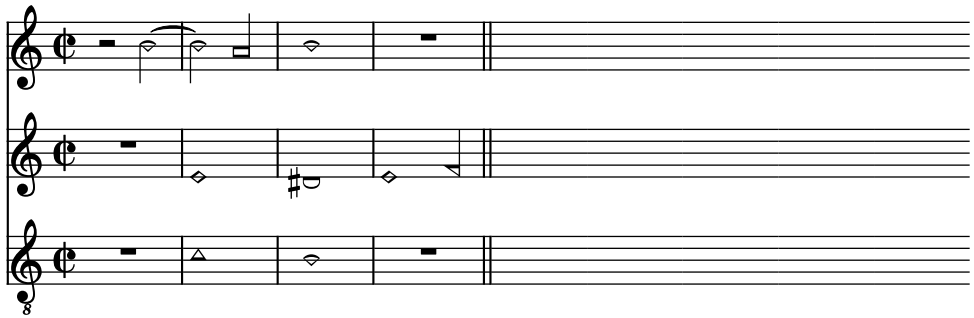
Aloyſ. Mitte timorem; noſcis enim patientiam meam; nequc ego ignoro hujus ſtudii in tyronibus difficultatem.

Modus E.



Joseph. Why is it that at the end of the fugue I see the note-values of the subject somewhat altered?

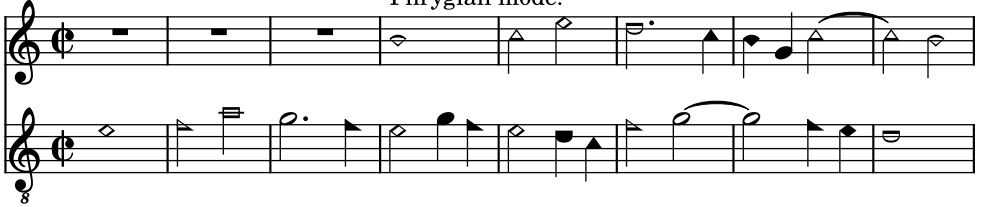
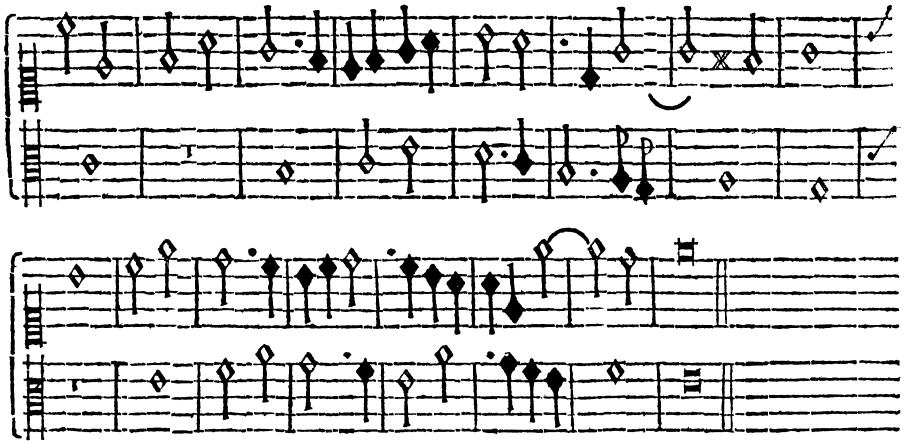
Aloys. I say that it is not only permitted occasionally to tie notes that were previously free, but that something pleasing is often added to the composition by doing so; indeed, when the subject cannot otherwise be contracted into a narrow space, this breaking up of the note-values becomes necessary. Now take up Mode E yourself, and, choosing a subject that fits no other mode, attempt to compose a fugue according to the pattern of the previous example; with this difference, however: that just as in the previous mode, and in all others, the first cadence is on the fifth, so in this one it must be on the sixth. The reason is that, with the entrance of a third part requiring a major third, the modulation would be too foreign to this mode; and the harsh modulation arising from mi contra fa, because of the F that immediately follows, would offend the ear. For example:



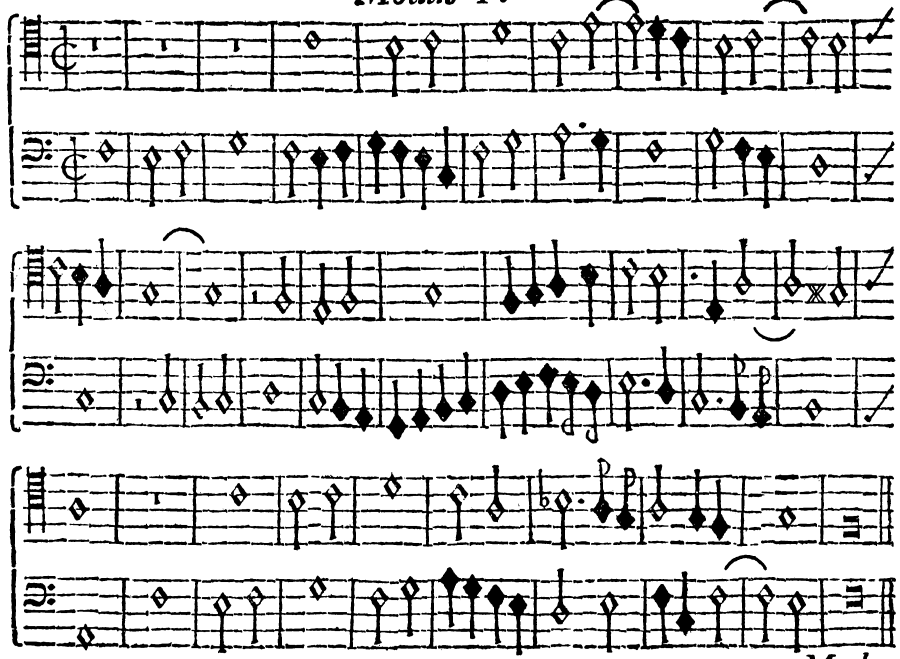
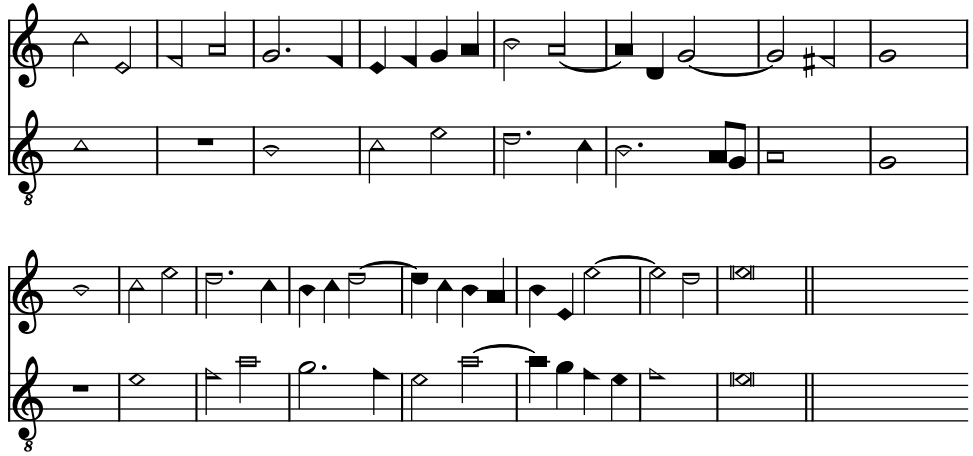
Now that all things necessary for this have been, I think, sufficiently explained, set to work.

Joseph. If my first attempt does not turn out well, do not be displeased, I beg you, Master.

Aloys. Put aside your fear; you know my patience, and I am not unaware of the difficulty of this study for beginners.

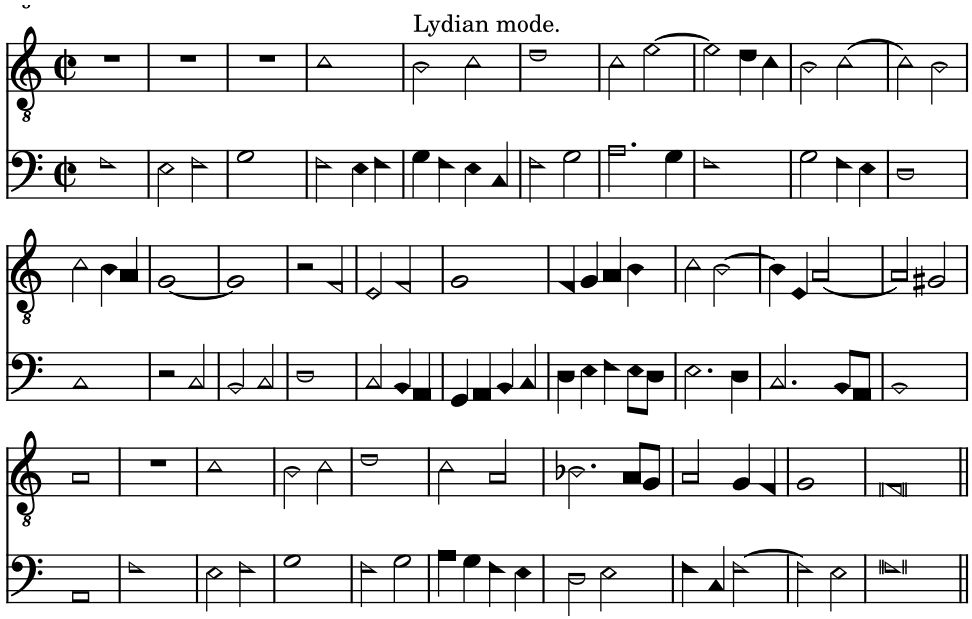
Phrygian mode.

Joſeph. Subjectum nulli alteri Modorum in genere Diatonico applicari poſſe confido; an autem tractus æquè reſpondeat, à te intelligere cupio.

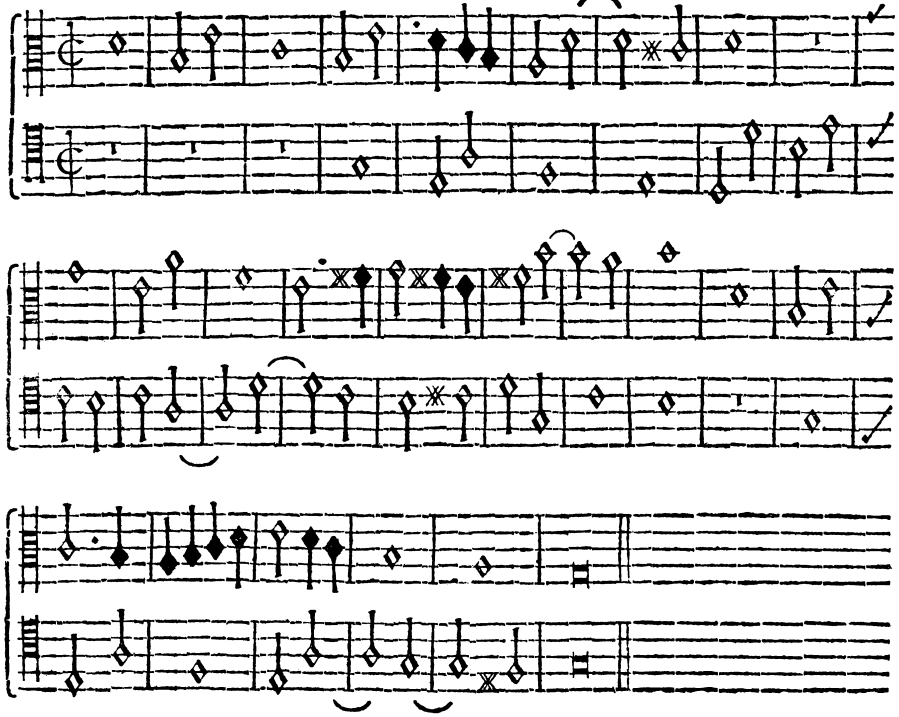
Aloyſ. Mactè animo, Joſephe, & ſubjectum tam ſcitè inventum, ejusdemque tractûs fermè Tyrocinium tuum excedunt; ſpemque facis haud vulgaris hac in arte futuri progreſsûs. Perge nunc ad Modum F. & cæteras ſuperſtitum Modorum claves.

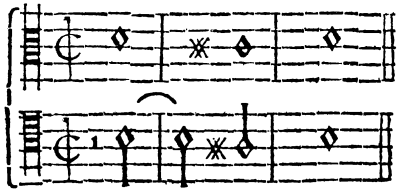
Modus F.

Modus

Joseph. I am confident that the subject cannot be applied to any other mode in the diatonic genus; but whether the treatment corresponds equally well, I wish to learn from you.

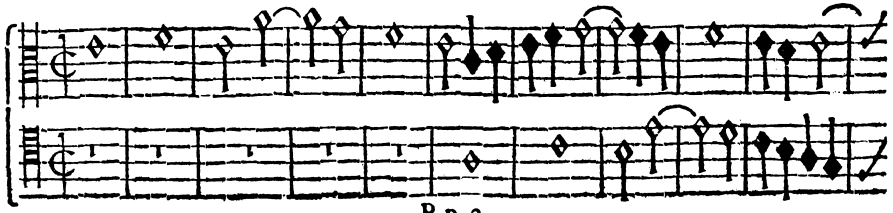
Aloys. Well done, Joseph! The subject is so cleverly invented, and its treatment so nearly exceeds your beginner's level, that you give hope of no ordinary progress in this art. Now proceed to Mode F and the remaining keys of the modes.

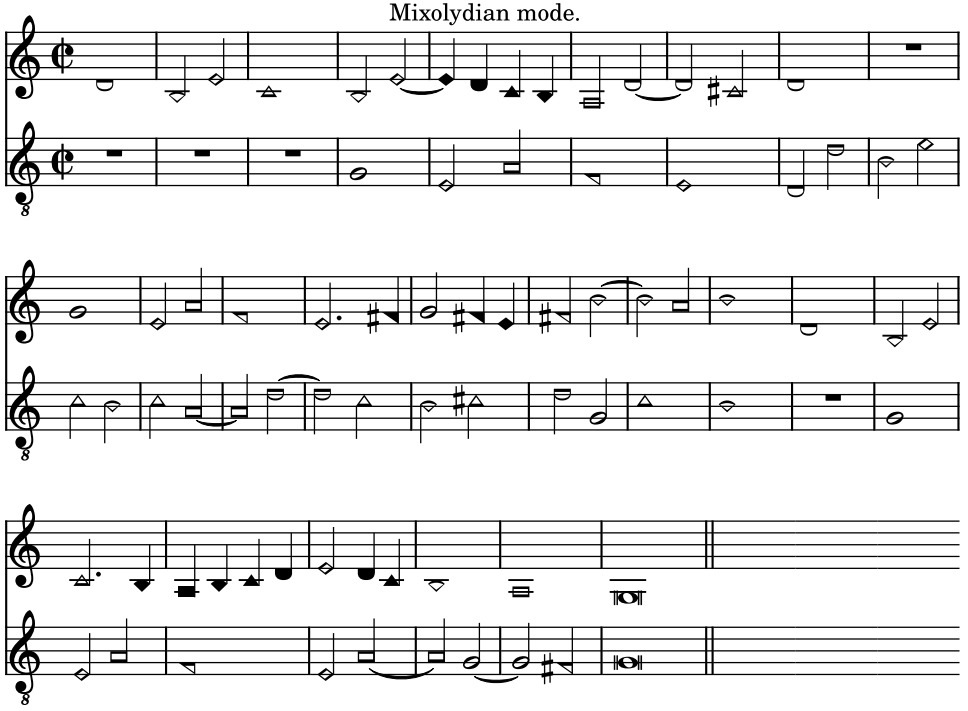
Lydian mode.



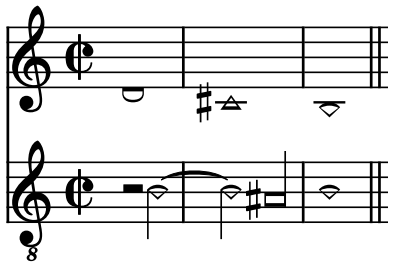
Modus G.

Joſeph. In clauſula ſecunda conficienda nonnihil hæſitabam, an ſic, vel ſequenti modo facienda ſit. E. G.



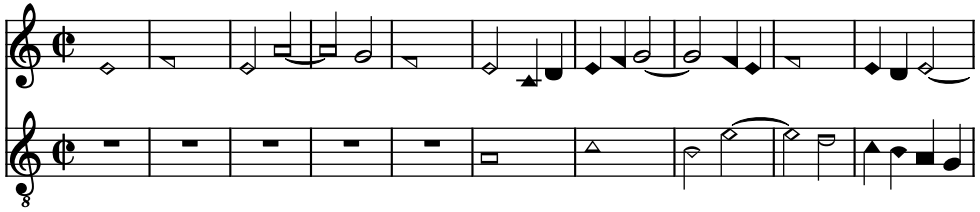
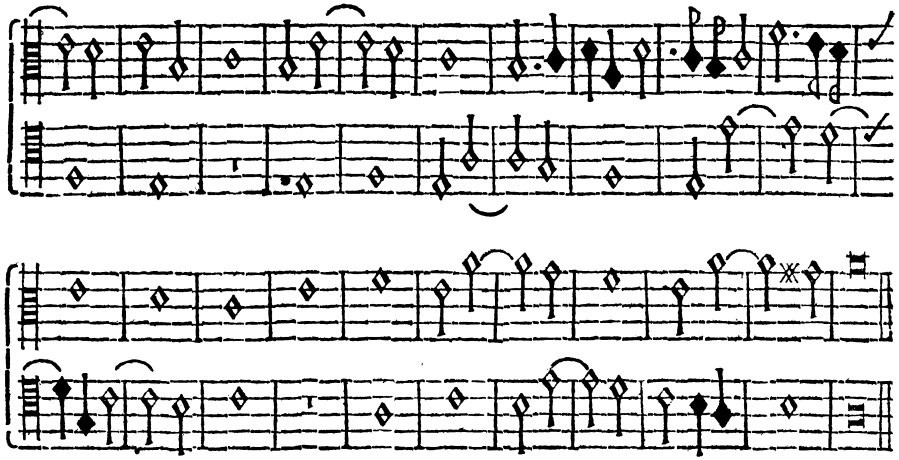
Aloyſ. Clauſula iſta, cui duabus opus eſt Dieſibus, imò tribus, accedente parte tertiâ, nimis ab hujus Modi natura recedit; magisque conveniens eſt Modo transpoſito, de quibus infrà dicetur. Valet itaque clauſula per Septimam, & Sextam à te uſurpata. Jam ad Modum A. ordine noſtro quintum procedamus, cujus, quum aliquantò diſcrepet ab aliis Modis, exemplum ego tibi ante oculos ponam. E. G.

Mixolydian mode.



Joseph. In forming the second cadence I hesitated somewhat whether it should be done thus, or in the following way. For example:



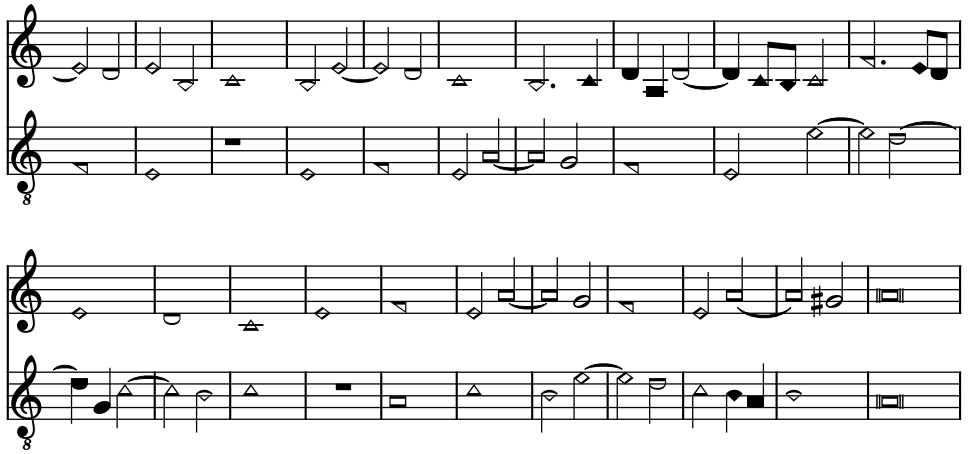
Aloys. That cadence, which requires two sharps—indeed three, if a third part is added—departs too far from the nature of this mode, and is more suitable to a transposed mode, of which we shall speak later. Therefore the cadence you used, through the seventh and sixth, is valid. Now let us proceed to Mode A, the fifth in our order, which differs somewhat from the others; I will place an example before your eyes.



Joſeph. Cur de ſecundæ partis prima Nota ad ſecundam per ſaltum Tertiæ proceſſeris, comprehendo; quia *fa* illud in primæ partis ſecunda Nota expreſſum aliàs haberi non licuiſſet. Sed quare poſt primam Clauſulam reaſſumptio ſubjecti in Alto in *B. mi*, itaque in chorda Modo non convenienti facta fuerit, ignoro.

Aloyſ. Hanc neceſſitatem cauſat partis fundamentalis in E. exiſtentia, quæ impedit introitum in A. utpote diſſonantia reſpectu E. infrapoſiti: verùm nihil perfectioni Compoſitionis derogatur, imò in meliùs vertitur, ſi id in medio Fugæ fiat; quia hoc pacto partes ſubjecti ubique æquantur. Aliud judicium ferendum eſt de principio Fugæ, ubi ratio Modi, & chorda introitioni ſubjecti debita, de qua jam dictum eſt, accuratè tenenda eſt.

Joſeph. Magnam animadverti difficultatem in ſubjectis magìs, magìsque contrahendis, & videtur, ſubjecta de induſtria ad id eligenda fore.

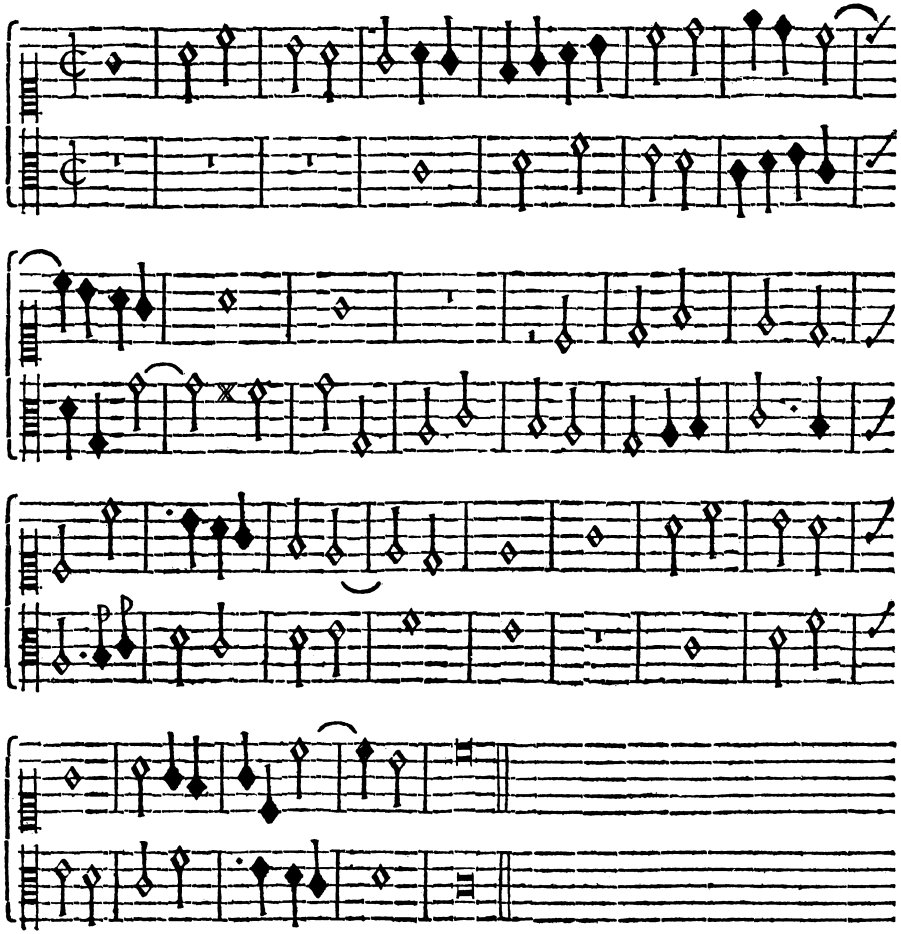
Aloyſ. Ità eſt. Probanda itaque primò, inquirendaque ſubjecta ſunt, hac ratione tractabilia: ideoque ante ſelectum ad id reflectendum erit. Cæterùm nonneminem fortaſſis audies cachinando tibi objicientem: quàm trita, jejuna ſunt hæc ſubjecta; nulliusque ſaporis? Rem verò ità ſe habere, neutiquam inficias ivero. Non enim ad conceptûs ſublimitatem hîc reſpicitur, ſed ſolùm ad genuinam in Genere Diatonico Modorum naturam, quod Genus limitibus ſatìs anguſtis includitur: major excurrendi, ſuccoſioraque capiendi ſubjecta libertas erit, ubi mixti Generis uſus concedetur, de quo ad finem Libri Primi

Joseph. I understand why you proceeded from the first note of the second part to the second by a leap of a third—because that fa expressed in the second note of the first part could not otherwise have been obtained. But why, after the first cadence, the subject was taken up again in the Alto on B-mi, and thus on a pitch not belonging to the mode, I do not understand.

Aloys. This necessity is caused by the presence of the fundamental part on E, which prevents an entry on A, since A would be a dissonance with respect to the E below it. But nothing is lost to the perfection of the composition; indeed, it is improved if this happens in the middle of the fugue, because in this way the entries of the subject are balanced everywhere. A different judgment must be made about the beginning of the fugue, where the nature of the mode and the proper pitch for the entrance of the subject—of which we have already spoken—must be observed with care.

Joseph. I have noticed great difficulty in contracting subjects more and more, and it seems that subjects must be chosen deliberately with this in mind.

Aloys. So it is. Therefore subjects must first be tested and examined to see whether they can be handled in this manner; and so, before choosing one, you must reflect on this point. Moreover, you may perhaps hear someone laughing and saying to you: "How worn-out and barren these subjects are—without any flavor!" I will by no means deny that the matter stands thus. For here we are not concerned with sublimity of invention, but only with the genuine nature of the modes in the genus diatonicum (diatonic genus), which is confined within rather narrow limits. A greater freedom to range more widely and to choose richer subjects will be granted when the use of the mixed genus is

Primi abundanter dictum eſt. Nunc ad ultimum Modum C. progredere.

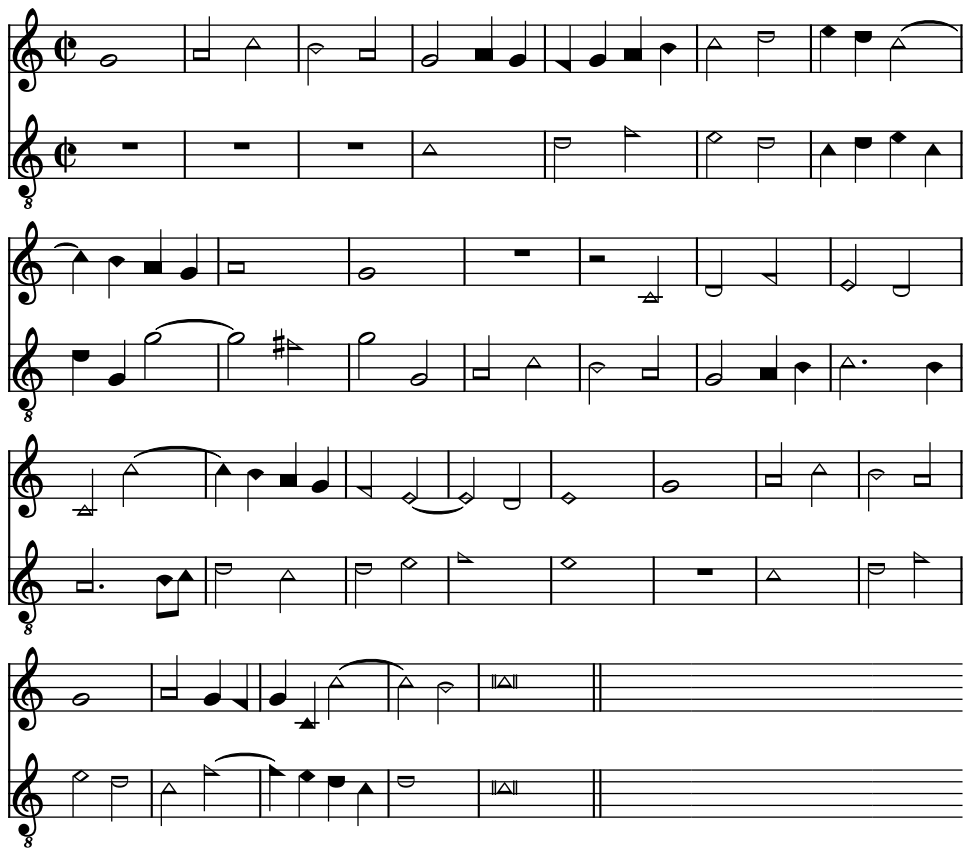


Hæc itaque eſt adumbratio, veſtibulumque, quâ patet ingreſſus ad Fugas.

Quemadmodum ars pingendi nisì præviâ frequenti delineatione non aquiritur, ità quoque ad uberiorem Fugarum ſcientiam opus eſt haud brevi ad formulam præſcriptam exercitatione, quam tibi ſummopere commendo, alia ſemper, atque alia inveſtigando.

Joſeph. Inops mihi videtur hoc Genus, atque magnâ laborare penuriâ, inde exiguæ pervium eſſe inventioni.

Aloyſ. Reſtrictiùs eſſe Genere mixto, neque adeò patentem excurrendi campum habere, non contradico: verùm

allowed, of which much was said at the end of Book I. Now proceed to the last mode, C.

This, then, is the sketch and the vestibule through which the entrance to fugues is opened.

Just as the art of painting is not acquired without frequent preliminary drawing, so also for a fuller knowledge of fugues one needs no short amount of practice according to the prescribed pattern—practice which I strongly recommend to you, always seeking out one subject after another.

Joseph. This genus seems poor to me, and to suffer from great scarcity, so that it offers little room for invention.

Aloys. I do not deny that it is more restricted than the mixed genus, nor that it does not offer so wide a field for roaming. But it is most suitable for recognizing the nature of the modes and their

aptiſſimum ad dignoſcendam Modorum naturam, eorumque differentias, & ad Compoſitiones à Capella absque Organo decantandas magnopere neceſſarium : neque tamen inopiæ tantæ eſt, quin varia ſubjectorum genera reperiendi copia ſit. Relicto itaque Bicinii ſtudio privatæ exercitationi tuæ, ad trium partium Fugas procedamus.

EXERCITII V.

LECTIO TERTIA,

De trium partium Fugis.

QUid muneris Tricinio ſit, ſeu interveniente Cantu firmo, ſeu absque eo, ex hucusque dictis jam notum tibi fore ſuppono : magnopere nempe perfectæ Triadis rationem habendam eſſe. Nunc videndum occurrit, quo modo trium partium Fuga inſtituenda, & quid in ea obſervandum ſit. Quod de Fuga duarum partium dictum eſt, hîc quoque ſubſiſtit, uſque dum parti tertiæ intrandi tempus ſit ; id quod ſit abſoluto in utraque partium ſubjecto, vel poſt quandam modulationem, vel extemplò ſinè modulatione, prout partium conſtitutio aut patietur, aut poſtulabit, quod æquo Componentis judicio ponderandum erit. Et ne pars illa fruſtrà, & absque novæ harmoniæ acceſſione intrâſſe videatur, adnitendum eſt, ut introitu ſuo aut Triadem harmonicam efficiat, aut in regulata per Ligaturam diſſonantia (quod elegantius eſt) intrare detur.

Joſeph. Quam utrarum partium quoad intervallum pars tertia ſequi debet?

Aloyſ. Illam plerumque, à qua ſubjectum primò inceptum eſt, ut varietati, cujus permagna ratio hadenda eſt, conſultum ſit. Quod ſi partium conſtitutio feret, ut alio intervallo intrare conſultiùs tibi viſum fuerit, judicium tuum ſequi licebit, ad quod præceptum hoc flectendum erit.

Joſeph. Eſtne eadem clauſularum in Bicinio paulò ante dicta?

Aloyſ. Minimè. Imò nulla clauſula formalis, aut abſoluta (quæ tertiâ majore clauditur) facienda, in qua ſubjecto poteſtatem intrandi non eſſe perſpexeris. Econtra, ſi ſubjectum intrare poſſe cognoveris, non ſolùm in Quinta, Tertia,

differences, and is greatly necessary for compositions sung a cappella without organ. Nor is the poverty so great that there is not a plentiful supply of various kinds of subjects to be repeated. Therefore, leaving the study of the bicinium to your private practice, let us proceed to fugues in three parts.

EXERCISE V.
LESSON THREE,
On Three-Part Fugues.

What the office of the tricinio is—whether with a cantus firmus (fixed melody) interposed or without it—you already know from what has been said: namely, that the perfect triad must be carefully considered. Now we must see how a three-part fugue is to be established, and what must be observed in it. What was said about the two-part fugue applies here as well, until the time comes for the third part to enter. This happens when the subject has been completed in both parts, or after some modulation, or immediately without modulation, as the constitution of the parts either permits or requires—something that must be weighed with the composer's fair judgment. And so that the third part may not seem to have entered in vain and without adding new harmony, one must strive that at its entrance it either forms a harmonic triad, or enters upon a regulated dissonance through a ligature (which is more elegant).

Joseph. Which of the two parts, in terms of interval, should the third part follow?

Aloys. Most often the one in which the subject was first begun, so that variety—which must be greatly regarded—may be served. But if the constitution of the parts suggests that entering at another interval would be more advisable, you may follow your own judgment, to which this rule must bend.

Joseph. Are the cadences the same as those described earlier in the bicinium?

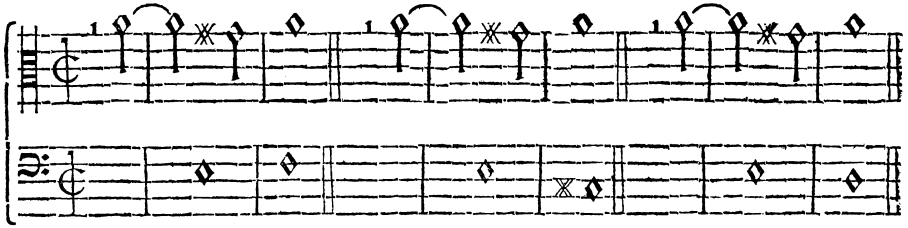
Aloys. By no means. Indeed, no formal or perfect cadence (which ends with a major third) is to be made if you perceive that the subject cannot enter there. On the contrary, if you see that the subject can enter, I assert that a cadence—both perfect and ficta (feigned)—may rightly be made not only on the fifth or third, but also on other intervals

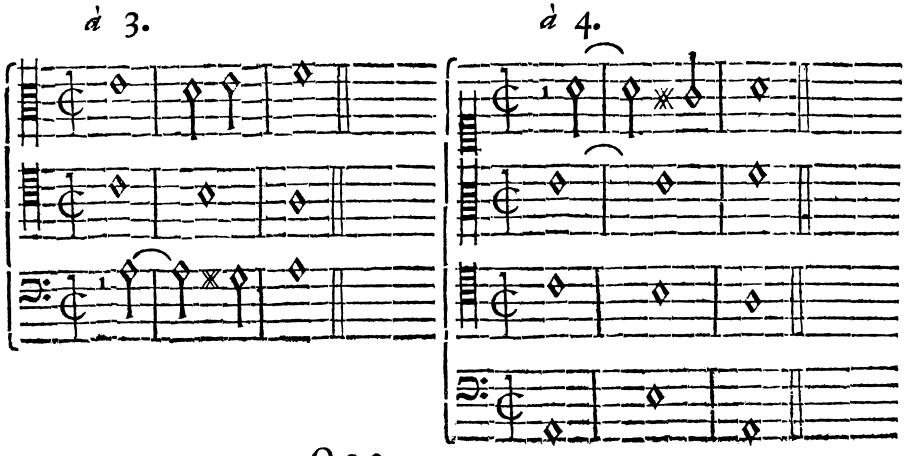
tia, verùm etiam aliis intervallis non nimis averſis ab natura Modi, clauſulam, & abſolutam, & fictam rectè fieri aſſero.

Joſeph. Quid intelligis, ſuaviſſime Magiſter, per clauſulam fictam?

Aloyſ. Clauſulam formalem, ſive abſolutam, Tertiâ majore concludi, inde in Octavam labi, non ignoras. Clauſula verò ficta loco Tertiæ majoris, minore utens, clauſulam formalem auribus expectatam evitando eludit, atque audientis opinionem fallit. Unde ab Italis *Inganno* nuncupata eſt. Videantur exempla.

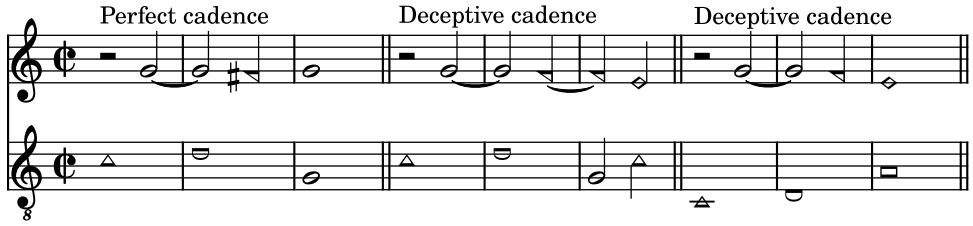
Poteſt etiam Clauſula formalis evitari ſervatâ in parte ſuperiori Tertiâ majore, cùm pars fundamentalis Octavam effugiens aliam aſſumit Conſonantiam. E. G.

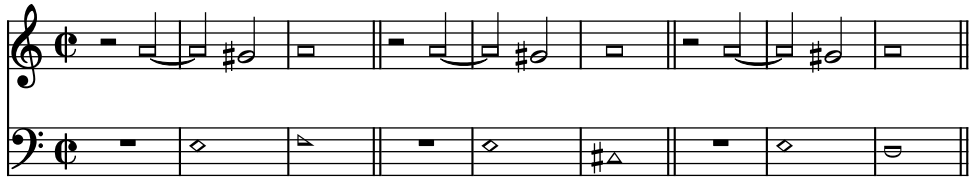


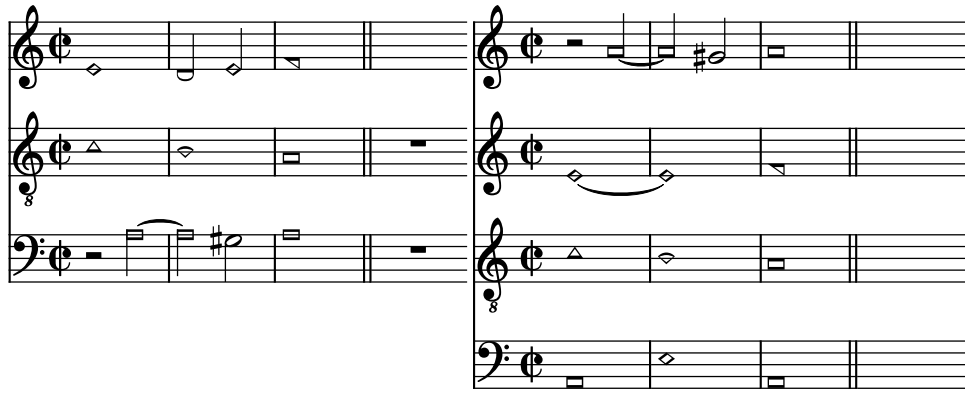
Id quod in plurium partium concentu multò elegantiùs uſuvenit. E. G.

not too foreign to the nature of the mode.

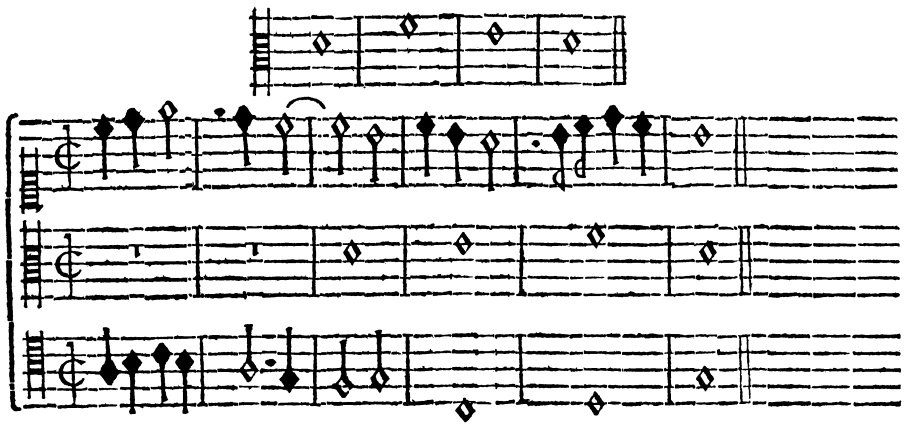
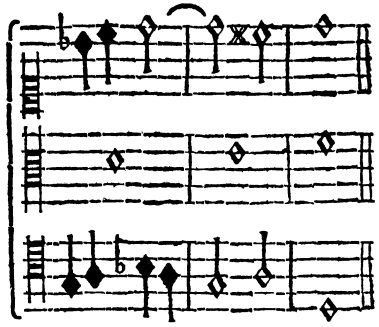
Joseph. What do you mean, most gentle Master, by a clausula ficta (feigned cadence)?

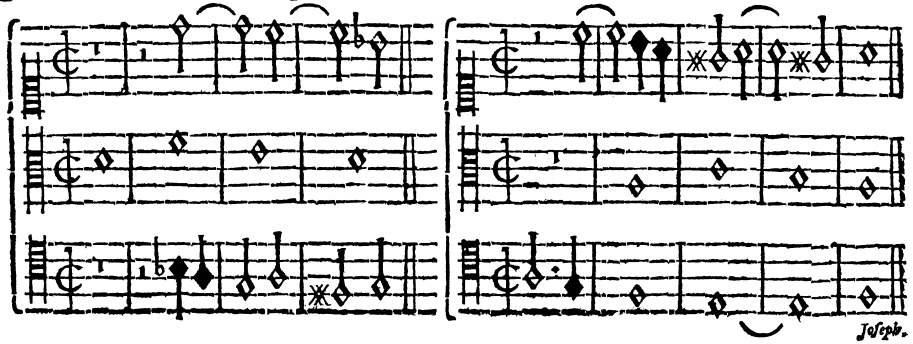
Aloys. You know that the formal, or perfect, cadence ends with a major third and then falls to the octave. A clausula ficta, however, using the minor third in place of the major, avoids the formal cadence expected by the ear and deceives the listener's expectation. Hence it is called inganno (deception) by the Italians. Let the examples be seen.

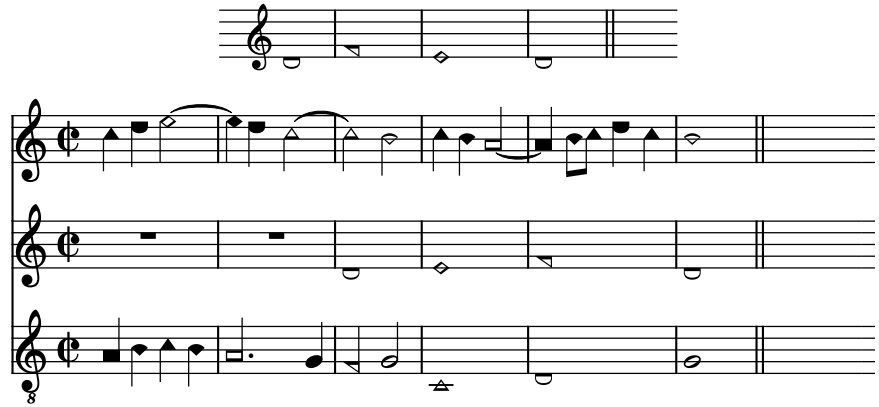
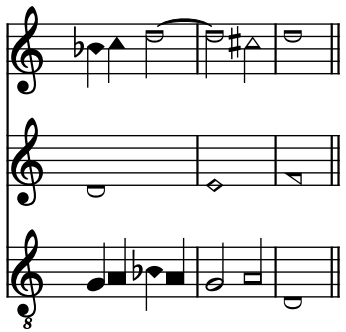


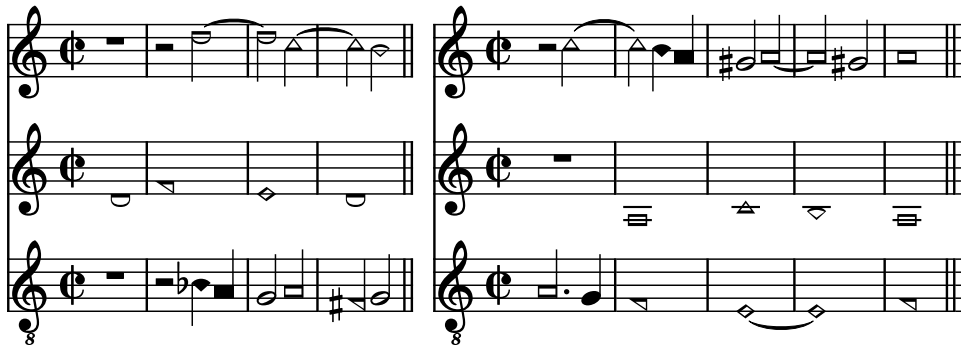
The formal cadence can also be avoided by keeping the major third in the upper part while the bass, avoiding the octave, takes another consonance. For example:

This occurs much more elegantly in compositions of many parts.



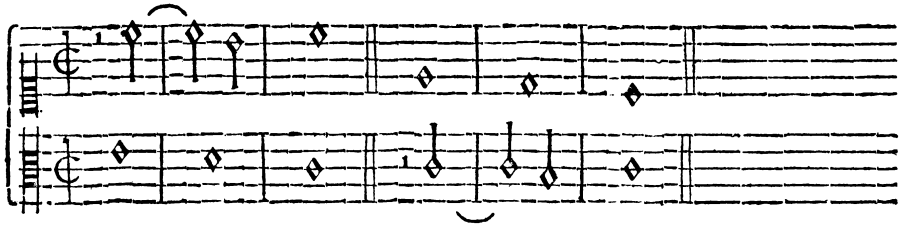
Videamus nunc, quo modo Clauſula formalis ſubſiſtere poſſit beneficio ſubjecti intrantis etiam in intervallo, modo non multùm conſueto. E. G. Subjecti ſequentis uſu:

Hoc exemplo demonſtratur non ſolùm, quo modo Clauſula formalis in prima ſubjecti Nota formari poſſit, verùm etiam, quâ ratione reliquæ partes eò diſponendæ ſint. Videamus nunc clauſulam in ſecunda ſubjecti Nota exſtructam:

Ejusmodi clauſulas formales non ſolùm permitti pronuntio, verùm etiam per eas haud parùm ornamenti Compoſitioni accedere. Hæc de clauſula formali ſubjectis innectenda. Nunc exemplis docebo, quo modo clauſulæ ſubjectis immixtæ, vel Tertiæ majoris adjumento, in parte fundamenti, effugiendæ ſint. Accipiamus ſubjectum primæ Fugæ in Bicinio aſſumptum.

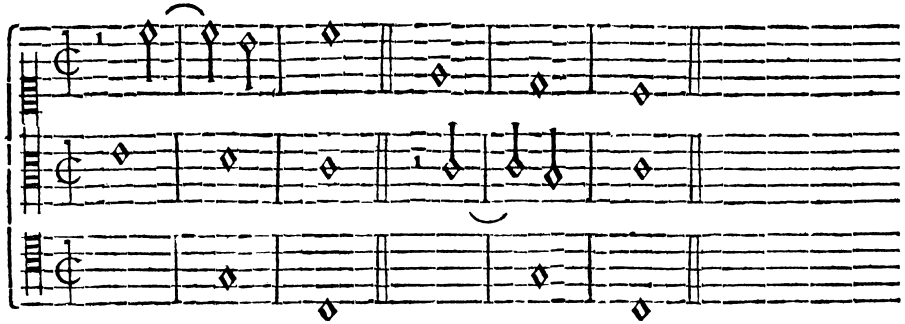


Let us now see how a formal cadence can stand even with the subject entering at an interval not commonly used. For example, using the following subject:

This example shows not only how the formal cadence can be formed on the first note of the subject, but also how the remaining parts must be arranged. Let us now see a cadence constructed on the second note of the subject:

Such formal cadences, I declare, are not only permitted, but add no small ornament to the composition. So much for attaching the formal cadence to subjects. Now by examples I will show how cadences mixed with subjects, or aided by the major third in the bass, may be avoided. Let us take the subject used in the first fugue of the bicinium.



Joſeph. Quare tot clauſulas in Bicinio ſtatuiſti, & hîc in Tricinio reſtringis?

Aloyſ. Clauſulas in Bicinio conceſſas naturâ differentes eſſe ab clauſulis formalibus, haud ignoras: utpote Septimâ, & Sextâ, vel Secundâ, & Tertiâ conſtantes, minusque quietis habentes. Unde hac re accuratiùs inſpectâ, compertum eſt, clauſulas Septimâ, & Sextâ, vel Secundâ, & Tertiâ nixas, eſſe potiùs diſpoſitiones ad clauſulas formales, quàm ut ipſæ tales appellandæ ſint: quia accedente parte tertiâ, cæteris nihil immutatis tales efficiuntur. E. G.

Acceſſione Partis tertiæ.



Joſeph. Obſecro te ſuaviſſime Magiſter, ne excandeſcas ad tot quæſtiunculas tibi forte tædium parituras.

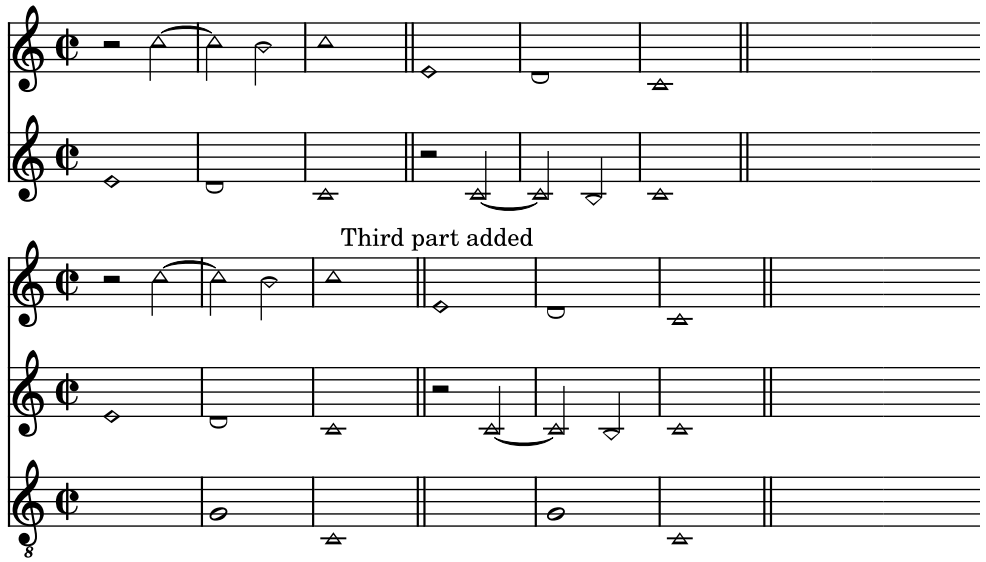
Aloyſ. Audacter expone; in integrum enim hunc offenſionis lapidem amovere cupio, in quem tot Profeſſores tam frequenter impingere comperio.

Joſeph. Quare Clauſula formalis cum ſubjecto intrante valet, & absque eo non admittitur?

Aloyſ. Clauſula formalis, quietis indicium eſt; ideoque non nisi in fine, aut finito quodam articulo, novi cujusdam ſubjecti aſſumendi causâ adhibenda. Subjectum autem in ipſa Clauſula formali intrans, quietis opinionem abſtrahendo, motumque in hoc Compoſitionis genere uſque ad finem deſideratum promovendo, finem nondum adeſſe ſignificat.

Joseph. Why did you establish so many cadences in the bicinium, and here in the tricinio you restrict them?

Aloys. You know that the cadences allowed in the bicinium are by nature different from formal cadences: being formed by the seventh and sixth, or the second and third, they have less repose. Thus, upon closer inspection, it is clear that cadences built on the seventh and sixth, or the second and third, are rather dispositions toward formal cadences than cadences themselves; for when a third part is added, with nothing else changed, they become true cadences. For example:



Joseph. I beg you, most gentle Master, do not grow angry at so many little questions, which may perhaps cause you annoyance.

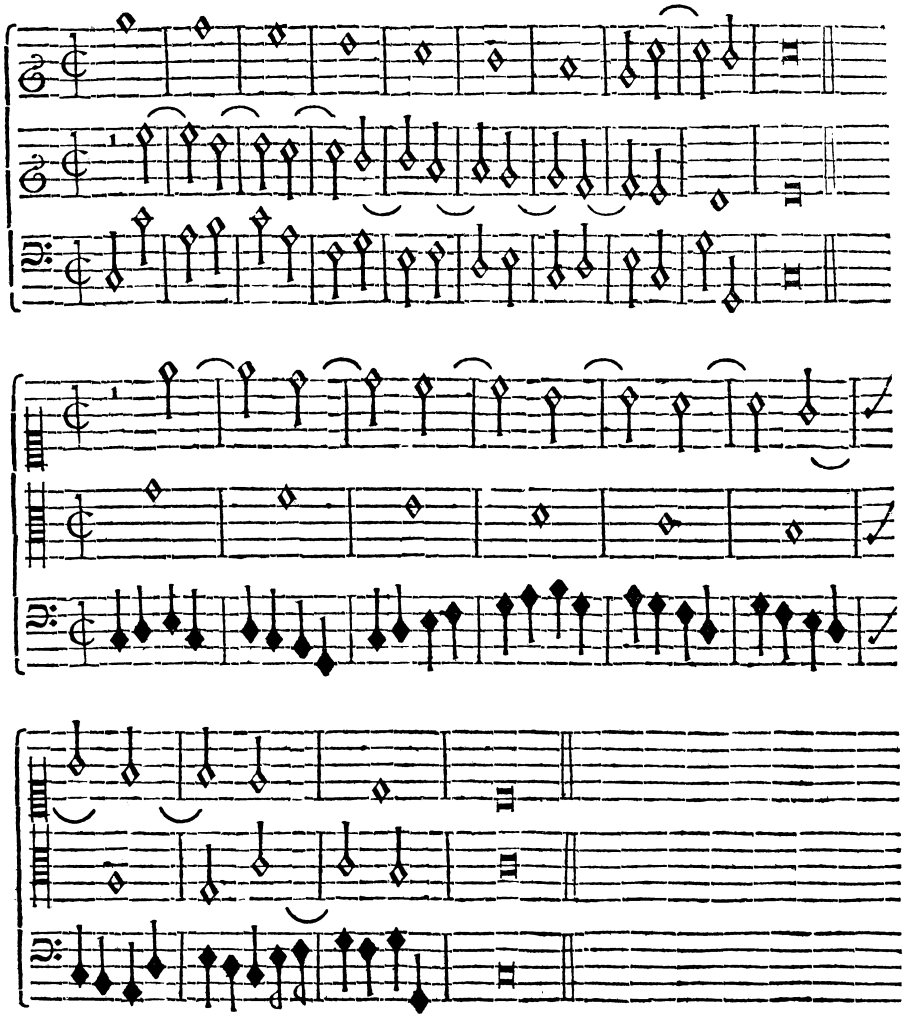
Aloys. Speak boldly; for I wish to remove entirely this stumbling block, into which I find so many teachers frequently fall.

Joseph. Why is it that the formal cadence is valid when the subject enters, but is not allowed without it?

Aloys. The formal cadence is a sign of repose; therefore it should be used only at the end, or at the close of some section, for the purpose of taking up a new subject. But when the subject enters within the formal cadence, removing the impression of repose and carrying forward the motion—so necessary in this kind of composition—until the desired end, it signifies that the end has not yet arrived.

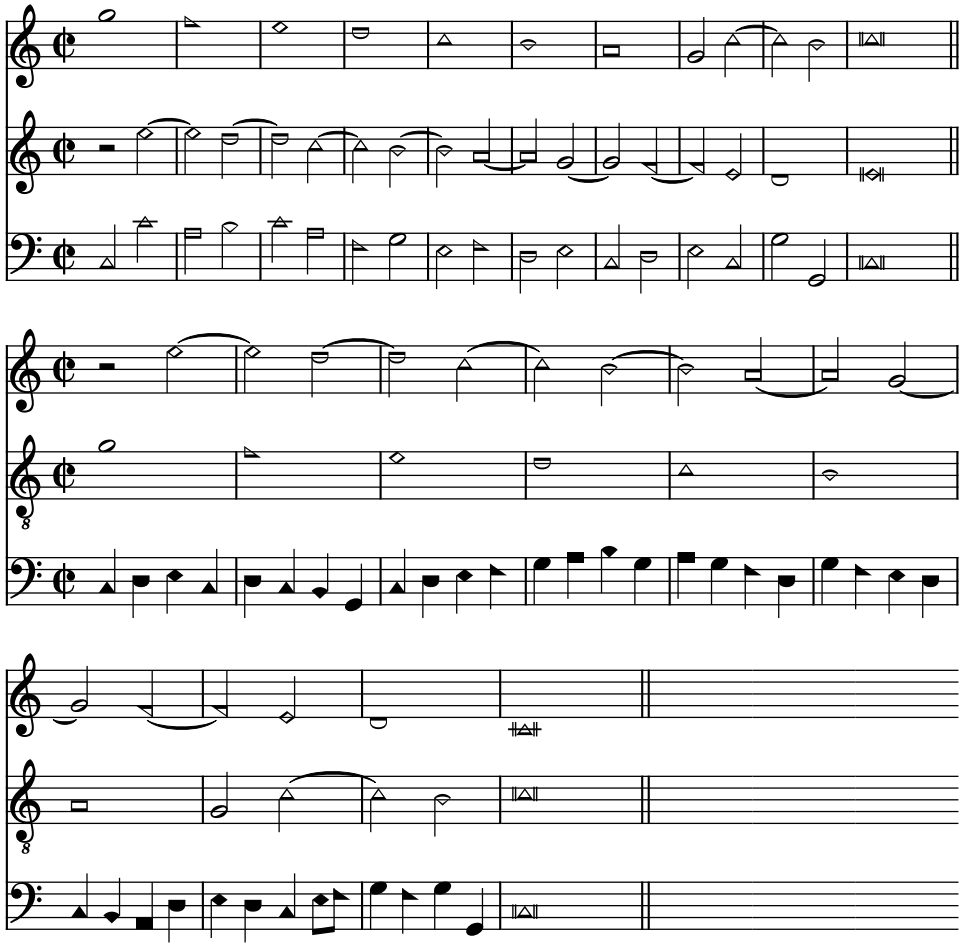
Joſeph. Soluto hoc dubio, perceptisque hucuſque dictis ſatis inſtructum me puto ad Fugas trium partium aggrediendas.

Aloyſ. Fiat. Poſteaquàm antè demonſtravero, qua ratione duabus partibus per Septimam, & Sextam, perque Secundam, & Tertiam deſcendendo progreſſis pars fundamentalis accommodanda ſit; quo haud parva modulationi facilitas accedet.

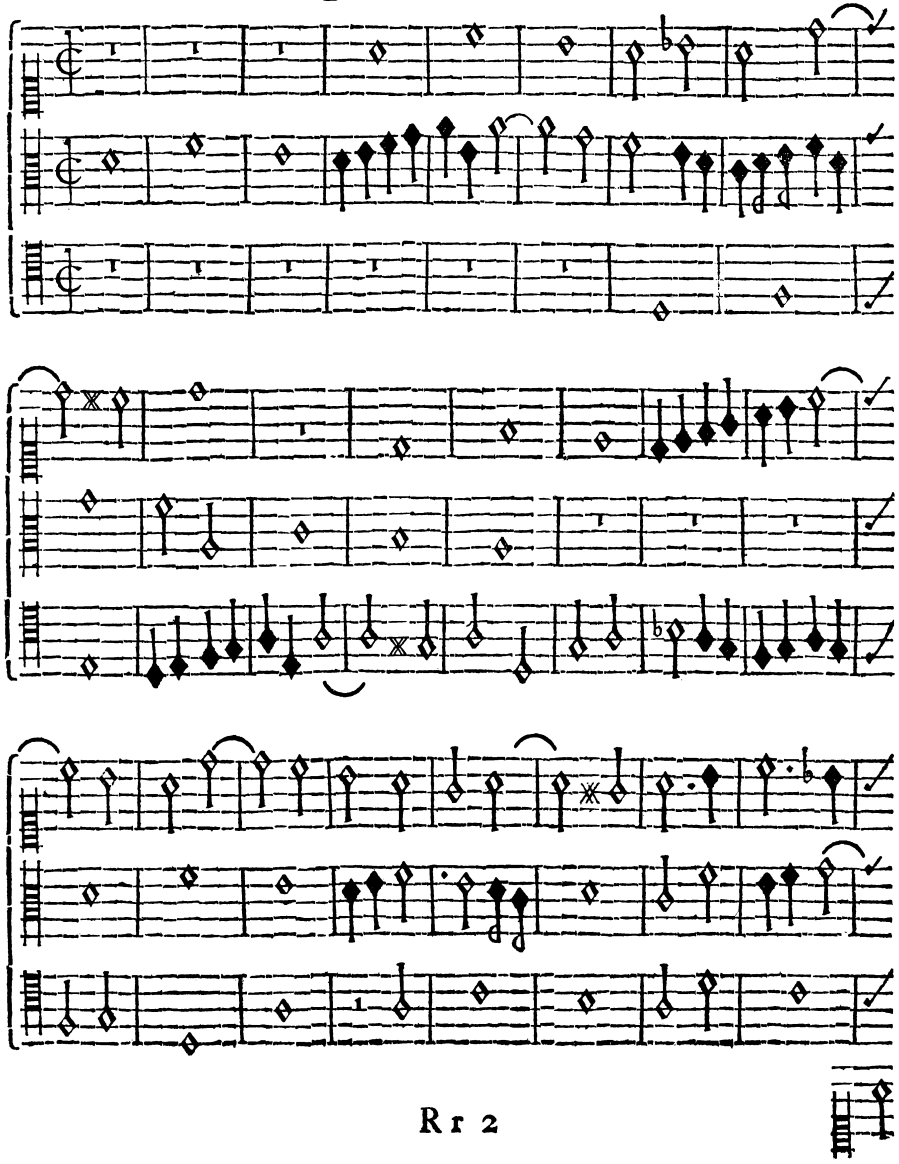


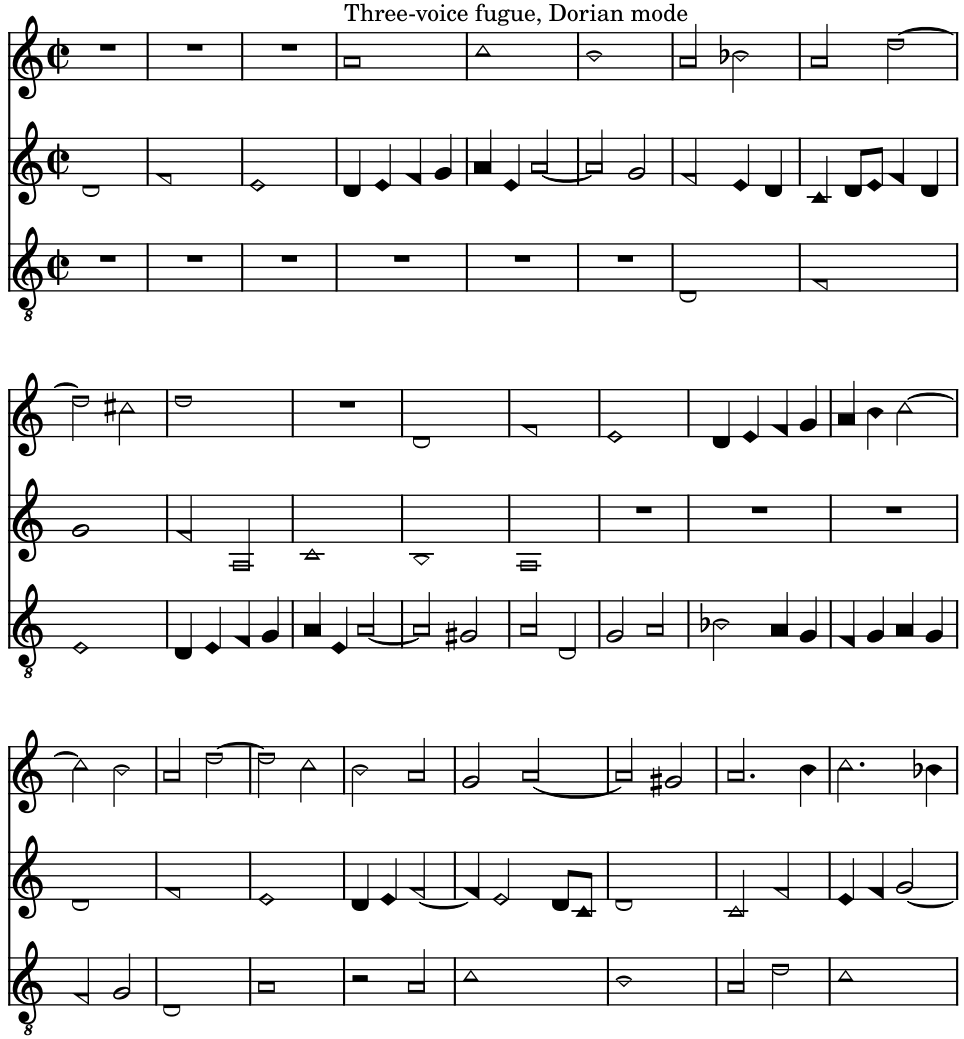
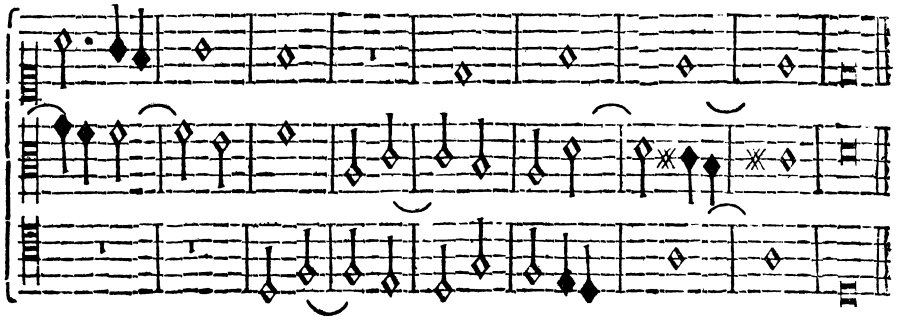
Ex quibus exemplis colligendum eſt, ut Quintæ, & Sextæ conjunctim poſitis locus ſit, inter partes ſuperiores Septimam in Sextam, vel Secundam in Tertiam reſolutam reperiri debe-

Joseph. With this doubt resolved, and with what has been said up to now clearly understood, I think myself equipped to undertake three-part fugues.

Aloys. Let it be so. But first I will show you in what manner, when two parts descend through a Seventh and a Sixth, or through a Second and a Third, the fundamental part must be accommodated—by which no small ease of modulation is gained.

From these examples it must be gathered that, when a Fifth and a Sixth are placed together, there must be found between the upper parts a Seventh resolved into a Sixth, or a Second resolved into a Third.

debere. Id quod etiam de Quarta, & Quinta aggregatim exiſtentibus intelligendum puta. Hujus, ſimiliſque modulationis aliqua parte, vel extra, vel intra ſubjectum te juvare, & ingreſſuro ſubjecto viam ſternere poteris. Pergamus itaque ad Tricinium, Bicinii ſuperioris ſubjecta ſervantes, trium partium texturâ deducenda. Primum exemplum ego ſubjiciam, per cæteros deinde modos tibi imitandum.

Fuga à 3. Modi D.

The same must be understood of a Fourth and Fifth occurring together. By means of this kind of modulation—whether outside the subject or within it—you will be able to assist yourself and prepare the way for the subject that is about to enter. Let us therefore proceed to the tricinio, preserving the subjects of the preceding bicinium, and drawing out the texture for three parts. I will supply the first example, to be imitated by you in the remaining modes.

Three-voice fugue, Dorian mode



Principium hujus Fugæ nihil diverſi continet ab illa in Bicinio poſitâ, uſque ad tertiæ partis ingreſſum, ubi durante ſubjecto, ſuperiorum partium congruâ canendi modulatione Contrapunctum interim inſtructum eſt. Quo facto Altus ſubjectum repetit, alio, quàm incepit intervallo, modulante interim parte fundamentali. Sub id tempus Cantus adjumento pauſæ quieſcit, atque ad novum ingreſſum ſe præparans, diverſo, quàm in principio, intervallo intrat, idque in diſſonantia valida ad ſubjecti præſentiam indicandam: clauſulaque, ſextæ minoris ope, devitatur.

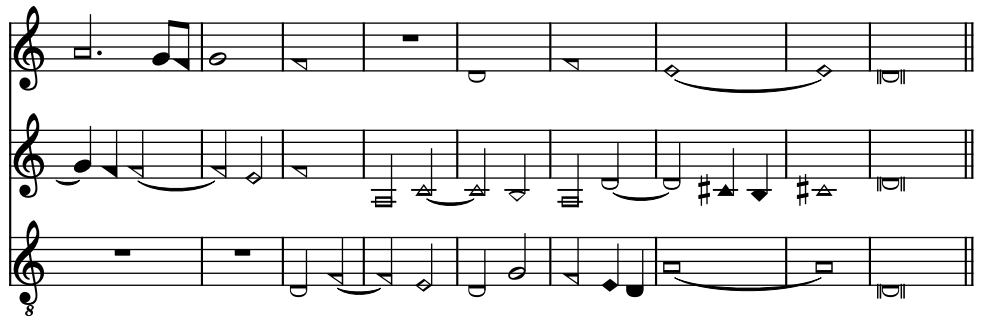
Obſerva prætereà rationem, ob quam ibi ſubjecto finito, poſitâ trium tactuum pausâ protinùs Alto quies data ſit: ideò nempe primò, quia parti ſuperiori cum parte inferiore tam cominùs exiſtere contingit, ut in medio locus opportunus Alto non relinquatur. Secundò, quia Alto cum ſubjecto proximè intrandum eſt. Neque incuriâ præteriri vellem hunc Alti introitum in Quinta, & Sexta: maximam enim addit vim Compoſitioni, ejusmodi intrandi modus. Animadverte porrò mox ſequentem Tenoris introitum, modulationemque ſuper exſtructam.

Joſeph. Haud videtur introitus iſte Tenoris ad præſcriptum modum eſſe factus; nec Triade harmonicâ, neque diſſonantiâ ſtipatus, utpote Sextâ duntaxat, & Octavâ conſtans.

Aloyſ. Summopere delector mente tuâ ad omnia circumſpectâ. Verùm ſcias oportet, modum illum intrandi, de conſilio potiùs, quàm de præcepto eſſe datum. Deinde, præterquàm quòd Nota ſubjecti prima parvi ſit momenti, utpote minima, Sexta ſuperimpoſita per ſe fortis, auditúque facilis ingreſſum ſubjecti haud obſcurè prodit, atque defectum Harmoniæ ſupplet. Eſtne tibi ſatisfactum?

Joſeph. Omnino.

Aloyſ.



The beginning of this fugue contains nothing different from that set in the bicinium, up to the entrance of the third part. There, while the subject continues, a suitable melodic modulation in the upper parts provides the counterpoint. When this is done, the Alto repeats the subject at an interval different from the one in which it began, while the fundamental part modulates beneath. At that moment the Cantus, aided by a rest, falls silent, and preparing for a new entrance, enters at an interval different from the first—and that in a strong dissonance, to mark the presence of the subject. The cadence is avoided by means of the minor sixth.

Observe also the reason why, when the subject ends there, a rest of three beats is immediately given to the Alto: first, because the upper part lies so close to the lower that no suitable place in the middle remains for the Alto; second, because the Alto must soon enter with the subject. Nor would I wish you to overlook this entrance of the Alto on the Fifth and Sixth: such a manner of entering adds great force to the composition. Notice also the Tenor's entrance that follows, and the modulation built above it.

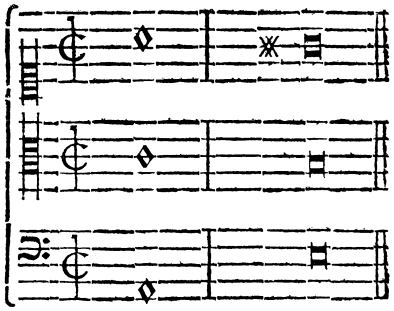
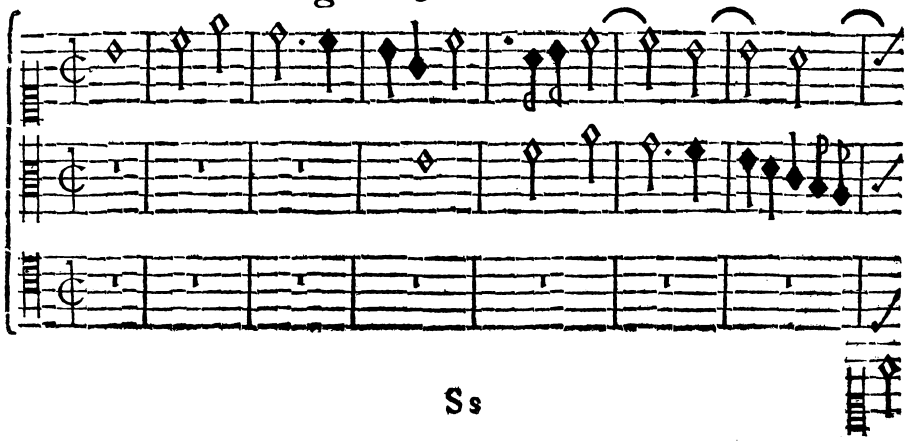
Joseph. That entrance of the Tenor does not seem to have been made according to the prescribed manner; it is supported neither by a harmonic triad nor by a dissonance, consisting only of a Sixth and an Octave.

Aloys. I am greatly delighted by your careful attention to everything. But you must know that this manner of entering is given as a matter of advice, not of rule. Moreover, since the first note of the subject is of little weight—being a half note—the Sixth placed above it is strong in itself and easily heard, clearly revealing the entrance of the subject and supplying what harmony is lacking. Are you satisfied?

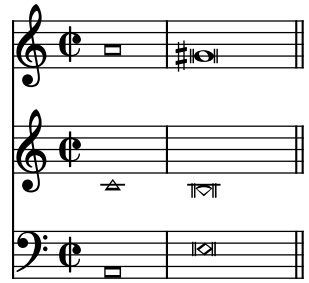
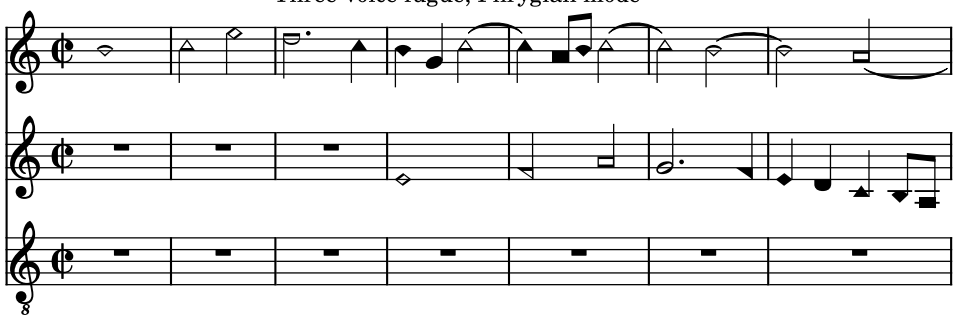
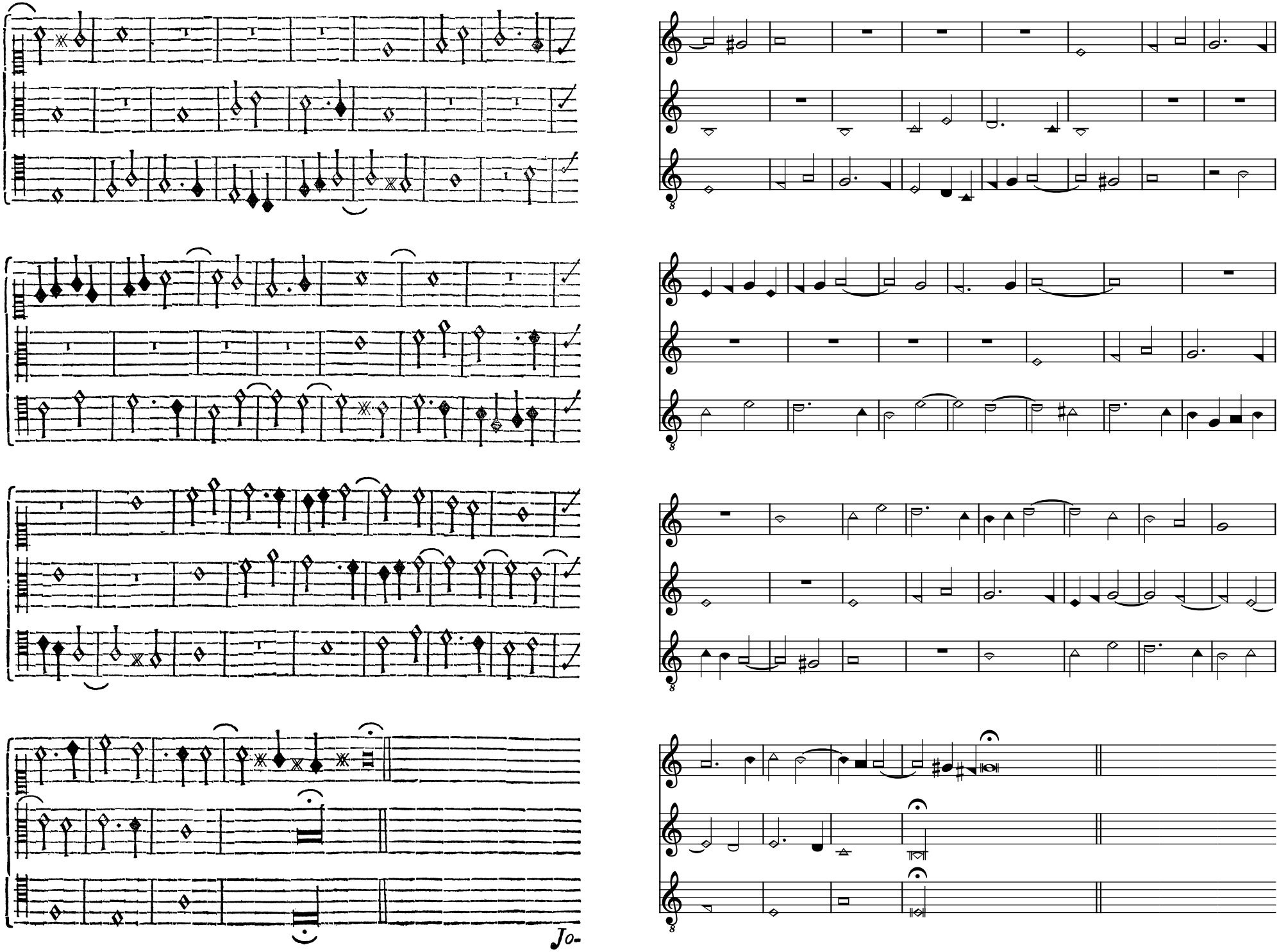
Joseph. Entirely.

Aloyſ. Reliquum igitur examinaturi progrediamur. Conſidera deinde modulationem ſupra ſubjectum exſtructam, quâ in ſuperioribus partibus continuatâ, clauſulam vides in F. factam, eluſam tamen per ſubjectum in Tenore poſt pauſas intrans. Deinde figuris Notarum nonnihil immutatis, ut ſingulis in tactibus cuilibet parti cum ſubjecto intrandi poteſtas ſit; tandem Clauſulâ finali Fugæ Finis imponitur.

Joſeph. Ex dictis, factisque colligo, nunquam pauſam ponendam eſſe, nisì immediatè ſubjectum ſequatur.

Aloyſ. Ità ſanè. Aut ſubjectum vetus, aut certè novum, idque cæteris partibus deinceps repetendum, poſt pauſam induci neceſſe eſt: nisì objurgationem in te experiri velis Evangelicam; Math. 22. cap. *Amice, quomodo huc intrâſti, non habens veſtem nuptialem.* Jam modum E. ad manus accipe, Fugamque trium partium eo, vel alio haud multùm diſſimili modo conficere tenta: neque hæſitans, quo intervallo clauſulas exſtruas, dummodo in iis ſubjecto eleganter intrandi facultas detur. Juvabit autem antea clauſulam finalem hujus Modi monſtrare, quæ diverſa ab aliis Modis eſt.

Fuga à 3. Modi E.

Aloys. Let us then proceed to examine the rest. Consider next the modulation built above the subject, which, continued in the upper parts, forms a cadence in F—yet this is eluded by the subject entering in the Tenor after the rests. Then, with the note-values somewhat altered so that in each measure every part may have the opportunity to enter with the subject, the fugue is finally brought to its end with the final cadence.

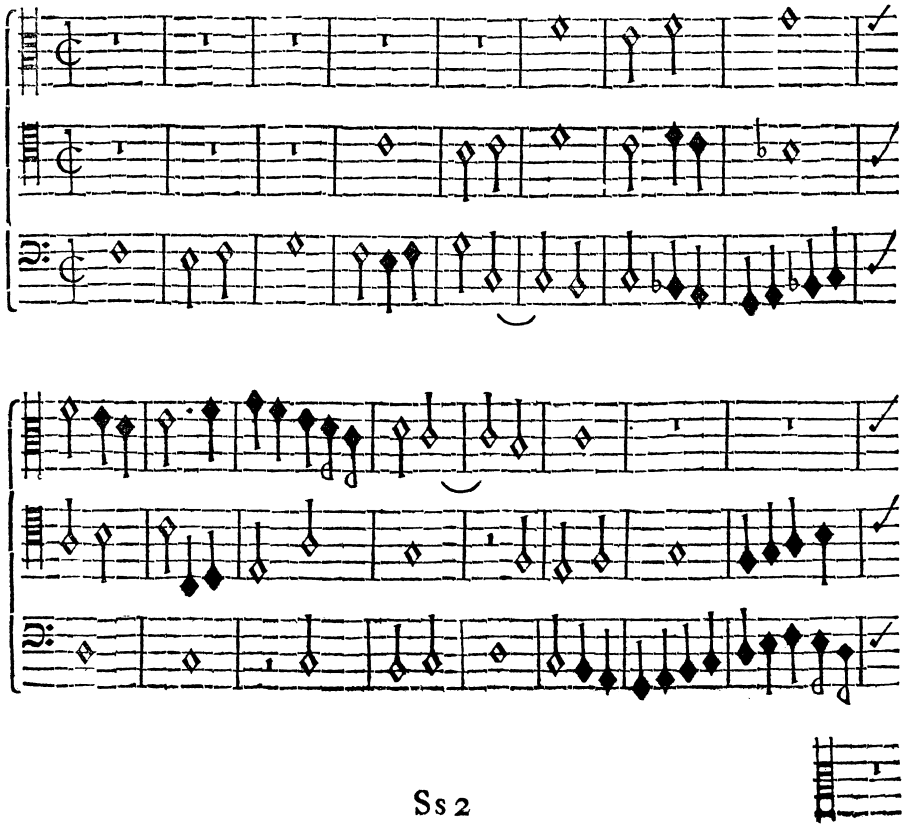
Joseph. From what has been said and done, I gather that a rest must never be placed unless the subject immediately follows it.

Aloys. Exactly. Either the old subject, or at least a new one—one that the other parts must then repeat—must be introduced after a rest, unless you wish to incur the Evangelical rebuke: "Friend, how did you come in here without a wedding garment?" (Matthew 22). Now take Mode E in hand, and attempt to compose a three-part fugue in it, or in another mode not much different. Do not hesitate over what interval to build your cadences, provided that in them the subject may enter gracefully. It will help you first to show the final cadence of this mode, which differs from that of the others.

Three-voice fugue, Phrygian mode

Jo-

Joſeph. Haud absque timore, Venerande Magiſter, coràm te cum hac Fuga compareo. Vereor etenim, ne clauſulam ſæpiùs, quàm par eſt, in A. factam animadvertas.

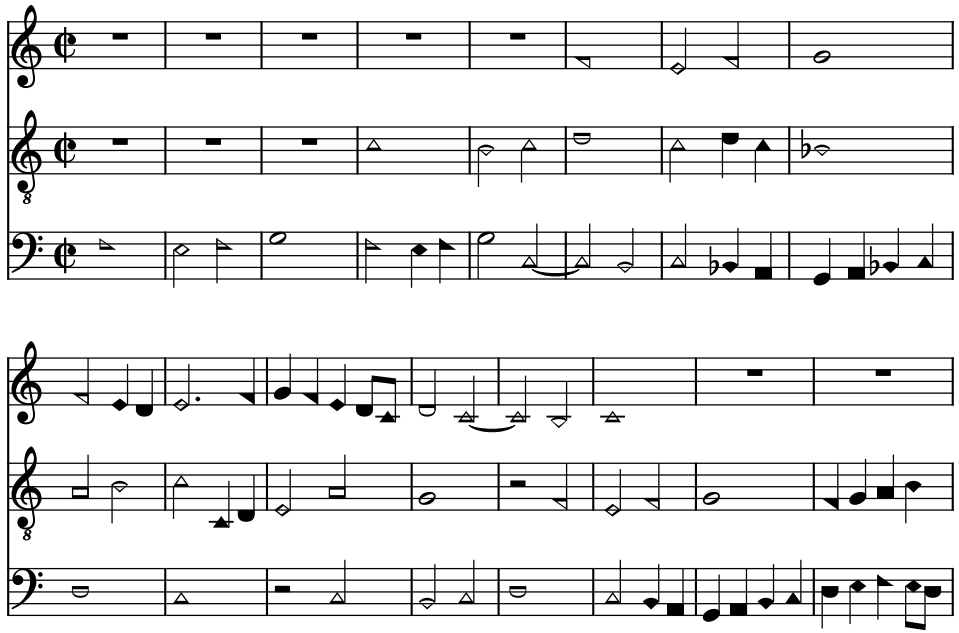
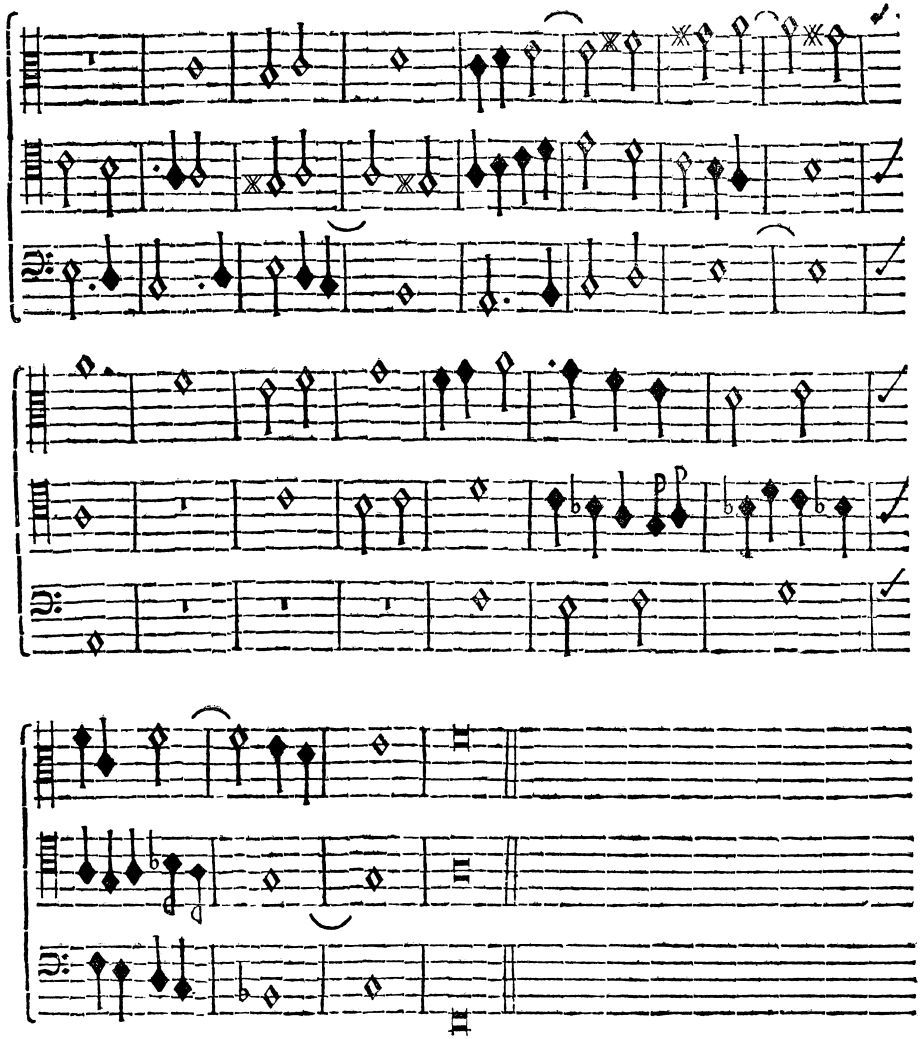
Aloyſ. Mitte timorem Joſephe. Longè ab reprehenſione miror, quòd in hac re Tyroni, & ſubjectum tam ſcitè intexere, & modulationem, Modo tam propriam reperire tibi contigerit. Ad clauſulam enim ſæpiùs in A. repetitam ipſa ſubjecti qualitas te induxit: quæ non obeſt varietati, utpote alio ſemper, atque alio modo per Notas comitantes variata. Et ſi quid lepôri in modulis Muſicis tantopere deſiderato deeſſe videtur, Generis Diatonici reſtrictioni, maximè autem hujus Modi egeſtati tribuendum eſt. Laudo prætereà concinnum partium Cantum, quarum neutra impedita ab altera liberrimè vagatur. Perge nunc proximè ad modum F. & inde ad reliquos.

Fuga à 3. Modi F.

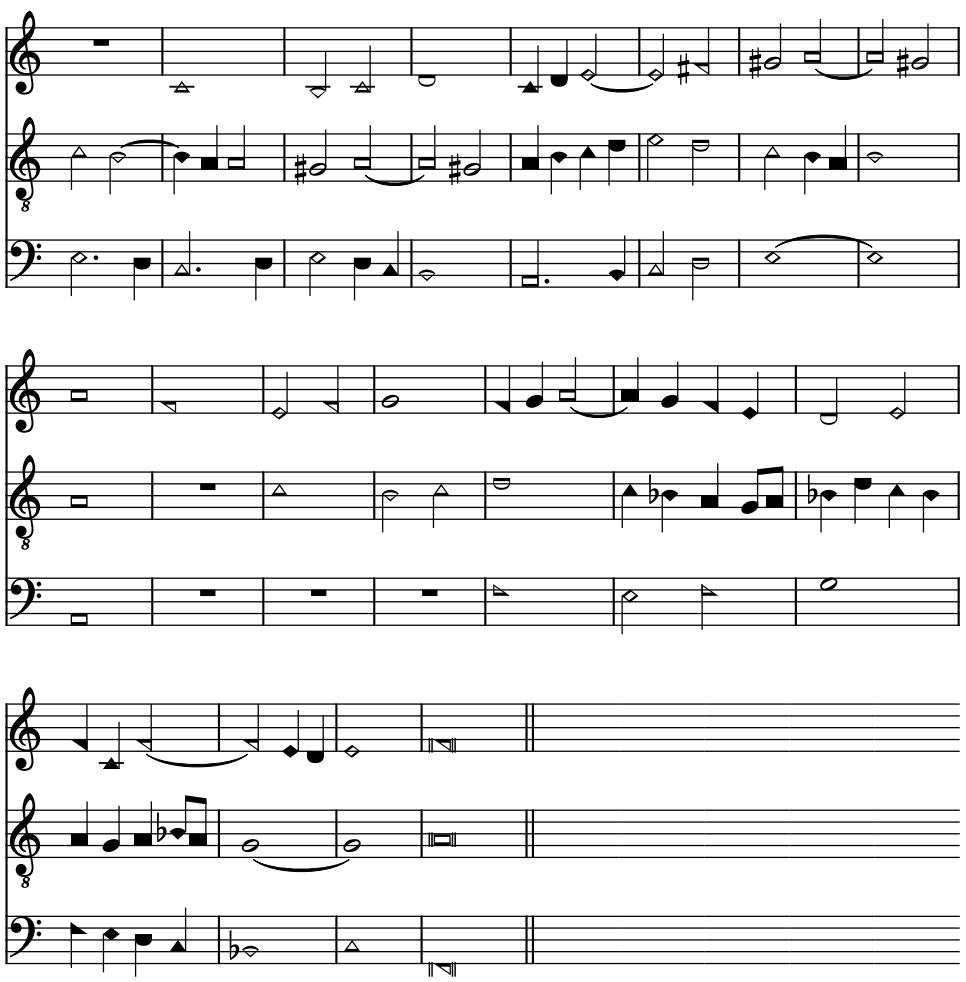


Joseph. Not without fear, venerable Master, do I present this fugue before you. I fear you may notice that I have made the cadence in A more often than is fitting.

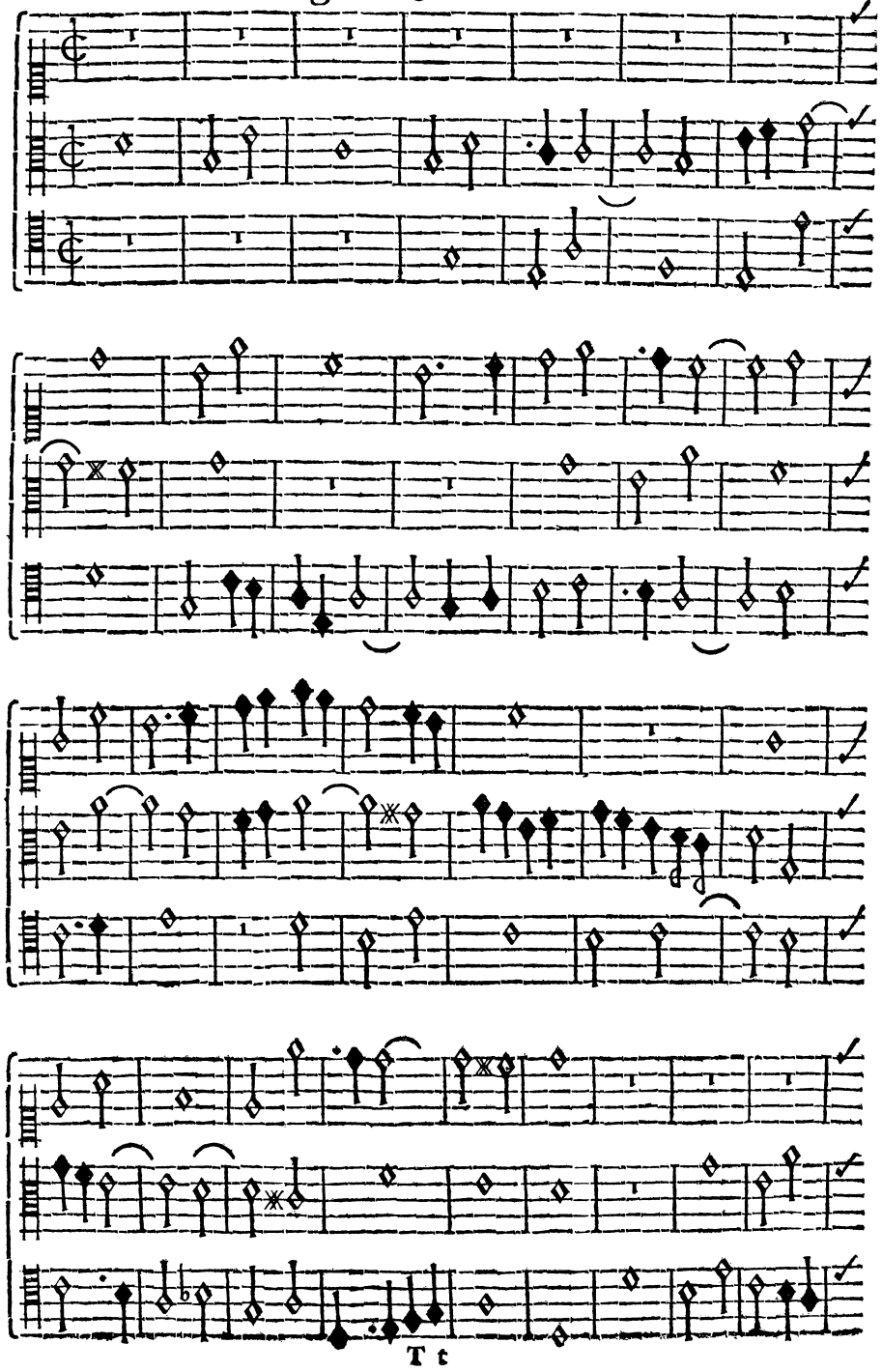
Aloys. Put aside your fear, Joseph. Far from reproaching you, I marvel that, as a beginner, you have managed both to weave the subject so skillfully and to find modulation so proper to the mode. As for the cadence repeated so often in A, the nature of the subject itself led you to it; nor does it harm variety, since it is always varied in one way or another by the accompanying notes. And if anything seems lacking in the charm so greatly desired in musical passages, it must be attributed to the restrictions of the diatonic genus, and especially to the poverty of this mode. I also praise the graceful movement of the parts, none of which is hindered by another, but all wander freely. Now proceed to Mode F, and then to the rest.

Three voice fugue, Lydian mode

Joſeph. Vix non mihi blandior, Fugam hanc non admodùm malè mihi ſucceſſiſſe, nisì in modulationem illam animadverteris immixtam ſubjecto, ubi ſecundâ vice Altus intrat.

Aloyſ. E contrariò valde vafrè, ingenioſeque modulationem illam inſtitutam eſſe ajo. Etenim cùm, & ob Generis Diatonici, & Modi rigorem omnia tritâ viâ incedere debeant, paululum peregrini, non nisì modulationis præſidio, ſubjecto immiſceri poſſe, perſpicuum eſt. Admiror prætereà ſubjectorum ingreſſum ubique opportuno tempore adhibitum, concomitantemque ad Contrapuncti regulas Concordantiam. Perge ad Modum G.

Fu-

Joseph. I almost flatter myself that this fugue has not turned out too badly, unless you notice that modulation mixed with the subject where the Alto enters the second time.

Aloys. On the contrary, I say that modulation was devised very cleverly and ingeniously. For since, owing both to the strictness of the diatonic genus and of the mode, everything must proceed along well-trodden paths, it is clear that anything slightly foreign can be introduced only with the help of modulation. I also admire the timely entrances of the subject everywhere, and the consonances accompanying it according to the rules of counterpoint. Proceed to Mode G.

Fuga à 3. Modi G.

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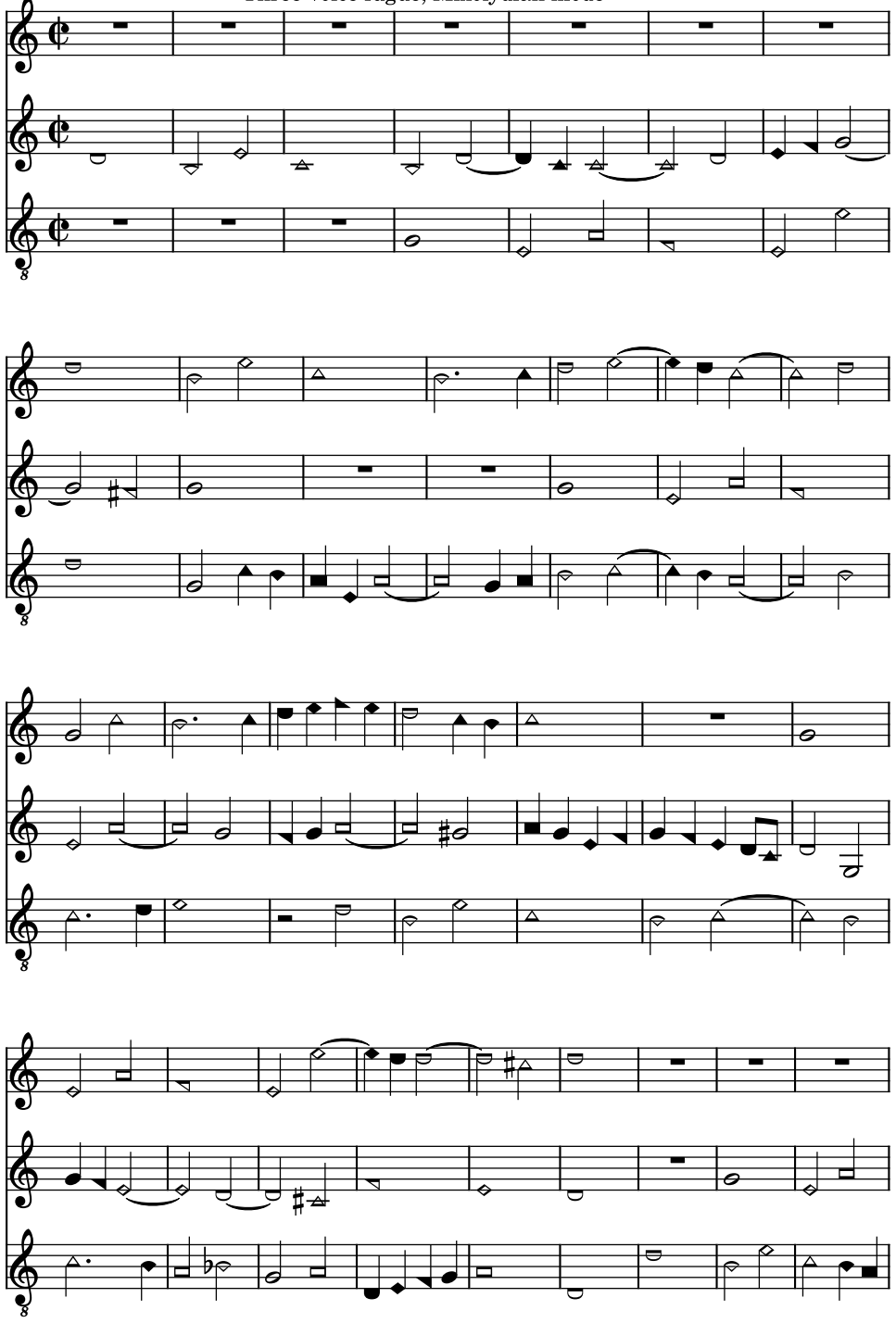
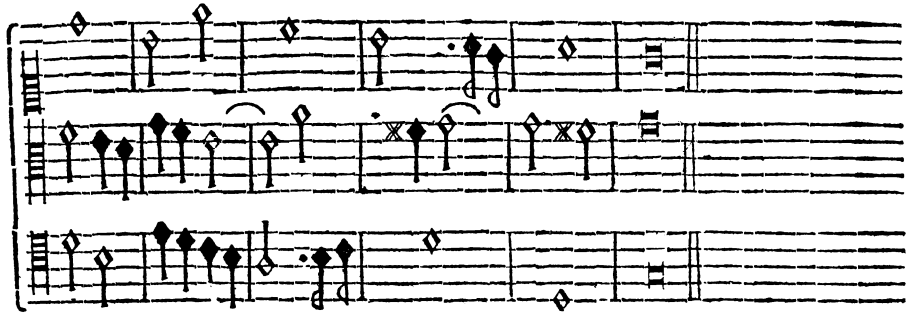
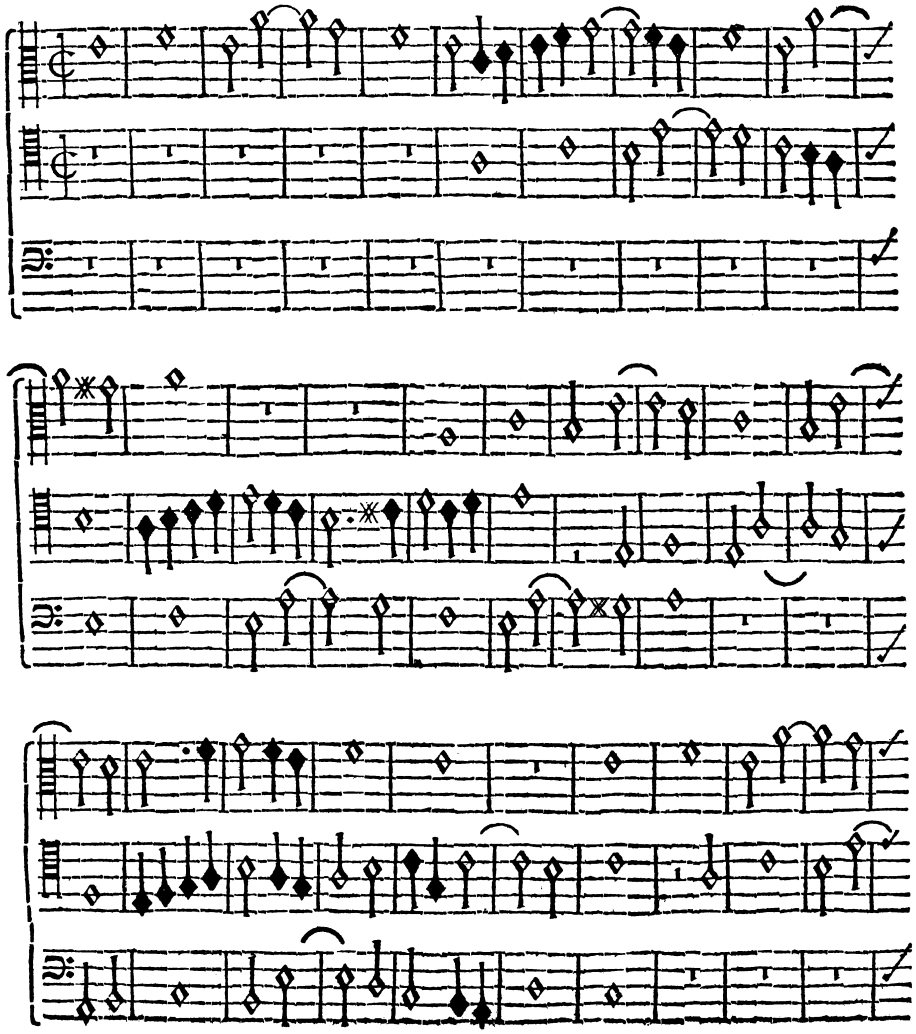
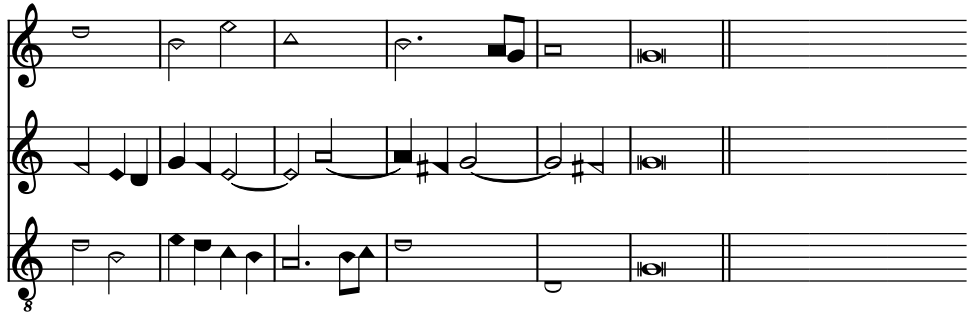
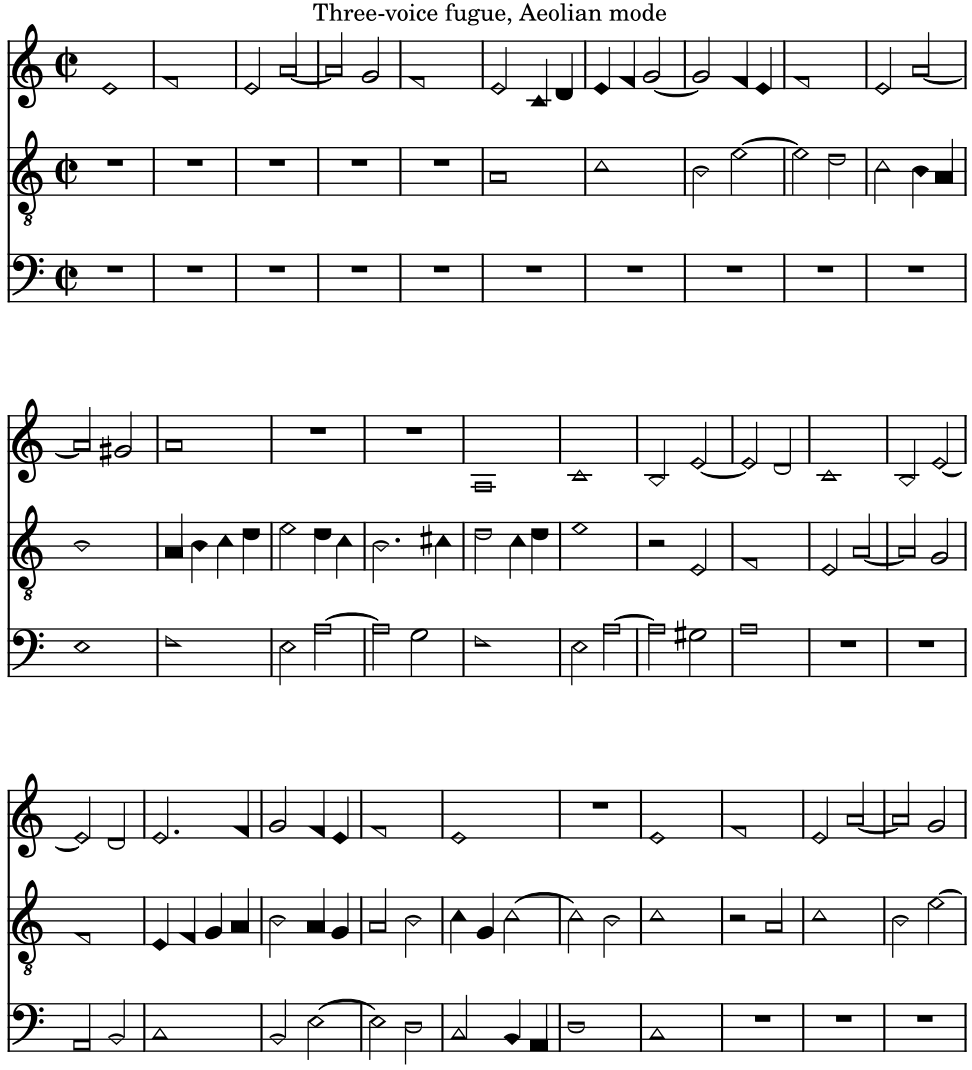
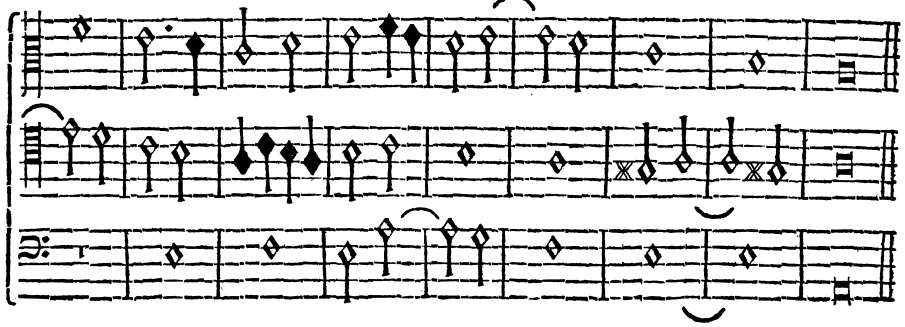
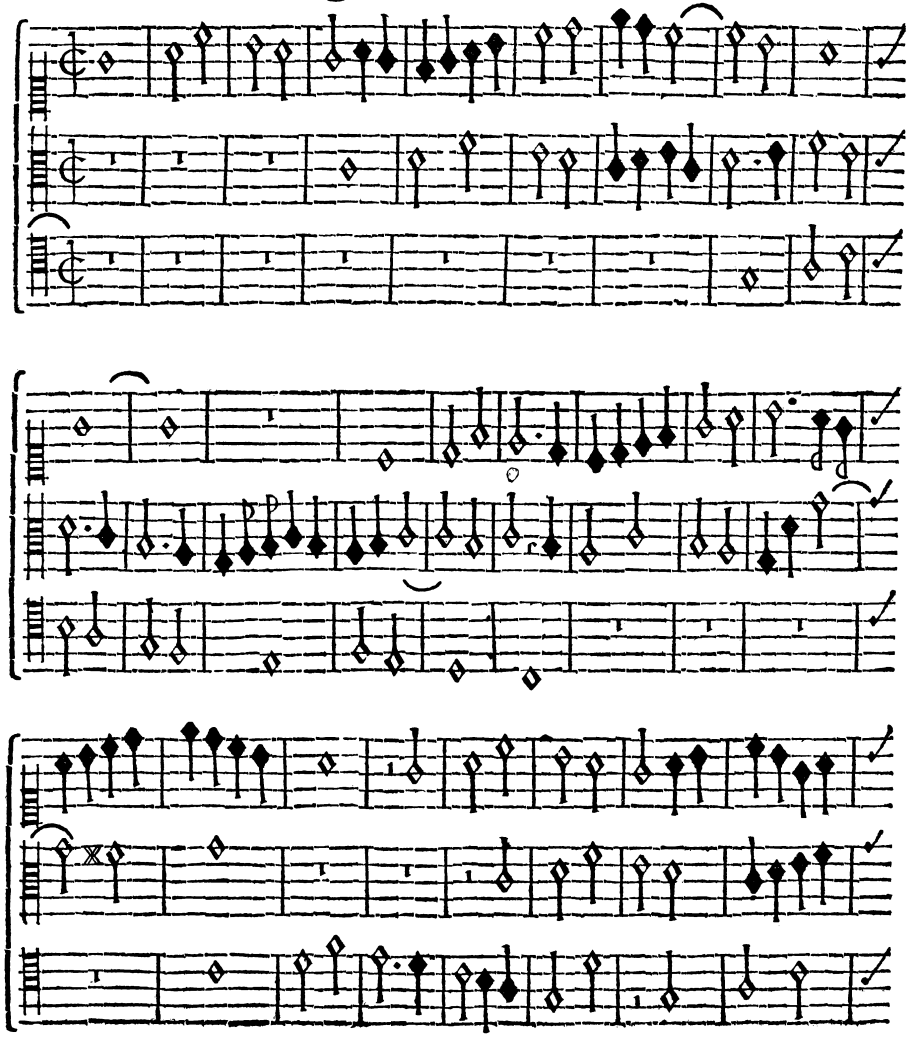
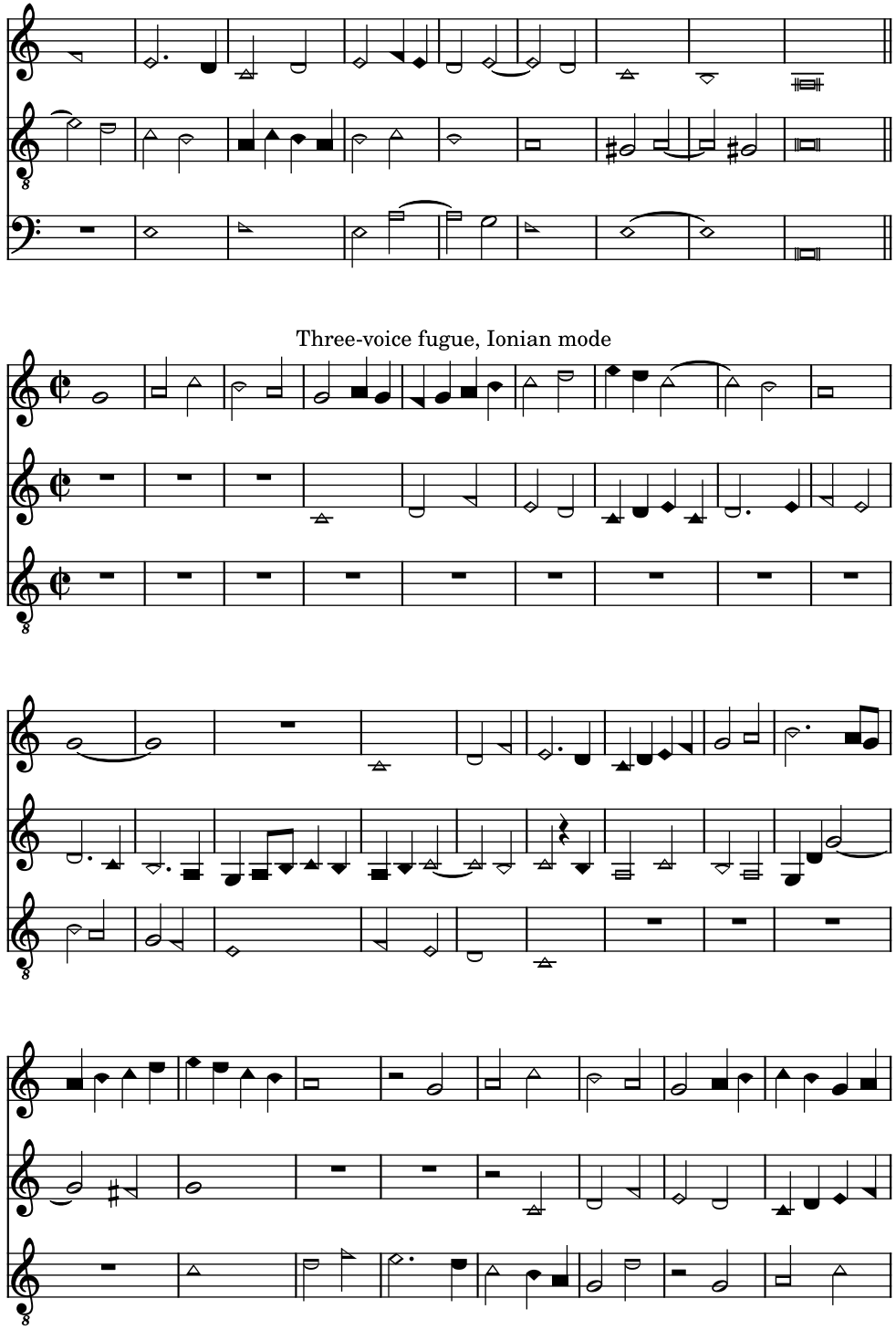
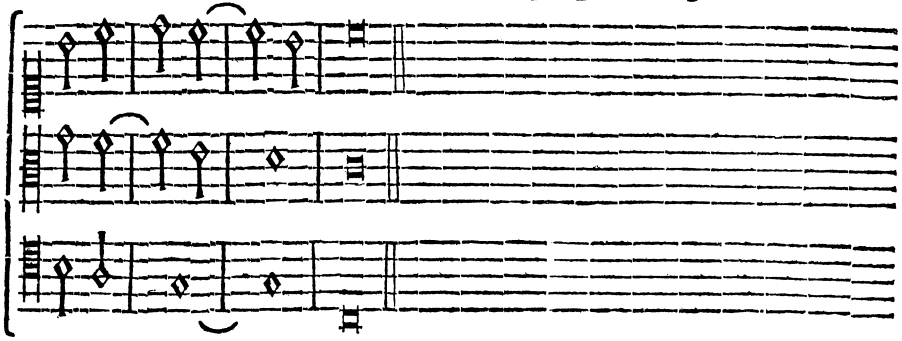
Three-voice fugue, Mixolydian mode

Fuga à 3. Modi A.

Three-voice fugue, Aeolian mode

Fuga à 3. Modi C.



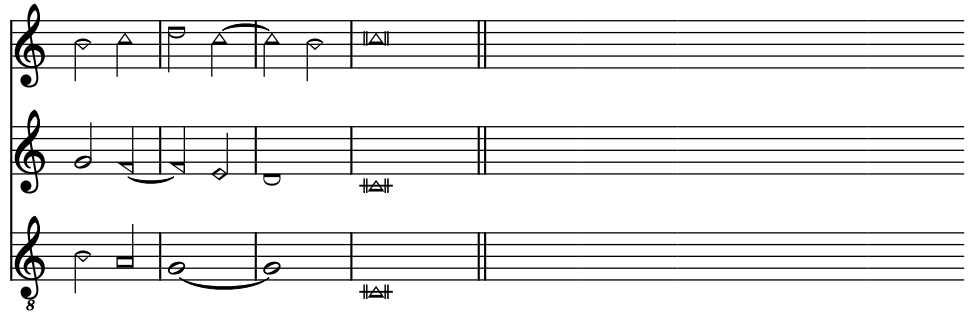
Three-voice fugue, Ionian mode

Aloyſ. Incredibiliter lætor, hac brevi ſimpliciſque Fugas componendi methodo te non modo leviter tinctum eſſe, verùm etiam longiùs jam proceſſiſſe ; id quod partes pulchro canendi modo incedentes, debitoque tempore quieſcentes, & opportunè intrantes haud obſcurè demonſtrant. Ut adeò in hoc Compoſitionis genere nihil tibi deeſſe videatur, quàm ut uſu per aliquod tempus continuato facilitatem adipiſcaris. Da prætereà operam, ut crebris ligaturis jam uni, modò alteri parti inſertis compoſitionem diſtinguas : vix enim credibile eſt, quantum condimenti per eas modulis accedat, quô fit, ut quælibet fermè pars, motum diſparem nacta, facili ratione perceptibilis auditui reddatur : id quod non ſolùm in hoc, verùm in omni Compoſitionis genere tibi dictum volo. Cæterùm, cùm in Trino perfectio exiſtat, te etiam, atque etiam exhortor, ut huic ſtudio privatim ampliùs vacare pergas, experturus inde, quantùm emolumenti ad plurium partium Compoſitionem, quantaque facilitas enaſcatur. Nunc ad Quatricinium.

EXERCITII V.

LECTIO QUARTA,

De Fugis quatuor partium.

QUid muneris ſit quartæ partis, quemque locum in ſupplementum Harmoniæ occupare debeat, ſuprà per totius Exercitii Tertii curſum abunde explanatum, ſatisque tibi notum eſſe confido. Supereſt itaque dicendum, quam ex tribus partibus in Fugis quarta pars intrando ſequi debeat. Tametſi hoc prudenti Compoſitoris judicio relinquendum eſſe videatur, cujus



Aloys. I am incredibly delighted that by this brief and simple method of composing fugues you have not only been lightly touched by the art, but have already advanced much further; as is clearly shown by the parts moving in a beautiful manner of singing, resting at the proper time, and entering at the right moment. Thus it seems that nothing is lacking to you in this kind of composition except the ease that comes from continued practice over some time. Give attention also to distinguishing the composition by inserting frequent ligaturae (ligatures), now in one part, now in another; for it is scarcely believable how much seasoning they add to the musical passages, so that each part, having a different motion, becomes easily perceptible to the ear — something I want you to remember not only here, but in every kind of composition.

Furthermore, since perfection exists in the number three, I urge you again and again to devote yourself privately to this study, and you will discover how much benefit and how much ease arises for the composition of more parts. Now to the Quatricinium (four-part writing).

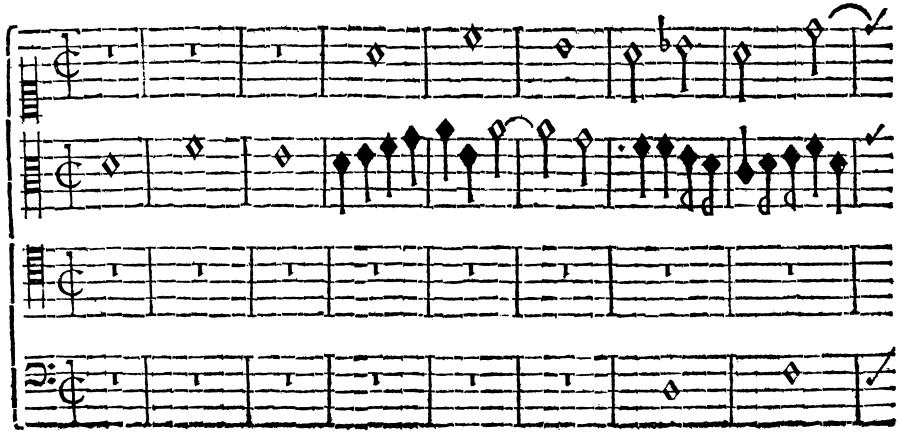
EXERCISE V.
LESSON FOUR,
On Four-Part Fugues.

What the office of the fourth part is, and what place it must occupy in supplementing the harmony, has been abundantly explained throughout the course of Exercise III, and I trust is well known to you. It remains to say which of the three parts the fourth part ought to follow when entering in a fugue. Although this seems to be left to the prudent judgment of the composer — whose task it is to determine how more

cujus eſt indagare, quo modo concentui, aut plus Harmoniæ, aut gratiæ aut varietatis acceſſurum ſit; plerumque tamen uſu receptum eſt, ut Tenor Cantum, Baſſus Altum intrando ſequatur. Cæterùm creſcente partium copiâ, haud exigua tibi cura ſit, ne partibus nimìs infarcitis, & in anguſtum contractis ab una alteri liberè vagandi facultas admittatur. Quemadmodum enim per confertam turbam tranſeunti à dextra, ſiniſtraque, & à fronte impingere contingit; ideoque tranſitus aut omninò non, aut difficulter ſuccedit: ità pari modo fit in Compoſitione, quando pars parti obſtans liberum incedendi campum non habet, bellèque concinendi facultas tollitur. Quapropter ſummopere adnitendum eſt, ut hunc ſcopulum evadas.

Joſeph. Quid verò remedii, quando præter opinionem tale quid contingeret?

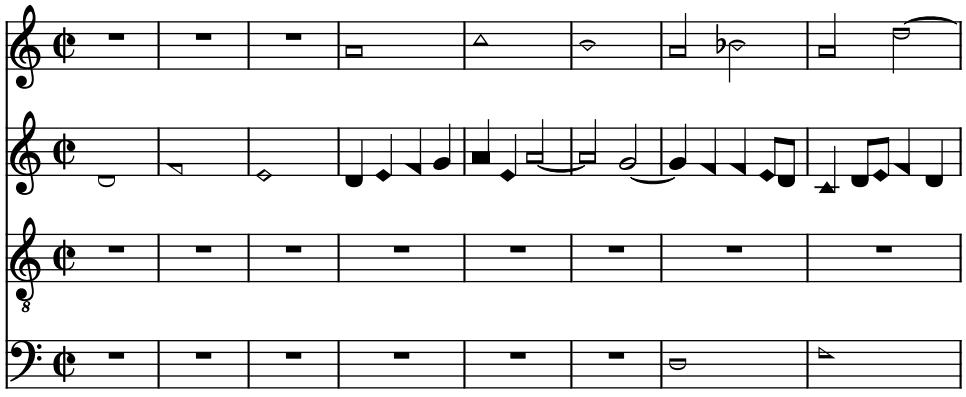
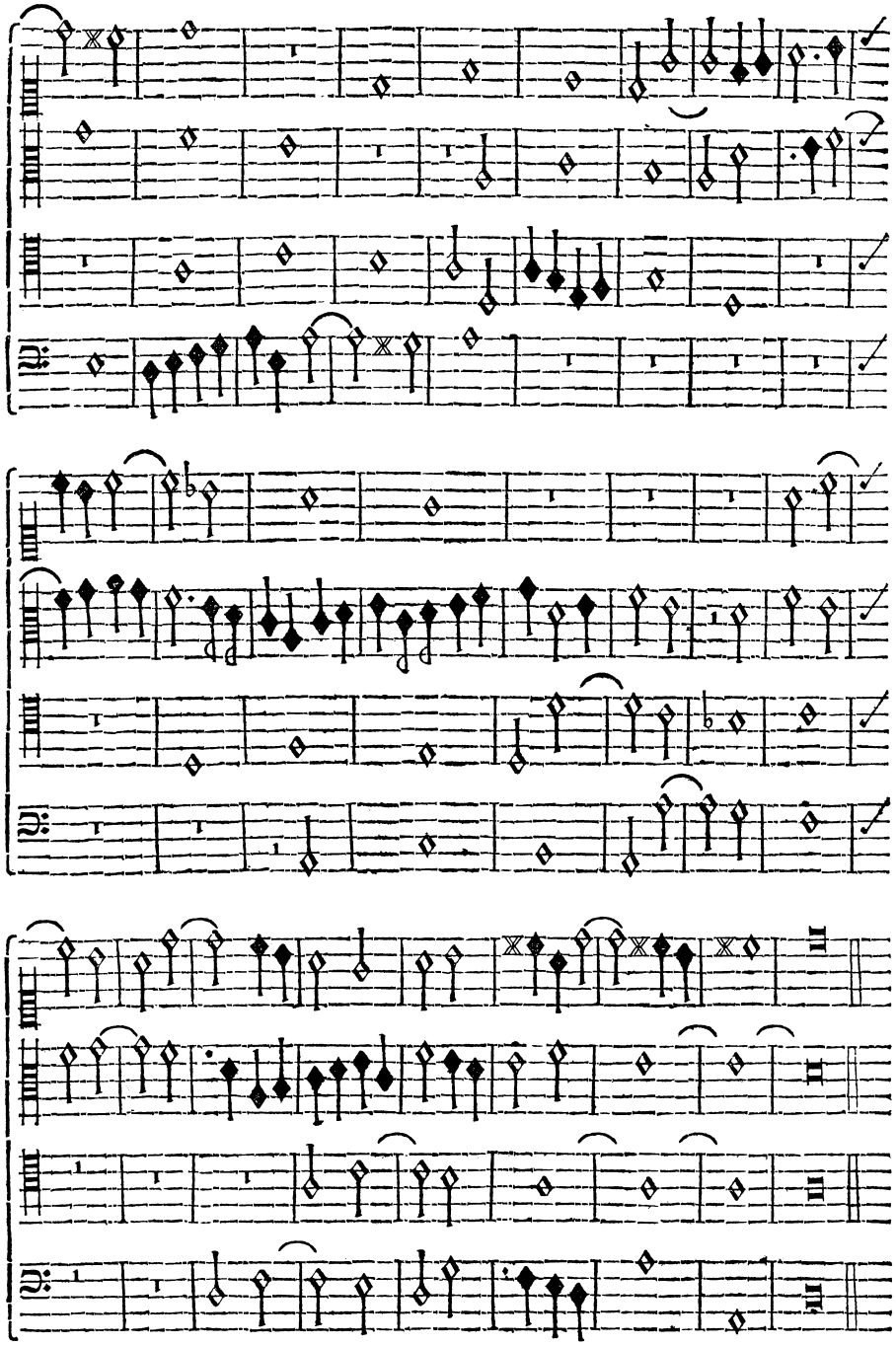
Aloyſ. Aut mutandum modulationis conſilium erit, aut parti, cui, nonniſi intrudendæ locus eſt, adminiculo pauſæ ſilentium impones, uſque dum reingrediendi cum ſubjecto ſpatium aperiatur. Satiùs tamen eſt, prævidendo, meditandoque partes ita diſponere, ne conſilii cœpti paulò poſt te pœniteat: præſtat enim jura intacta relinquere, quàm poſt vulneratam cauſam remedium quærere. Opus igitur eſt, ut, dum uni parti vacas, cæteræ ne memoriâ excidant, neque partibus aliquibus tantum tribuas, ut ſuperſtiti nihil aut ſpatii ad gradiendum, aut concinnitatis ad cantandum remaneat. Exemplar tibi conſiderandum, imitandumque ſubjiciam, ſervato primæ trium partium Fugæ themate, aliiſque ad quartæ partis introitum.

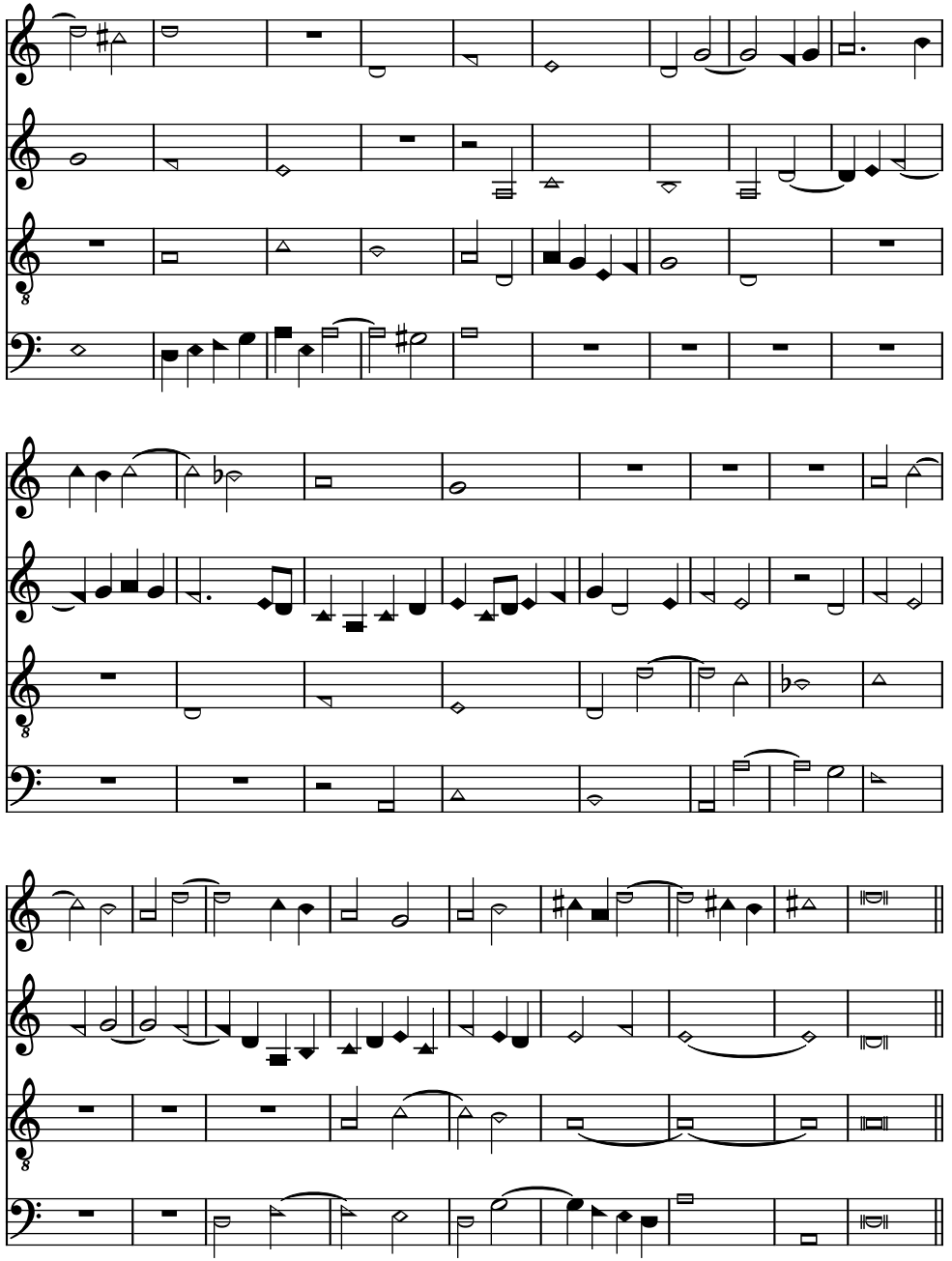


harmony, grace, or variety may be added to the ensemble — nevertheless it is commonly accepted that the Tenor follows the Cantus, and the Bass follows the Alto.

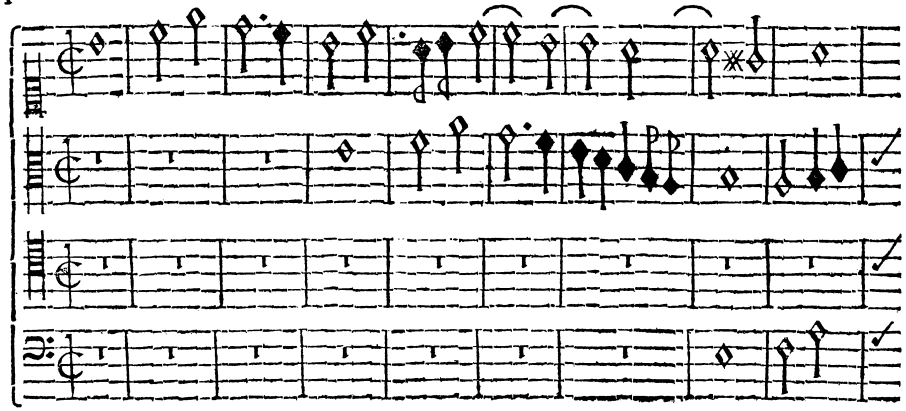
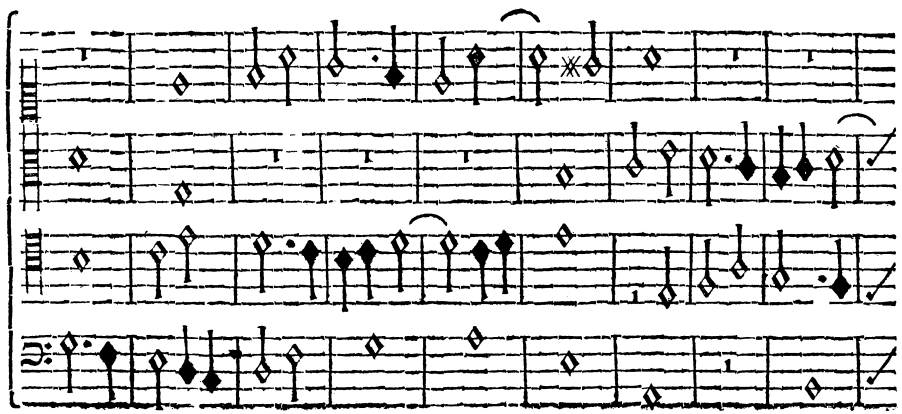
Moreover, as the number of parts increases, you must take great care that the parts are not overcrowded and squeezed together, so that each may have the freedom to move without hindrance. For just as one passing through a dense crowd is jostled on the right, left, and front, and therefore cannot proceed at all or only with difficulty, so too in composition, when one part blocks another, the freedom to move is lost, and thus the ability to sing together is destroyed. Therefore you must strive with all your might to avoid this danger.

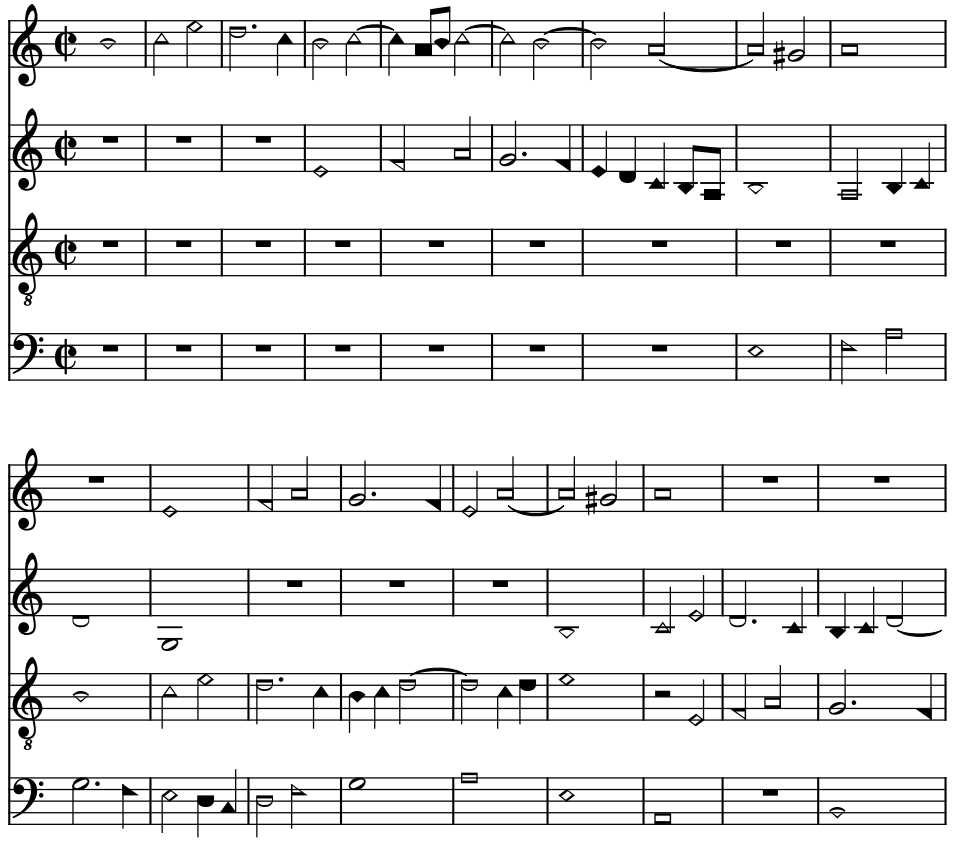
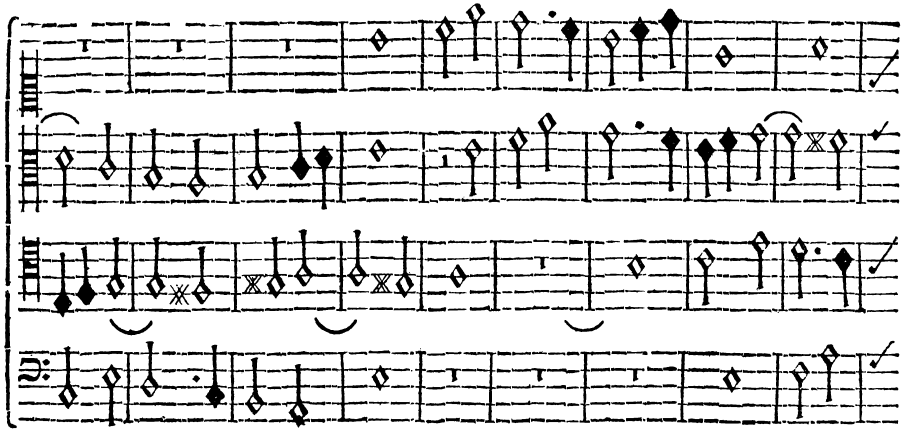
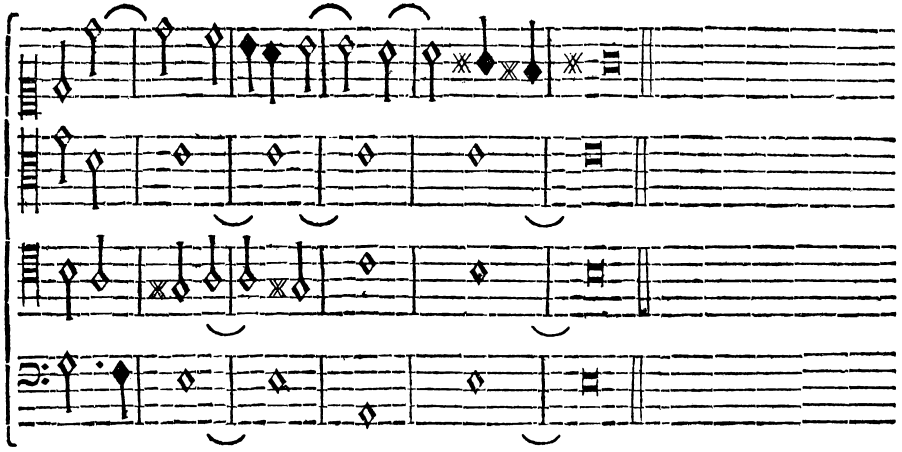
Joseph. But what remedy is there when such a thing happens unexpectedly?

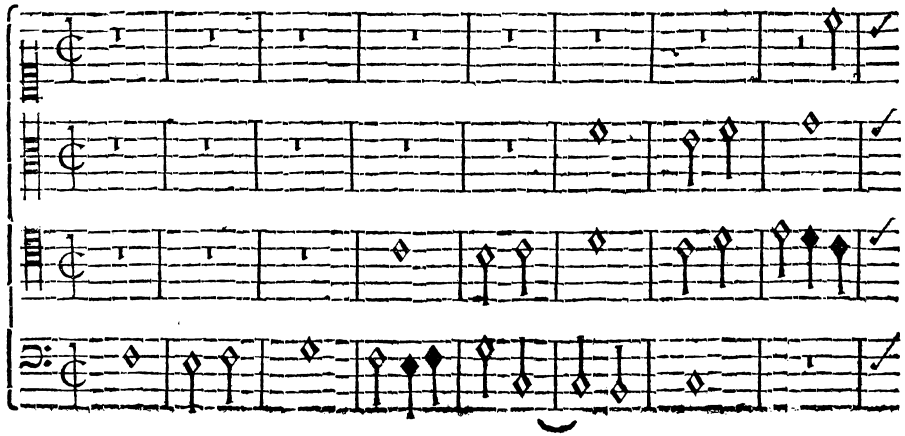
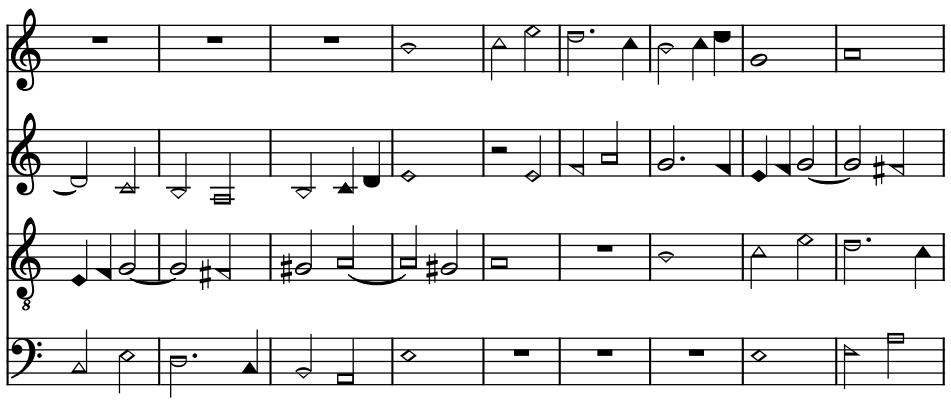
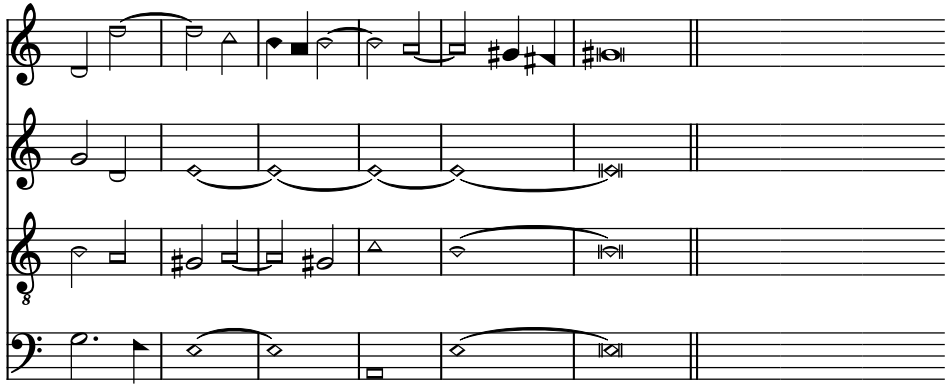
Aloys. Either the plan of modulation must be changed, or you must impose a brief silence on the part that needs room to enter, until space opens for it to re-enter with the subject. Yet it is better to foresee and arrange the parts so that you do not soon regret the plan you began; for it is better to leave the laws intact than to seek a remedy after the case has been wounded. Therefore, while you are occupied with one part, take care that the others do not slip from your mind, and do not give so much to some parts that nothing remains for the rest — neither space to move nor elegance in singing. I will place before you an example to consider and imitate, preserving the subject of the first three-part fugue and adding what is needed for the entrance of the fourth part.



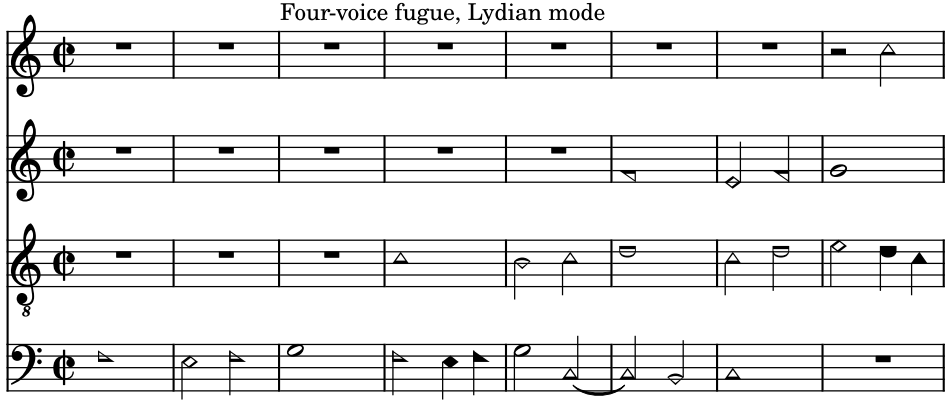
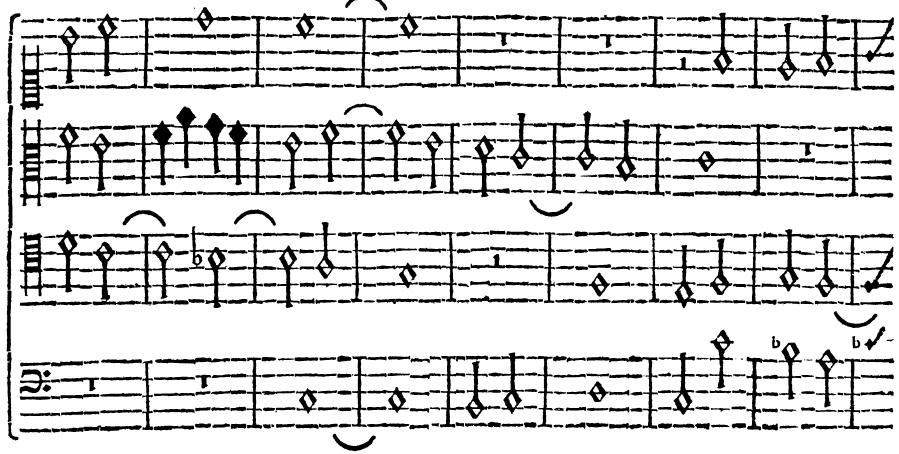
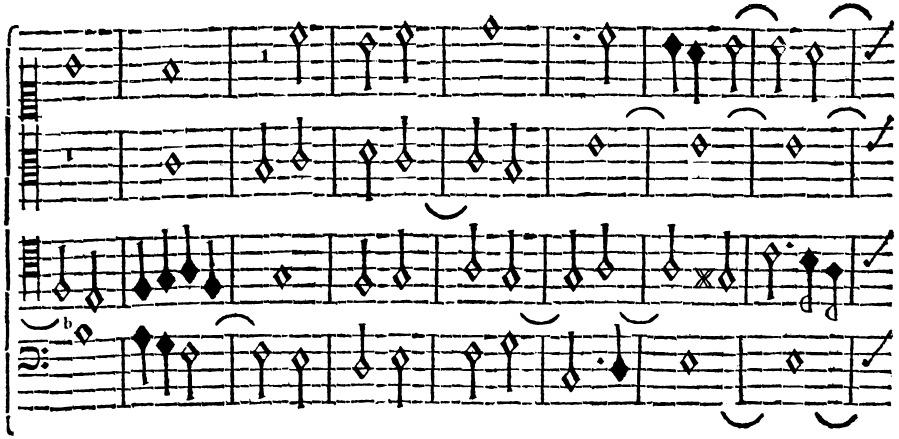
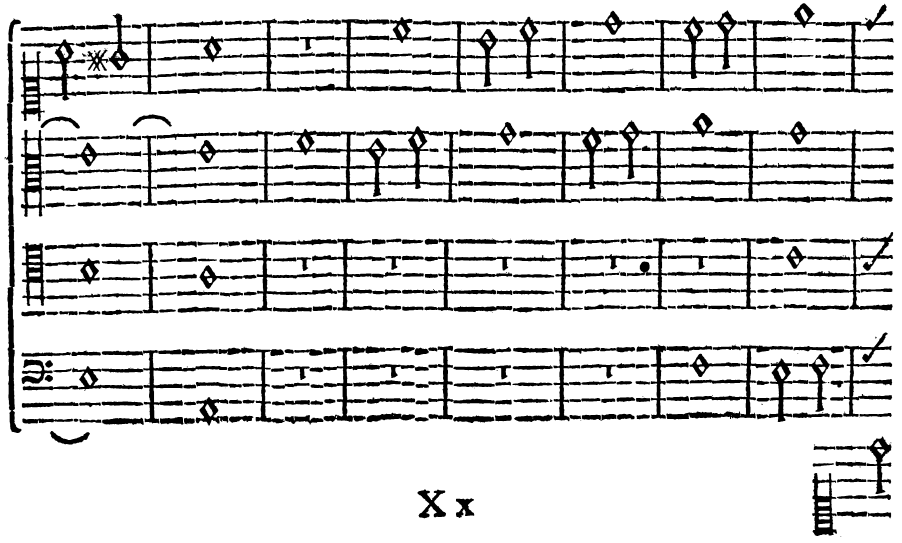
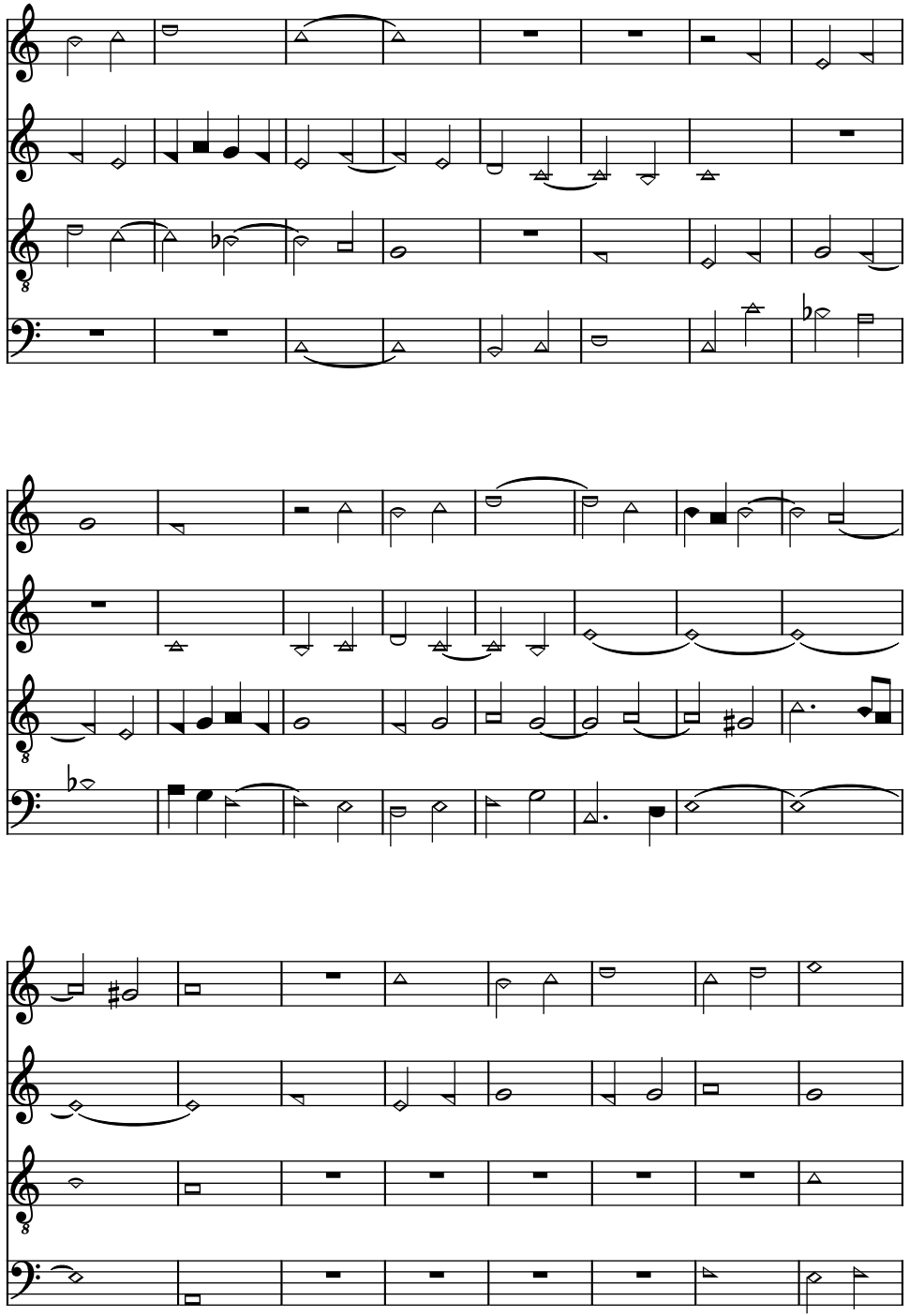
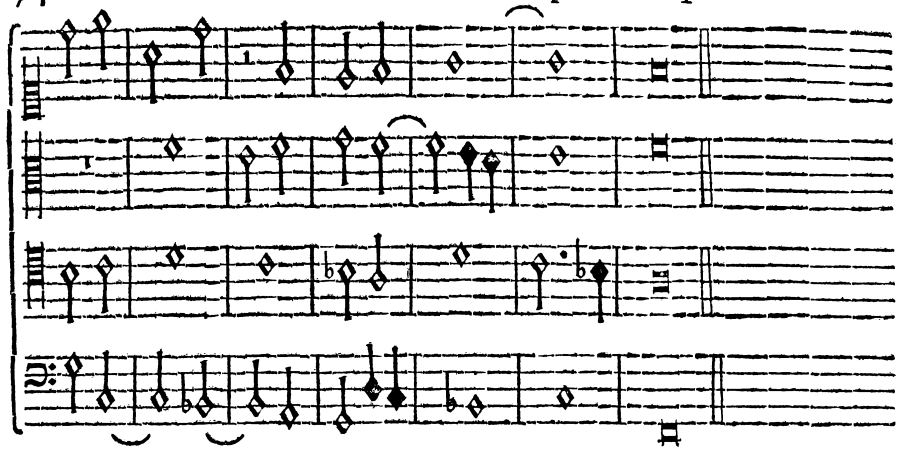
Joſeph. Videtur haud magnum eſſe diſcrimen inter hanc, & trium partium Fugam: rarò enim quatuor partes, niſi ad finem,

Joseph. It seems there is not much difference between this and the three-part fugue; for I rarely see all four parts together except at the end,

finem, convenire conſpicio; obtruncatâ confeſtim unâ vel alterâ, ubi quarta pars intrat.

Aloyſ. Similitudinem, quam dicis, ex uno eodemque Subjecto reſultare conſtat. Quòd autem in hac quatuor partium Fuga ad eundem ferè modum, quô in Tricinio, ſubjectum ductum ſit, ideò factum puta, ut facilior tibi cedat Quatricinii compoſitio: ob quam cauſam tantopere Tricinii ſtudium paulò antè tibi commendavi. Neque mirare raram quatuor partium conjunctionem. Præterquàm enim quod non multò ante præmonuerim, ut à partibus importunè intrudendis abſtineres, ad Quatricinium non requiritur, ut ſemper quatuor partium Harmoniâ compleatur, cui ſufficit, ſi cæteris partibus in actione exiſtentibus, una vel altera interim quieſcens ſubjectum denuò reaſſumere, & ſubjecti fini aliquot Notarum apendicem innectendo ſuplementum Harmoniæ protrahere poſſit. Cætera Modorum ſubjecta ad hunc modum pertractanda tibi committo.

one or another being immediately cut off when the fourth part enters.

Aloys. The similarity you mention arises from the same subject being used. And the fact that in this four-part fugue the subject is treated almost in the same way as in the tricinio is done so that the composition of the quatricinium may be easier for you — for which reason I so strongly recommended the study of the tricinio earlier. Nor should you be surprised at the rare conjunction of all four parts. Besides the warning I gave not long ago to avoid forcing parts in at unsuitable times, the quatricinium does not require that the harmony of all four parts be complete at all times; it is enough if, while the other parts are active, one or another rests for a time, then resumes the subject, and by adding a small appendix of notes to the end of the subject, prolongs the harmony. I leave the subjects of the remaining modes to be treated by you in this manner.



Fuga à 4. Modi F.

Four-voice fugue, Lydian mode



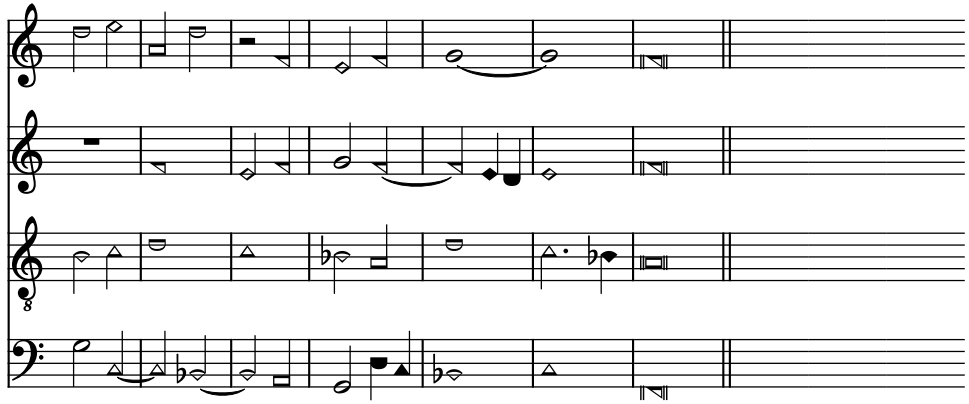
Aloyſ. Simplici hac methodo ſatìs imbutus mihi videris, id quod exempla commodè facta demonſtrant. Relictis itaque reliquis Modis privato ſtudio tuo, ut tempori conſulatur, ad Fugam, plurium ſubjectorum ornamento diſtinctam, progrediamur. Quum autem abſque cognitione Contrapuncti Duplicis id effici nequeat, primò de eo diſſerendum occurrit.

EXERCITII V.

LECTIO QUINTA.

De Contrapuncto duplici.

PEr Contrapunctum duplex intelligitur Compoſitio artificioſa, eoque modo conſtructa, ut partes ejuſdem inter ſe convertibiles eſſe poſſint, & pars, quæ modò ſuperior, nunc per inverſionem inferior exiſtat: ſic meâ ſententiâ dictum, quòd præter inverſionem partium, cæterùm re nullâ immutatâ, duplicem & ratione acuminis, gravitatisque differentem exhibeat Melodiam. Quàm præſtans, elegansque Contrapuncto huic ſit uſus, cùm in omni Compoſitionis genere, tum maximè in Fugis plurium ſubjectorum ligamine combinandis, paulò poſt experientiâ comperies; idcircò uberiori quoque explicatione illuſtrandum. Variæ Generis hujus ſpecies à nonnullis ſtatuuntur; ùt: Contrapunctum duplex in Tertia, Quarta, Quinta, Sexta, Octava, Decima, Duodecima, &c. Nos autem relictis, quorum ob anguſtias aut exiguus uſus eſt, aut cum aliis fermè coincidunt, eas tantummodo



Aloys. By this simple method you seem to me sufficiently instructed, as the well-made examples show. Leaving the remaining modes to your private study, so as to save time, let us proceed to the fugue adorned with multiple subjects. But since this cannot be done without knowledge of contrapunctus duplex (double counterpoint), we must first discuss that.

EXERCISE V.
LESSON FIVE.

On Double Counterpoint.

By contrapunctus duplex (double counterpoint) is understood an artful composition constructed in such a way that its parts can be exchanged with one another, and the part that is now above may, by inversion, become the lower part; hence, in my opinion, the name, because besides the inversion of the parts, everything else remaining unchanged, it presents a double melody, differing in sharpness and gravity. How excellent and elegant the use of this counterpoint is — in every kind of composition, but especially in combining fugues of multiple subjects — you will soon learn by experience; therefore it must be explained more fully.

Various kinds of this technique are described by some: double counterpoint in the Third, Fourth, Fifth, Sixth, Octave, Tenth, Twelfth, etc. But leaving aside those whose use is narrow or which coincide with others, we will pursue only those species that are more useful and of greater importance in composition — for example, double counterpoint in the Octave, in the Tenth, and in the Twelfth; in which either part, by

modo proſequemur ſpecies, quæ & uſitatiores, & in Compoſitione majoris momenti ſunt. V.G. Contrapunctum in Octava, in Decima, in Duodecima: quorum alterutra pars abſtinentiâ quarundam Conſonantiarum, Diſſonantiarumque pro cujusvis qualitate de loco ſuo proprio in aliud intervallum transferri poteſt. Antequàm de hujus Contrapuncti ſpeciebus diſſerere exordiar, generalia quædam præmittenda ſunt. Primò curandum eſt, ut ſubjecta differentes inter ſe ſortiantur motus, quibus facili negotio diſtingui poſſint; id quod diſparitate figurarum efficitur, tribuendoque alteri ſubjecto minutiores, alteri majoris valoris figuras: ſic diſcriminis perceptio eminebit, atque confuſio evitabitur. Secundò: Ità inſtituenda ſubjecta ſunt, ne ſimul, eodemque tempore incipiant; ſed alterutrum poſitione cujusdam pauſæ tardiùs intret, neceſſe eſt. Tertiò: Non excedantur limites cuilibet Contrapuncti ſpeciei infra præſcribendi. His ſuppoſitis fiat initium à Contrapuncto in Octava, utpote & facilitate, & uſu præſtantiori.

Contrapunctum igitur in Octava eſt Compoſitio, ita inſtituta, ut inversâ alterutrâ parte in Octavam acutam, aut gravem, variam reddat Harmoniam; nihilominùs rectè regularum rationi conſentaneam. Ut autem hoc evenire poſſe, extra dubium ſis Joſephe, Primò: à Quinta abſtinendum eſt. Secundò: Non procedatur in Octavam ſaltando. Tertiò: infra limites Octavæ te contineas oportet. Ad clariorem hujus rei intelligentiam, & in quasnam Conſonantias, diſſonantiasque, factâ inverſione, immutentur priores, ſequentes contra ſe poſiti numeri declarabunt.

1. 2. 3. 4. 5. 6. 7. 8.
8. 7. 6. 5. 4. 3. 2. 1.

Patet itaque, Uniſonum per inverſionem Octavam fieri: Secundam in Septimam inverti: ex inverſa Tertia Sextam reſultare: ex Quinta Quartam: & ſic de cæteris. Unde ratio, ob quam hoc in Contrapuncto Quinta prohibeatur, in promptu eſt: quia inverſa diſſonantiam, nempe Quartam efficit. E. G.

abstaining from certain consonances or dissonances, can be transferred from its proper place to another interval.

Before beginning to discuss the species of this counterpoint, certain general principles must be stated. First: the subjects must be arranged so that they have different motions, by which they can easily be distinguished; this is achieved by giving one subject smaller note-values and the other larger ones, so that the distinction stands out and confusion is avoided. Second: the subjects must be arranged so that they do not begin at the same time; one must enter later by means of a rest. Third: the limits prescribed for each species of counterpoint must not be exceeded.

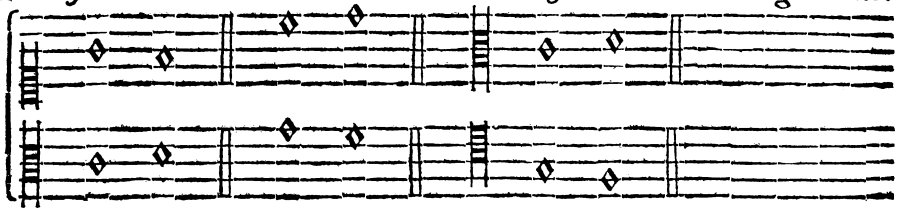
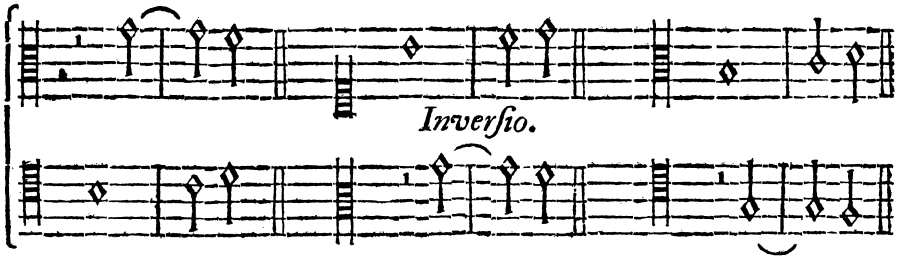
With these things established, let us begin with double counterpoint in the Octave, as being both easier and more useful.

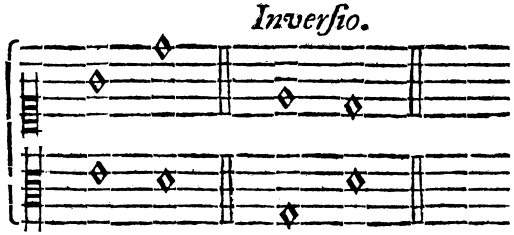
Double counterpoint in the Octave is a composition arranged so that, when either part is inverted into the higher or lower octave, it yields a different harmony, yet one that still agrees with the rules. But so that this may happen, be assured of this, Joseph: first, the Fifth must be avoided; second, one must not proceed to the Octave by a leap; third, you must keep yourself within the limits of the Octave.

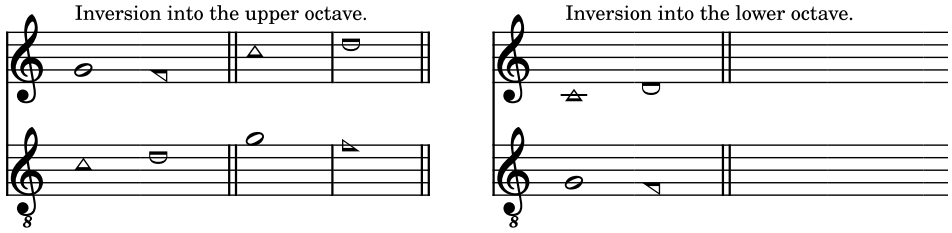
For clearer understanding of this matter, and into what consonances and dissonances the original intervals are changed by inversion, the following numbers placed opposite each other will show:

1 2 3 4 5 6 7 8
8 7 6 5 4 3 2 1

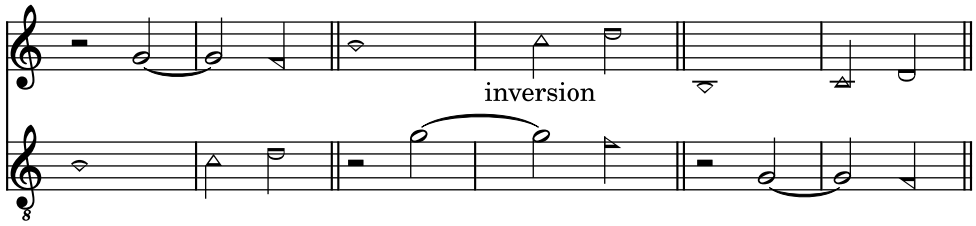
Thus it is clear that the unison becomes the octave by inversion; the second becomes the seventh; the third becomes the sixth; the fifth becomes the fourth; and so on. Hence the reason why the Fifth is forbidden in this counterpoint is obvious: because when inverted it becomes a dissonance, namely the Fourth. For example:

Inverſio in Octavam acutam. *Inverſio in Octavam gravem.*

Uſurpata tamen cum ligatura locum habet. E. G.

Joſeph. Comprehenſis hucusque dictis, reſtat explanare cauſam, ob quam in Octavam per ſaltum illicitus ſit progreſſus, & cur Octavæ limites excedere non liceat: Ut citò facias, te Venerande Magiſter etiam atque etiam rogo: omnis enim mora patientiam meam tentat; quòd hoc Contrapuncti genus permagno crebriùs encomio efferri audiverim, ideoque cupido videndi exempla pridem jam animum inceſſerit.

Aloyſ. Faciam absque mora. Ex Octava per inverſionem Uniſonum fieri, ſuprà contra ſe poſiti numeri ante oculos ponunt: in Uniſonum verò per ſaltum, niſi per modum clauſulæ, non ritè procedi, ſæpiùs jam dictum eſt. In exemplo:

Quòd Octavam in Theſi prorsùs evitaveris, rectiùs feciſti; nam hæc per inverſionem Uniſonus fit, qui niſi per Syncopen minùs rectè, ùt jam dictum eſt, uſurpatur. Non excedendorum Octavæ limitum ratio eſt, quia Contrapuncti duplicis munus eſt, efficere variam, differentemque per inverſionem

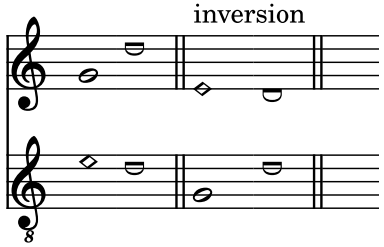


Nevertheless, it has a place when used with a ligature. For example:

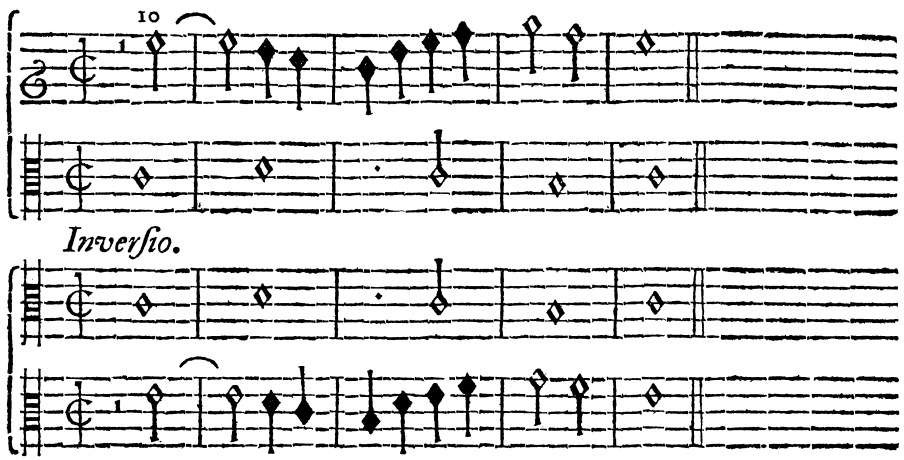


Joseph. Having understood what has been said so far, it remains to explain why it is forbidden to proceed to the octave by a leap, and why one may not exceed the limits of the octave. I beg you, venerable Master, to do this quickly; every delay tries my patience, for I have often heard this kind of counterpoint praised most highly, and therefore a desire to see examples has long stirred my mind.

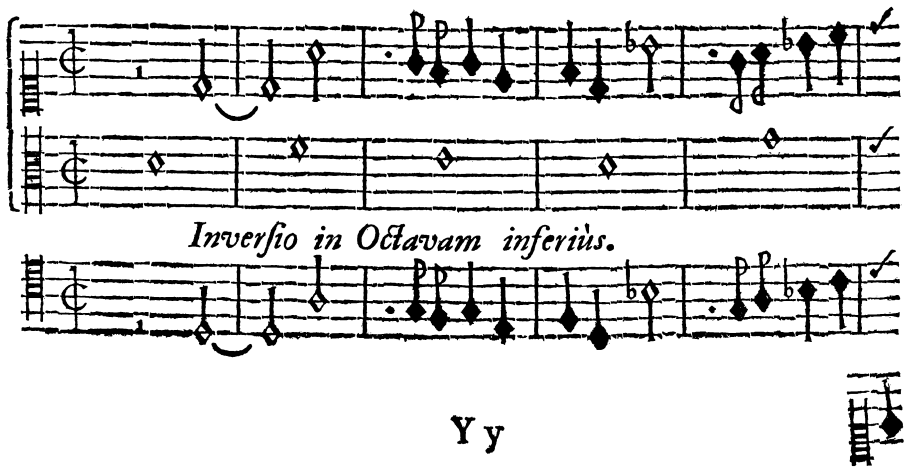
Aloys. I will do so without delay. The numbers placed opposite each other show that the octave becomes the unison by inversion; but to proceed to the unison by a leap, unless in the manner of a cadence, is not correct, as has been said many times. For example:

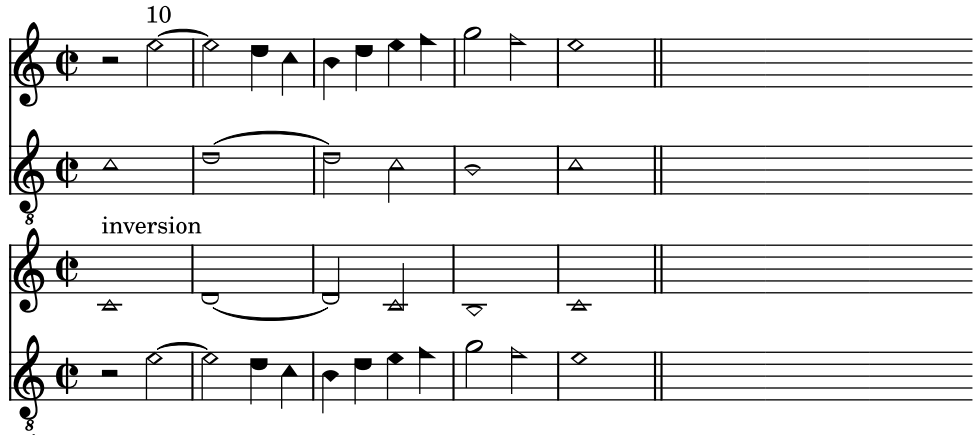


You did rightly in avoiding the octave in the thesis (downbeat), for by inversion it becomes the unison, which, unless used through syncopation, is less proper, as has already been said. The reason for not exceeding the limits of the octave is that the function of double counterpoint is to produce varied and different harmony through inversion; but if you go beyond the bounds of the octave,

ſionem Harmoniam : ſi autem Octavæ terminos egreſſus fueris, licet Conſonantiæ compoſitæ in ſimplices mutentur, eadem reſultabit Harmonia, nec tam naturâ, quàm ſitu tantùm diſcrepans : ùt ſequens exemplum demonſtrat. E. G.

Ubi vides, ex Decima, utpote Tertiâ compoſitâ per inverſionem fieri Tertiam ſimplicem : ex Nona, Secundâ compoſitâ oriri ſecundam ſimplicem, & ſic de cæteris. Inter Conſonantias enim compoſitas, & ſimplices haud alia, quàm loci differentia intercedit.

Joſeph. Jam omnibus animo perceptis, quantum quidem mihi videtur, ne Paradigmata oculis ſubducere, diutiùs morêre, te iterum, atque iterum obteſtor.

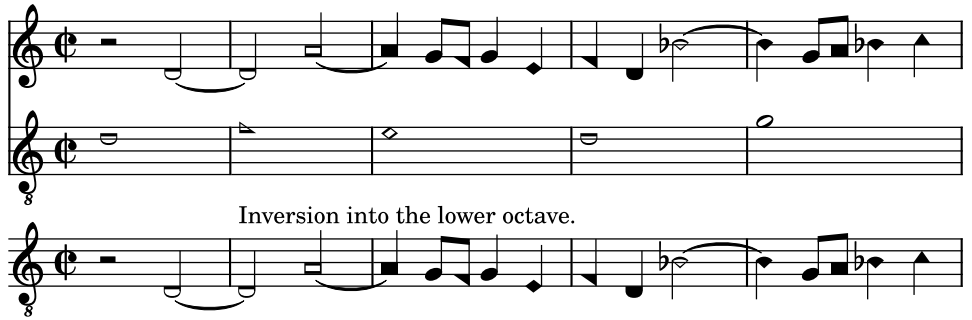
Aloyſ. En exemplum primum obligatione Cantûs firmi obſtrictum :

although composite consonances may change into simple ones, the resulting harmony will be the same, differing not in nature but only in position — as the following example shows. For example:

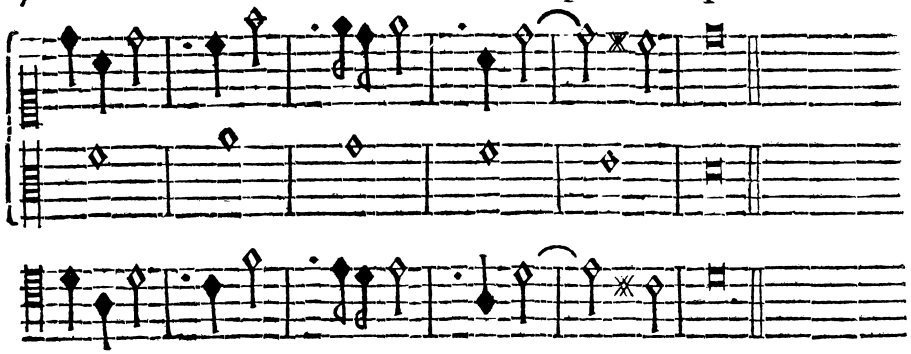


Here you see that from the Tenth — a composite Third — inversion produces a simple Third; from the Ninth — a composite Second — a simple Second arises; and so on. For between composite and simple consonances there is no difference except that of position.

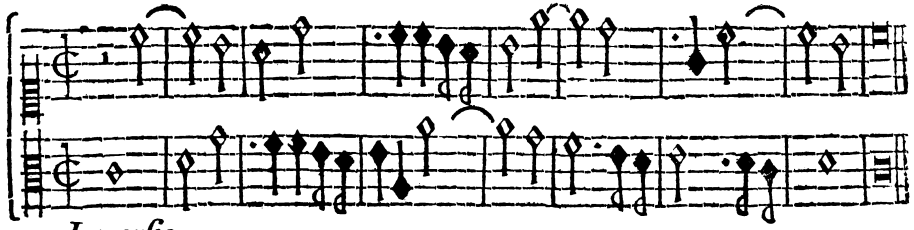
Joseph. Now that I have taken all this in with my mind—at least as far as it seems to me—do not delay any longer in removing these paradigms from my sight, I beg you again and again.

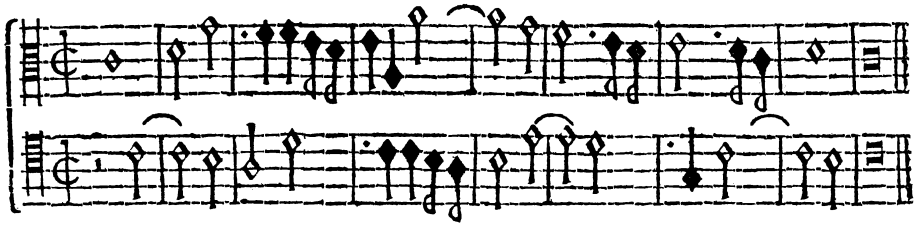
Aloys. Here is the first example, bound by the obligation of the cantus firmus.

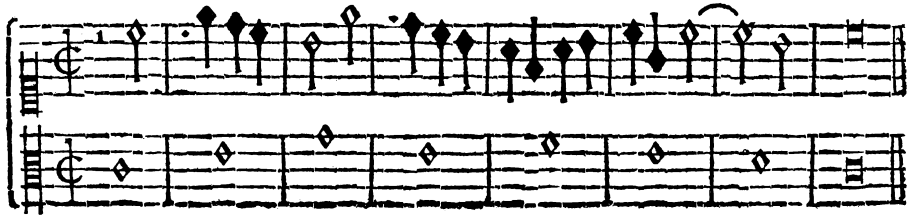




Sequitur Paradigma absque Cantûs firmi obligatione expreſſum:



Inverſio.

Ex quibus Paradigmatibus perſpicuum eſt, obſervatis hucuſque dictis inverſionem infallibilem eſſe, neceſſarioque rectæ Compoſitionis præceptis conſentaneam evenire.

Primi exempli Contrapunctum, ſi ità inſtituatur, ut omnis Theſis aut motum contrarium, aut obliquum obtineat, etiam tribus vocibus decantari poteſt, accedente parte tertiâ cum translatione Contrapuncti in Decimam inferiùs. E. G.

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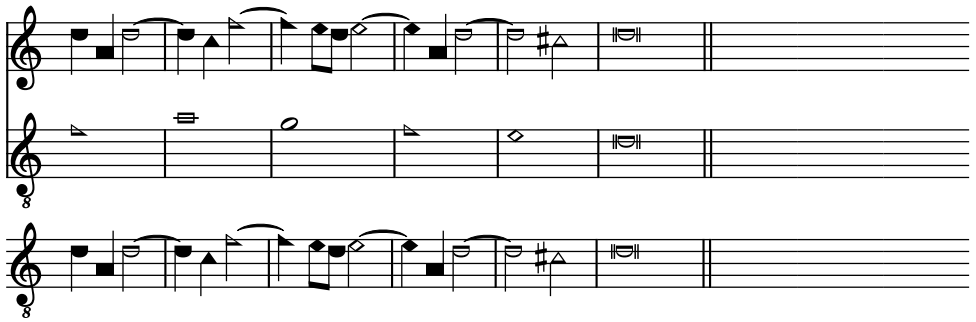
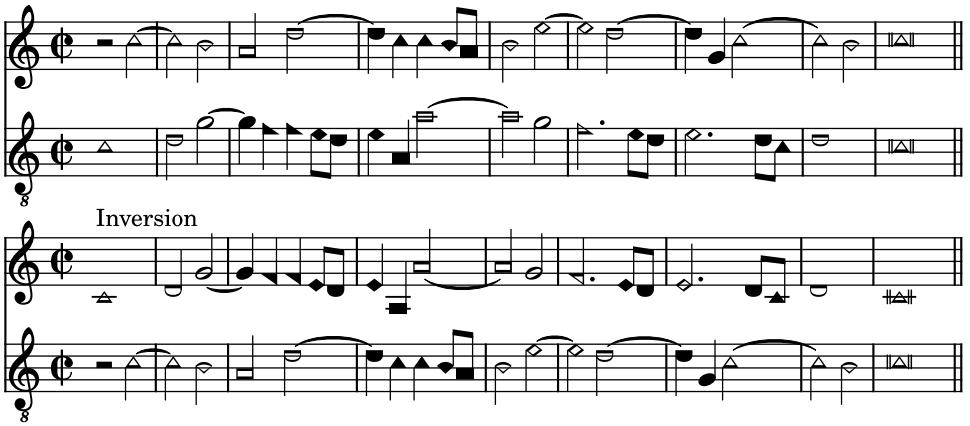
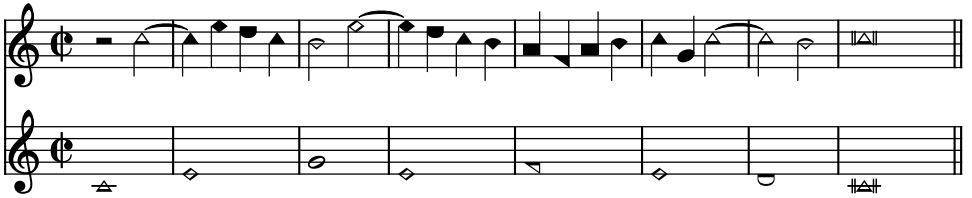
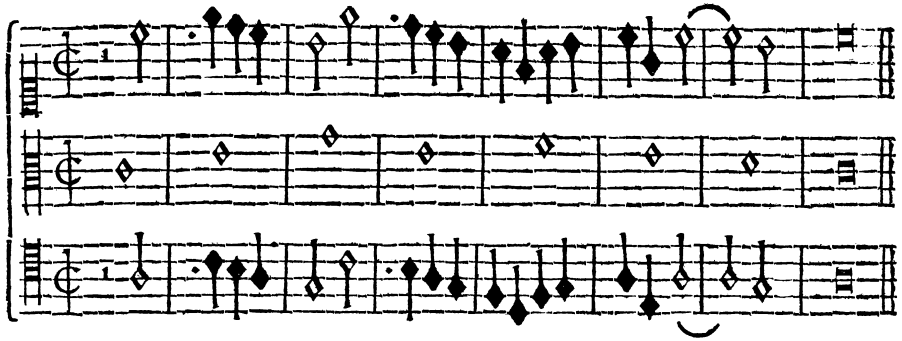


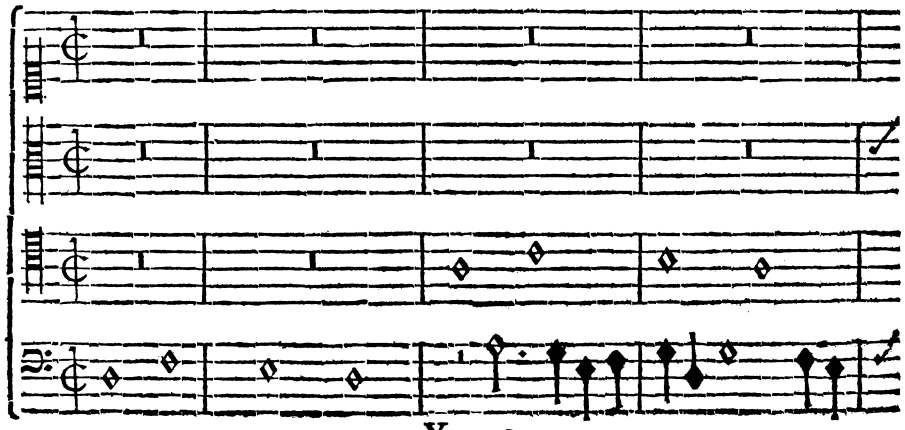
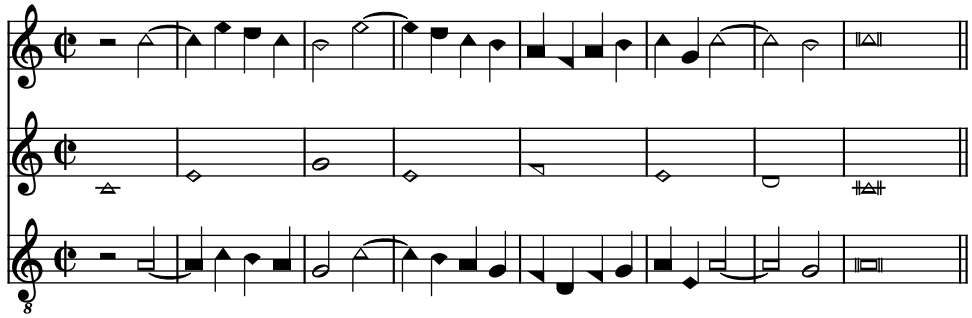
Next follows a paradigm presented without the obligation of the cantus firmus:

From these paradigms it is clear that, if what has been said up to now is observed, the inversion is infallible and necessarily turns out in accordance with the rules of correct composition.

The counterpoint of the first example, if arranged so that every thesis (downbeat) has either contrary motion or oblique motion, can also be sung by three voices, with a third part added by placing the counterpoint a tenth lower. For example:

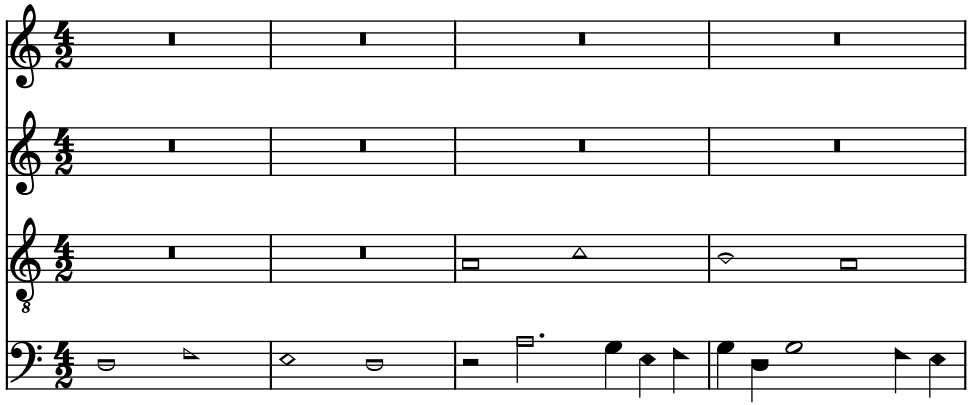
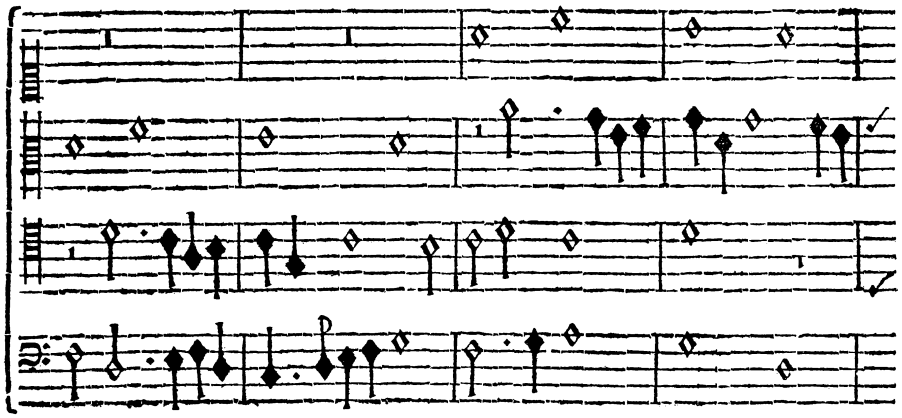
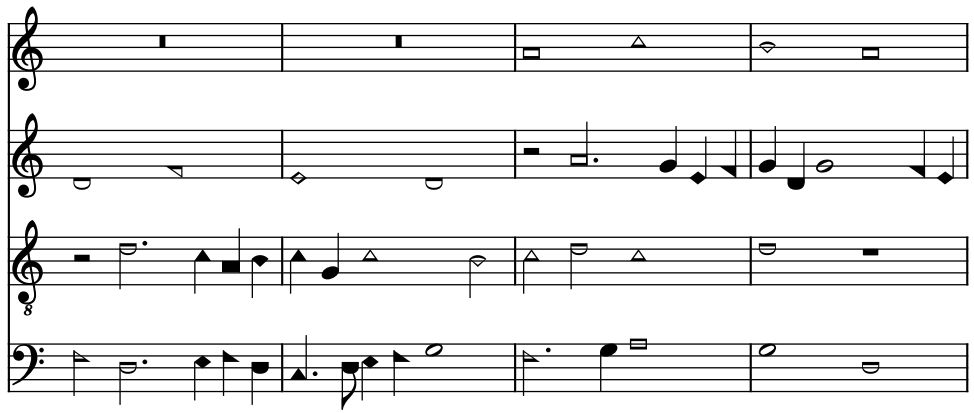
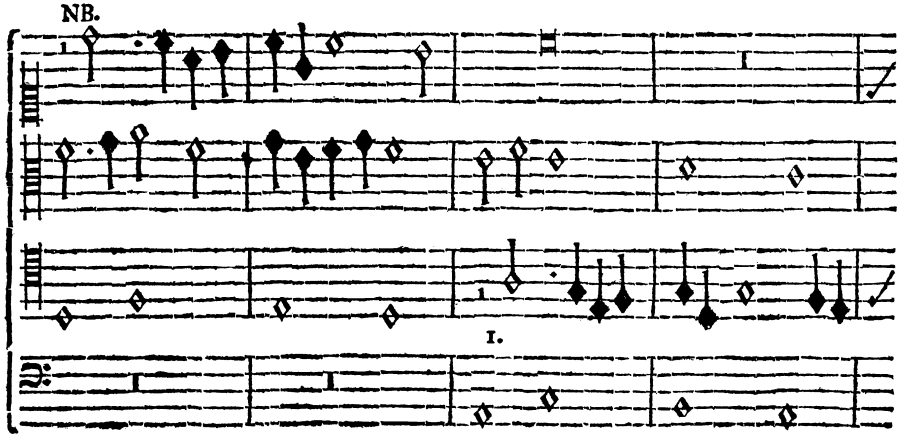


In hujus Contrapuncti exemplo, omnis Theſis, ſive principium tactûs aut motum contrarium, aut motum obliquum obtinet, ideò in Tricinium verti poteſt deſcribendo de Nota ad Notam Contrapunctum, idque collocando in Decima inferiùs.

Joſeph. Mirificè delector Contrapuncti hujus artificio, maximaque jam pridem cupiditate ſciendi impulſus à te flagito, quo modo Paradigmata hæc in actum practicum redigenda ſint?

Aloyſ. Quanquam & reliquas hujus generis ſpecies primò pertractandi mihi animus fuit, tamen, ut deſiderio tuo morem geram, accipiam Fugæ primæ ſubjectum in Modo primo uſurpatum, illique contraſubjectum intexendo demonſtrabo, qua ratione id inſtituendum, ac per totum Fugæ curſum perducendum ſit.

Fuga à 4. Contraſubjecti artificio expreſſa: fundata in Contrapuncto duplici, cum inverſione in Octavam.

In this example of counterpoint, every thesis—that is, the beginning of each measure—has either contrary or oblique motion; therefore it can be turned into a tricinium (three-part texture) by writing the counterpoint note-against-note and placing it a tenth lower.

Joseph. I am wonderfully delighted by the artifice of this counterpoint, and driven by a long-standing desire to know, I beg you to show me how these paradigms are to be put into practical use.

Aloys. Although I had intended first to treat the remaining species of this kind, yet to satisfy your desire I will take the subject of the first fugue used in the first mode, and by weaving a contrasubjectum (countersubject) into it, I will show how it must be set up and carried through the entire course of the fugue.

A four-part fugue expressed by the artifice of the countersubject, founded on double counterpoint with inversion in the octave.

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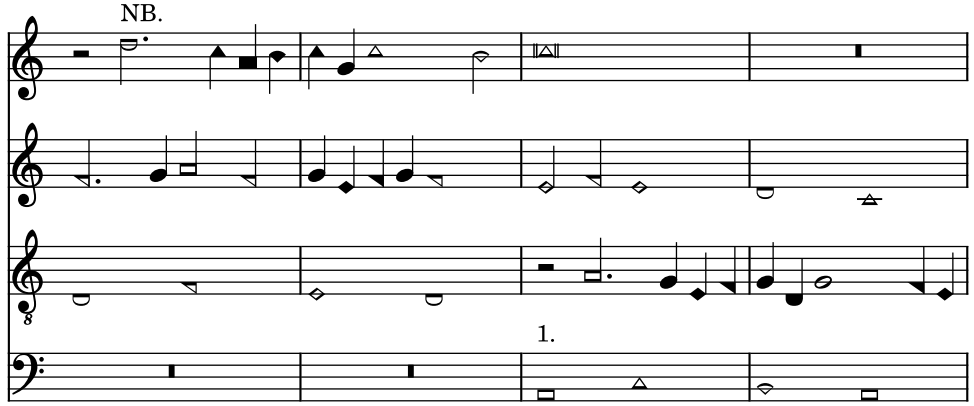
NB.
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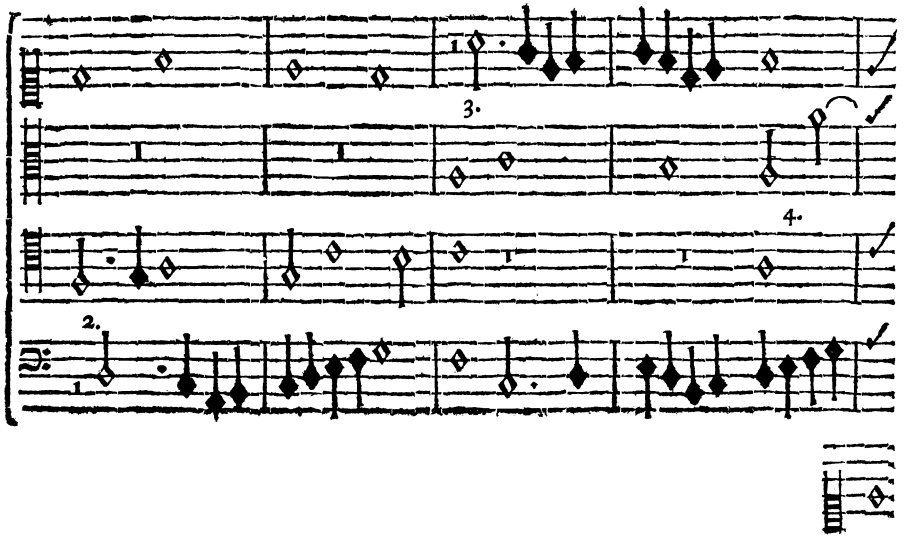
3.
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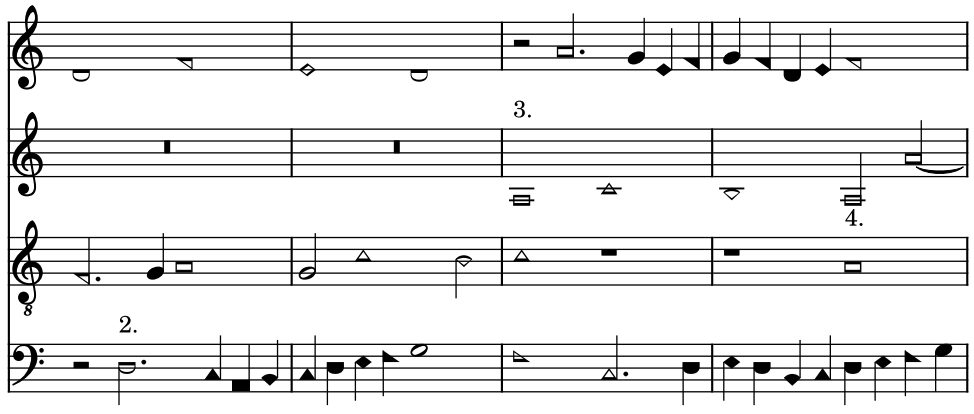
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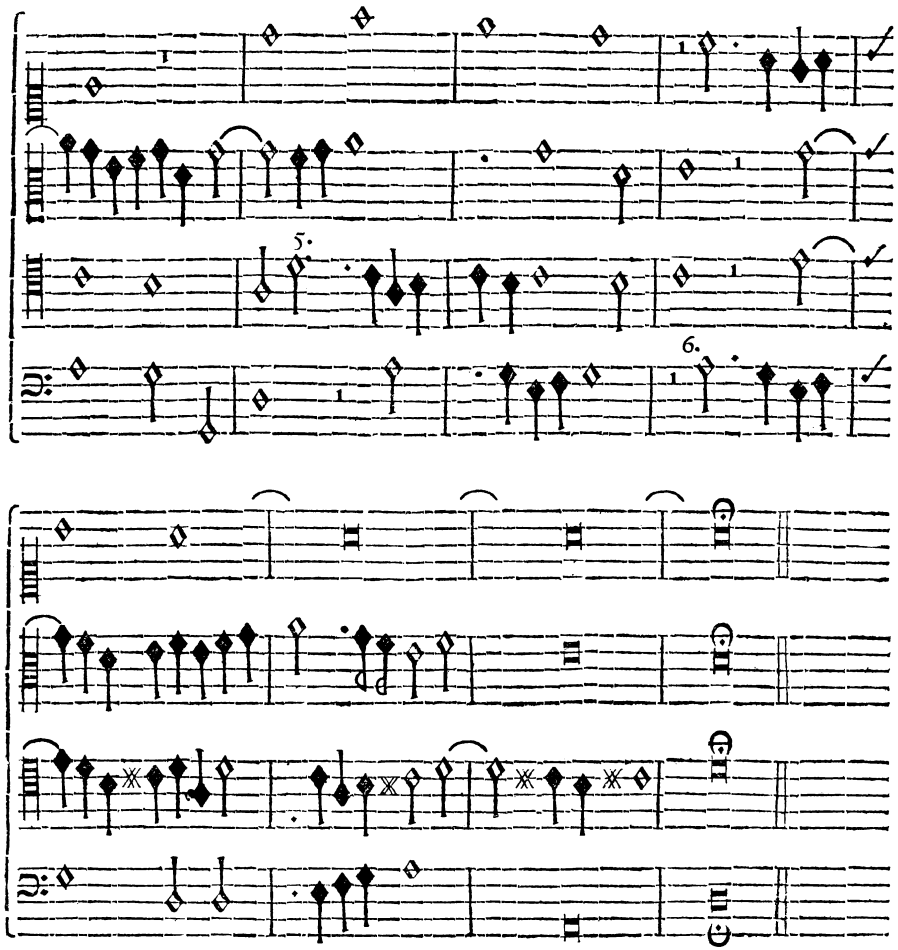
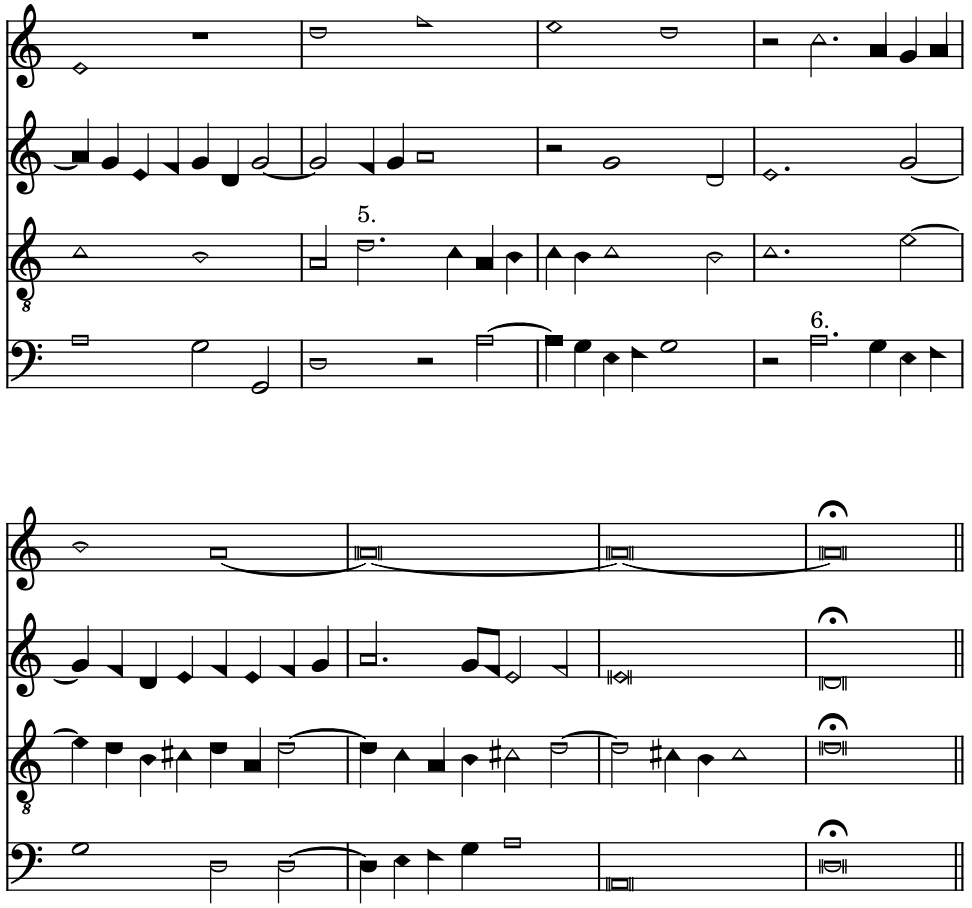


Aloyſ. En uſum hujus Contrapuncti duplicis, & exemplar impatientiâ tuâ extortum. Obſerva Primò, Contraſubjectum poſitâ dimidiæ menſuræ pausâ in Uniſono inſtructum; atque per ſubjectorum inverſionem in Octavam reſolvi, ut N.° 1. 2. 3. 4. 5. videre licet, ubi Contraſubjectum modò in partibus extremis repertum, modò in parte media, ſemper in Octava reſpondet ſubjecto ſuo principali, ex qua immutatione diverſa ſemper oritur Harmonia.

Joſeph. Hîc loci NB. Uniſonum non in Octavam, verùm in Quintam decimam inverti apparet.

Aloyſ. Eandem eſſe rationem invervallorum compoſitorum quoad uſum cum intervallis ſimplicibus jam antè dictum

Aloys. Behold the use of this double counterpoint, and the example drawn from your impatience. Observe first: the countersubject, after a half-measure rest, is set in the unison; and through the inversion of the subjects it is resolved into the octave, as you may see at Nos. 1, 2, 3, 4, 5, where the countersubject is sometimes found in the outer parts, sometimes in the middle part, always answering its principal subject at the octave, from which alteration a different harmony always arises.

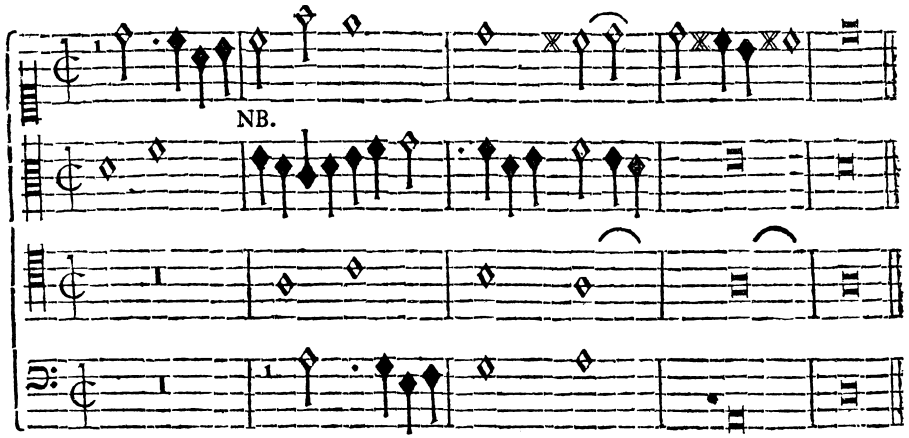
Joseph. Here I read "NB": it appears that the unison is not inverted into the octave, but into the fifteenth.

Aloys. It has already been said that the nature of composite intervals is the same in use as that of simple intervals.

eſt. Unde hæc translatio, ut partibus mediis locus detur, haud abſque induſtria nonnunquam conceditur. Porrò animadvertes, quomodo varietatis gratiâ, ſub N.° 6. relicto ſubjecto principali partes cum Contraſubjecto ſolo artificiosè ludant, idque in arctum contrahant.

Joſeph. Cum ſtupore hanc partium combinationem admiror: verùm non potuiſſet fieri hæc partium coarctatio cum utroque ſubjecto, & ita clauſula finalis concludi.

Aloyſ. Potuiſſet, mutato unius, alteriuſve figuræ valore, hoc modo:



Aloyſ. Obſerva hoc ſignum: NB. ubi locò duarum Semibrevium, duæ Semiminimæ poſitæ, quod aliàs uſuvenire non potuiſſet, Tenori, & Baſſo in tertio ſubjecti tactu intrandi poteſtatem faciunt. Similis itaque figurarum fractio, hanc ob cauſam adhibita non ſolùm conceditur, verùm etiam haud parvam Compoſitori induſtriæ pariet exiſtimationem. Hæc igitur ſubjectorum convertibilitas Contrapuncti duplicis beneficio in acceptis referenda eſt, cujus operâ, ſi ſubjecta ritè inſtituta ſint, facili negotio Fuga inſtrui, ac in longum protrahi poteſt. Tuum nunc eſt, Joſephe, cæteras ordine Modorum Fugas pari ratione tractando perſequi. Verùm ut varietati conſultum ſit, non ſemper uno modo, id eſt, in primo ſubjecti tactu Contraſubjectum introducendum eſt: ſed pro qualitate ſubjecti principalis, in ſecundo, vel tertio tactu Contraſubjecti introitus fieri poteſt: ùt ſequens principium Fugæ ſecundi Modi à te ad finem perducendum demonſtrabit.

Hence this transposition, so that room may be given to the middle parts, is sometimes allowed not without skill. You will also notice how, for the sake of variety, at No. 6, the principal subject being left aside, the parts play artfully with the countersubject alone and compress it into a narrow space.

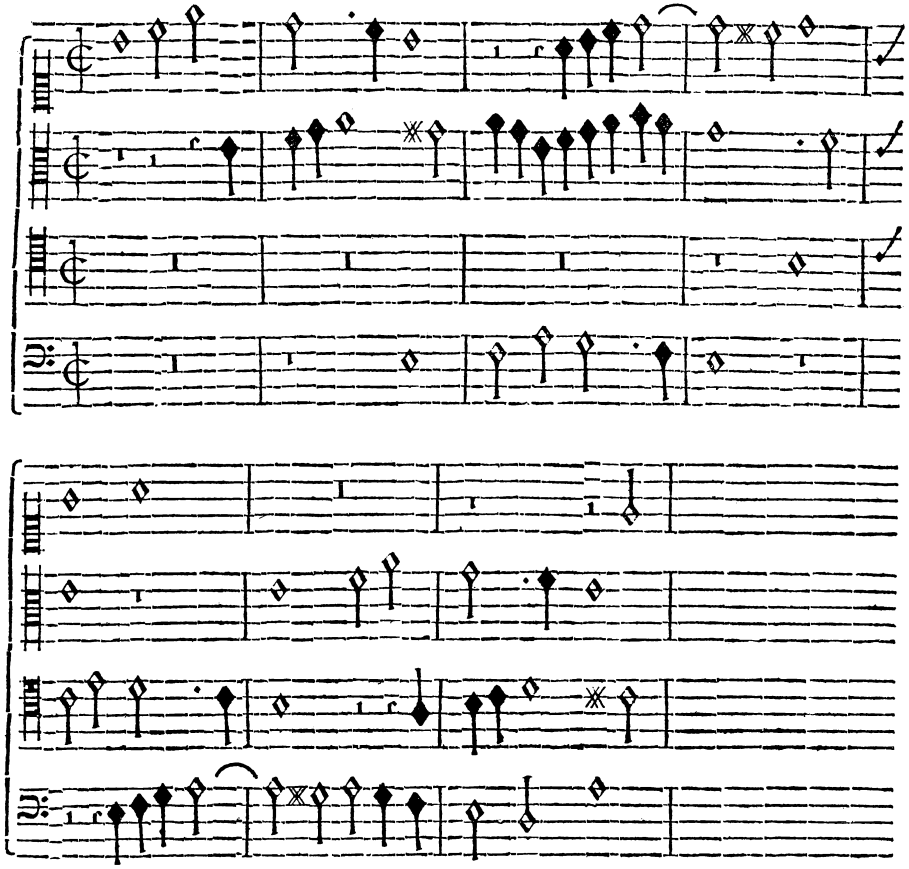
Joseph. I admire with astonishment this combination of the parts; but could not this compression of the parts have been done with both subjects, and so the final cadence concluded?

Aloys. It could have been, by changing the value of one or another note, in this way:

Aloys. Observe this sign "NB": where, in place of two whole notes, two quarter notes are set—something that otherwise could not occur—giving the Tenor and Bass the ability to enter in the third measure of the subject. A similar breaking-up of note-values, used for this reason, is not only permitted but also brings no small esteem to the composer's skill.

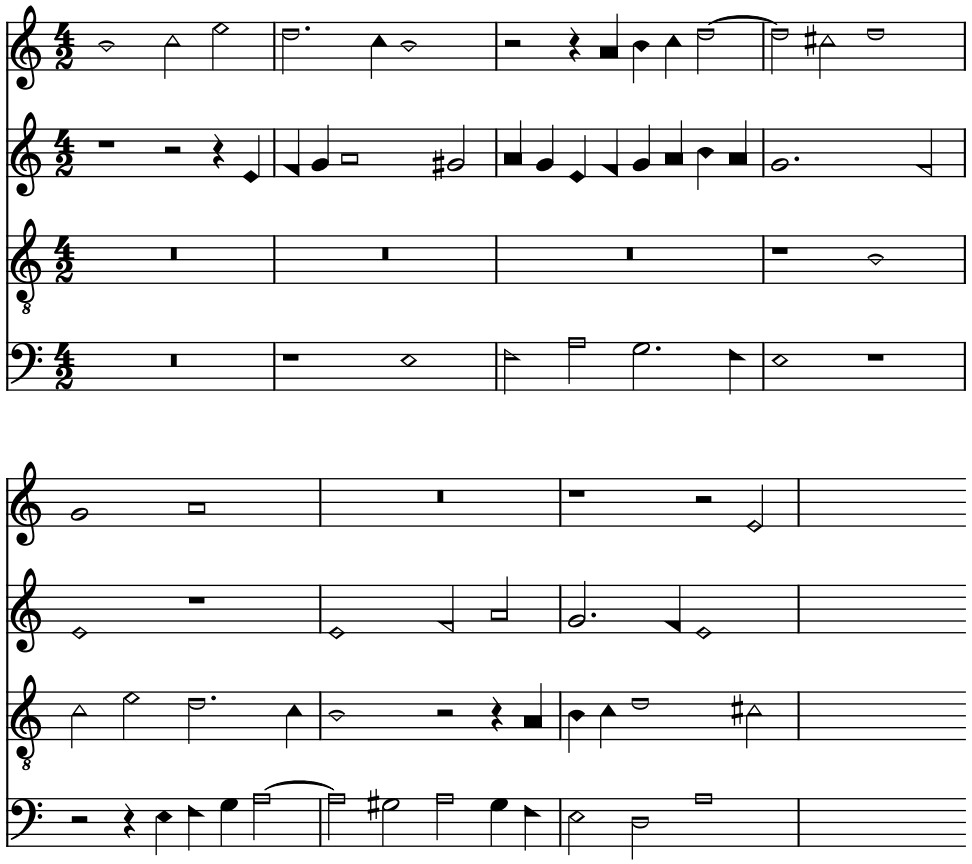
This convertibility of subjects is therefore to be credited to the benefit of double counterpoint, by whose aid, if the subjects are properly constructed, a fugue can be set up and extended at length with little difficulty.

It is now your task, Joseph, to go through the remaining fugues in order of the modes, treating them in the same way. But so that variety may be provided for, the countersubject is not always to be introduced in the first measure of the subject; rather, depending on the nature of the principal subject, the countersubject may enter in the second or third measure, as the following beginning of a fugue in the second mode —left to you to complete—will show.



Hanc itaque Fugam, cæterasque Contrasubjectis proprio ingenio inveniendis elaboratas ad corrigendum mihi feres.

Joseph. Nihil præterea peculiare in hoc Compositionis genere observandum occurrit?

Aloys. Cætera ex dictis de Fugis simplicibus, deque communibus Contrapuncti regulis petenda sunt. Peculiarem autem esse cuilibet Modo & modulandi, & protrahendi rationem, propter varium Semitonii situm, ex sæpius jam dictis tibi constare suppono. Porrò relicto pro nunc hoc Contrapuncto, domi tuæ per reliquos Modos te exercendo, ad Contrapunctum duplex cum translatione in Decimam pergamus.

Bring me this fugue, and the others elaborated with countersubjects invented by your own ingenuity, for correction.

Joseph. Is there nothing else peculiar to observe in this kind of composition?

Aloys. Everything else must be drawn from what has been said about simple fugues and the common rules of counterpoint. But each mode has its own particular manner of modulation and continuation, because of the varying placement of the semitone, as I suppose is already clear to you from what has been said so often. Now, leaving this counterpoint aside for the moment, and exercising yourself at home through the remaining modes, let us proceed to double counterpoint with transposition into the tenth.

EXERCITII V.

LECTIO SEXTA,

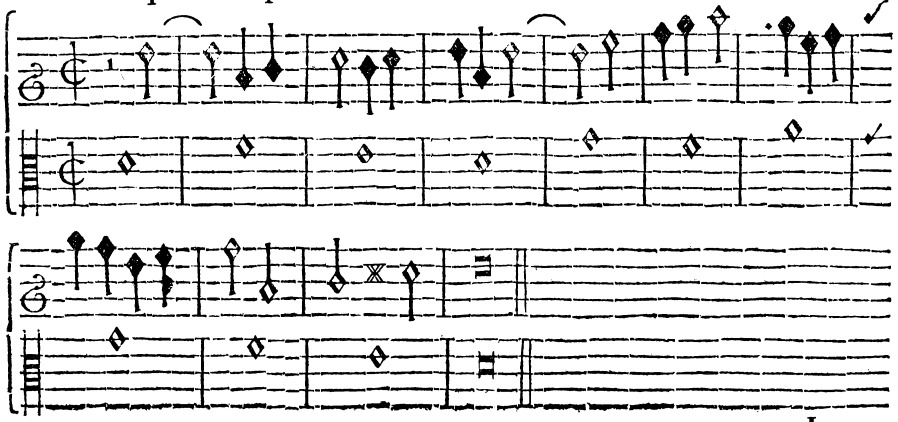
De Contrapuncto Duplici cum Translatione in Decimam.

CONtrapunctum duplex ut ſic, Genus eſſe, proinde complures complecti Species, earumque differentias exoriri pro varia partium translatione ex paulò antè dictis colligendum eſt.

De prima Specie: nempe de Contrapuncto in Octava ſatìs actum puto.

Porrò ad ſecundam Speciem, nempe Contrapunctum in Decima progrediendum eſt; ità dictum, quia omiſſis quibusdam Conſonantiis, & Diſſonantiis pars alterutra in Decimam vel acutam, vel gravem tranſponi poteſt: manentibus iisdem prorsùs figuris. A quibus autem intervallis abſtinendum ſit, ex numeris ſequentibus contra ſe poſitis perſpicuum eſt.

1. 2. 3. 4. 5. 6. 7. 8. 9. 10.
10. 9. 8. 7. 6. 5. 4. 3. 2. 1.

Inde colligitur, duas Tertias, duasque Decimas, per motum rectum ſe invicem ſequi non poſſe: quia ex illis duæ Octavæ: ex iſtis duo Uniſoni per diverſionem reſultarent. Pari modo prohibentur duæ Sextæ, eò quòd earum inverſio in Decimam duas Quintas exhiberet. Porrò abſtinendum eſt à Quarta ligata in parte ſuperiori; quia inverſa efficit Septimam uſu non receptam, ùt alibi jam demonſtratum eſt. Poſtremò non excedatur Decima. Vide ſis exemplum, Cantu firmo hucuſque uſurpato obſtrictum.

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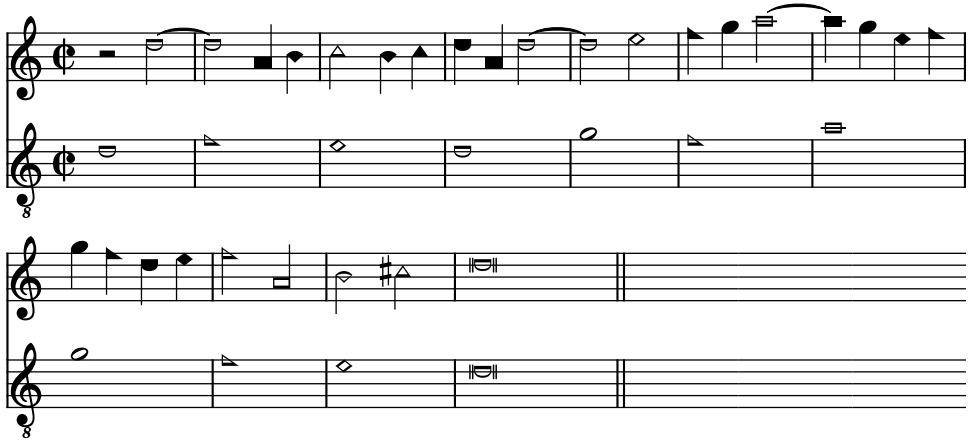
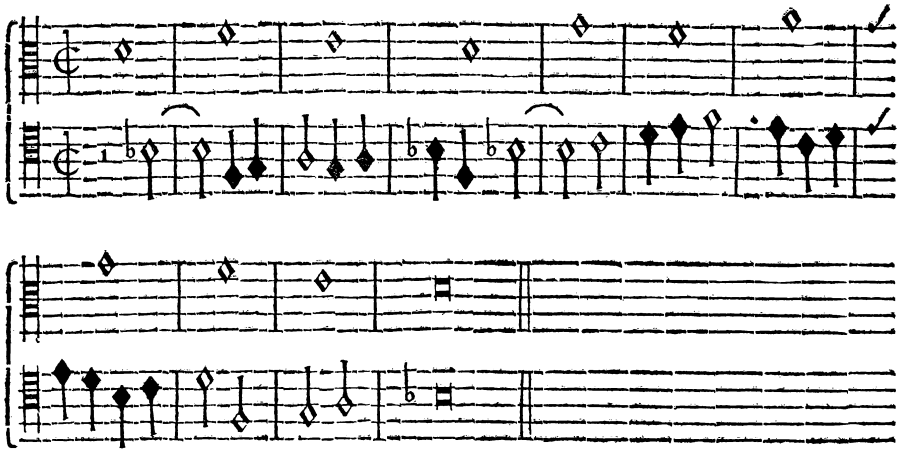
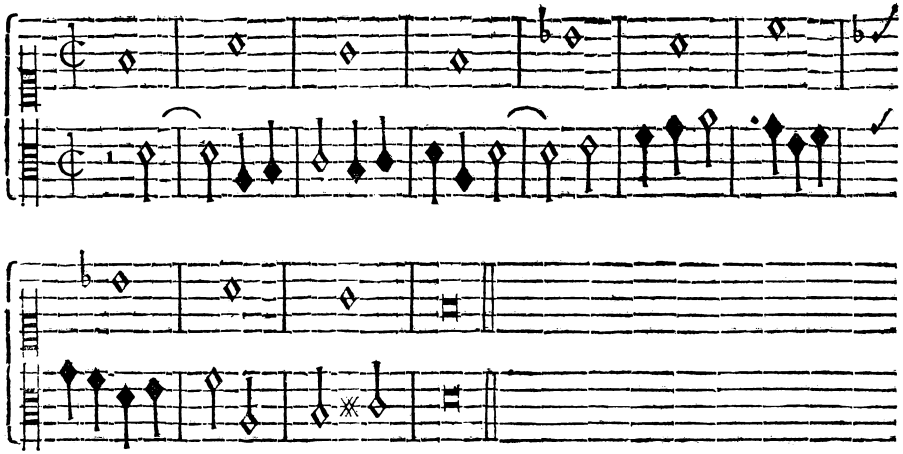
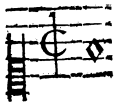
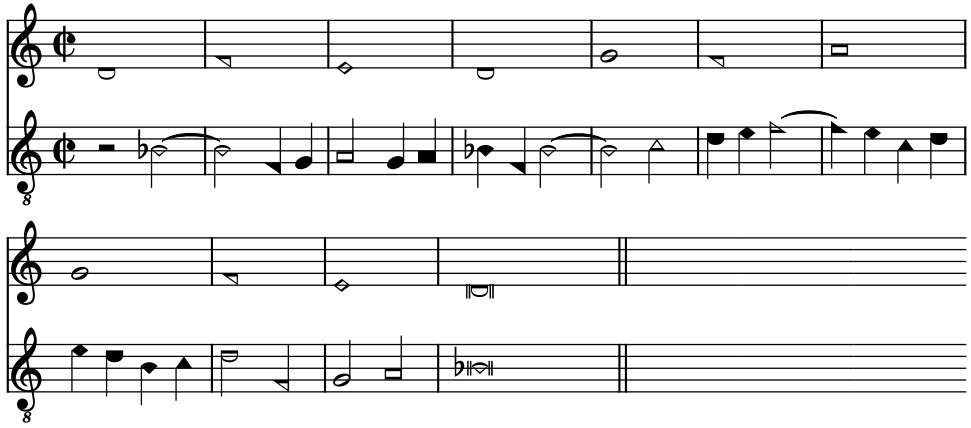
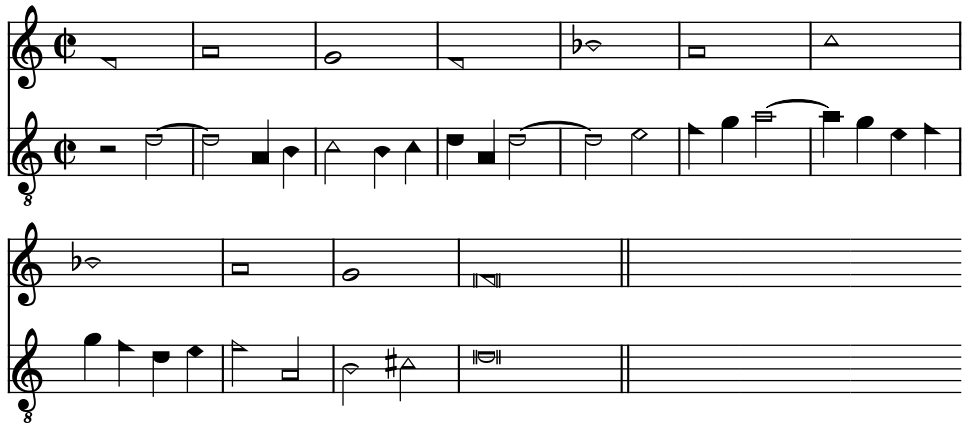
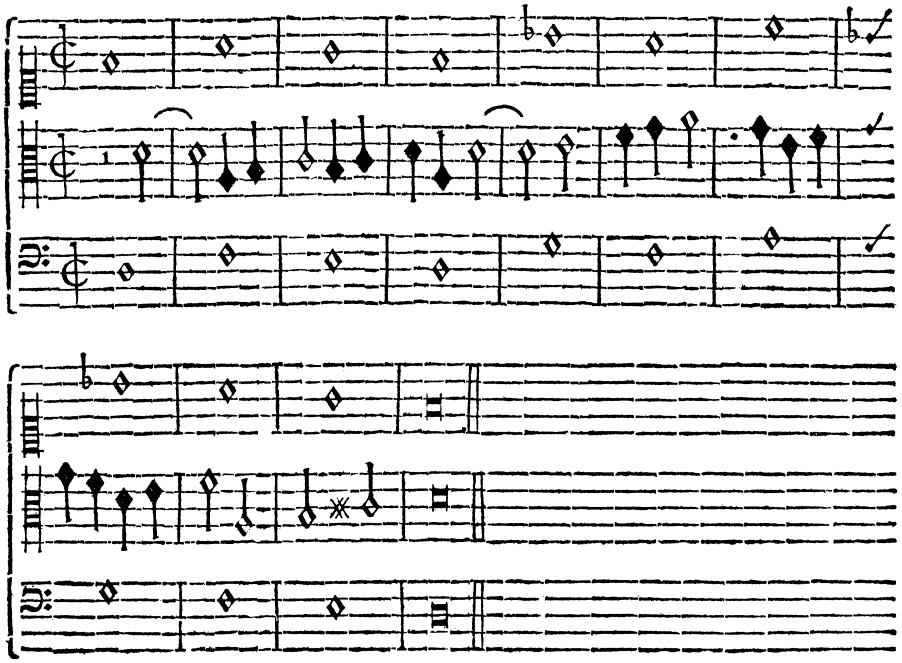
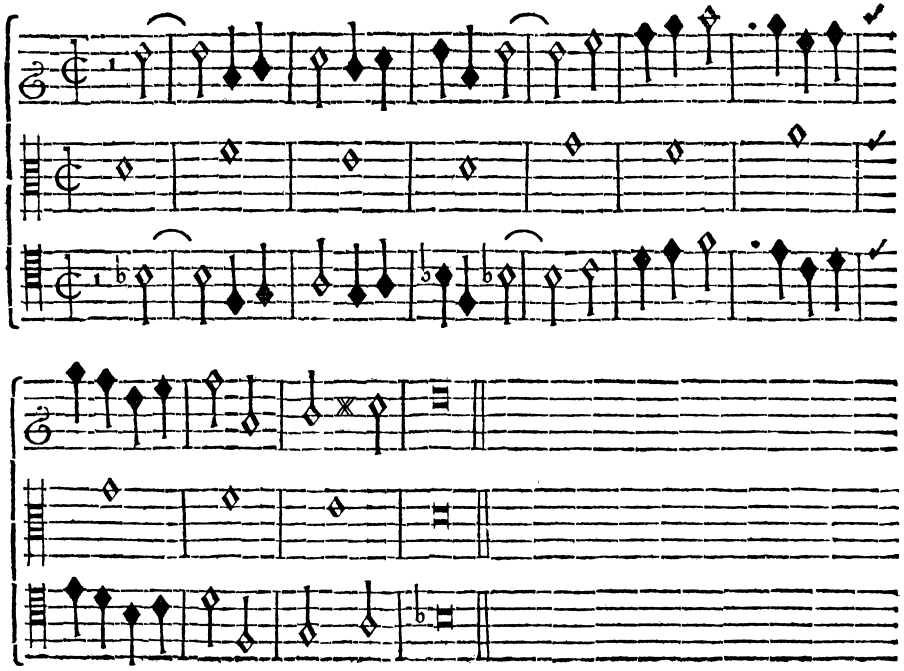
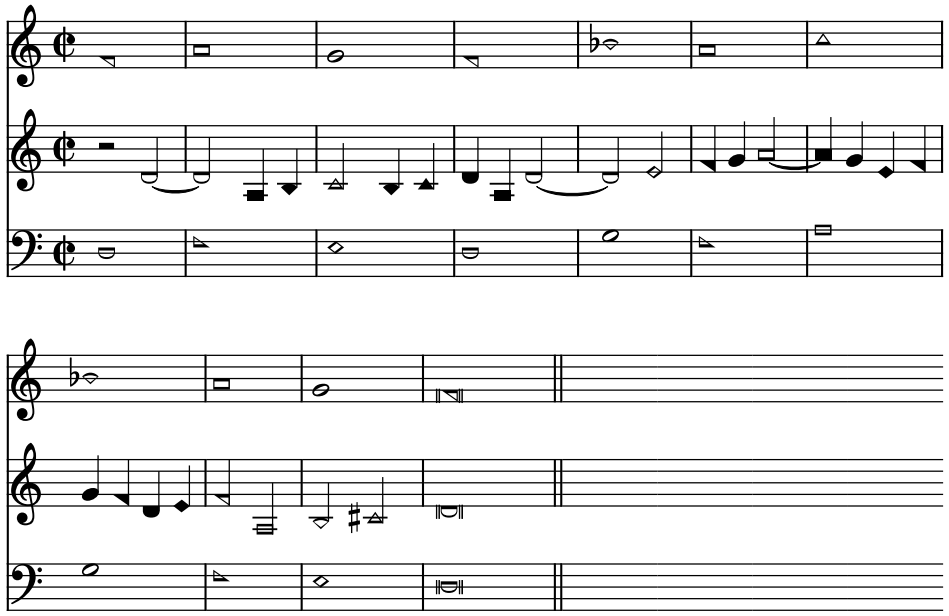
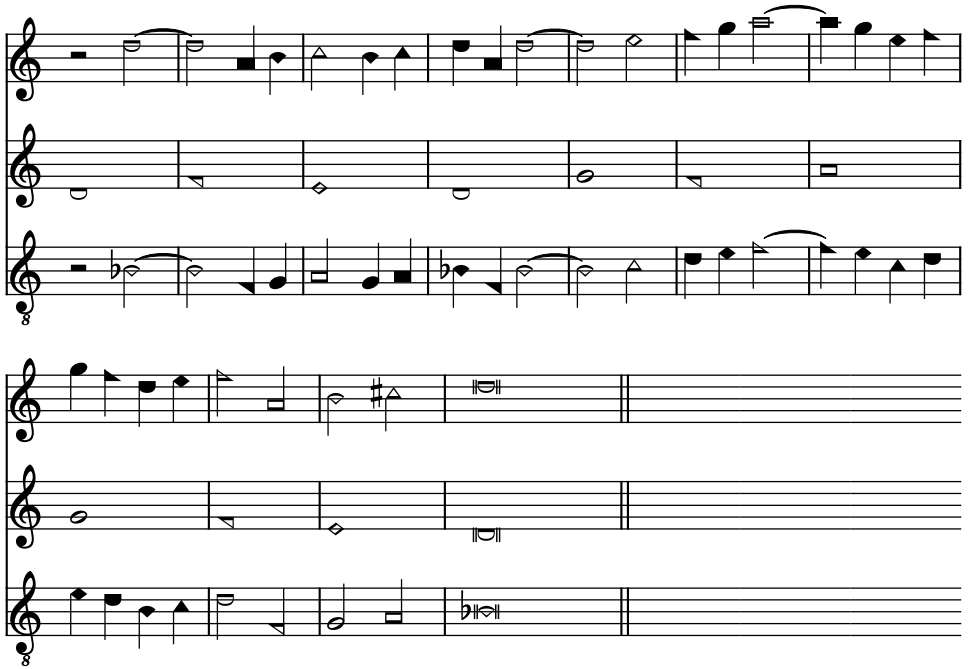
EXERCISE V.
LESSON SIX,
On Double Counterpoint with Transposition into the Tenth.

Since double counterpoint is a genus (class) and therefore includes several species (types), their differences arise from the various transpositions of the parts, as may be gathered from what was said a little earlier.

Concerning the first species, namely double counterpoint in the octave, enough has been said.

Now we must proceed to the second species, namely double counterpoint in the tenth, so called because, with certain consonances and dissonances omitted, either part can be transposed into the tenth—either above or below—while the note-values remain entirely the same. From which intervals one must abstain is clear from the following numbers placed opposite each other:

1 2 3 4 5 6 7 8 9 10
10 9 8 7 6 5 4 3 2 1

From this it is gathered that two thirds and two tenths cannot follow one another in direct motion, because from the former two octaves would result, and from the latter two unisons. Likewise two sixths are forbidden, because their inversion into the tenth would produce two fifths. Moreover, one must abstain from a tied fourth in the upper part, because when inverted it produces a seventh not accepted in practice, as has already been shown elsewhere. Finally, the tenth must not be exceeded. See this example, bound by the cantus firmus used up to now:

Inverſio in Decimam, manente eodem Cantu firmo in loco ſuo.

Idem uſuvenit, ſi Cantus firmus in una Tertia exaltetur, Contrapuncto in Octavam gravem translato. Ratio perſpicua eſt: quia duo toni ad Octavam additi æquè Decimam efficiunt.

Hoc idem Contrapunctum à tribus vocibus decantari poteſt, ſi Cantus firmus de Nota ad Notam deſcriptus in Decimam inferiùs ponatur, manentibus duabus partibus.

Inversion into the tenth, with the cantus firmus remaining in its place.

The same happens if the cantus firmus is raised by a third, with the counterpoint transposed into the lower octave. The reason is clear: two tones added to the octave make the tenth.

This same counterpoint can be sung by three voices if the cantus firmus, written note-against-note, is placed a tenth lower, with the other two parts remaining.

Quod pari modo intelligendum eſt de primo hujus Contrapuncti exemplo, tranſcribendo Contrapunctum in Decimam, inferiùs manentibus reliquis partibus.

The same is to be understood of the first example of this counterpoint, by transcribing the counterpoint a tenth lower, with the remaining parts unchanged.